



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

4025/1

PAPER 1 Multiple Choice

NOVEMBER 2021 SESSION

1 hour

Additional materials:

- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C and D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

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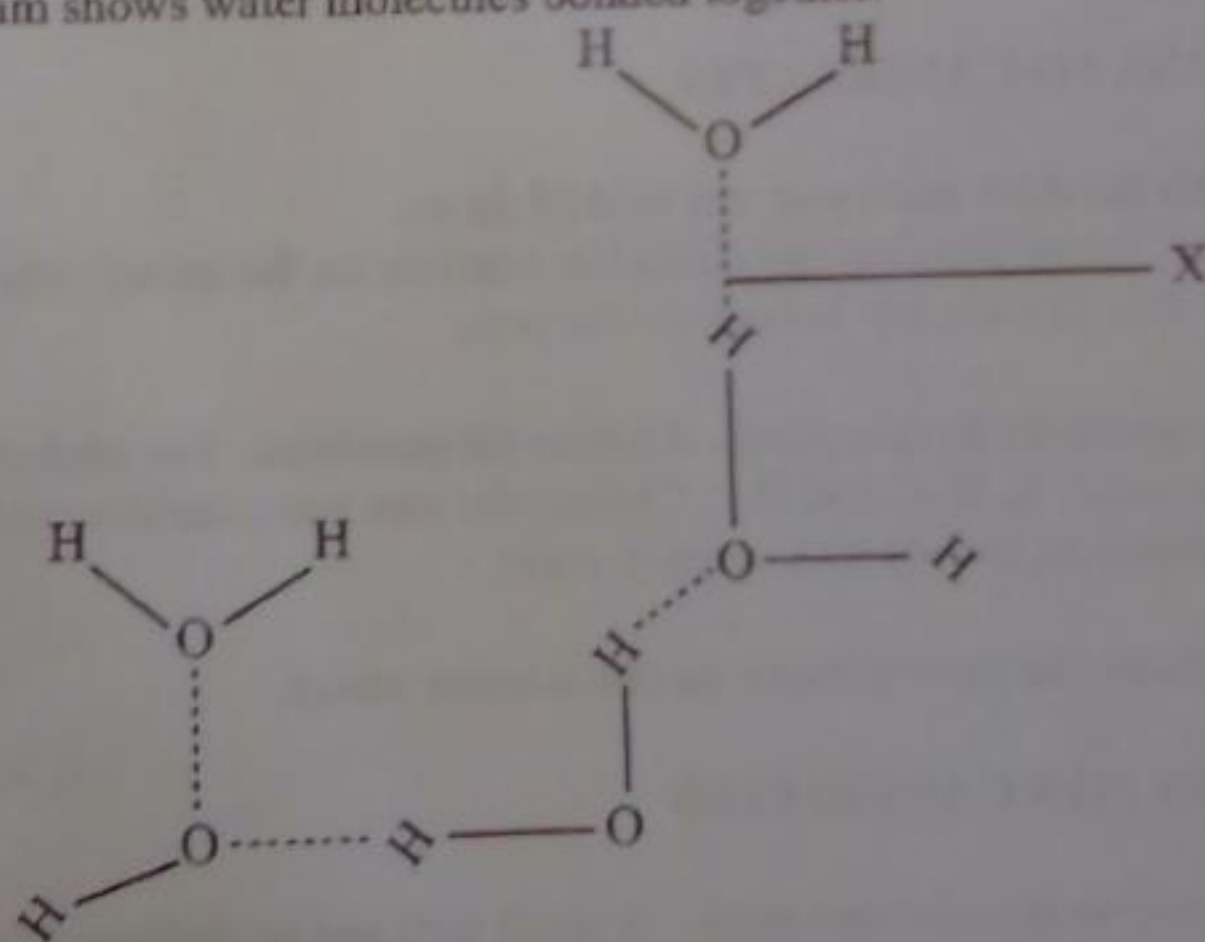
INFORMATION FOR CANDIDATES

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This question paper consists of 13 printed pages and 3 blank page.

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- 1 Cytology is the study of
- A cell structure and function.
 - B molecular structure and function.
 - C animal structure and function.
 - D plant structure and function.
- 2 What is used for protection from fumes in the laboratory?
- A goggles
 - B gloves
 - C respirator
 - D helmet
- 3 The diagram shows water molecules bonded together.



The bond X contributes to the

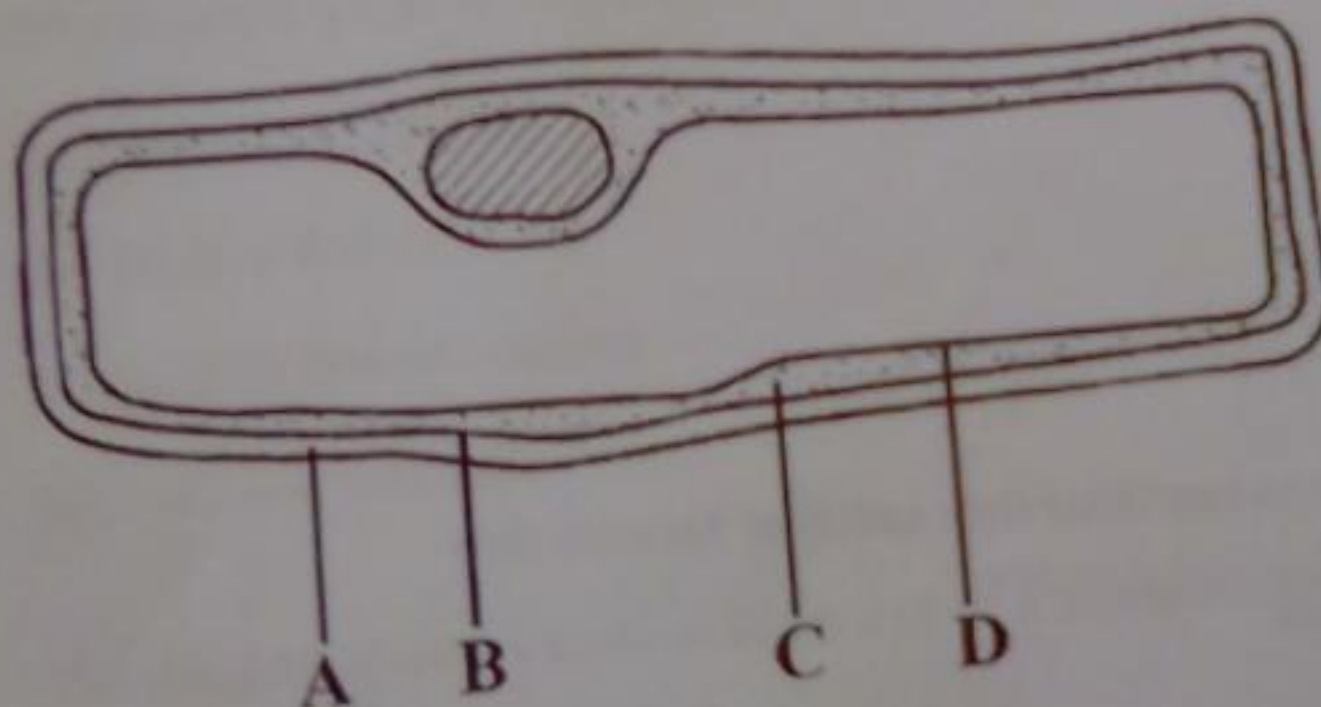
- A polarity of water.
- B solvent property of water.
- C reagent property of water.
- D heat capacity of water.

4 Which one is **not** a polysaccharide?

- A amylose
- B amylopectin
- C glycogen
- D sucrose

5 The diagram shows a plant cell.

Which part, A, B, C or D, is the site for anaerobic respiration?



6 Which organelle is for the production of energy in living cells?

- A nucleus
- B cytoplasm
- C vacuole
- D mitochondrion

- 7 Which specialised animal cell **does not** have a nucleus?
- A red blood cell
 - B white blood cell
 - C neurone
 - D ovum
- 8 A mixture of protease and albumin solution was placed in a Visking tubing. The Visking tubing was then placed into a water bath maintained at 37°C . Which molecules would be found in the Visking tubing after 20 minutes?
- A fatty acids
 - B amino acids
 - C glycerol
 - D glucose
- 9 Enzymes are specific to reactions they catalyse because they
- A are sensitive to pH.
 - B are protein in nature.
 - C have precisely shaped active sites.
 - D lower the activation energy of the reaction.
- 10 Which combination of enzymes, in a biological washing powder, is suitable for removing a beef stew stain?
- A amylase and protease
 - B amylase and lipase
 - C maltase and sucrase
 - D protease and lipase

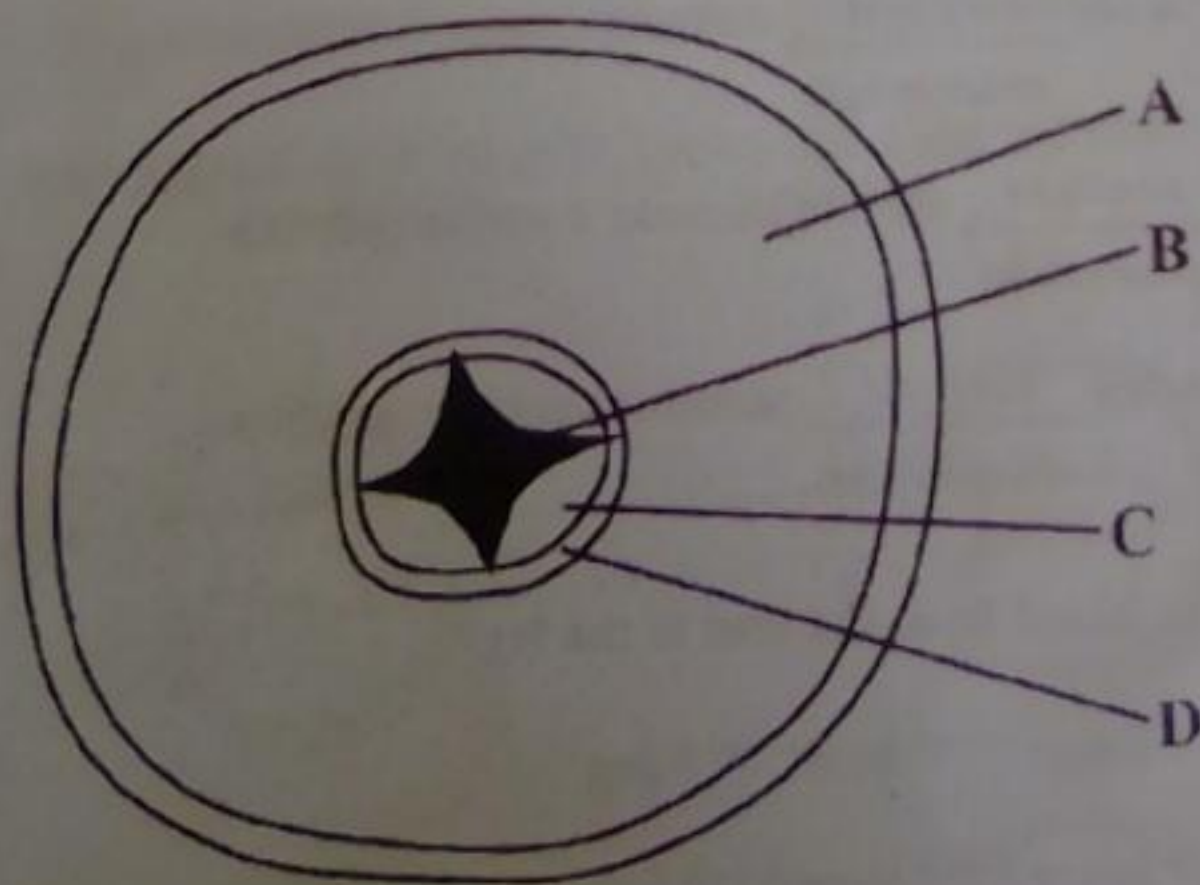
- 11 A tobacco plant shows browning of shoots and produces a slimy ooze from a cut stem.

This is due to the effect of

- A tissue eating pests.
 - B sap sucking pests.
 - C bacterial wilt disease.
 - D fungal rust disease.
- 12 Which one is a role of ethylene in plants?
- A promotes flowering
 - B inhibits fruit ripening
 - C inhibits flower opening
 - D promotes leaf fall

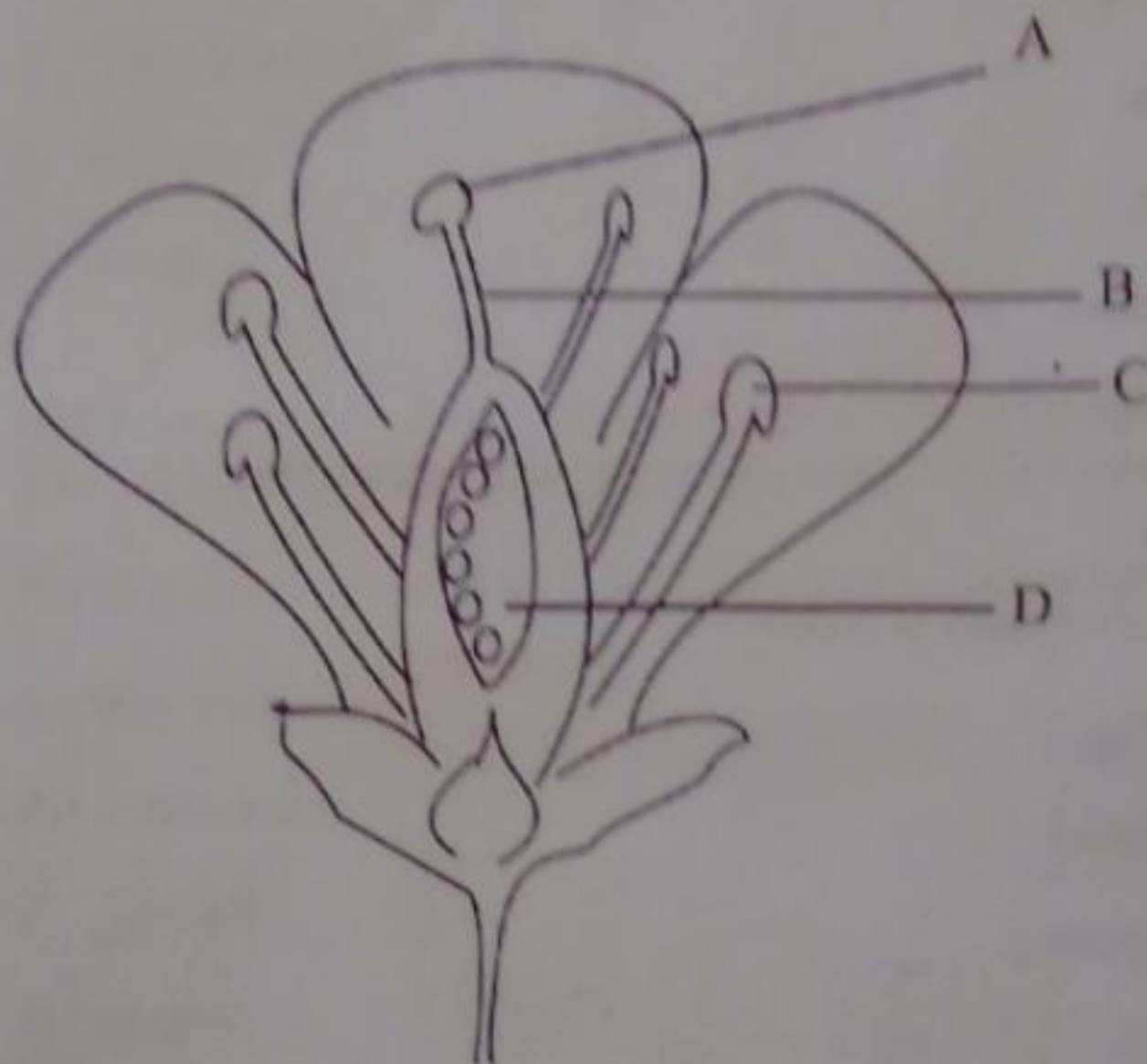
- 13 The diagram shows a transverse section of a dicotyledonous root.

Which tissue, A, B, C or D, is the phloem?



- 14 The diagram shows the structure of an insect pollinated flower.

Which part, A, B, C or D, is the site of fertilisation?



- 15 Which word equation represents the process of photosynthesis?

- A carbohydrates + oxygen $\xrightarrow[\text{chlorophyll}]{\text{sunlight}}$ carbon dioxide + water
- B carbon dioxide + water $\xrightarrow[\text{chlorophyll}]{\text{sunlight}}$ carbohydrates + oxygen
- C water + oxygen $\xrightarrow[\text{chlorophyll}]{\text{sunlight}}$ carbon dioxide + carbohydrates
- D carbohydrates + water $\xrightarrow[\text{chlorophyll}]{\text{sunlight}}$ carbon dioxide + oxygen

- 16 Which route is taken by blood from the arm to the leg?

- A arm $\rightarrow$ heart $\rightarrow$ lungs $\rightarrow$ heart $\rightarrow$ leg
- B arm $\rightarrow$ heart $\rightarrow$ lungs $\rightarrow$ liver $\rightarrow$ leg
- C arm $\rightarrow$ lungs $\rightarrow$ heart $\rightarrow$ lungs $\rightarrow$ leg
- D arm $\rightarrow$ liver $\rightarrow$ heart $\rightarrow$ lungs $\rightarrow$ leg

17 Which part of the brain detects blood temperature changes?

- A hypothalamus
- B medulla
- C cerebrum
- D cerebellum

18 A patient has these signs and symptoms:

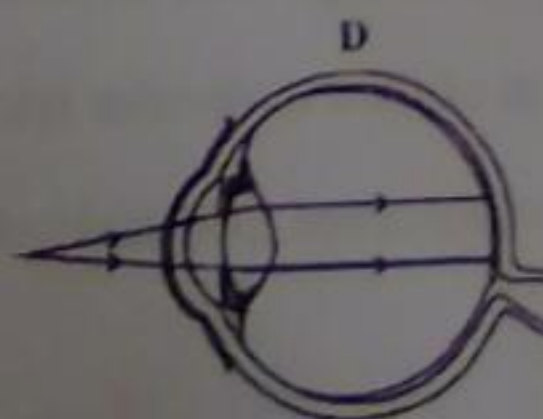
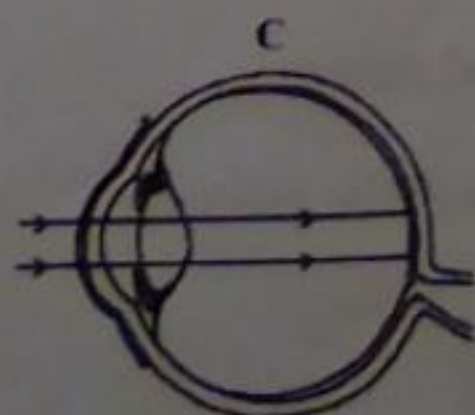
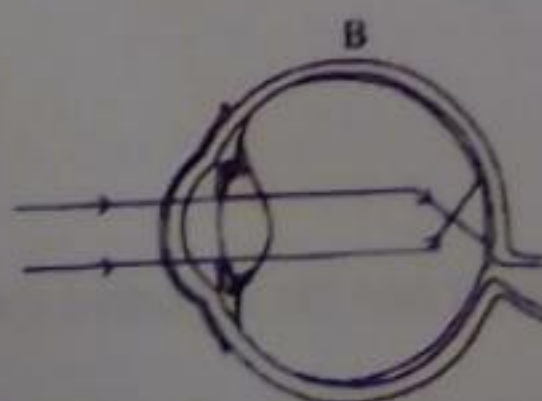
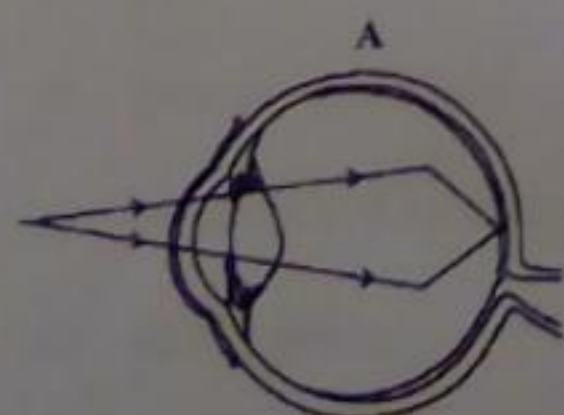
reduced elasticity of air sacs
shortness of breath
a wheezing when breathing

The patient has

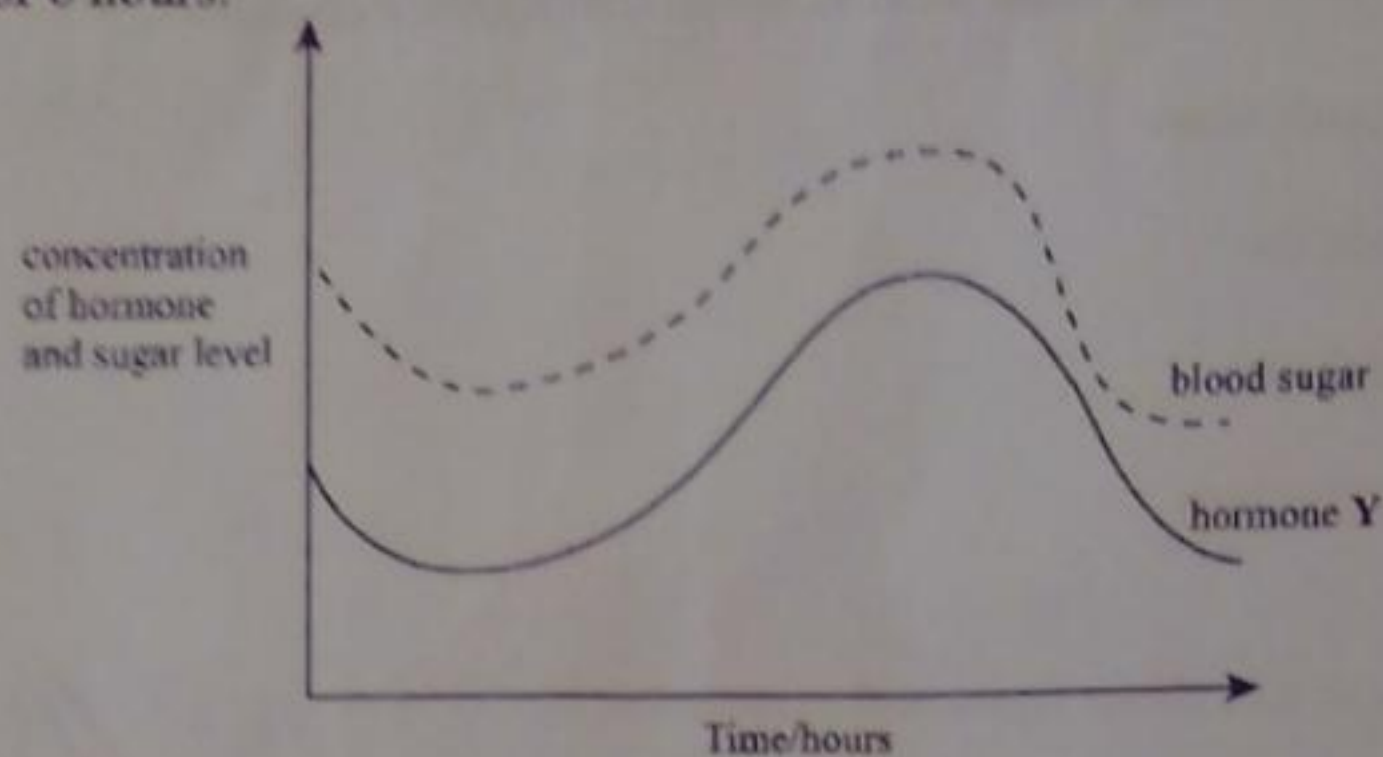
- A tuberculosis.
- B emphysema.
- C bronchitis.
- D asthma.

19 The diagrams show the accommodation of light in the eye.

Which diagram, A, B, C or D, shows short sightedness?



- 20 The graphs show changes in concentration of blood sugar and hormone Y over a period of 6 hours.



Hormone Y is

- A thyroxine.
 - B adrenaline.
 - C insulin.
 - D glucagon.
- 21 A yellow discharge and pain when urinating are symptoms of
- A syphilis.
 - B AIDS.
 - C chancroid.
 - D gonorrhoea.
- 22 Assimilation is
- A taking in of food into the digestive system.
 - B the break down of large food molecules.
 - C the absorption of food molecules into the blood stream.
 - D making use of absorbed food molecules in metabolic processes.

- 23 The sternum protects the
- A brain.
 - B heart.
 - C liver.
 - D kidneys.
- 24 Which one is a hormonal birth control method?
- A pill
 - B IUD
 - C rhythm
 - D vasectomy
- 25 The movement of food along the alimentary canal is by
- A assimilation.
 - B absorption.
 - C peristalsis.
 - D swallowing.
- 26 Viruses are **not** considered cells because they lack
- A protein molecules.
 - B genetic materials.
 - C metabolic processes.
 - D chitin walls.

- 27 Human manipulation of genes outside the reproductive process is genetic
- A selection.
 - B profiling.
 - C mutation.
 - D engineering.
- 28 Which one **does not** contribute to an increase in the rate of mutation?
- A tar
 - B x-rays
 - C gamma rays
 - D carbon monoxide
- 29 Which characteristic is controlled by genes **only**?
- A intelligence
 - B complexion
 - C earlobes
 - D mass
- 30 The phenotype of an organism is its
- A genetic make up.
 - B physical appearance.
 - C combination of alleles.
 - D number of chromosomes.

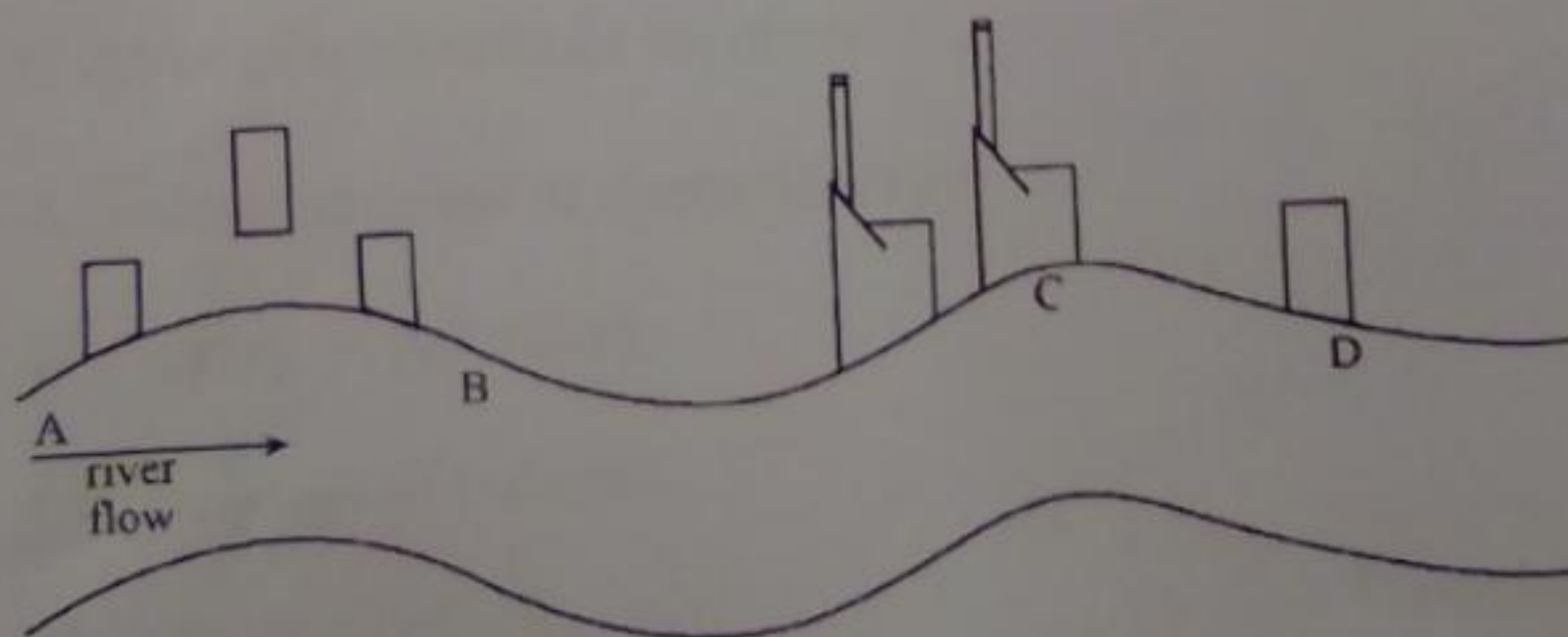
- 31 The **expected** genotypic ratio from a cross involving heterozygous plants is
- A 1:1
 - B 1:2:1
 - C 1:2
 - D 3:1

- 32 Which blood group is a universal donor of blood?

- A A
- B B
- C AB
- D O

- 33 The diagram shows human settlements located along a river.

At which point, A, B, C or D, is biodiversity lowest?



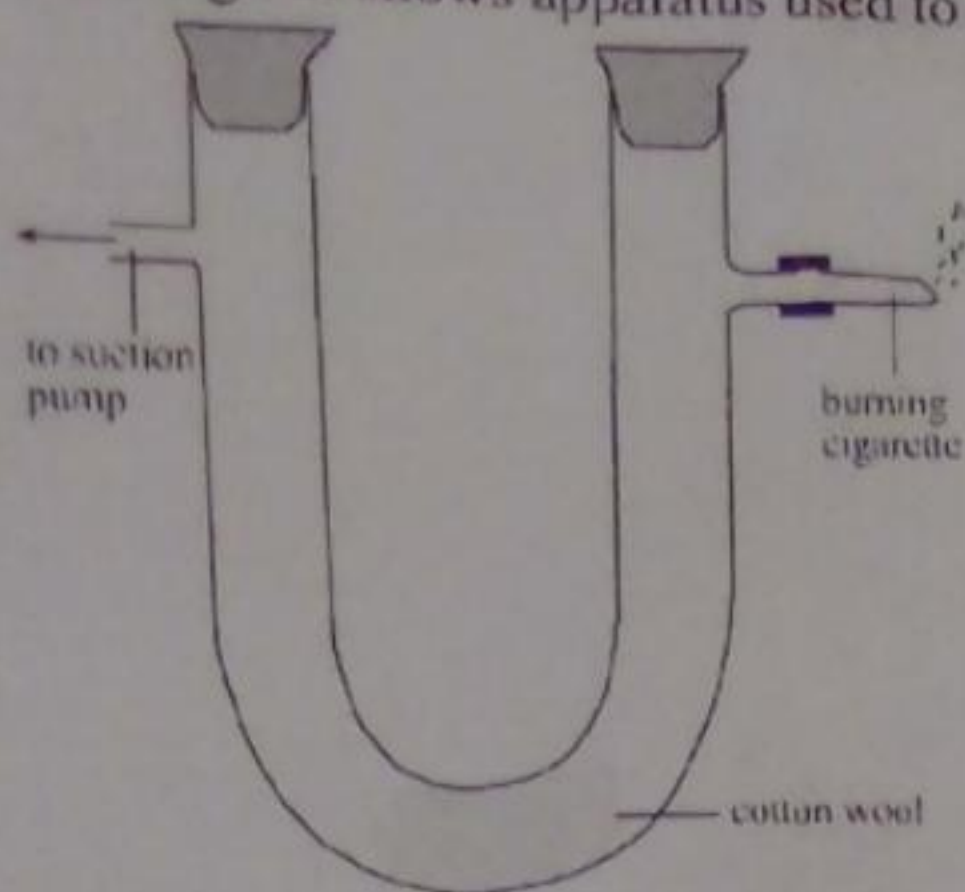
- 34 The similarity between a plant and fungus is that they both have

- A cell walls.
- B chlorophyll.
- C hyphae.
- D sporangia.

- 35 The role of nitrogen fixing bacteria is to convert
- A ammonia to nitrogen.
 - B nitrogen to nitrates.
 - C nitrites to ammonia.
 - D nitrites to nitrogen.
- 36 Which chain has the lowest energy transfer efficiency?
- A grass $\longrightarrow$ goat $\longrightarrow$ hyena
 - B wheat $\longrightarrow$ sparrow $\longrightarrow$ hawk $\longrightarrow$ vulture
 - C pawpaw $\longrightarrow$ baboon $\longrightarrow$ leopard $\longrightarrow$ bacteria
 - D maize grain $\longrightarrow$ weevil $\longrightarrow$ wasp $\longrightarrow$ small bird $\longrightarrow$ eagle
- 37 Which human activity least contributes to acid rain?
- A mining
 - B tourism
 - C farming
 - D industry
- 38 Which drug is an analgesic?
- A penicillin
 - B paracetamol
 - C chloroquine
 - D amoxicillin

39

The diagram shows apparatus used to collect substances in cigarette smoke.



After some time the cotton wool turns brown.
Which substance causes the cotton wool to turn brown?

- A carbon dioxide
 - B carbon monoxide
 - C nicotine
 - D tar
- 40 Control of tuberculosis involves
- A administration of chemotherapy.
 - B taking of paludrine.
 - C isolation of patients.
 - D use of repellents.



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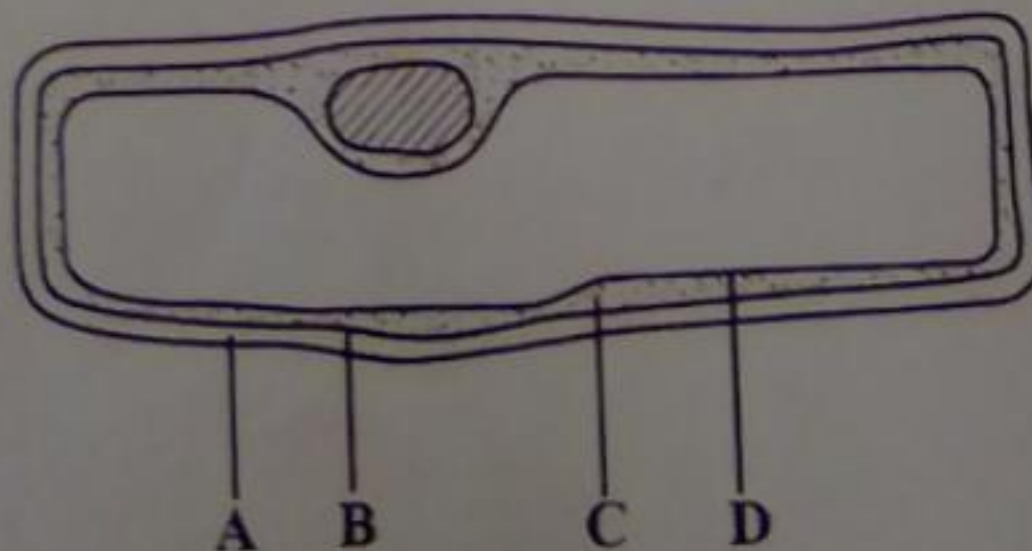
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This question paper consists of 10 printed pages and 2 blank pages.

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1. Productivity in farming can be improved by
 - A cytology.
 - B ecology.
 - C biotechnology.
 - D microbiology.
2. Cytology is a branch of Biology which deals with
 - A modification of genes.
 - B cell structure and function.
 - C classification of animals.
 - D parts of the body and their function.
3. Ice protects aquatic life by
 - A allowing light to pass through.
 - B acting as a conductor of heat.
 - C insulating the water below.
 - D keeping predators out.
4. The sub units of genetic molecules are
 - A amino acids.
 - B fatty acids.
 - C glucose.
 - D nucleotides.
5. The part of a eukaryotic cell that contains DNA is the
 - A cytoplasm.
 - B membrane.
 - C nucleus.
 - D vacuole.
6. The diagram shows a plant cell.

Which part, A, B, C or D, controls the turgidity of the cell?

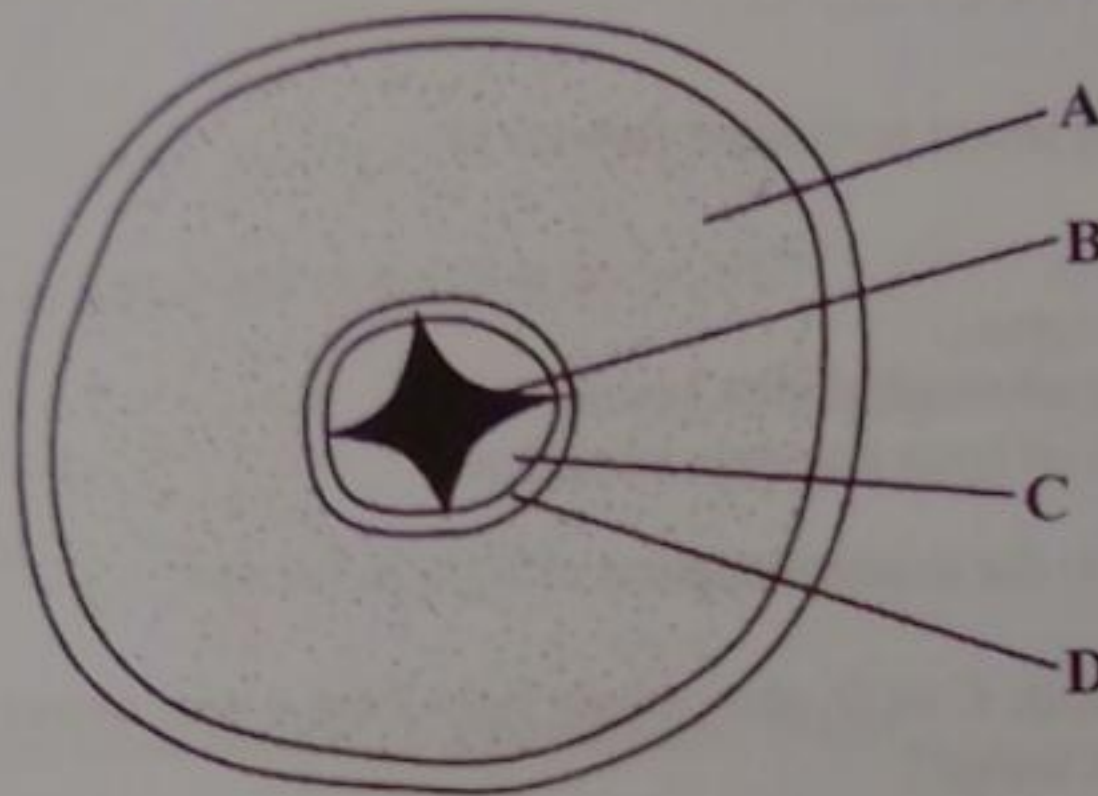


7. The sub units of sucrose are

- A galactose and fructose.
- B galactose and glucose.
- C glucose and ribose.
- D glucose and fructose.

8. The diagram shows a transverse section of a dicotyledonous root.

Which tissue, A, B, C or D, is the cortex?



9. In plants, carbohydrates are translocated in the form of

- A maltose.
- B glucose.
- C fructose.
- D sucrose.

10. Different sized pustules on plant organs are caused by

- A bacterial wilt.
- B fungal rust.
- C sap-sucking pests.
- D tissue-eating pests.

11. Which mineral element is deficient in plants showing poor flowering and fruiting?

- A magnesium
- B nitrogen
- C phosphorus
- D potassium

12. Which pair of factors reduces the rate of water loss in plants?

- A low humidity and high wind speed
- B high humidity and low temperature
- C high temperature and high wind speed ✗
- D low humidity and high light intensity ✗

13. Which is an artificial method of vegetative propagation?

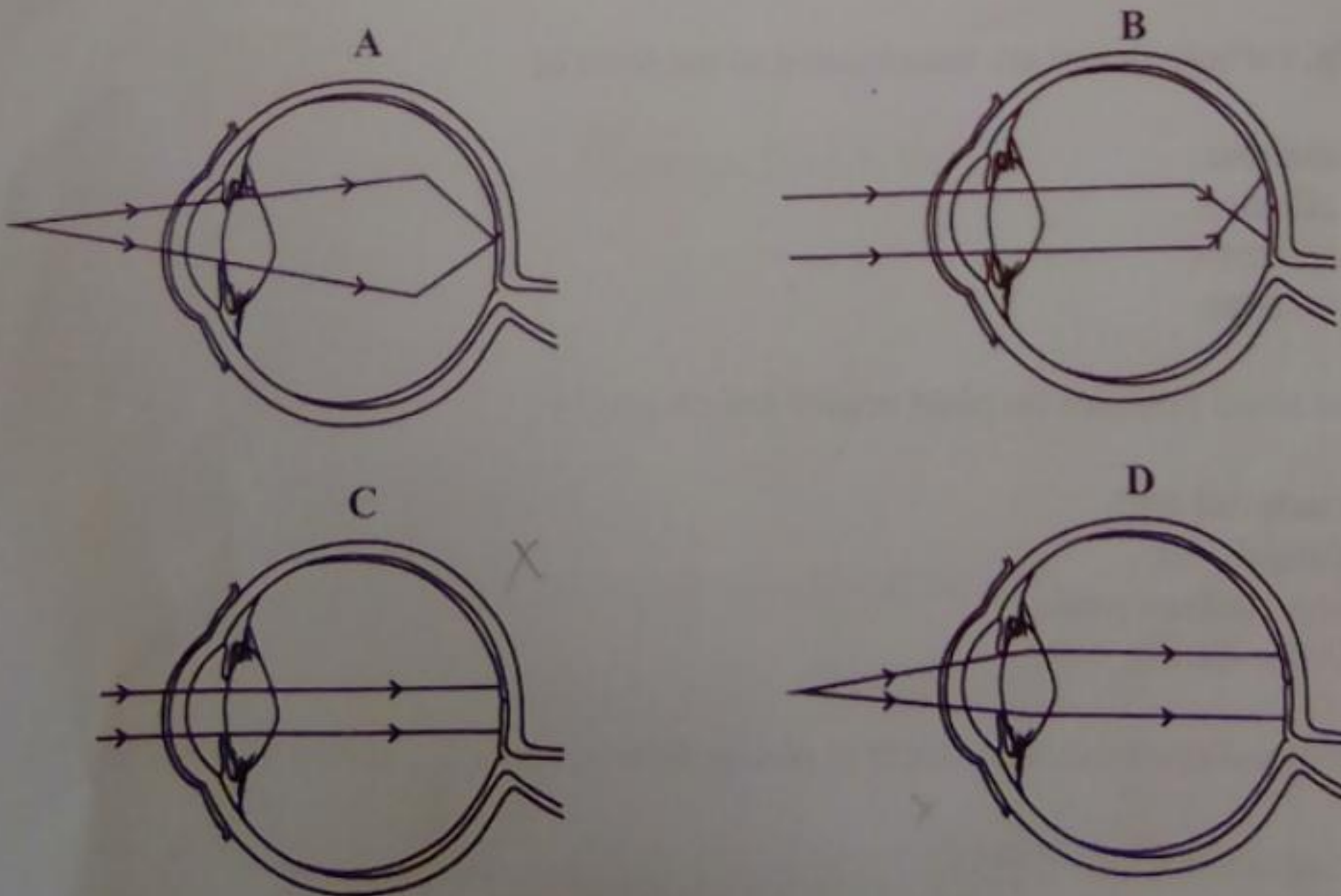
- A cuttings
- B rhizomes
- C suckers
- D tubers

14. A characteristic of insect pollinated flowers is

- A dull petals. ✗
- B feathery stigma. ✗
- C production of rough pollen grains.
- D production of large amounts of pollen. ✗

15. The diagrams show the accommodation of light in the eye.

Which diagram, A, B, C or D, shows how light from a near object falls on the back of the eye to form a clear image?



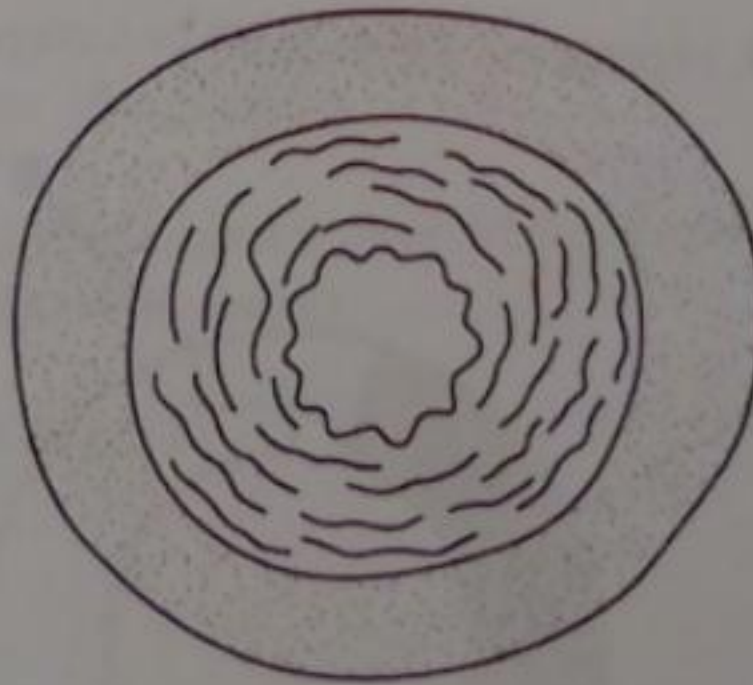
16. During inhalation,

- A chest pressure increases.
- B chest volume increases.
- C the rib cage is lowered. ✗
- D the diaphragm relaxes. ✗

17. Which animal parasite is controlled by dipping?
- A tape worm
 - B round worm
 - C tick
 - D fluke
18. Which hormone is released when an animal is frightened?
- A thyroxine
 - B insulin
 - C glucagon
 - D adrenaline

19. Which one is the site of anaerobic respiration in the cell?
- A chloroplast
 - B cytoplasm
 - ☒ C mitochondrion
 - ☒ D nucleus

20. The diagram shows the cross section of a mammalian vessel.



The vessel is

- ☒ A a capillary.
 - B an artery.
 - C a vein.
 - ☒ D a lymph vessel.
21. What is the action of the extensor and flexor muscles when stretching the arm?
- A extensor contracts and flexor contracts
 - B extensor relaxes and flexor relaxes
 - C extensor contracts and flexor relaxes
 - D extensor relaxes and flexor contracts

22. Which two bones form a ball and socket joint?

- A femur and scapula
- B humerus and scapula
- C pelvis and humerus
- D pelvis and tibia

23. Which bone is part of the hind limb of a mammal?

- A femur
- B humerus
- C radius
- D scapula

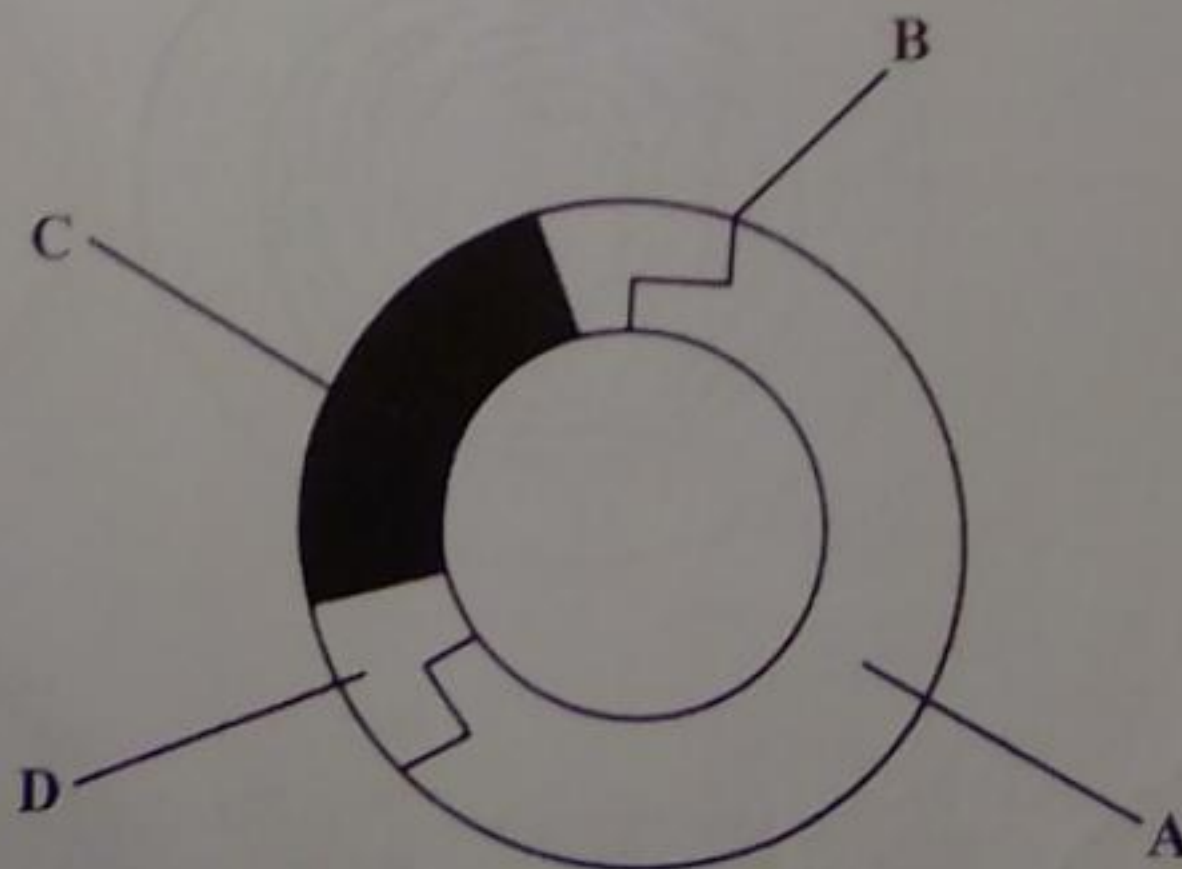
24. Immunity can be acquired by transmission of antibodies from mother to foetus.

This type of immunity is

- A natural active.
- B natural passive.
- C artificial active.
- D artificial passive.

25. The diagram shows recombinant DNA.

Which part, A, B, C or D, represents the vector?



26. Bacteria is used in the production of

- A alcohol.
- B bread.
- C penicillin.
- D yoghurt.

22. Which two bones form a ball and socket joint?

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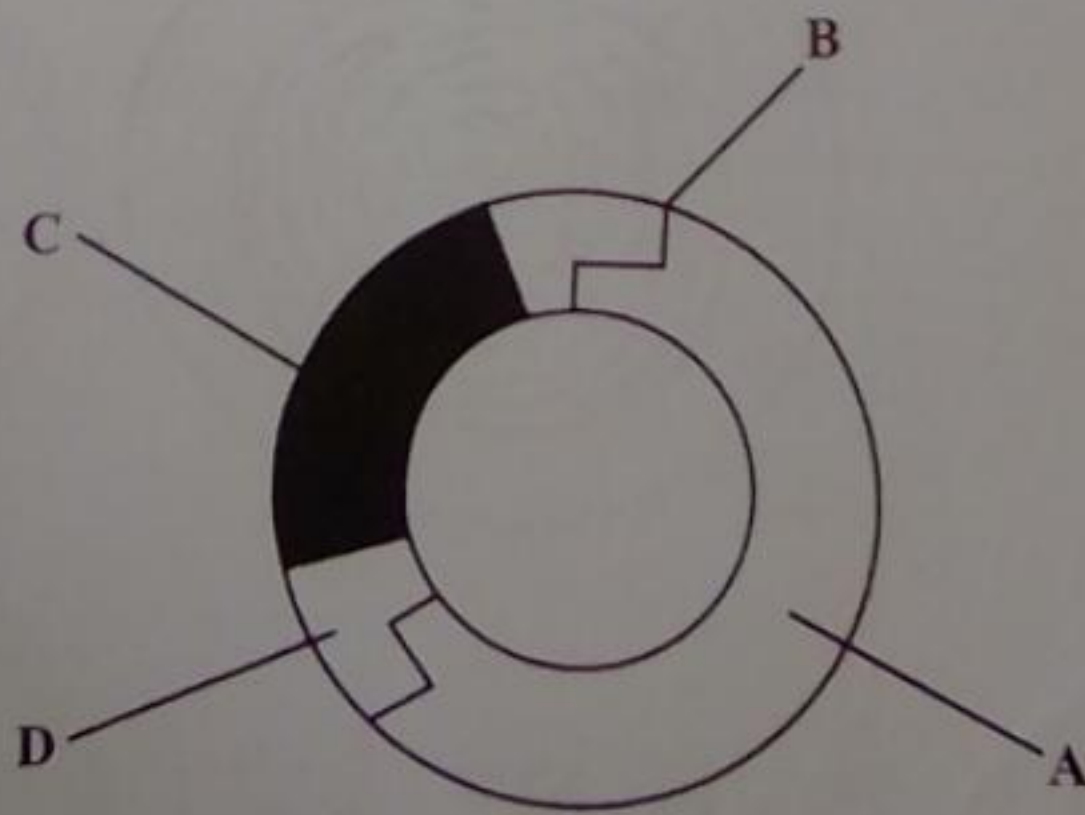
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27. Viruses can be classified as living organisms because they
- A excrete.
 - B feed.
 - C reproduce.
 - D respire.
28. The full set of genes in an organism is its
- A genotype.
 - B genome.
 - C phenotype.
 - D alleles.
29. Which condition is a result of chromosomal mutation?
- A albinism
 - B Down's syndrome
 - C haemophilia
 - D sickle cell anaemia
30. An allele is
- A a section of DNA coding for a single characteristic.
 - B the set of genes in the nucleus of an individual.
 - C the external appearance of an organism.
 - D the alternative form of a gene.
31. In certain plants, red petals is dominant over white petals. When two heterozygous plants were crossed, 85 plants had white petals.
- How many plants had red petals?
- A 21
 - B 64
 - C 85
 - D 255
32. The allele which influences the appearance of the phenotype even in the presence of an alternative allele is
- A dominant.
 - B heterozygous.
 - C homozygous.
 - D recessive.

33. Which one is a balanced ecosystem?

- A orchard
- B forest
- C garden
- D pond

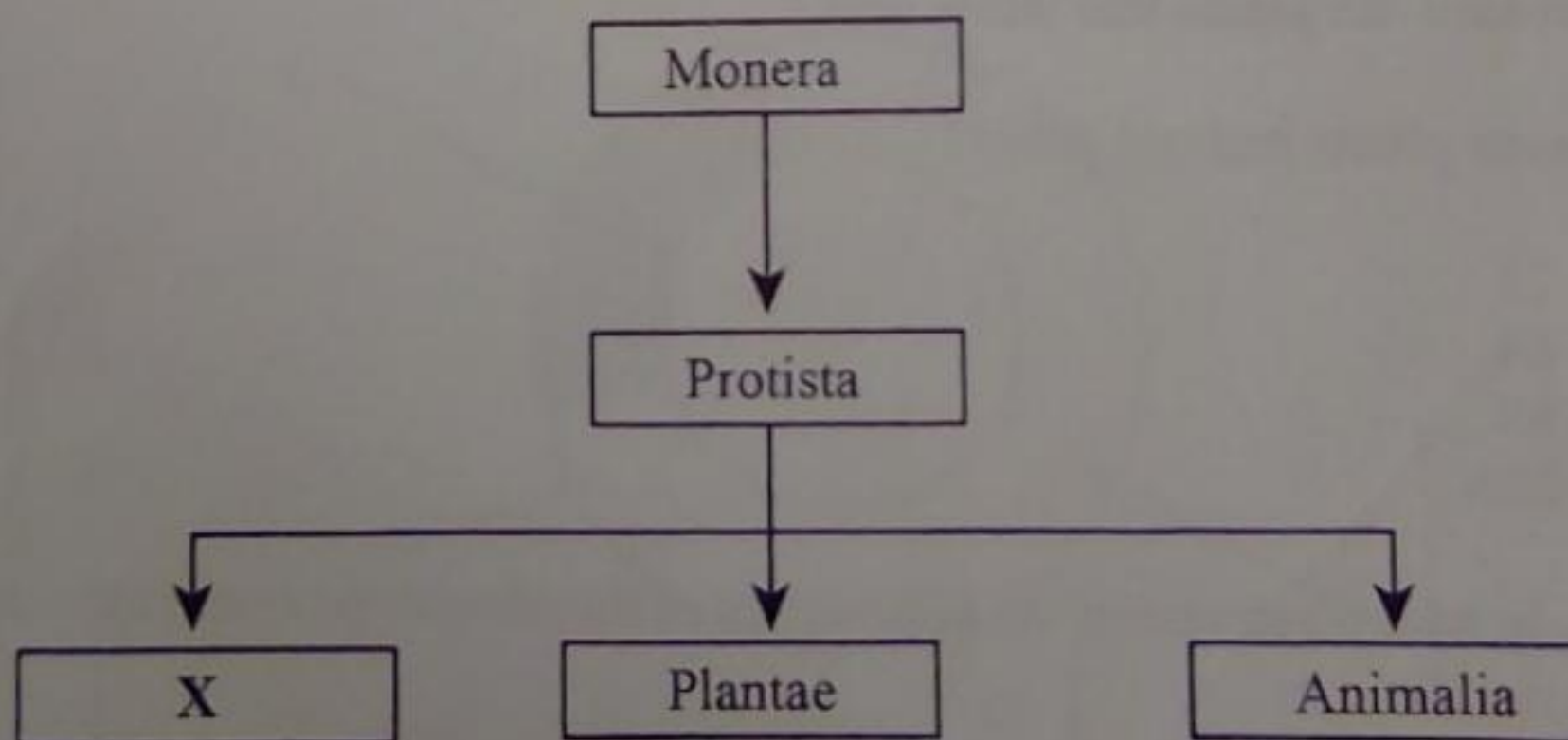
34. The diagram shows an organism.



The organism belongs to kingdom

- A animalia.
- B fungi
- C plantae.
- D protista.

35. The diagram shows the ancestry of the five kingdoms.



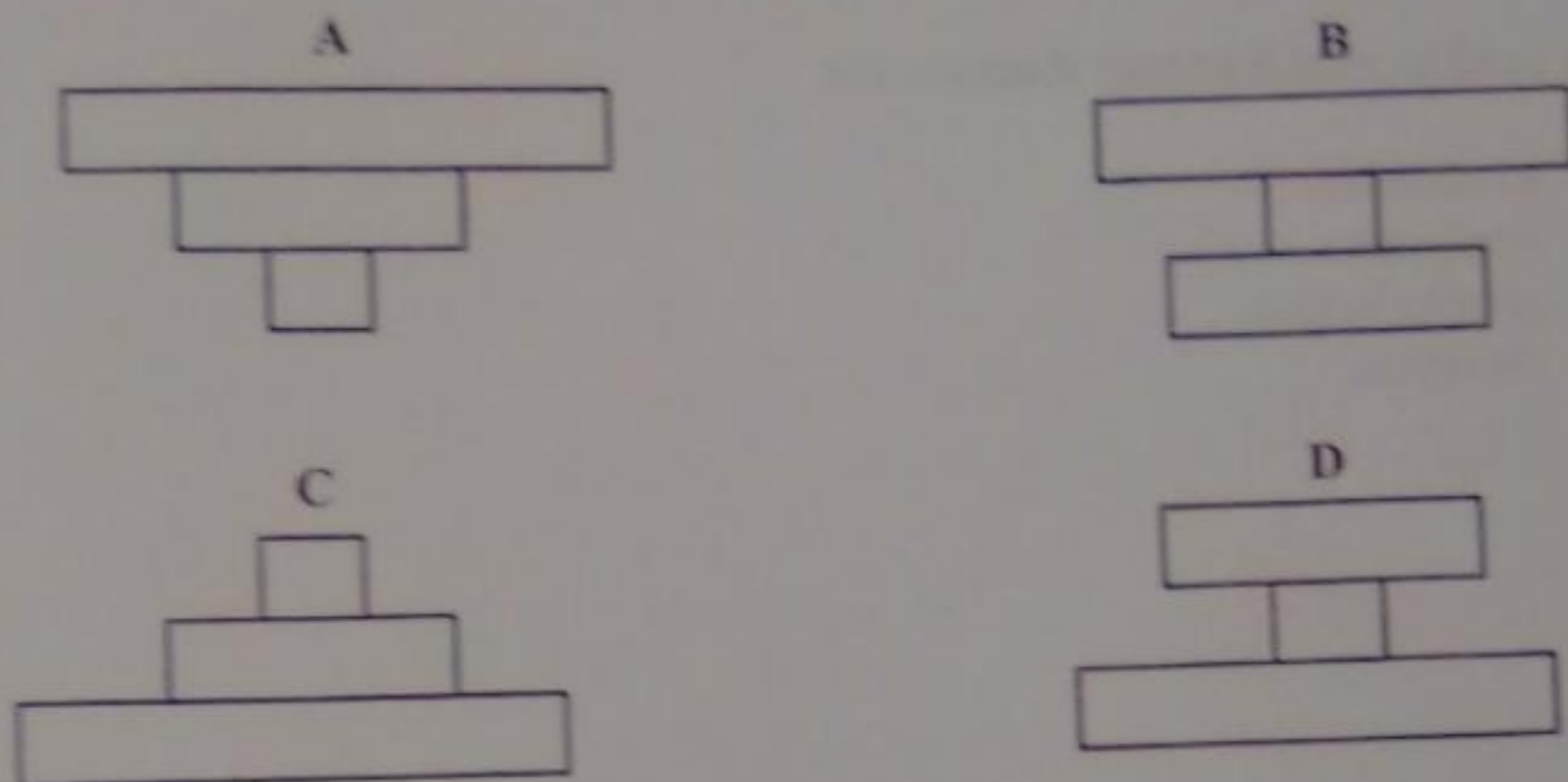
An example of an organism in kingdom X is

- A *E. coli*.
- B *Penicillium*.
- C *Plasmodium*.
- D *Vibrio cholerae*.

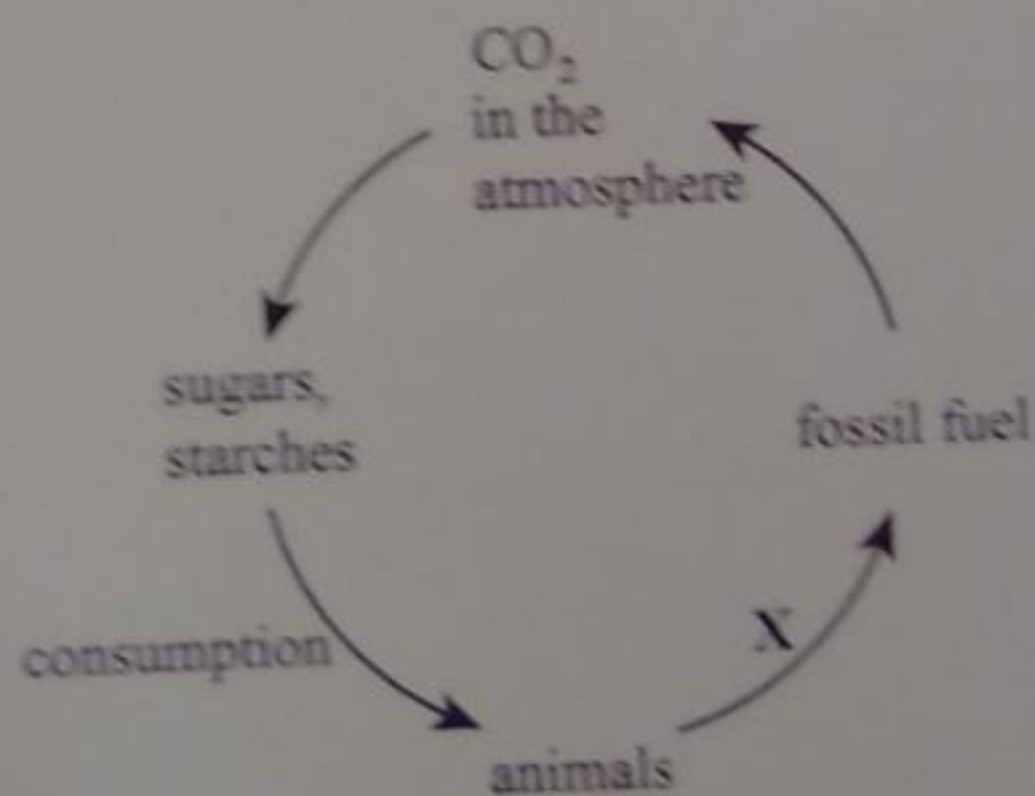
36. Three living organisms in an ecosystem are represented by the food chain:

grass $\longrightarrow$ caterpillars $\longrightarrow$ birds

Which pyramid of biomass represents this food chain?



37. The diagram shows the carbon cycle.



What is process X?

- A respiration
 - B transpiration
 - C decomposition
 - D digestion
38. Which disease is caused by a worm?

- A bilharzia
- B cholera
- C syphilis
- D typhoid

39. Which one is a short term effect of drinking alcohol?

- A heart disease
- B liver cirrhosis
- C damaged blood vessels
- D increased reaction time

40. Drugs used to treat bacterial illnesses are

- A antibiotics.
- B analgesics.
- C prophylactics.
- D diuretics.



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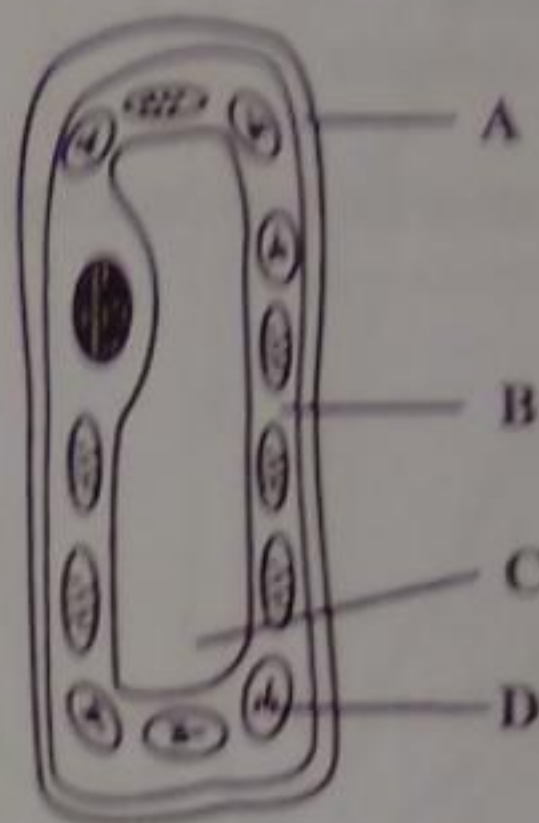
- 1 The study of body parts and their functions is called
 - A anthropology.
 - B osteology.
 - C cytology.
 - D physiology.
- 2 What is the function of glucose in living organisms?
 - A storage compound
 - B structural material
 - C respiratory substrate
 - D growth material
- 3 An example of a disaccharide is
 - A cellulose.
 - B galactose.
 - C maltose.
 - D starch.
- 4 Which cell changes shape to perform its function?
 - A neurone cell
 - B root hair cell
 - C red blood cell
 - D white blood cell
- 5 Which branch of Biology involves the production of genetically engineered organisms?
 - A mycology
 - B biotechnology
 - C cytology
 - D microbiology
- 6 A root hair cell is specialized for its function in that it
 - A lacks a nucleus.
 - B lacks mitochondria.
 - C has a large surface area.
 - D has numerous chloroplasts.

7 Which polysaccharide is stored in animals?

- A cellulose
- B chitin
- C glycogen
- D starch

8 The diagram shows the structure of a plant cell.

In which part, A, B, C or D is the most energy produced?



9 Which enzyme, in a bio-detergent, would remove a fatty stain?

- A amylase
- B cellulase
- C lipase
- D protease

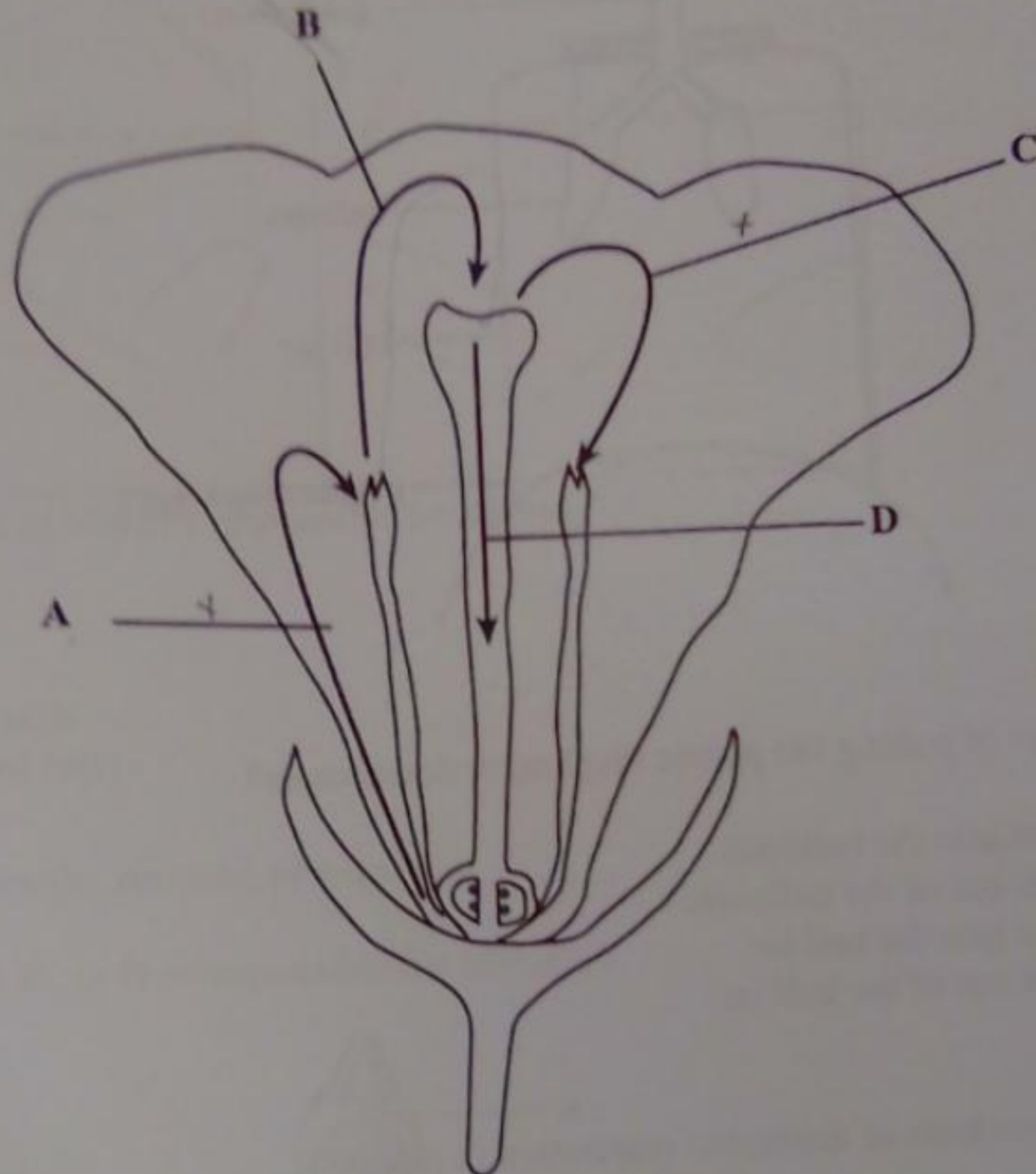
10 Which is an adaptation of a leaf for photosynthesis?

- A shiny surface
- B presence of stomata
- C presence of hairs
- D waxy cuticle

- 11 Which element is essential for osmotic balance in plants?
- A calcium
 - B nitrogen
 - C phosphorus
 - D potassium
- 12 Which factor reduces plant productivity?
- A high light intensity
 - B low water availability
 - C presence of mineral ions
 - D absence of pests and diseases
- 13 Which plant tissue is responsible for translocation?
- A cortex
 - B epidermis
 - C phloem
 - D xylem
- 14 Which feature of a leaf minimises water loss?
- A large surface area
 - B presence of hairs
 - C thin leaf blade
 - D thin waxy cuticle
- 15 Which plant can be propagated by rhizomes?
- A banana
 - B runner grass
 - C sugar cane
 - D sweet potato

16 The diagram shows a section through a flower.

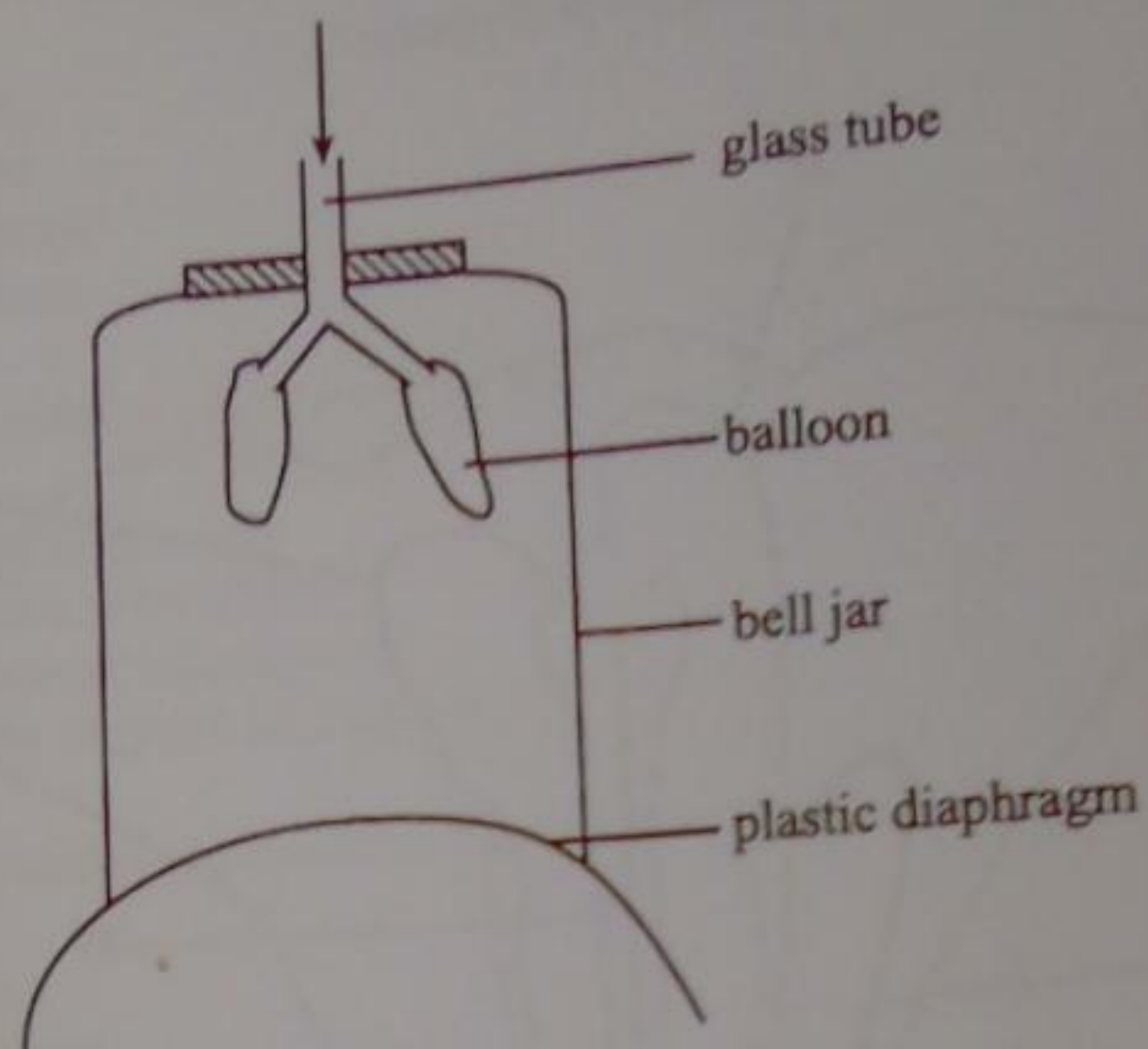
Which arrow, A, B, C, or D represents pollination?



17 Which disease is a result of iron deficiency?

- A scurvy
- B rickets
- C kwashiokor
- D anaemia

- 18 The diagram shows a model of the chest organs illustrating breathing mechanism.



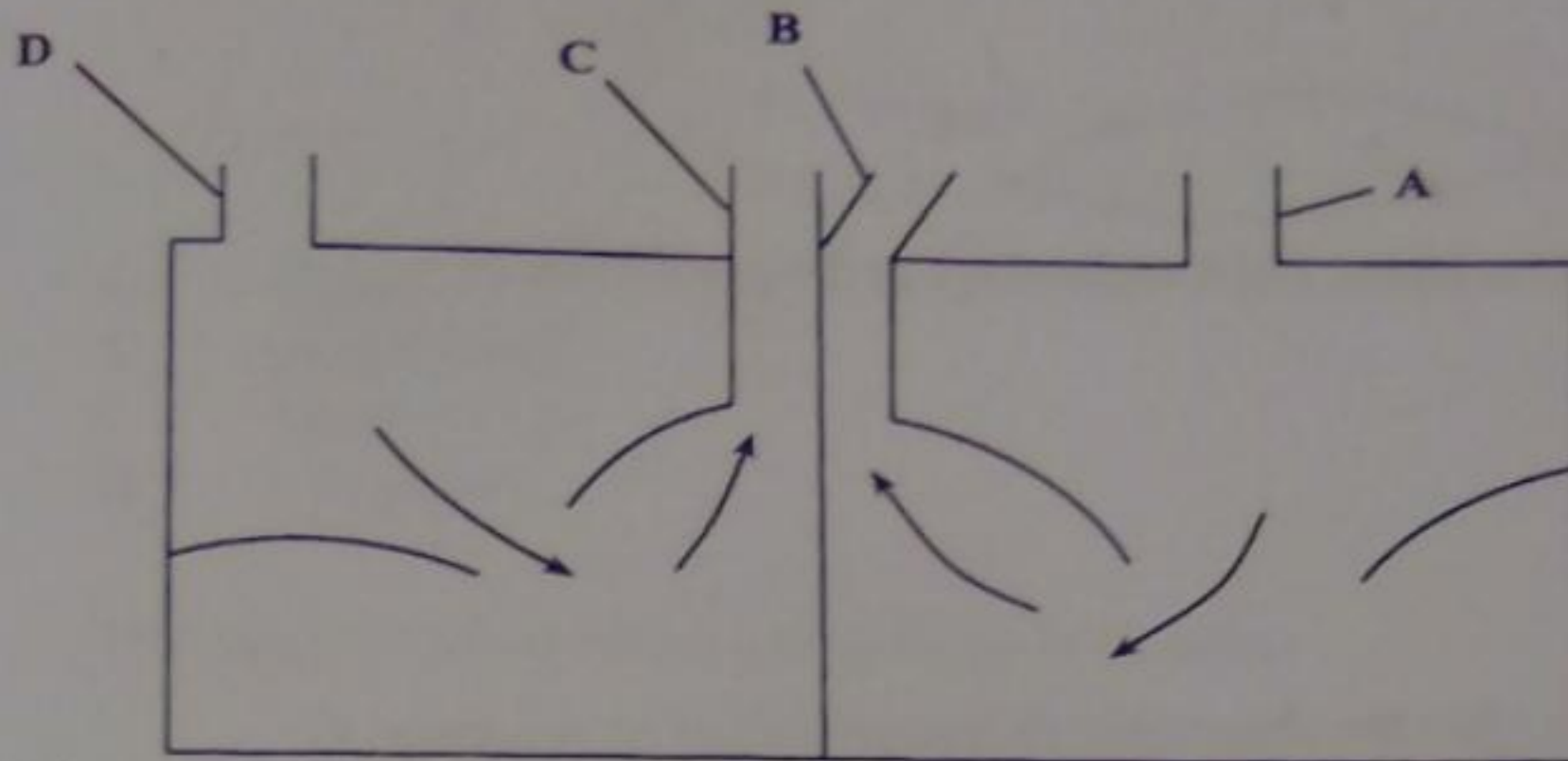
What is the effect of pulling the plastic diaphragm downwards?

- A Air moves into the balloons.
- B Air moves out of the balloons.
- C Air moves into the bell jar.
- D Air moves out of the bell jar.

- 19 What are the by-products of anaerobic respiration in plants?

- A carbon dioxide and lactic acid
- B carbon dioxide and water vapour
- C ethanol and carbon dioxide
- D ethanol and water vapour

- 20 The diagram is a model mammalian heart.
Which part, A, B, C, or D represents the pulmonary vein?

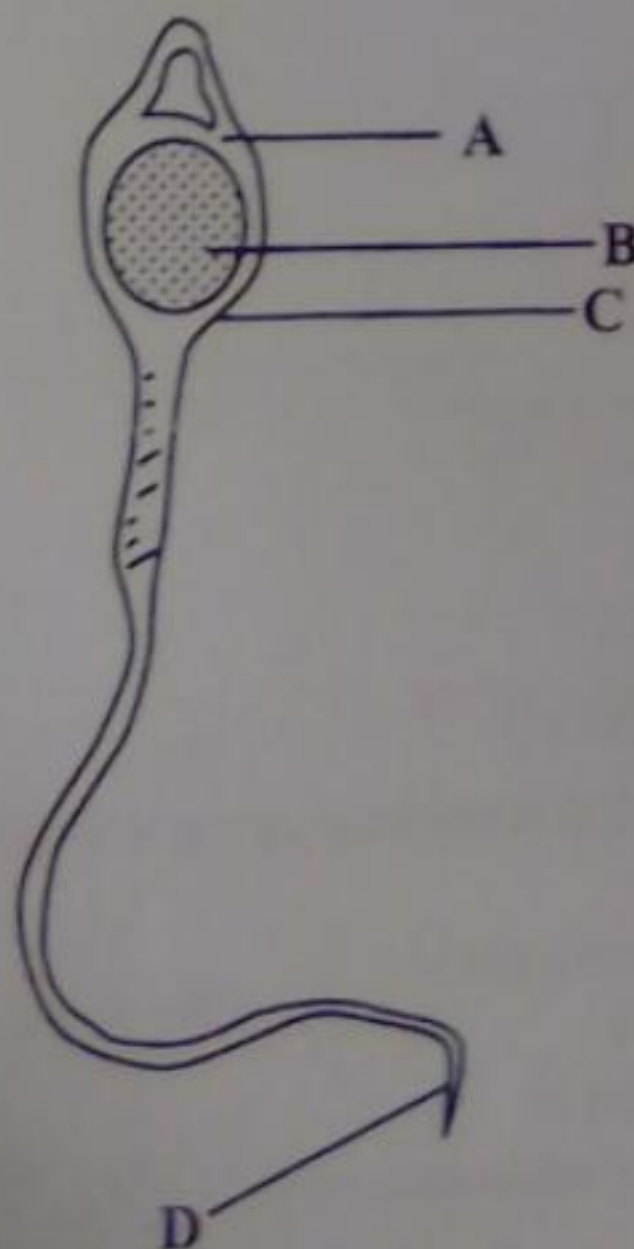


- 21 Which component of blood is responsible for clotting?

- A plasma
- B platelets
- C red blood cells
- D white blood cells

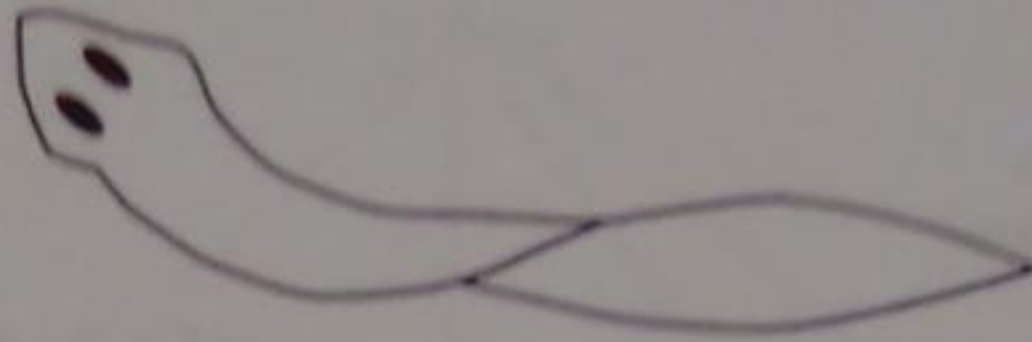
- 22 The diagram shows the structure of a sperm.

Which part, A, B, C, or D is responsible for mobility?



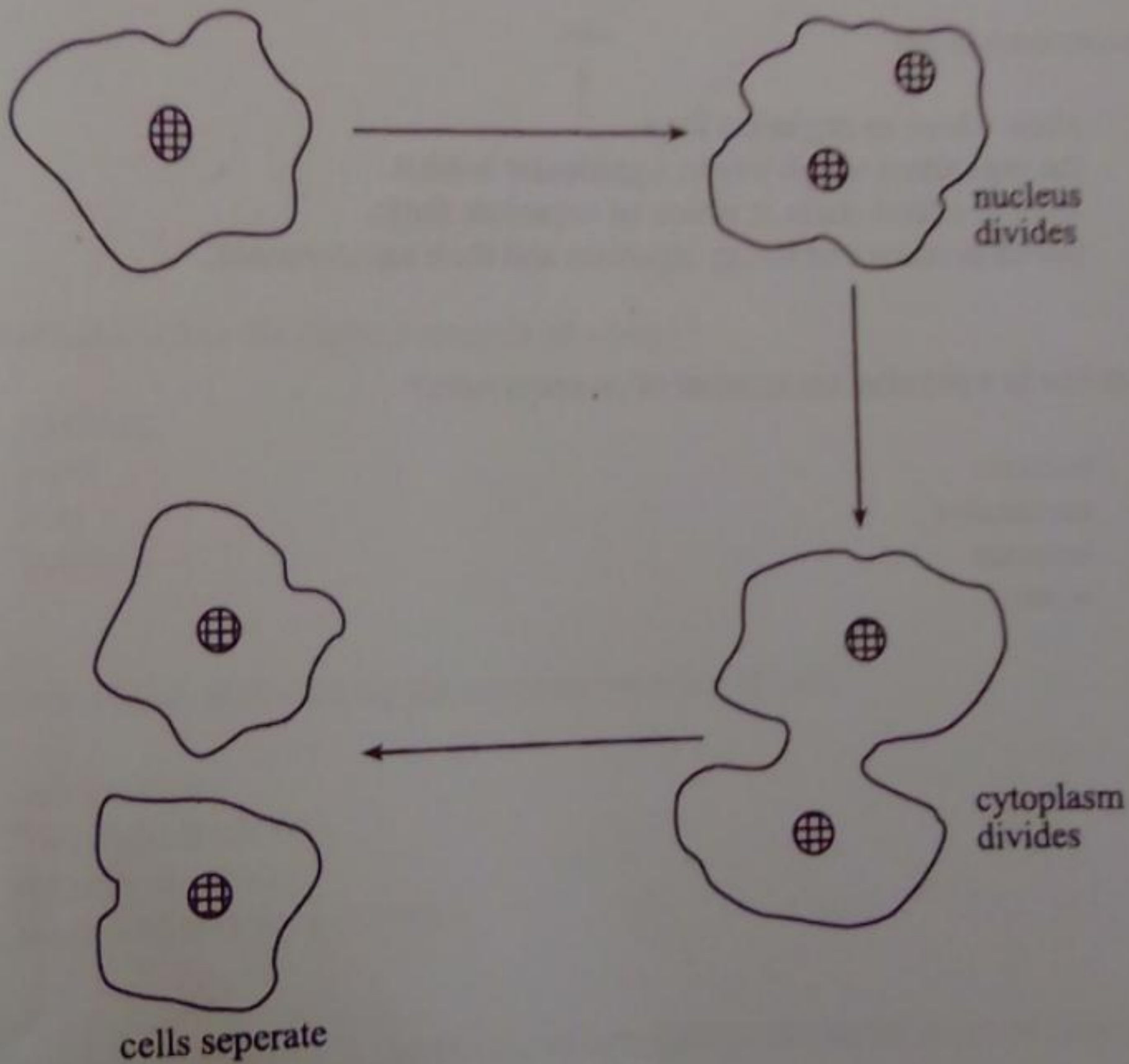
- 23 Which hormone is secreted in response to low blood glucose level?
- A adrenaline
 - B glucagon
 - C insulin
 - D oestrogen
- 24 One benefit of recombinant DNA technology is the production of
- A bread.
 - B cheese.
 - C insulin.
 - D yoghurt.
- 25 The word that describes the observable characteristics of an organism is
- A allele.
 - B chromosome.
 - C genotype.
 - D phenotype.
- 26 Two parents are heterozygous for albinism.
- What is the probability of the parents, producing an albino child?
- A 25 %
 - B 50 %
 - C 75 %
 - D 100 %
- 27 Down's syndrome is caused by
- A carcinogens.
 - B radiation.
 - C gene mutation.
 - D chromosome mutation.
- 28 Which statement is correct about discontinuous variation?
- A results from effects of genes only x
 - B it has many intermediates
 - C gives a normal distribution x
 - D results from effects of the environment x

- 29 Which kingdom does the organism shown belong to?



- A plantae
- B prokaryotae
- C fungi
- D animalia

- 30 The diagram shows reproduction stages in amoeba.



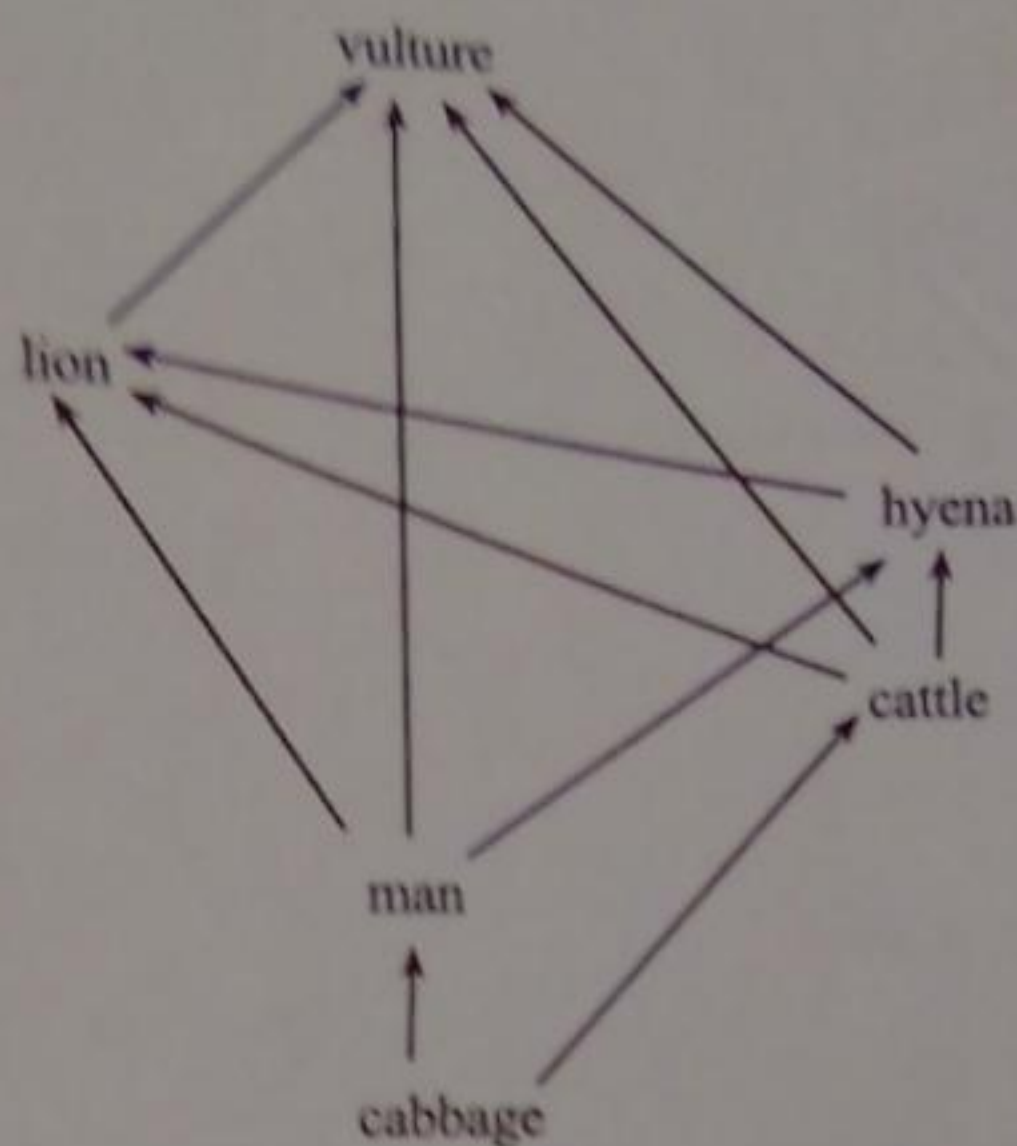
What type of reproduction is shown?

- A spore production
- B binary fission
- C budding
- D meiosis

- 31 In which organisms are eggs fertilised externally?
- A birds
 - B reptiles
 - C mammals
 - D amphibians
- 32 An organism which digests dead organic matter belongs to the kingdom
- A animalia.
 - B fungi.
 - C plantae.
 - D protocista.
- 33 An ecosystem is the
- A place where an organism lives.
 - B the organisms which live in a particular habitat.
 - C level in a food chain at which an organism feeds.
 - D interdependence of living organism and their environment.
- 34 Which one is a physical component of an ecosystem?
- A bacteria
 - B nematodes
 - C termites
 - D water.

35

The diagram shows a food web.



Which organism has the highest amount of energy?

- A cabbage
- B cattle
- C lion
- D vulture

36 What is the effect of exceeding the carrying capacity of land?

- A soil erosion
- B increased birth rate
- C high productivity
- D increased species diversity

37 Which method is effective in destroying mosquitoes at the larval stage of development?

- A burning mosquito coils
- B killing by hand
- C spreading oil on water
- D using mosquito repellents

1. The following are the four main types of business organizations:

- a. Sole proprietorship
- b. Partnership
- c. Corporation
- d. Limited liability company

2. The following are the four main types of business organizations:

- a. Sole proprietorship
- b. Partnership
- c. Corporation
- d. Limited liability company

3. The following are the four main types of business organizations:

- a. Sole proprietorship
- b. Partnership
- c. Corporation
- d. Limited liability company

38 Use of prophylactic drugs is one method of preventing

- A bilharzia.
- B cholera.
- C malaria.
- D typhoid.

39 The organism that causes cholera is a

- A bacterium.
- B fungus.
- C protozoan.
- D worm.

40 Typhoid is transmitted by

- A sexual contact.
- B droplet infection.
- C sharing bath towels.
- D drinking contaminated water.

- 1 The diagram shows a spongy mesophyll cell of a dicotyledonous leaf.



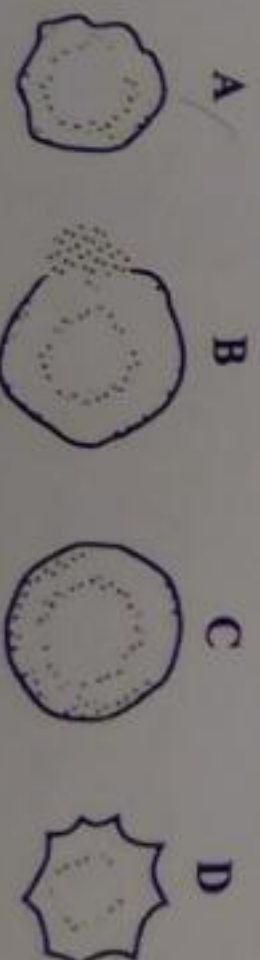
Which row correctly identifies the structures labelled Y and Z?

	Y	Z
A	cytoplasm	cell membrane
B	vacuole	cell wall
C	cytoplasm	cell wall
D	vacuole	cell membrane

- 2 The root hair cell is specialised for

- A translocation.
- B transport.
- C support.
- D absorption.

- 3 Which diagram, A, B, C or D shows the appearance of a red blood cell that had been placed in a concentrated salt solution?



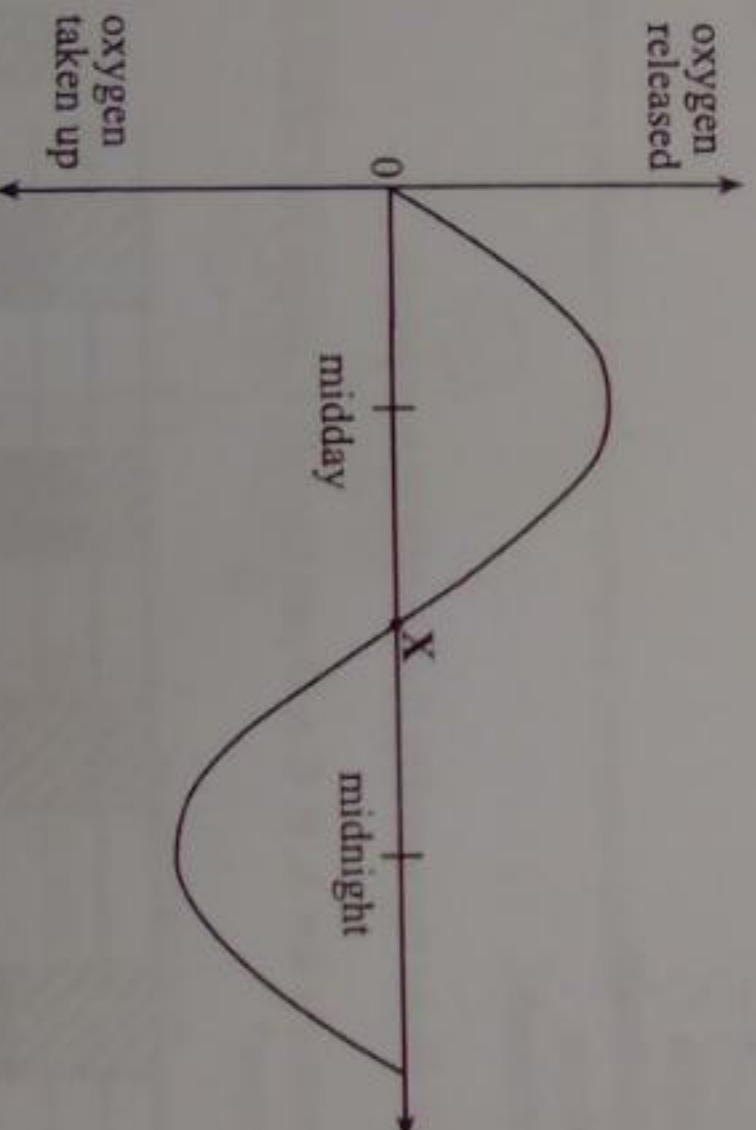
4 Which statement correctly describes the biological control of a crop pest?

- A crop rotation to break the life cycle of the pest
- B planting the crop when the pest is not present
- C introducing predators of the pest to the crop
- D spraying the crop with an organic pesticide

5 Which products of photosynthesis are essential to animals?

- A oxygen and water
- B oxygen and carbohydrate
- C carbon dioxide and water
- D carbohydrate and carbon dioxide

6 The graph shows how the amount of oxygen released and taken up by a plant changes during a 24 hour period.



Which statement explains what is happening at point X?

- A rate of respiration is highest
- B photosynthesis stops
- C respiration and photosynthesis have stopped
- D rates of respiration and photosynthesis are the same

- 7 Which pair correctly gives products of digestion of fats?

A glucose and fatty acids
 B glucose and glycerol
 C glycerol and fatty acids
 D amino acids and glycerol

- 8 Which nutrient should be included in the diet of a person to prevent slow healing of wounds?

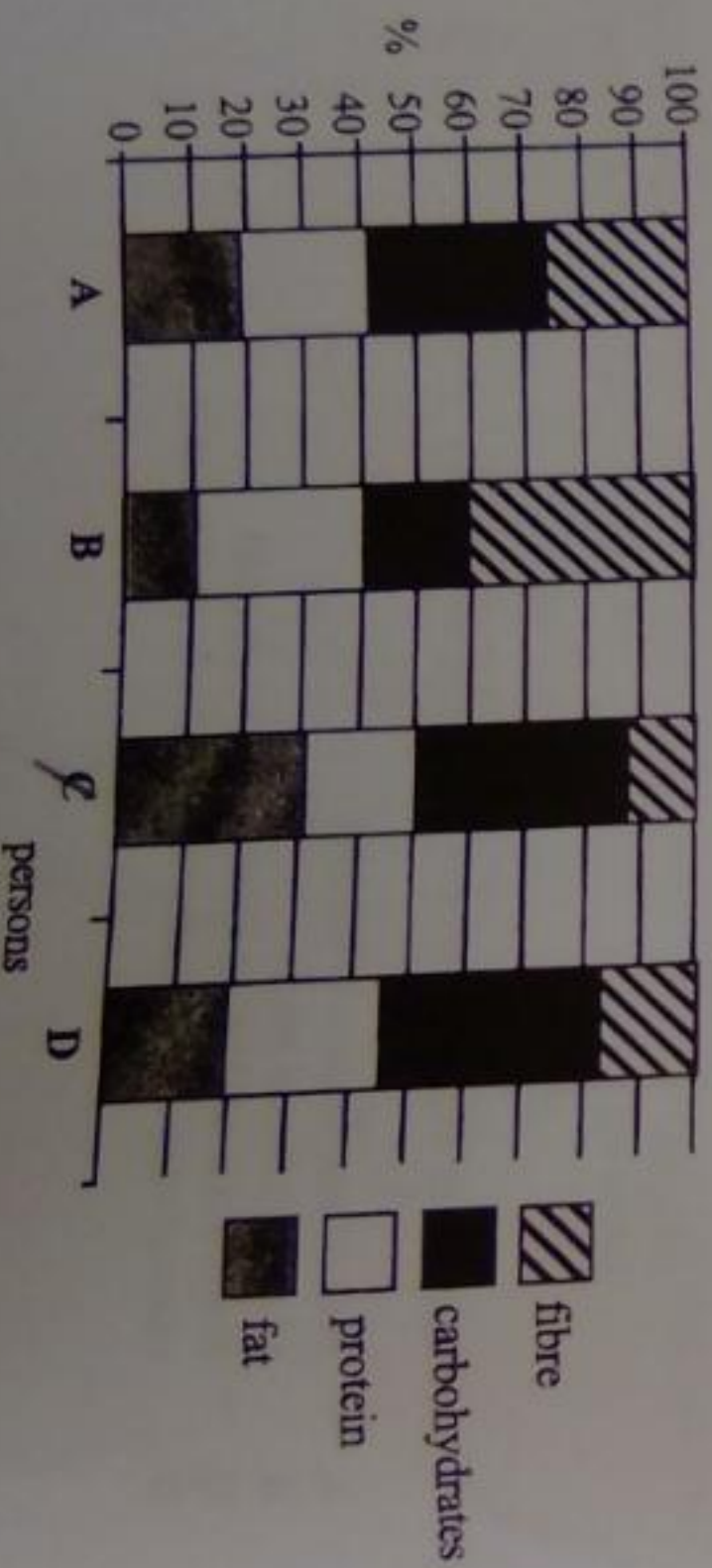
A vitamin A
 B vitamin C
 C iron
 D iodine

- 9 In which part of the alimentary canal does absorption of end products of digestion take place?

A colon
 B duodenum
 C ileum
 D stomach

- 10 The graph shows the monthly food intake of four people.

Which person's diet A, B, C or D puts him/her at greater risk of developing coronary heart disease?



11 Which process does **not** need mitochondria?

- A diffusion of oxygen
- B** maintenance of body temperature
- C contraction of muscles
- D transmission of a nerve impulse

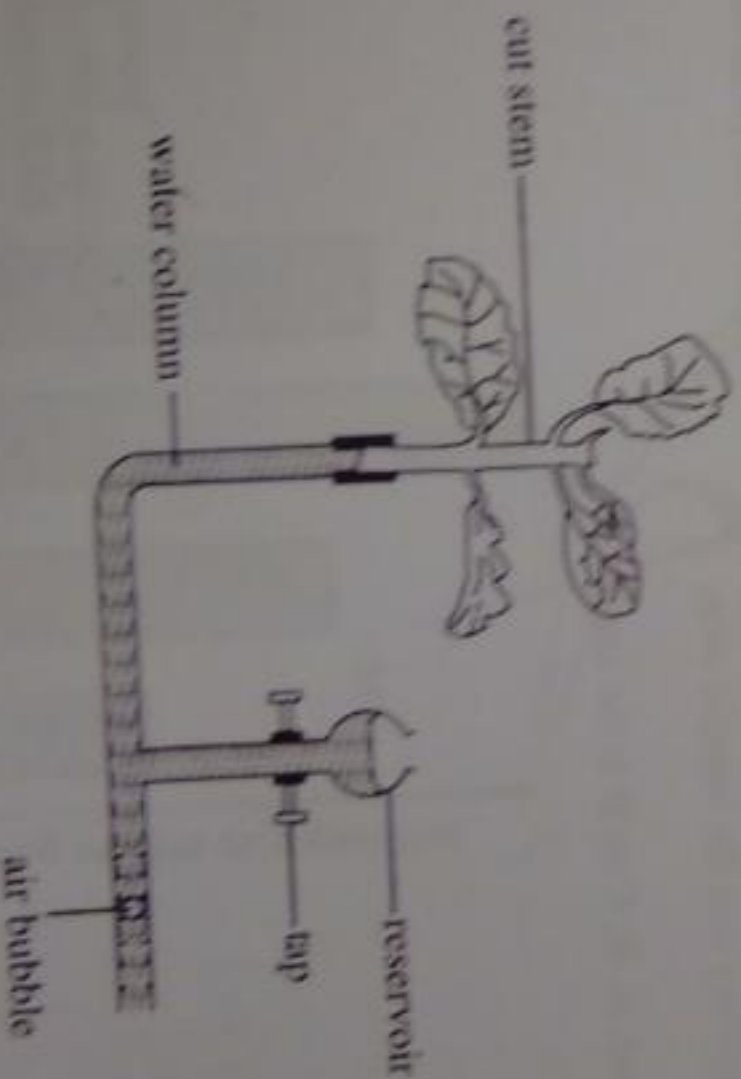
12 What is the volume of carbon dioxide in 20 cm^3 of exhaled air?

- A 16 cm^3
- B** 4 cm^3
- C 3.2 cm^3
- D 0.8 cm^3

13 Which component of cigarette smoke reduces the oxygen carrying capacity of blood?

- A carbon monoxide
- B** smoke particles
- C nicotine
- D tar

14 The diagram shows a potometer used to investigate transpiration rate.



When the tap is opened the

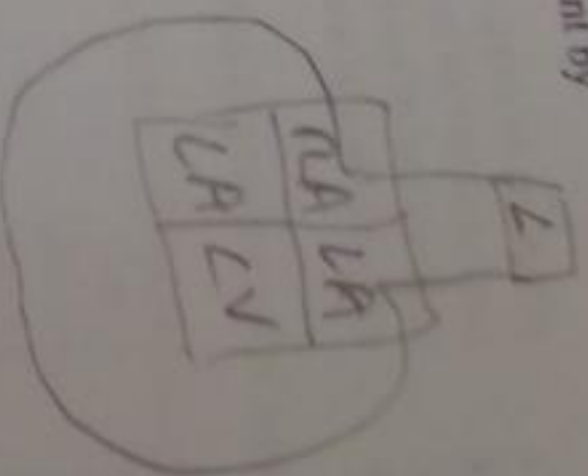
- A air bubble moves to the right.
- B** air bubble moves to the left.
- C water level in the reservoir rises.
- D water column level drops.

- 15 Sucrose is transported from the leaf to other parts of a plant by

A evaporation.
 B osmosis.
 C ☒ translocation.
 D transpiration.

- 16 The liquid from the xylem of a plant is likely to contain

A starch.
 B ☒ nitrates.
 C proteins.
 D sucrose.

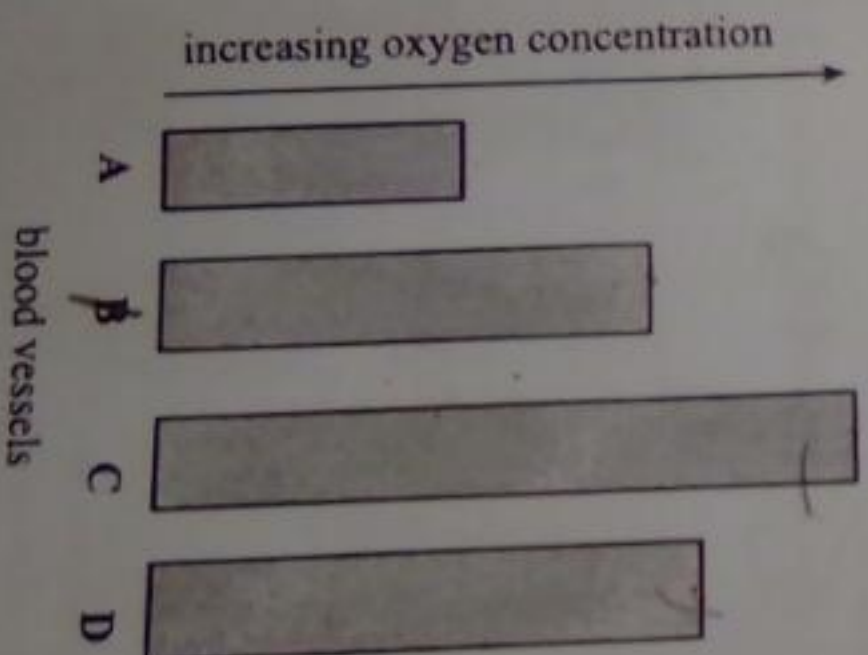


- 17 Which chamber of the heart is the first to receive nicotine absorbed from cigarette smoke?

A right atrium
 B ☒ left atrium
 C right ventricle
 D left ventricle

- 18 The bar chart shows the relative concentration of oxygen in the pulmonary vein, renal artery, a capillary and the hepatic vein.

Which vessel, A, B, C or D is the capillary?



19 Which one is **not** a characteristic of all arteries?

- A have narrow lumen
- ☒ B have thick walls
- C carry oxygenated blood
- D have a pulse

20 The main function of lymphocytes in the blood is to

- A aid in clotting.
- ☒ B form antigens.
- C produce antibodies.
- D digest foreign particles.

21 The diagram shows a section through a joint.



What is **M**?

- A ligament
- ☒ B synovial membrane
- C synovial fluid
- D cartilage

- 22 Which substances are absent in the urine of a healthy person?

~~A~~ amino acids and glucose
 B amino acids and salts
 C glucose and salts
 D salts and urea

- 23 The diagram shows the front view of the eye.

Which part, A, B, C or D contains muscles involved in the pupil reflex?



- 24 The heart beat of a frightened mammal increases considerably due to a secretion from the

A pancreas.
 B liver.
 C thyroid gland.
 D adrenal gland.

- 25 The diagram shows a section through the skin.

Which part, A, B, C or D detects changes in skin temperature?

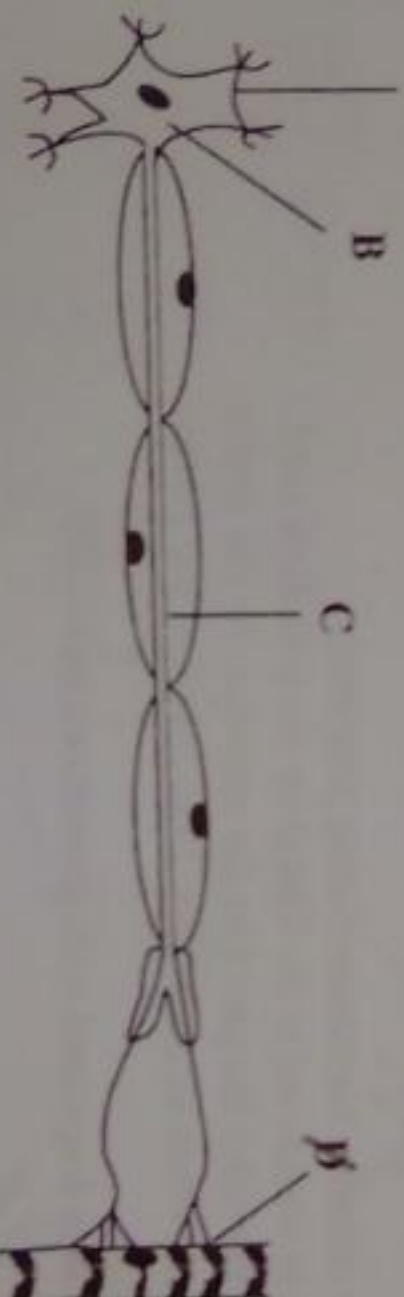


- 26 Which row correctly shows the set of conditions that would result in a person producing a small volume of concentrated urine?

	environmental temperature	volume of water taken	amount of exercise
A	high	high	low
B	high	low	high
C	low	low	high
D	low	high	low

- 27 The diagram shows a motor neurone.

In which part, A, B, C or D, are electrical impulses converted to kinetic energy?



- 28 Plants can reproduce naturally using

- A runners
B cuttings
C grafting
~~D budding.~~

- 29 How are the pollen grains of insect-pollinated plants adapted for pollination?

- ~~A They are smooth and light.~~
B They are produced in large numbers.
C They are sticky and spiky.
D They contain nectar.

- 30 Different parts of flowers were removed from four groups of plants and insects were allowed to visit these flowers freely. The table shows how flowers in each group of the plants were treated.

Which row correctly describes the group which would produce the largest number of seeds?

Group of plants	stigma	anthers	petals
A	not removed	removed	not removed
B	not removed	not removed	removed
C	removed	not removed	removed
D	removed	removed	not removed

- 31 Which statement about the placenta is **not** true?
- ~~A~~ It releases hormones during pregnancy.
 B It is connected to the foetus by the umbilical cord.
 C It provides the pathway for excretion by the foetus.
 D It prevents all pathogens from reaching the foetus.

- 32 A patient was diagnosed with gonorrhoea and AIDS.

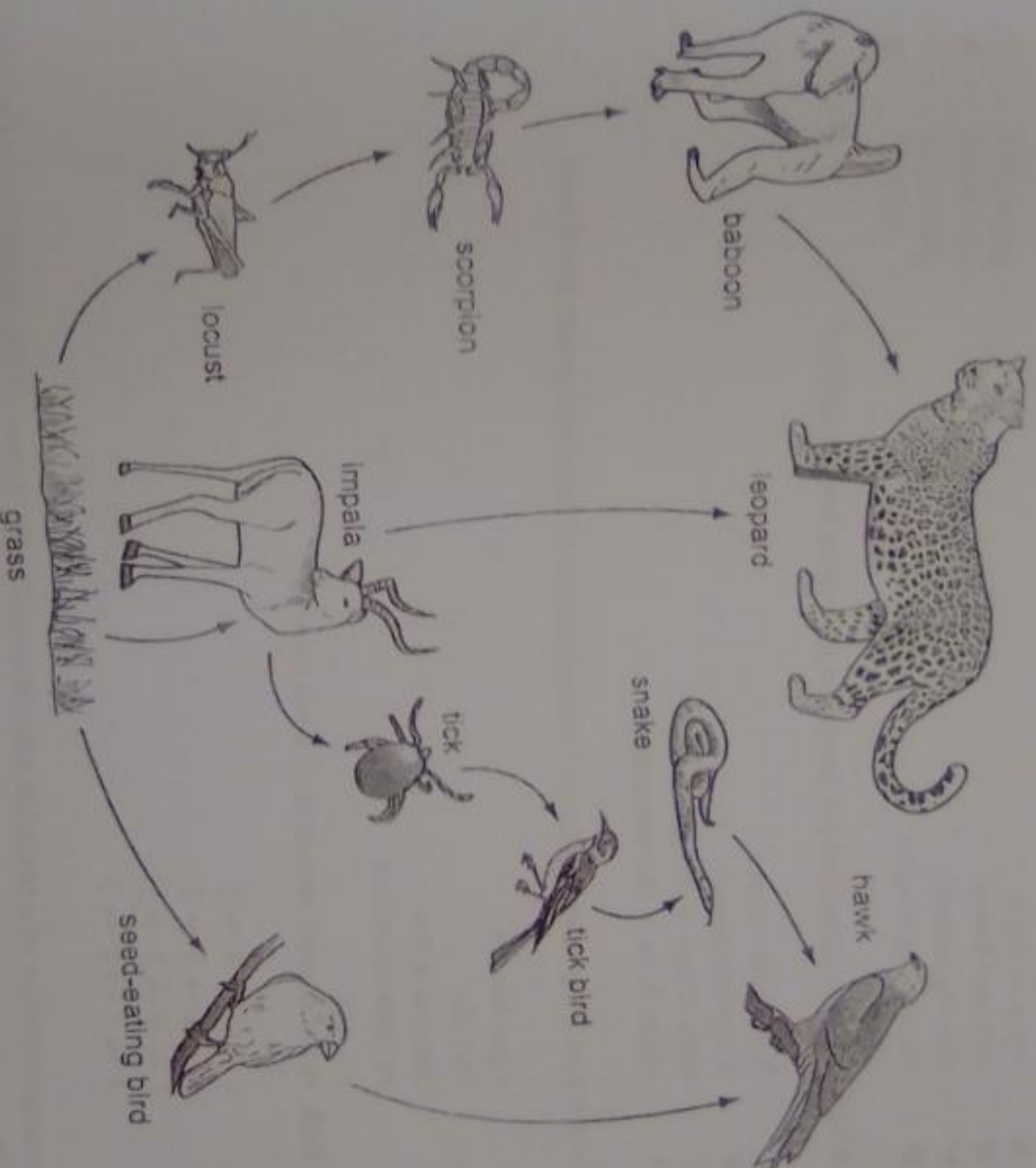
Which groups of micro-organisms were present in his blood?

- A bacteria and protozoa
 B fungi and bacteria
 C bacteria and viruses
~~D~~ protozoa and viruses

- 33 How does Oral Rehydration Therapy assist in the treatment of cholera?

- A prevents vomiting and reduces fever
~~B~~ maintains osmotic balance of tissue fluid
 C kills the vibrio cholerae which causes the disease
 D stops diarrhoea and the stomach pains

34 The diagram shows a food web.



Which food chain in the food web is the most efficient?

- A grass → locust → scorpion → baboon → leopard
- B grass → impala → leopard**
- C grass → impala → tick → tick bird → snake → hawk
- D grass → seed-eating bird → hawk

35 The malaria parasite, *Plasmodium*, causes fever by

- A releasing toxins into the blood.**
- B bursting red blood cells.
- C damaging liver cells.
- D bursting white blood cells.

36 HIV is **not** spread through

- A open wounds,
- B infected blood.
- ☒ C sharing utensils.
- D sexual intercourse.

37 Which sex chromosomes are present in all mature human sperms?

- A Y chromosomes
- ☒ B X chromosomes
- C both X and Y chromosomes
- D either X or Y chromosomes

38 Events resulting in the process of eutrophication in lakes and ponds are shown

1. algal bloom
2. fish die
3. bacteria grow
4. oxygen decreases

Which is the correct sequence of these events?

- ☒ A 1 → 3 → 4 → 2
- B 1 → 4 → 3 → 2
- C 3 → 4 → 2 → 1
- D 4 → 1 → 2 → 3

39 The concentration of a pesticide was measured in the tissues of the organisms in the food chain:

plant → small fish → large fish → eagle

Which organisms had the lowest concentration of the pesticide?

- ☒ A eagle
- B large fish
- C small fish
- D plants

The carrying capacity of a given area is determined by

- ☒ A resources.
- ☐ B death rate.
- ☐ C birth rate.
- ☐ D fertility.



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

PAPER 1 Multiple Choice

5008/1

NOVEMBER 2016 SESSION

1 hour

Additional materials:

Multiple choice answer sheet

Soft pencil (type B or HB is recommended)

Soft clean eraser

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are forty questions in this paper. Answer **all** questions. For each question, there are four possible answers, A, B, C and D. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

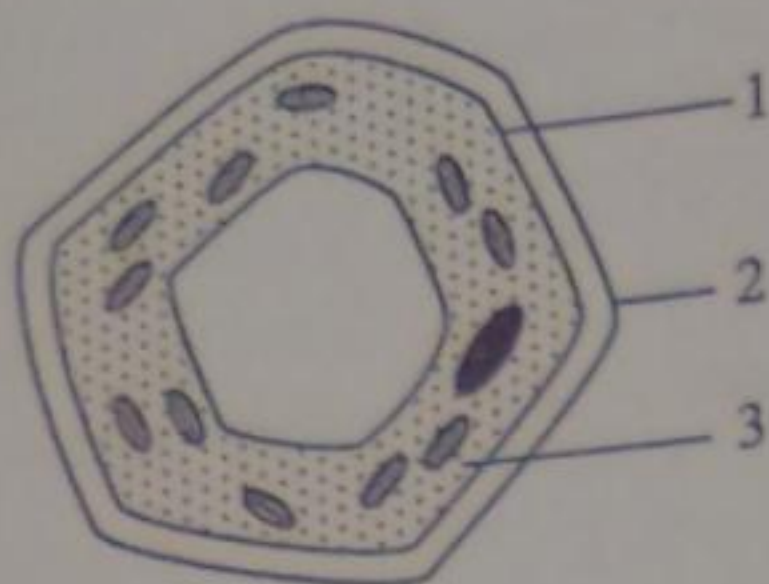
Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

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- 1 The diagram shows a cell.



Which row correctly identifies the numbered parts?

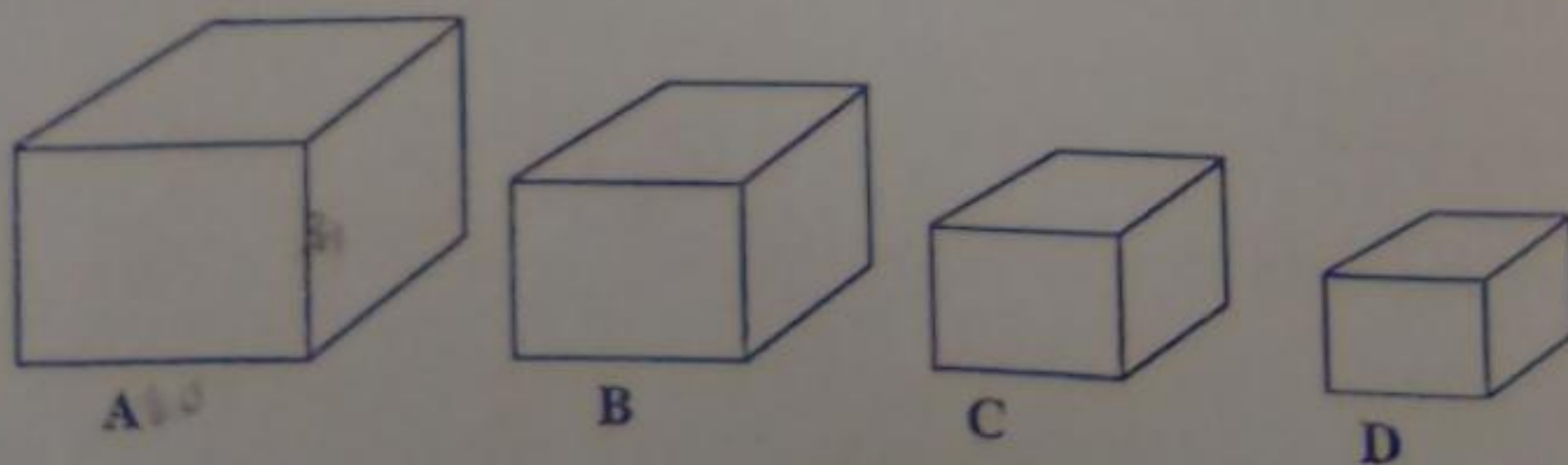
	cell membrane	cell wall	cytoplasm
A	3	2	1
B	1	2	3
C	2	1	3
D	3	1	2

- 2 Why is the palisade layer a tissue?

- A All the cells have the same function.
- B All the cells have a similar structure.
- C All the cells have a similar structure and function.
- D All the cells allow water to enter them by osmosis.

- 3 The diagram shows cuboids of different sizes.

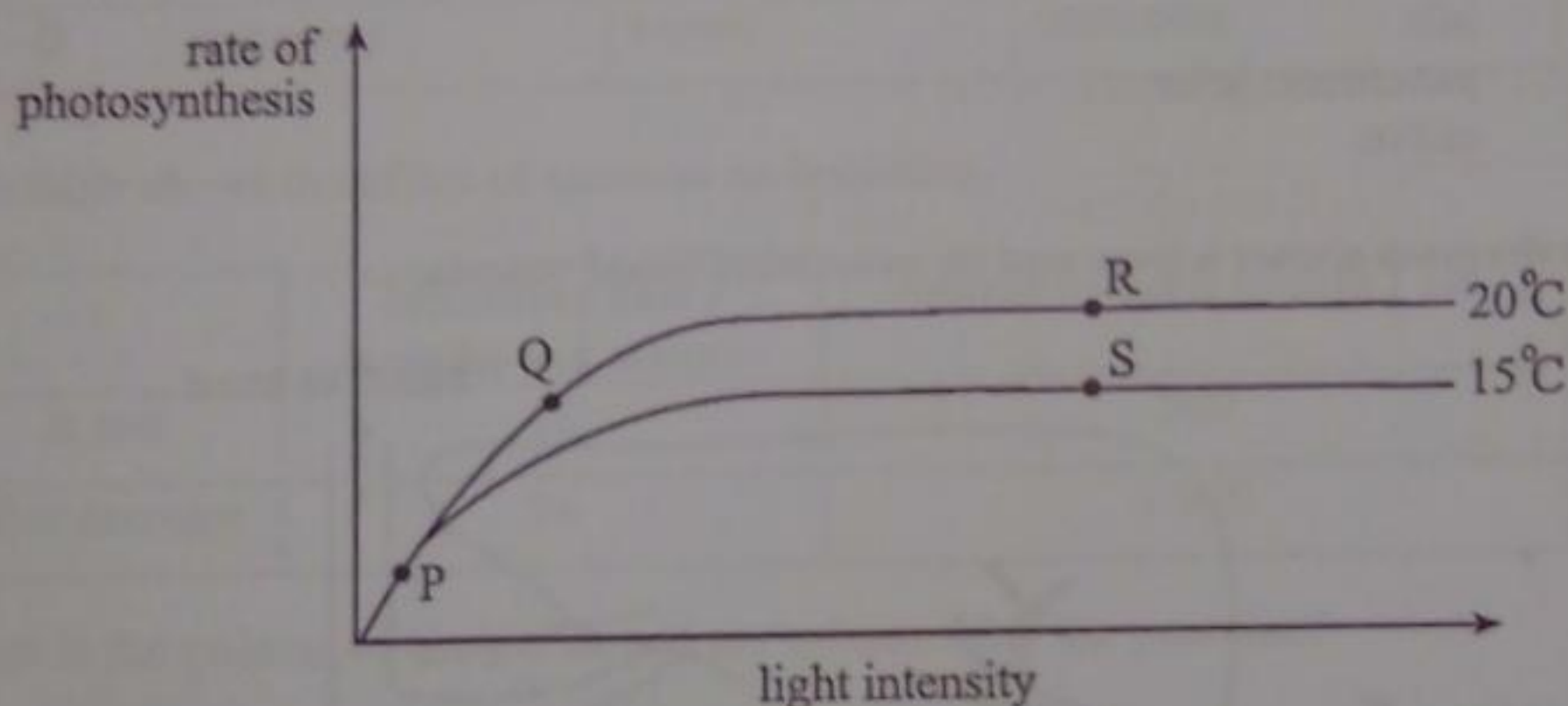
Which of the cuboid A, B, C or D has the lowest surface area to volume ratio?



4 Which statement is true about enzymes?

- A Enzymes are made of carbohydrate molecules.
- B One enzyme catalyzes several reactions.
- C Enzymes are denatured by high temperatures.
- D Enzymes catalyze reactions in all pH ranges.

5 The graph shows how the rate of photosynthesis varies with light intensity at two different temperatures.



At which points is light intensity limiting the rate of photosynthesis?

- A P and Q
- B P and R
- C Q and R
- D R and S

6 In a crop field, a farmer observes that most crop plants have pale green leaves and show poor growth.

What is likely to be deficient in this soil?

- A potash
- B phosphate
- C sulphate
- D nitrates

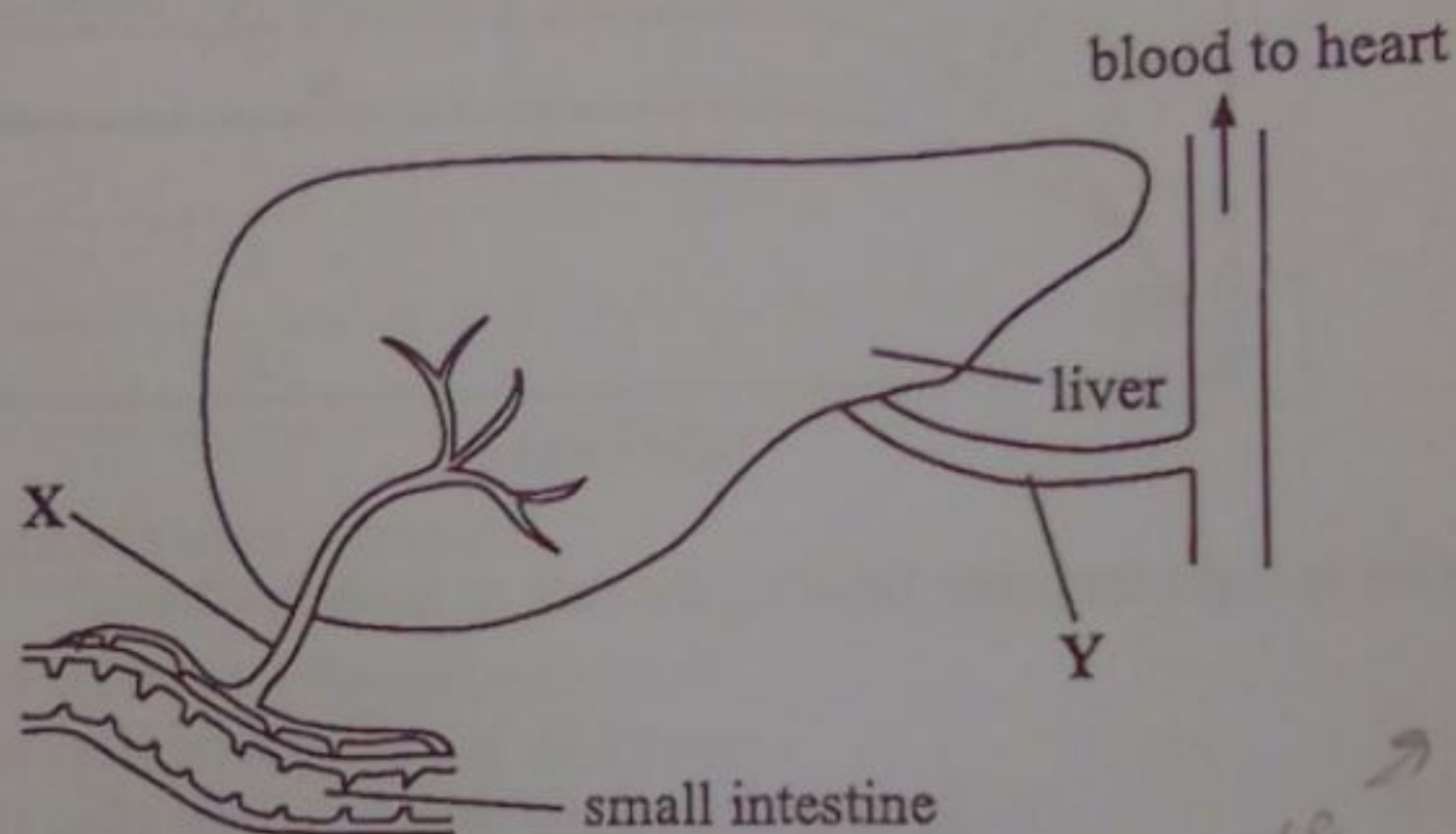
7 Which one is a method of biological pest control?

- A applying a pesticide
- B catching pests in a trap
- C digging out pests with a hoe
- D introducing a pest predator

8 Which secretion does **not** contain digestive enzymes?

- A gastric juice
- B bile
- C pancreatic juice
- D saliva

9 The diagram shows a liver and its associated blood vessels.



What are the blood vessels X and Y?

	X	Y
A	hepatic portal vein	hepatic vein
B	mesenteric artery	renal vein
C	hepatic artery	hepatic vein
D	aorta	vena cava

10 Which order is correct for the processes in human nutrition?

- A ingestion → digestion → assimilation → absorption
- B digestion → ingestion → assimilation → absorption
- C digestion → ingestion → absorption → assimilation
- D ingestion → digestion → absorption → assimilation

*glucose → mouth
fats → stomach*

- 11 Which row correctly shows the changes that occur in the thorax, rib cage and diaphragm when breathing out?

	volume of thorax	rib cage	diaphragm
A	decreases	lowered	relaxes
B	decreases	raised	contracts
C	increases	lowered	relaxes
D	increases	raised	contracts

- 12 The table shows the effect of exercise on breathing.

	breathing rate / breaths per minute	volume of each breath / cm ³
at rest	12	500
after exercise	24	1 000

What is the volume of air exchanged per minute after the exercise?

- A 1 000 cm³
 B 6 000 cm³
 C 1 800 cm³
 D 24 000 cm³

$$24 \times \frac{1000}{1} = 24000$$

Handwritten calculation: $24 \times 1000 = 24000$

- 13 Which substances are produced when yeast cells respire anaerobically?

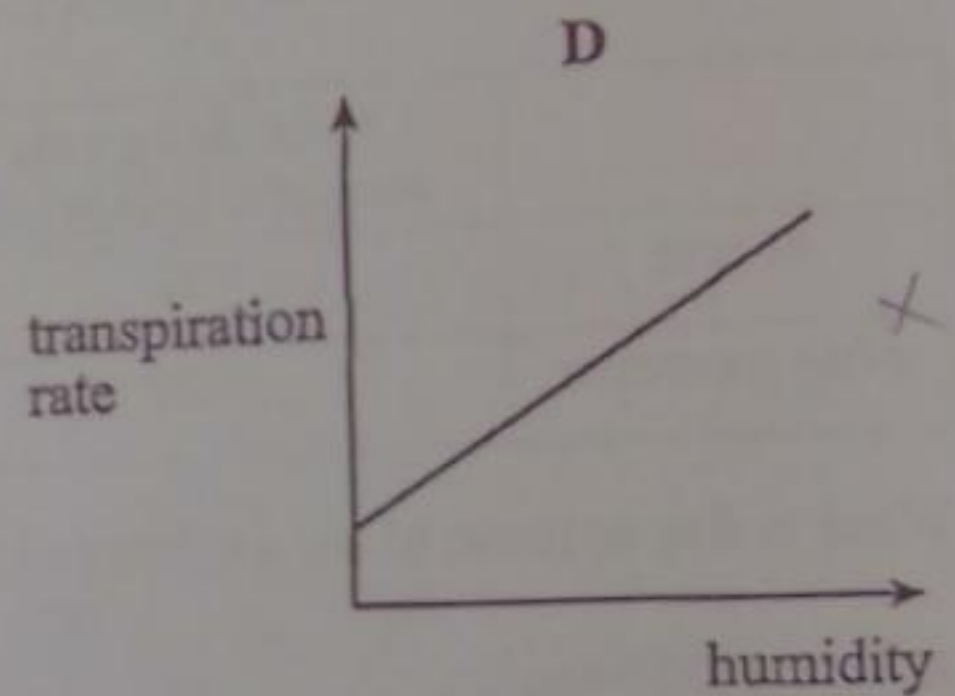
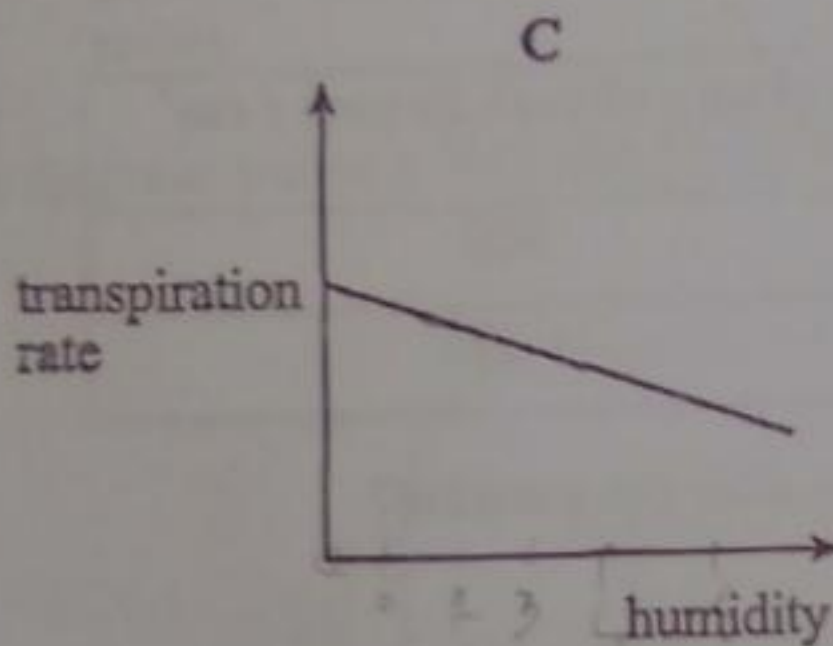
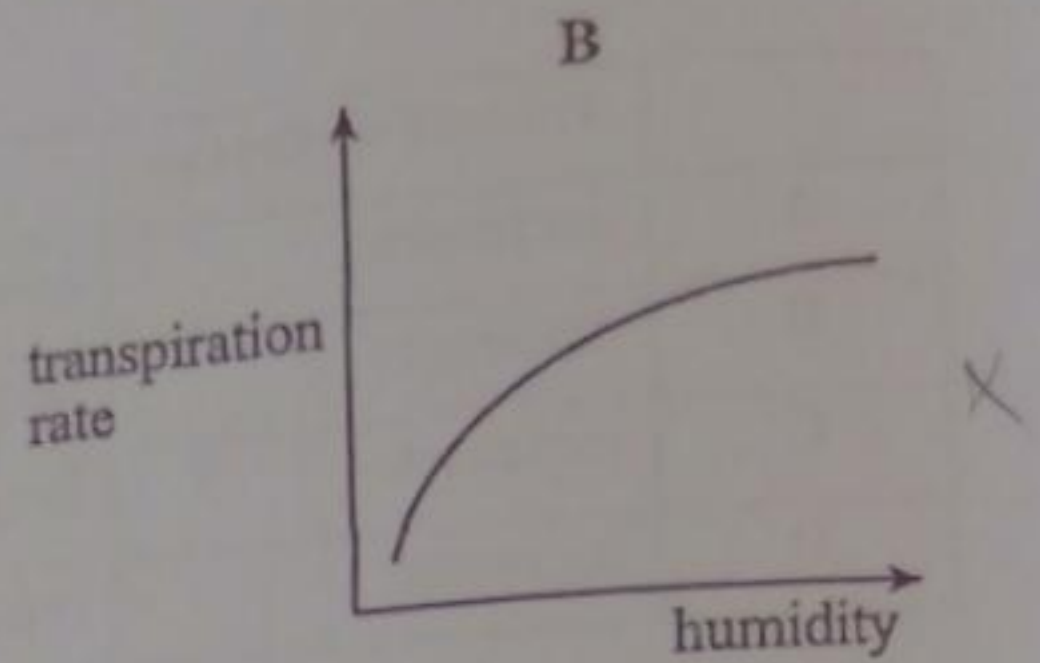
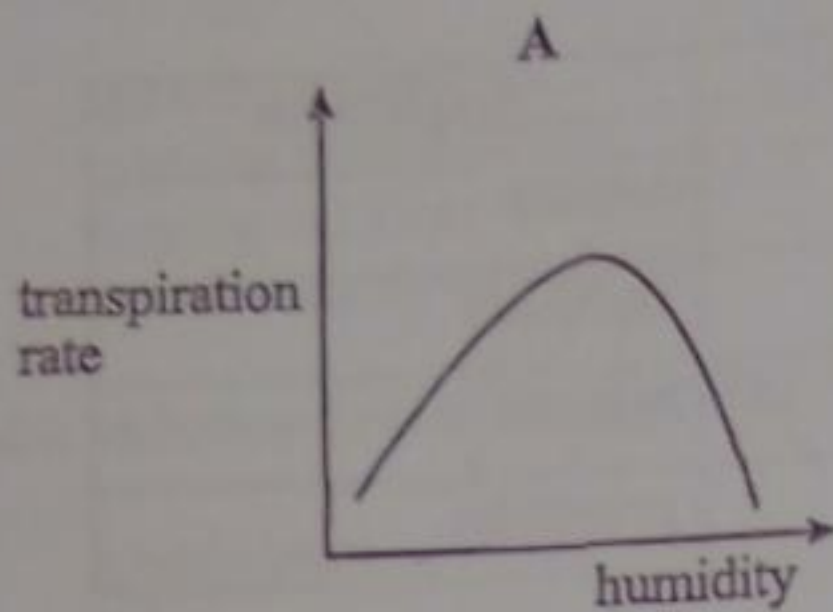
- A lactic acid and water
 B carbon dioxide and water
 C alcohol and carbon dioxide
 D carbon dioxide and lactic acid

- 14 Which row correctly shows substances transported in the phloem and xylem?

	phloem	xylem
A	nitrate ions	glucose
B	sucrose	water
C	glucose	mineral ions
D	water	amino acids

Handwritten note: starch / carbohydrates

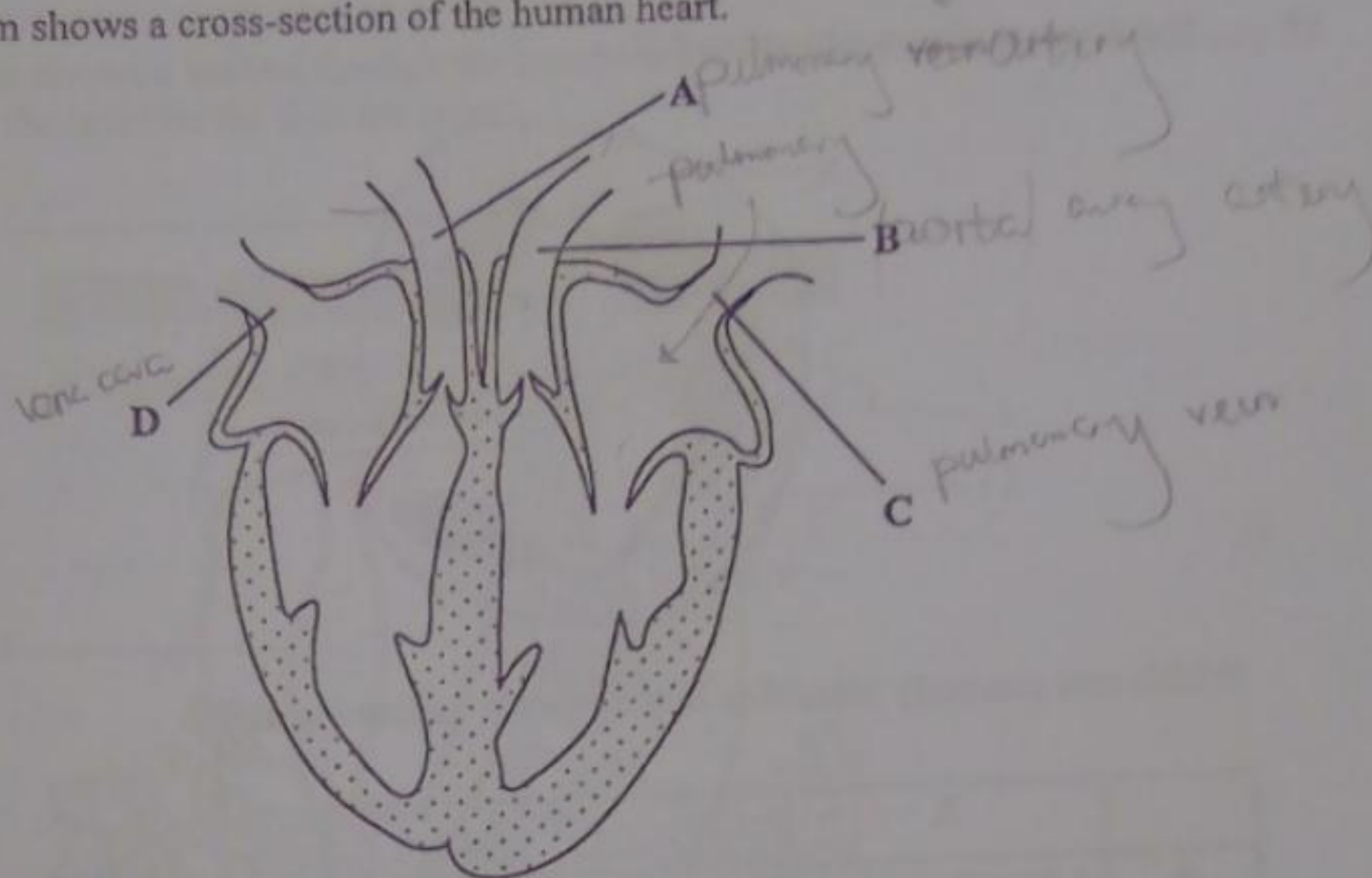
- 15 Which graph, A, B, C or D shows the effect of increasing humidity on the rate of transpiration in plants?



- 16 A young plant may wilt when transplanted in another place because the
- A surface area of the root is reduced.
 - B roots cannot take up mineral salts.
 - C rate of transpiration is greater than the rate of photosynthesis.
 - D the plant is adjusting to new conditions.

17

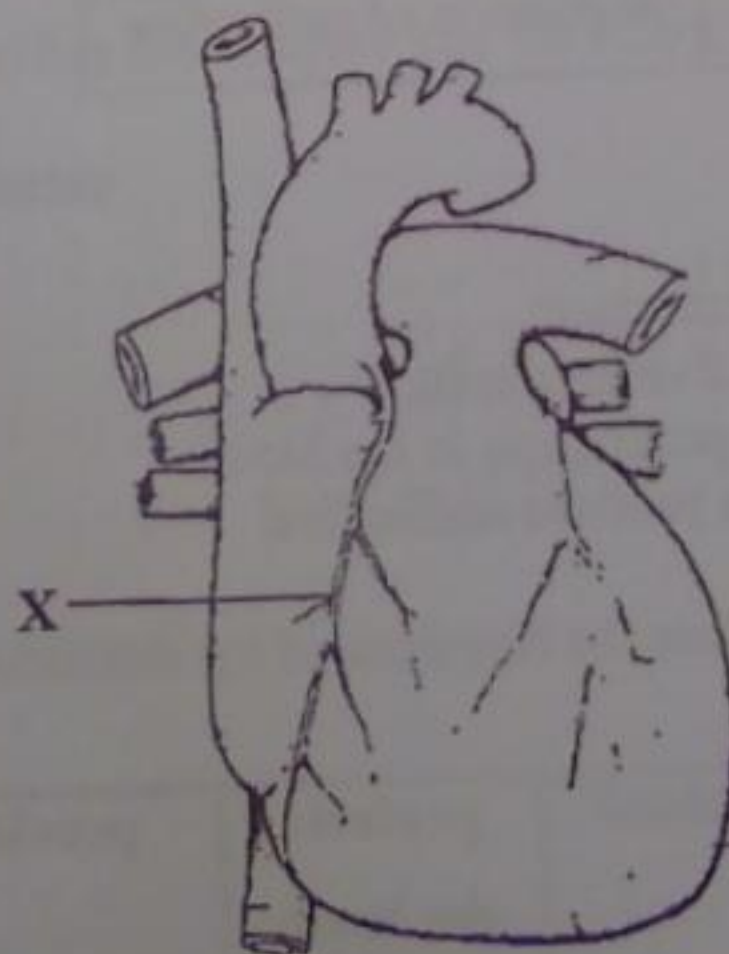
The diagram shows a cross-section of the human heart.



Which vein carries oxygenated blood?

18

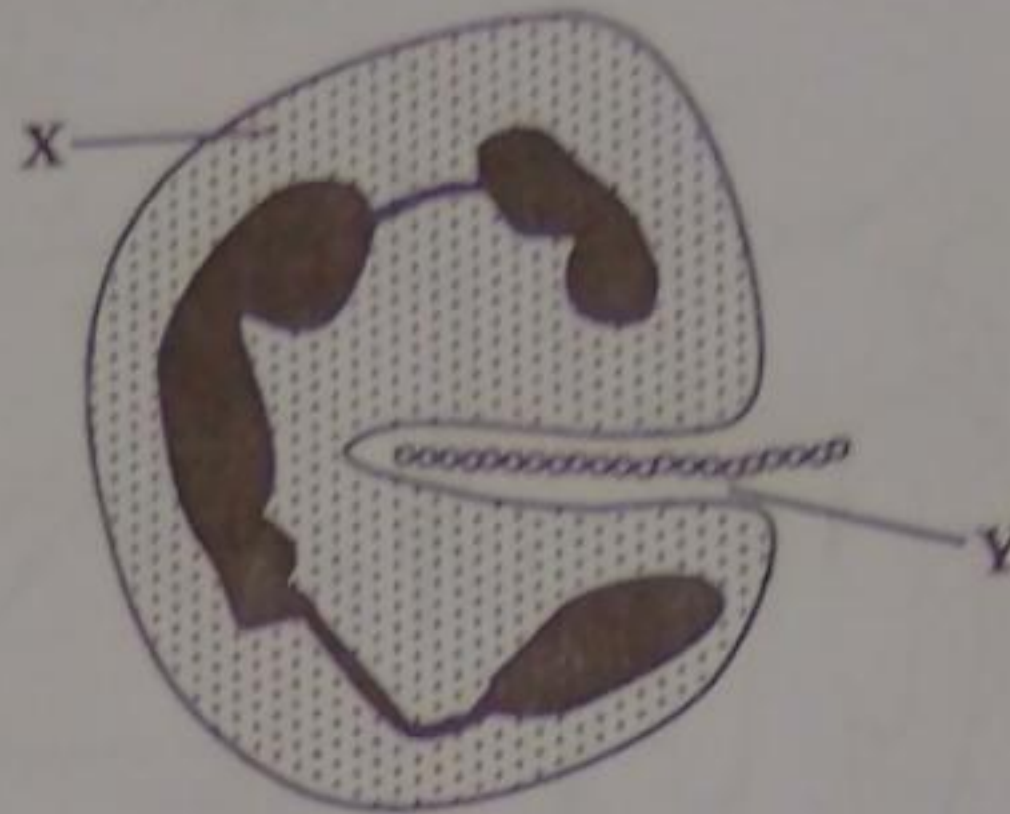
The diagram shows the external view of a human heart.



What is the effect of a blockage in vessel X?

- A stroke
- B heart failure
- C high blood pressure
- D low blood pressure

- 19 The diagram shows a blood cell X.



Which row correctly identifies X and what is happening at Y?

	X	Y
A	a lymphocyte	ingestion of bacteria
B	a lymphocyte	production of antibodies
C	a phagocyte	ingestion of bacteria
D	a phagocyte	production of antibodies

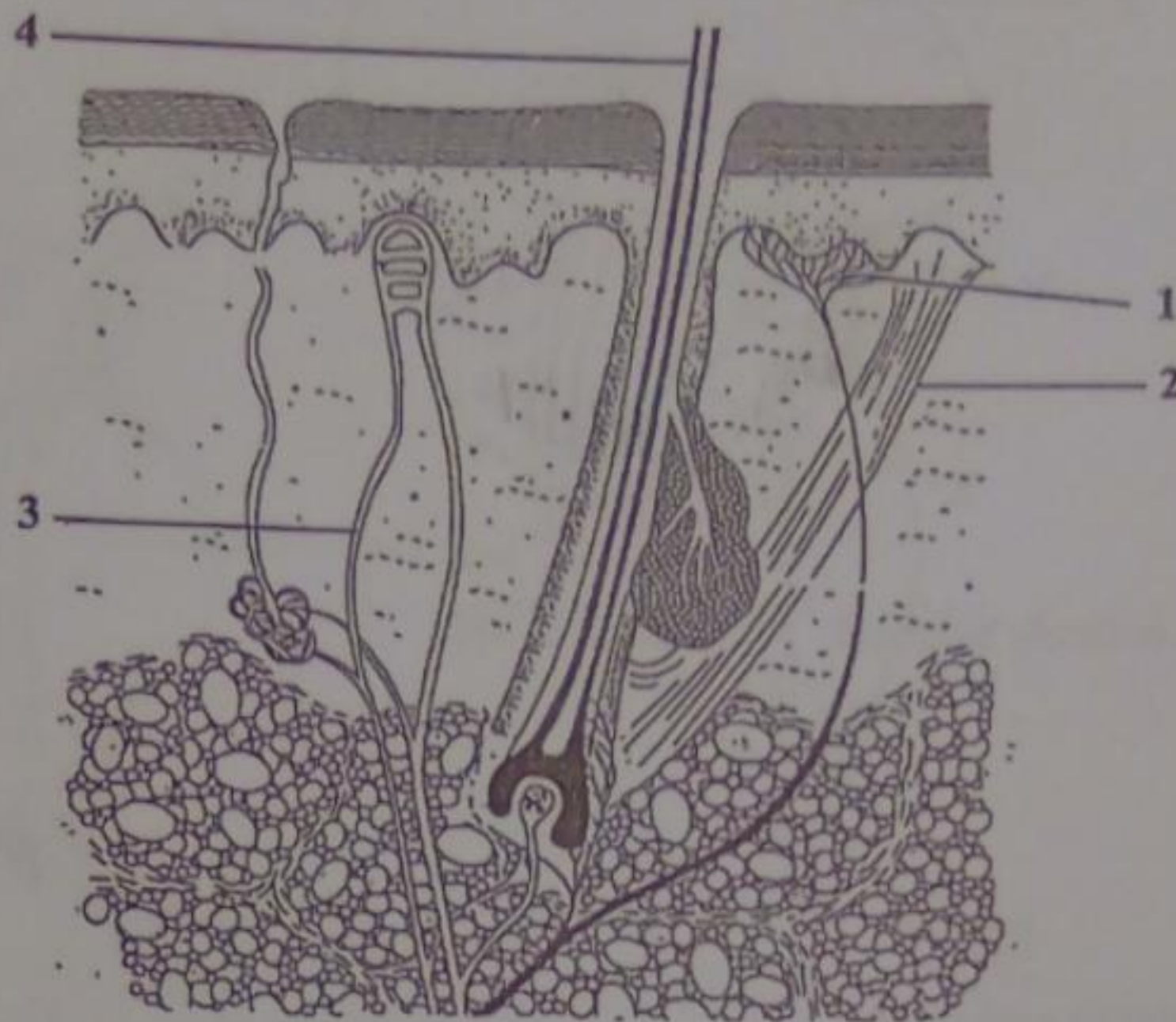
- 20 Passive immunity

- A gives immediate protection.
- B lasts longer than active immunity.
- C increases antibody production in the blood.
- D makes lymphocytes produce antibodies.

- 21 Which row correctly describes the functions of the skeleton?

	produces red blood cells	protects the brain	provides support for muscles
A	yes	no	yes
B	no	yes	no
C	no	no	no
D	yes	yes	yes

- 22 The diagram shows a section through the human skin. When a person walks into a very cold room, the hairs on the skin are raised.



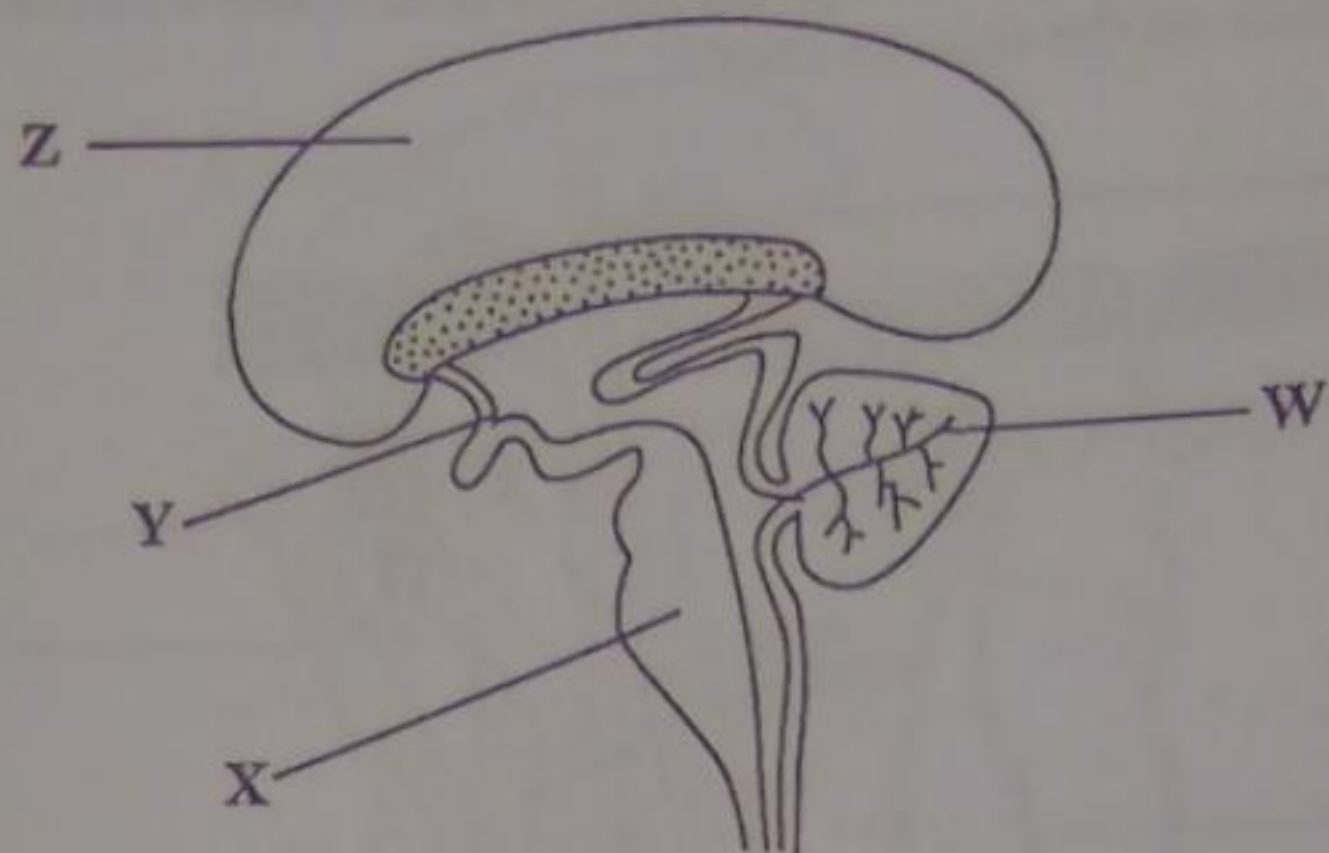
Which row correctly identifies the receptor and the effector for this reflex action?

	receptor	effector
A	1	2
B	1	3
C	2	3
D	4	3

- 23 In which organ is urea produced?

- A kidney
- B liver
- C skin
- D pancreas

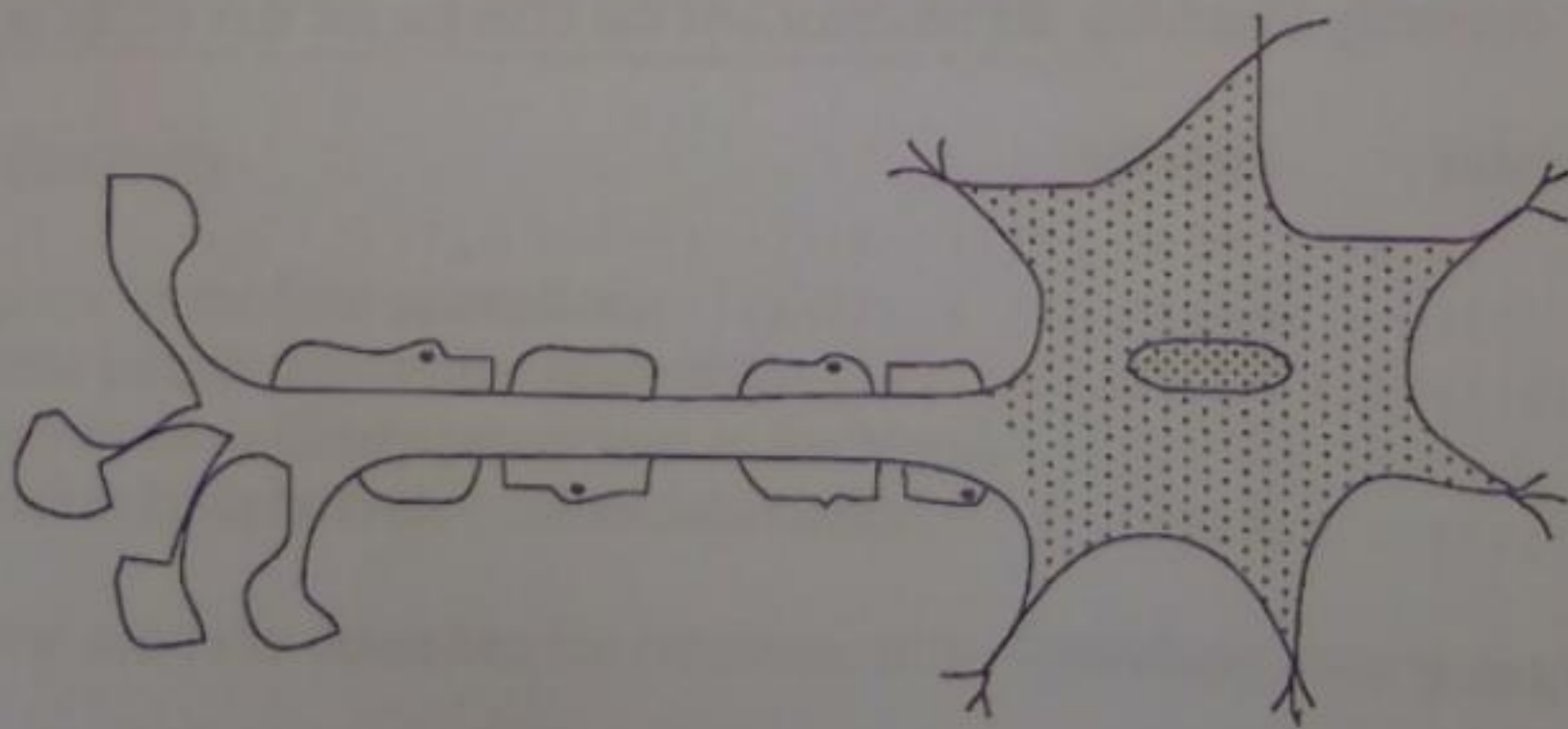
- 24 The diagram shows a section through the human brain.



Which region controls balance?

- A W
- B X
- C Y
- D Z

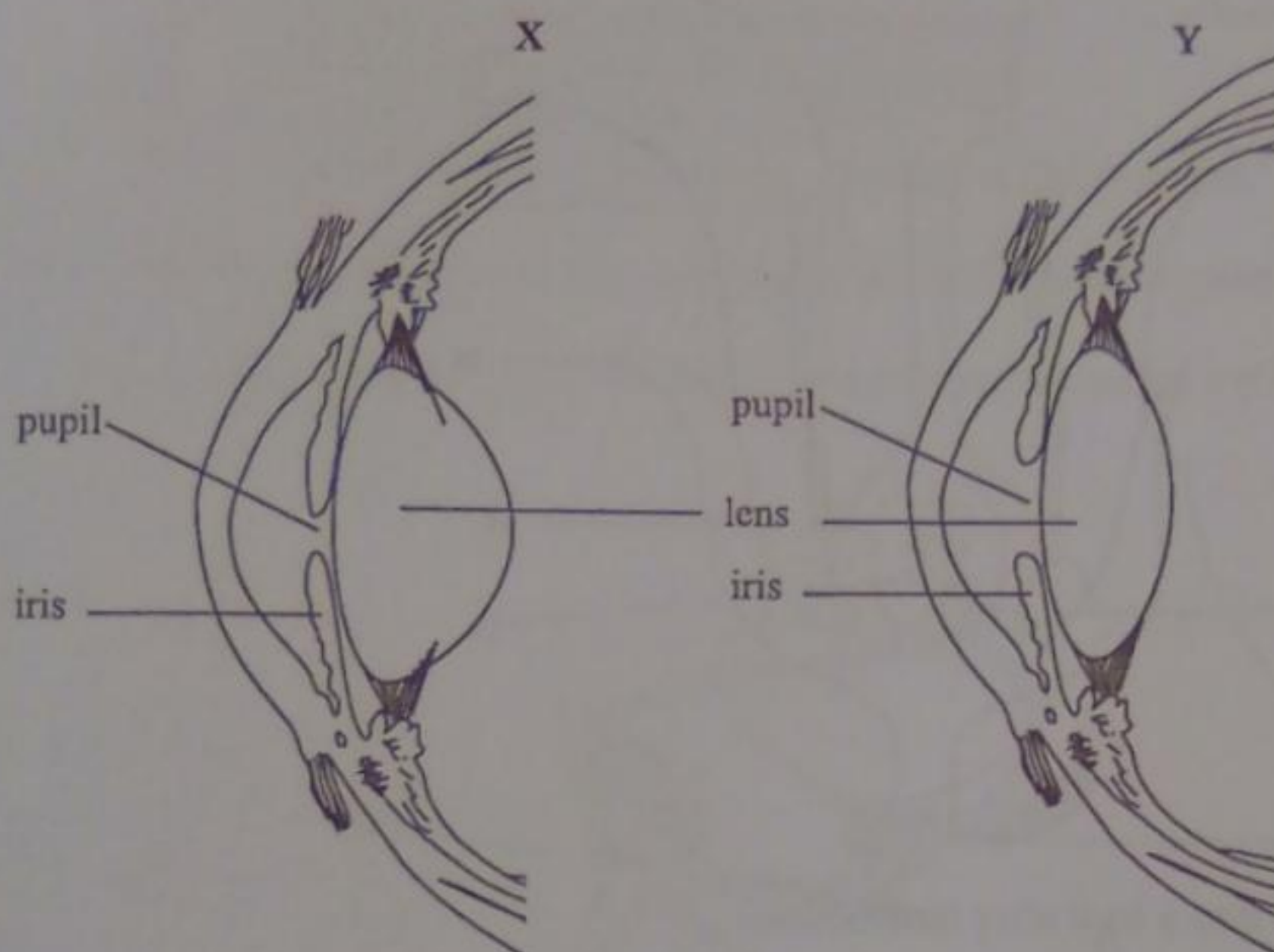
- 25 The diagram shows a neurone.



Which row correctly identifies this neurone and its function?

	neurone	function
A	motor	connecting with muscle fibres
B	sensory	connecting with muscle fibres
C	motor	connecting with receptors
D	sensory	connecting with receptors

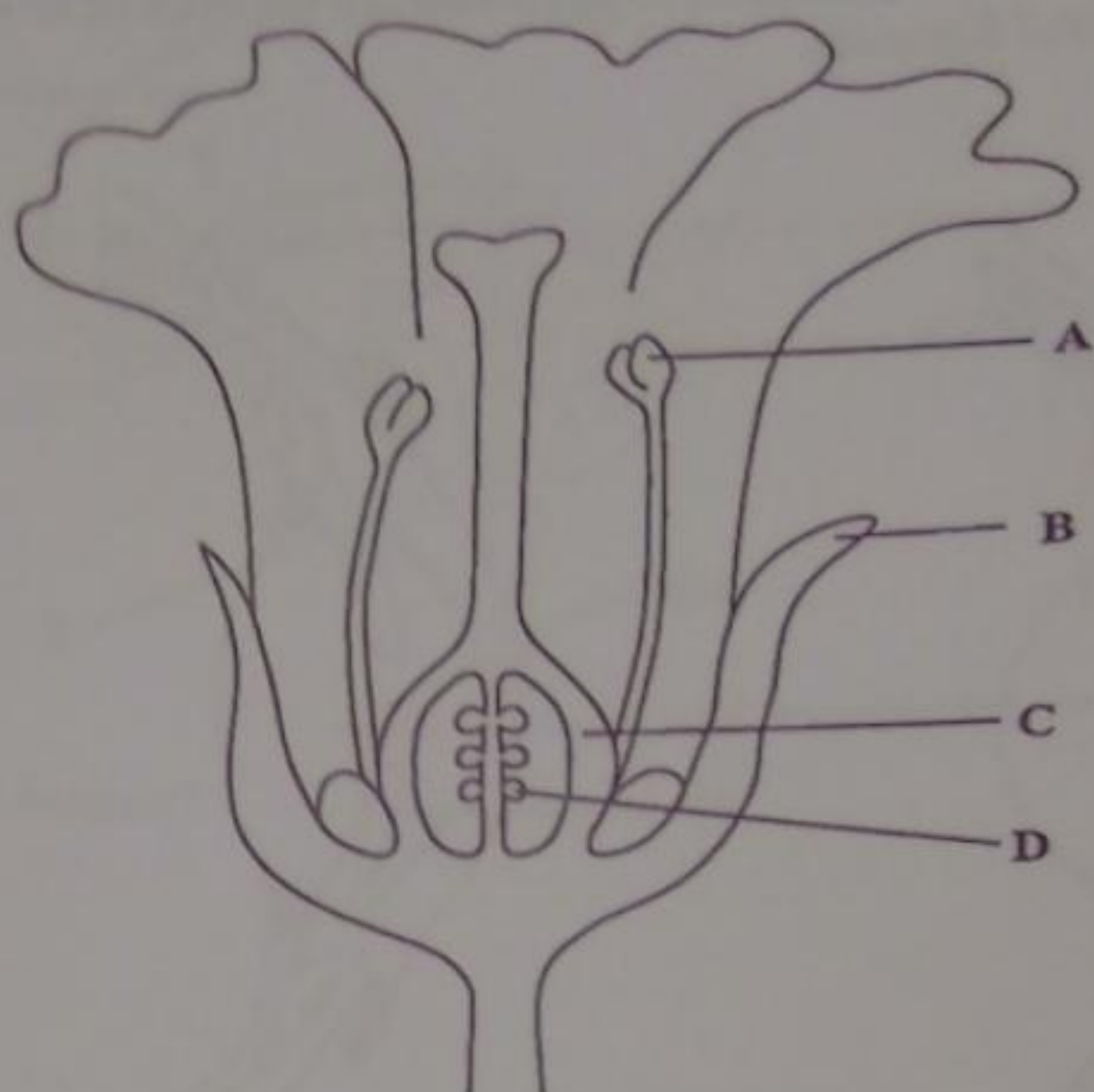
- 26 The diagrams show the horizontal sections through the front of an eye focussing on objects at different distances.



Which statement is correct?

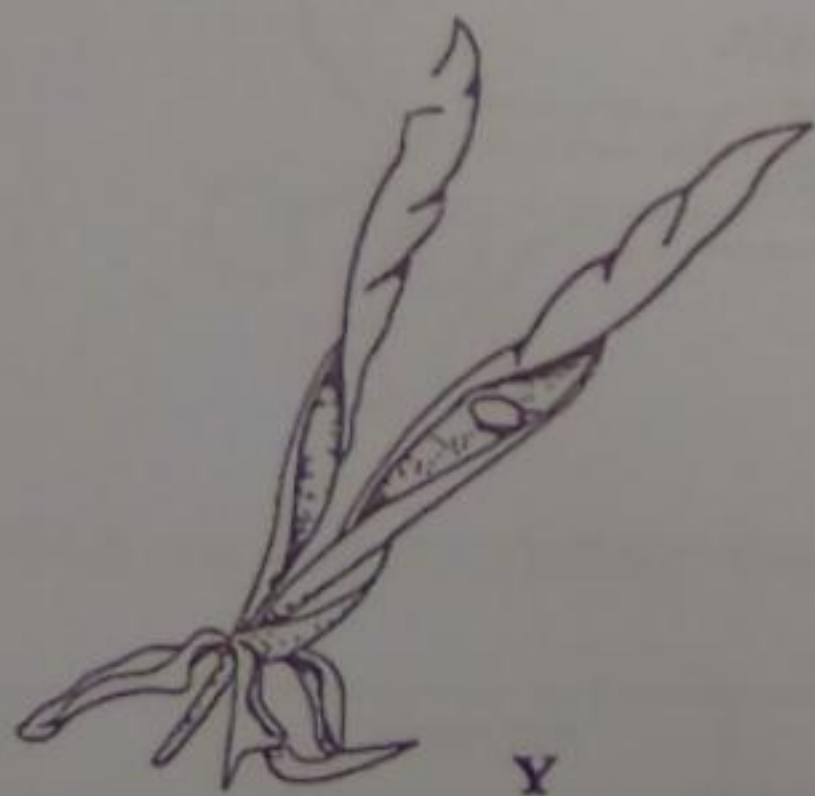
- A X is adapted to focus near objects in dim light.
 - B X is adapted to focus distant objects in bright light.
 - C Y is adapted to focus near objects in bright light.
 - D Y is adapted to focus distant objects in dim light.
- 27 Which symptom is **not** a sign of diabetes mellitus?
- A tiredness
 - B fever
 - C frequent urination
 - D glucose in urine
- 28 What is the short term effect of drinking alcohol?
- A Body temperature decreases.
 - B Liver cells are destroyed.
 - C Rate of impulse transmission is slowed down.
 - D Reaction time is reduced.

29 The diagram shows the structure of a flower.



Which part develops into a fruit after fertilisation?

30 The diagram shows a fruit, Y.



What is the method of seed dispersal for fruit Y?

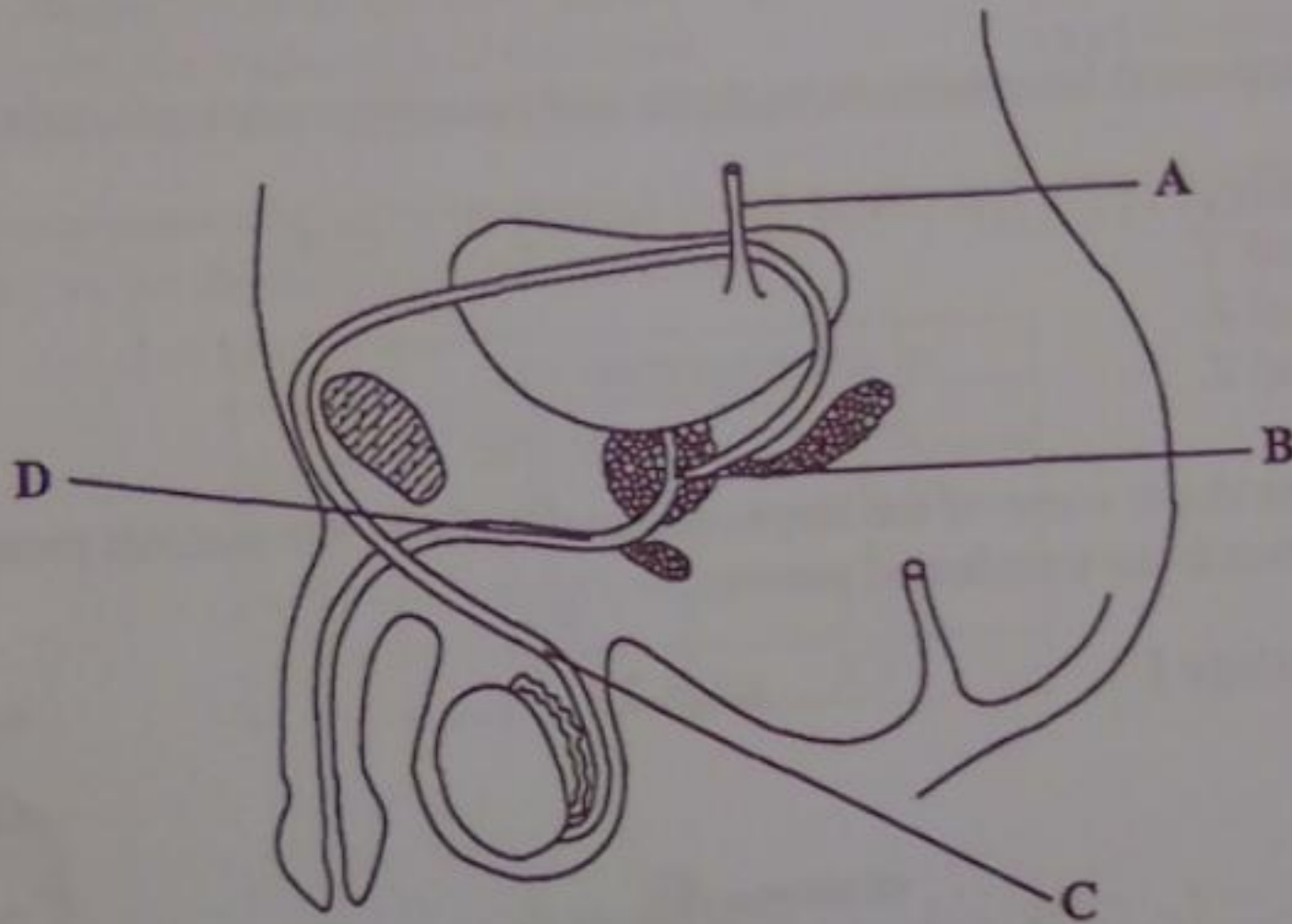
- A animal
- B self
- C water
- D wind

31 Which statement is true about asexual reproduction in plants?

- A Insects are always involved in the process.
- B Plants complete their life cycle in one year.
- C Survival rate of the plants is low.
- D The offspring are genetically identical to their parents.

32 The diagram shows part of the human male reproductive system.

At which point, A, B, C or D should vasectomy be carried out?



33 How does the intra-Uterine-Device (UDI) contraceptive prevent pregnancy in women?

- A It removes semen from vagina.
- B It prevents the ovum from leaving the fallopian tube.
- C It blocks entrance of sperms through the cervix.
- D It prevents implantation.

- 34 The diagram shows three types of microorganisms X, Y and Z.



non cellular
X



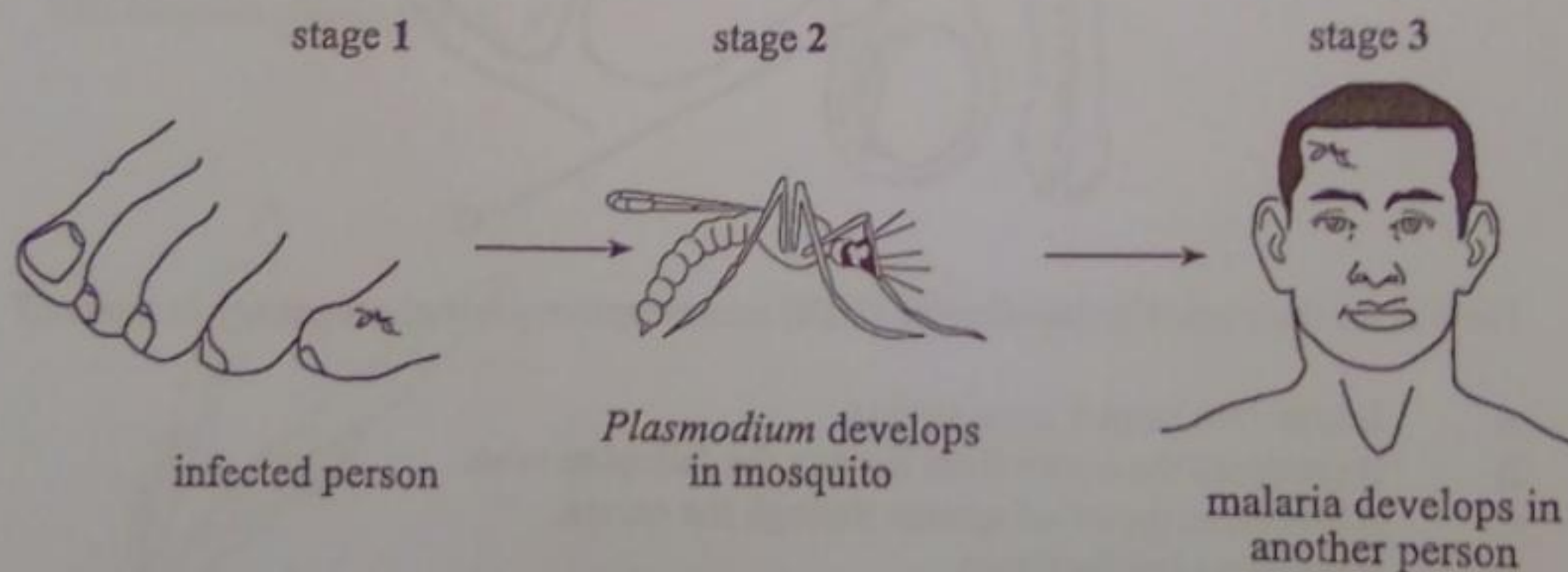
cell wall
Y



mycelium
Z

Which organism(s) have both saprophytic and parasitic modes of nutrition?

- A X only.
B X and Y.
C X and Z.
D Y and Z.
- 35 The diagrams show some of the stages in the transfer of the malaria parasite from an infected person to an uninfected person.



Which methods of control will help to reduce the spread of malaria at stages 1, 2 and 3?

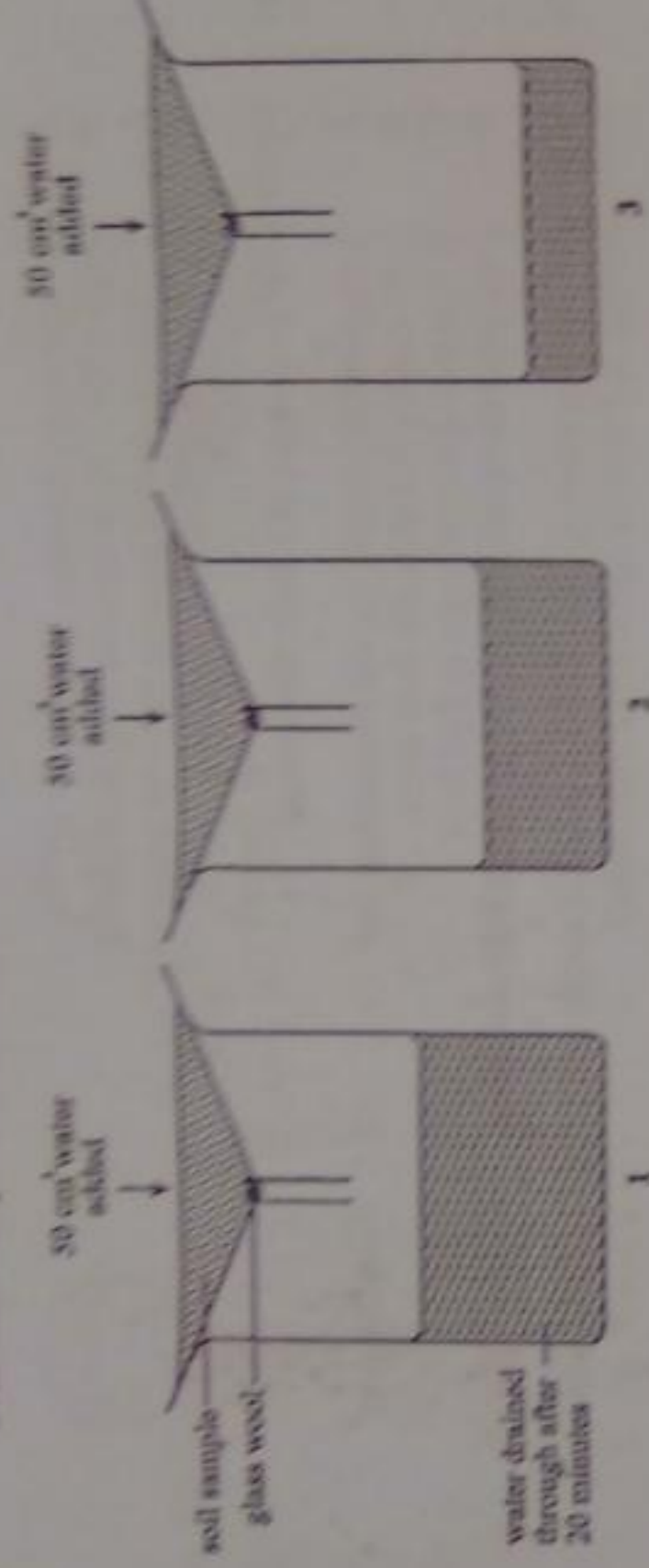
	stage 1	stage 2	stage 3
A	spray insecticide	take chloroquine	sleep under a mosquito net
B	sleep under a mosquito net	spray insecticide	take chloroquine
C	pour oil on water	take an antibiotic	spray insecticide
D	spraying insecticide	take an antibiotic	sleep under a mosquito net

- 36 Why are cholera patients given a preparation of salt and sugar solution?
- A to prevent dehydration of the body
 B to prevent cholera bacteria from multiplying
 C to remove cholera bacteria from the intestines
 D to increase antibody production
- 37 What is the advantage of a short food chain?
- A Less food is required by the consumers.
 B Less energy is lost in the food chain.
 C It is easier for consumers to find food.
 D Few producers can support a large number of consumers.
- 38 Which sex chromosome(s) must be present in the sperm and ovum for fertilisation to occur, leading to the development of a male child?

	sex chromosomes in	
	sperm	ovum
A	X	X
B	Y	X
C	XY	XX
D	XX	XY

Eukaryotes

- 39 The diagrams show the results of an experiment in which 50 cm³ of water was poured over dried samples of three different soils and allowed 20 minutes to drain.

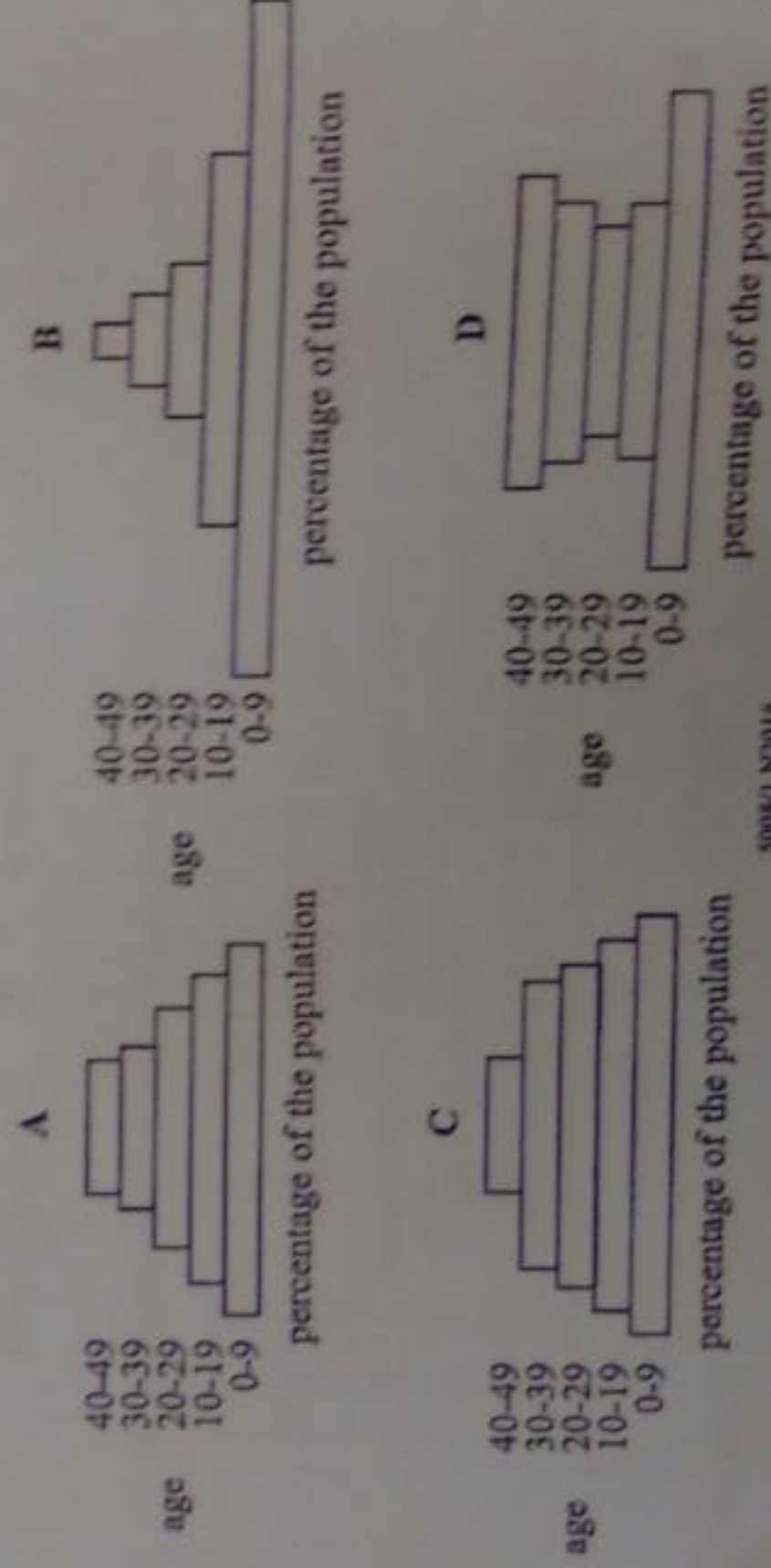


Which row correctly identifies the three soil samples?

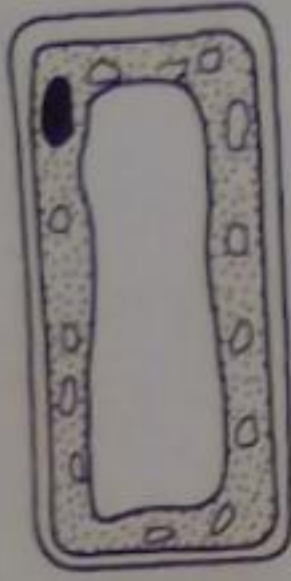
	1	2	3
A	clay	loam	sand
B	sand	clay	loam
C	sand	loam	clay
D	clay	sand	loam

- 40 The diagrams show the age-pyramid of four different countries.

Which country is likely to have the largest population of the 60-70 year age group after 20 years?



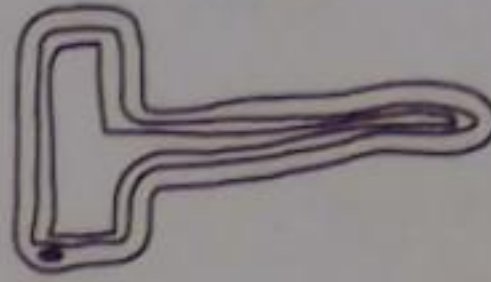
- 1 The diagrams show four different specialised cells.
Which cell functions by changing its length?



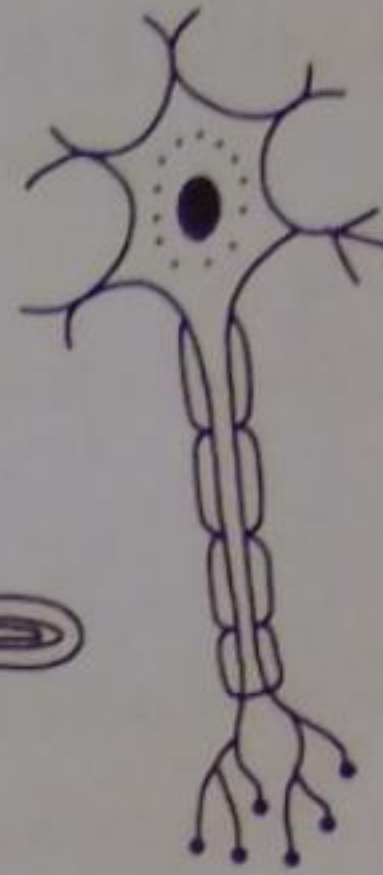
A



B



C



D

- 2 The information refers to the functions of some cells.

Which one is a red blood cell?

Key: ✓ ability to carry out function
 x inability to carry out function

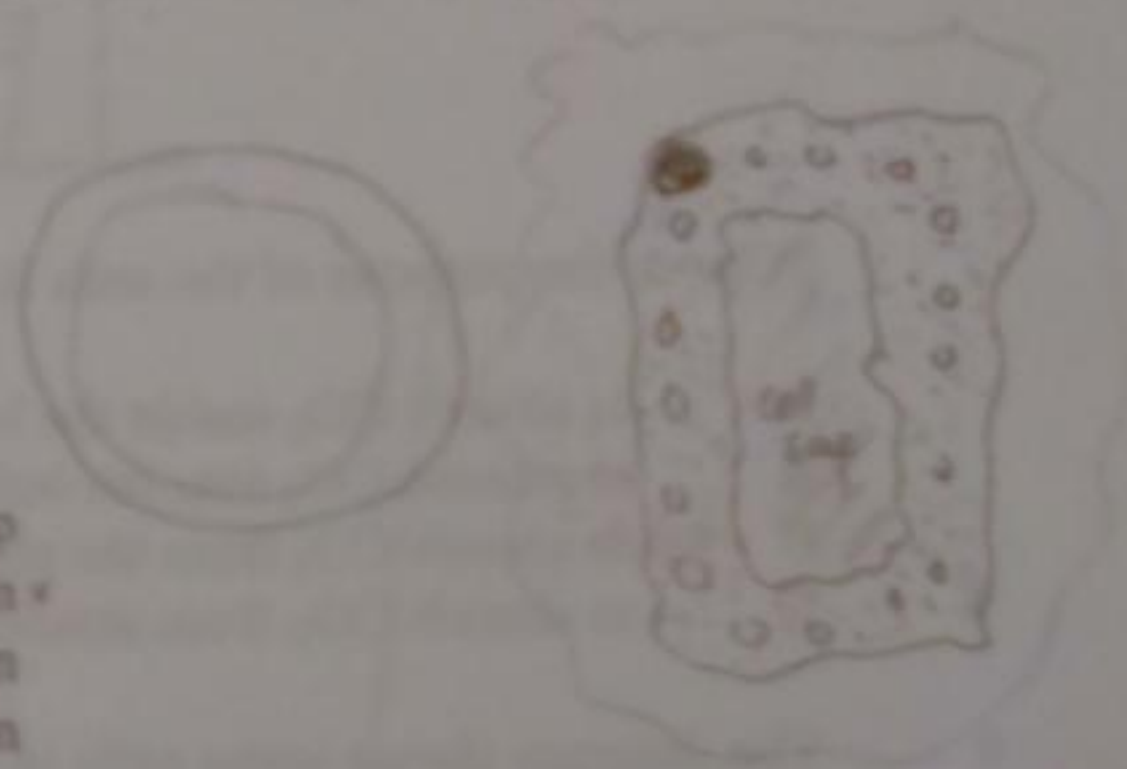
	absorption	transport	contraction
A	✓	x	✓
B	✓	✓	x
C	x	x	✓
D	x	✓	x

- 3 Which is correct about the concentration of water molecules in the cytoplasm, cell sap and the external solution during plasmolysis?

Key

> greater than
 = equal to

- A external solution > cytoplasm > cell sap
 B cytoplasm = cell sap > external solution
 C cell sap > cytoplasm > external solution
 D cell sap > cytoplasm = external solution



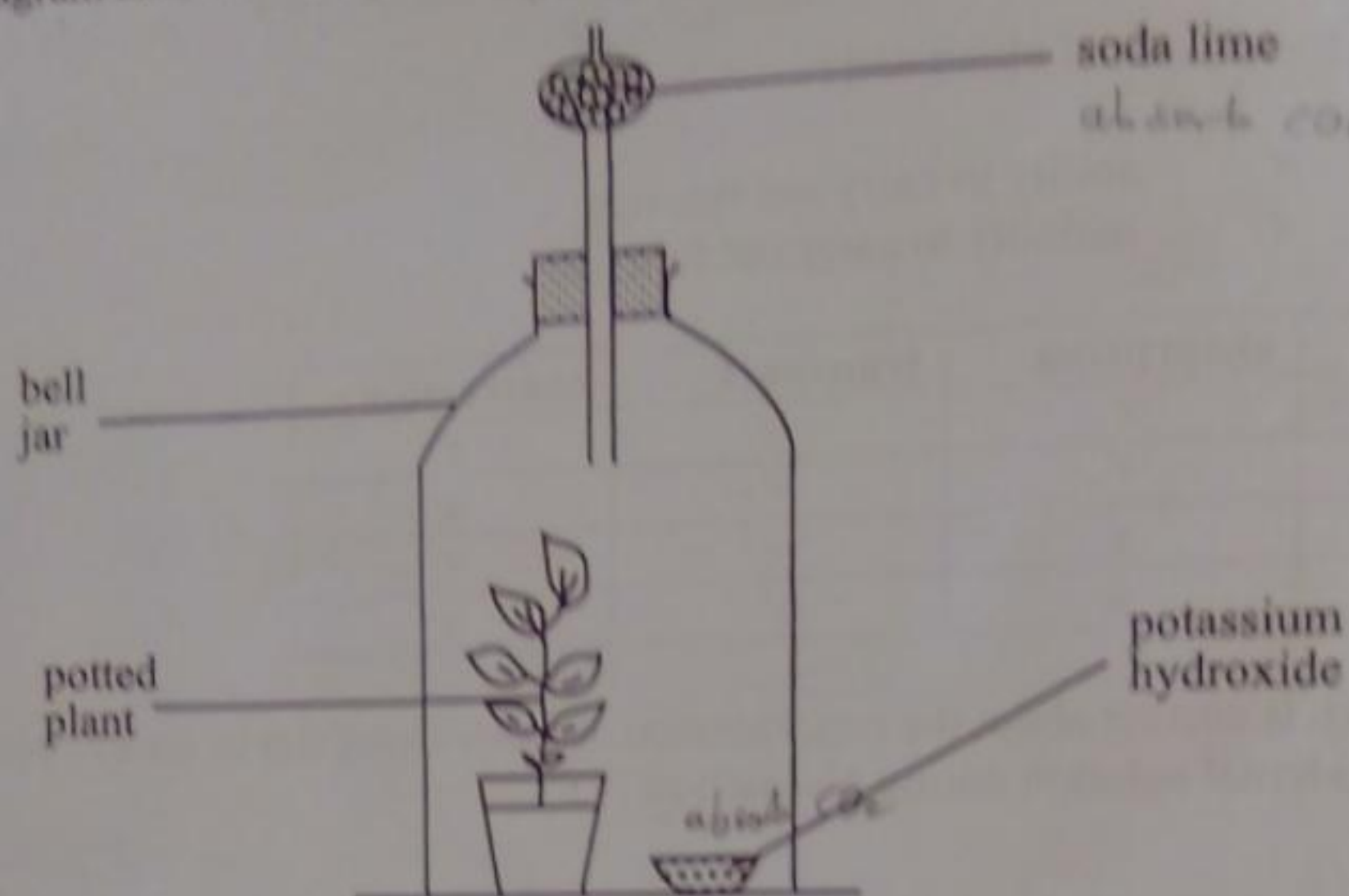
- 4 Which row of elements is correct for the given deficiency symptoms in plants?

	poor root growth	yellowish leaves	premature death
A	potassium	phosphorus	nitrogen
B	phosphorus ✓	nitrogen ✓	potassium ✓
C	phosphorus ✓	potassium	nitrogen
D	nitrogen	phosphorus	potassium

N yellow leaves
 P purple roots
 K brown fl. drop

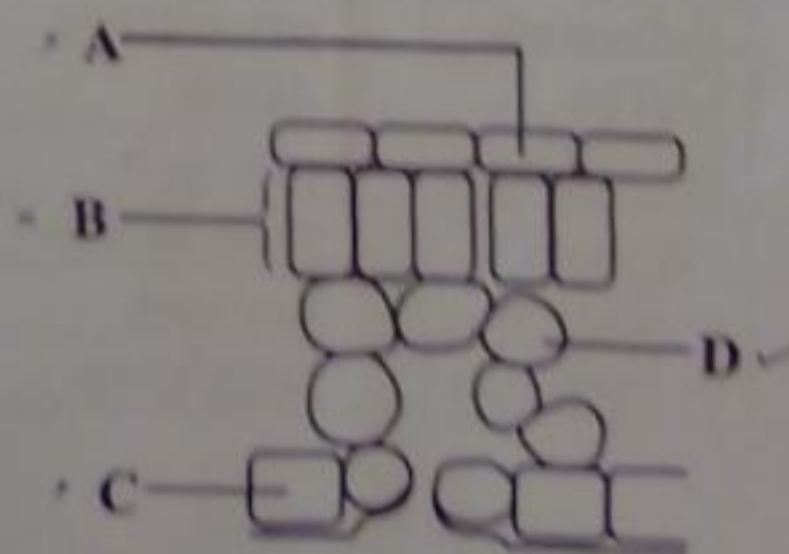
5

The diagram shows an experiment to investigate a requirement for photosynthesis.



What is the purpose of the soda lime?

- A to absorb CO₂ inside the bell jar ✗
 - B to release CO₂ into the bell jar ✗
 - C to absorb CO₂ from the plant ✗
 - D to absorb CO₂ from atmospheric air ✓
- 6 The diagram shows some cells in a leaf which was exposed to bright light.
- Which cell contains most starch after a period of exposure to bright light?



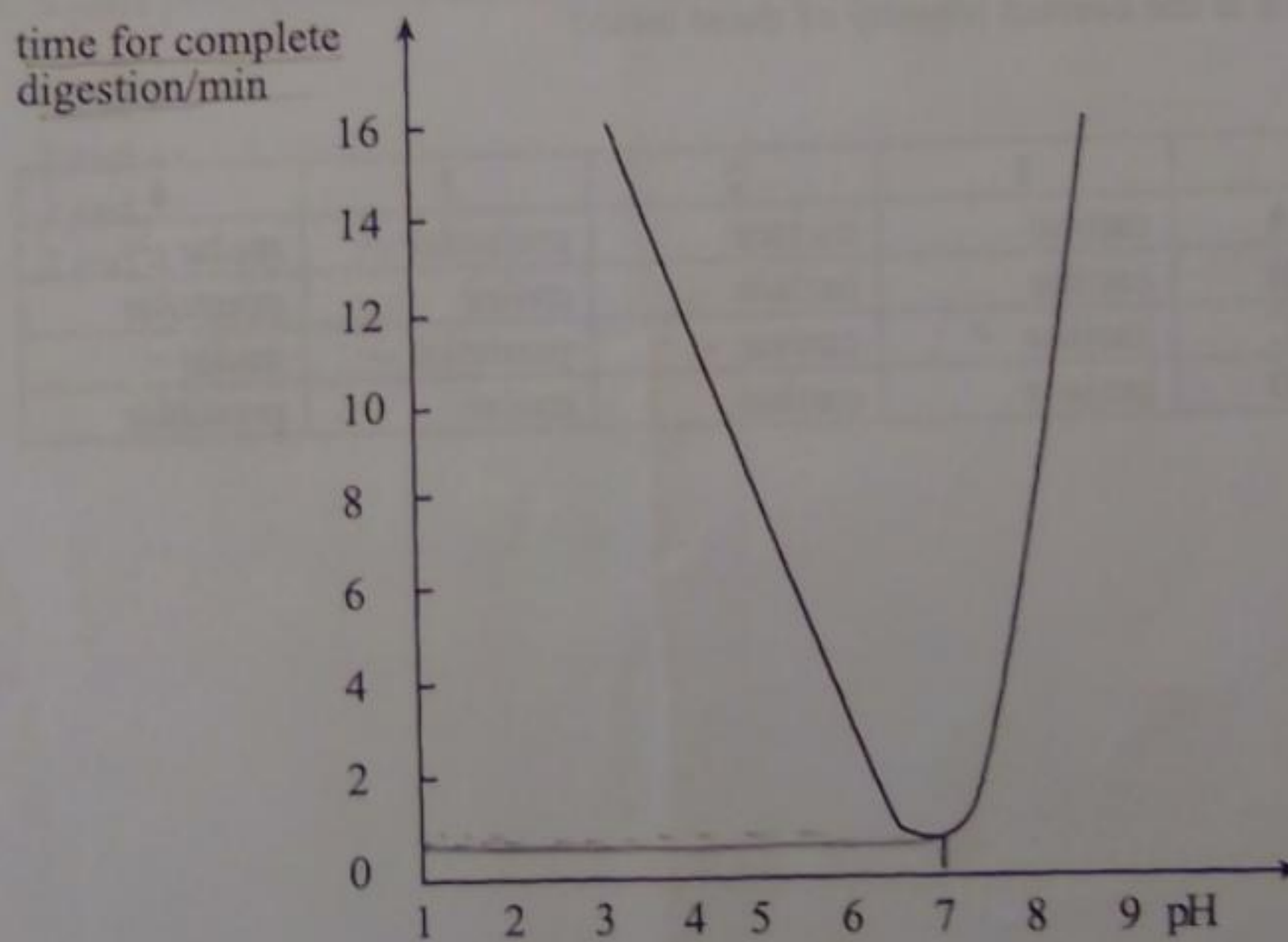
- 7 The table shows chemical elements present in four substances, A, B, C and D.

Which substance could be sucrose? $\rightarrow$ SUGARS CHO

Key ✓ present x absent

substance	carbon	hydrogen	nitrogen	oxygen
A	✓ ✓	✓ ✓	x	✓
B	✓	✓	✓	x
C	✓	✓	✓	✓
D	✓	x	✓	✓

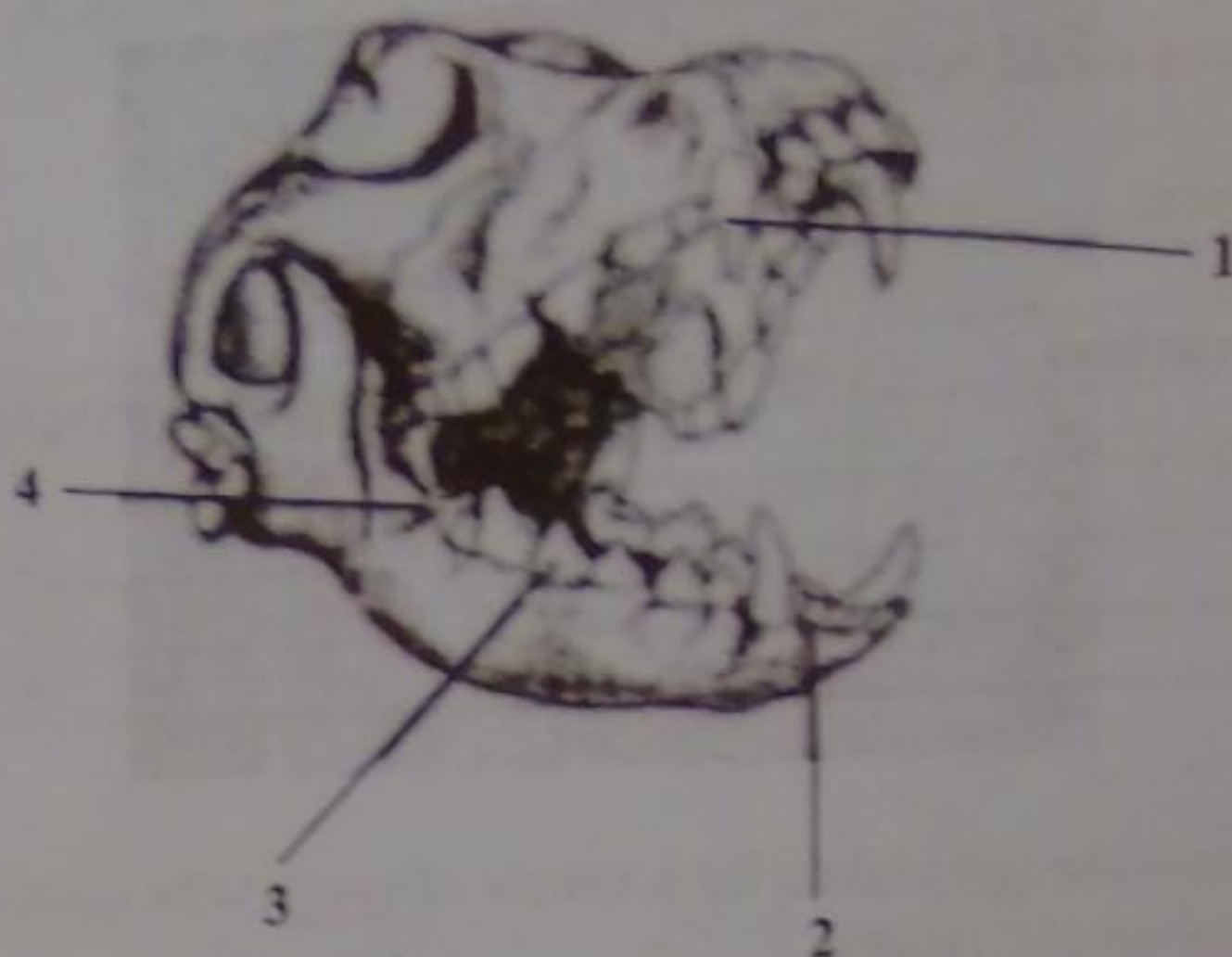
- 8 The graph shows the time taken for a sample of starch to be completely digested by amylase at various pH values.



What is the optimum pH for amylase?

- A 2
- B 4
- C 7
- D 8

The diagram shows a skull and teeth of a dog.

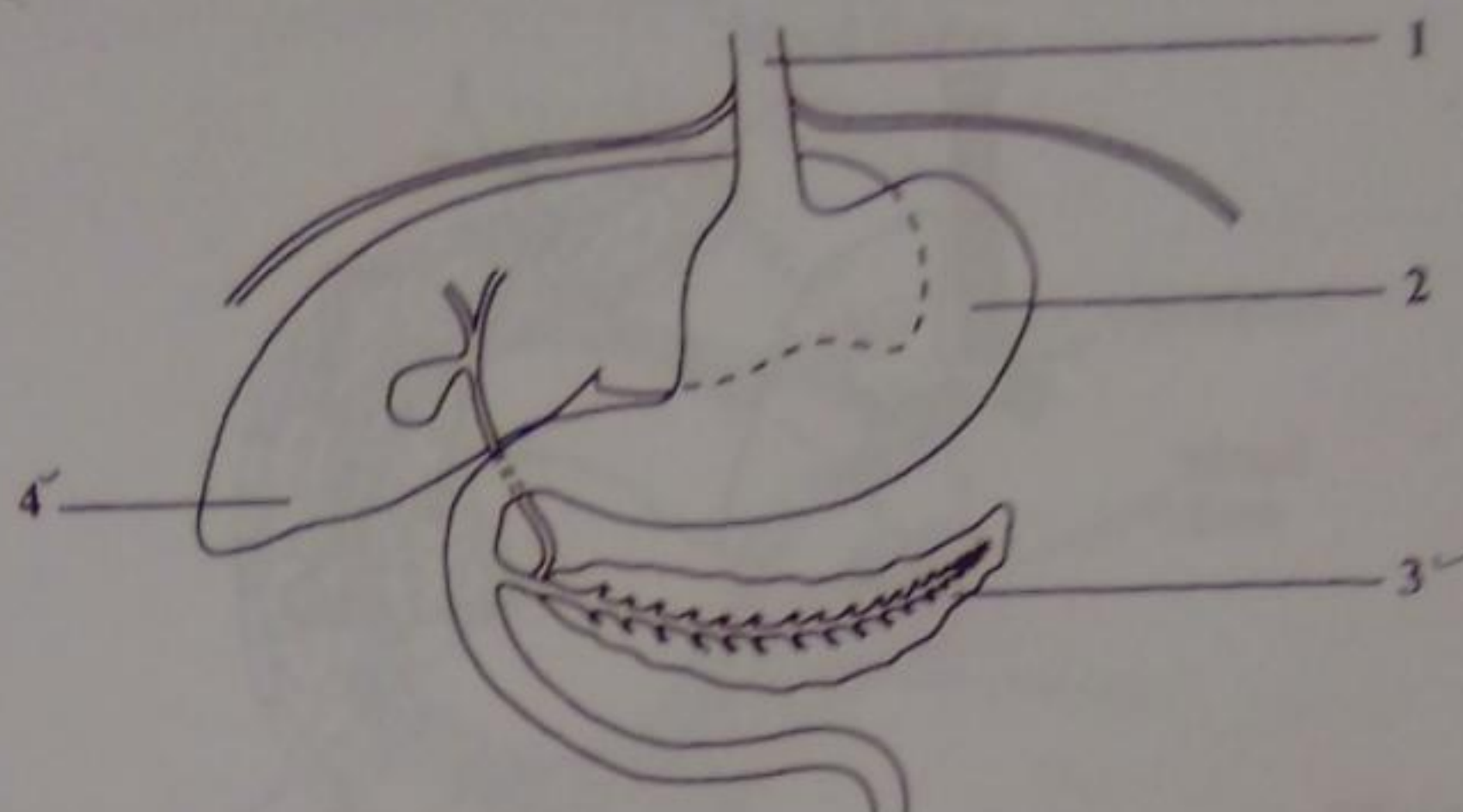


Which is the correct identity of these teeth?

	1	2	3	4
A	canine	incisor	premolar ✓	molar ✓
B	canine	incisor	molar	premolar
C	incisor ✓	canine ✓	premolar ✓	molar ✓
D	incisor	canine	molar	premolar

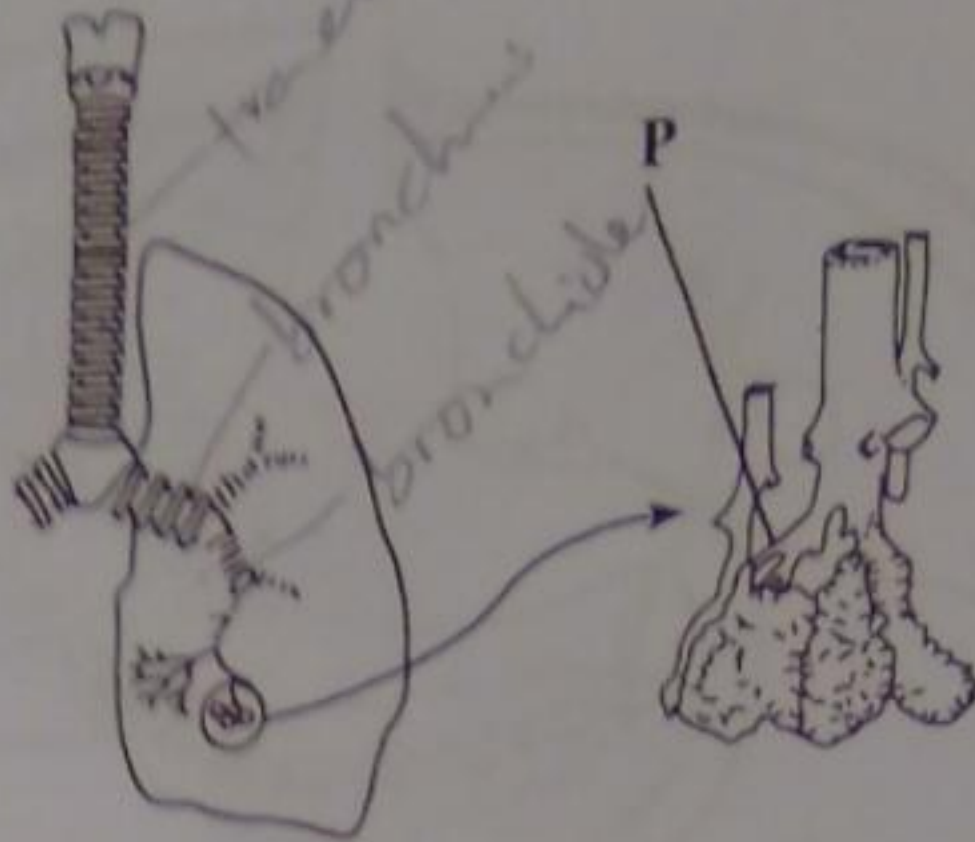
10

The diagram shows part of the human digestive system.



Which organs produce substances involved in fat digestion?

- A 1 and 3 ✓
- B 3 and 4 ✓
- C 2 and 3 ✓
- D 2 and 4 ✓

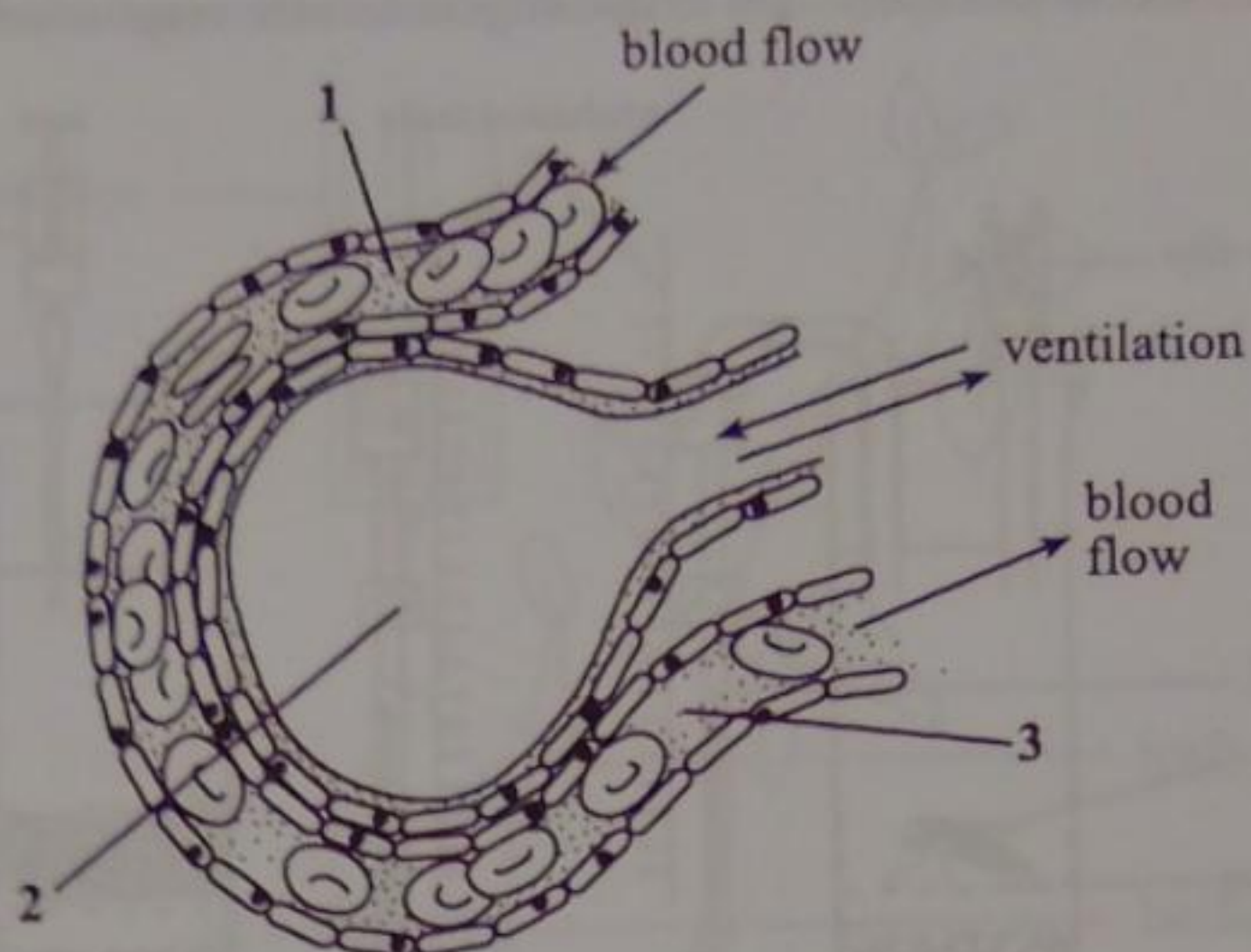


What is P?

- A bronchus *
- B alveolus *
- C alveolar wall *
- D bronchiole *

12

The diagram shows a section of an alveolus.



Which are the relative oxygen concentrations in 1, 2 and 3?

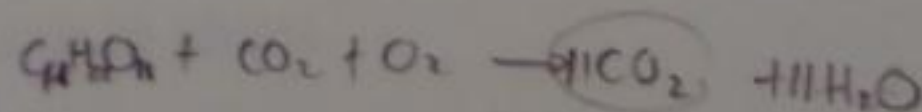
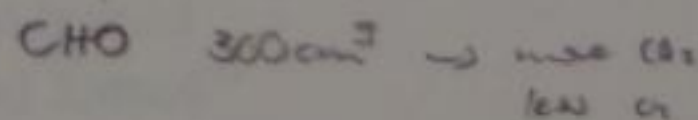
	1	2	3
A	low ✓	low ✗	high ✓
B	low ✓	high ✓	high ✓
C	high ✗	low ✗	low ✗
D	low ✓	high ✓	low ✗

13

Excess sugar was burnt in 300 cm^3 of exhaled air.

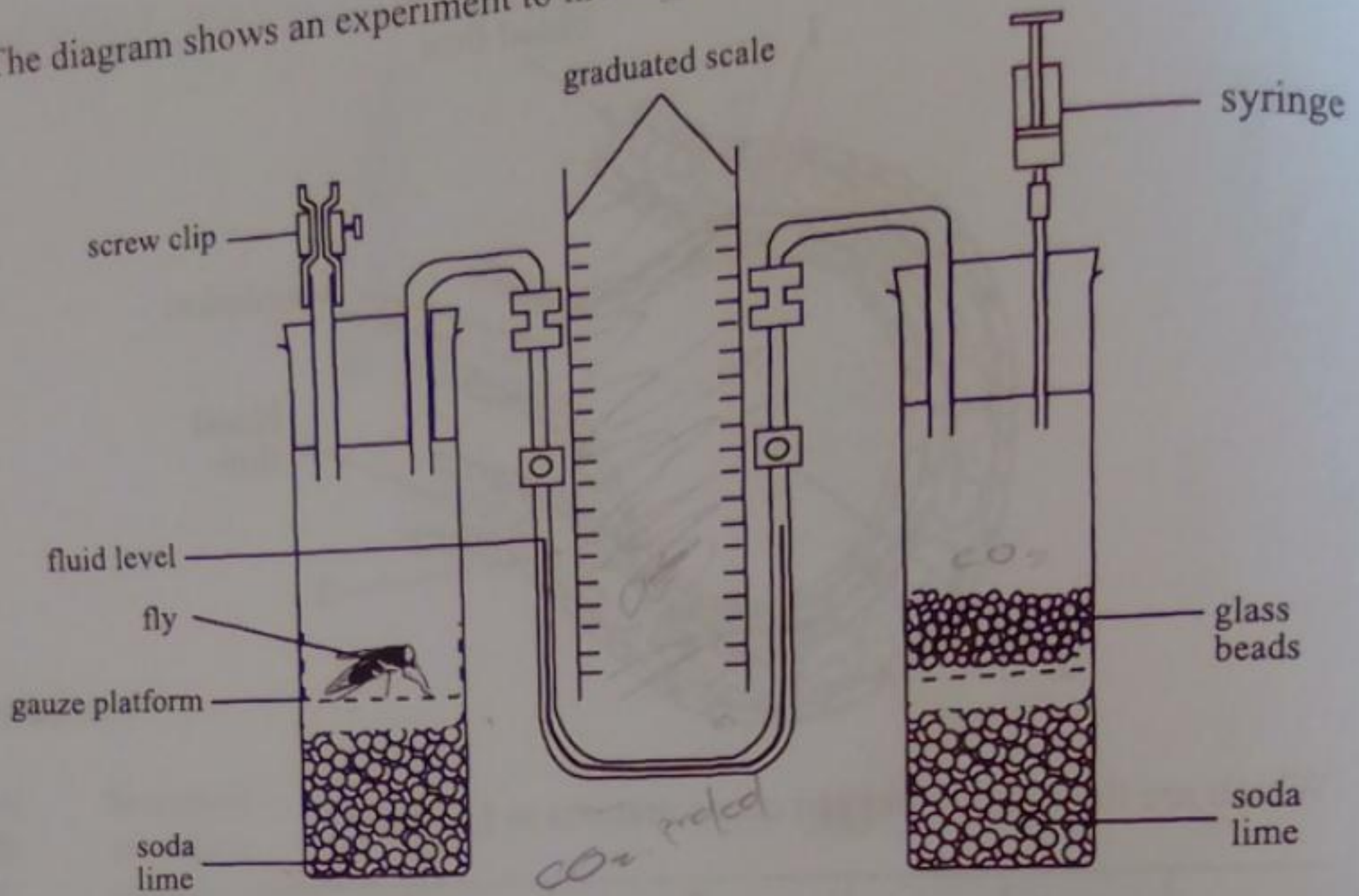
What volume of air remained at the end of the experiment?

- A 300 cm^3
 B 150 cm^3
 C 252 cm^3
 D 284 cm^3



14

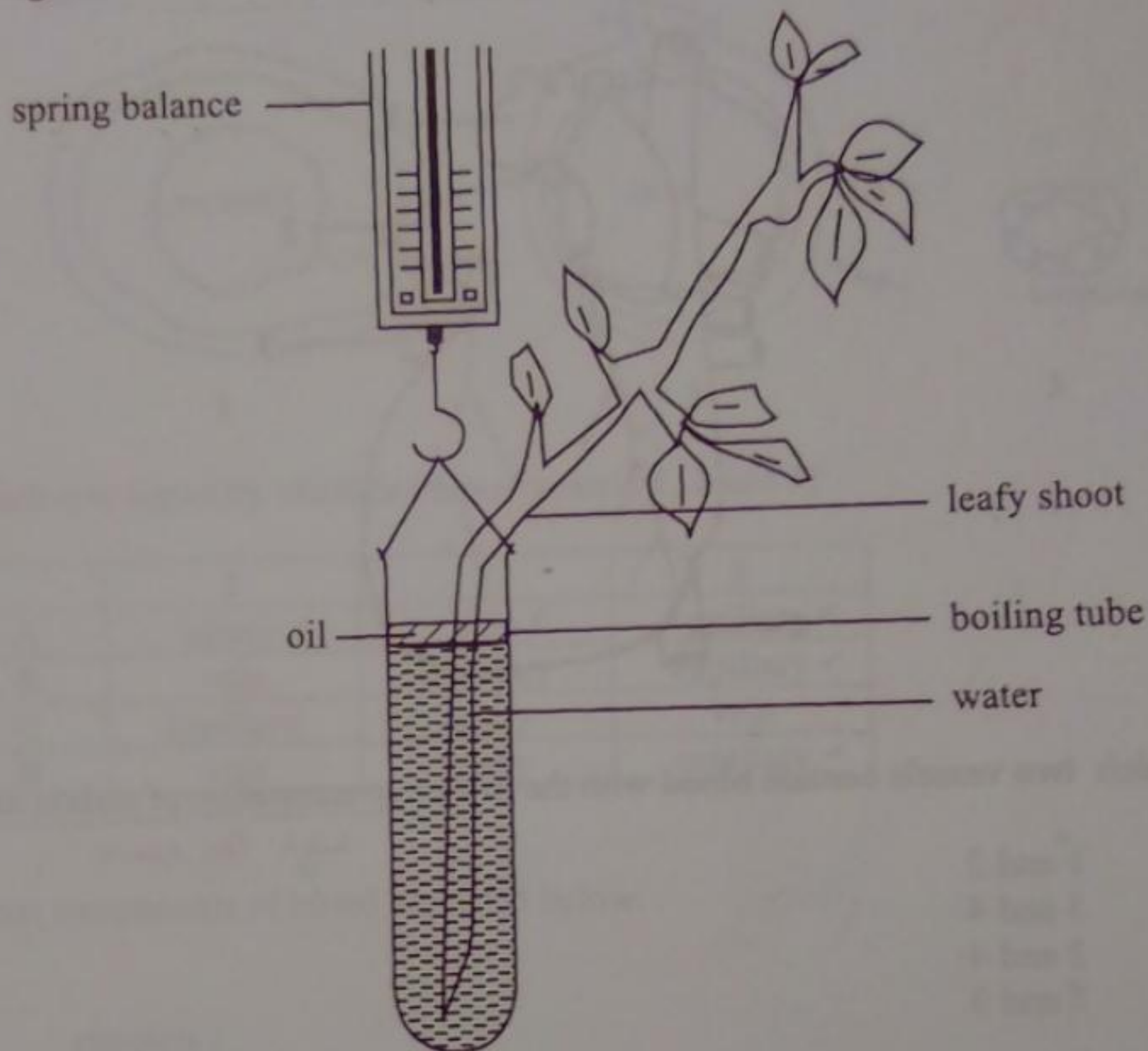
The diagram shows an experiment to investigate aerobic respiration.



What can be measured using this apparatus?

- A oxygen absorbed ✓
- B carbon dioxide released ✓
- C oxygen released ✗
- D carbon dioxide absorbed ✗

- 15 The diagram shows apparatus used to measure the rate of transpiration.



Which set of conditions causes the lowest decrease in mass per unit time?

- A cold, sunny and dry
- B warm, cloudy and dry
- C cold, cloudy and humid
- D warm, sunny and humid

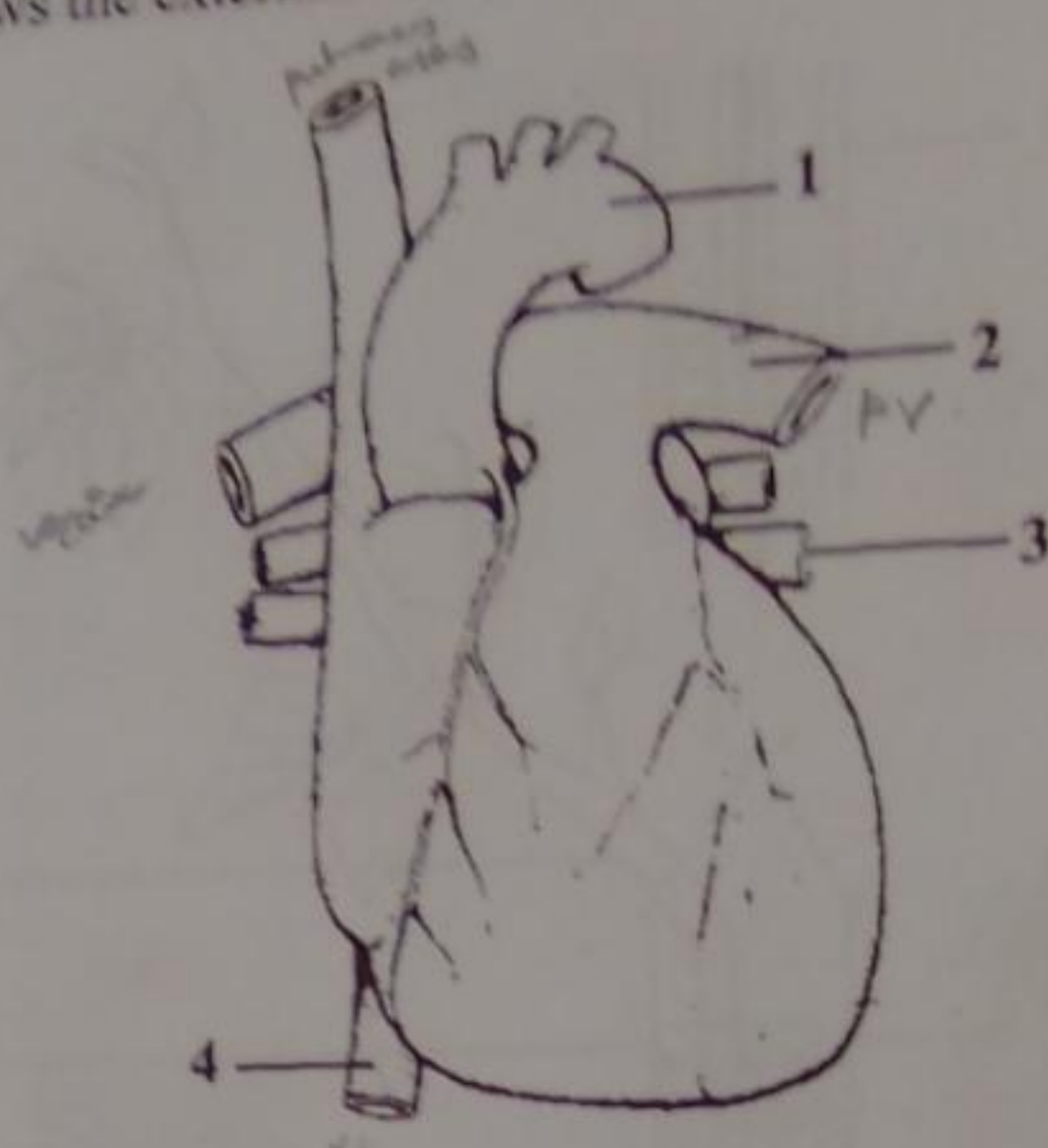
- 16 The following processes occur in plants.

1. movement of minerals from soil to root hair cells ✓ A.T
2. movement of water from xylem to mesophyll cells ○
3. movement of minerals from xylem to mesophyll cells △
4. movement of sucrose from phloem to developing fruits A.T

Which processes involve active transport?

- A 1 and 2 *
- B 2 and 3
- C 1 and 3 ✓
- D 1 and 4 ✓

- 17 The diagram shows the external view of a mammalian heart.



Which **two** vessels contain blood with the lowest concentration of carbon dioxide?

- A 1 and 2
B 3 and 4
C 2 and 4
D 1 and 3

high O₂ conc

18 The diagram shows three types of blood vessels.



1



2



3

Which row correctly identifies blood vessels 1, 2 and 3?

	1	2	3
A	artery ✓	vein ✓	capillary ✓
B	vein	capillary	capillary ✓
C	capillary	artery	vein
D	vein	artery	capillary ✓

19

Some components of blood are listed below.

1. platelets ✓
2. antibodies ✓
3. fibrinogen *
4. white blood cells ✓

Which two components protect the body from attack by bacteria?

- A 1 and 2
 B 1 and 3
 C 2 and 4
 D 3 and 4

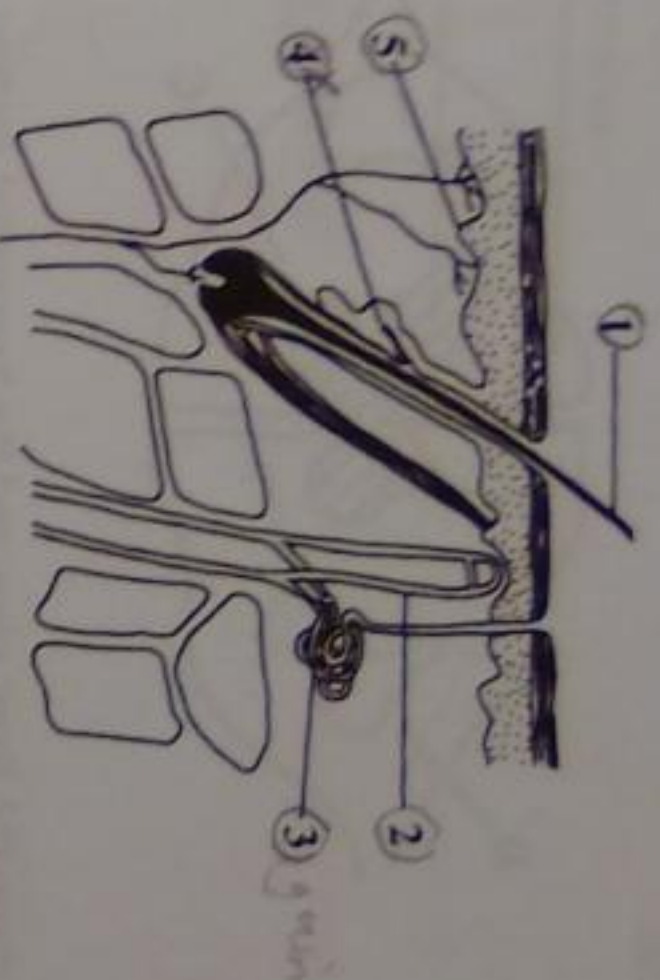
The diagram shows an elbow joint.



Which row correctly identifies X, Y and Z?

	X	Y	Z
A	tendon	ligament	cartilage
B	tendon	cartilage	ligament
C	ligament	cartilage	tendon
D	ligament	tendon	cartilage

- 21 The diagram shows a section of the human skin.



Which parts are involved in the control of heat loss?

- A 1, 2 and 3 ✓
 B 1, 2 and 4 ✓
 C 2, 3 and 4 ✓
 D 2, 3 and 5 ✓

- 22 Which are the two functions of the kidney?

function 1

function 2

- | | | |
|---|-------------------------------------|--------------------------------------|
| A | removal of glucose from blood * | control of water content of blood |
| B | removal of urea from blood ✓ | control of blood sugar * |
| C | removal of urea from blood ✓ | removal of excess water from blood ✓ |
| D | control of water content of blood ✓ | control of blood sugar * |

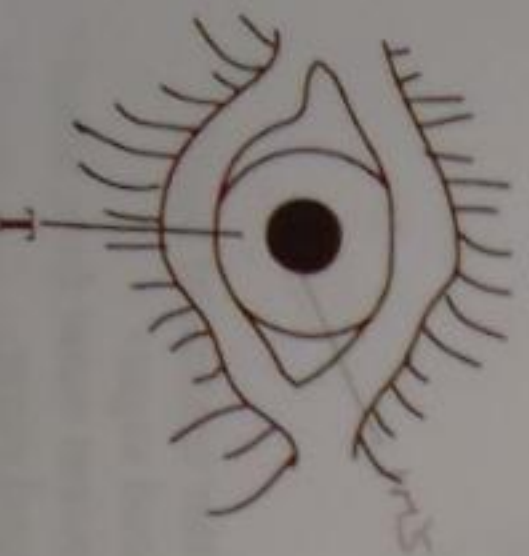
- 23 The diagram shows a cross-section of the spinal cord. Which part contains cell bodies of motor neurones?



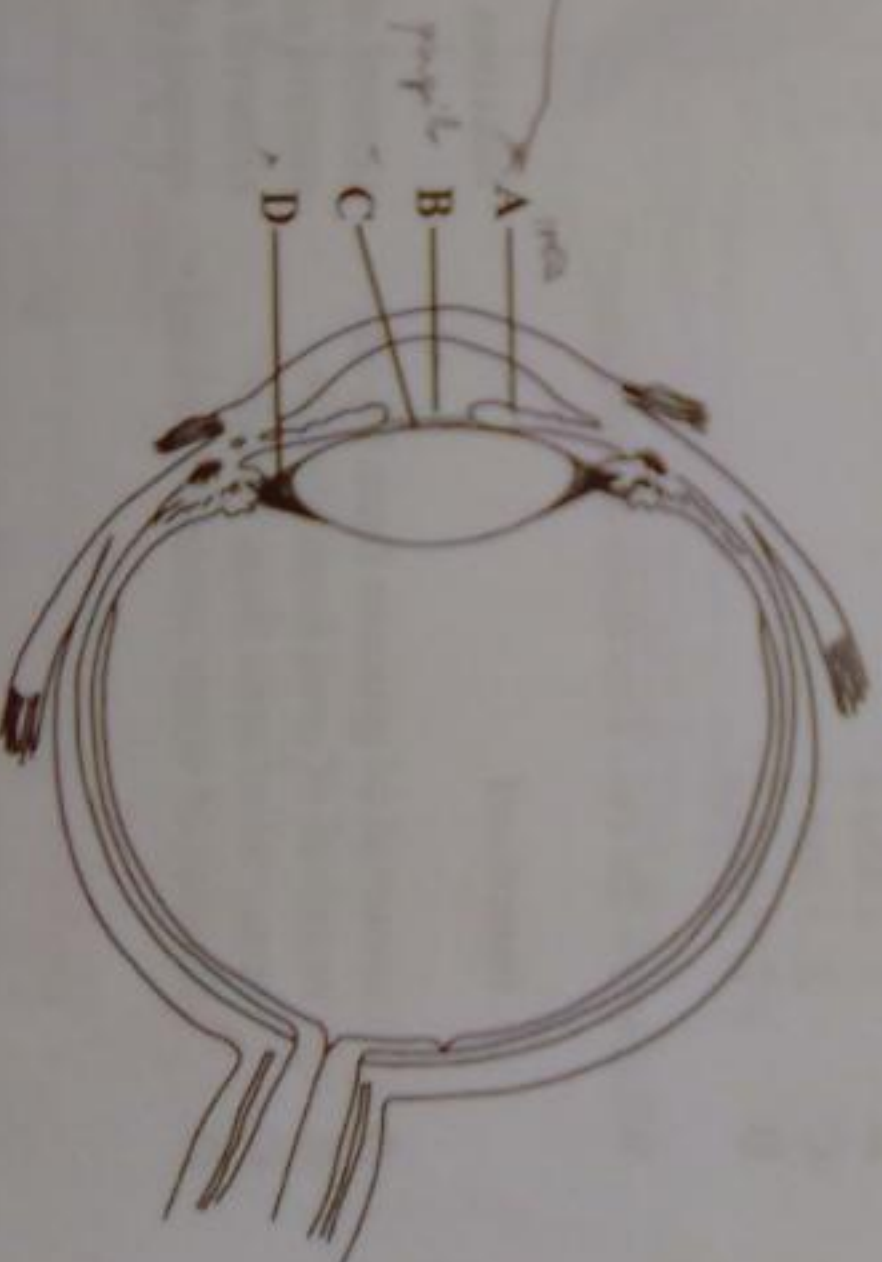
- 24 The diagrams show the front view and the horizontal section of the eye.

Which labelled structure on the horizontal section is the same as the part labelled T?

front view



horizontal section



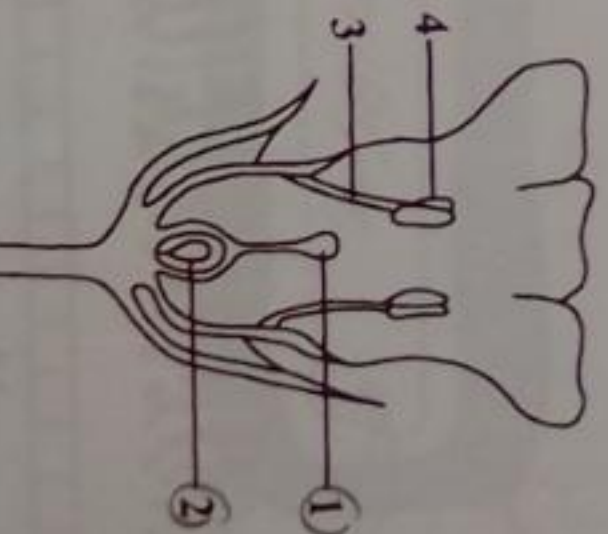
25 Which hormone raises glucose level in blood?

- A anti-diuretic hormone *
- B insulin ☒
- C adrenaline ☒
- D thyroxine *

26 Which drug is the least addictive?

- A heroin *
- B cocaine *
- C nicotine *
- D alcohol *

27 The diagram shows a section through a flower.



Which two structures are part of the pistil or carpel?

stamen

female reproductive organ

- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 3 and 4

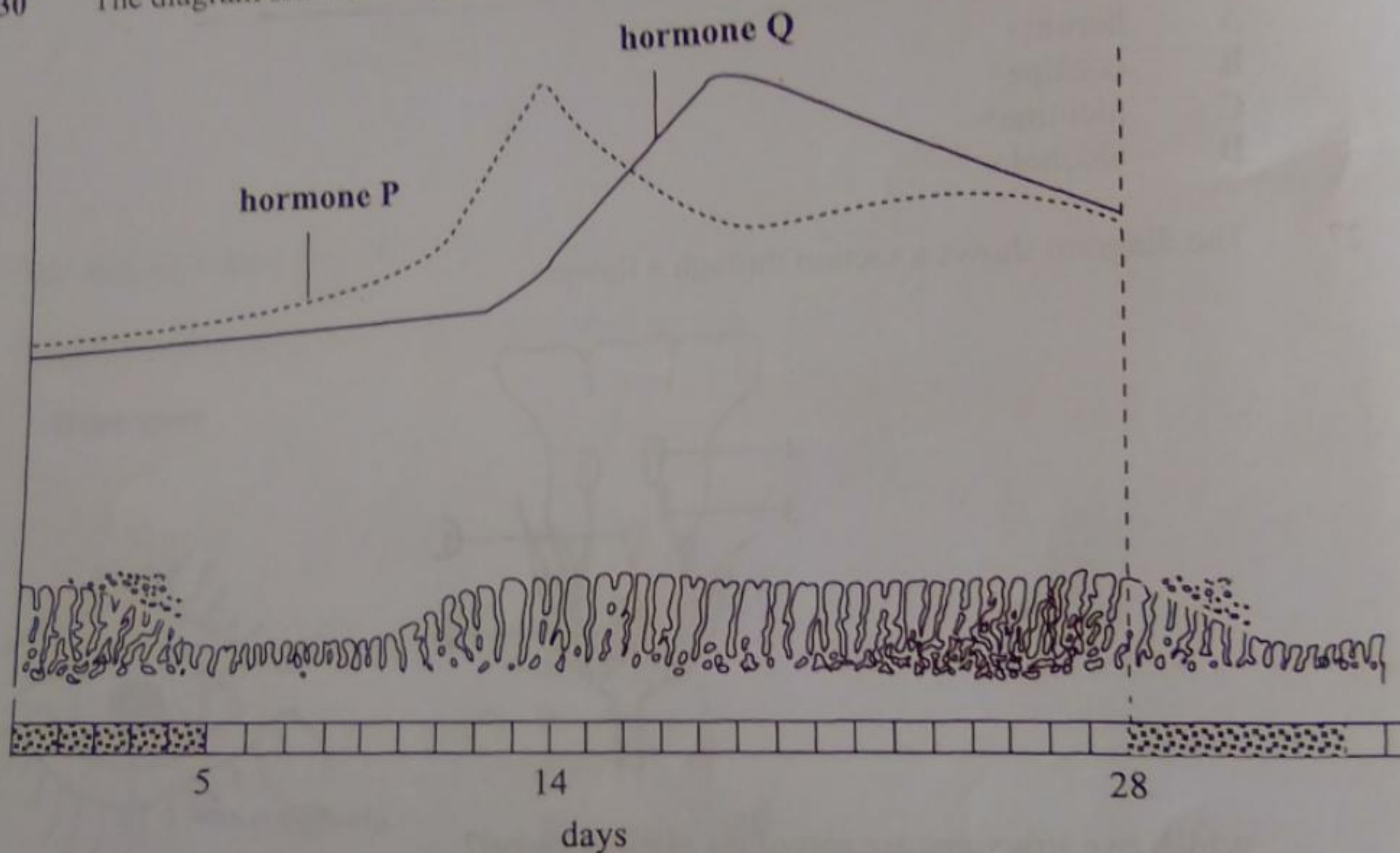
28 Which process produces the greatest variety of offsprings?

- A pollination between flowers on the same plant *
- B pollination between flowers of different plants of the same species *
- C pollination between flowers of different plants of different species
- D pollination within the same flower *

29 Which of the following would you find in a grass flower?

- A small nectaries
- B anthers with short filaments
- C small, smooth pollen grains ✓
- D sticky stigma ✓

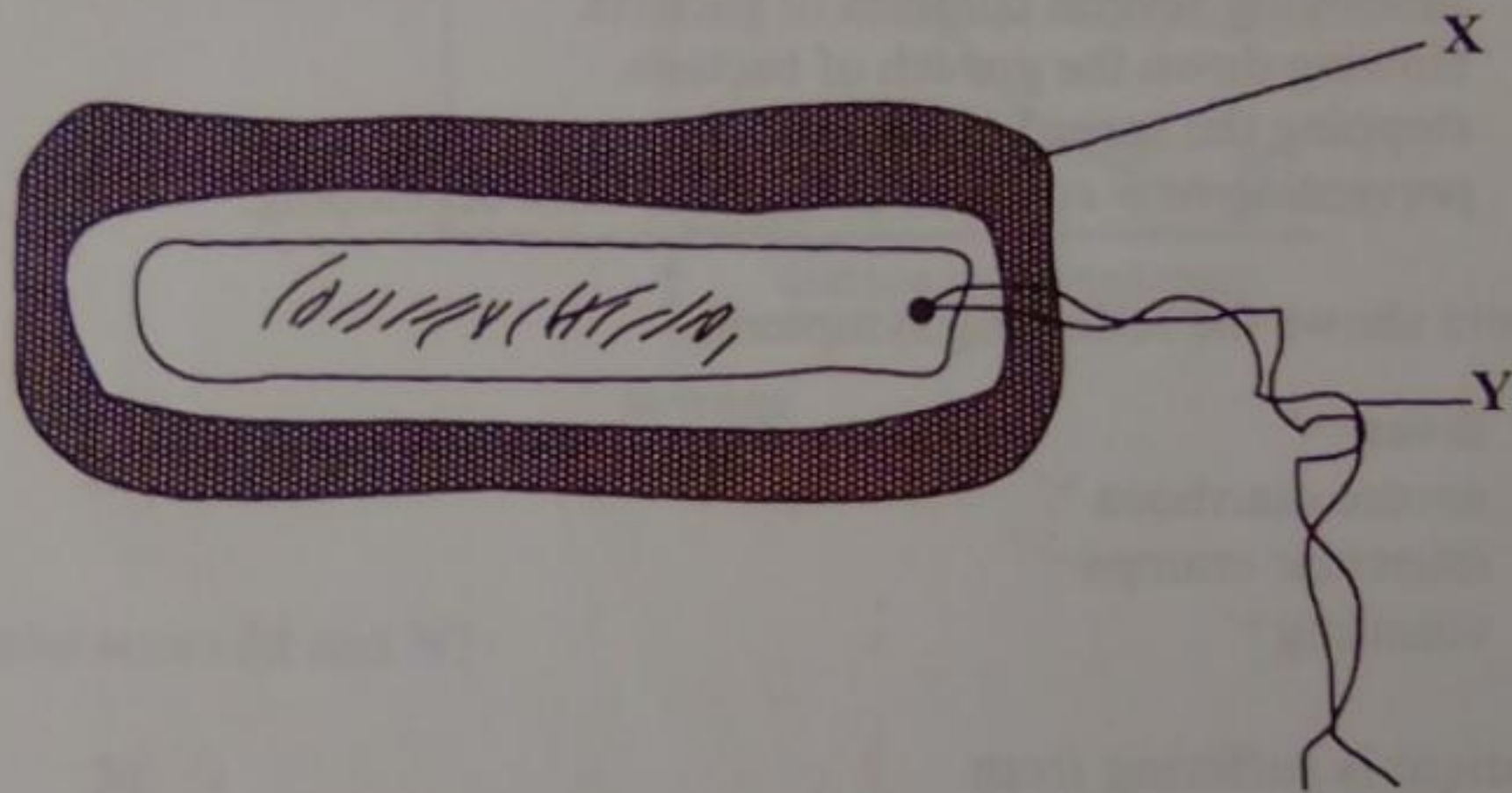
30 The diagram shows changes during the human menstrual cycle.



Which row correctly identifies hormones **P** and **Q**?

	P	Q
A	oestrogen	FSH
B	oestrogen	progesterone
C	progesterone	FSH
D	progesterone	oestrogen

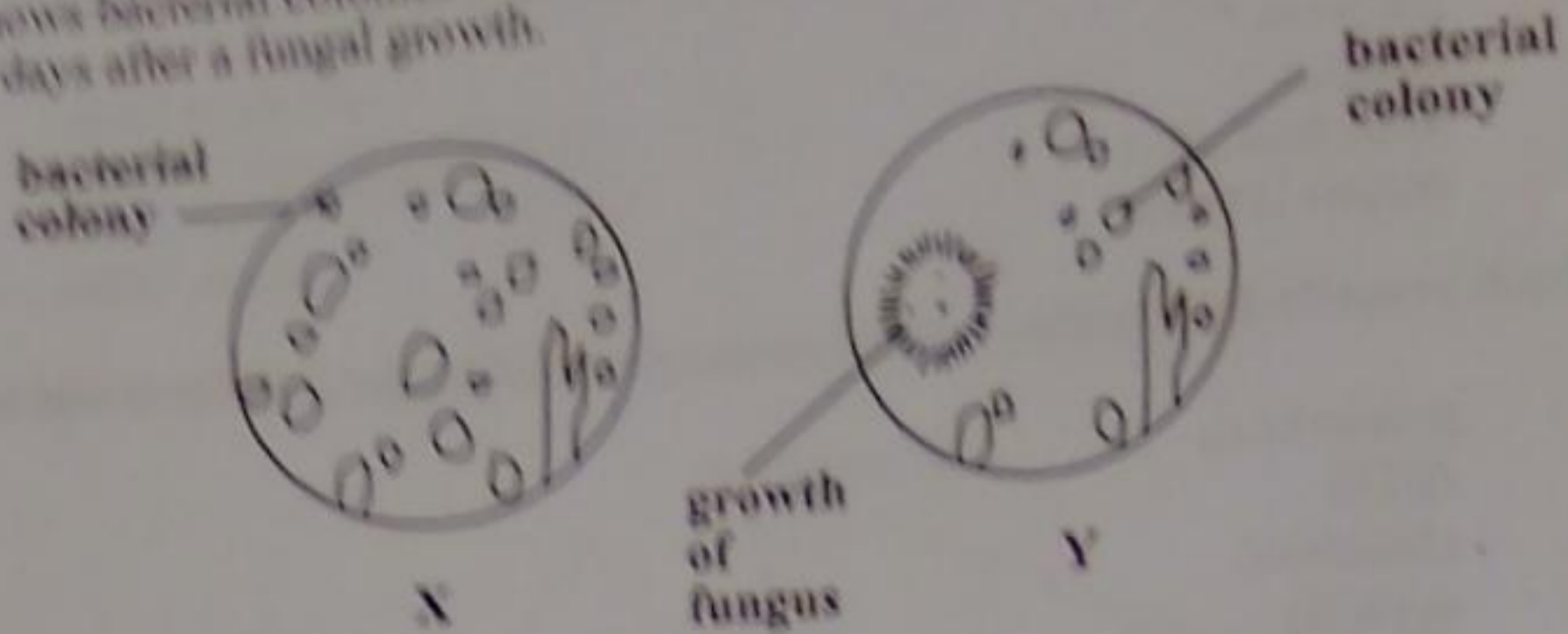
- 31 Which organ consists of both maternal and foetal tissues?
- A placenta ✓
 B amnion ×
 C umbilical cord ×
 D corpus luteum
- 32 Which sexually transmitted disease causes mental disorder, blindness and heart damage?
- A gonorrhoea
 B AIDS
 C chancroid
 D syphilis ✓
- 33 The diagram shows the structure of a bacterium.



What are the functions of **X** and **Y**?

- | | X | Y |
|---|------------|---------------------|
| A | locomotion | attachment |
| B | absorption | sensitivity |
| C | protection | <u>locomotion</u> ✓ |
| D | attachment | feeding |

- 34 X shows bacterial colonies which had grown on nutrient jelly. Y shows the same culture five days after a fungal growth.



The result in Y suggests that the fungus is

- A destroying several colonies of bacteria.
 B slowing down the growth of bacteria.
 C stopping the spread of bacteria.
 D preventing new colonies of bacteria from developing.
- 35 A patient shows the following symptoms.

1. fever ✓
2. severe diarrhoea ✓
3. muscular cramps ✓
4. vomiting ✓

This patient is suffering from

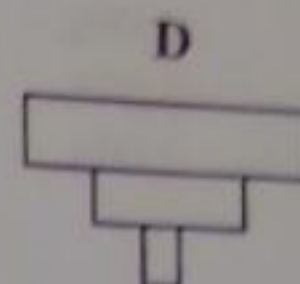
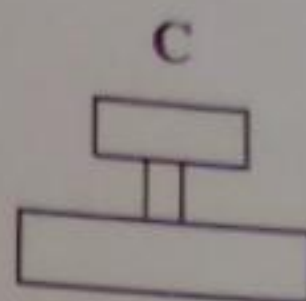
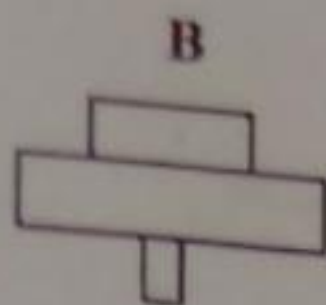
- A malaria. ✗
 B AIDS. ✗
 C typhoid ✗
 D cholera. ✗
- 36 In certain pea plants, yellow flower is dominant to white flowers. Two heterozygous plants were crossed.

How many plants had yellow flowers if 207 of the offspring were white flowered?

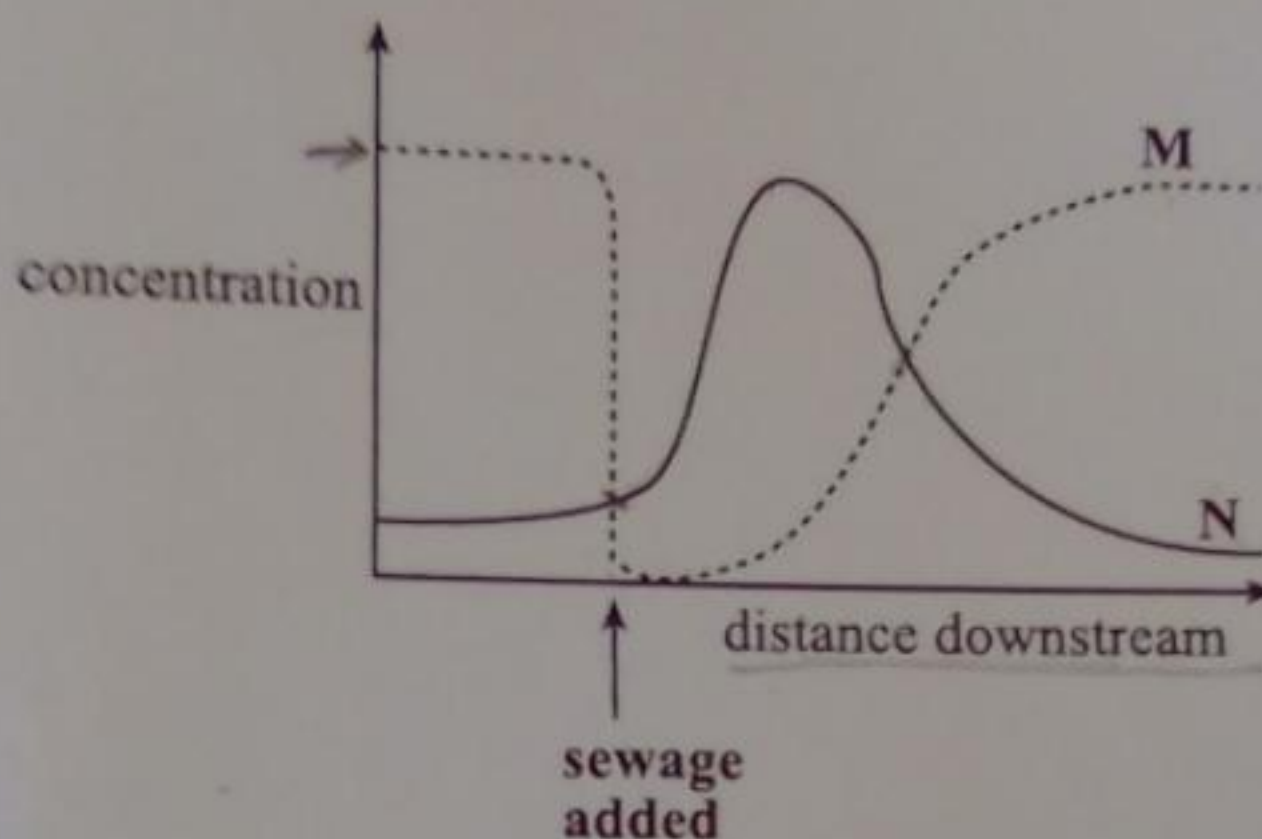
- A 52
 B 155
 C 207
 D 620



which pyramid of biomass correctly represents the food chain:
 grass → donkey → hyena?



- 38 The diagram shows changes in the concentration of substance **M** and **N** in a heavily polluted river.



What are substances **M** and **N**?

M

N

- | | | |
|----------|-----------------------|-----------------------|
| A | carbon dioxide | nitrogenous compounds |
| B | nitrogenous compounds | carbon dioxide |
| C | oxygen | nitrogenous compounds |
| D | nitrogenous compounds | oxygen |

- 39 A soil sample was weighed and placed in an oven at 105°C until the mass was constant. It was then strongly heated and reweighed.

Which soil properties can be determined from this experiment?

- | | |
|----------|--|
| A | water content ✓ and organic content ✓ |
| B | air content ✓ and water content ✓ |
| C | water retention and mineral content |
| D | organic content and water holding capacity |

40 The growth rate of a human population that doubles in 20 years is

- A 1,0%.
- B 5,0%.
- C 3,5%.
- D 2,0%.

- not grow



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

PAPER 1 Multiple Choice

5008/1

NOVEMBER 2013 SESSION

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

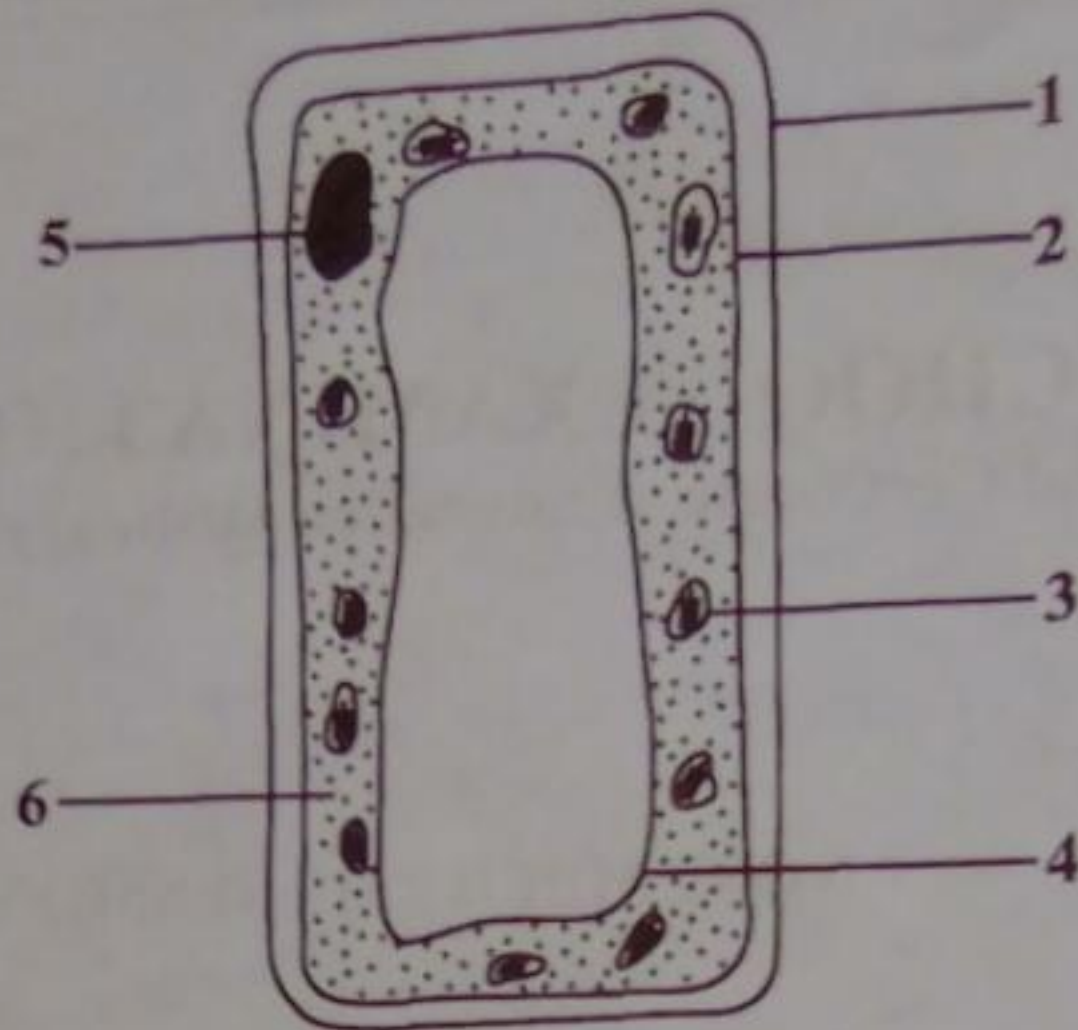
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

Copyright: Zimbabwe School Examinations Council, N2013.

1 The diagram shows a plant cell.



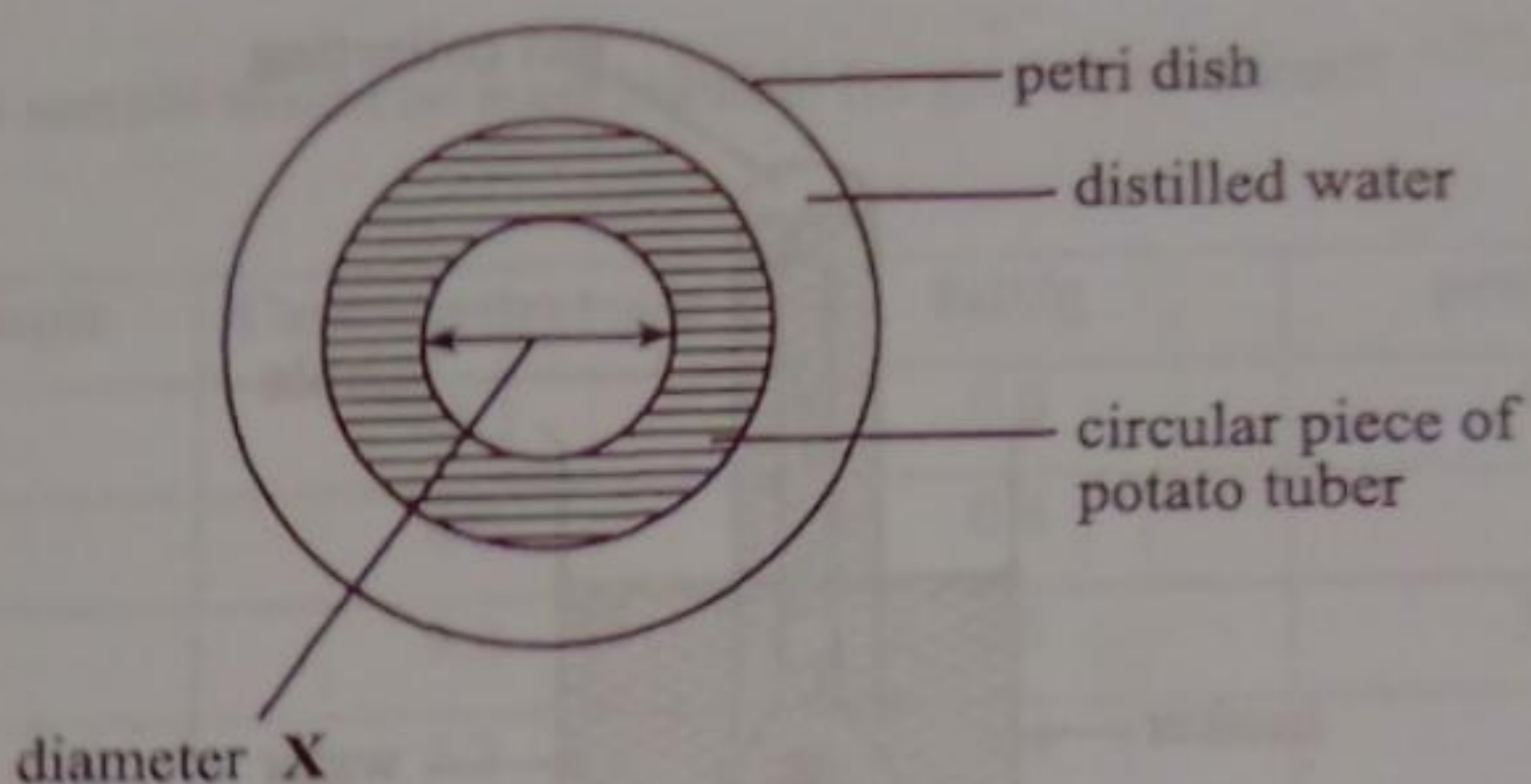
Which features are absent in an **animal** cell?

- A 1, 2 and 5
- B 1, 3 and 5
- C 3, 4 and 6
- D 1, 3 and 4

2 Which specialised animal cell functions by conducting electrical energy?

- A muscle cell
- B neurone cell
- C red blood cell
- D epithelial cell

The diagram shows a peeled potato tuber in distilled water.



Which are the changes that occur in 20 minutes?

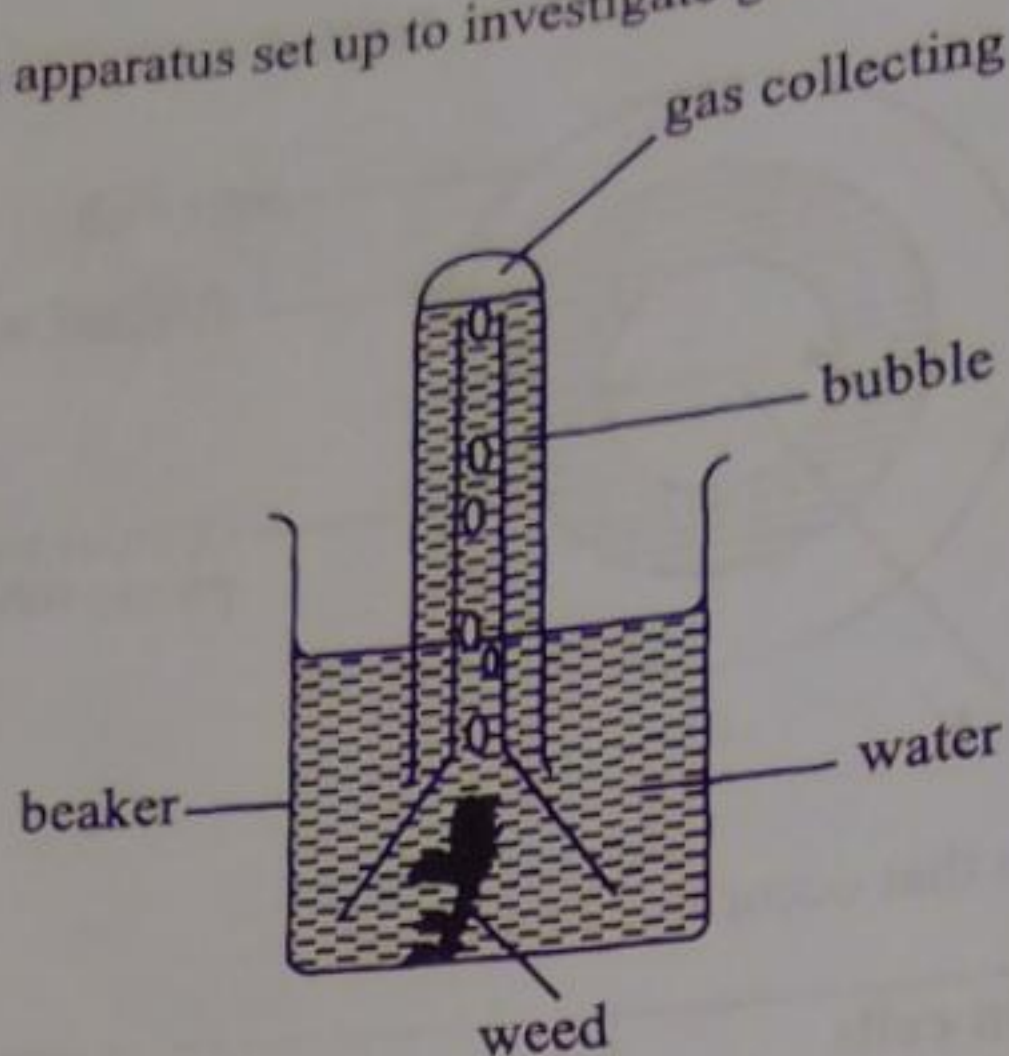
	turgor in cells	diameter X
A	increases	decreases
B	increases	increases
C	decreases	increases
D	decreases	decreases

4 Which plant tissue is involved in translocation?

- A epidermis
- B cortex
- C phloem
- D xylem

5

The diagram shows apparatus set up to investigate gas production by a pond weed.



Which action will decrease gas production by the pond weed?

- A maintaining a constant temperature
- B adding sodium hydrogen carbonate
- C adding water snails to the water
- D placing the apparatus in a dark cupboard

6

What is the function of phosphorus in plants?

- A protein synthesis
- B promotes flowering and fruiting
- C synthesis of energy carrier molecules
- D promotes leaf growth

7

Which food substances could be identified using 1% copper sulphate solution, 5% potassium hydroxide solution and Benedict's solution.

- A protein and fats
- B glucose and starch
- C fats and starch
- D protein and glucose

- 8 The table shows the composition of four different food samples in grams per 100 g samples.

Which food sample would be **least** suitable for growth to occur?

food sample	Carbohydrates/g	fats/g	protein/g
A	68.1	0.0	0.6
B	0.0	0.8	16
C	9.8	46	30.2
D	5.2	4.4	5.6

- 9 Some deficiency symptoms are listed below.

1. night blindness
2. poor skin development
3. bleeding of gums
4. slow healing of wounds
5. drying of cornea

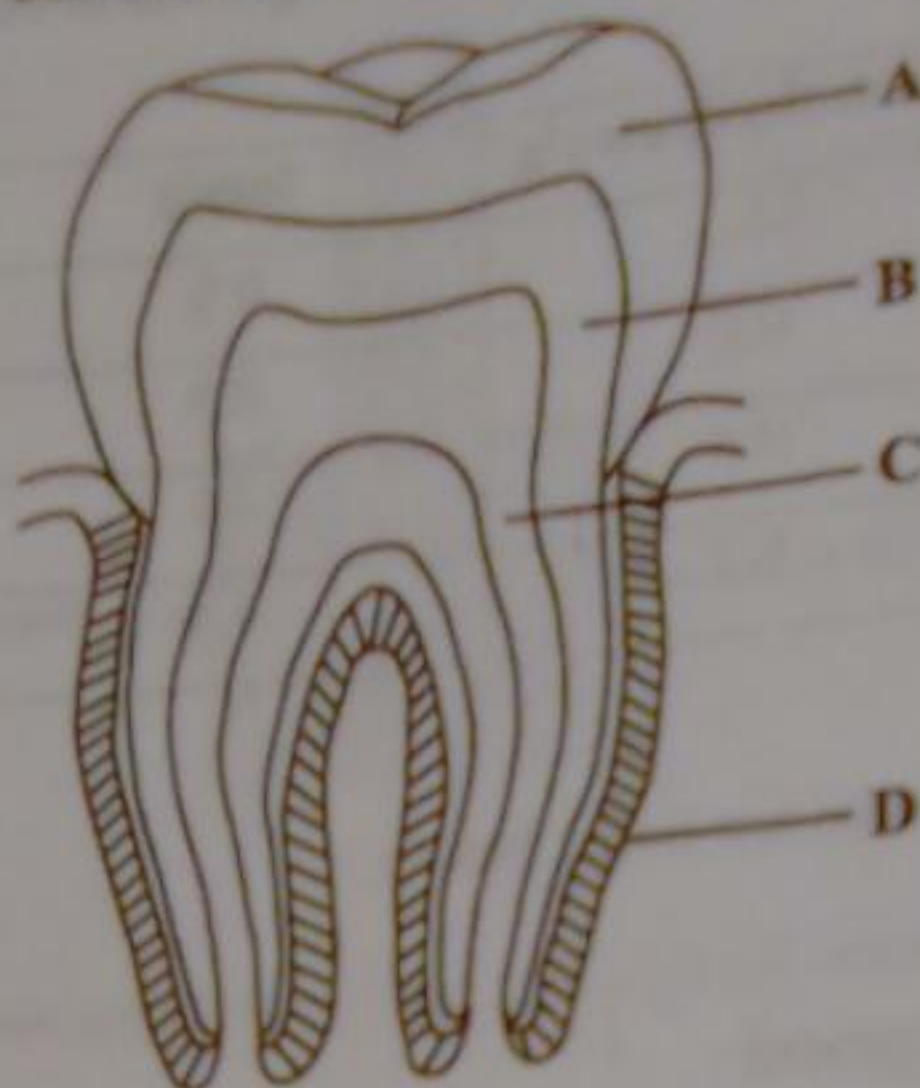
Which symptoms are caused by lack of vitamin A?

- A** 1, 2 and 3
B 2, 3 and 4
C 3, 4 and 5
D 1, 2 and 5

10

The diagram shows a tooth.

Which part of the tooth contains nerve endings and blood vessels?



11

Which chemical substance in cigarette smoke is addictive?

- A nicotine
- B tar
- C carbon monoxide
- D vapour

12

The list shows products of respiration.

1. carbon dioxide
2. ethanol
3. lactic acid
4. water

Which are the products of fermentation?

- A 1 and 3
- B 1 and 2
- C 2 and 4
- D 3 and 4

13

The following are features shown by some plant leaves.

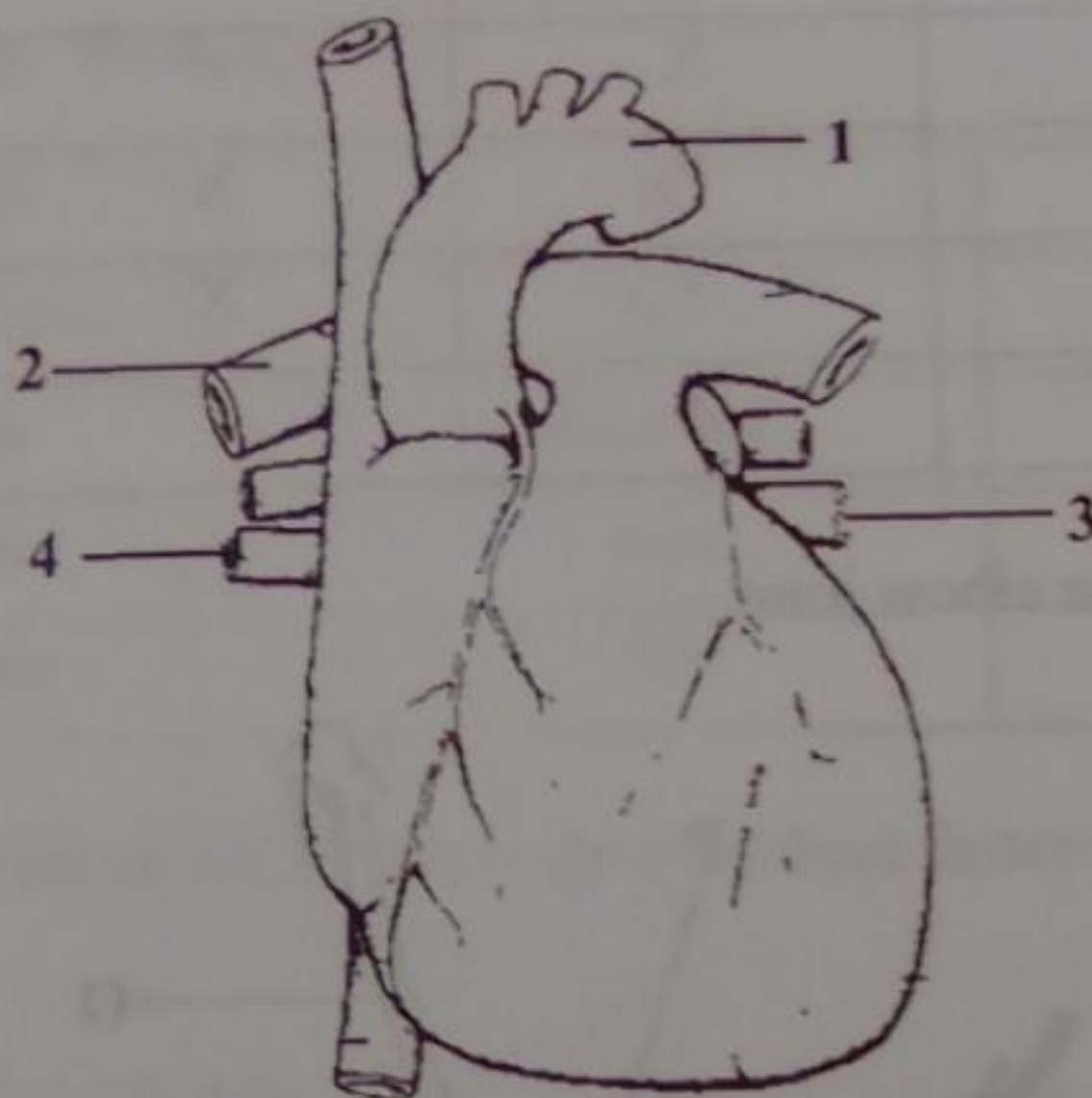
1. waxy cuticle
2. sunken stomata
3. large surface area

Which features reduce water loss?

- A 1 and 3
- B 1 and 2
- C 2 and 3
- D 3 only

14

The diagram shows an external view of the human heart.



Which blood vessels carry deoxygenated blood?

- A 1 and 2
- B 2 and 4
- C 3 and 4
- D 1 and 3

15

Which is a function of the blood plasma?

- A transportation of excretory products
- B transportation of oxygen
- C production of antibodies
- D destruction of bacteria

16

Some characteristics of blood vessels are listed

1. thick elastic muscle fibres
2. porous wall
3. small lumen
4. presence of semi-lunar valves

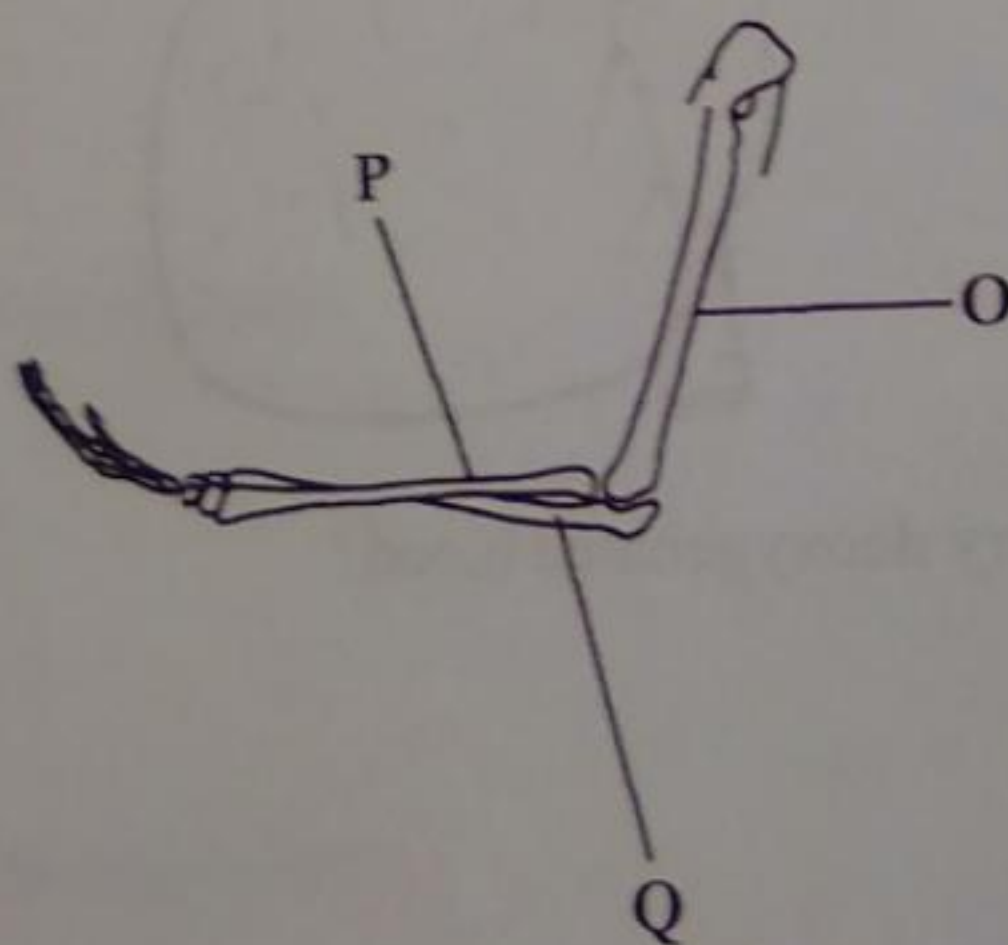
Which set identifies an artery?

Key √ present
 X absent

	1	2	3	4
A	X	X	√	X
B	√	√	X	√
C	X	√	√	X
D	√	X	√	

17

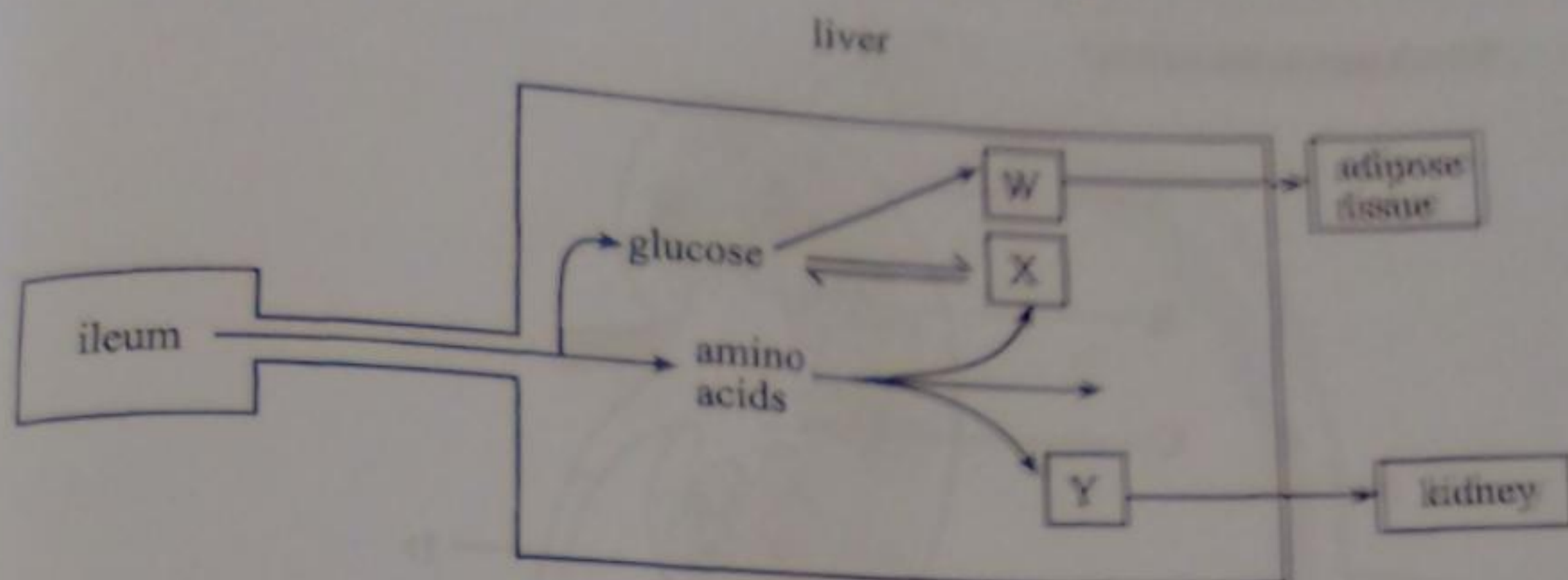
The diagram shows the elbow joint.



The parts labelled O, P and Q are

	O	P	Q
A	ulna	radius	humerus
B	humerus	ulna	radius
C	humerus	radius	ulna
D	radius	ulna	humerus

The flow diagram shows the role of the liver in metabolism.



What is represented by W, X and Y?

	W	X	Y
A	glycogen	fat	urea
B	fat	glycogen	urea
C	fat	urea	glycogen
D	urea	fat	glycogen

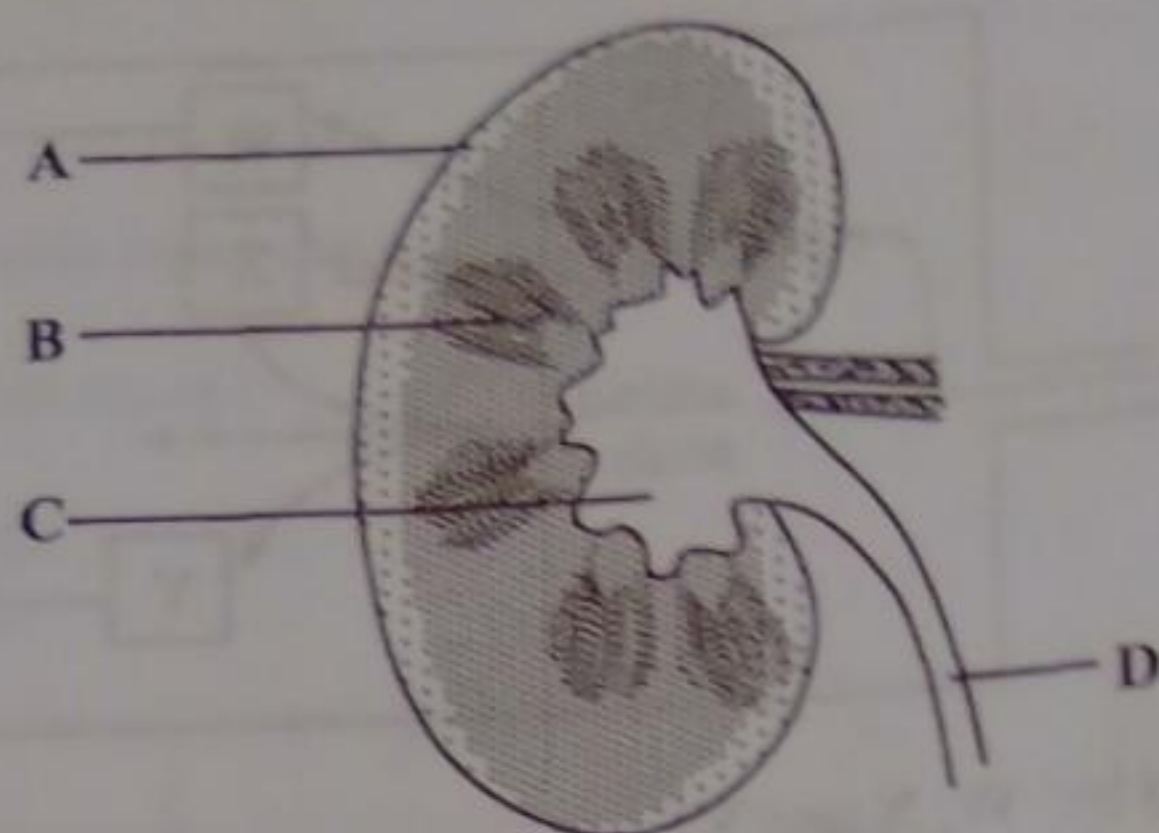
Which process occurs when the human body loses heat faster than it can produce?

- A vasodilation
- B sweat production increases
- C vasoconstriction
- D hair erector muscle relaxes

20

The diagram shows a cross section of the kidney.

Which part is the pelvis?



21

Which components are present in urine of a diabetic person?

Key √ present
 X absent

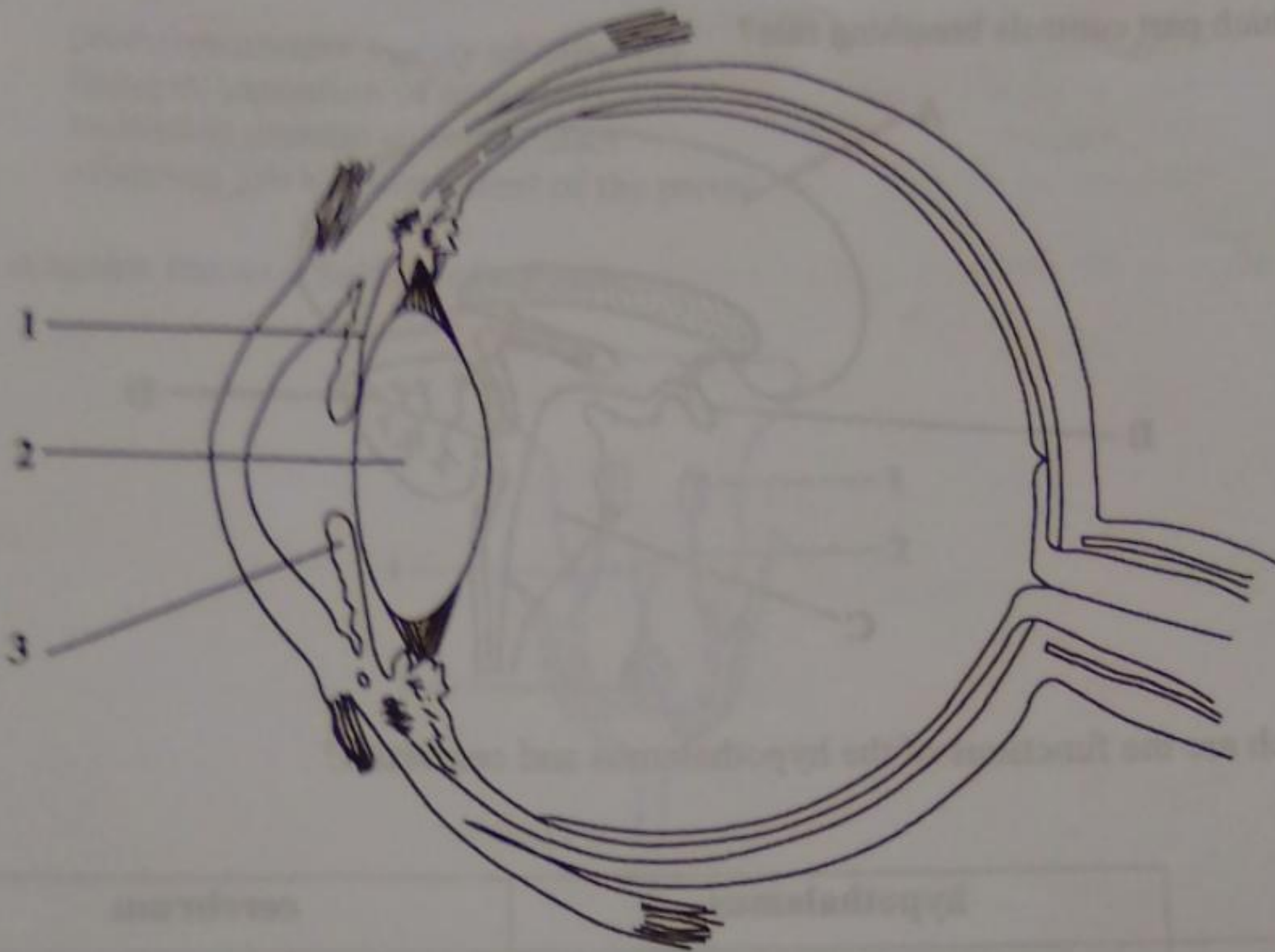
	salts	urea	glucose	amino acids
A	√	√	X	X
B	X	X	√	X
C	√	√	X	√
D	√	√	√	X

22

What is the effect of adrenaline on the amount of glucose released by the liver and on pulse rate?

	glucose released by liver	pulse rate
A	reduced	increased
B	reduced	reduced
C	increased	reduced
D	increased	increased

The diagram shows a section through the human eye.



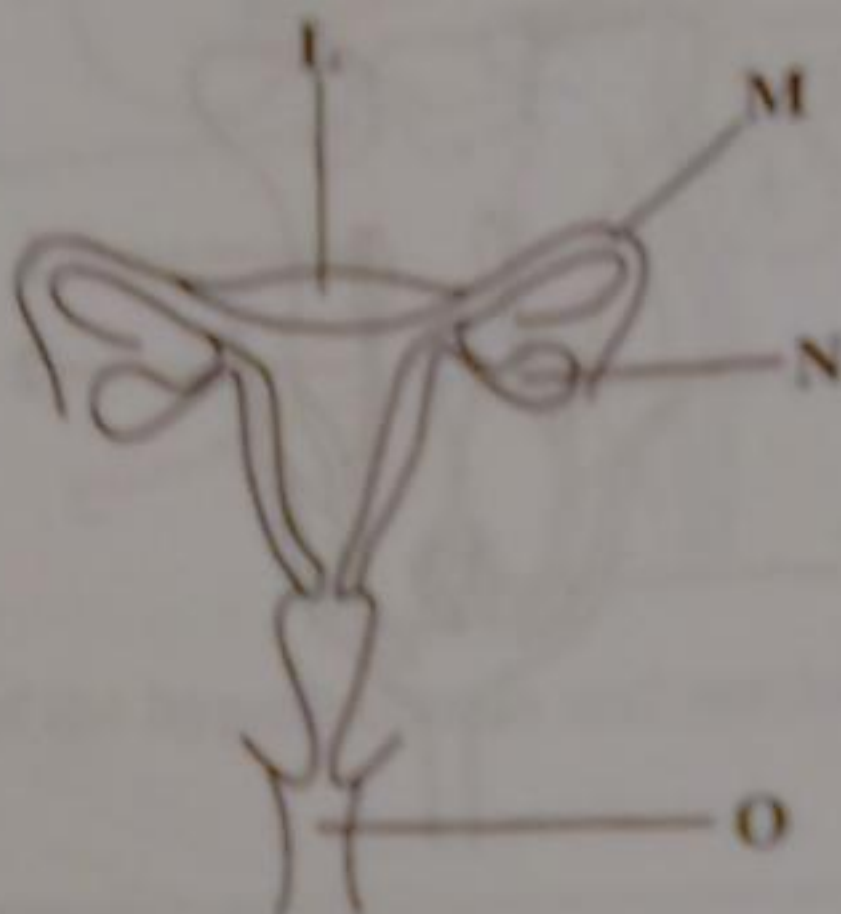
What are the functions of the numbered structures?

	1	2	3
A	accommodation	focuses light	control light entrance
B	accommodation	controls light entrance	focuses light
C	controls light entrance	focuses light	accommodation
D	focuses light	accommodation	controls light entrance

31 Which method of family planning prevents implantation?

- A spermicide
- B oral contraceptive pill
- C tubal ligation
- D intra-uterine device

32 The diagram shows a part of the female reproductive system.



Where does ovulation, fertilisation and implantation take place?

	ovulation	fertilisation	implantation
A	M	N	L
B	L	M	O
C	O	L	M
D	N	M	L

33 Which disease has the following symptoms?

1. discharge of yellowish pus
2. pain during urination
3. inflammation of urethra

- A chaneroid
- B gonorrhoea
- C syphilis
- D measles

34 Which hormone increases in concentration soon after ovulation?

- A thyroxine
- B oestrogen
- C testosterone
- D progesterone

35 Which pathogen causes cholera?

- A fungus
- B protozoan
- C bacterium
- D virus

36 A man of blood group AB had a child with a mother of blood group B.

Which **cannot** be a possible genotype of their child?

- A $I^A I^B$
- B $I^A I^O$
- C $I^B I^B$
- D $I^O I^O$

37 In an energy flow chart of an ecosystem which is a secondary consumer?

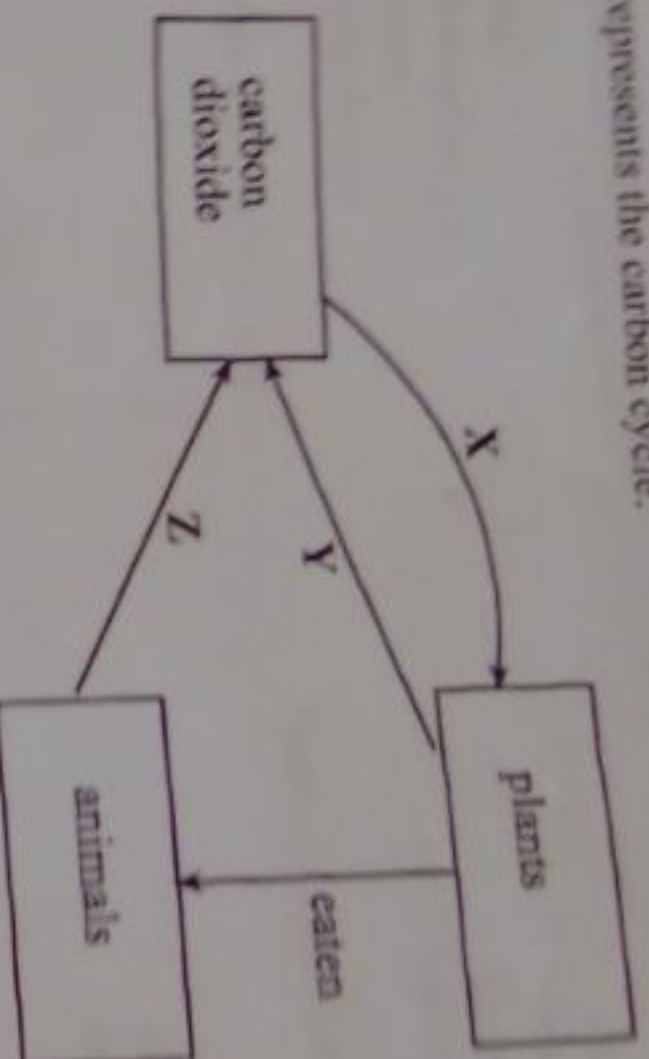
trophic level	energy flowing in kJ
---------------	----------------------

A	2
B	28
C	250
D	4 500

38 Denitrifying bacteria change

- A ammonia compounds to nitrates.
- B ammonium compounds to nitrites.
- C nitrates to nitrogen.
- D nitrogen to nitrates.

- 39 The diagram represents the carbon cycle.



Which processes are represented by letters X, Y and Z?

	X	Y	Z
A	photosynthesis	respiration	respiration
B	absorption	photosynthesis	respiration
C	respiration	photosynthesis	absorption
D	absorption	respiration	photosynthesis

- 40 The infants mortality rate changed from 86 per 1 000 live births in 1980 to 23 per 1 000 live birth in 2000.

What is the percentage change in mortality over this period?

- A 2,7 %
 B 6,3 %
 C 7,3 %
 D 27,4 %



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple Choice

NOVEMBER 2012 SESSION

1 hour

Additional materials:
Multiple choice answer sheet
Soft clean eraser
Soft pencil (Type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are forty questions in this paper. Answer all questions. For each question there are four possible answers, A, B, C and D. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

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Any rough working should be done in this booklet.

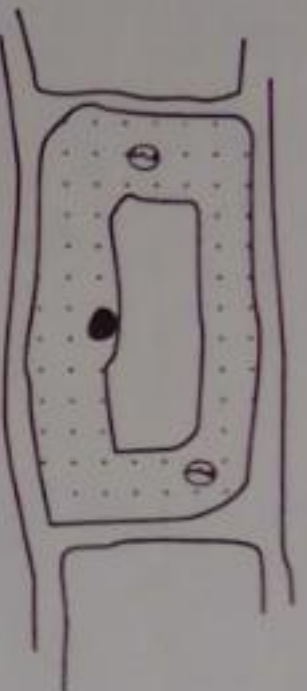
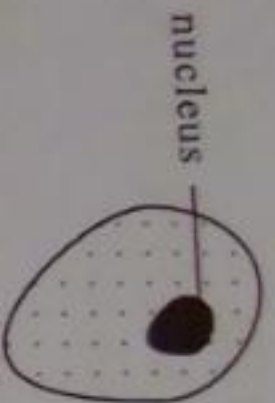
This question paper consists of 16 printed pages.

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Turn over

2

- 1 The diagrams show plant and animal cells as seen under a light microscope.



3

4

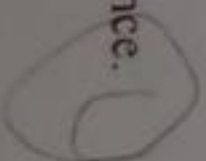
Which are plant cells?

- A 1 and 2
B 1 and 3
C 2 and 3
D 2 and 4

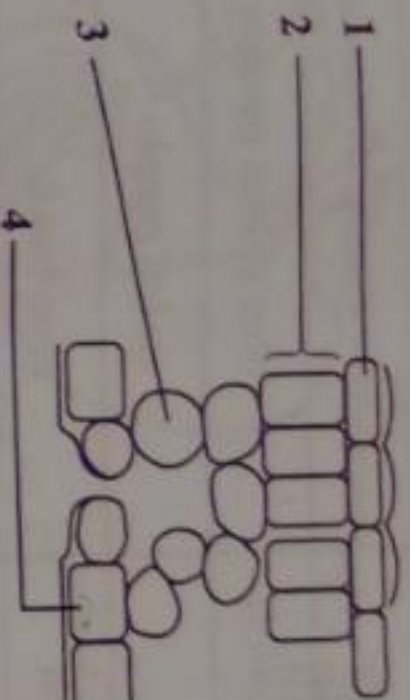


- 2 What would be the effect of putting red blood cells in distilled water for thirty minutes?

- A Cells swell and then burst.
B There is no change in the cells' appearance.
C Cells shrink slightly.
D Cells lose their smooth rounded shape.



- 3 The diagram shows cells in a leaf.



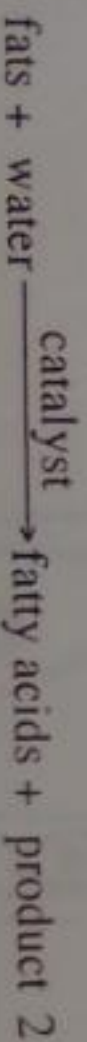
Which cells can photosynthesise?

- A 1 and 2
B 2 and 3
C 2 and 4
D 3 and 4

- 4 Effects of deficiency of potassium in young plants are

- A stunted growth and chlorosis.
B stunted growth of roots, sometimes purplish leaf colouring.
C poor flowering and fruiting.
D stunted growth of shoots and purplish leaf colouring.

- 5 Fats are hydrolysed as shown by the equation:



Identify the catalyst and product 2.

- | | | |
|----------|-----------------|------------------|
| <u>A</u> | catalyst | product 2 |
| <u>B</u> | amylase | glycerol |
| <u>C</u> | lipase | glycogen |
| <u>D</u> | lipase | glycerol |
| | protease | glucose |

- 6 The table shows results of tests carried out on a food sample.

test	result
Benedict's solution	blue colour persists
Biuret	violet/ purple
iodine solution	blue-black
emulsion	no colour change

12 Sugars
Protein
Starch
Fats

What does the food sample contain?

- A protein and fat
B protein and starch
C fat and starch
D starch and reducing sugar
- 7 Which is a role of hydrochloric acid in digestion?

- A catalysing digestion
B emulsification of fats
C destroys bacteria in food
D activates amylase

8

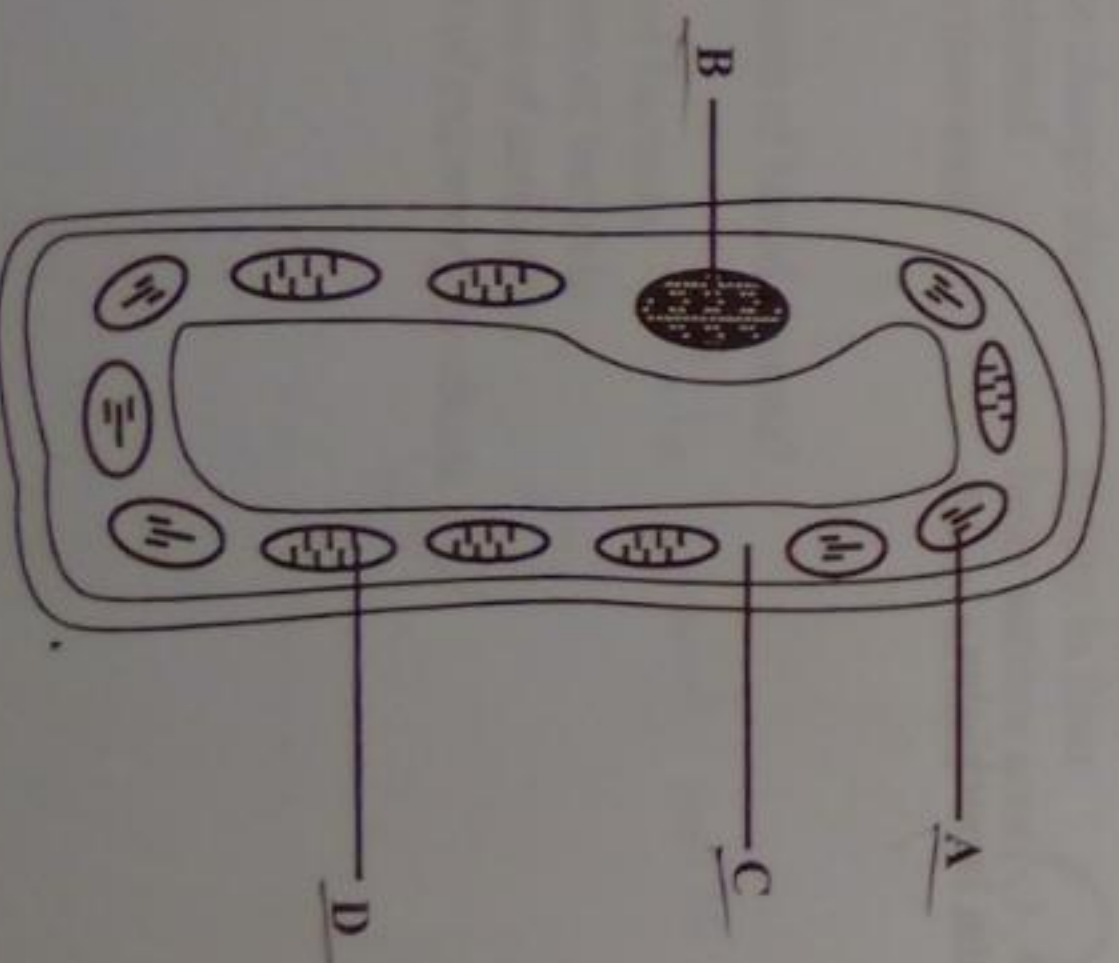
The diagram shows part of the human digestive system.



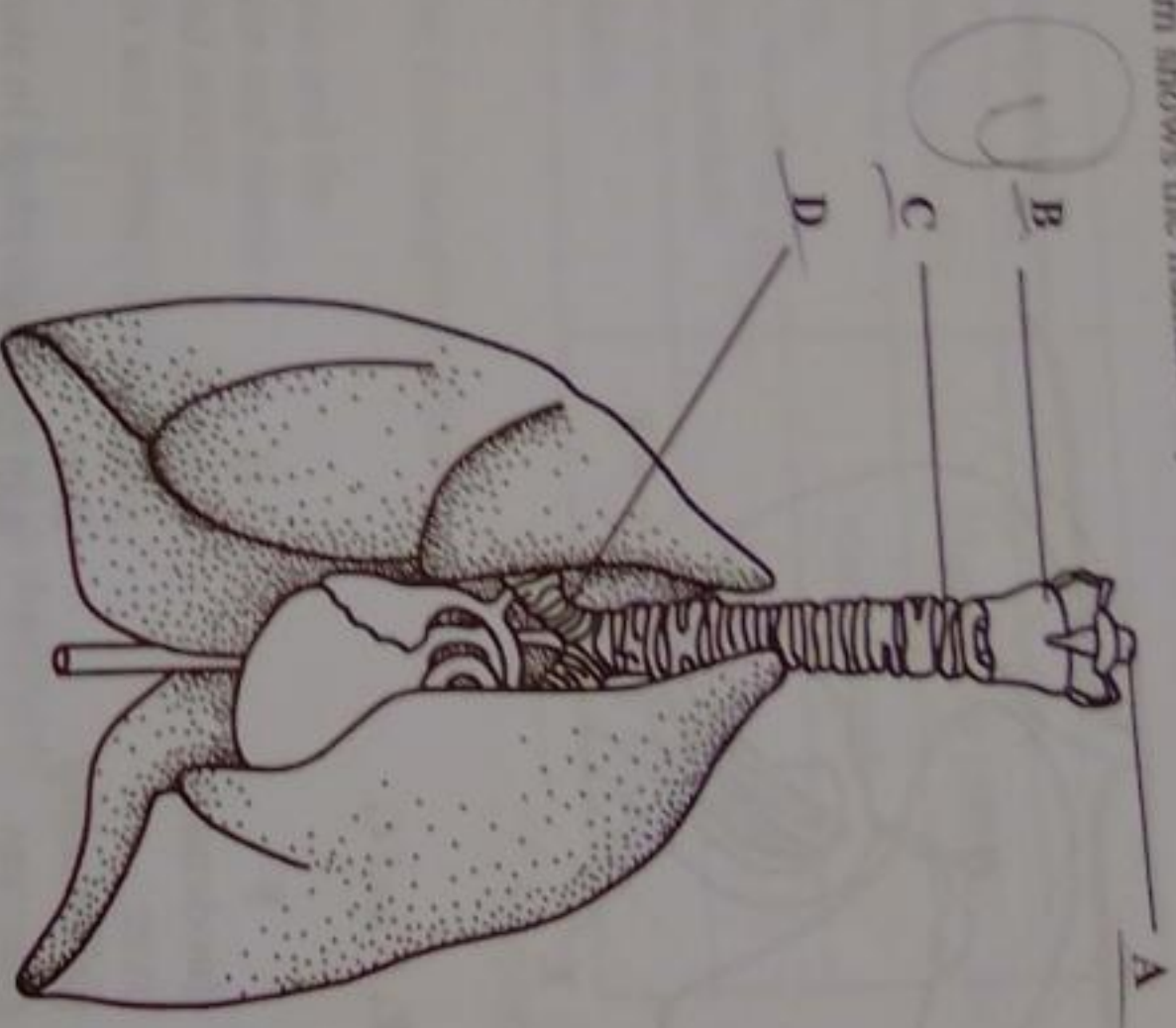
Name the structures E, F and G.

- | | | | | | |
|----------|----------|----------|--------------|----------|--------------|
| <u>A</u> | liver | <u>E</u> | duodenum | <u>F</u> | stomach |
| <u>B</u> | ileum | | liver | | pancreas |
| <u>C</u> | ileum | | gall bladder | | liver |
| <u>D</u> | duodenum | | pancreas | | gall bladder |

9 The diagram shows a plant cell. In which structure does aerobic respiration take place?



- 10 The diagram shows the human respiratory system. Which part is the larynx?



- 11 Which is the most severe effect of prolonged smoking?

A irritation of alveoli
B bronchitis
C emphysema
D damage of the bronchioles

- 12 The diagram represents a cross-section through a root.



What is **H**?

- A phloem
B xylem
C pith
D cortex

- 13 What are the adaptations of red blood cells for absorption and transport of oxygen?

- | | absorption | transport |
|----------|-------------------------|--------------------|
| <u>A</u> | large surface area | non-nucleated |
| <u>B</u> | ability to change shape | ability to float |
| <u>C</u> | non-nucleated | large surface area |
| <u>D</u> | large surface area | ability to float |

- 14 Which is a function of blood plasma?

- A clotting of bacteria
B destruction of bacteria
C production of antibodies
D transport of excretory products

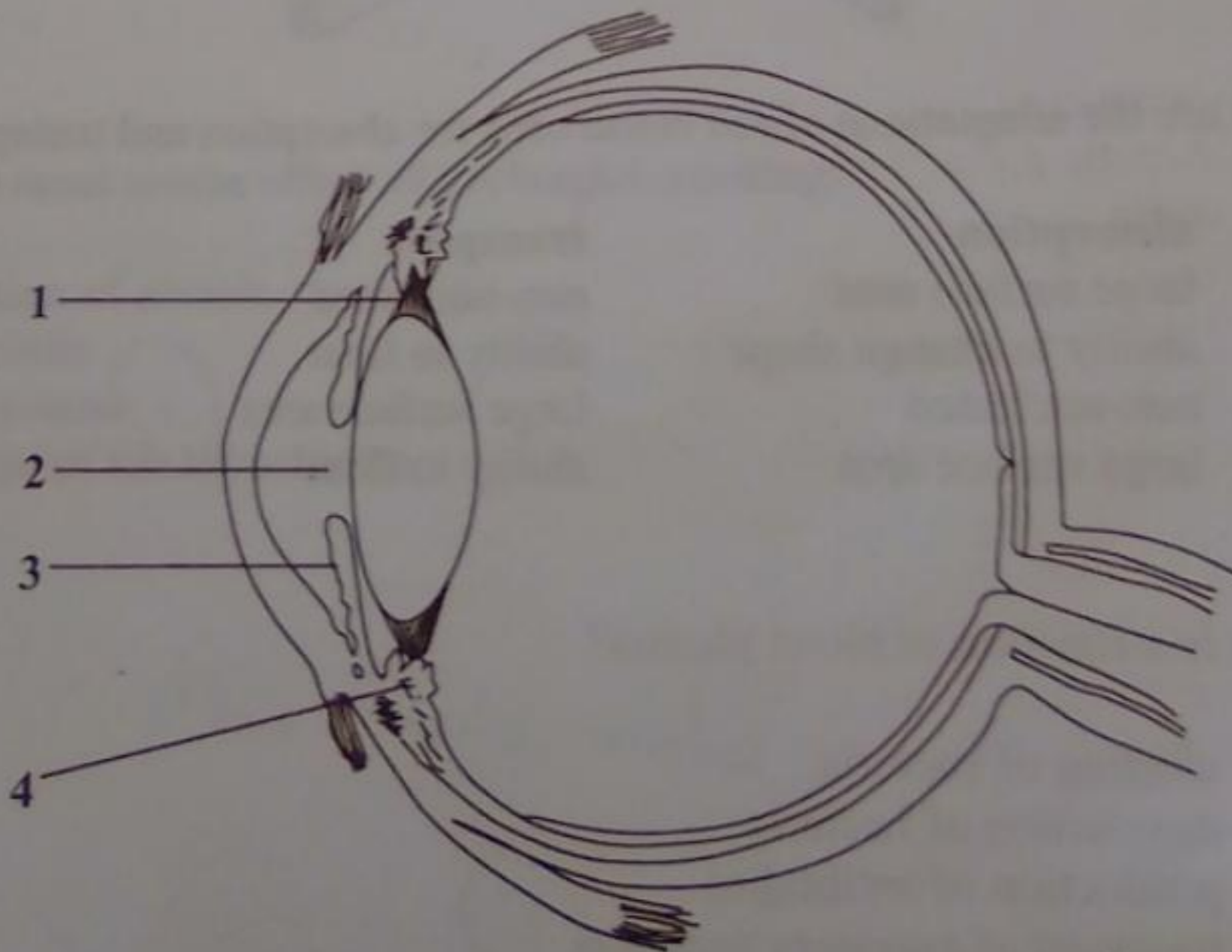
phule blood cells

- 15 The table shows the results of an experiment in which a bone was immersed in an acid for several hours.

mass of bone at the beginning of experiment	mass of bone after immersion in acid	appearance after experiment
50g	5g	same shape, bends easily

Which part of the bone reacted with the acid?

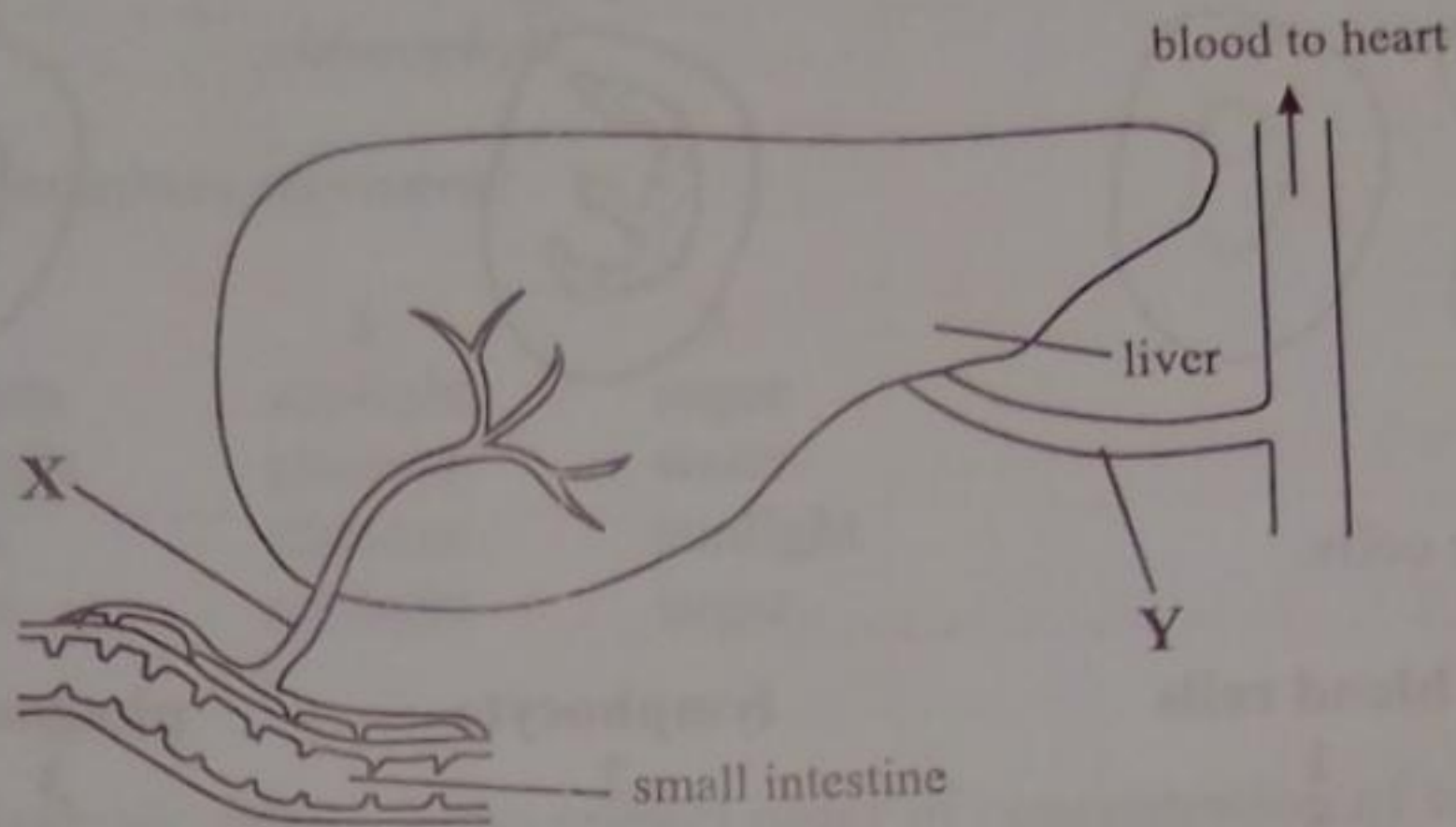
- A collagen
 - B calcium
 - C cartilage
 - D protein
- 16 The diagram shows a section through a human eye.



Which parts change in size in bright light?

- A 1 and 2
- B 2 and 3
- C 1 and 3
- D 2 and 4

- 17 The diagram shows the liver and its associated blood vessels.



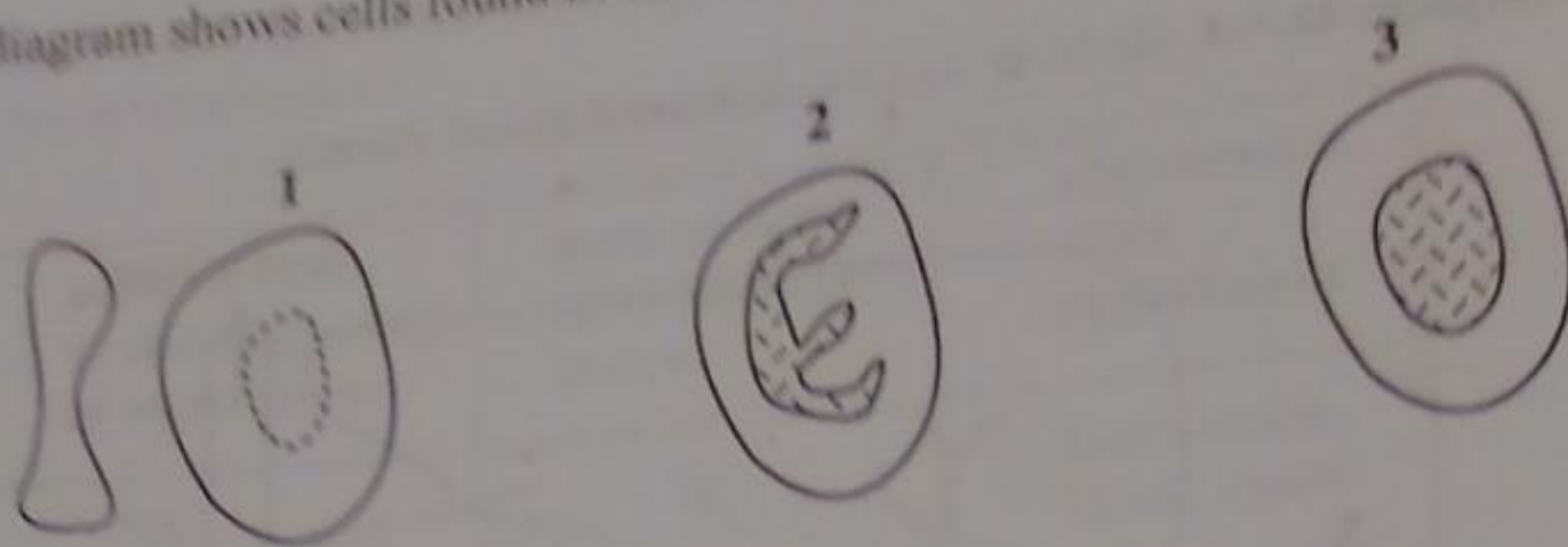
Identify blood vessels X and Y.

	X	Y
<u>A</u>	hepatic artery	hepatic vein
<u>B</u>	hepatic portal vein	hepatic vein
<u>C</u>	aorta	vena cava
<u>D</u>	mesenteric artery	renal vein

- 18 From which blood vessels do white blood cells escape into at a wound?

<u>A</u>	arterioles
<u>B</u>	capillaries
<u>C</u>	veins
<u>D</u>	venules

The diagram shows cells found in blood.



Identify the cells.

	red blood cells
<u>A</u>	1
<u>B</u>	2
<u>C</u>	3
<u>D</u>	1

lymphocyte

2
3
1
3

phagocyte

3
1
2
2

Below is a list of drugs.

- 1 penicillin
- 2 aspirin
- 3 chloroquine
- 4 paludrine
- 5 paracetamol

Which are analgesics?

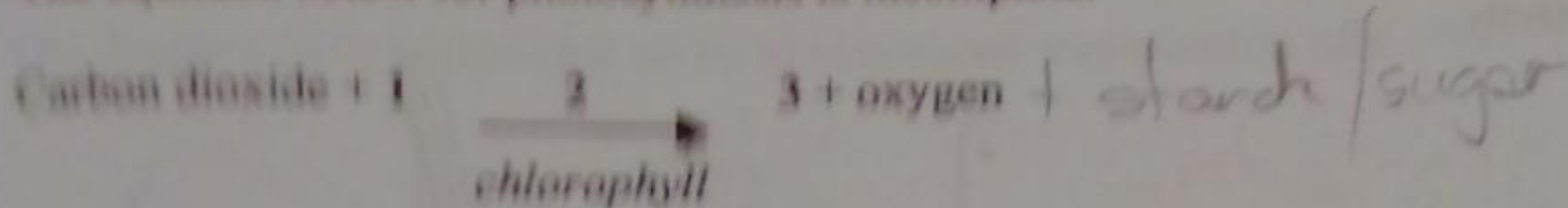
- A 1, 2 and 5
- B 2 and 4 only
- C 2 and 5 only
- D 4 and 5 only

What is the effect of adrenaline on pulse rate and amount of glucose released by the liver?

	pulse rate	glucose released
<u>A</u>	increased	increased
<u>B</u>	increased	reduced
<u>C</u>	reduced	increased
<u>D</u>	reduced	reduced

22

The equation below for photosynthesis is incomplete.



What do the numbers represent?

	1	2	3
A	oxygen	sunlight	sugar
B	sunlight	glucose	water
C	water	glucose	sunlight
D	water	sunlight	sugar

23

During plasmolysis, which is the correct order of concentration of water molecules: cytoplasm, cell sap and the external solution?

KEY

> greater than

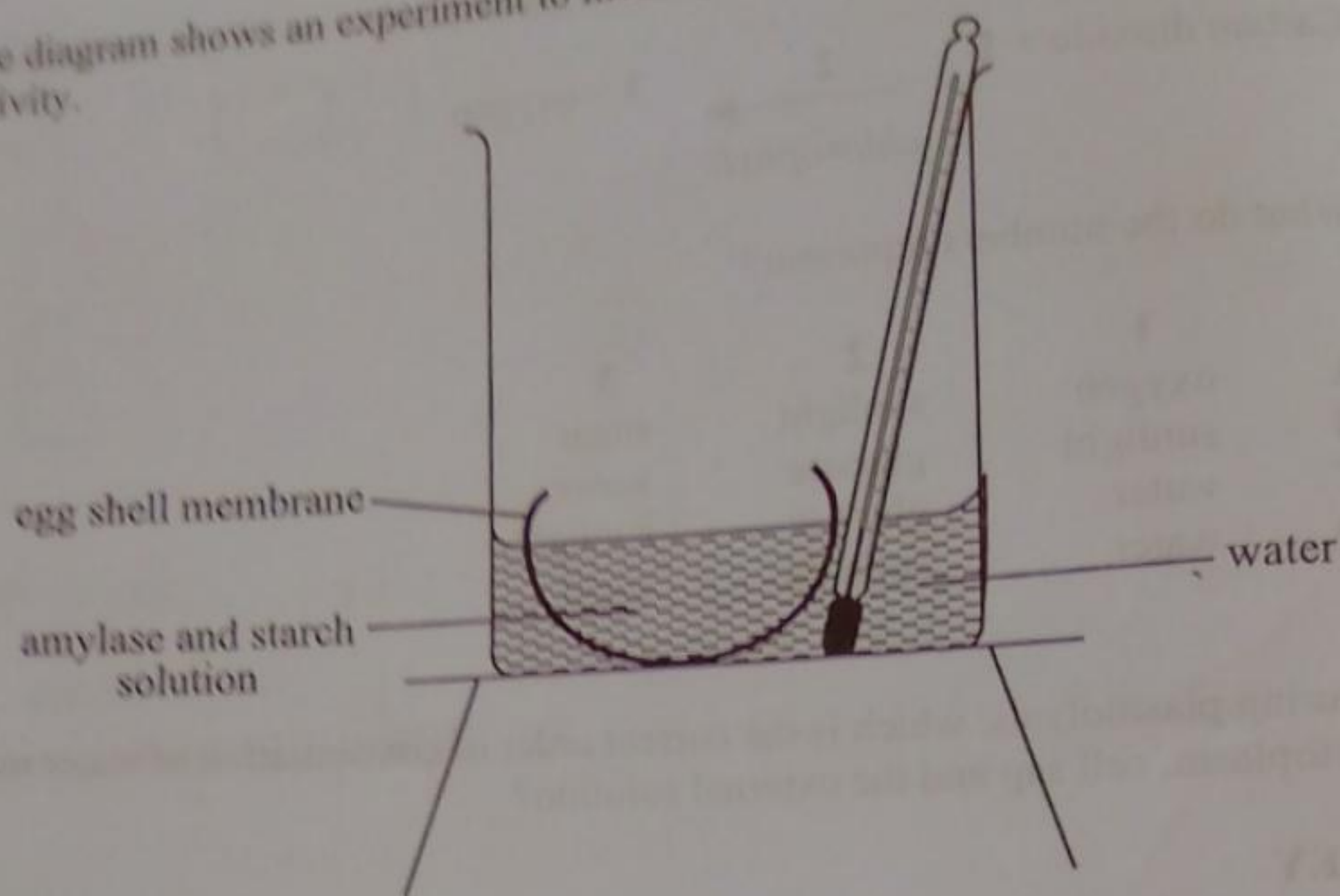
= equal to

- A external solution > cytoplasm > cell sap
- B cell sap > cytoplasm > external solution
- C cell sap > cytoplasm = external solution;
- D cytoplasm = cell sap > external solution;

24

Where is insulin produced and where does it act?

	site of production	site of action
A	adrenal glands	body muscles
B	pituitary glands	liver
C	pancreas	liver
D	liver	pancreas



The water was tested for the presence of reducing sugar at one minute intervals. At which temperature would the concentration of reducing sugar rise most rapidly?

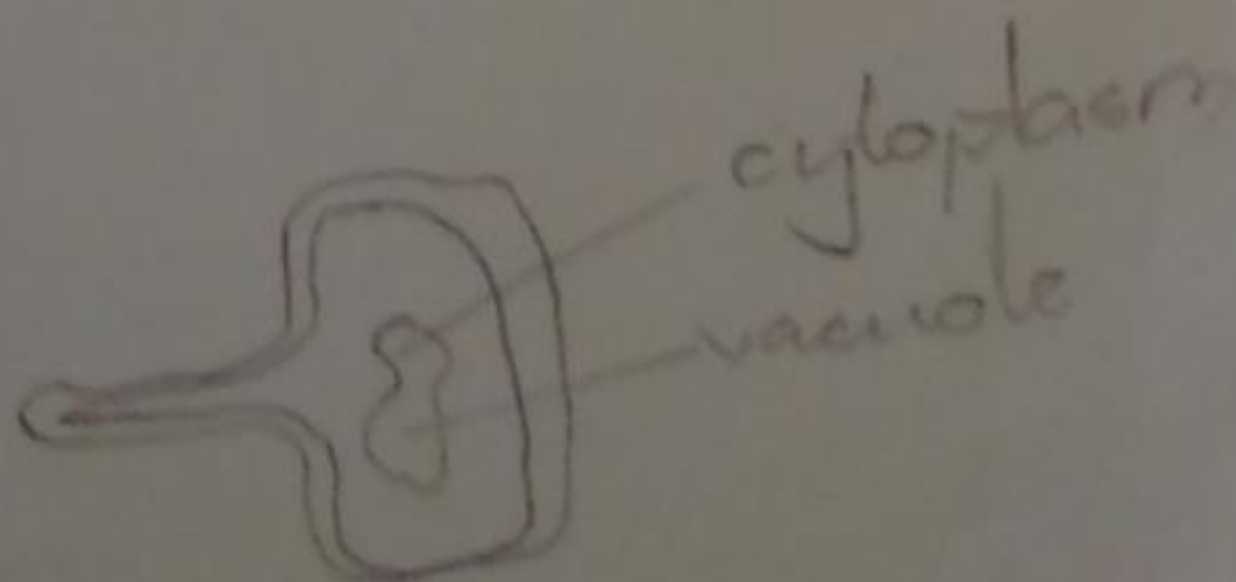
- ☒ A 20°C
☐ B 30°C
☐ C 40°C
☐ D 50°C

26 Which are the products of anaerobic respiration in plants?

- ☐ A alcohol and lactic acid
☐ B alcohol and water
☒ C carbon dioxide and alcohol
☐ D carbon dioxide and lactic acid

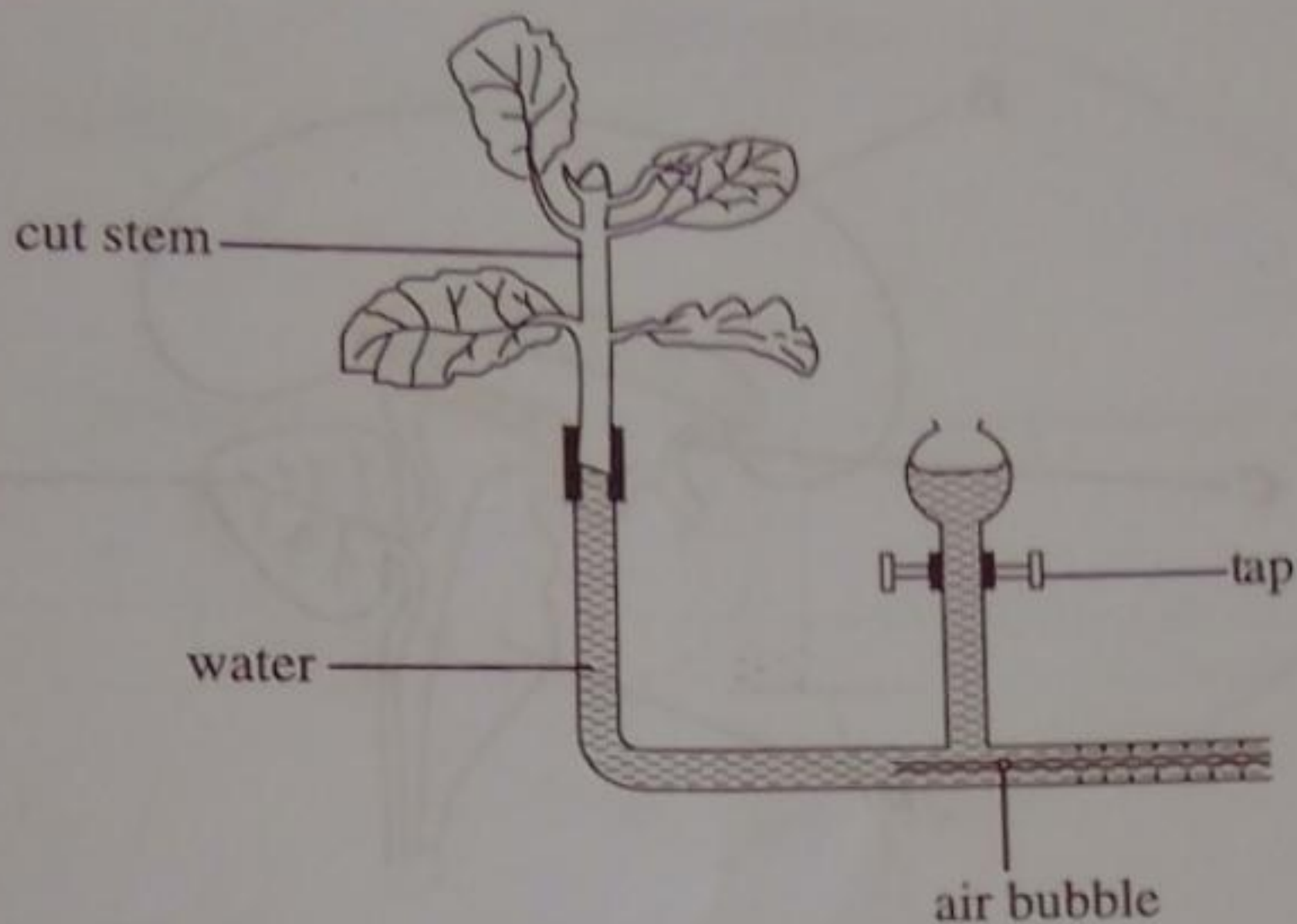
27 Water enters the vacuole of the root hair cell by

- ☐ A active transport.
☐ B diffusion.
☒ C osmosis.
☐ D transpiration.



28

The diagram shows apparatus set up to investigate transpiration rate.



Why is it necessary to keep the cut end of the stem in water the moment it is cut from the plant?

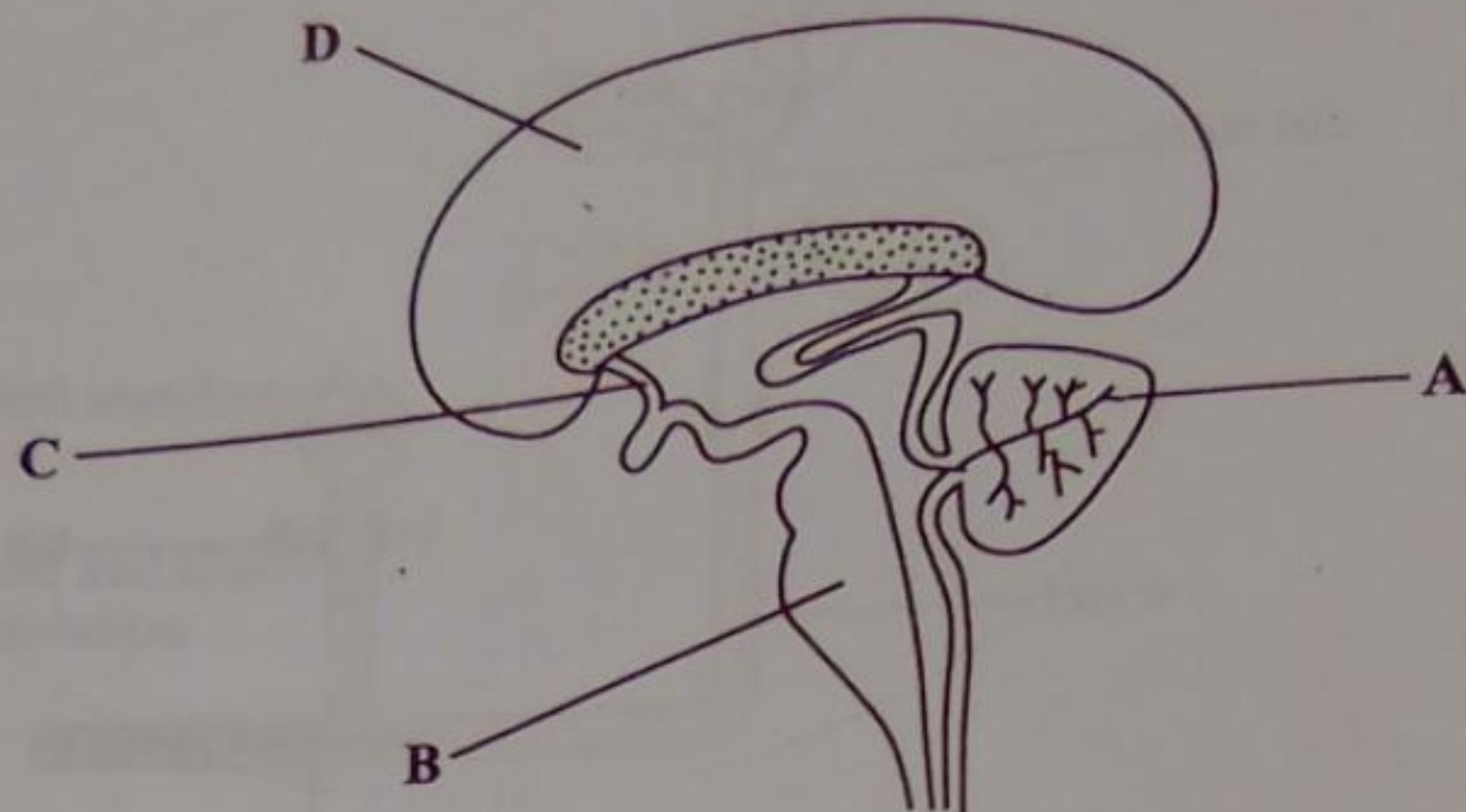
- A to keep the twig alive
- B to introduce an air bubble into the capillary tube
- C to prevent wilting
- D to prevent air entering its xylem vessels

29

When the right atrium contracts, blood flows into the

- A aorta.
- B left atrium.
- C pulmonary artery.
- D right ventricle.

Questions 30 and 31 refer to the diagram of the human brain.



- 30 Which part is the cerebrum?
- 31 Which structure controls posture, balance and coordinated movement?
- 32 Which form of drug abuse exposes an individual to the highest risk of infection with the HIV (AIDS) virus?

- ☒ A cigarette - smoking
- ☐ B drinking alcohol
- ☐ C injection of heroin
- ☐ D inhaling solvents

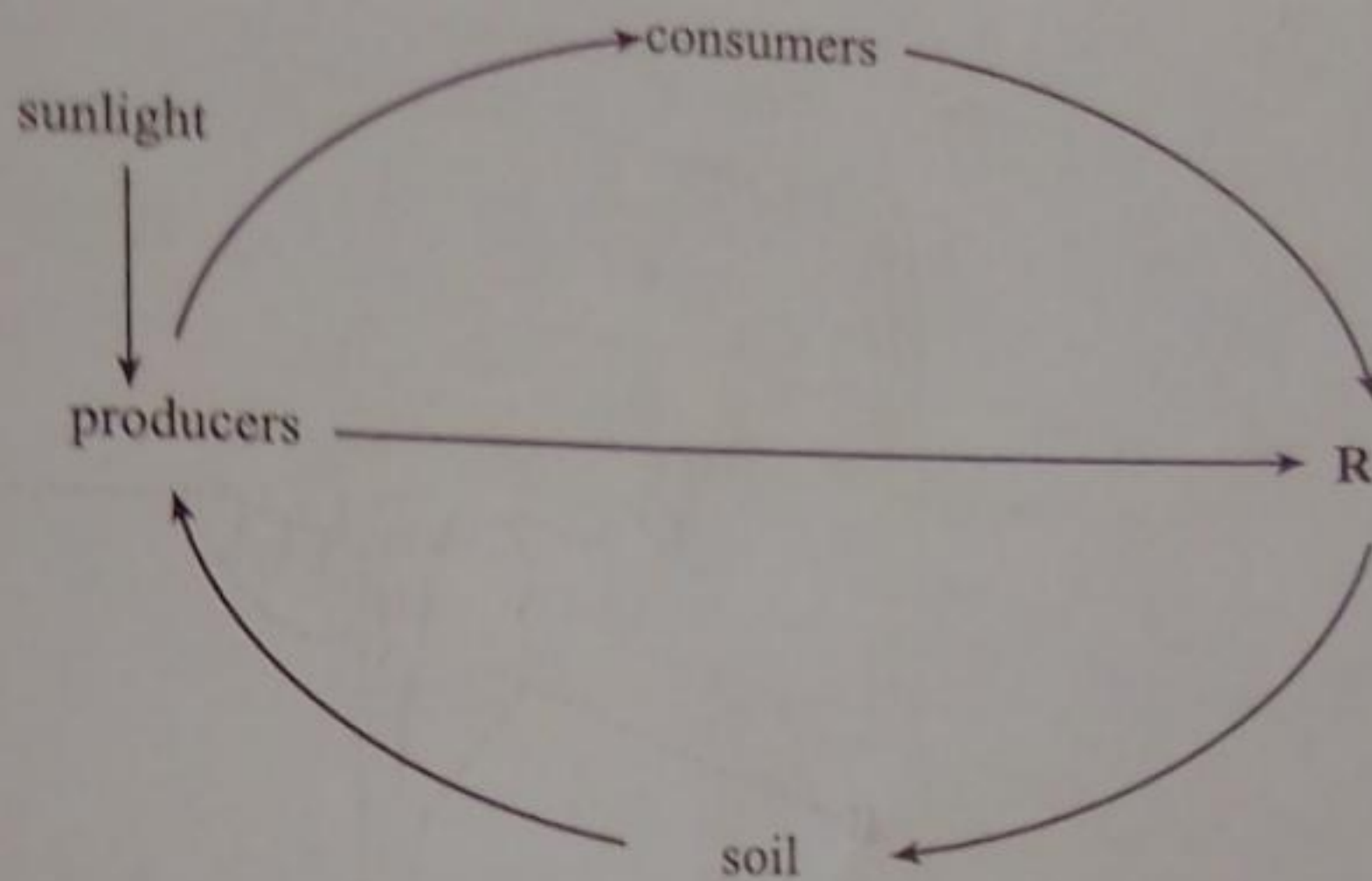
- 33 What causes fever in a patient suffering from malaria?

- ☒ A bursting of red blood cells
- ☐ B destruction of liver cells
- ☐ C multiplication of plasmodium
- ☐ D toxins released from plasmodium

- 34 The term *allele* refers to

- ☒ A alternative forms of the same gene.
- ☐ B genes produced as a result of mutation.
- ☐ C genes of true breeding organisms.
- ☐ D two genes.

35 The diagram shows the flow of nutrients in an ecosystem.



What does **R** represent in the cycle?

- A decomposers
 B plants
 C secondary consumers
 D tertiary consumers

36 Which is the correct set of the sex chromosomes in humans?

	female	male
A	XX	XY
<u>B</u>	XY	XX
C	YY	XY
D	XY	XY

37 The following are symptoms of a sexually transmitted disease;

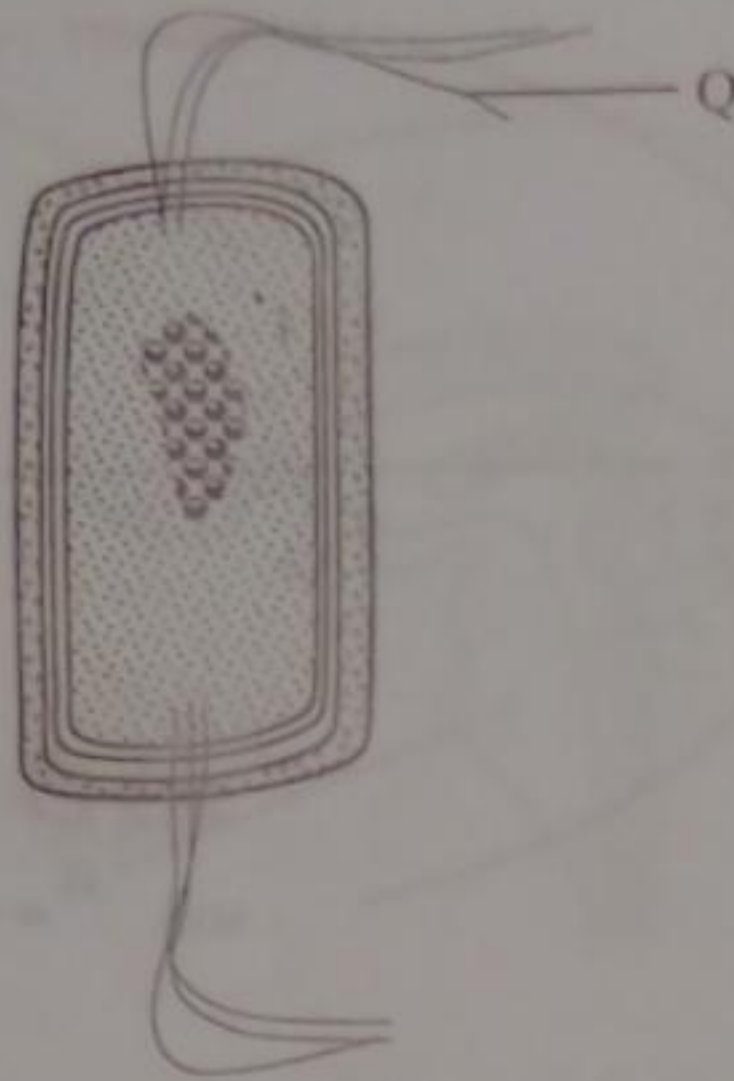
pus from penis,
 sterility,
 burning sensation when urinating.

Which disease causes these symptoms?

- A chancroid
 B gonorrhoea
 C genital herpes
 D syphilis

38

The diagram shows the structure of a bacterium.



What is the function of the part labelled Q?

- ☒ A for attachment to host
☐ B for feeding
☐ C for movement
☐ D for sensitivity

39 Nitrifying bacteria change

- ☐ A ammonium compounds to nitrates.
☐ B organic remains to ammonia.
☒ C nitrates to nitrogen.
☐ D nitrogen to nitrates.

40 The formula for determining the time it takes for human population to double is

A $\frac{\text{dependency ratio}}{\% \text{ growth}}$

B $\% \text{ growth} \times 70.$

☒ C $\frac{70}{\% \text{ growth}}$

D $\frac{\% \text{ growth}}{70}$

010184



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple Choice

NOVEMBER 2010 SESSION

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

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This question paper consists of 20 printed pages.

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- 1 The diagram shows an animal cell.



In which of the labelled structures do the following processes take place?

	mutation	respiration	active transport
A	1	2	3
B	2	3	1
C	2	1	3
D	3	1	2

- 2 Which function is related to a specialised animal cell which is long, multinucleated and has many mitochondria?

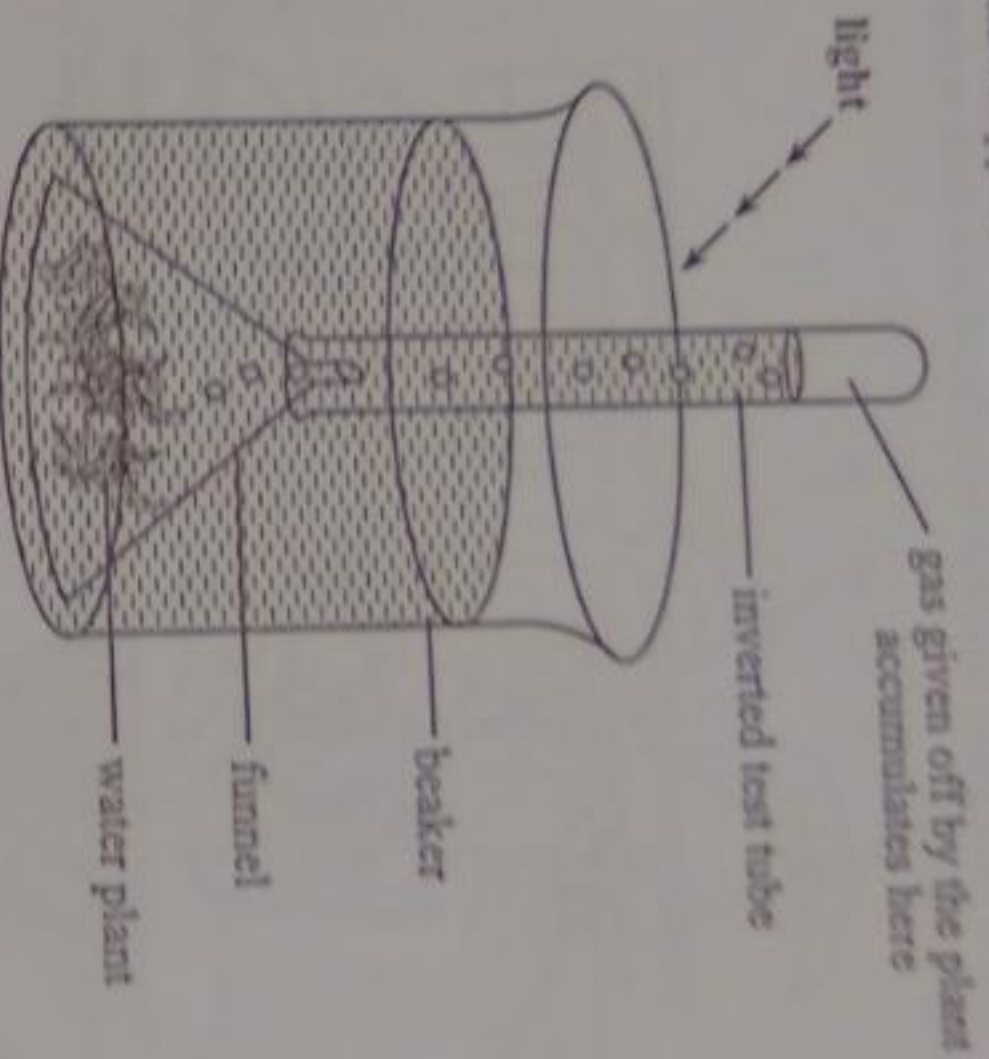
- A contraction
B impulse transmission
C oxygen transport
D secretion

- 3 Which specialised plant cell has the following features?

1. elongated and fibrous
2. thin
3. no chloroplasts
4. many mitochondria

- A palisade cell
B epidermal cell
C root hair cell
D guard cell

- 4 The diagram shows apparatus set up in an experiment.

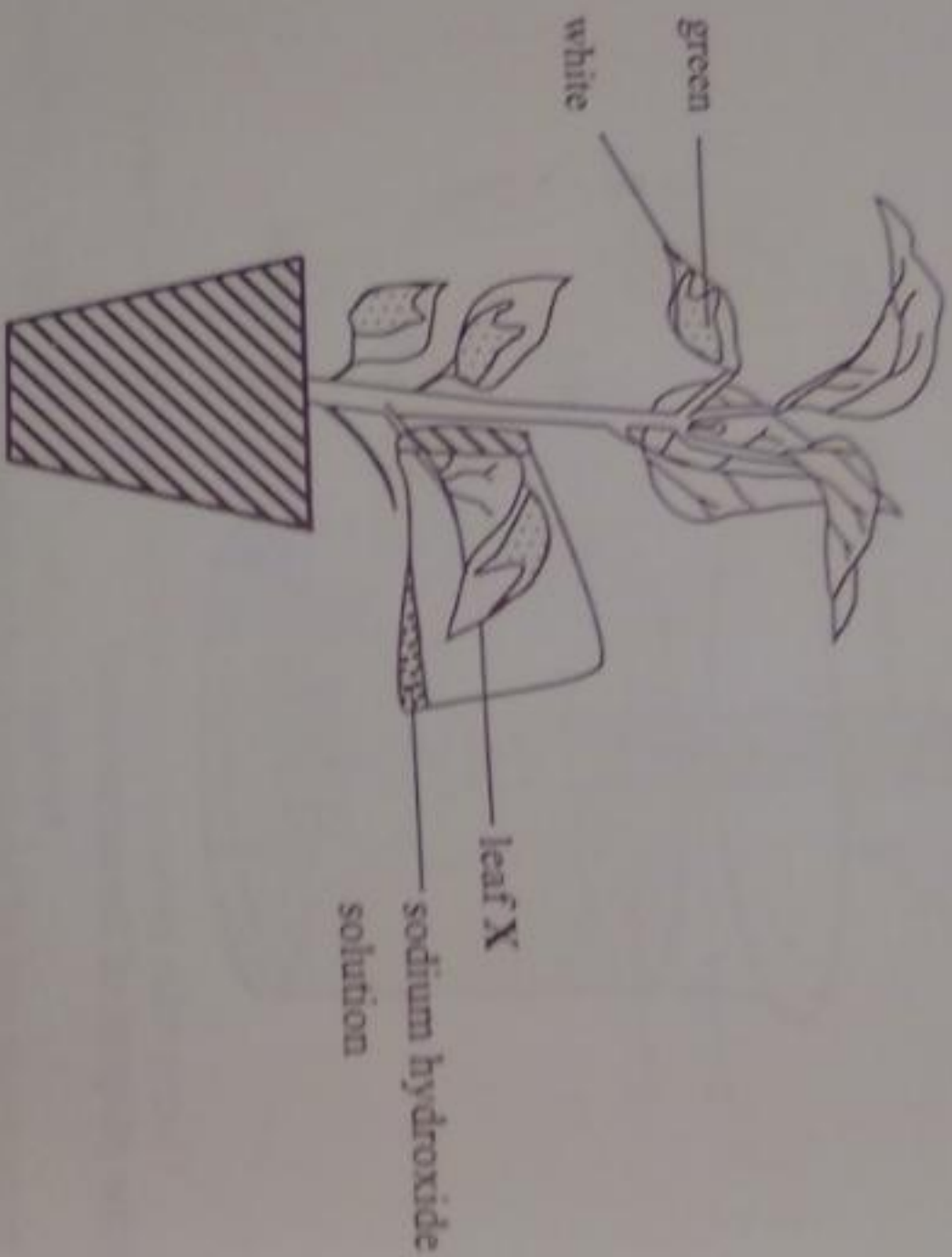


What is the purpose of the experiment?

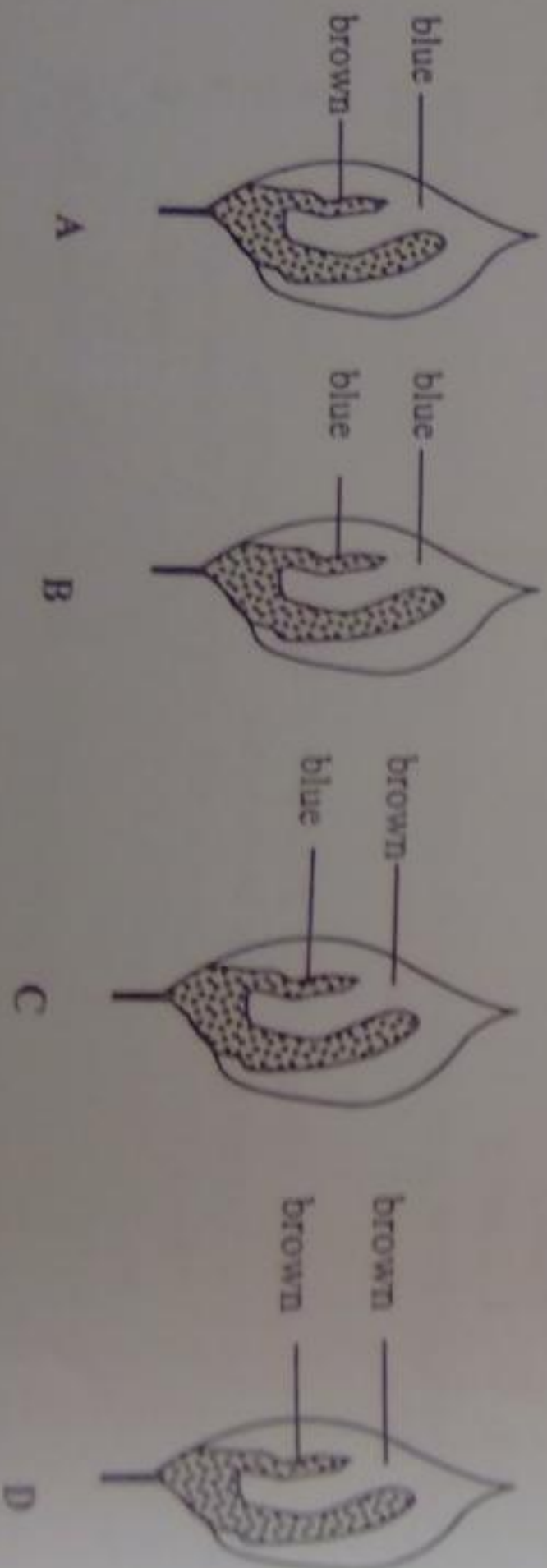
- A to measure rate of breakdown of carbohydrates
- B to show that CO_2 is necessary for photosynthesis
- C to show that water is necessary for photosynthesis
- D to measure rate of photosynthesis

5

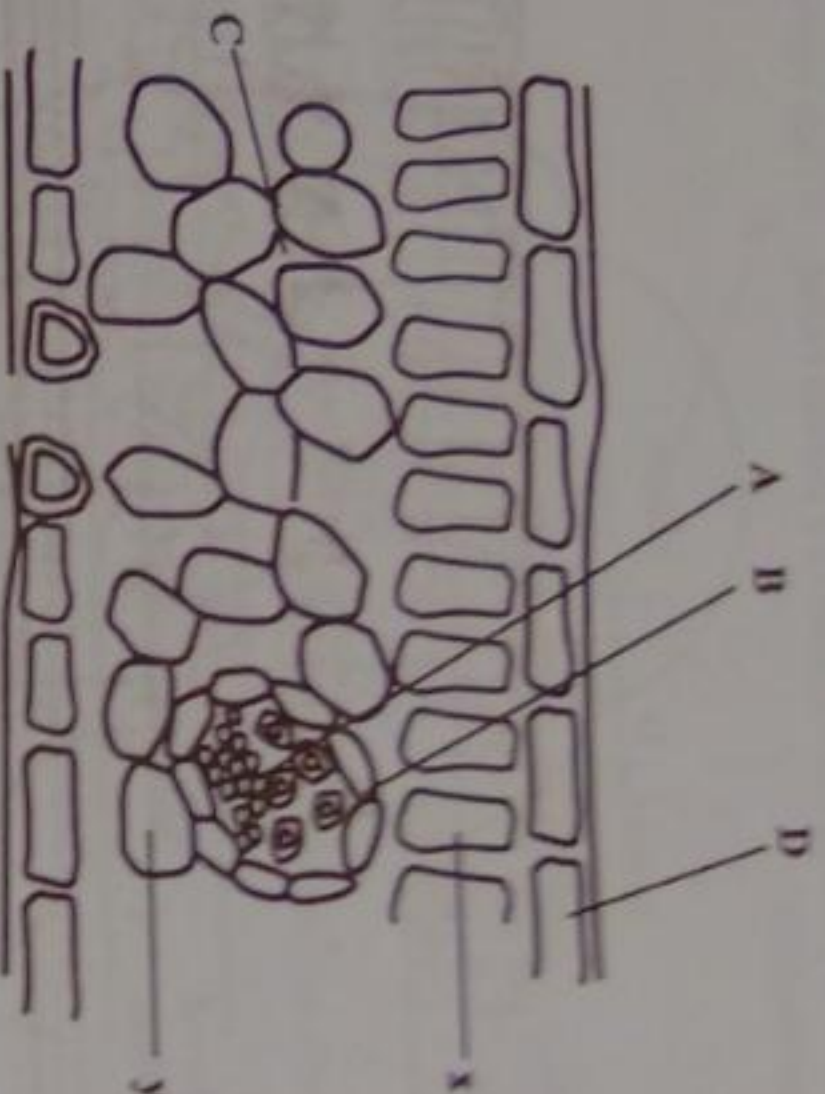
The diagram shows a variegated leaf that was placed in a flask containing concentrated sodium hydroxide solution.



Which leaf shows the correct colour after testing X with iodine?



- 6 The diagram shows the cross section of a leaf. Where do cells **x** and **y** obtain water for photosynthesis?

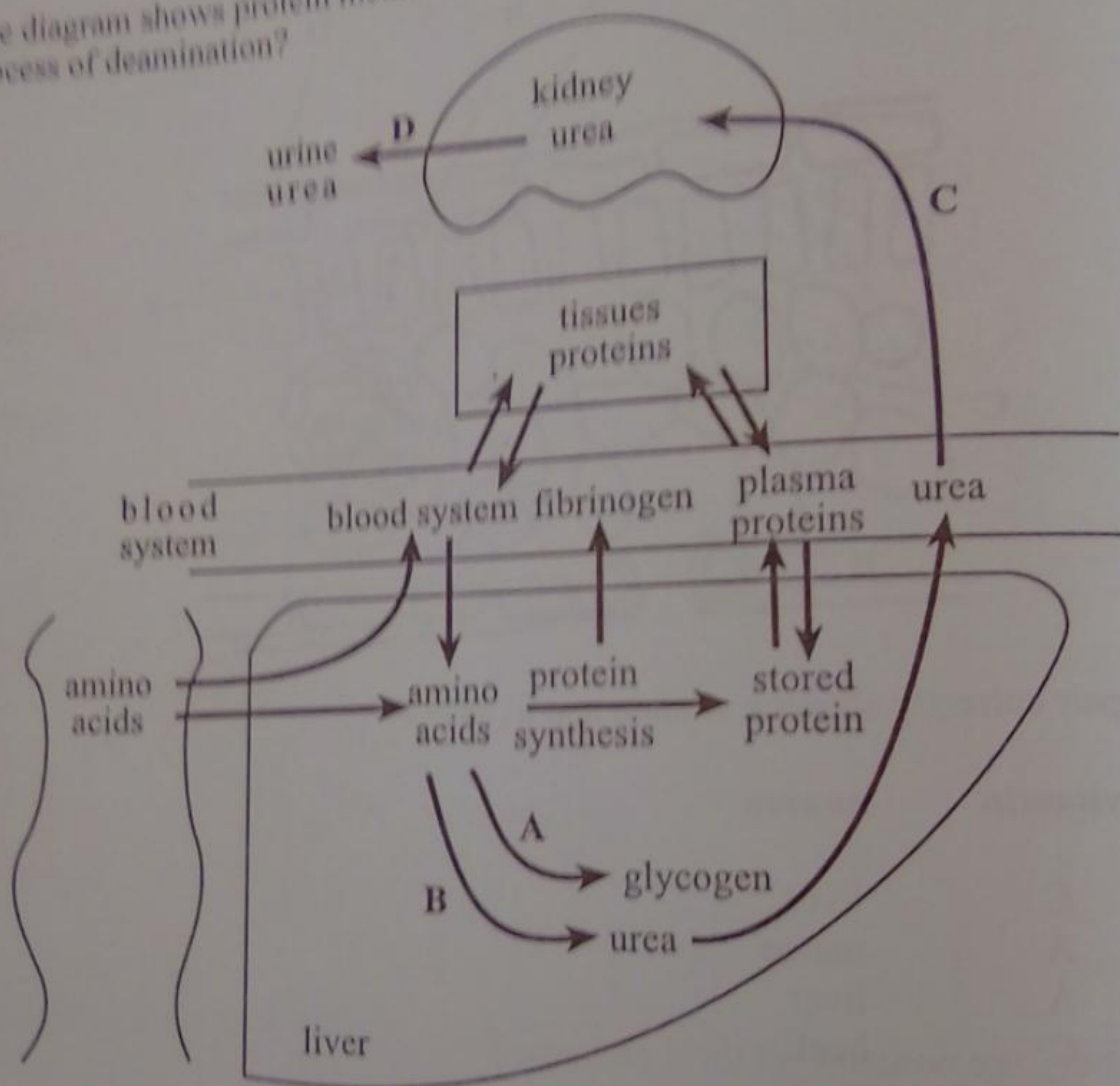


- 7 Which pair correctly shows a vitamin and its source

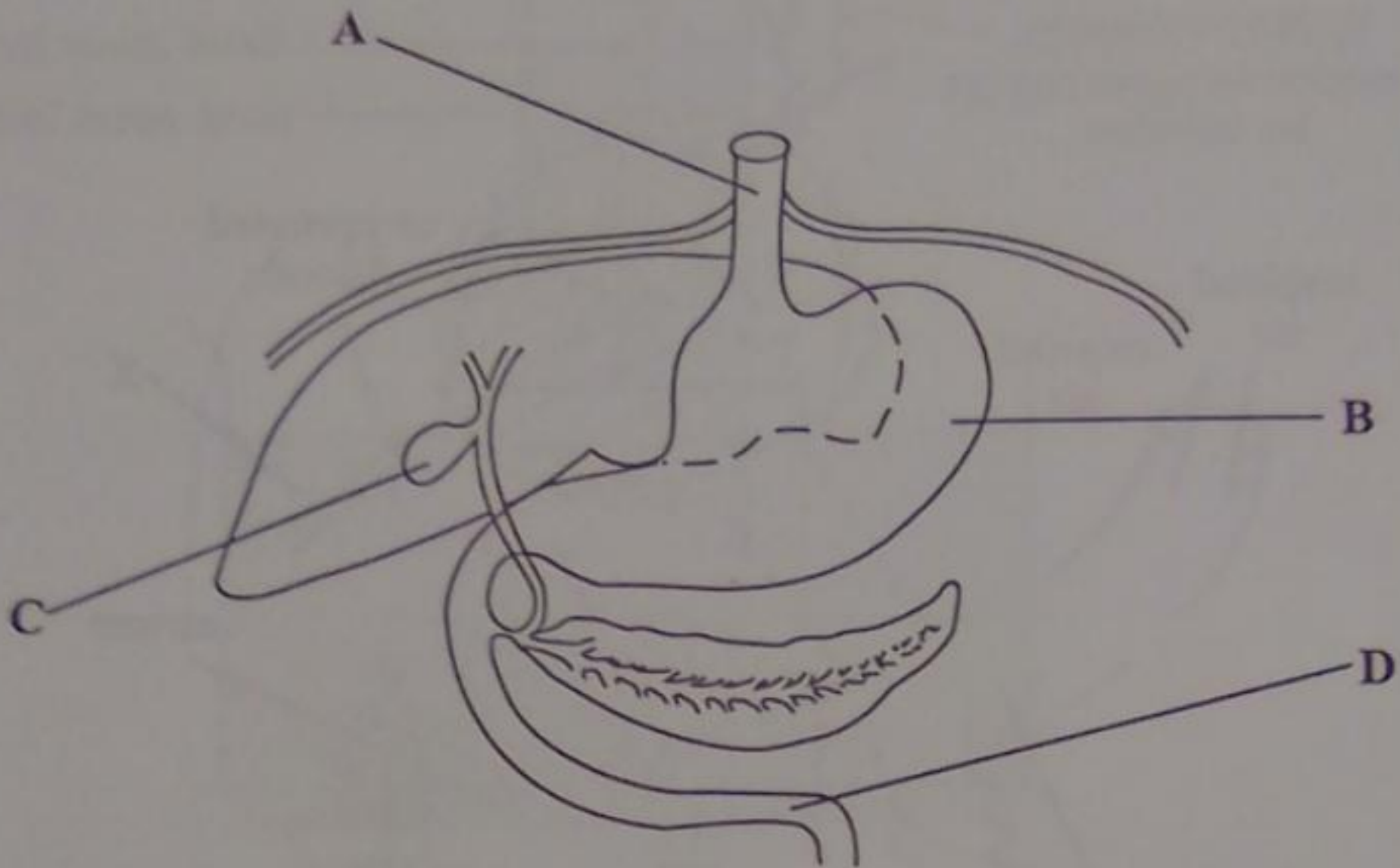
vitamin	source
A	C
B	A
C	A
D	C
	liver
	maize
	liver
	beef

8

The diagram shows protein metabolism in a human being. Which letter represents the process of deamination?



- 9 The diagram shows part of the human alimentary canal. Which part has contents with the lowest pH?



- 10 The following is a list of elements that are found in a food molecule.

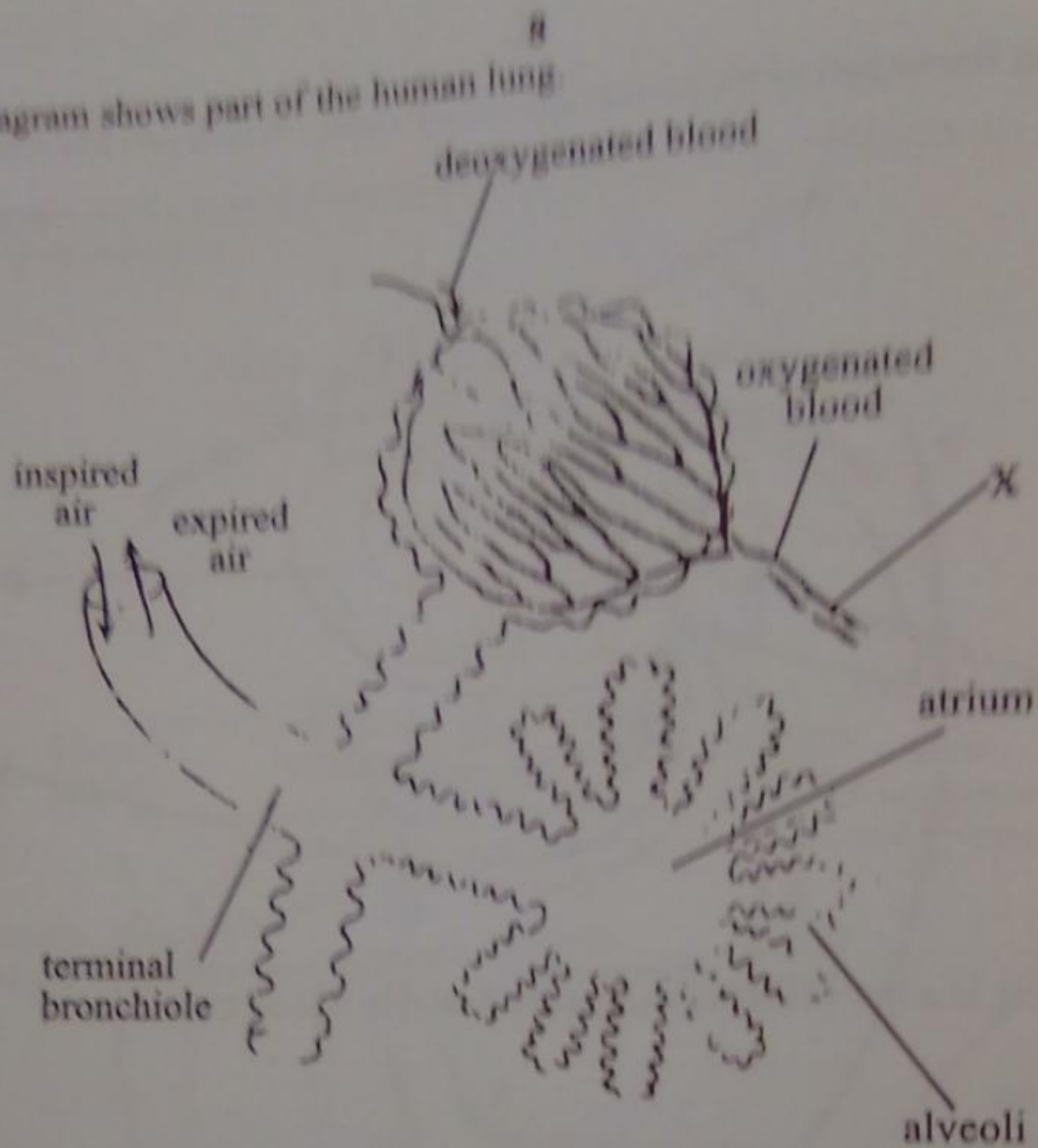
1. carbon
2. hydrogen
3. oxygen
4. nitrogen

Which molecule contains all these elements?

- A cellulose
- B glycogen
- C starch
- D albumin

11

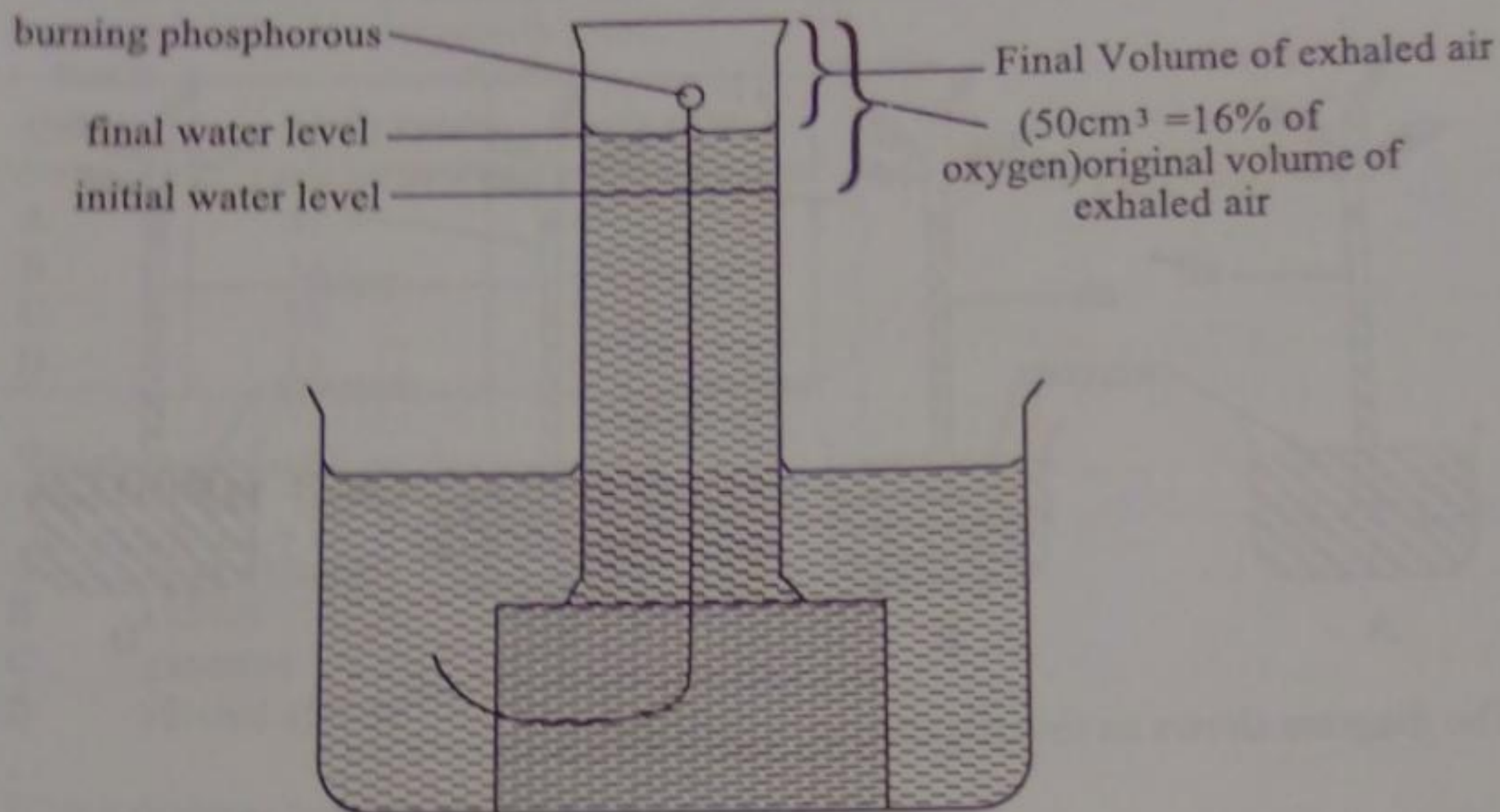
The diagram shows part of the human lung.



What is vessel X?

- A artery
- B arteriole
- C venule
- D capillary

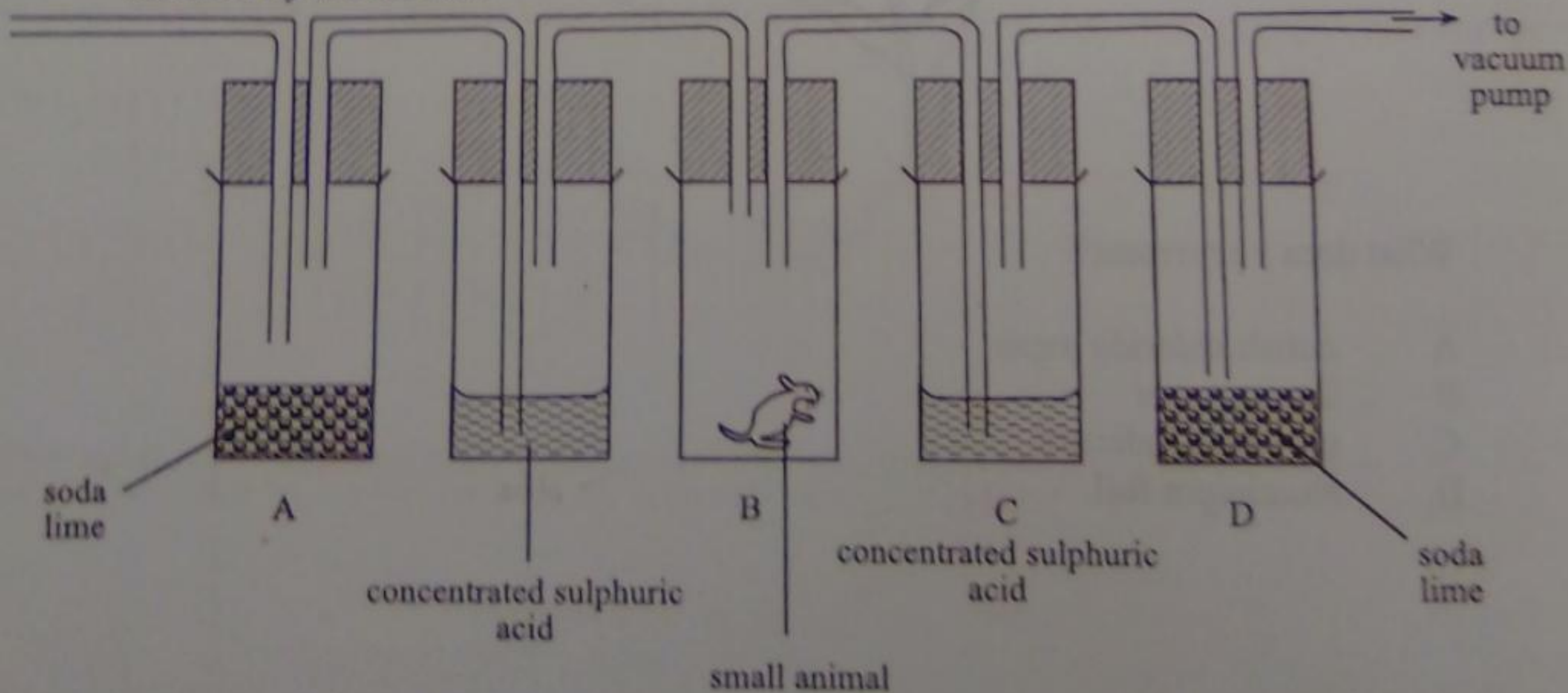
- 12 The diagram shows an experiment to demonstrate the percentage of oxygen in exhaled air.



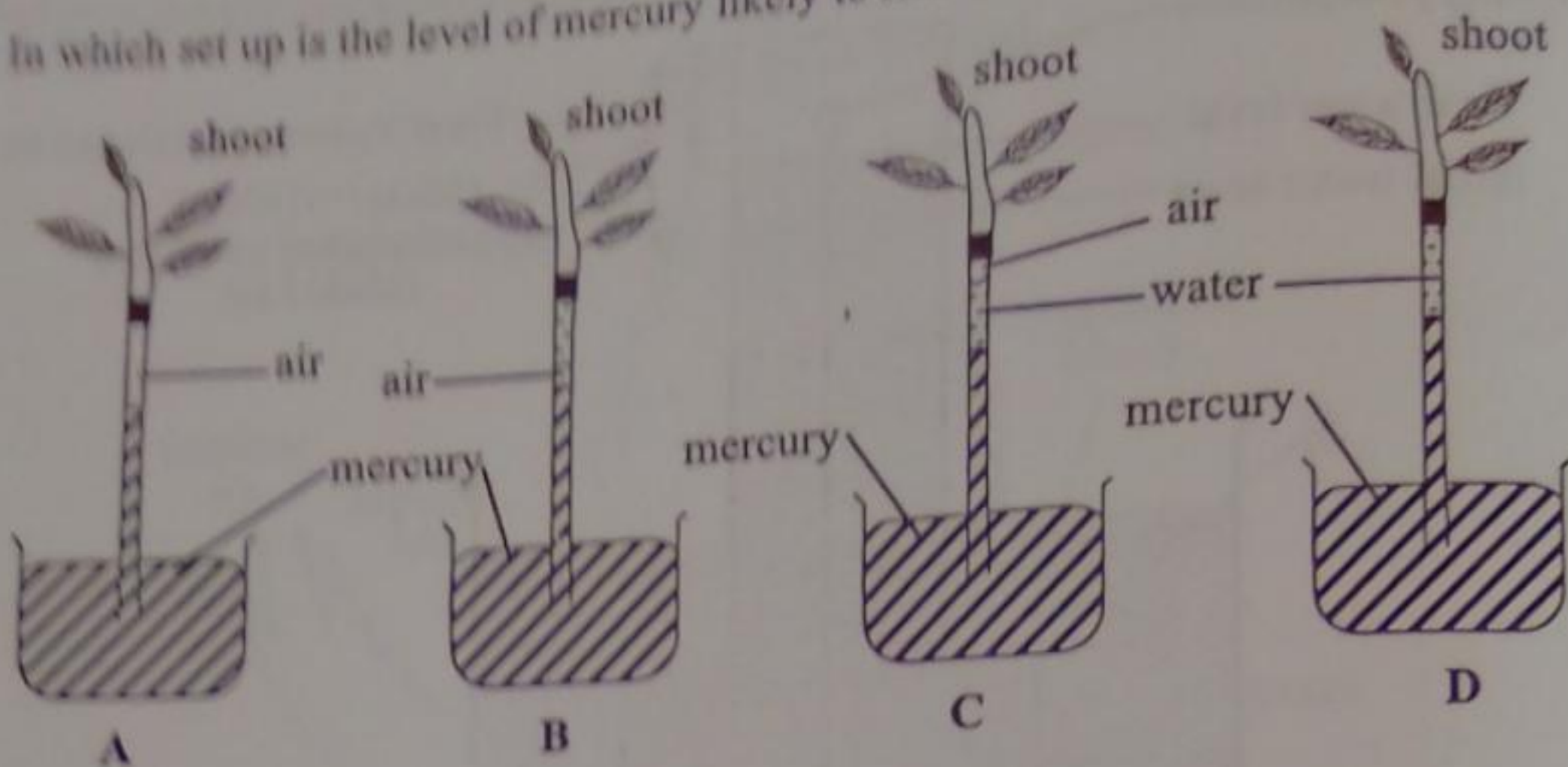
What is the final volume of exhaled air?

- A 8 cm³
- B 40 cm³
- C 42 cm³
- D 47,5 cm³

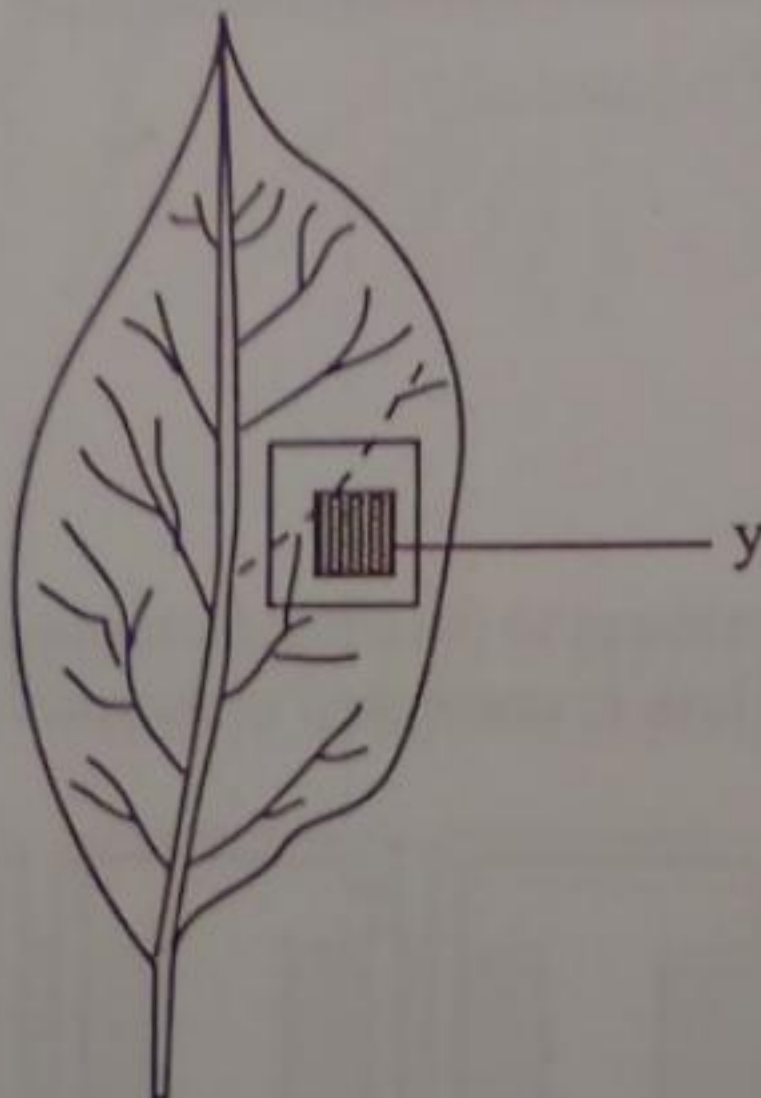
- 13 The diagram shows an experiment to determine consumption and carbon dioxide release by a small animal. Which flask is reweighed to determine the amount of carbon dioxide released by the animal?



- 14 In which set up is the level of mercury likely to rise furthest?



- 15 The diagram shows an investigation on transpiration.



What does y represent?

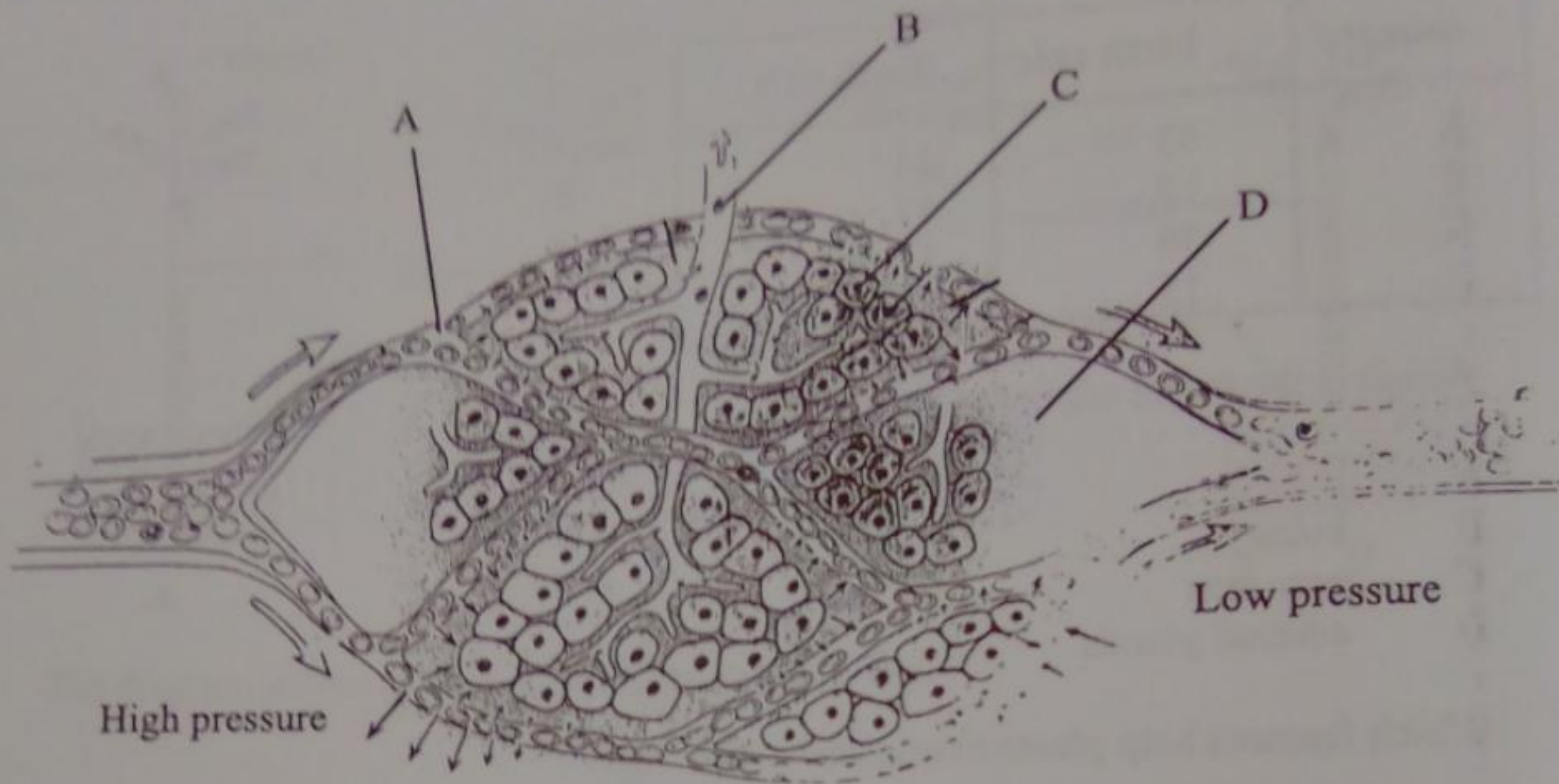
- A cobalt chloride paper
- B litmus paper
- C universal indicator
- D aluminium foil

- 16 The table shows birth and death rate statistics (1981 world population data sheet). Which country had the lowest growth rate?

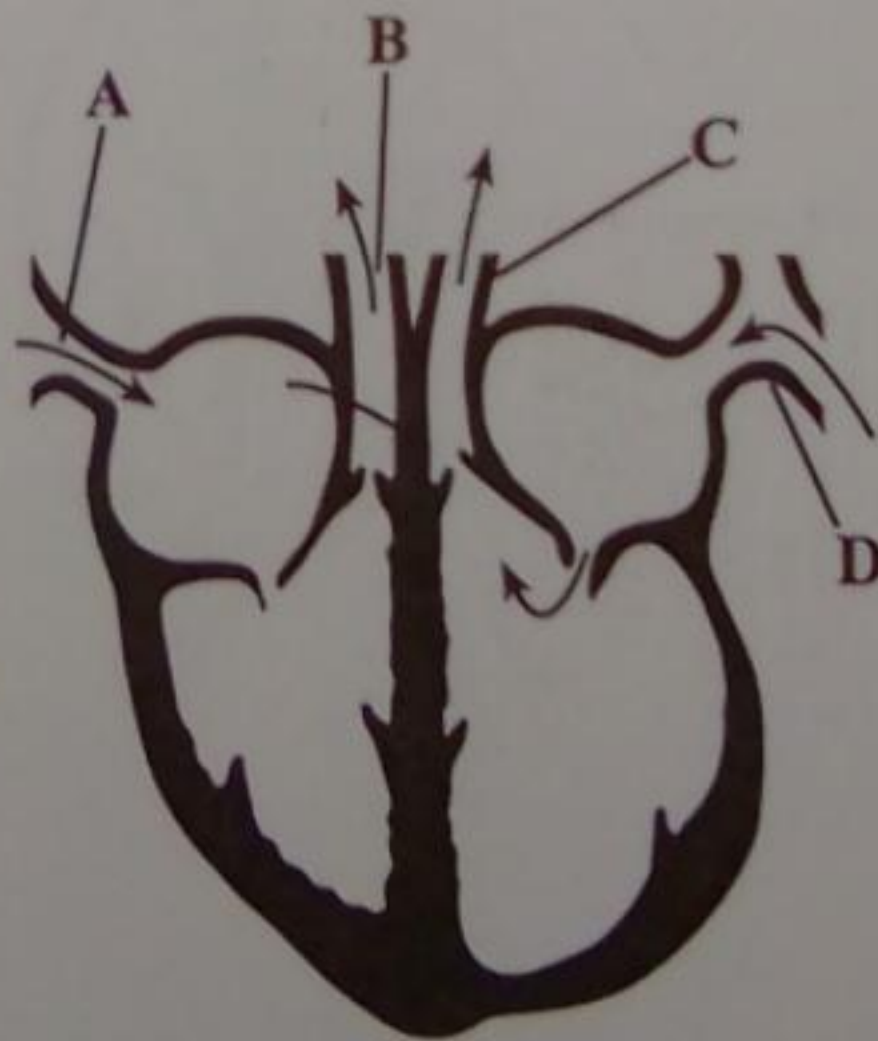
country	birth rate	death rate
A	53	14
B	13	12
C	36	12
D	15	6

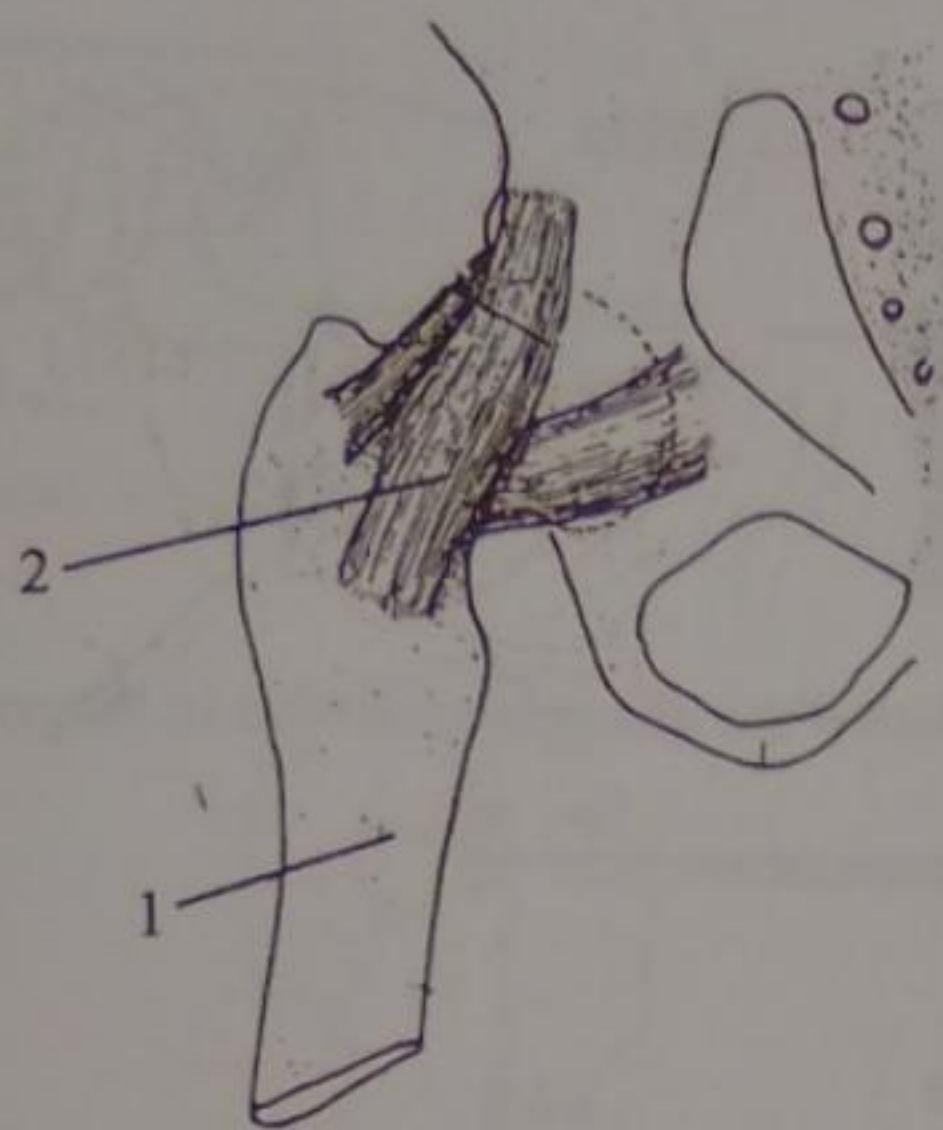
- 17 Which is the target organ for insulin?
- A liver
 - B kidney
 - C pancreas
 - D adrenal glands
- 18 Which features help plants to minimise water loss?
- 1. sunken stomata
 - 2. thin cross section
 - 3. broad surface
 - 4. presence of hairs
- A 1 and 4
 - B 2 and 4
 - C 1 and 2
 - D 1, 2 and 3

- 19 The diagram shows a capillary bed. Which label shows a tissue fluid?



- 20 The diagram shows the human heart. Which blood vessel has the thickest walls?





What are the structures 1 and 2?

	1	2
A	femur	tendon
B	femur	ligament
C	humerus	tendon
D	humerus	ligament

22 Which blood vessel carries blood from the liver to the vena cava?

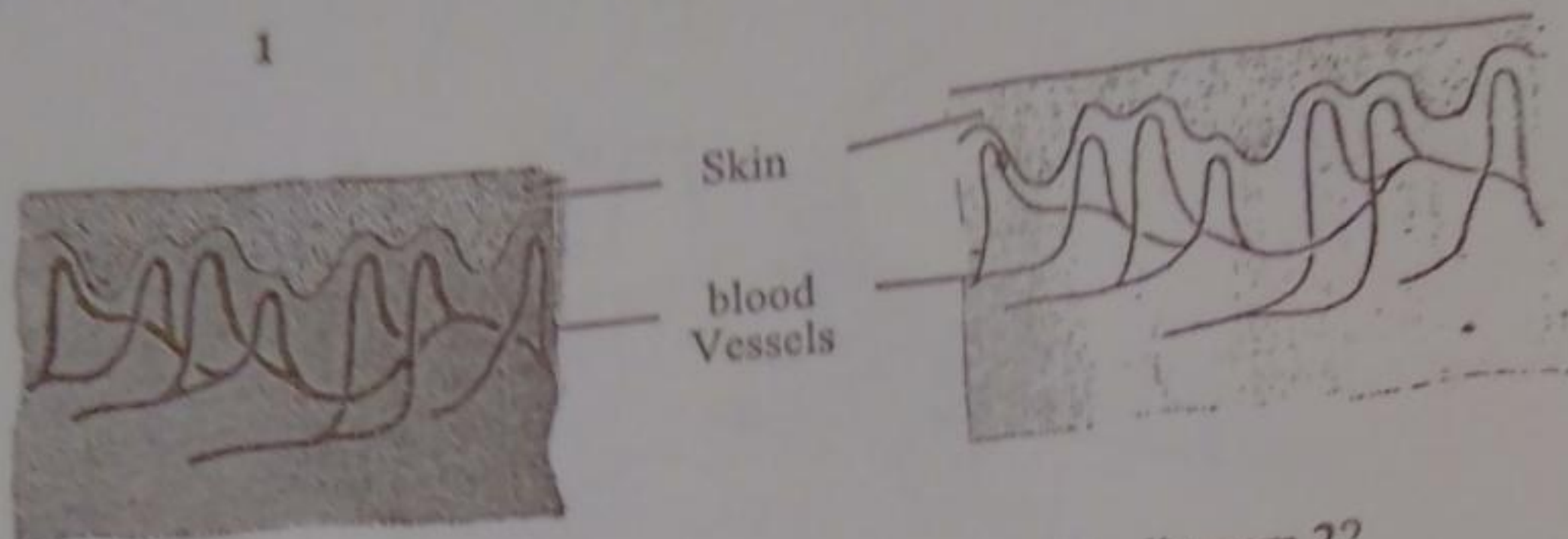
- A hepatic vein
- B hepatic portal vein
- C renal vein
- D pulmonary vein

23 Which is the correct sequence of vessels through which urea moves from blood to the external environment.

- A renal tubules → urethra → bladder → ureter
- B renal tubules → ureter → bladder → urethra
- C bladder → renal tubules → ureter → urethra
- D bladder → renal tubules → urethra → ureter

24

14
Diagrams 1 and 2 show the same skin under different conditions.



What condition could have caused the changes observed in diagram 2?

- A increase in external temperature
- B excessive sweating
- C taking a cold bath
- D reduced metabolic rate

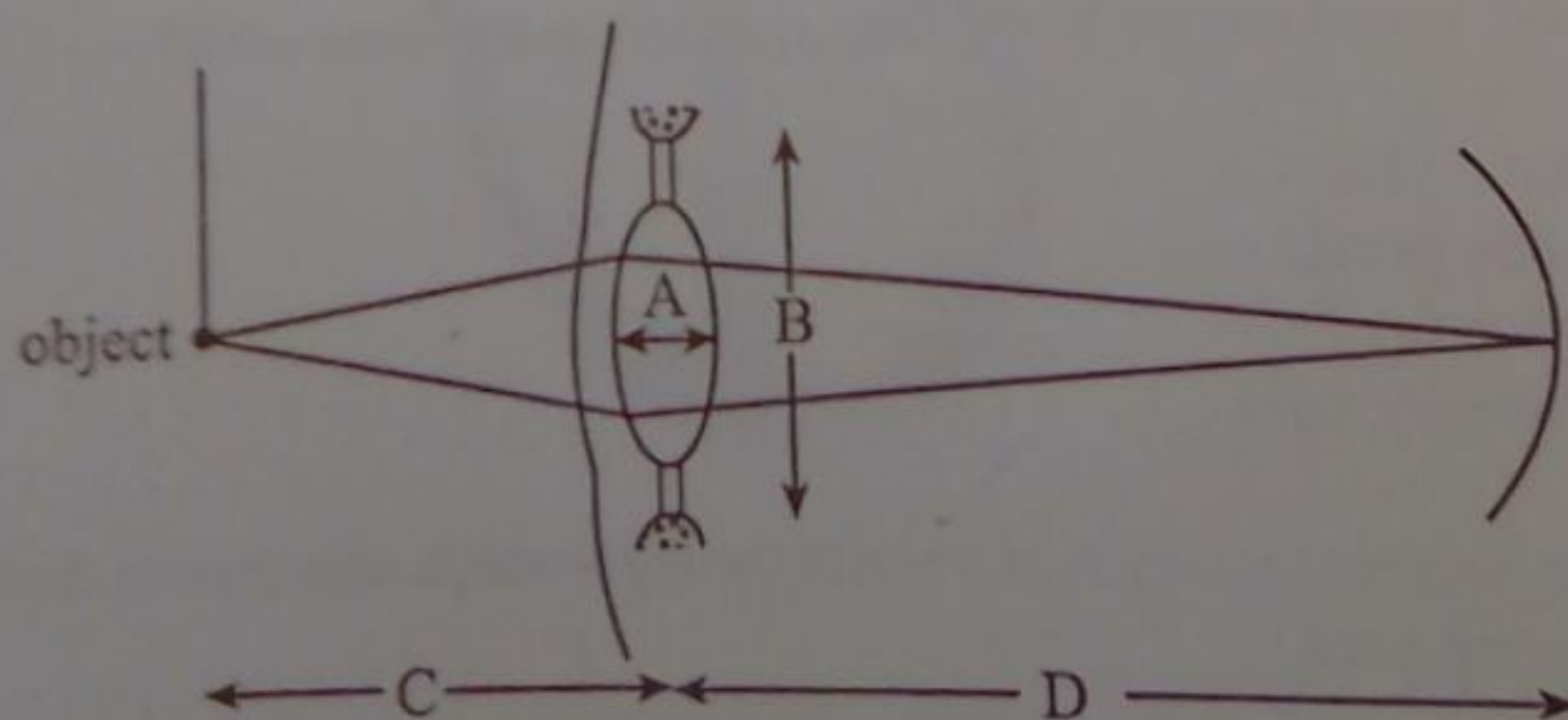
25

Which nerve fibre is directly linked to a receptor?

- A dendron of sensory neurone
- B axon of sensory neurone
- C dendron of motor neurone
- D axon of motor neuron

26

The diagram shows the human eye focusing on a distant object. Which distance remains unchanged when the same eye focuses on a nearby object?



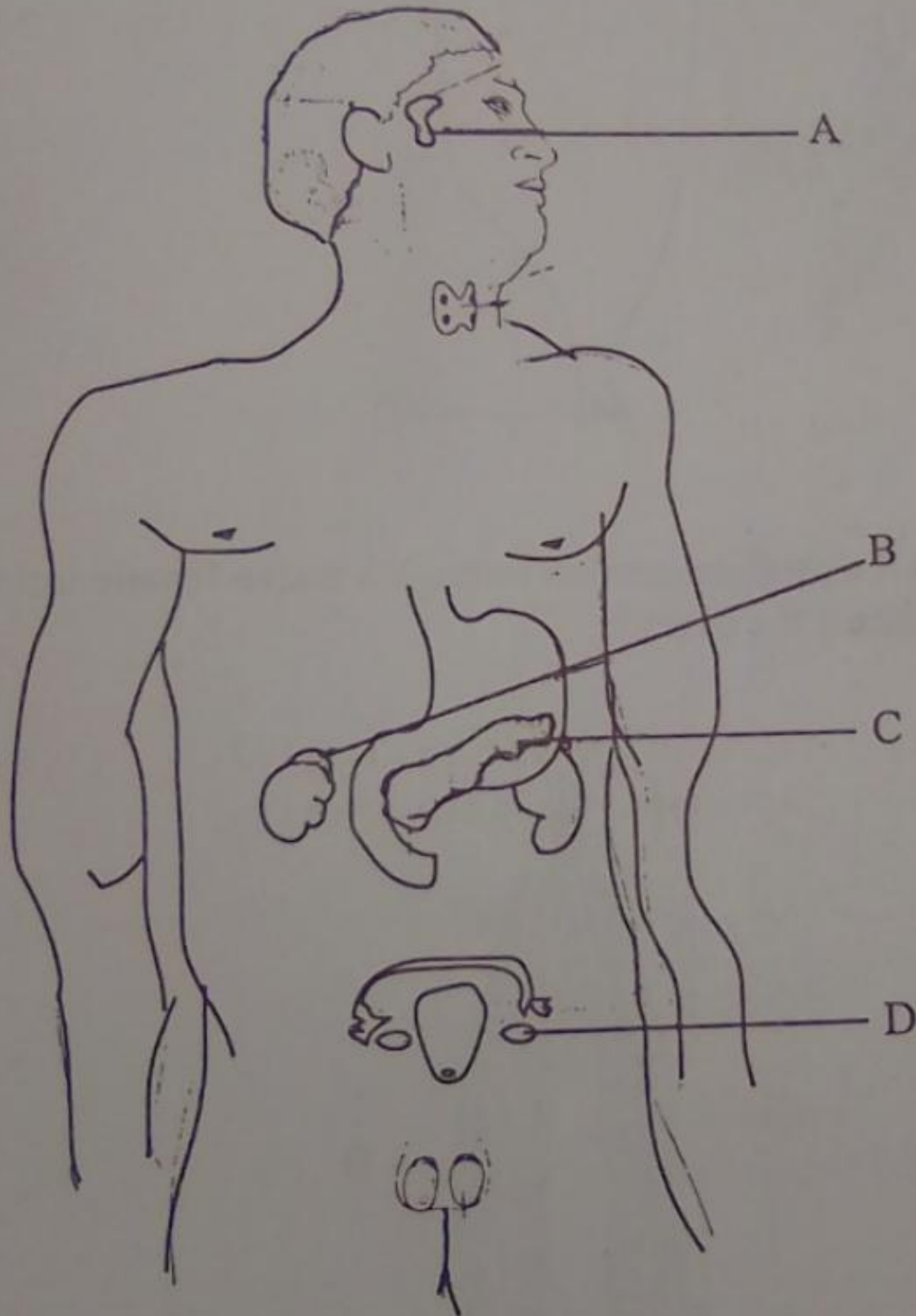
27

Which drug is a stimulant when taken in small quantities but a depressant when taken in large quantities?

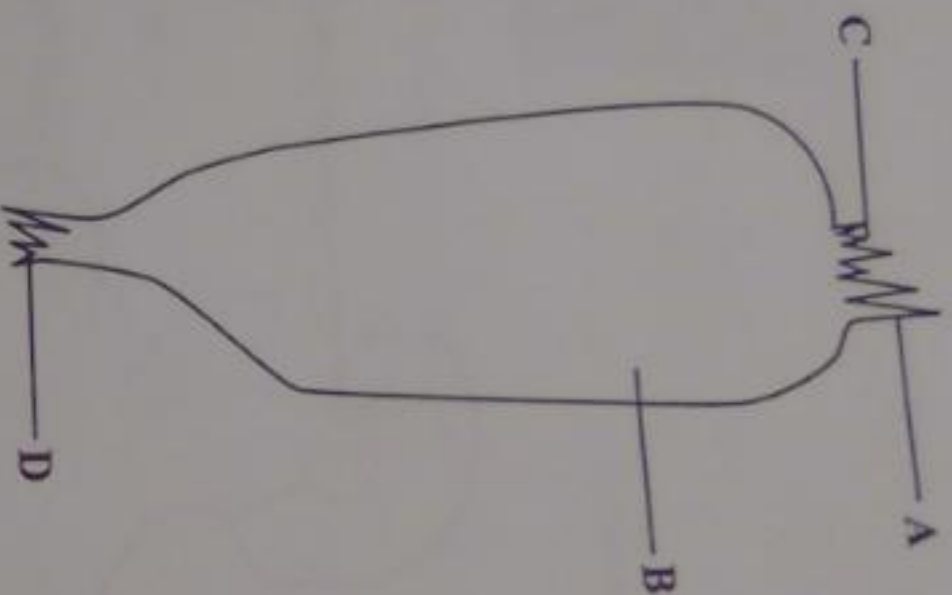
- A nicotine
- B cannabis
- C heroine
- D alcohol

28

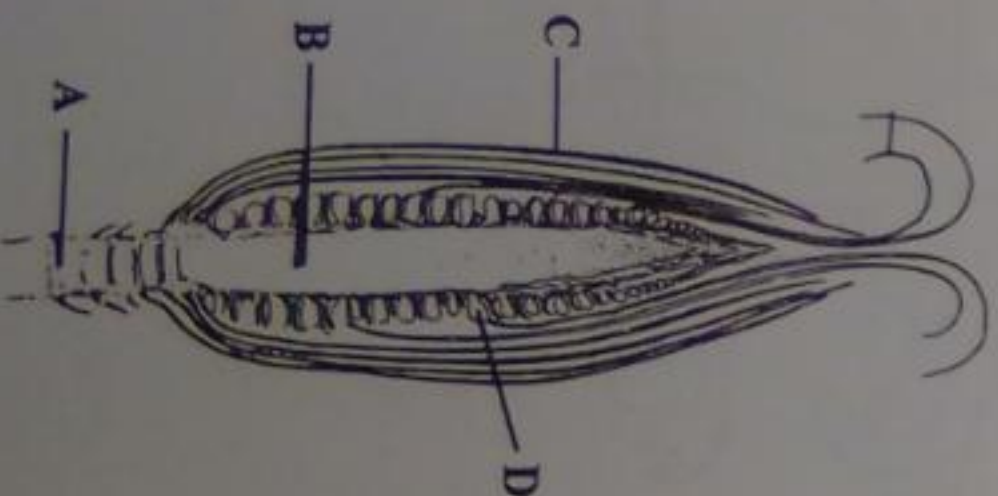
Which gland secretes adrenaline?



- 29 The diagram shows a section through a bulb. Which structure stores food for growth in the next rainy season?



- 30 The diagram shows a longitudinal section through a maize female inflorescence (cob). Which structure represents the ovary?



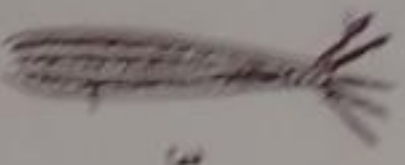
41 The diagram shows four different fruits. Which fruits have the same method of dispersal?



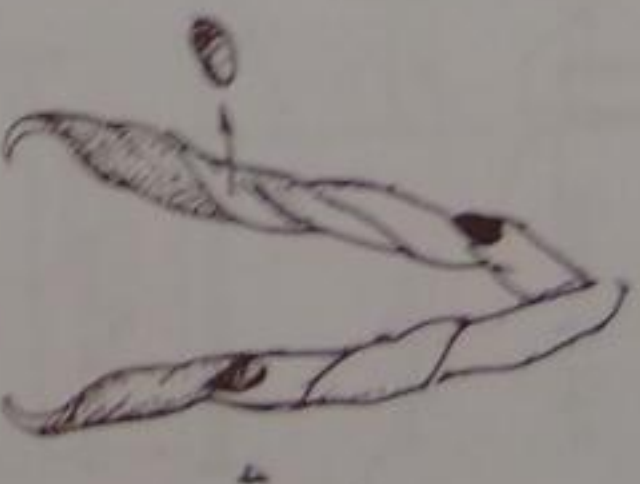
1



2



3



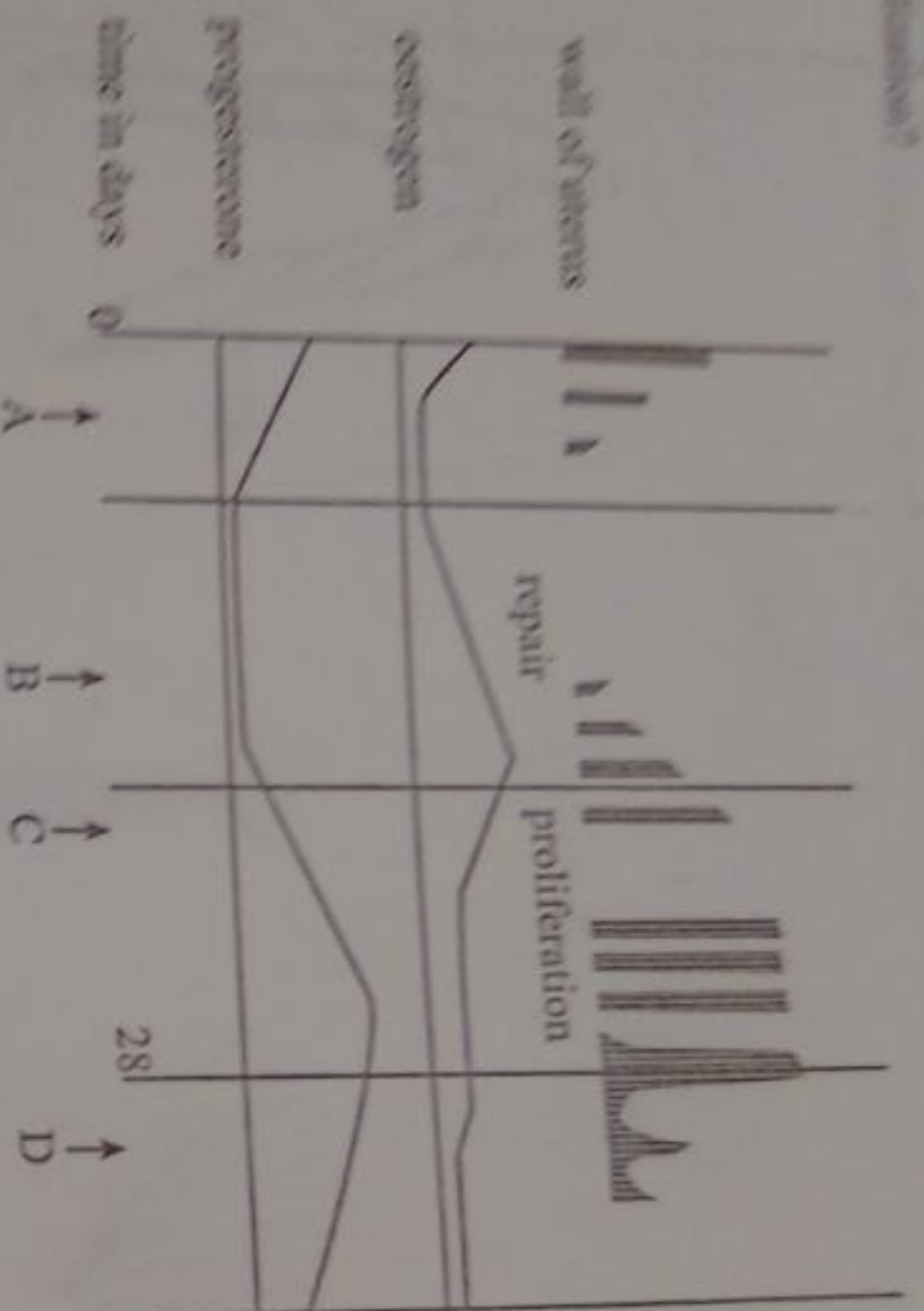
4

- A 1 and 4
B 1 and 2
C 2 and 3
D 1, 2 and 3

32 Which statement about the placenta is true?

- A It always forms on the upper part of the uterus.
B It is formed just before implantation.
C It allows foetal blood to mix with maternal blood.
D It protects foetal circulation from maternal blood pressure.

33 The diagram shows the menstrual cycle. During which day is copulation likely to result in fertilisation?



34 Which characteristics refer to viruses?

1. They are all parasitic.
2. Their nuclei contain DNA/RNA.
3. They are non-cellular.

- A 1 and 2
B 2 and 3
C 1 and 3
D 1, 2 and 3

35 Which is the least frequent method of spread of HIV in Zimbabwe?

- A blood transfusion
B mother to child via placenta or breast milk
C use of unsterilised needles by drug addicts
D sexual intercourse

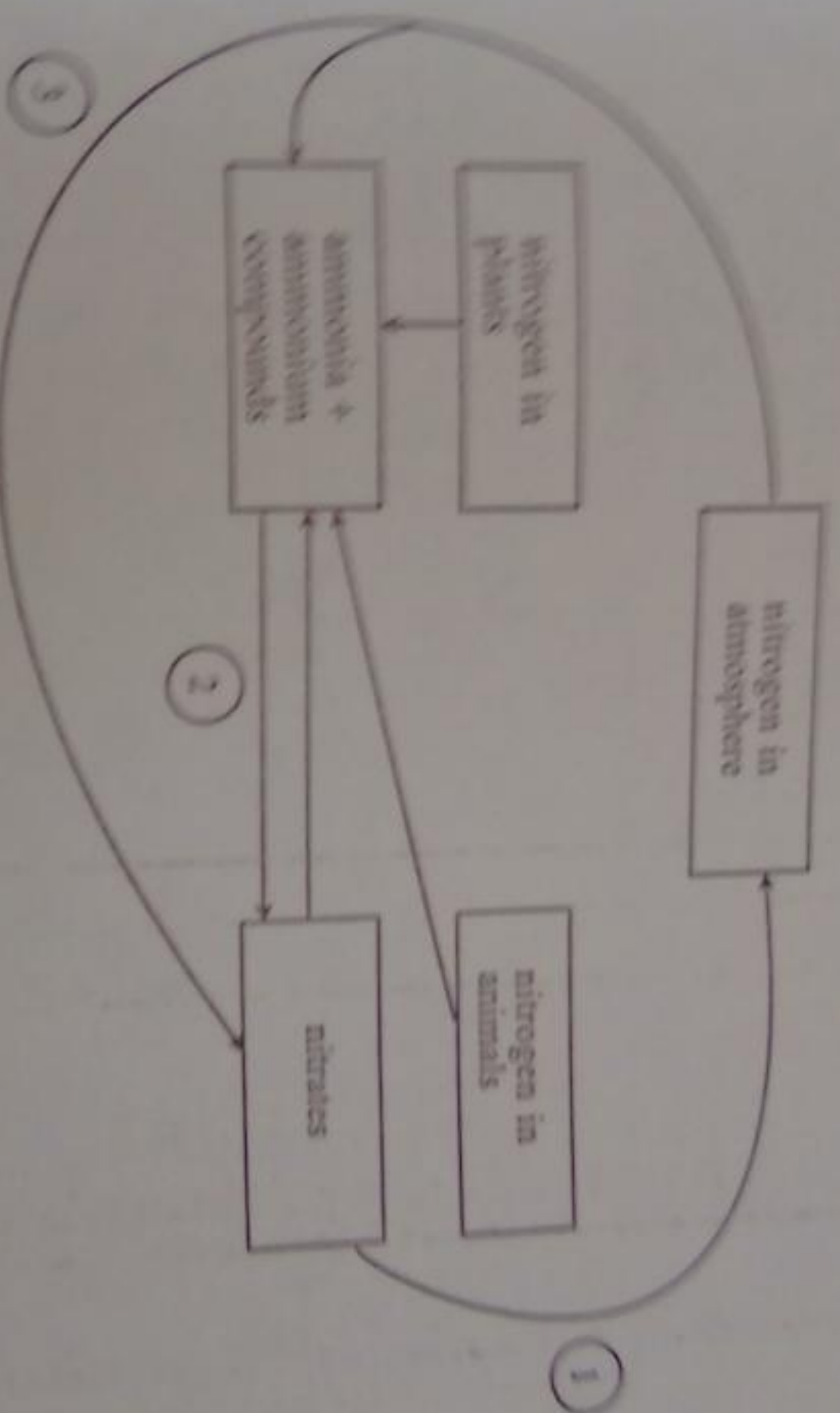
36 What causes Malaria?

- A virus
B bacterium
C plasmodium
D amoebae

19 The term allele refers to

- A two different genes.
- B genes found on the same chromosome.
- C alternative forms of the same gene.
- D genes that occur in pairs.

20 The diagram shows part of the nitrogen cycle.

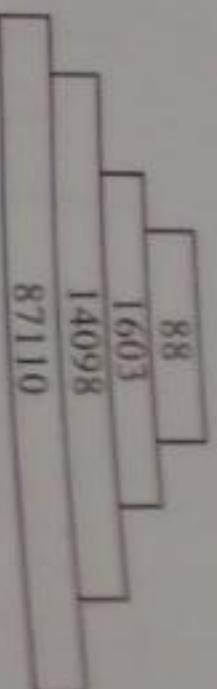


What are processes 1, 2 and 3?

nitrogen fixation nitrification denitrification

- A 1 3 2
- B 3 2 1
- C 2 3 1
- D 3 1 2

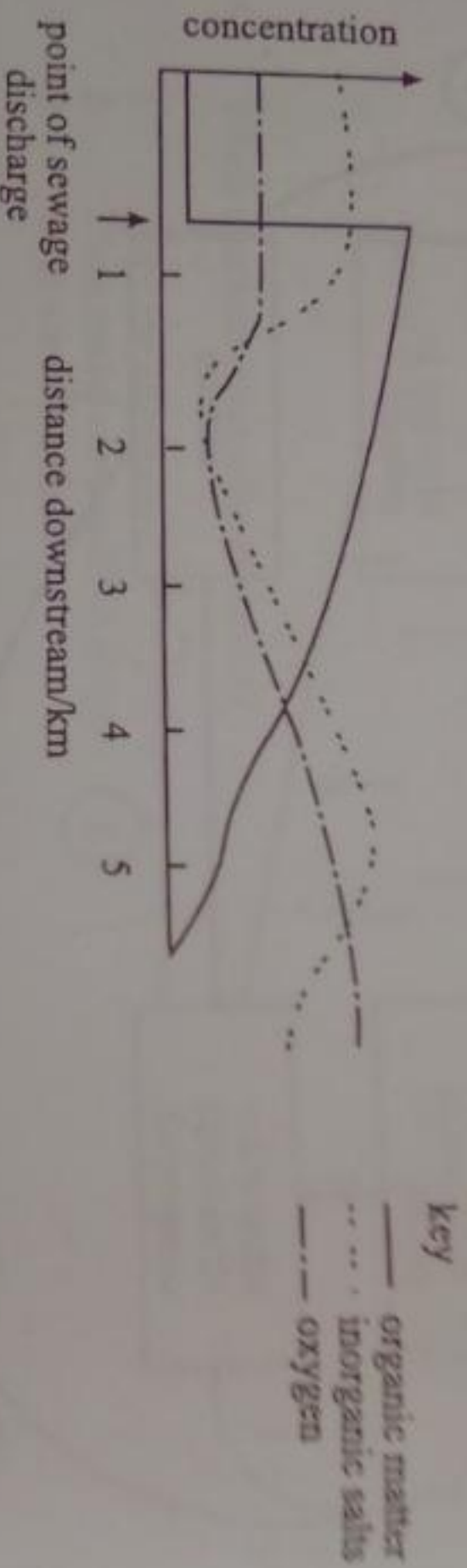
- 39 The diagram shows the energy content of each trophic level per unit area (kJ/m^2).



Which process contributes to the decrease in energy content?

- A secretion
- B excretion
- C digestion
- D photosynthesis

- 40 The diagram shows changes in the content of river water downstream from a discharge of sewage.



What causes the increase in oxygen concentration after 2 km?

- A decomposition of organic matter
- B algal bloom
- C decrease in animal population
- D increase in dissolved carbon dioxide



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

5008/1

BIOLOGY

PAPER 1 Multiple Choice

NOVEMBER 2009 SESSION

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

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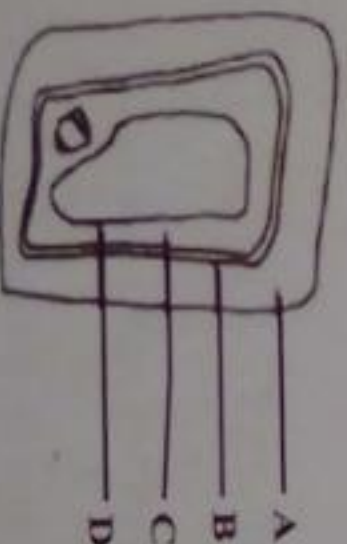
1 Cells have the following characteristics:

1. presence of starch grains ✓
2. presence of a cell membrane ✓
3. capable of dividing ✓
4. presence of chloroplasts ✓

Which characteristics distinguish plant cells from animal cells?

- A 4 only
 B 1 and 4
 C 2 and 3 +
 D 1, 3 and 4 ✓

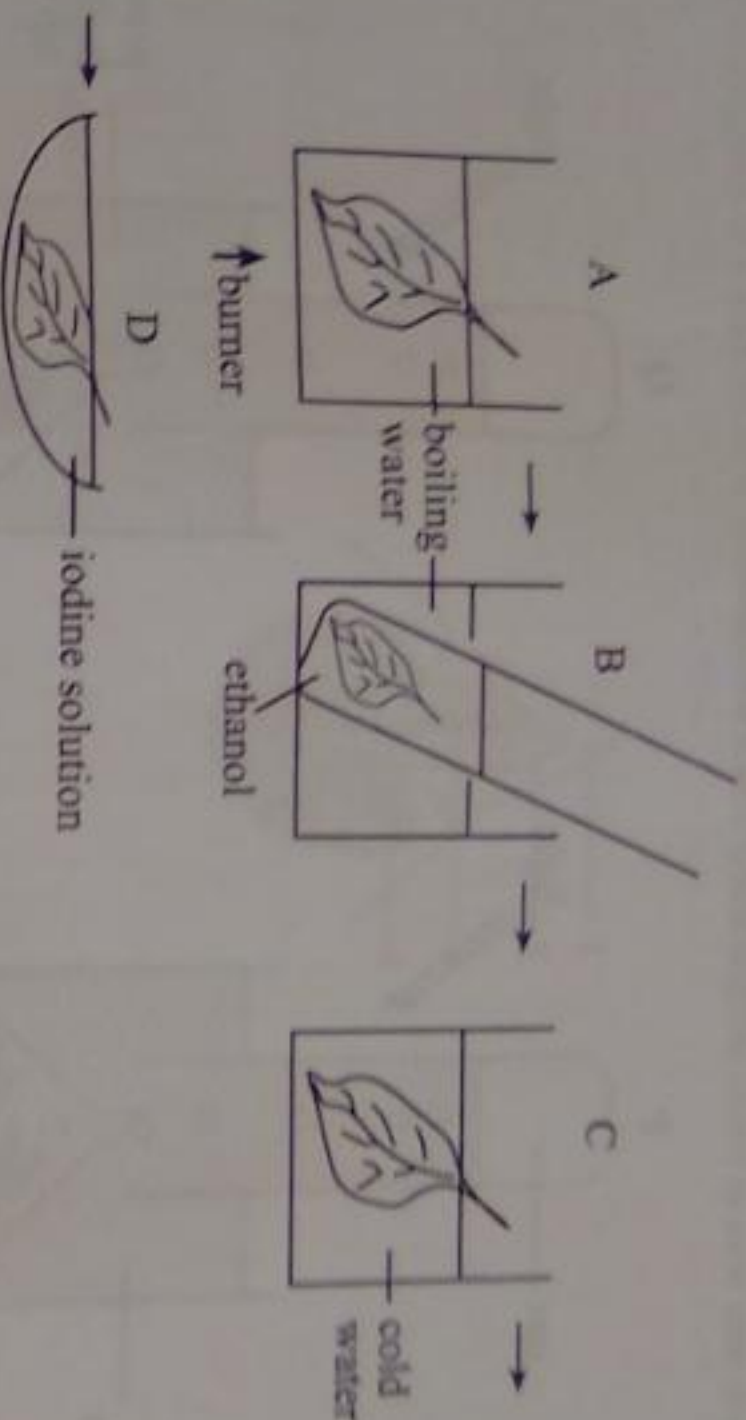
2 The diagram shows a plant cell. Which part of the cell is freely permeable?



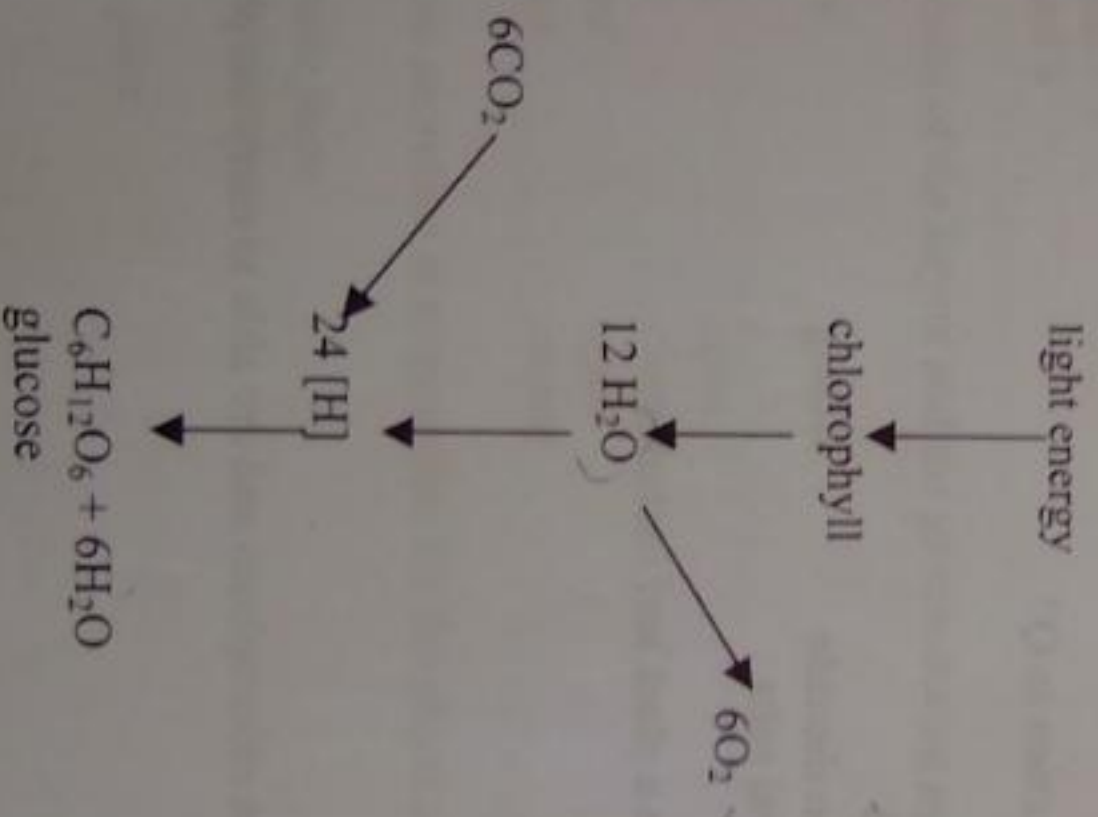
3 Unlike plant cells, red blood cells burst when put in distilled water. What is the reason for this?

- A They do not have cell walls.
 B They are not capable of osmoregulation.
 C They are more concentrated than the water.
 D They do not have nuclei.

- 4 The diagrams show a test carried out on a leaf. At what stage does the leaf become brittle?



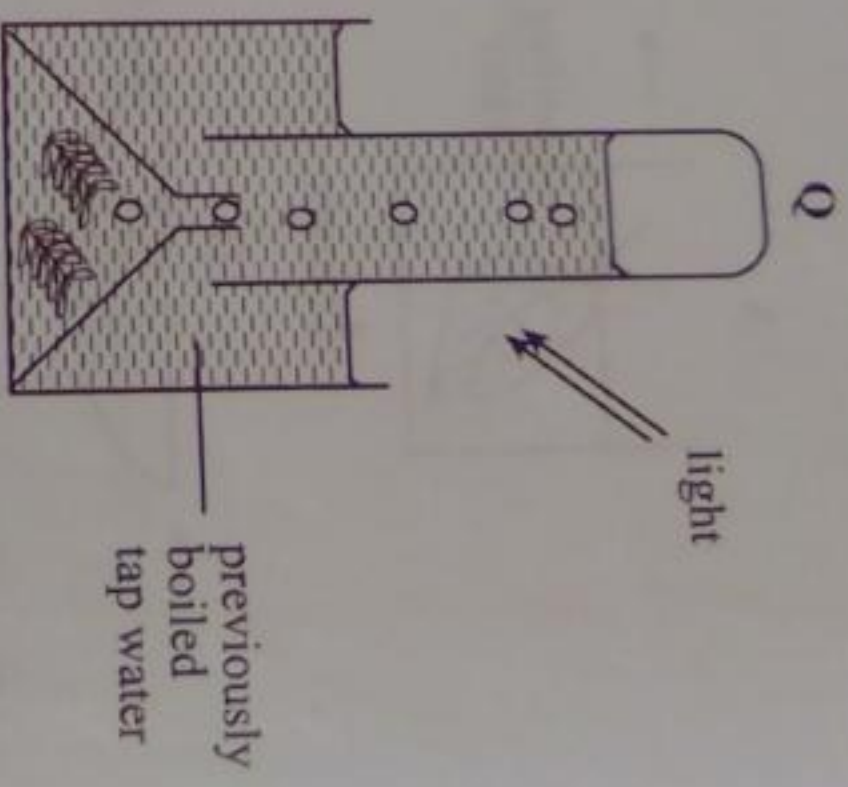
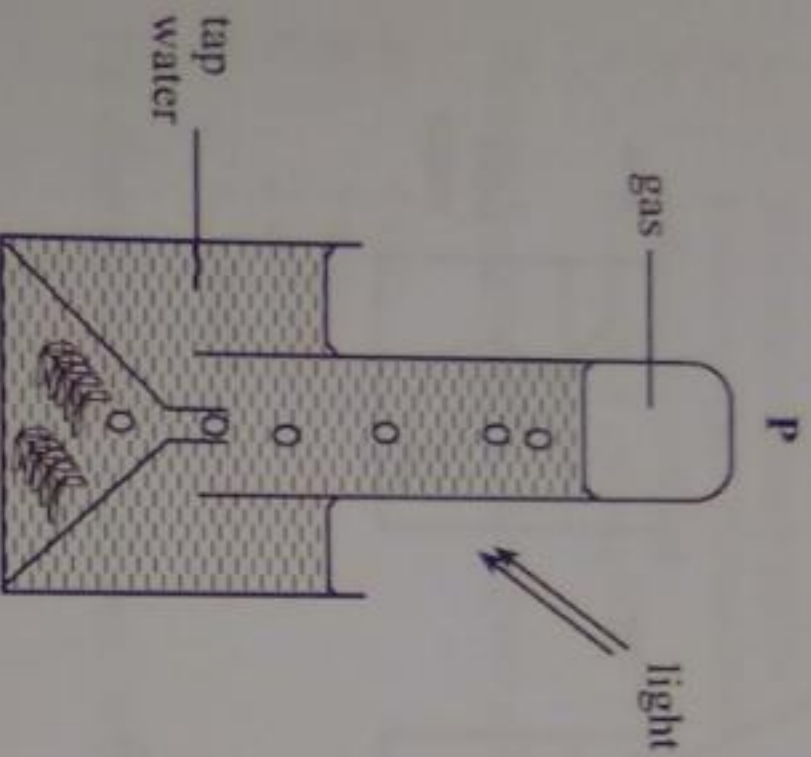
- 5 The diagram shows a summary of the process of photosynthesis.



Which substance is both a reactant and a product?

- A O_2
 B CO_2
 C H_2O
 D $\text{C}_6\text{H}_{12}\text{O}_6$

- 6 An experiment on photosynthesis was set up as in diagrams **P** and **Q**. The water in the two sets was at room temperature when set up.



Why is there no gas production in **Q**?

- A** no dissolved oxygen
- B** enzymes denatured
- C** no dissolved carbon dioxide
- D** no dissolved mineral salts

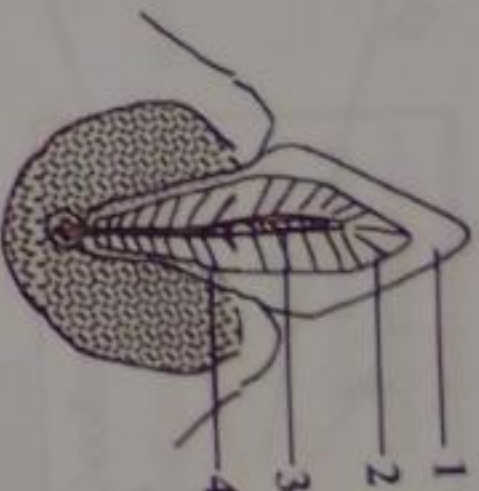
- 7 The types of cells found in a plant leaf include:

- 1. xylem vessels
- 2. phloem tube cells
- 3. guard cells
- 4. mesophyll cells

Which types of cell contain chloroplasts and are able to carry out photosynthesis?

- A** 4 only
- B** 3 and 4
- C** 1, 2 and 3
- D** 2, 3 and 4

- 8 The diagram is a section of a human tooth.



Which structures contain living material?

- A 2 and 3
- B 3 and 4
- C 1, 2 and 3
- D 2, 3 and 4

- 9 What is the status of the larynx and the glottis during swallowing?

larynx

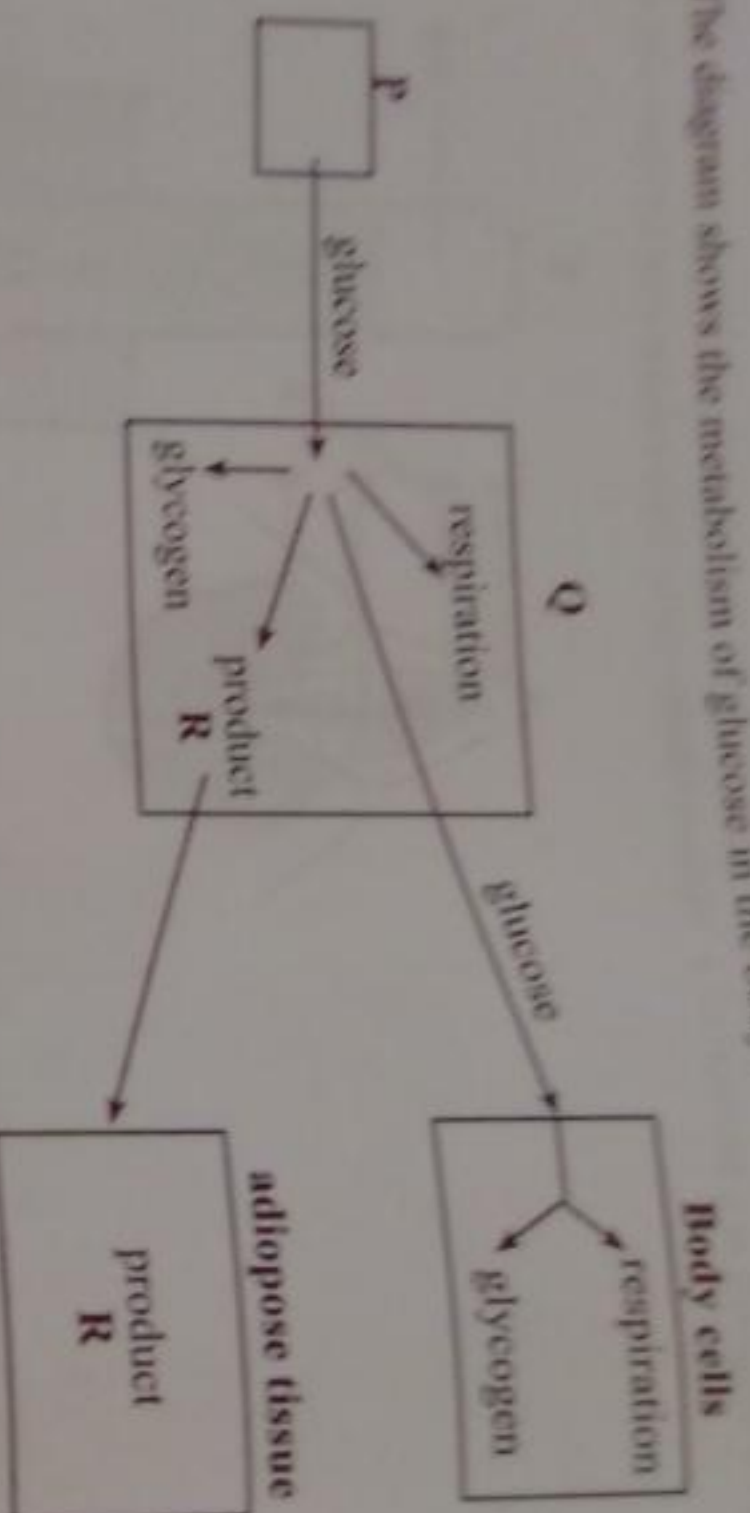
glottis

- A raised open
- B raised closed
- C lowered open
- D lowered closed

- 10 Which digestive secretion is responsible for the physical digestion of fats?

- A pancreatic juice
- B intestinal juice
- C bile
- D gastric juice

- 11 The diagram shows the metabolism of glucose in the body.



Name the organs P and Q and the product R.

P Q Q R

- | | | | |
|---|-------|----------|----------|
| A | ileum | liver | fat |
| B | liver | pancreas | fat |
| C | ileum | liver | glycogen |
| D | ileum | pancreas | glycogen |

- 12 What is the importance of ventilation of a gas exchange surface?

- A to keep it moist
 B to keep it alive
 C to maintain blood circulation
 D to maintain concentration gradients

- 13 Which is an example of a short term effect and of a long-term effect of tobacco smoking?

short-term

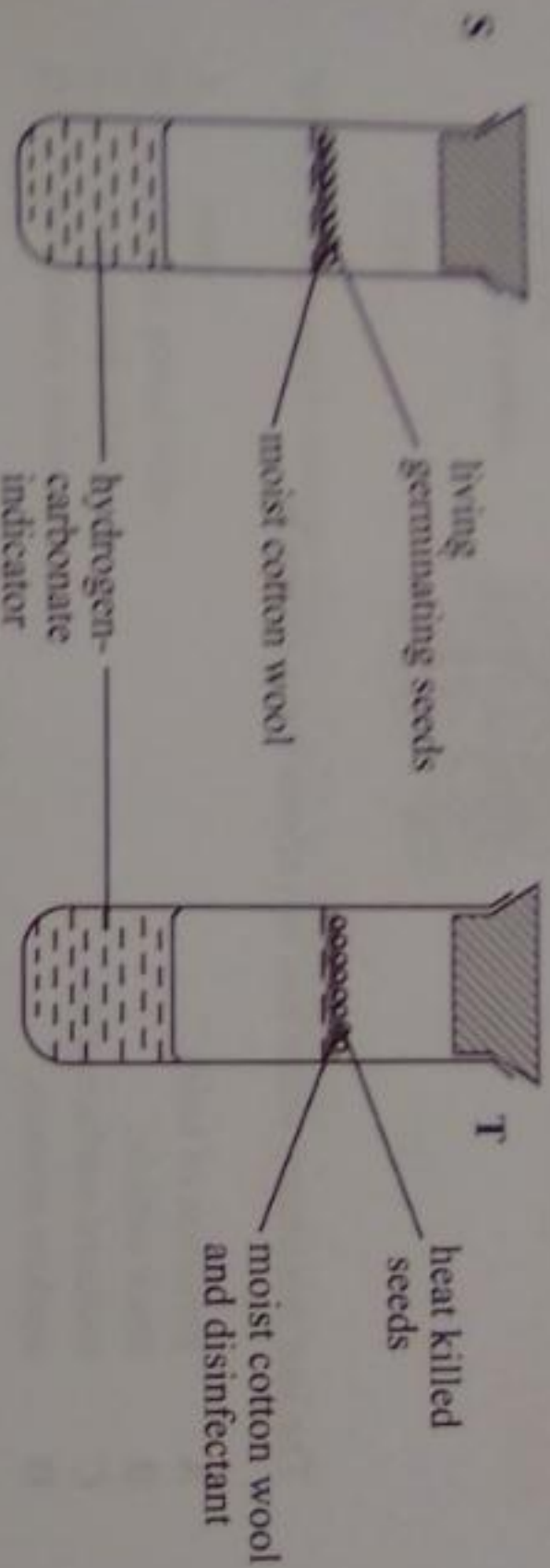
long-term

- | | | |
|---|----------------------|----------------|
| A | emphysema | lung cancer |
| B | cough | emphysema |
| C | destruction of cilia | cough |
| D | lung cancer | mucus build up |

14 What are the sites of anaerobic and aerobic respiration in a cell?

anaerobic respiration	aerobic respiration
A mitochondria	mitochondria
B cytoplasm	cytoplasm
C cytoplasm	mitochondria
D mitochondria	cytoplasm

15 The diagrams show an experiment on respiration of germinating seeds. The colour of the indicator was initially red.



The indicator is:

yellow in acidic conditions
red in neutral condition
purple in alkaline conditions

After two days, what would be the colour of the indicator in the two tubes?

	Tube S	Tube T
A yellow	red	
B yellow	purple	
C purple	red	
D purple	yellow	

16 Which list shows the features of a legume such as a bean?

- A fibrous root system, parallel venation, two cotyledons
- B tap root system, net venation, two cotyledons
- C fibrous root system, net venation, one cotyledon
- D tap root system, parallel venation, one cotyledon

- 17 The diagram shows a vertical section of a Venus's leaf with a small area highly magnified.



The leaf minimizes water loss in various ways except by

- A presence of hairs
- B thick cuticle
- C reduced surface area
- D sunken stomata

- 18 The diagram shows an external view of a human heart. Which blood vessel is obstructed in coronary heart disease?



19 Which component of blood contains cell fragments?

- A plasma
- B red blood cells
- C white blood cells
- D platelets

20 The table lists the characteristics of the blood found in one blood vessel in the circulation of a mammal.

oxygen concentration	carbon dioxide concentration	pressure
high	low	low

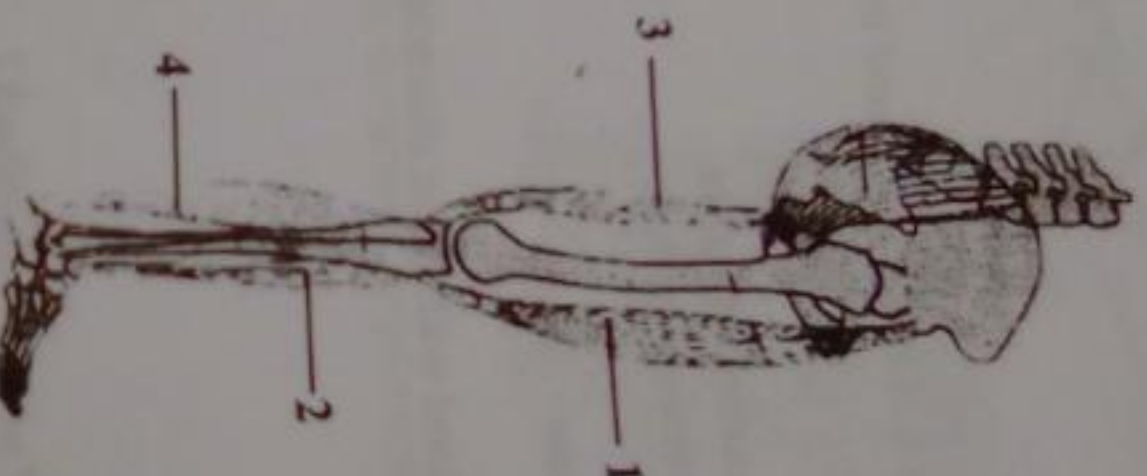
Which blood vessel contains blood with these characteristics?

- A aorta
- B hepatic portal vein
- C pulmonary artery
- D pulmonary vein

21 Which process is an example of negative feedback?

- A rise in blood glucose after a meal
- B secretion of adrenaline when frightened
- C contraction of hair extensor muscle when it is cold
- D suppression of pituitary activity by oestrogen

- 22 The diagram shows muscles used for standing and walking.



Which muscles are flexor?

- A 1 only
- B 2 and 3
- C 1 and 4
- D 3 and 4

- 23 Which substance(s) is(are) produced by the pancreas?

- 1 amylase
- 2 lipase
- 3 insulin
- 4 progesterone

- A 1, 2 and 3
- B 2, 3 and 4
- C 1, 2 and 4
- D 3 only

24

The diagram shows a sensory neurone. In which structure are stimuli converted to electrical impulses?



25 Which statement about the change from far vision to near vision is correct?

- A circular ciliary muscles relax
- B pupil becomes smaller
- C suspensory ligaments become taut
- D focal length of lens increases

26 Which organ has the least specific response to stimuli?

- A eye
- B tongue
- C skin
- D nose

- 27 Which hormone produces a response in the shortest time after secretion?
- progesterone
 - oestrogen
 - insulin
 - adrenaline
- 28 Which of the terms describes the condition in which the body only functions normally in the presence of a drug?
- tolerance
 - withdrawal symptoms
 - addiction
 - psychological dependence
- 29 Fertilisation in plants involves fusion of
- pollen grain and ovule.
 - pollen tube and ovule.
 - male nucleus and egg cell in ovule.
 - male nucleus and ovule.
- 30 The diagram shows a flower.



What is true about the flower?

- | type of flower | reason |
|---------------------|---------------------------------|
| A wind pollinated | anthers above stigma |
| B wind pollinated | hairy stigma |
| C insect pollinated | stigma and anthers were exposed |
| D insect pollinated | small petals |

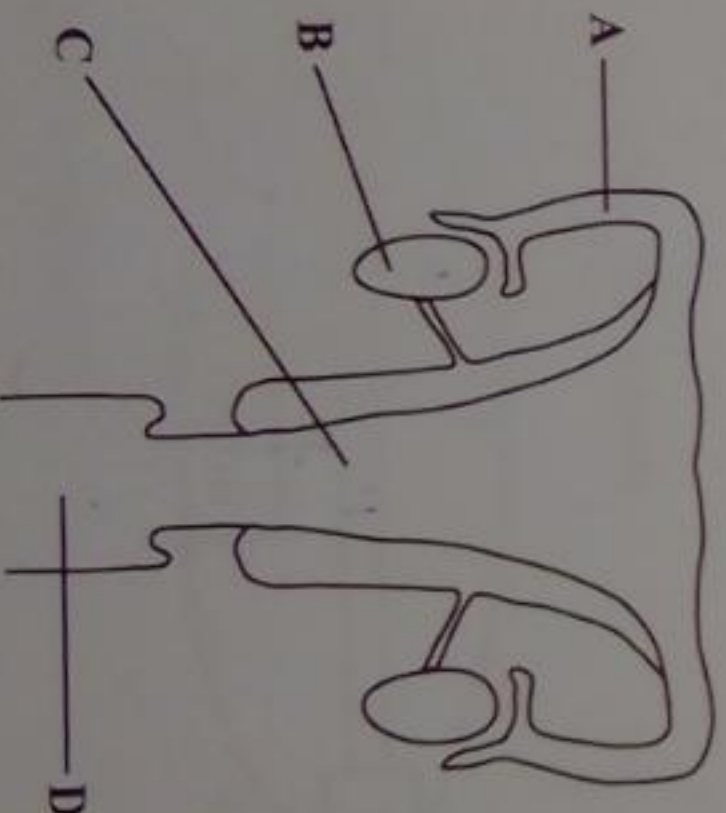
31 The diagram shows three fruits.



What are the methods of dispersal for these fruits.

	1	2	3
A	self	wind	wind
B	wind	wind	animal
C	animal	animal	self
D	wind	self	wind

32 The diagram shows the female reproductive system. In which structure does fertilisation take place?



33 Which contraceptive method is administered to the male human being?

- A tubal ligation
- B vasectomy
- C use of the loop
- D use of spermicides

34 Which is the correct relative sizes of the cells listed below?

- 1 ovum
- 2 ordinary/unspecialized body cell
- 3 sperm

- A $1 = 2 > 3$
- B $1 < 2 = 3$
- C $1 < 2 > 3$
- D $1 > 2 > 3$

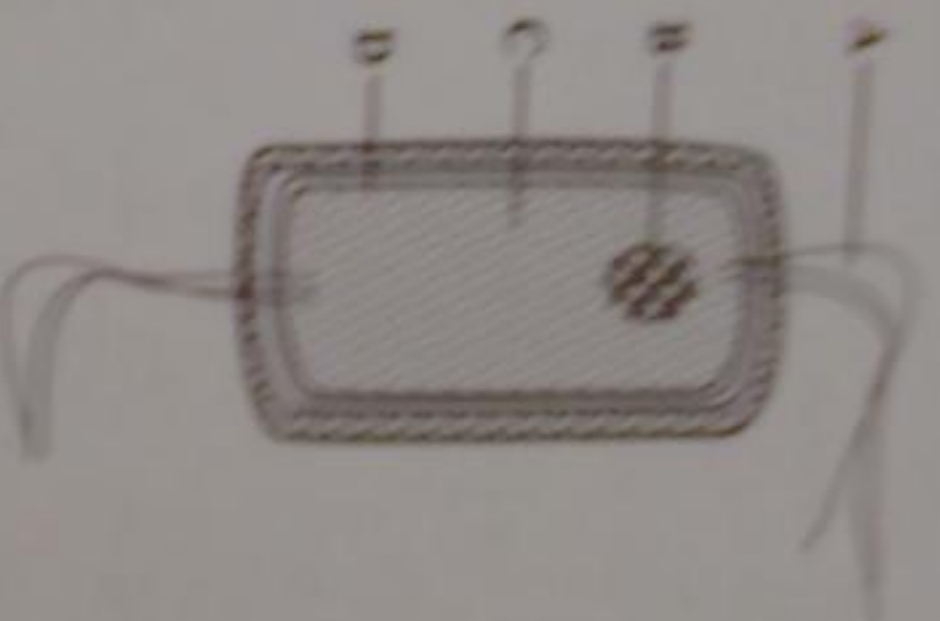
35 The following are disease symptoms.

- 1 acheing of joints
- 2 high fever
- 3 severe headache

Which disease causes these symptoms?

- A cholera
- B malaria
- C tuberculosis
- D gonorrhoea

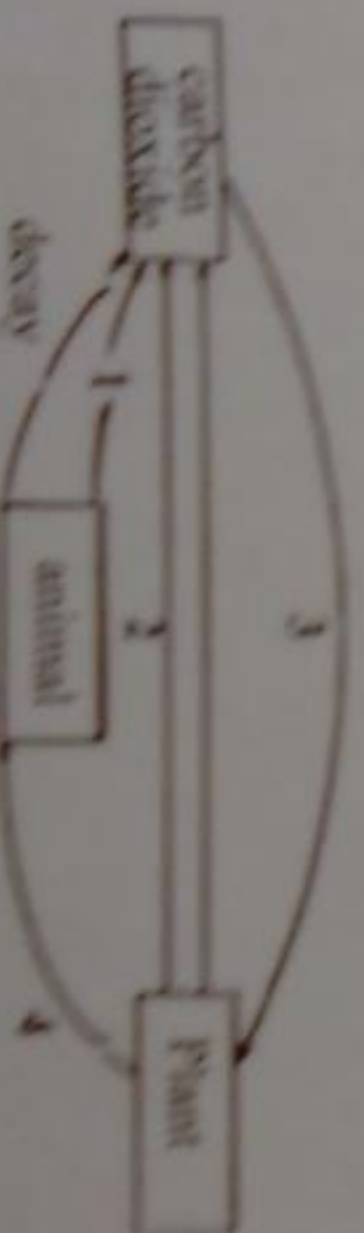
36 The diagram shows a bacterium. Which structure is absent in some bacteria?



37 Nitrogen fixing bacteria change

- A ammonium compounds to nitrates
- B ammonium compounds to nitrites
- C nitrogen to nitrates
- D nitrates to nitrites

38 The diagram shows part of the carbon cycle.



Which arrows represent respiration?

- A 1 and 3
- B 1 and 2
- C 3 and 4
- D 1, 2 and 4

39 Which food chain is least efficient for providing energy to humans?

- A water weed $\rightarrow$ fish $\rightarrow$ human
- B grass $\rightarrow$ cow $\rightarrow$ human
- C beans $\rightarrow$ human
- D leaves $\rightarrow$ caterpillar $\rightarrow$ bird $\rightarrow$ human

40 The diagram shows population pyramids for India and U.S.A.



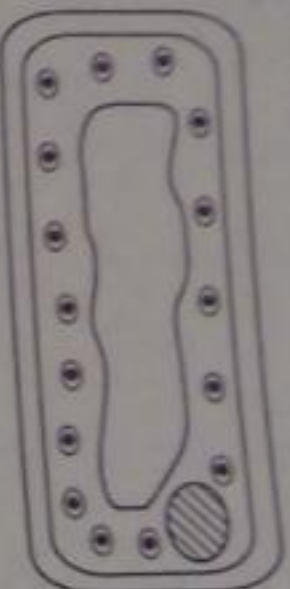
Which statement about the pyramids is true?

- A India has a higher life expectancy.
- B India has a higher growth rate.
- C U.S.A. has a higher infant mortality.
- D U.S.A. has a higher dependency ratio.

1 Which feature is true for all living cells?

- A a single nucleus
- B cytoplasm
- C a cell wall
- D the ability to divide

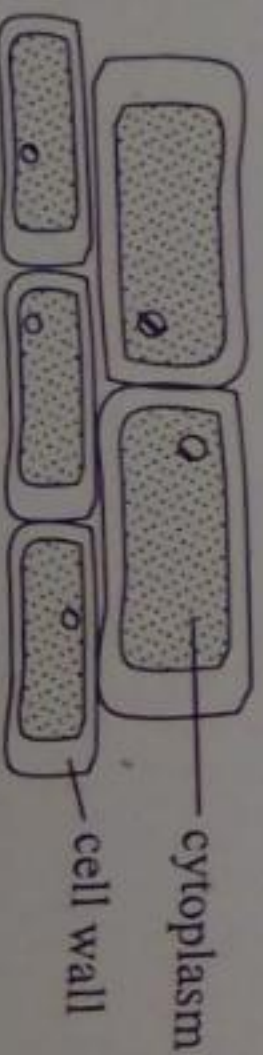
2 The diagram shows a cell as seen under a student microscope.



Is the cell a plant or animal cell and why?

type of cell	reason
A plant	present chloroplast
B animal	nucleus present
C plant	cytoplasm present
D animal	cytoplasm

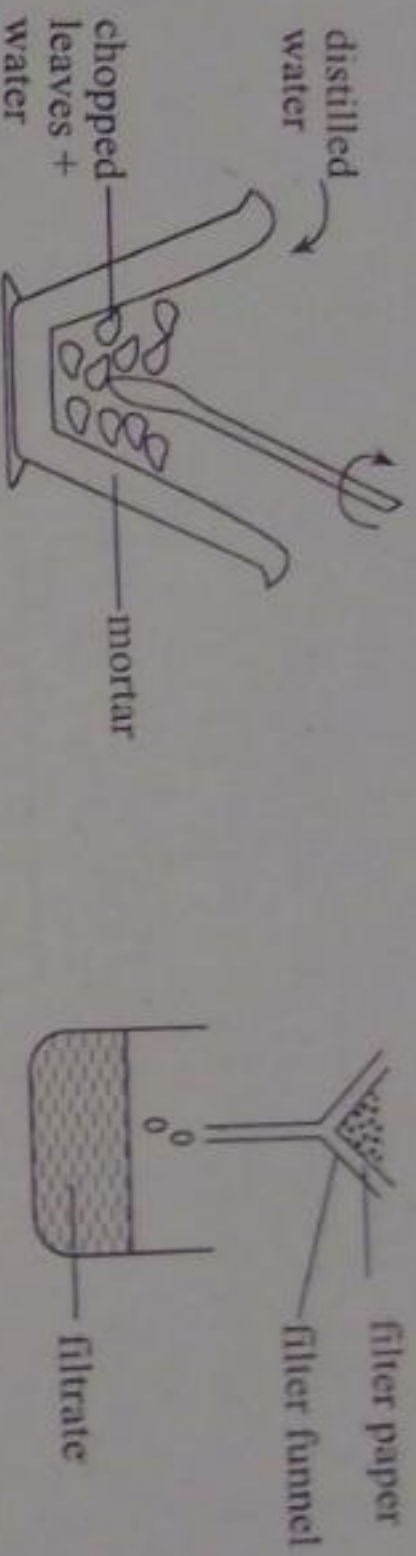
3 The diagram shows plasmolysed cells.



Why do the shapes of the cells remain unchanged?

- A Cell walls are rigid.
- B Cell walls are freely permeable.
- C The size of the vacuole does not change.
- D Water only is lost from the cells.

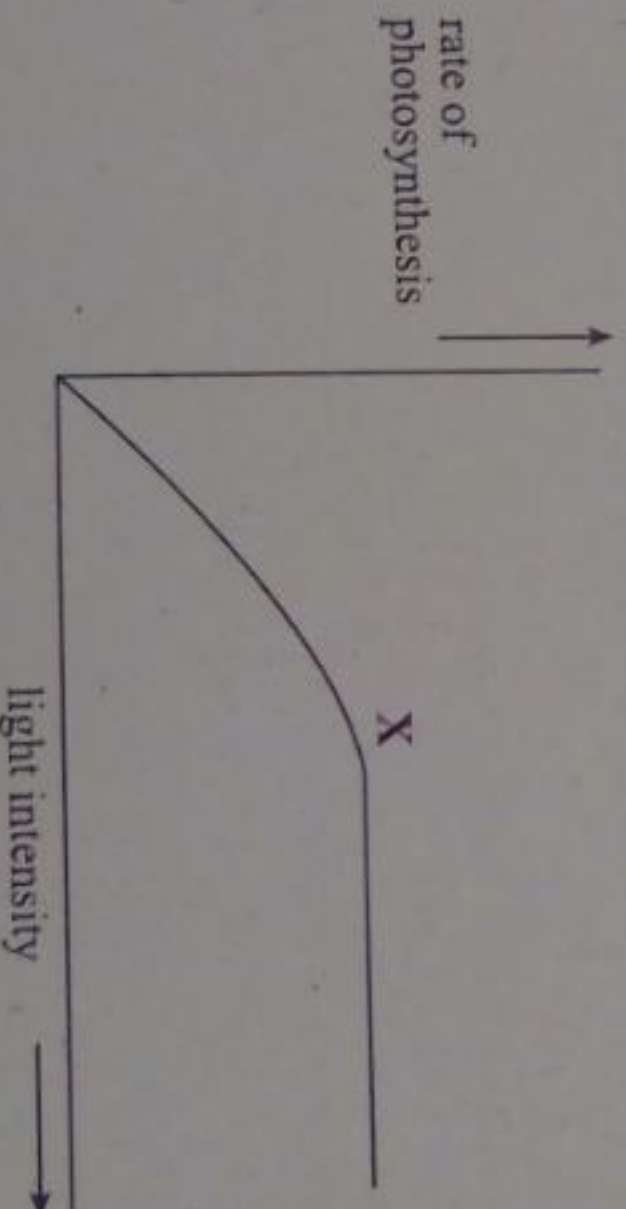
- 4 Leaves from a plant, which had been in sunlight, were treated as shown.



Which carbohydrates are likely to be found in the filtrate?

- A cellulose and starch
- B glucose and starch
- C starch only
- D glucose only

- 5 The graph shows how the rate of photosynthesis of a pondweed varies with light intensity.



Which is likely to be a limiting factor of photosynthesis at X?

- A carbon dioxide concentration in water
- B light intensity
- C concentration of mineral salts in water
- D pH of the water

- 6 The table shows time taken for a pondweed to release 20 bubbles of oxygen in light of different colours.

colour of light	time taken/seconds
blue	38
green	132
red	70

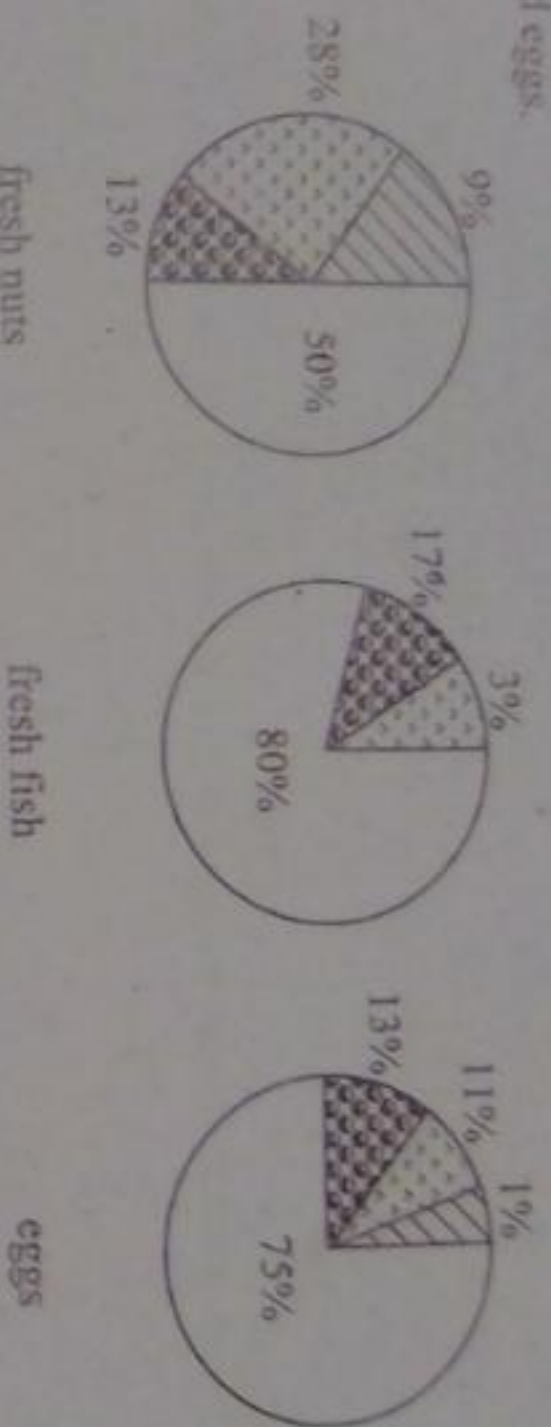
What deduction can be made from these results?

- A Blue light inhibits photosynthesis.
- B Green light inhibits photosynthesis.
- C Photosynthesis is fastest in blue light.
- D Photosynthesis is fastest in red light.

- 7 What is the effect of the absence of phosphorus on the growth and development of seedlings?

- A yellowing of leaves and poor growth
- B root hairs not formed
- C fibrous roots formed instead of tap root system
- D stunted roots and purplish leaves

Questions 8 and 9 refer to the diagrams which show proportion of nutrients in nuts, fish and eggs.



8 What is the major nutrient in each of the food items?

	nuts	fish	eggs
A	protein	protein	protein
B	lipid	protein	protein
C	lipid	protein	lipid
D	carbohydrate	protein	protein

9 What is the amount of protein in 100 g dry mass of each food item?

	nuts	fish	eggs
A	26 g	85 g	52 g
B	13 g	17 g	13 g
C	26 g	34 g	26 g
D	26 g	100 g	52 g

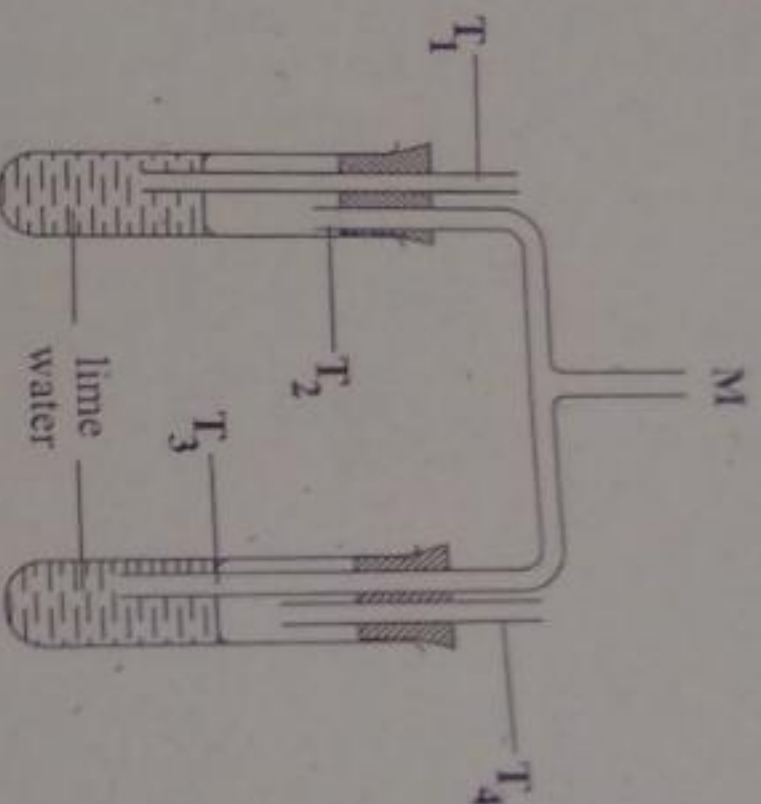
51/100 x 100

10 Which statement is **not** true of digestive secretions from the liver?

- A contain excretory products
- B convert fat to fatty acids and glycerol
- C neutralise acid from the stomach
- D reduce the surface tension of fat droplets

- 11 What is the advantage of using plants as a source of protein from plants?
- A high nutritional value
 - B high proportion of protein
 - C cheaper to produce
 - D easy to digest

- 12 The diagram shows apparatus used to compare inhaled and exhaled air.



When a person places tube **M** in his mouth and breathes out, air flows steadily through

- A T_1 and T_2 .
- B T_1 and T_3 .
- C T_2 and T_3 .
- D T_3 and T_4 .

- 13 Which component of tobacco smoke reduces the oxygen-carrying capacity of haemoglobin?

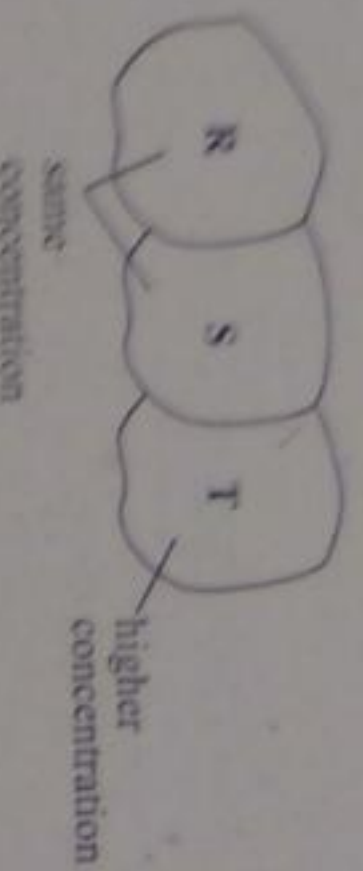
- A carbon dioxide
- B carbon monoxide
- C nicotine
- D tar

- 14 What is the main function of respiration?

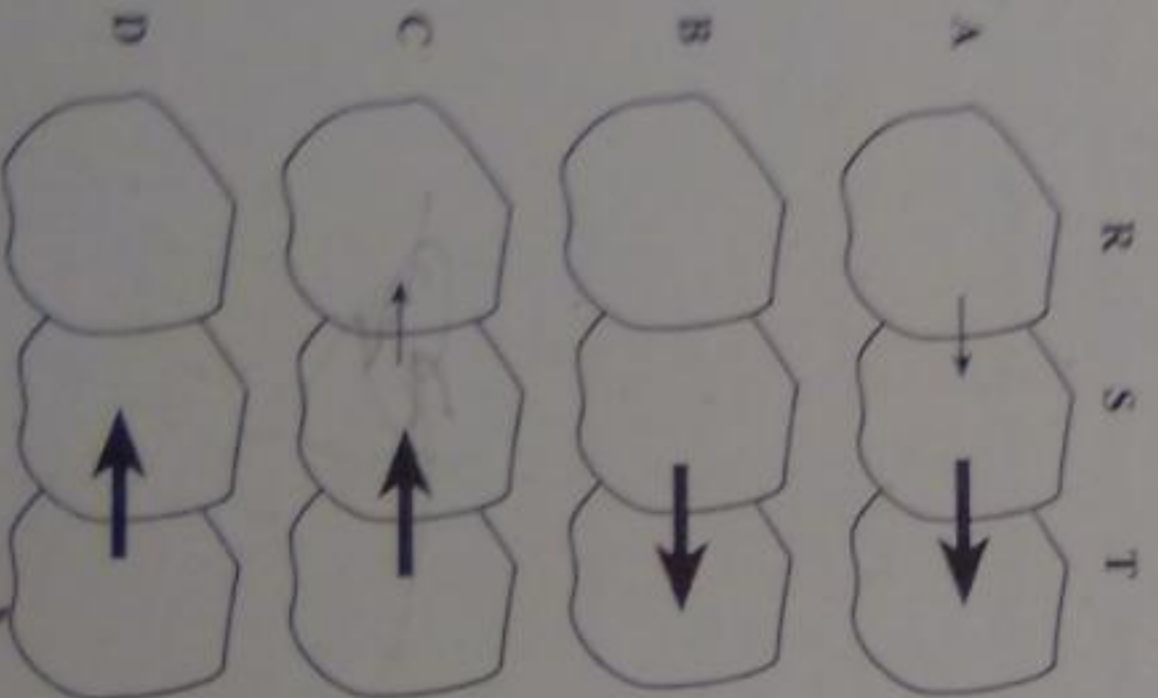
- A to produce heat by oxidising glucose
- B to release carbon dioxide to the environment
- C to release energy by oxidising glucose
- D to take oxygen in from the environment

- A It causes muscle fatigue.
- B It is converted to glycogen in muscles.
- C It is produced during aerobic respiration.
- D It is stored in the liver.

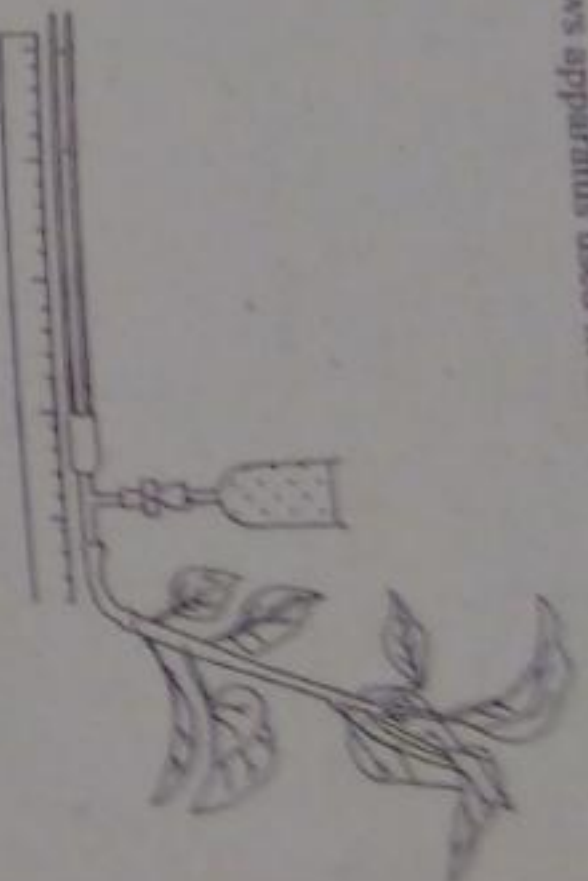
The diagram shows three cells R, S and T. Cells R and S have the same solute concentration but cell T has a higher concentration than the other cells.



Which diagram shows the correct movement of water between the cells?



- 17 The diagram shows apparatus used in an experiment to investigate transpiration.



What is measured by the apparatus?

- ~~A~~ amount of water used by the plant
- ~~B~~ rate at which the plant takes in water
- C rate of evaporation of water from the plant
- D amount of water loss from the plant

- 18 What mechanism is responsible for the transport of mineral salts from the roots to the shoots?

- A active transport
- B osmosis
- C translocation
- D transpiration pull

- 19 Which blood vessel carries oxygenated blood?

- A hepatic portal vein
- B pulmonary vein
- C renal vein
- D umbilical artery

- 20 The table shows the diameters of two blood vessels X and Y.

Blood vessel	external diameter (arbitrary units)	internal diameter (arbitrary units)
X	30	3.5
Y	36	30

What are blood vessels X and Y?

- X Y
- A artery artery
B artery vein
C vein artery
D vein vein

- 21 What are the roles of white blood cells?

- A antibody production and oxygen transport
B antibody production and phagocytosis
C antibody production and blood clotting
D blood clotting and phagocytosis

- 22 The table shows the concentration (arbitrary units) of substances X, Y and Z in blood, glomerular filtrate and urine.

substances	blood	glomerular filtrate	urine
X	80	0	0
Y	1	1	0
Z	0.3	0.3	20

Identify X, Y and Z.

- X Y Z

- A glucose protein urea
B protein glucose urea
C glucose urea protein
D protein urea glucose

14 Which cell in temperature regulation is not carried out by the hypothalamus?

- A increase the convection of blood
- B vasodilation of skin blood
- C control vasodilation of skin arterioles
- D produce heat during cold conditions

21 What does the ciliary muscle respond to stimuli?

- | | |
|----------|---------------------|
| depress | relaxant |
| contract | relaxant |
| relaxant | muscle constrictors |
| relaxant | relaxant |
| relaxant | muscles |

22 The diagram shows a section of the human eye.



What is the function of muscles M?

- A change the size of the pupil
- B change the shape of the eye
- C narrow the eye lid
- D move the eyeball within the socket

- 26 Some important drugs are listed below.

1	alcohol
2	aspirin
3	heroin
4	nicotine
5	penicillin

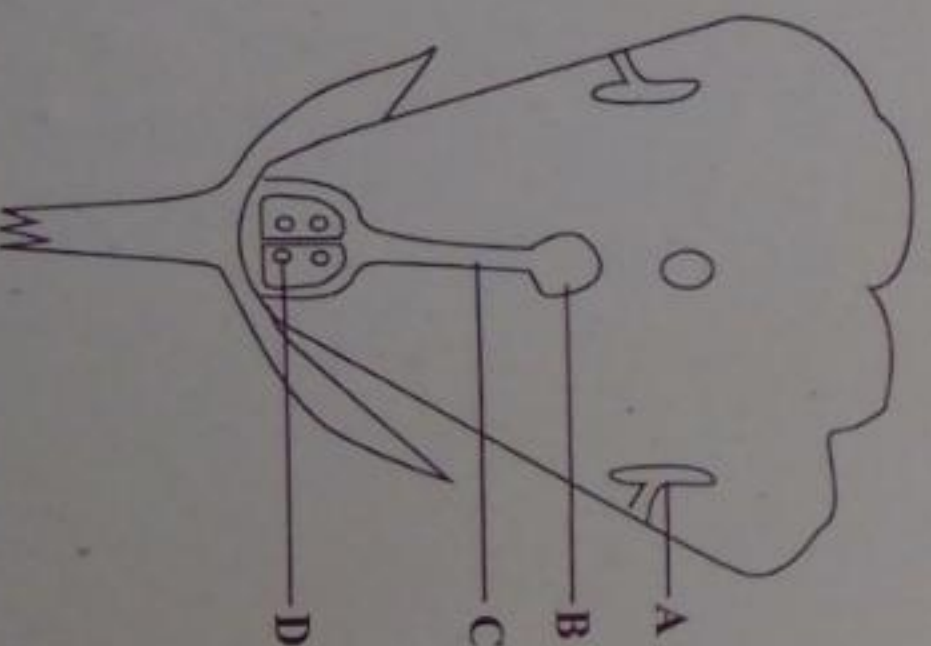
Which of these drugs are addictive, leading to withdrawal symptoms, when their use is discontinued?

- A 1 and 2 only
 B 2 and 3 only
 C 1, 3 and 4
 D 2, 4 and 5

- 27 Which hormone enables the body to react to a dangerous situation?

- A insulin
 B adrenalin
 C oestrogen
 D progesterone

- 28 The diagram shows a flower cut in half. At which position does a pollen grain start to germinate?



An experiment was set up using four groups of insect pollinated flowers in a field. In each group, different parts of the flowers were removed as shown in the table, and insects were allowed to visit all the flowers freely. Which group of flowers would produce most seeds?

group of flowers	stigmas	anthers	petals
A	left	removed	left
B	left	left	removed
C	removed	left	removed
D	removed	removed	left

Which statement can be regarded as a disadvantage of asexual reproduction?

- A Offspring are genetically identical to the parent.
- B Only one individual is needed.
- C An external agent is never required.
- D More offspring are produced.

Questions 31 and 32 refer to the diagrams of the reproductive system of a human female at different stages in the menstrual cycle.



31 In what order would the diagrams be placed to show changes during the menstrual cycle?

- A 1 → 2 → 3 → 4
 B 1 → 2 → 4 → 3
 C 1 → 3 → 4 → 2
 D 1 → 4 → 2 → 3

32 Which diagram shows the state of the uterus when ovulation would be most likely to occur?

- A 1
 B 2
 C 3
 D 4

Questions 31 and 32 refer to the diagrams of the reproductive system of a human female at different stages in the menstrual cycle.



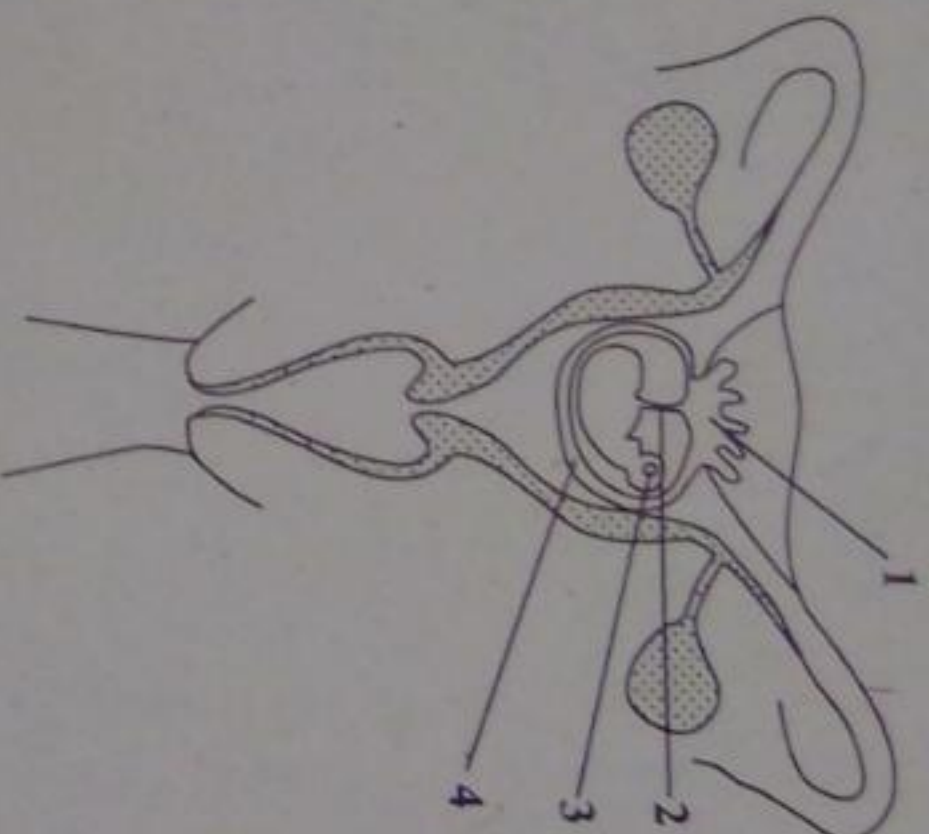
31 In what order would the diagrams be placed to show changes during the menstrual cycle?

- A 1 → 2 → 3 → 4
- B 1 → 2 → 4 → 3
- C 1 → 3 → 4 → 2
- D 1 → 4 → 2 → 3

32 Which diagram shows the state of the uterus when ovulation would be most likely to occur?

- A 1
- B 2
- C 3
- D 4

- 33 The diagram shows a developing mammalian embryo in the uterus.



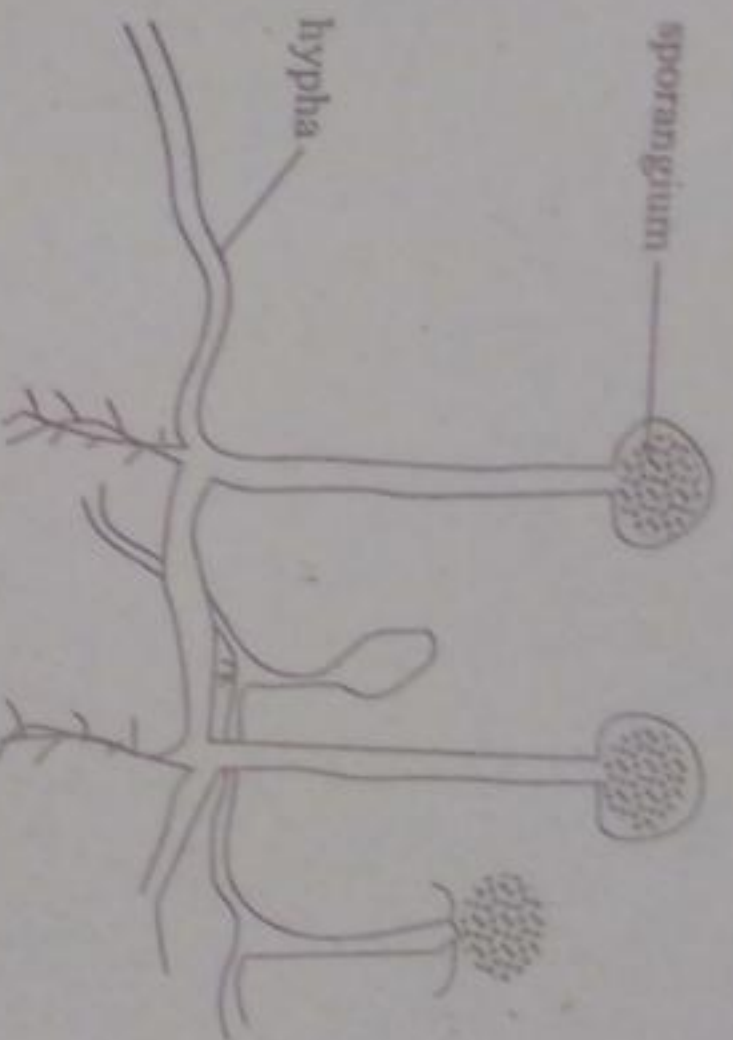
Which structures are temporary organs?

- A 1, 2 and 4
- B 1, 2 and 3
- C 2, 3 and 4
- D 1 only

- 34 Which characteristic is true for both viruses and bacteria?

- A contain cytoplasm
- B have cellular structure
- C all are parasitic
- D capable of reproduction

35 What are the characteristics of the organism shown in the diagram?



	chlorophyll	cellwall	storage carbohydrate
A	absent	absent	starch
B	absent	absent	glycogen
C	absent	present	glycogen
D	present	present	starch

36 What is true of *Plasmodium*?

	type of organism	number of host organism(s) in life cycle
A	bacterium	1
B	bacterium	2
C	protozoan	1
D	protozoan	2

37 What type of mutations cause sickle cell anaemia and Down's syndrome?

	sickle cell anaemia	Down's syndrome
A	change in chromosome number	change in chromosome number
B	change in gene structure	change in gene structure
C	change in chromosome number	change in gene structure
D	change in gene structure	change in chromosome number

38

The table shows the results of an experiment on a soil sample.

mass of fresh soil sample	mass of soil sample after being in an oven at 90°C for:		
	1 day	2 days	3 days
50 g			
	38 g	35 g	35 g

What is the percentage of water in the soil sample?

- A 12%
 B 15%
 C 24%
 D 30%

39 What is the role of bacteria found in root nodules of leguminous plants?

- A convert nitrogen to nitrates
 B increase surface area for absorption of nitrates
 C convert nitrites to nitrates
 D return nutrients to the soil by causing decay

40 There are fewer mosquitoes in the Eastern Highlands than in the Zambezi Valley. What is the likely limiting factor for mosquito population in the Eastern Highlands?

- A food
 B host population
 C temperature
 D water

$$\frac{50 - 38}{50} \times 100 = 24\%$$



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple Choice

NOVEMBER 2007 SESSION

1 hour

Additional materials:

- Multiple choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

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Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

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Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

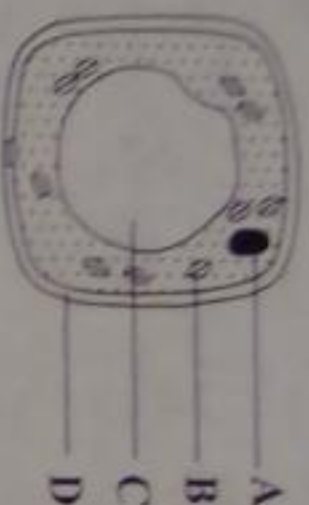
This question paper consists of 14 printed pages and 2 blank pages.

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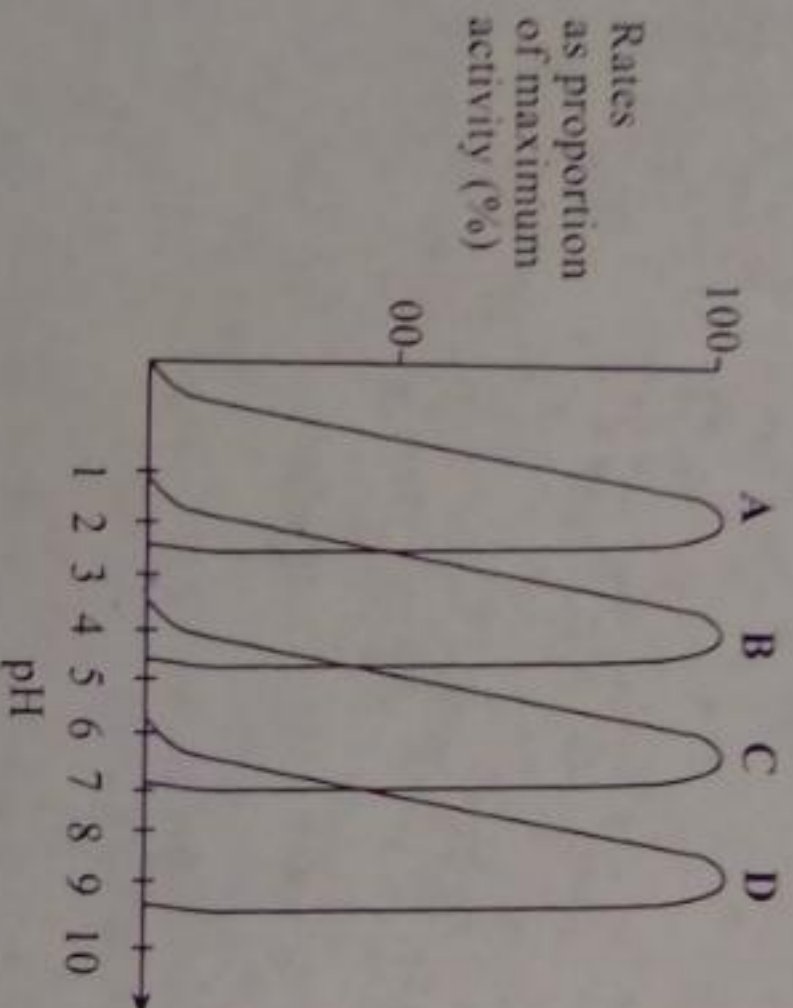
#ZIMSEC N2007

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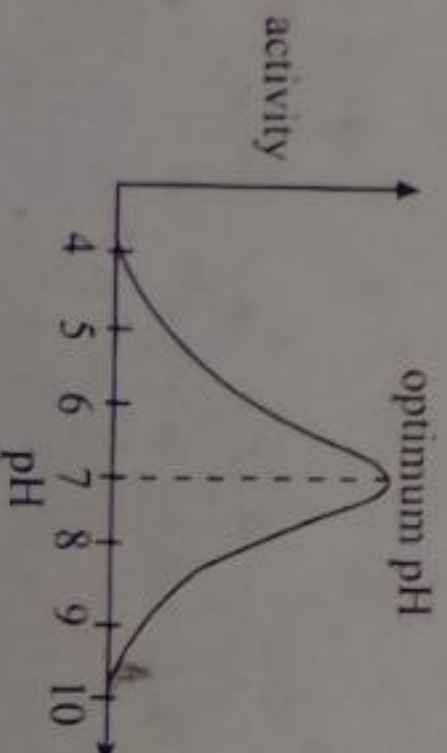
1 The diagram shows a plant cell in the cortex. In which structure does respiration occur?



2 The graph shows the activity of four enzymes plotted on the same axes. Which graph represents the activity of pepsin?



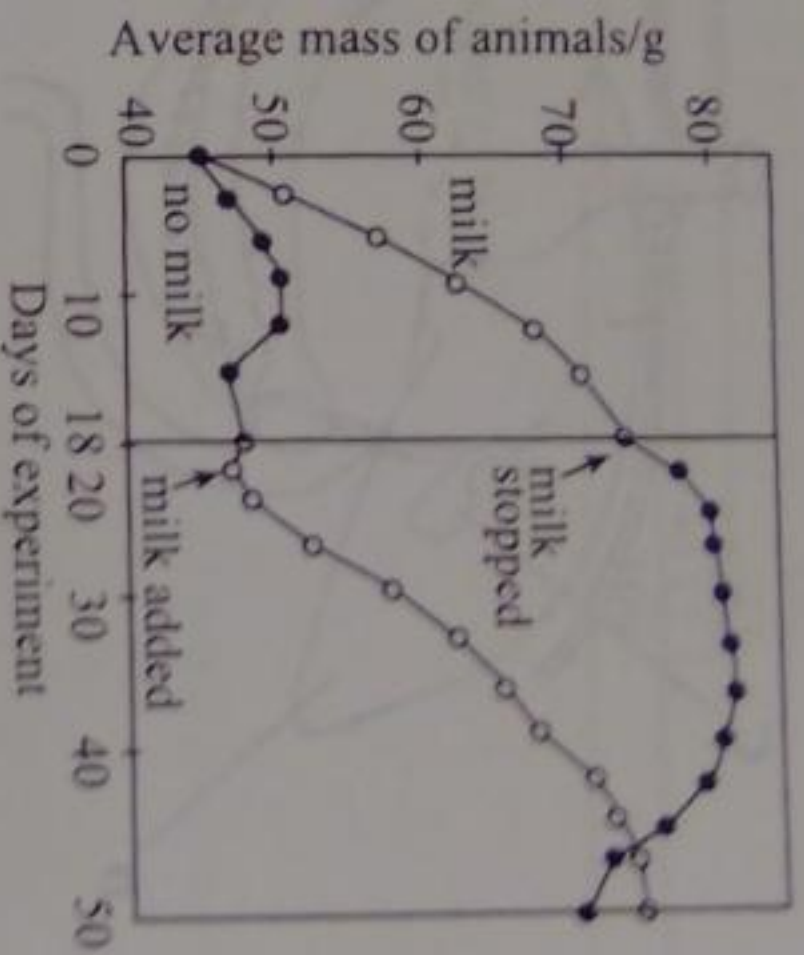
3 The diagram shows the activity/pH curve of an enzyme.



From which region of the alimentary canal was the enzyme extracted?

- A ileum
- B duodenum
- C stomach
- D mouth

4 Two sets of young mice were given 3 cm³ of milk per day in addition to their diet as illustrated in the graph.



Which component of milk was responsible for the changes observed?

- A vitamins
- B carbohydrates
- C proteins
- D fats

5 Which organ produces both hormones and enzymes?

- A liver
- B pancreas
- C spleen
- D salivary glands

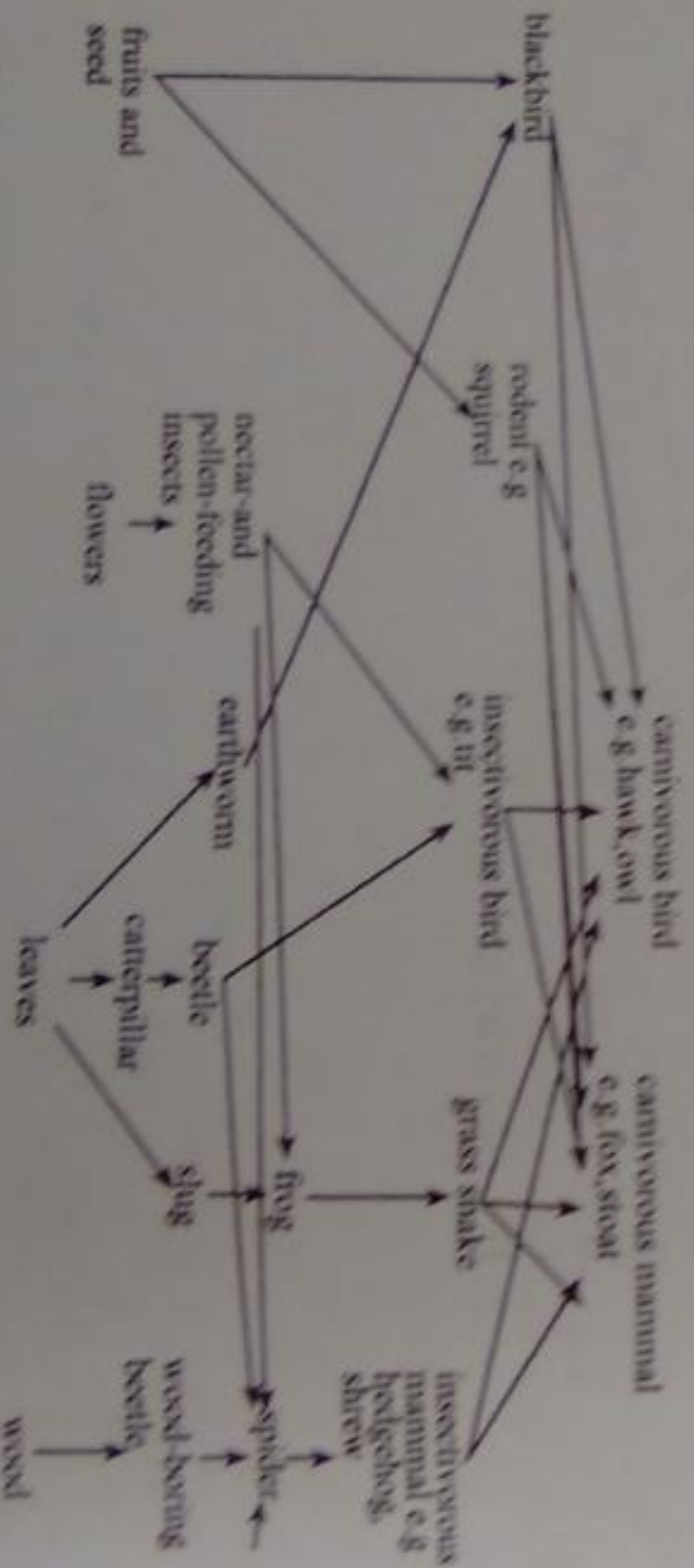
The diagram shows a human alimentary canal. Which labelled structure is the gall bladder?



Which type of teeth is used for tearing, grasping and piercing food?

- A canine
- B premolars
- C incisors
- D molars

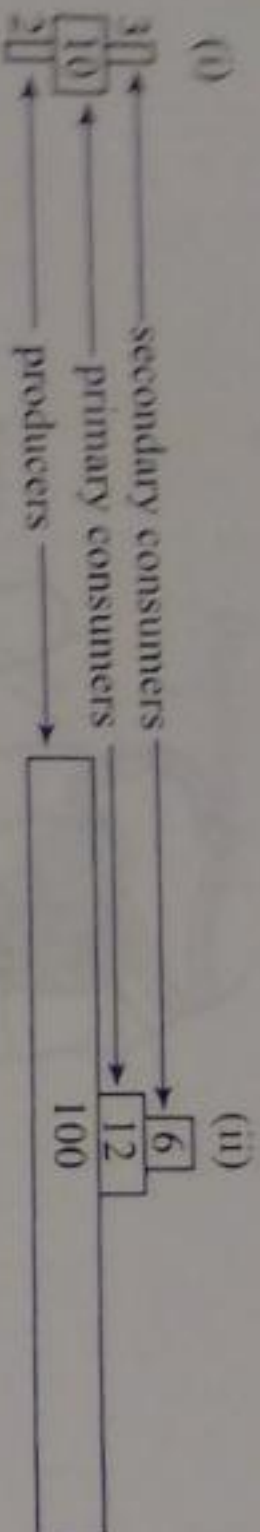
The figure shows a food web.



Which organism is a herbivore?

- A spider
- B frog
- C squirrel
- D hawk

The pyramids of biomass for lake Mutirikwi at different seasons are shown below.



During which months were the samples taken?

Sample (i)	Sample (ii)
A June	December
B December	June
C August	July
D July	August

10

The word equation summarises the fate of energy consumed by an animal.
Food consumed = growth + x + egestion + excretion.
 What does x represent?

- A absorption
- B respiration
- C ingestion
- D digestion

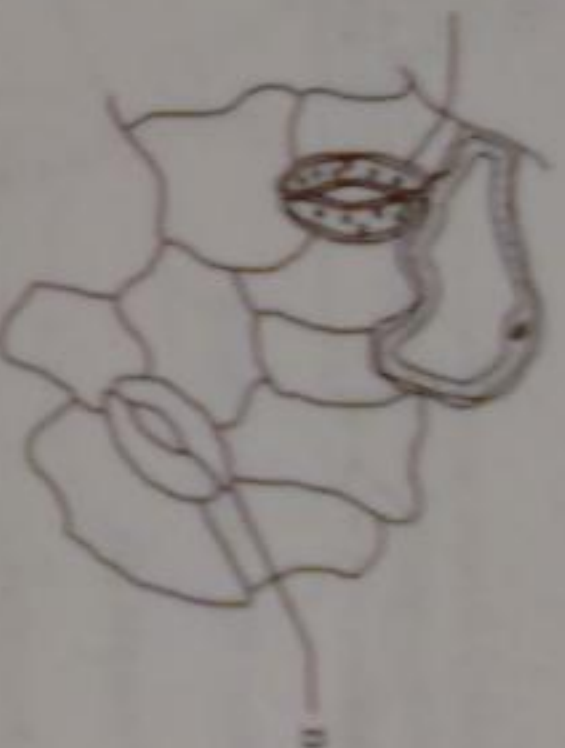
11

Which factor contributes to population increase?

- A emigration
- B ~~e~~ontrapeption
- C mortality
- D immigration

12

The diagram shows the surface view of a leaf.

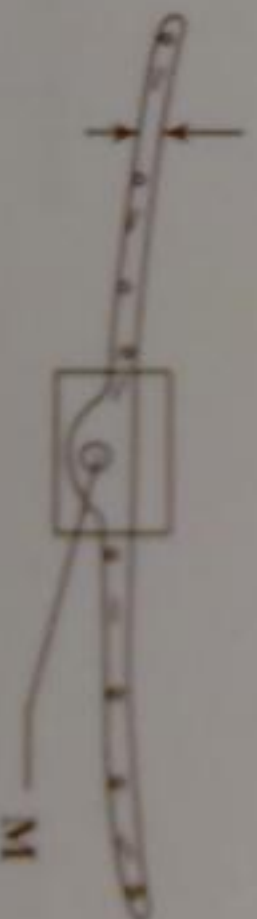


What is the part labelled **n**?

- A pathside cell
- B spongy mesophyll cell
- C guard cell
- D epidermal cell

13

The diagram shows the cross section of a leaf.



What is **M**?

- A vascular bundle
- B intercellular air space
- C midrib
- D pith

- 14 The diagram shows a leaf that was partly covered with aluminium foil during the day.



What was the colour of parts 1 and 2 when iodine was added to the leaf?

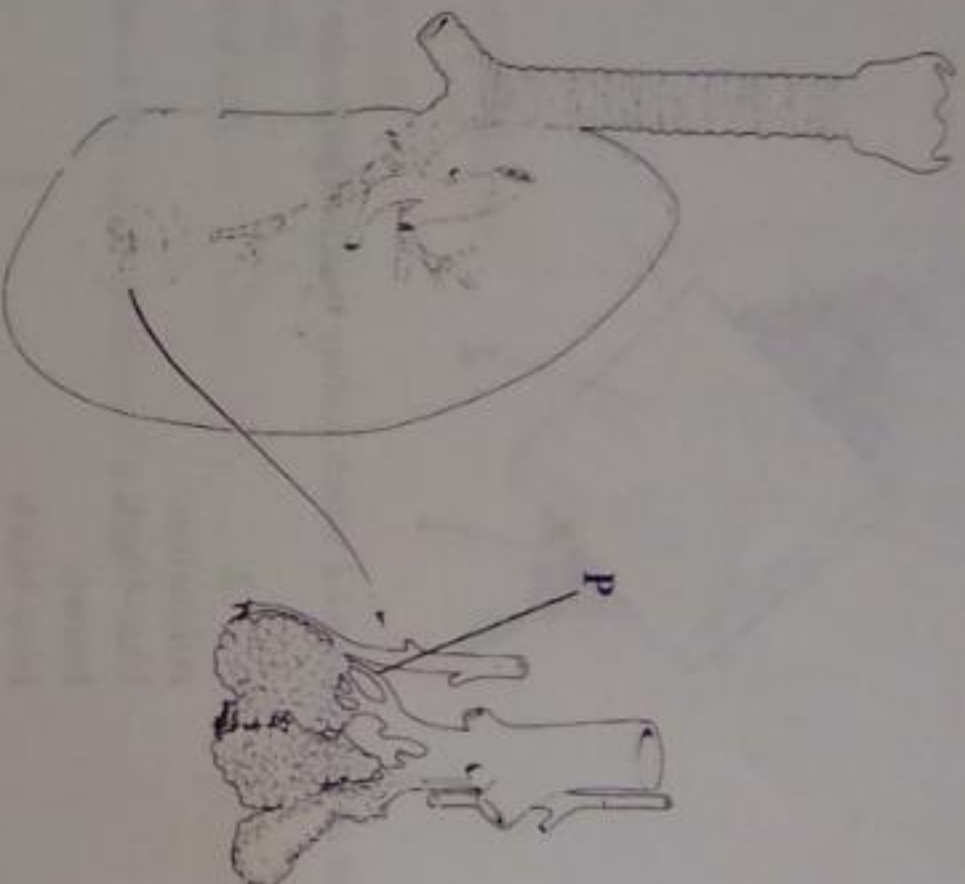
- | | 1 | 2 |
|---|------------|------------|
| A | blue-black | colourless |
| B | brown | blue-black |
| C | blue-black | brown |
| D | colourless | blue-black |

- 15 Which gas is equally concentrated in both inhaled and exhaled air?

- A carbon dioxide
 B nitrogen
 C water vapour
 D oxygen

OS

- 16 The diagram shows a human lung.



What is the part labelled P?

- A tracheole
- B bronchiole
- C capillary
- D verule

- 17 Which specialised cell has a biconcave shape?

- A erythrocyte
- B lymphocyte
- C root hair cell
- D palisade cell

- 18 Which is the source of lactic acid and the organ in which it is oxidised?

- | | source | organ |
|---|-------------|---------|
| A | amino acids | liver |
| B | amino acids | muscles |
| C | glucose | liver |
| D | glucose | muscles |

19 Which statement about respiration is correct?

- A CO_2 is the only product of tissue respiration
- B Alcohol is the product of anaerobic respiration in muscles ✓
- C Anaerobic respiration is sometimes referred to as external respiration
- D Aerobic and anaerobic respiration may occur simultaneously in muscles

20 Which cellular structure contains information that determines cell size?

- A cell membrane
- B nucleus ✓
- C mitochondrion
- D cytoplasm

21 The diagram shows the frontal view of the human eye.



What is the role of the part labelled Y?

- A to control the shape of the lens
- B to focus light onto the retina ✓
- C to control the amount of light that enters the eye
- D to act as the blind spot

22 Which part of the eye enables a person to see colour?

- A lens
- B iris ✓
- C retina
- D pupil

23 Which hormone raises glucose level in blood?

- A insulin
- B adrenaline ✓
- C thyroxine
- D progesterone

24 What is controlled by the kidneys?

- A concentration of glucose in blood
- B osmotic potential of blood plasma
- C carbon dioxide concentration in blood
- D concentration of lactic acid in muscles

25 Which drug is prophylactic?

- A penicillin
- B fentanyl
- C heroin
- D paludine

26 The following processes occur in plants

- 1. movement of sucrose from phloem to developing fruits
- 2. movement of water from xylem to leaf cells
- 3. movement of minerals from xylem to leaf cells
- 4. movement of water from phloem to xylem

Which processes involve active transport?

- A 1, 2 and 3 only
- B 2 and 3 only
- C 1 and 4 only
- D 1 and 3 only

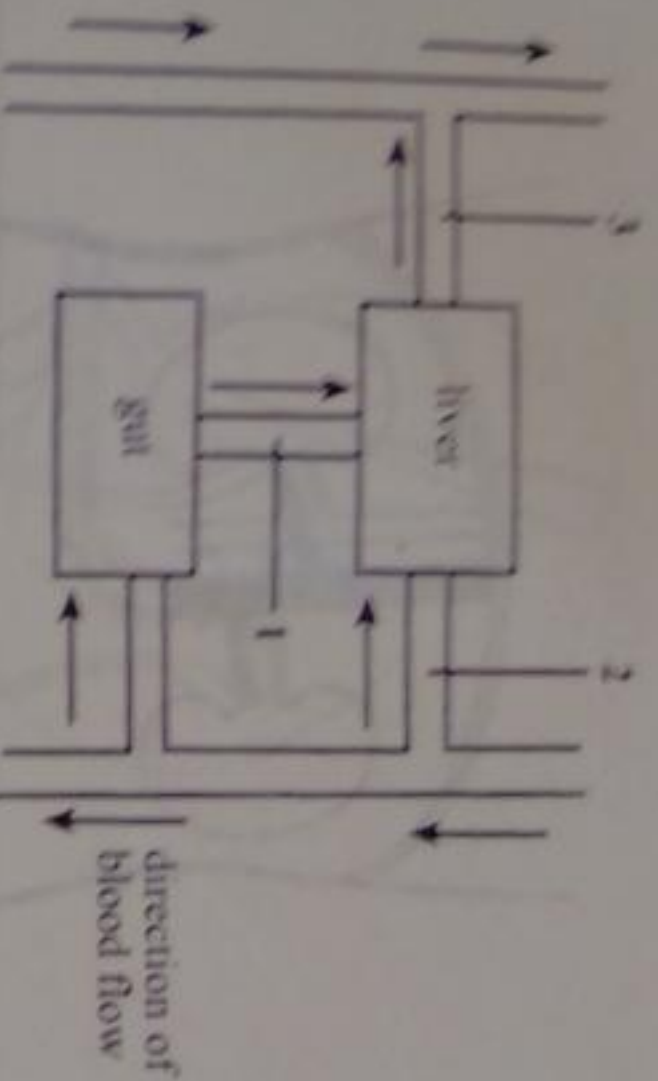
27 Which statement about wilting is **not** correct?

- A It reduces rate of water loss from leaves
- B It causes closing of stomata.
- C It only occurs when soil is completely dry.
- D It reduces the surface area of leaves.

28 Through which vessel does blood flow from the left side of the heart and to which part?

- A aorta to the body
- B pulmonary vein to the lungs
- C pulmonary artery to the lungs
- D vena cava to the body

The diagram shows part of the circulatory system.



Which is the order of increasing oxygen concentration in blood vessels 1, 2 and 3?

- A $1 > 3 > 2$
- B $2 > 1 > 3$
- C $3 > 1 > 2$
- D $3 > 2 > 1$

The diagram shows an elbow joint.

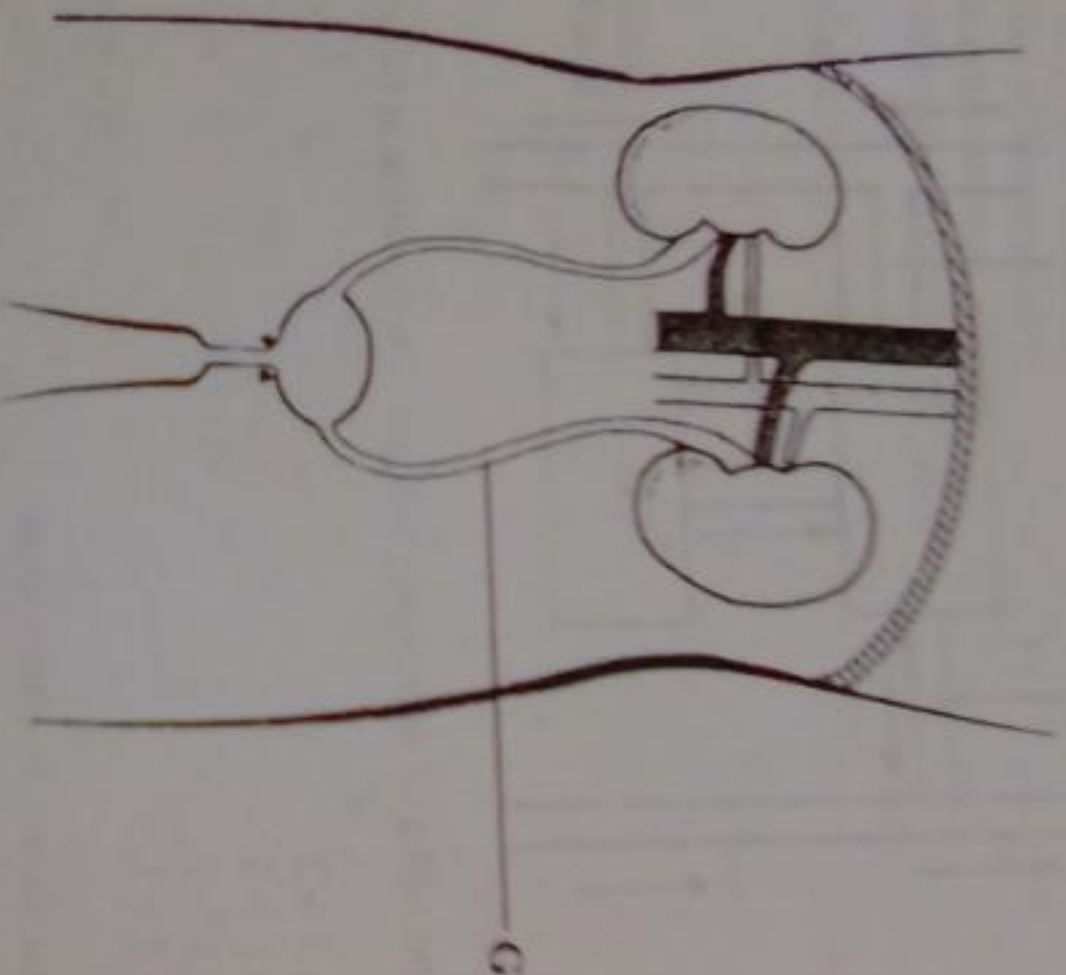


What is the part labelled J?

- A ligament
- B synovial fluid
- C tendon
- D muscle fibre

31

The figure shows the human urinary system. What is G and its function?



structure	function
A oviduct	allows ova to move to uterus
B renal tubule	site of urine formation
C urethra	allow flow of urine from kidney to bladder
D ureter	allows flow of urine from kidney to bladder

32

Which endocrine gland is responsible for the maintenance of pregnancy?

- A placenta
- B ovary
- C corpus luteum
- D uterus

- 33 The diagram shows a flower and part of the flower stalk.



Identify the class of plant and the type of pollination of the flower.

plant

flower

- | | | |
|---|------------------|-------------------|
| A | dicotyledonous | insect pollinated |
| B | dicotyledonous | wind pollinated |
| C | monocotyledonous | insect pollinated |
| D | monocotyledonous | wind pollinated |

- 34 Which feature is an adaptation for wind dispersal?

- A indigestible seeds
 B hooked fruit
 C succulent fruit
 D winged seed

- 35 Which of the listed diseases are caused by bacteria?

1. typhoid
 2. cholera
 3. malaria
 4. gonorrhoea

ST1
 1. typhoid - *Salmonella typhi*
 2. cholera - *Vibrio cholerae*
 3. malaria - *Plasmodium falciparum*
 4. gonorrhoea - *Neisseria gonorrhoeae*

- A 1, 2 and 3 only
 B 1 and 2 only
 C 3 and 4 only
 D 1, 2 and 4 only

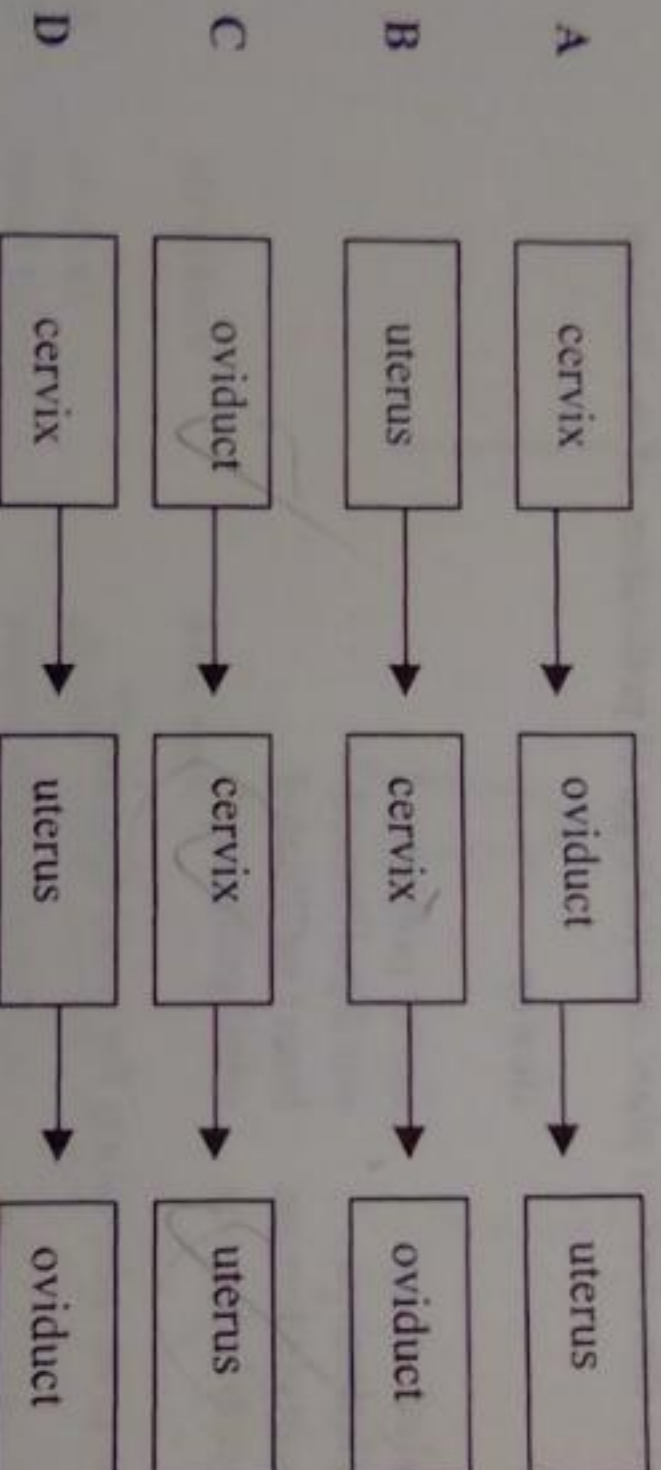
- 36 In a certain flowering plant yellow flower is dominant to white. When two heterozygous plants were crossed, 565 yellow flowered plants were obtained. A number of white flowered plants were also produced. How many white flowered plants were produced?

A 564
B 141
C 188
D 423

- 37 Which characteristic of fungi is of greatest importance to brewing industry?

A They can respire anaerobically.
B They are saprophytic.
C They are parasitic.
D They reproduce asexually.

- 38 Which is the correct path of sperms to the site of fertilization?



- 39 Which response is directly controlled by the medulla oblongata?

A release of saliva as food is tasted
B removal of hand from hot object
C sneezing
D breathing

- 40 Which sexually transmitted infection (STI) has the following symptoms?
painless on the vulva of sex organ or mouth followed by localised skin rash

A gonorrhoea
B AIDS
C chancroid
D syphilis

011042



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

5008/1

BIOLOGY

PAPER 1 Multiple Choice

NOVEMBER 2006 SESSION

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

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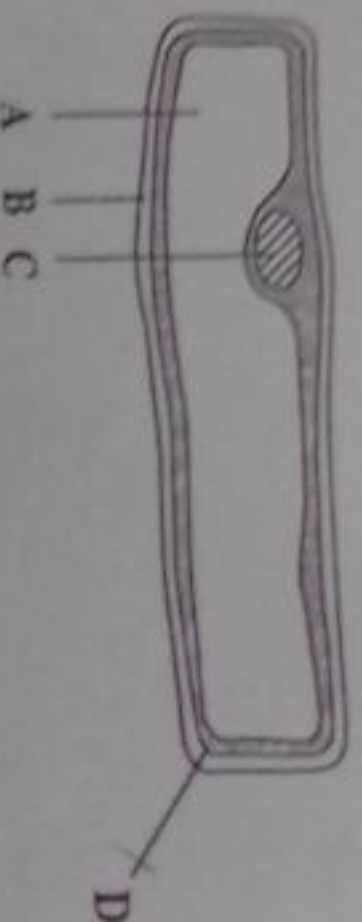
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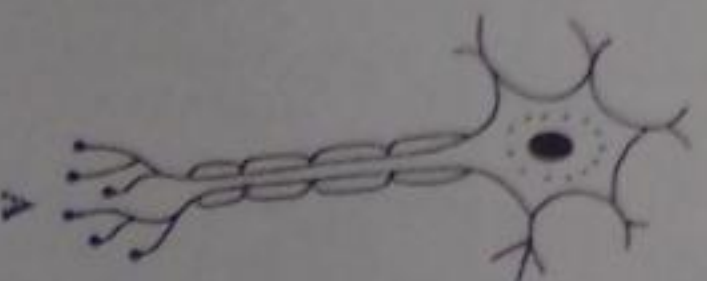
1 The table shows functions that are carried out by some cells. Which cell is the red blood cell?

	absorption	conduction	transport	contraction
A	x	✓	x	✓
B	✓	x	✓	x
C	✓	x	x	✓
D	x	✓	x	✓

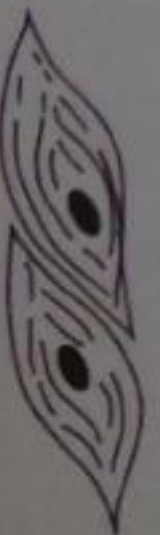
2 The diagram shows a plant cell. Which structure controls the transfer of substances into and out of the cell?



3 The diagrams show cells from different types of tissues. Which cell contracts when it is stimulated by nerve impulses?



A



B



C



D

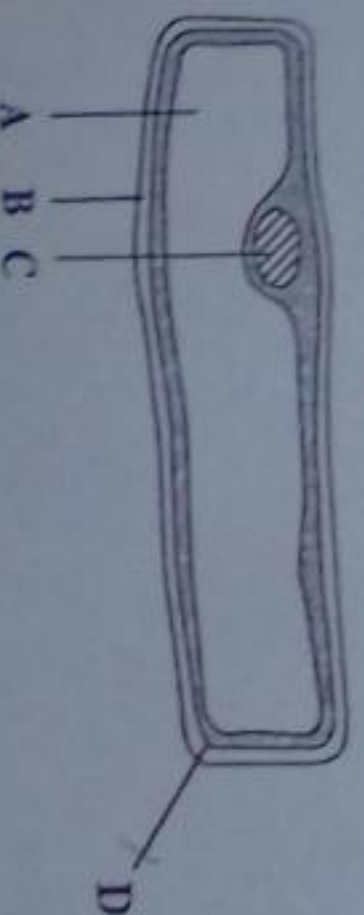
4 During the day a green plant uses carbon dioxide but does not seem to use oxygen. What is the explanation for this?

- A During the day the plant uses carbon dioxide.
- B During the day the amount of oxygen produced by photosynthesis is greater than the amount of oxygen required by respiration.
- C Respiration only takes place during the night without photosynthesis occurring.
- D Carbon dioxide and oxygen are required for respiration.

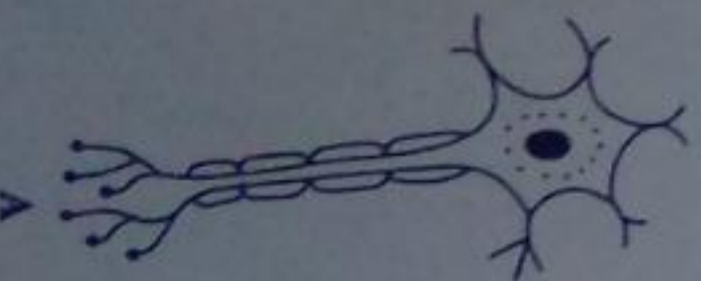
- 2
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A	x	✓	x	✓
B	✓	x	✓	✓
C	✓	x	x	✓
D	x	✓	x	

- 2 The diagram shows a plant cell. Which structure controls the transfer of substances into and out of the cell?



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A



B



C



D

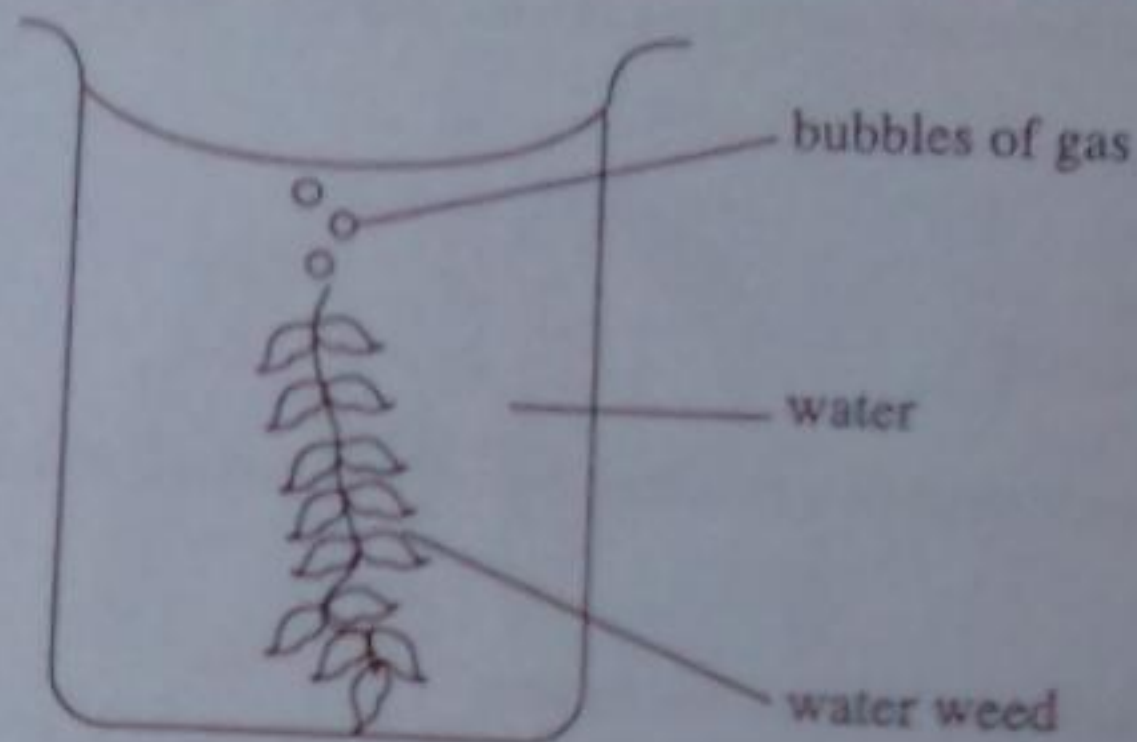
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- C Respiration only takes place during the night without photosynthesis occurring.
- D Carbon dioxide and oxygen are required for respiration.

5 Which mineral is essential for fruit formation?

- A phosphorus
- B iron
- C potassium
- D nitrogen

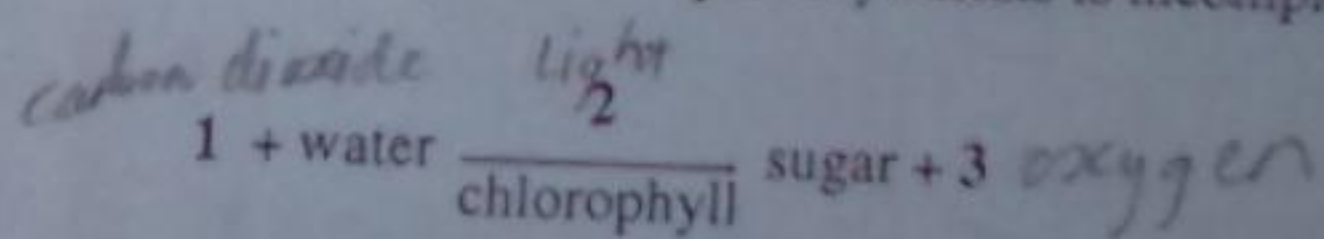
6 The diagram shows an experiment set up to investigate photosynthesis.



Which conditions would cause the plant to produce most bubbles?

	dissolved carbon dioxide			light		temperature	
A	high	✓		bright	✓	cool	✓
B	high	✓		bright	✓	warm	✓
C	low	✓		dim	✓	cool	✓
D	low	✓		dim	✓	warm	✓

7 The following equation for photosynthesis is incomplete.



What do the numbers represent?

	1	2	3
A	carbon dioxide ✓	oxygen ✓	sunlight ✓
B	carbon dioxide ✓	sunlight ✓	oxygen ✓
C	oxygen ✓	carbon dioxide ✓	sunlight ✓
D	sunlight ✓	oxygen ✓	carbon dioxide ✓

- 8 Which of the following nutrients release the most energy when oxidised per unit mass?

A protein
B fat
C starch
D sugar

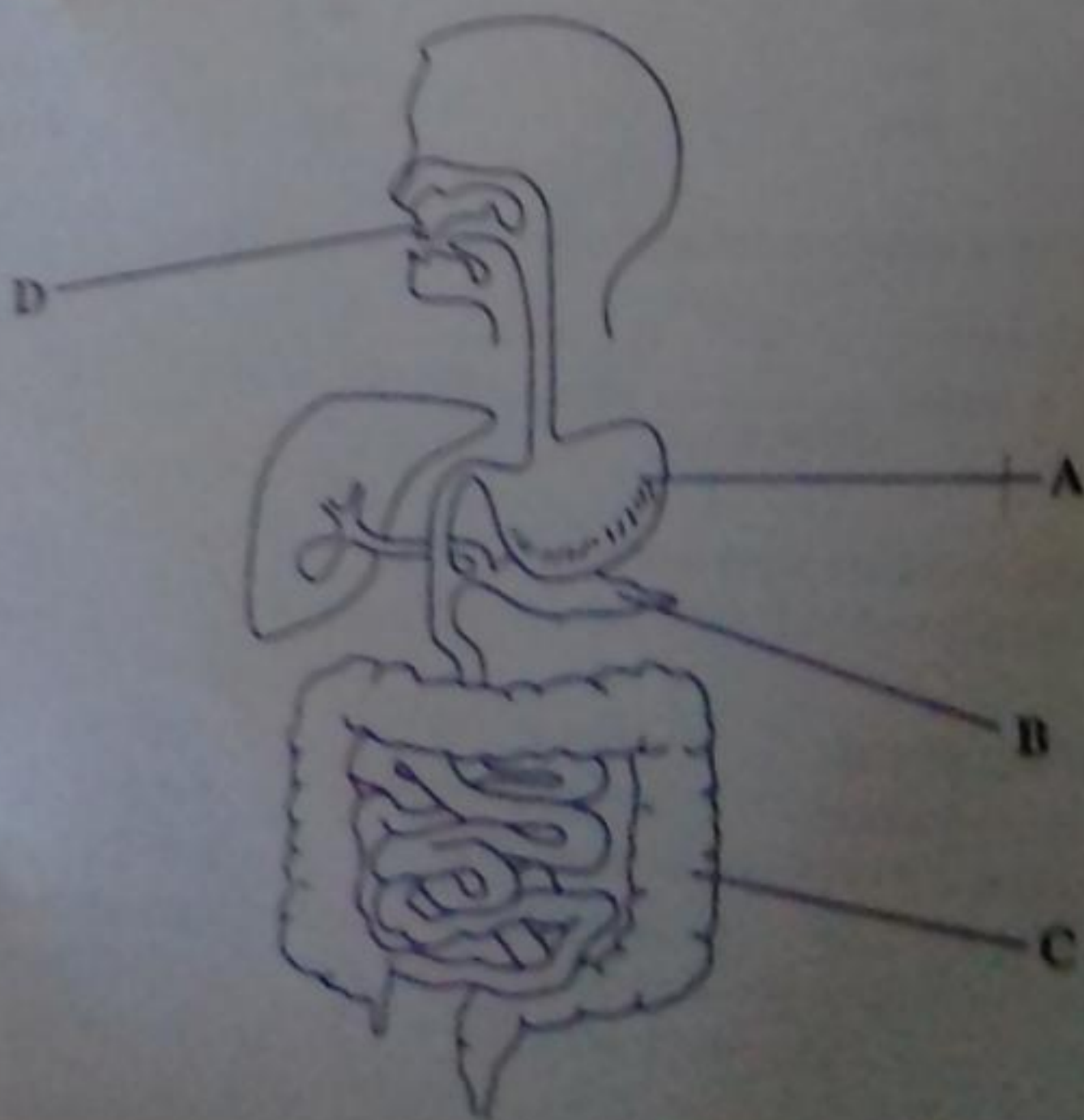
- 9 5 cm³ of 50% solution of human saliva was mixed in a test tube with an equal quantity of 0.1% starch solution. Several drops of iodine solution were added to samples of the mixture until there was no further colour change. What was the final colour?

A white
B brown
C green
D blue-black

- 10 If the pancreatic duct of a mammal becomes blocked, which symptom would the animal show?

A increased bile in the duodenum
B decreased blood insulin level
C increased blood sugar level
D decreased fat digestion

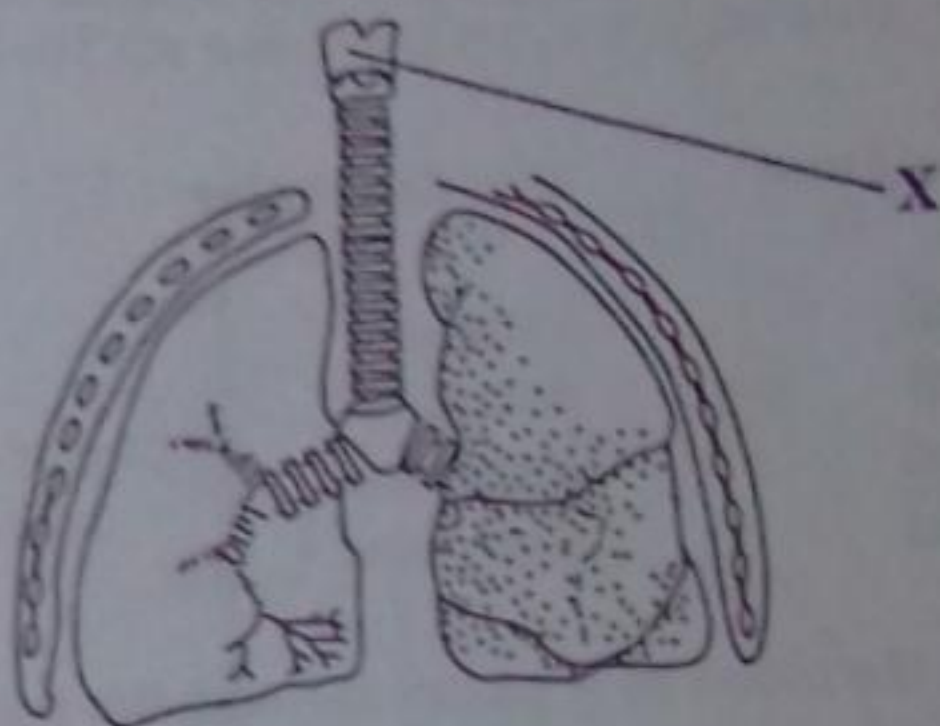
- 11 The diagram shows the human alimentary canal. Which region produces the enzyme responsible for the first stage of protein digestion?



12 What is the percentage of carbon dioxide in exhaled air?

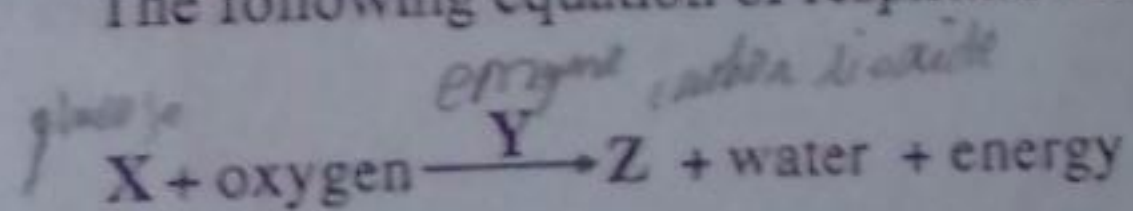
- A 16 %
 B 0.03 %
 C 21 %
 D 4 %

13 The diagram shows the human respiratory system. Identify part X.



- A cartilage
 B epiglottis
 C larynx
 D trachea

14 The following equation of respiration is incomplete.



What do the letters represents?

- | | X | Y | Z |
|---|----------------|----------------|----------------|
| A | carbon dioxide | glucose | enzyme |
| B | glucose | enzymes | carbon dioxide |
| C | enzymes | carbon dioxide | glucose |
| D | glucose | carbon dioxide | enzyme |

15 In anaerobic respiration yeast is used to produce

- A alcohol and carbon dioxide.
 B lactic acid and energy.
 C carbon dioxide and water.
 D alcohol only.

16 By which process does oxygen move into the cytoplasm of a root hair cell?

- A diffusion
- B osmosis
- C active transport
- D transpiration pull

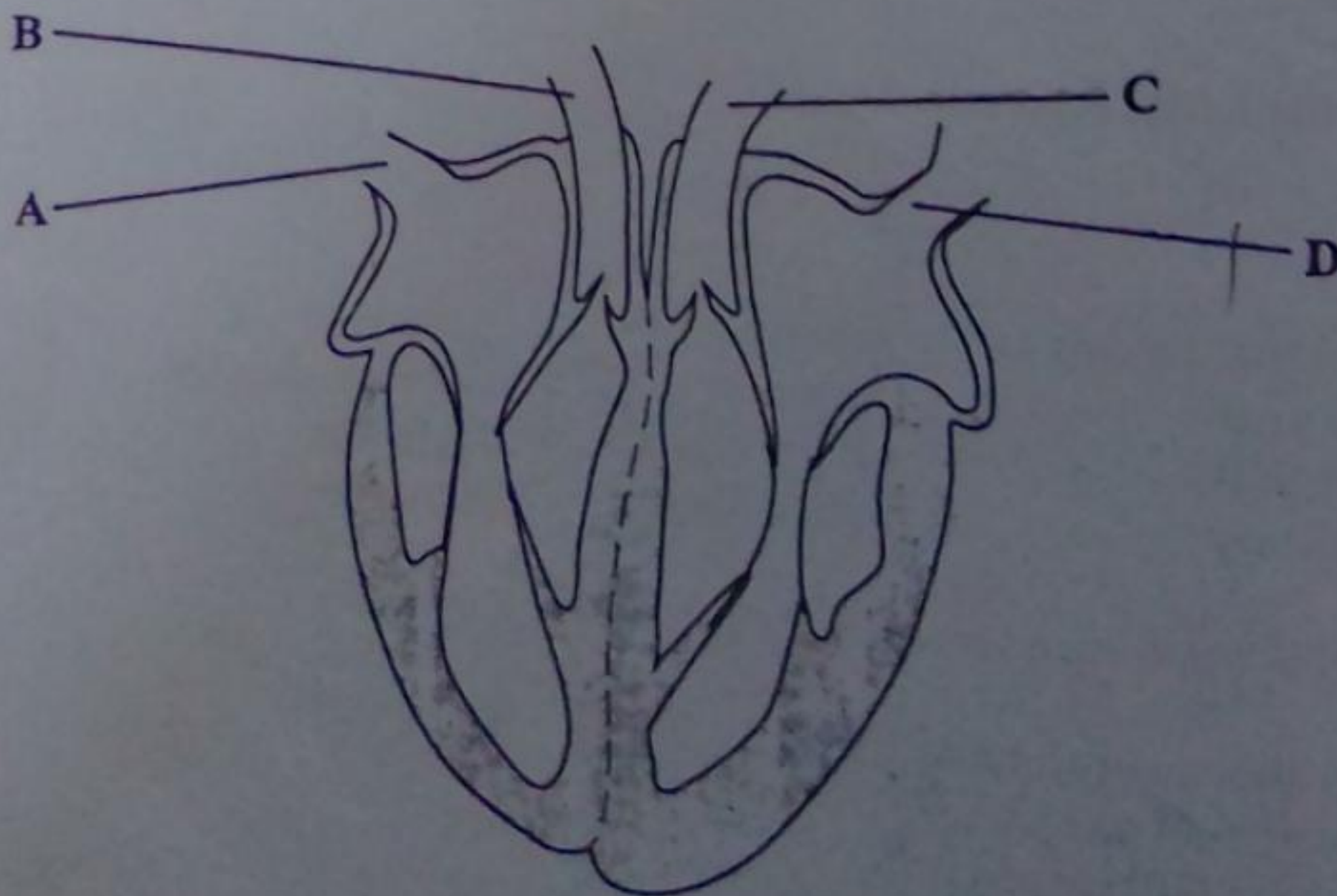
17 Cylinders were cut from a peeled raw potato. Some were then placed in sugar solution and others in distilled water. What happens to the lengths of the potato cylinders after 10 minutes?

	sugar solution	distilled water
A	increased	increased
B	increased	decreased
C	decreased	increased
D	decreased	decreased

18 Which substance enters the blood as it passes through the pancreas?

- A amino acids
- B adrenaline
- C insulin
- D lipase

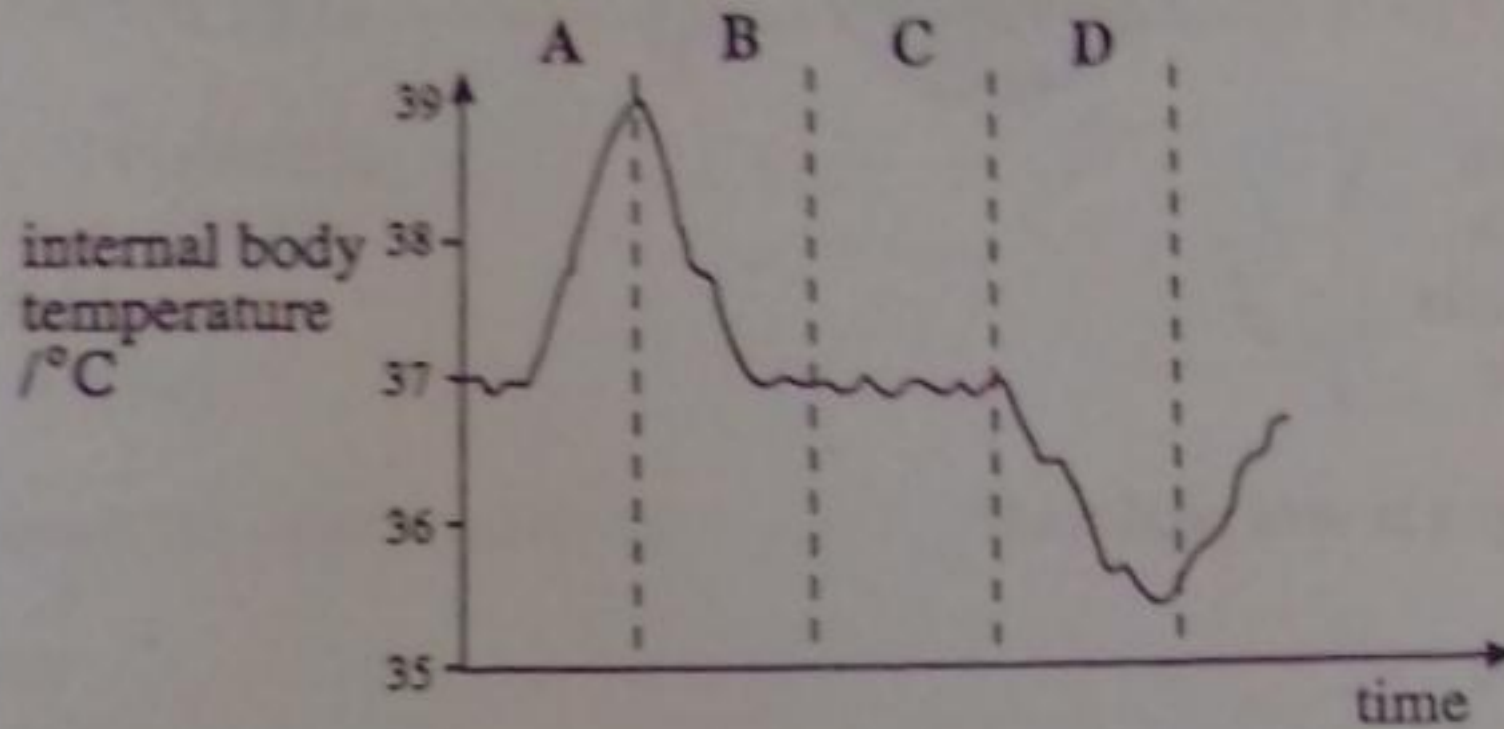
19 The diagram shows main blood vessels connected to the heart. Which vessel supplies the heart with blood to keep it active?



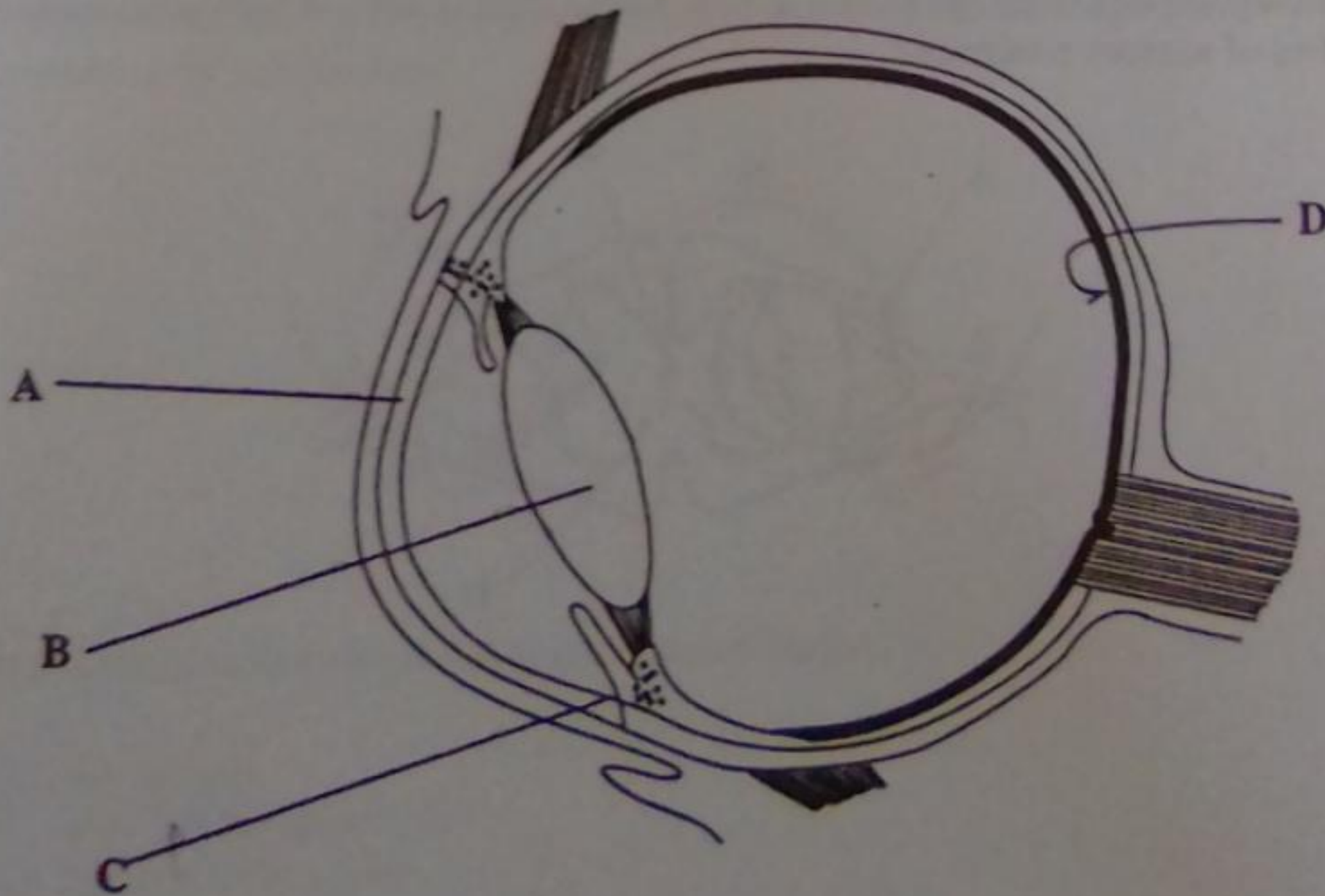
20 Which blood vessel carries absorbed food nutrients directly from the small intestine to the liver?

- A aorta
- B hepatic artery
- C hepatic portal vein
- D hepatic vein

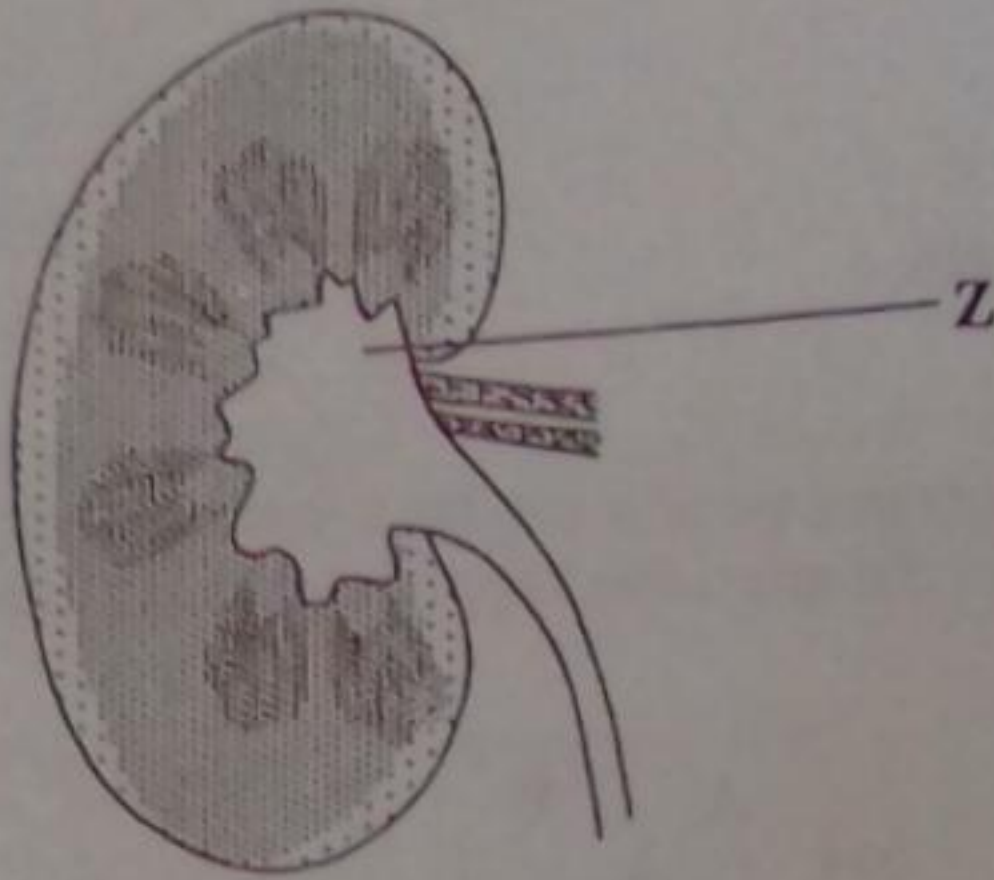
21 The graph shows changes in a person's internal body temperature over a period of time. During which period did the arterioles supplying blood to surface capillaries become constricted?



22 The diagram shows a section of the human eye. Which part detects changes in light intensity?



- 23 The diagram shows the section through the kidney. What is Z?

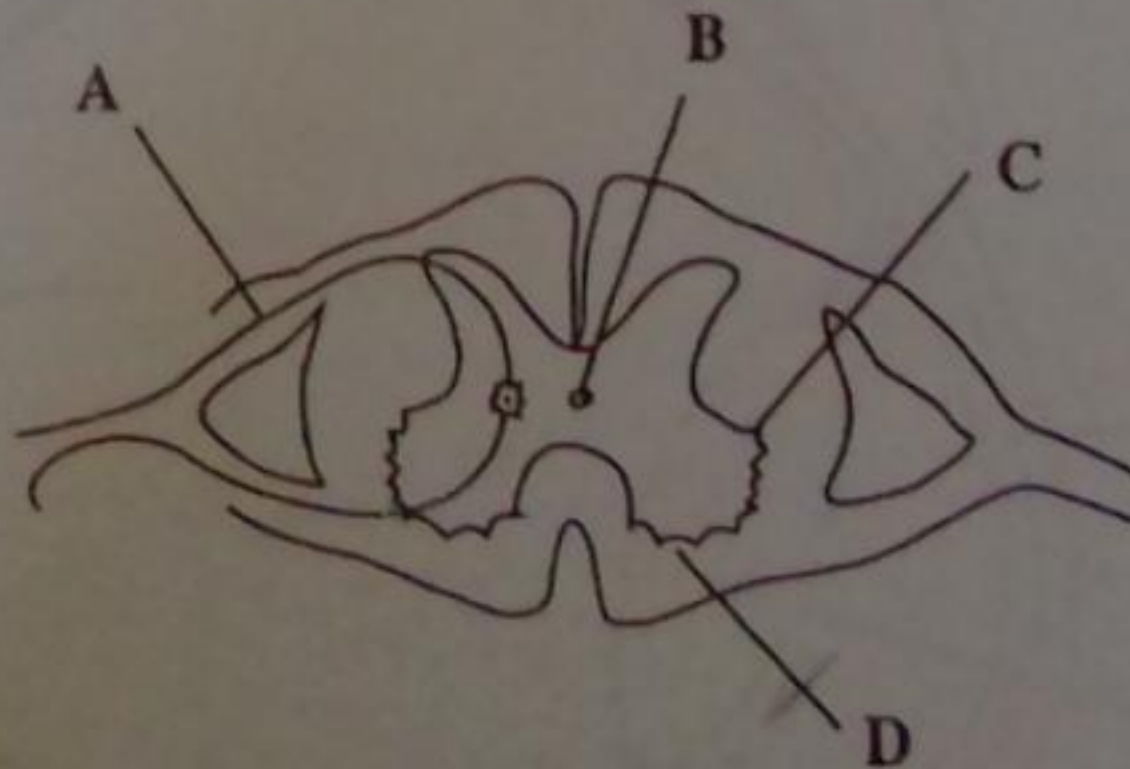


- A cortex
B pelvis
 C medulla
 D ureter

- 24 In which organ is urea produced?

- A kidney
 B liver
 C lung
 D skin

- 25 The diagram below shows a cross section of the spinal cord. Which part contains cell bodies of sensory neurones?



- 26 Which of the following describes sweat glands, skin temperature and blood vessels of the skin soon after exercise?

	sweat glands	skin temperature	blood vessels
A	inactive ✓	low ✓	constricted ✓
B	inactive ✓	high ✓	constricted ✓
C	active ✓	low ✓	constricted ✓
D	active ✓	high ✓	dilated ✓

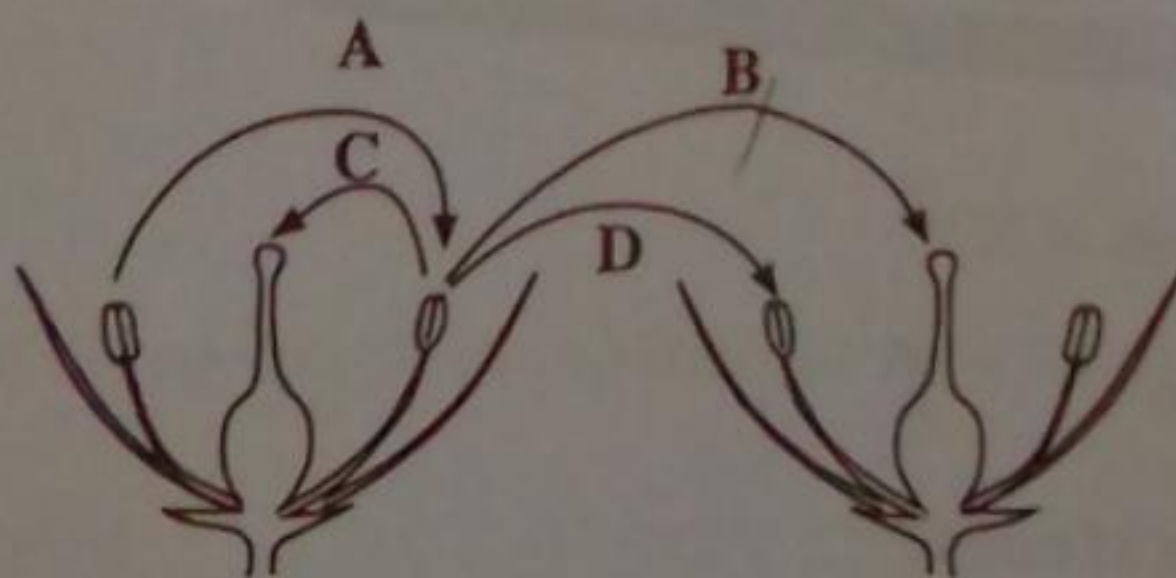
- 27 Which form of drug abuse is likely to result in the transmission of HIV?

- A marijuana smoking
 B injection of heroin
 C alcohol intake
 D glue sniffing

- 28 Which substance is a depressant drug?

- A aspirin
 B nicotine
 C heroin
 D penicillin

- 29 The diagram shows flowers from two different plants of the same species. Which arrow represents cross-pollination?



- 30 Which method of family planning prevents ovulation?

- A spermicide ✓
 B oral contraceptive pill ✓
 C intra-uterine device ✓
 D sheath ✓

31 Which disease has the following symptoms?

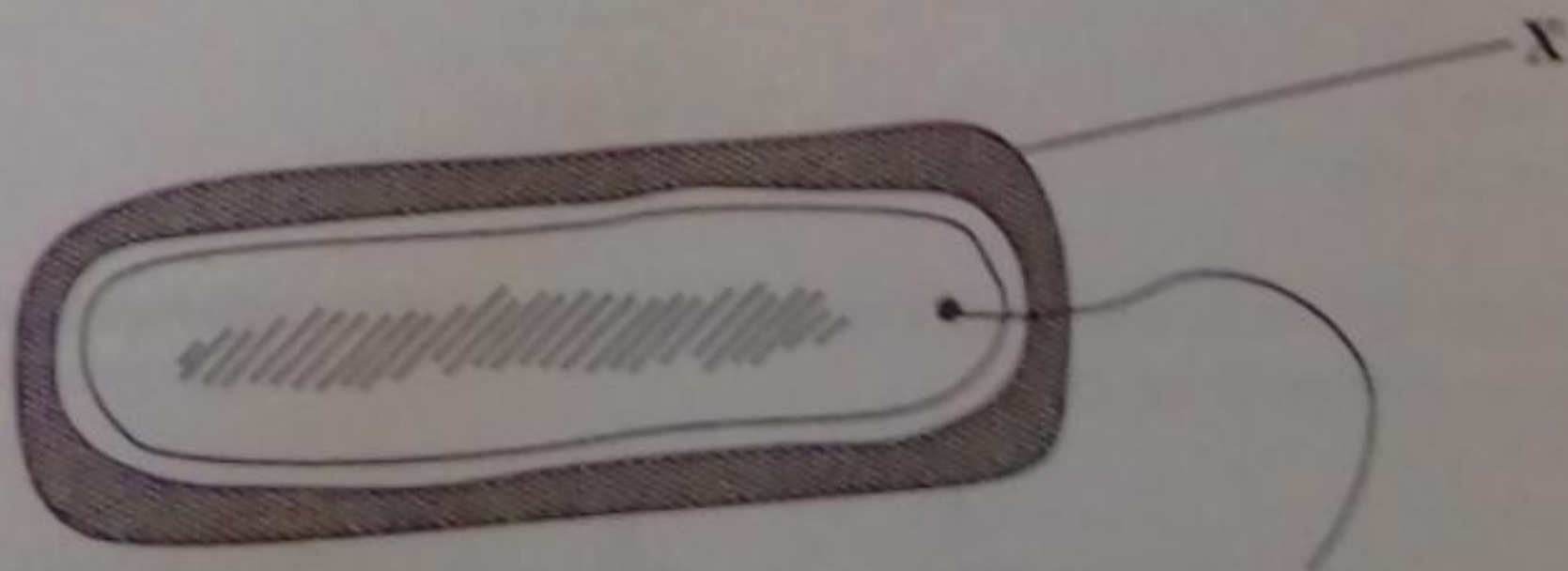
- 1 severe diarrhoea
- 2 fever
- 3 swollen glands
- 4 coughing

- A gonorrhoea
- B AIDS
- C syphilis
- D chancroid

32 What is the causative organism for cholera?

- A virus
- B bacterium
- C protozoan
- D housefly

33 The diagram shows the structure of a bacterium.



What is the function of X?

- A locomotion
- B absorption
- C production of toxins
- D protection

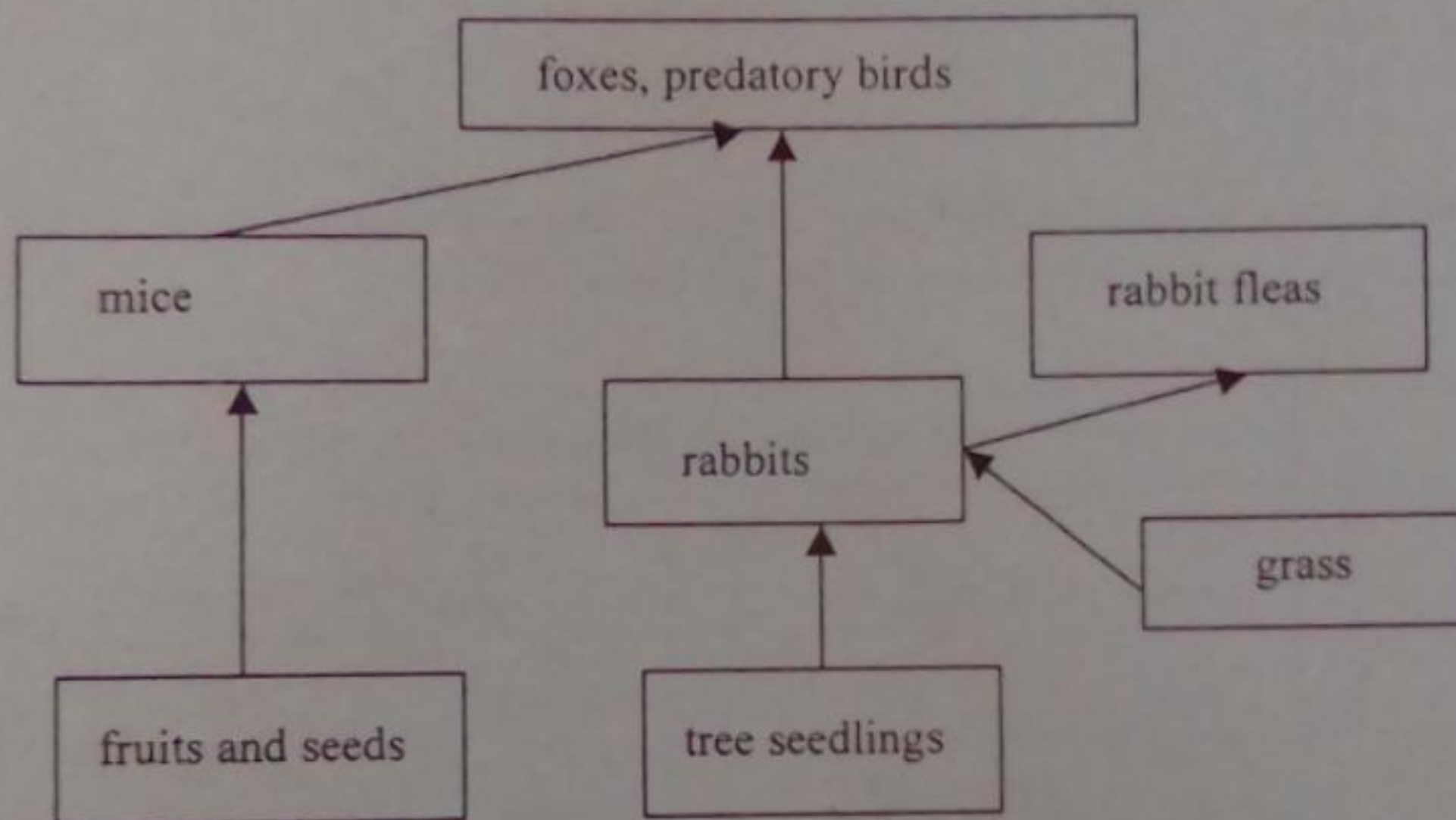
34 Which sample would be examined to find out whether a patient was infected by the malarial parasite?

- A faeces
- B sputum
- C blood
- D urine

35 What causes Down's syndrome?

- A X-rays
- B chromosome mutation
- C cancer
- D gene mutation

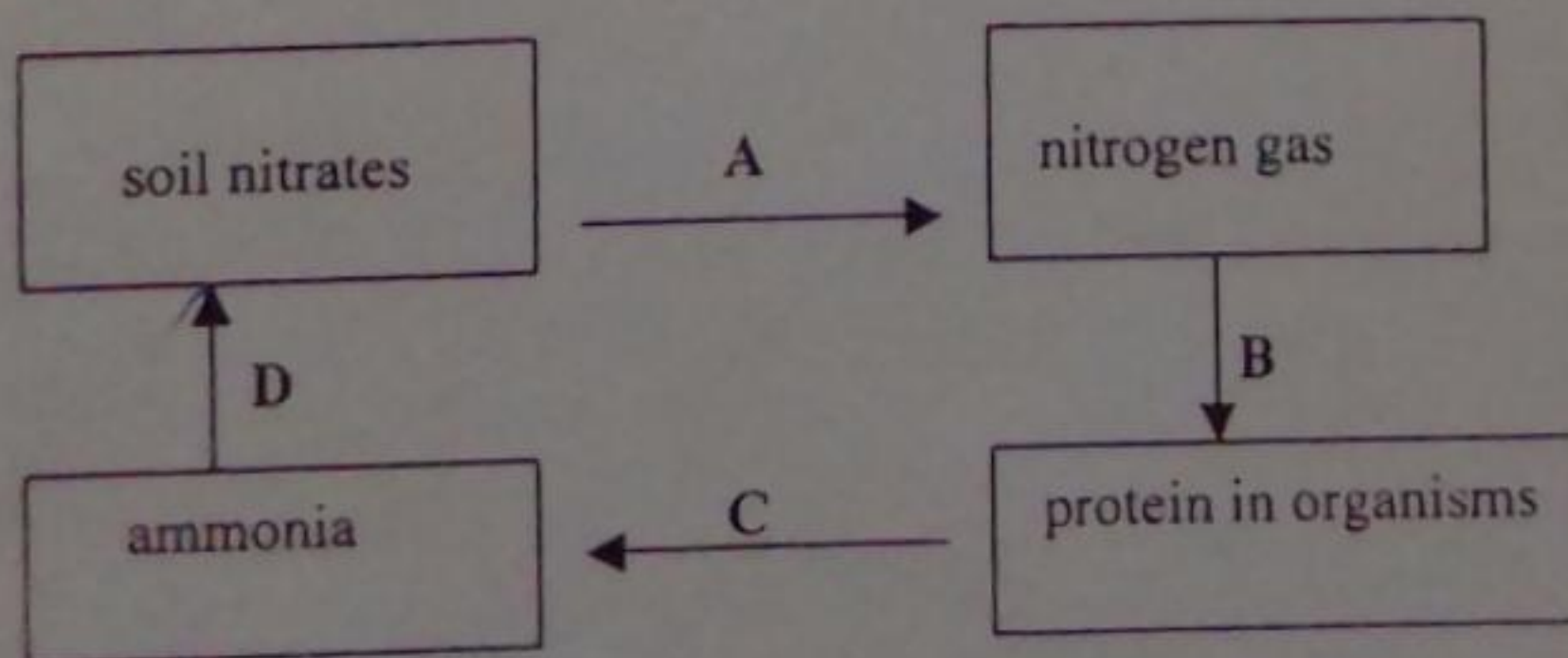
36 The diagram shows a food web.



Which organism is a primary consumer?

- A fox
- B rabbit
- C grass
- D rabbit flea

37 Which process in the nitrogen cycle shown depends upon the activity of nitrifying bacteria?



- 38 Which feature is an example of discontinuous variation?
- A body weight ✓
 B intelligence ✓
 C skin colour ✓
 D blood groups ✓
- 39 Which is **not** a biological component of an ecosystem?
- A litter
 B bacteria
 C water
 D humus
- 40 The table presents data on AIDS cases as cumulative numbers from 1987 to 1994 and as actual cases between October and December 1994.

Age group	Cumulative Numbers of cases		October – December 1994	
	1987 Female	1994 Male	Female	Male
0 – 4	2 594	2 976	178	203
20 – 29	5 992	4 902	391	327
40 – 49	1 475	3 357	135	252
60*	149	486	20	33

What is the difference in AIDS cases between the females and males of the most affected age-group between 1987 and 1994?

- A 1 090
 B 1 882
 C 64
 D 117

Handwritten calculations:

For age group 20-29:

$$\begin{array}{r} 5992 \\ - 391 \\ \hline 5601 \end{array}$$

For age group 40-49:

$$\begin{array}{r} 1475 \\ - 135 \\ \hline 1340 \end{array}$$

For age group 60*:

$$\begin{array}{r} 149 \\ - 20 \\ \hline 129 \end{array}$$

For age group 0-4:

$$\begin{array}{r} 2594 \\ - 178 \\ \hline 2416 \end{array}$$

For age group 20-29 (Male):

$$\begin{array}{r} 4902 \\ - 327 \\ \hline 4575 \end{array}$$

For age group 40-49 (Male):

$$\begin{array}{r} 3357 \\ - 252 \\ \hline 3105 \end{array}$$

For age group 60* (Male):

$$\begin{array}{r} 486 \\ - 33 \\ \hline 453 \end{array}$$

013997



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple Choice

NOVEMBER 2005 SESSION

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

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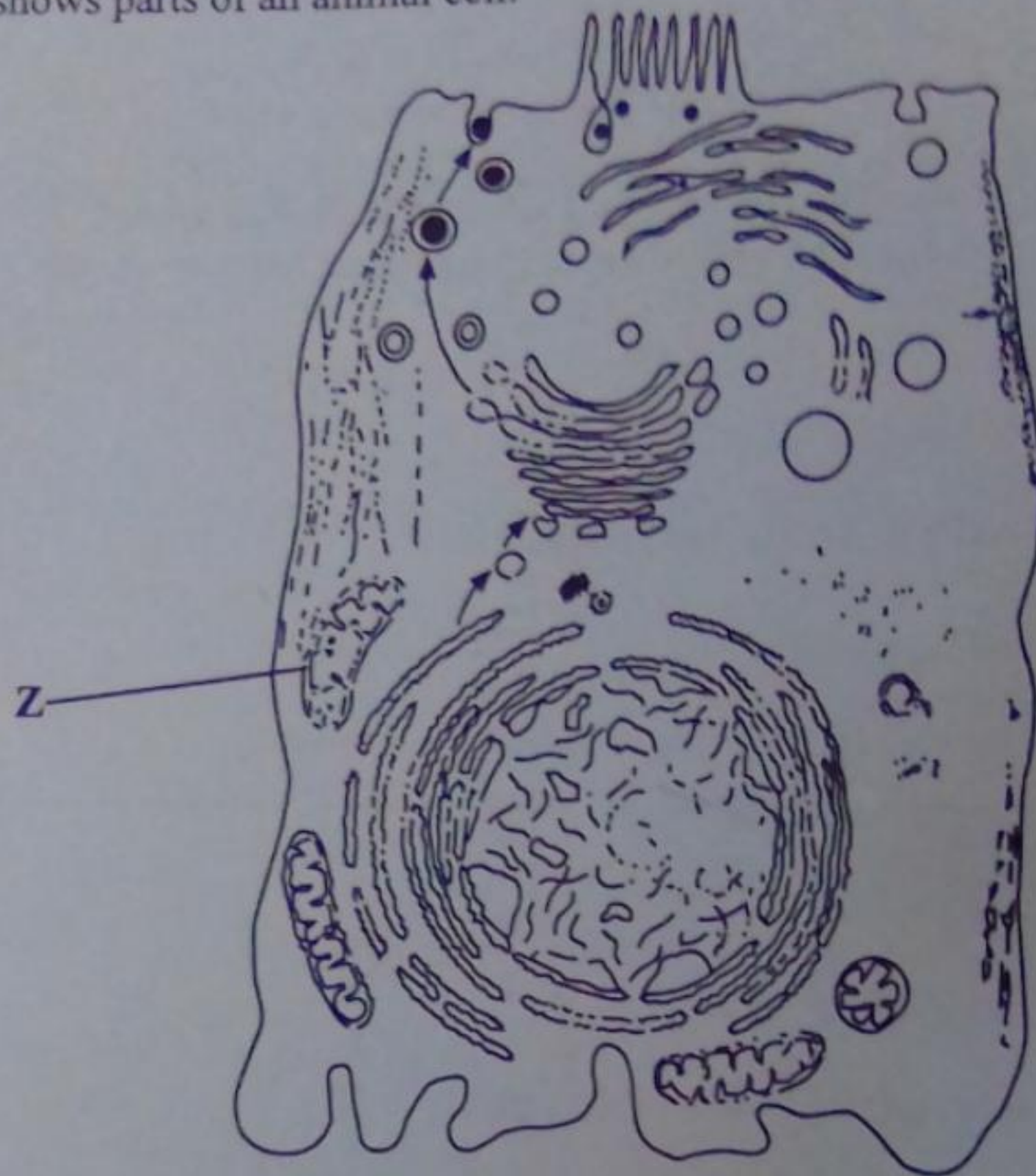
INFORMATION FOR CANDIDATES

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Any rough working should be done in this booklet.

This question paper consists of 18 printed pages and 2 blank pages.

- 1 The diagram shows parts of an animal cell.



What is Z?

- A starch grain
- B vacuole
- C nucleus
- D mitochondrion

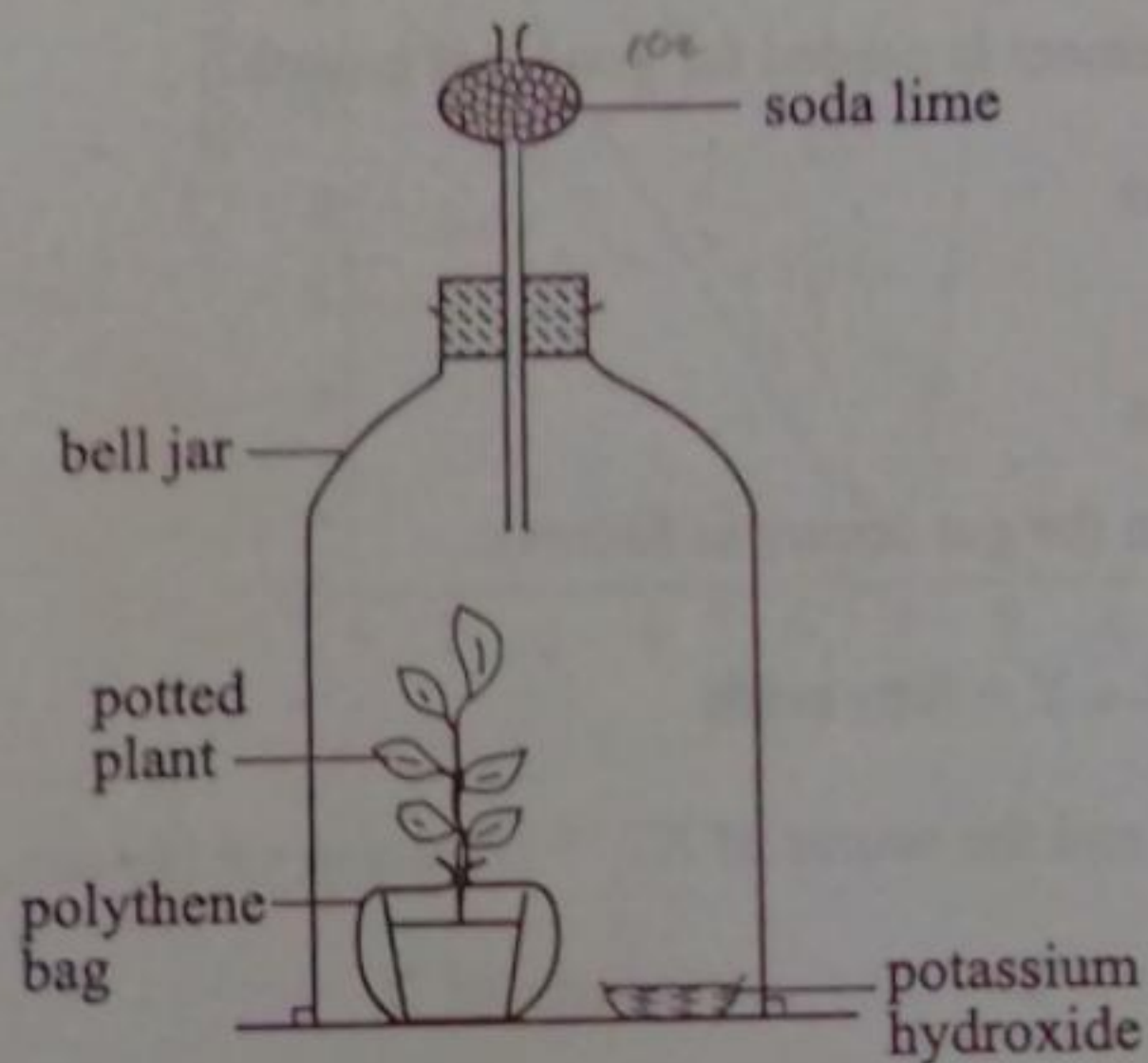
- 2 Which air pollutant causes acid rain?

- A sulphur dioxide
- B dust
- C insecticides
- D soot

3 Which equation represents the chemical changes which take place during photosynthesis?

- A carbon dioxide + oxygen $\longrightarrow$ glucose + water
 B glucose + water $\longrightarrow$ carbon dioxide + oxygen
 C carbon dioxide + water $\longrightarrow$ glucose + oxygen
 D glucose + oxygen $\longrightarrow$ carbon dioxide + water

4 The diagram shows an experiment to investigate a requirement for photosynthesis.



What is the purpose of potassium hydroxide?

- A to dry the air in the bell jar
 B to absorb CO_2 from atmospheric air ✓
 C to absorb oxygen from the plant
 D to absorb CO_2 from the plant ✓

In a water culture experiment to investigate effect of deficiency of ions on seedlings, the culture is covered by a black paper to prevent algae from growing. Why should algae be prevented from growing?

- A They would spread diseases to the seedlings.
- B They would deprive the seedlings of mineral salts.
- C They would deprive the seedlings of oxygen.
- D They would deprive the seedlings of water.

Which mineral element is needed for good leaf growth?

- A phosphorus
- B nitrogen
- C potassium
- D magnesium

Digestion of fats in the gut occurs as follows:

Fats $\xrightarrow{\text{enzyme X}}$ Y + fatty acids

What are X and Y and the source of X?

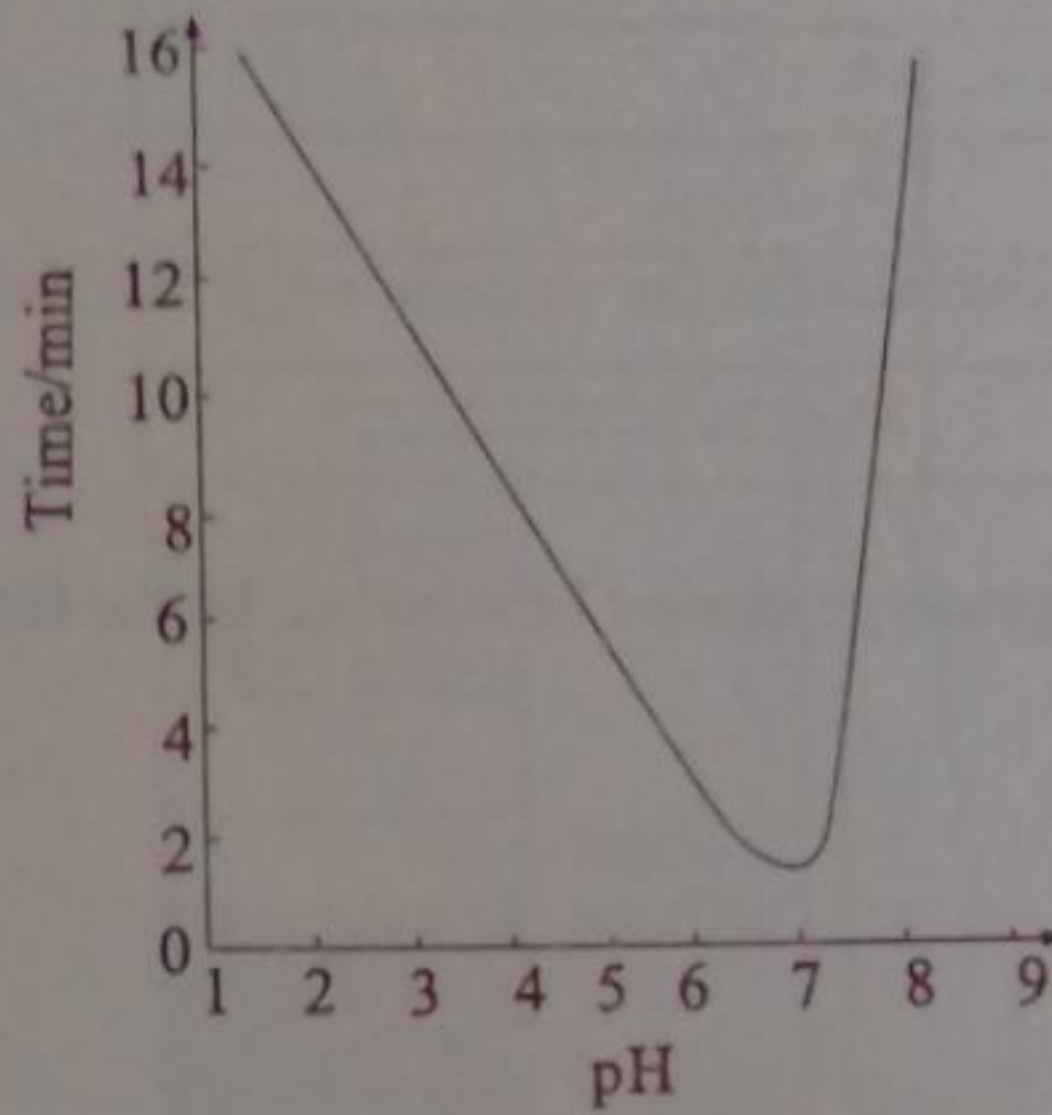
	X	Y	source of X
A	lipase	glycerol	pancreas
B	lipase	glycerol	liver
C	lipase	glycogen	gall bladder
D	amylase	glycogen	pancreas

Which digestive juice is concerned with mechanical digestion **only**?

- A saliva
- B bile ✗
- C gastric juice
- D pancreatic juice

9

The graph shows the time taken for a sample of starch to be completely digested by amylase at various pH values.



What is the optimum pH for amylase?

- A 4
- B 7
- C 8
- D 9

10 The table shows the blood groups of a family.

	Blood group
Mother	A
Father	B
First child	-
Second child	O

What blood group genotype **cannot** be inherited by the first child?

- A |A|A
- B |A|B
- C |A|O
- D |B|O

11 In which organ is urea formed?

- A pancreas
- B kidney
- C liver
- D gall bladder

12 Which is a result of too much fat in the diet?

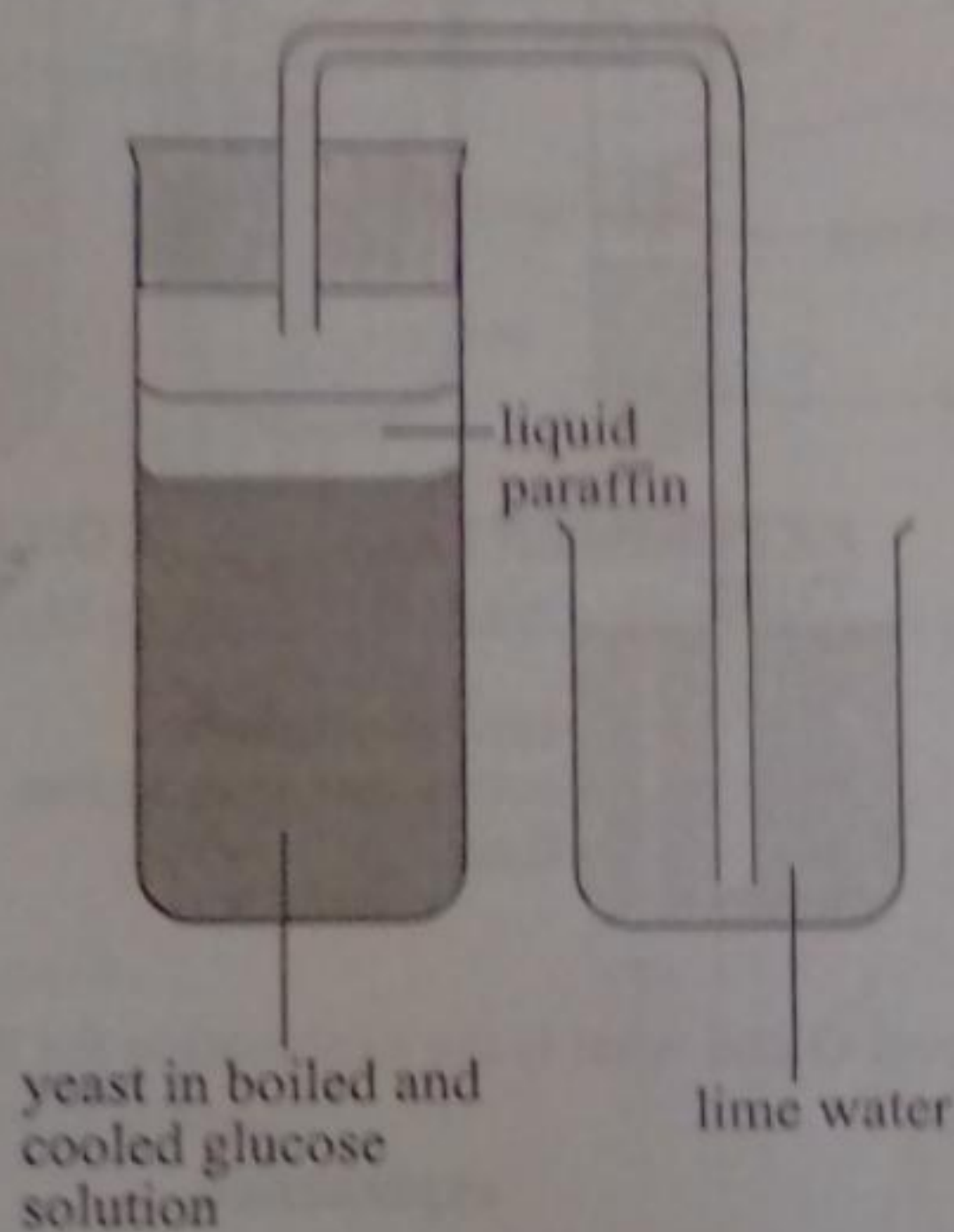
- A kwashiokor
- B sickle cell anaemia
- C stroke
- D diabetes mellitus

13 Which bacterial process converts ammonium compounds into nitrites?

- A denitrification
- B nitrification
- C nitrogen fixation
- D decomposition

- 14 A piece of phosphorous was allowed to burn in 200 cm^3 of exhaled air. What volume of air remained at the end of the experiment?
- A 168 cm^3
B 160 cm^3
C 158 cm^3
D 32 cm^3

- 15 The diagram shows an experiment to demonstrate anaerobic respiration in yeast.

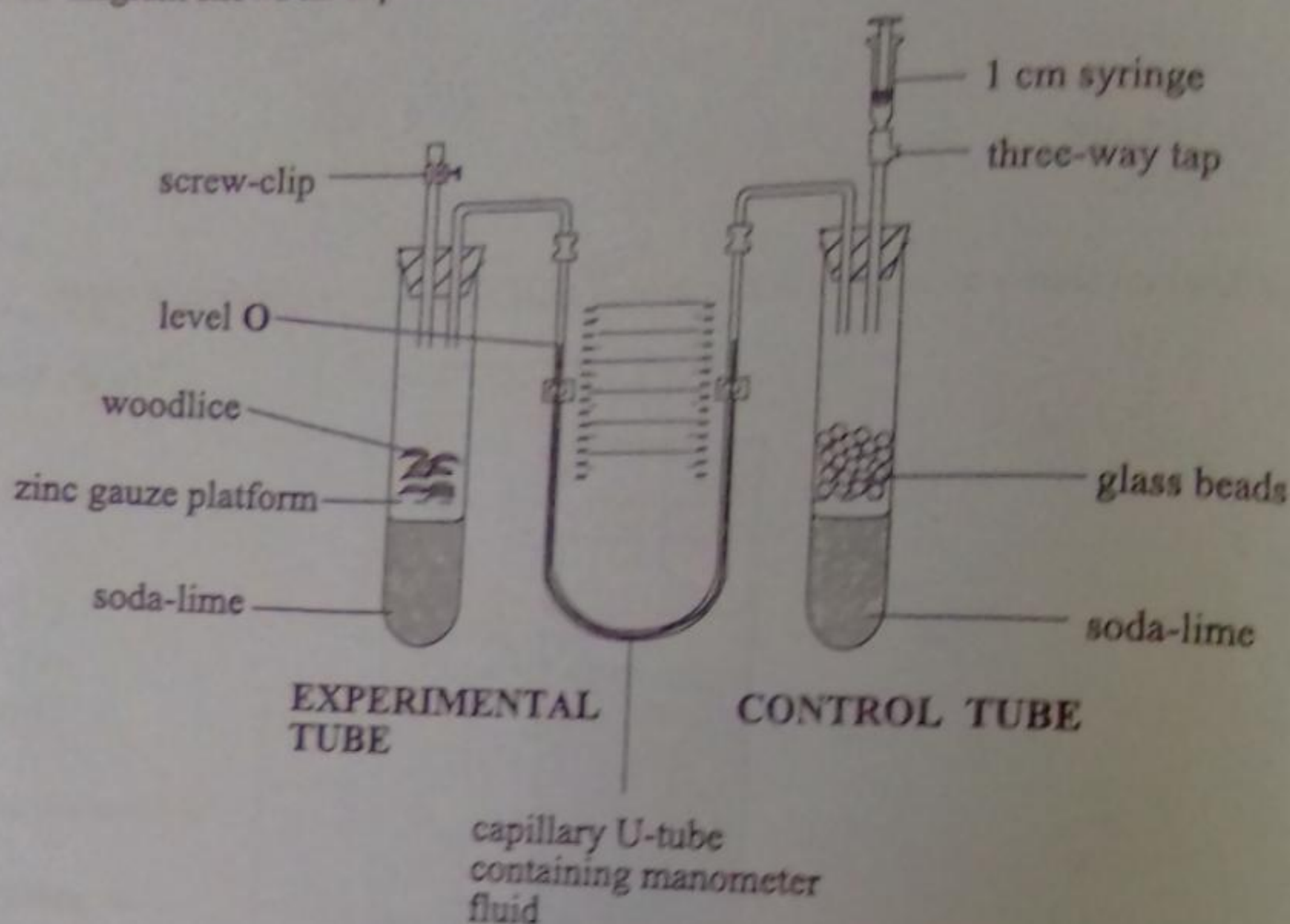


- Which substances are present below the liquid paraffin after 20 minutes?
- A glucose + water only
B glucose + water + alcohol
C glucose + water + carbon dioxide
D glucose + water + carbon dioxide + alcohol

- 6 What happens to lactic acid produced during anaerobic respiration?

- A It is broken down to carbon dioxide in the mitochondrion.
B It is excreted through the kidneys.
C It is broken down to carbon dioxide in the liver.
D It is oxidised to carbohydrate in the liver.

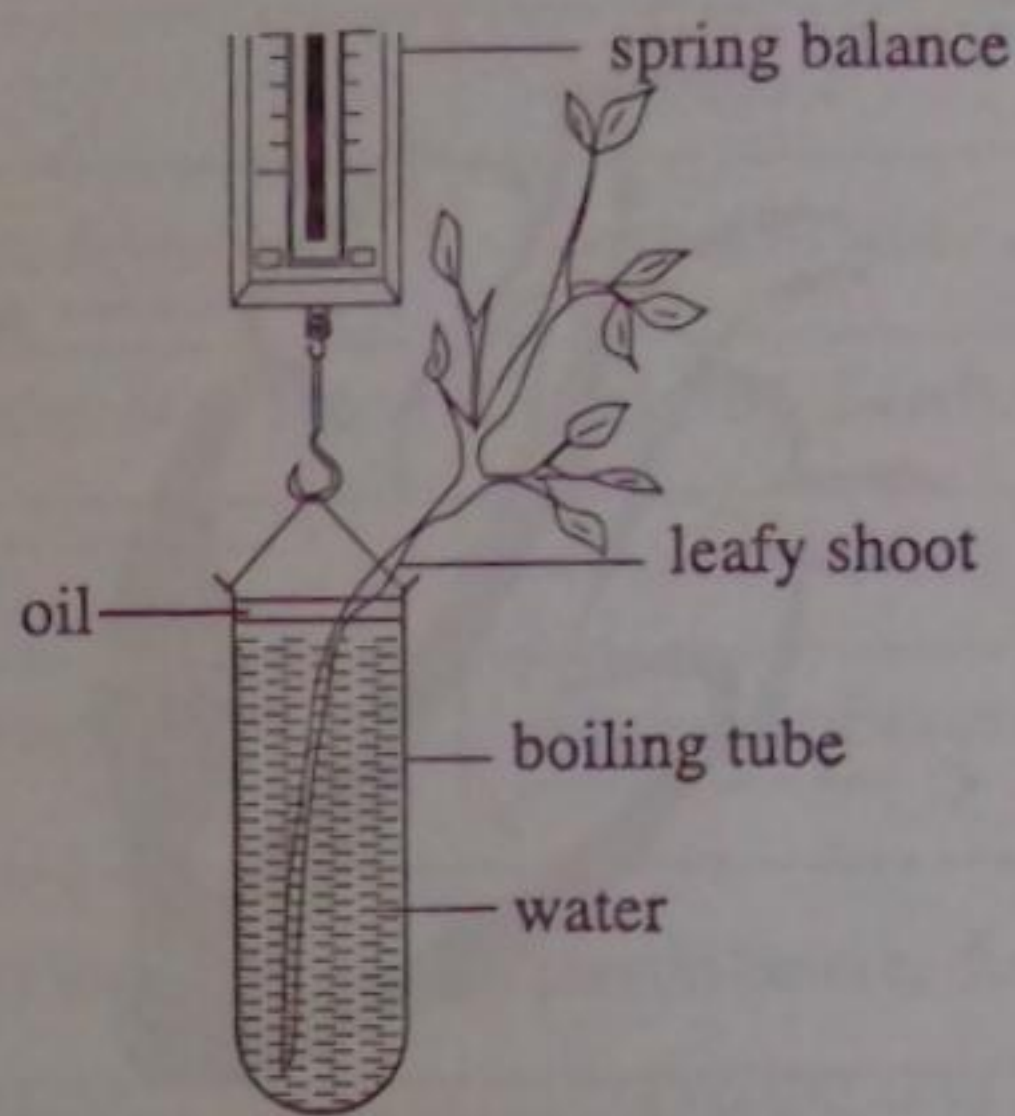
- 17 The diagram shows an experiment to investigate aerobic respiration.



What happens to level O and what is the explanation for this?

level O	explanation
A rises	Manometer fluid is pushed by syringe plunger.
B falls	Manometer fluid is pushed by CO_2 from woodlice.
C falls	Air is absorbed by glass beads.
D rises	Oxygen is used up by woodlice.

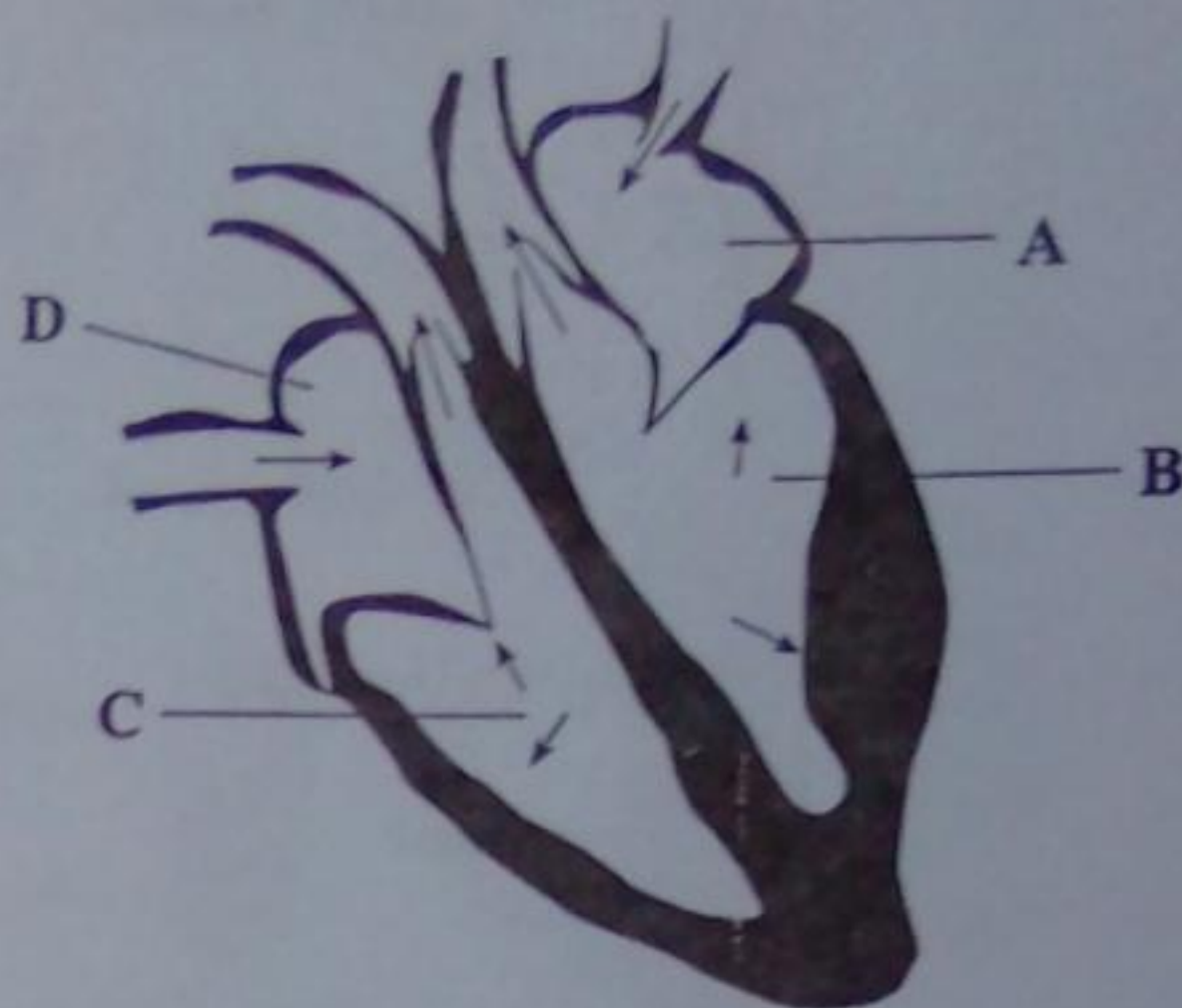
- 18 The diagram shows apparatus used to measure the rate of transpiration.



Which set of conditions would cause the **lowest** decrease in mass per unit time?

- A cold and cloudy
 - B warm and cloudy
 - C cold, sunny and dry
 - D warm, sunny and humid
- 19 What does a potometer directly measure?
- A rate of transpiration
 - B rate of photosynthesis
 - C rate of evaporation
 - D rate of absorption of water
- 20 Why is transpiration rate lowest at night?
- A There is no energy from photosynthesis.
 - B Stomata are closed.
 - C Temperature is too low.
 - D Plants become permanently wilted.

- 21 The diagram shows the heart during ventricular contraction. Which chamber has the highest pressure?



- 22 Which blood cells produce antibodies?
- A red blood cells
 - B blood platelets
 - C lymphocytes
 - D phagocytes
- 23 Which bones form a ball and socket joint?
- A humerus and scapula
 - B humerus and ulna
 - C femur and tibia
 - D femur and fibula

24 Which are **two** functions of the kidney?

function 1	function 2
A removal of glucose from blood	control of water content of blood
B control of water content of blood	removal of carbon dioxide from blood
C removal of urea	control of blood sugar
D removal of urea from blood	removal of excess water from blood

25 The graph shows the body temperature of a person before, during and after taking a cold bath.

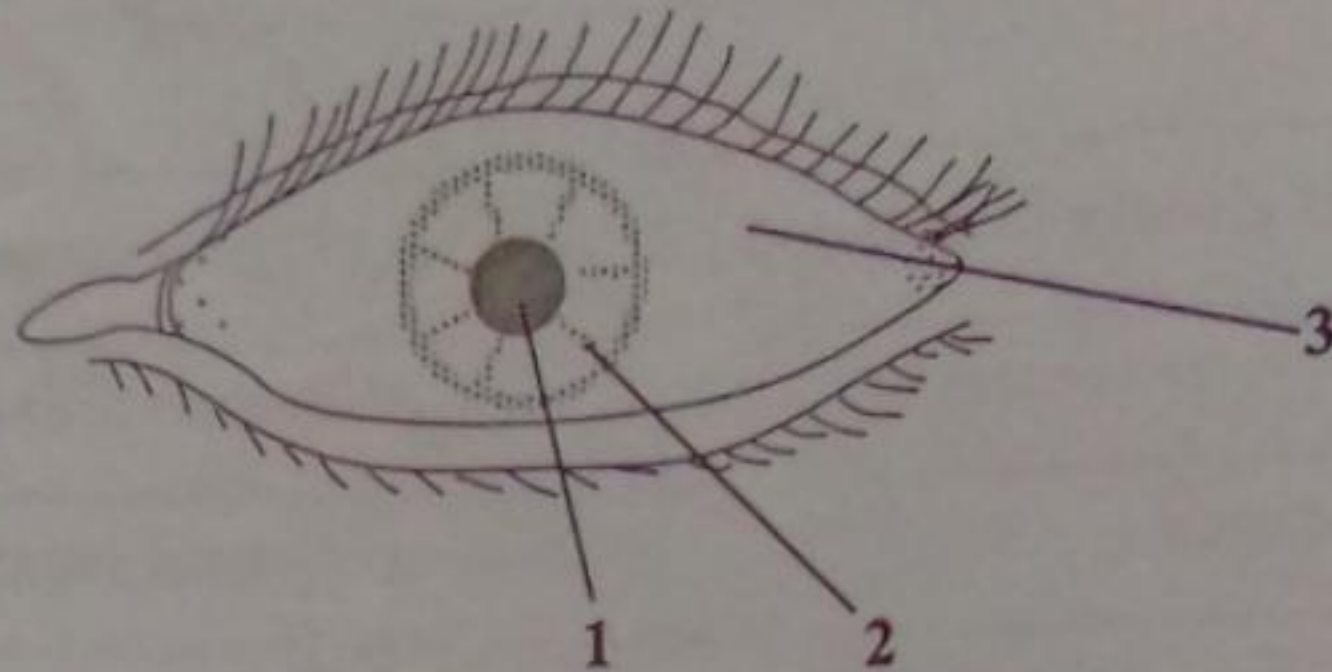


For how long was the person in the bath?

- A 10 minutes
- B 20 minutes
- C 30 minutes
- D 40 minutes

26

The diagram shows the frontal view of the human eye.



Identify 1, 2 and 3.

	1	2	3
A	pupil	iris	cornea
B	pupil	iris	conjunctiva
C	pupil	cornea	iris
D	pupil	iris	sclera

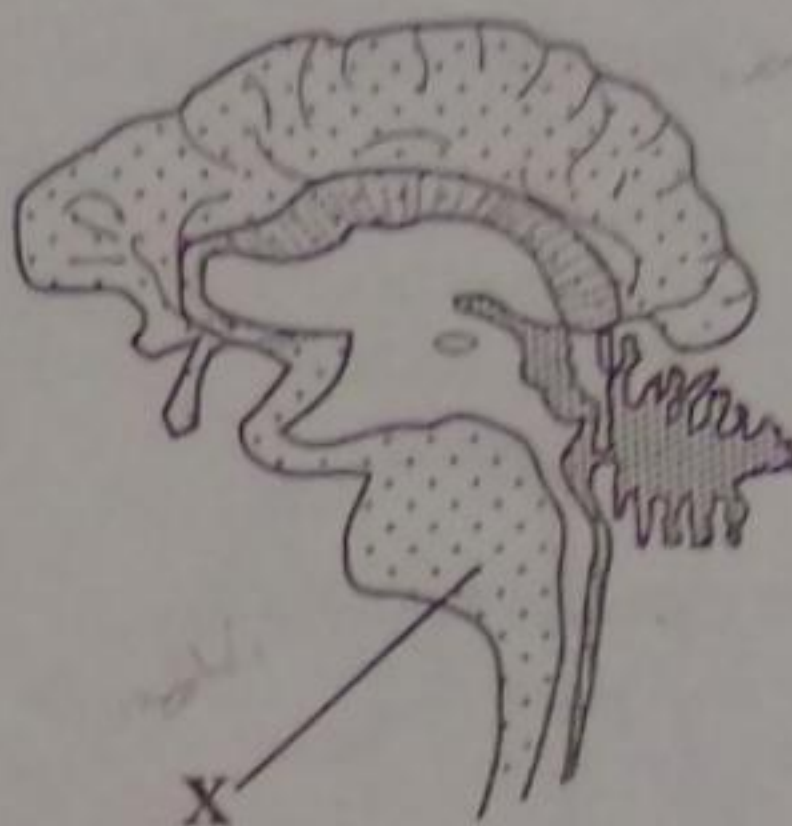
27

Which part of the eye controls the size of the pupil?

- A the ciliary body
- B the aqueous humour
- C the cornea
- D the iris

28

The diagram shows a section through the human brain.



X controls

- A respiratory movements.
- B muscular co-ordination.
- C body temperature.
- D appetite and emotion.

29

Which hormone controls development of secondary sexual characteristics and thickening of uterus lining before ovulation?

- A adrenalin
- B insulin
- C oestrogen
- D progesterone

30

The following are effects of hormones.

- 1 increase in mental awareness
- 2 increase in activity of the kidney
- 3 increase in diameter of the pupil
- 4 increase in activity of the pancreas

Which are the effects of a high level of adrenalin in blood?

- A 1 and 3 only
- B 2 and 3 only
- C 1, 2 and 3 only
- D 2, 3 and 4 only

31 What are the effects of deltaprim, paracetamol and tetracycline?

	deltaprim	paracetamol	tetracycline
A	prevents malaria	kills bacteria	relieves pain
B	prevents malaria	relieves pain	kills bacteria
C	relieves pain	prevents malaria	prevents malaria
D	relieves pain	kills bacteria	prevents malaria

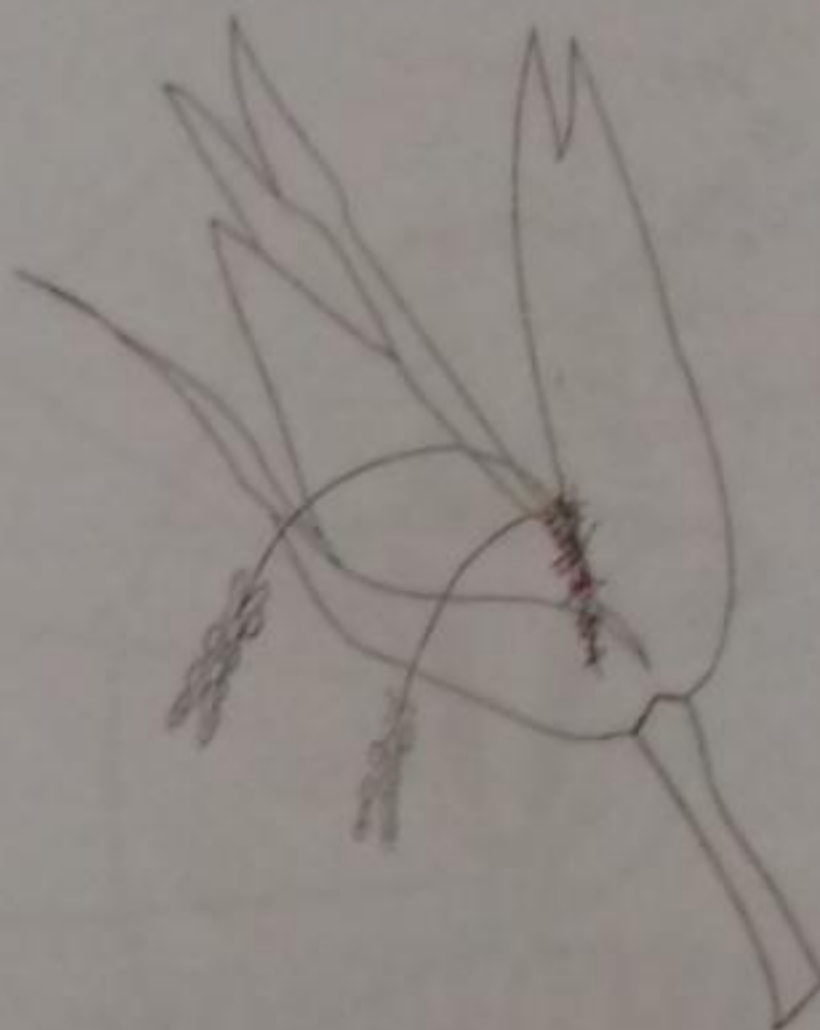
32 What is the effect of tar from cigarette smoking?

- A increase in heart beat ✓
- B vasoconstriction ✓
- C reduces surface area for gaseous exchange in lungs ✓
- D combines with haemoglobin in the red blood cell ✓

33 Which method of asexual reproduction occurs in bread mould?

- A budding
- B binary fission
- C spore formation
- D bulb formation

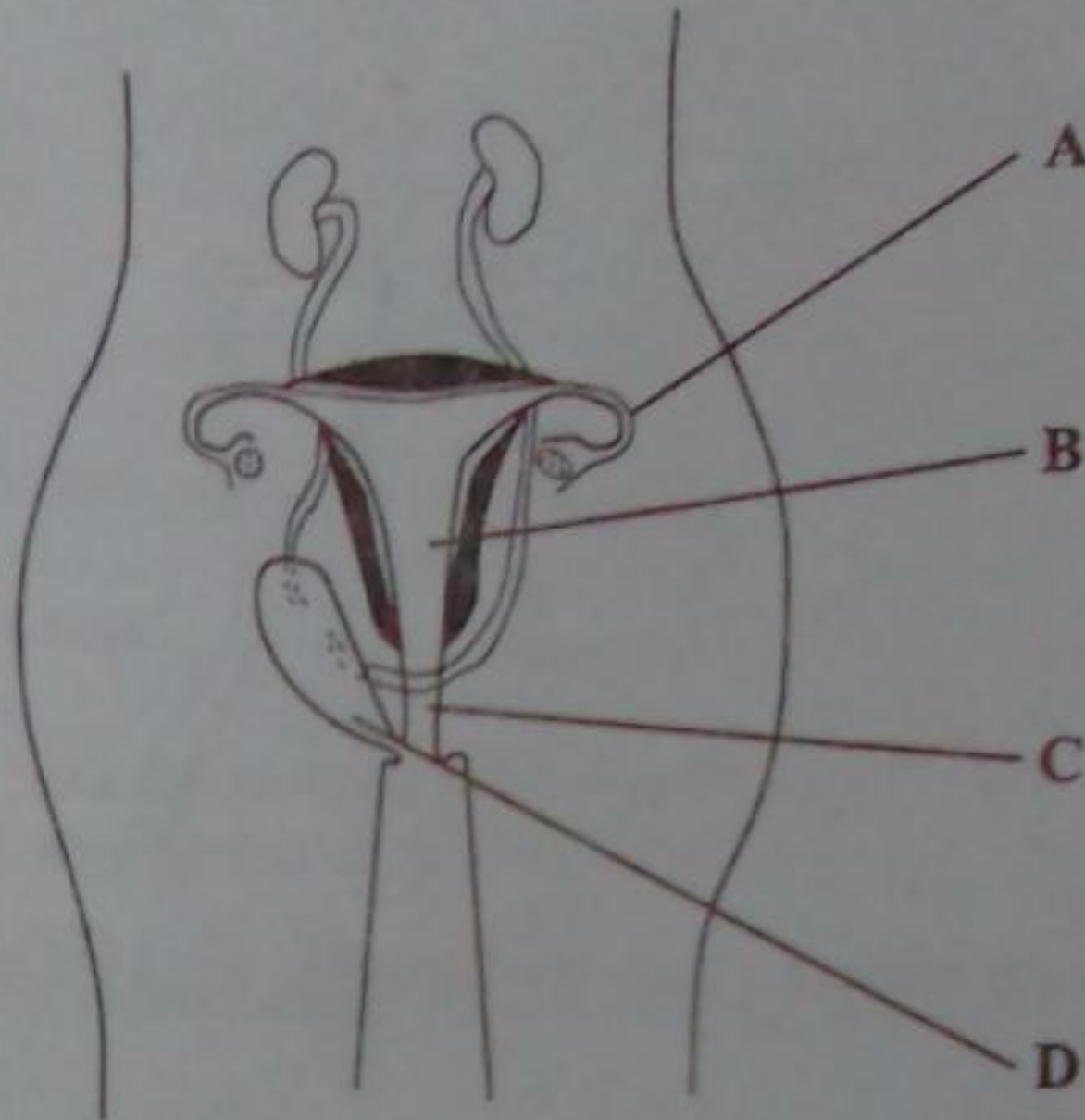
- 34 The diagram shows a wind pollinated flower.



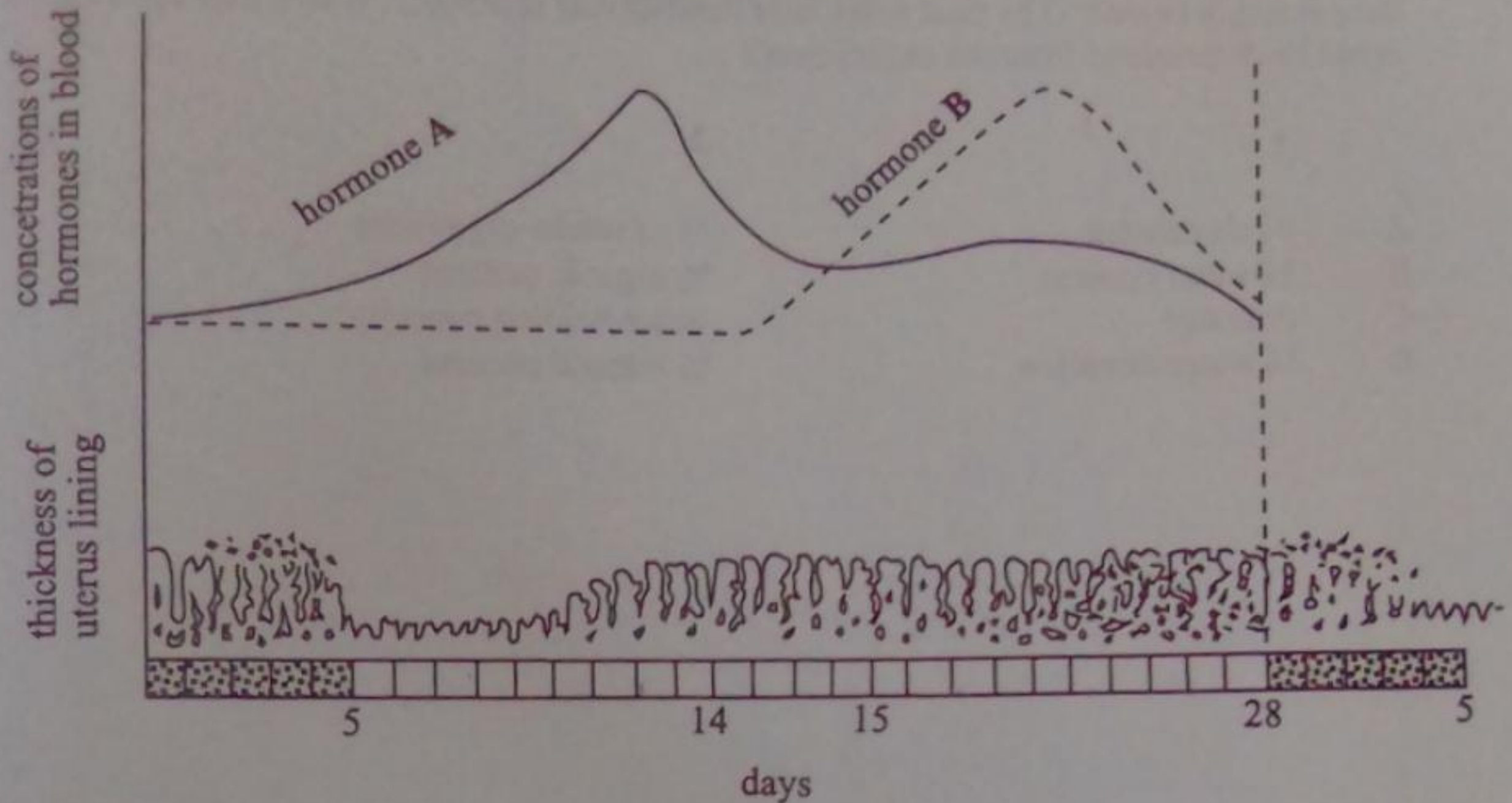
What observable feature indicates that the flower is wind pollinated?

- A dull petals
 B stigma branched into three parts
 C anthers dangling outside the flower
 D light pollen grains
- 35 Which condition produces the greatest variety of offspring in plants?
- A stigma pollinated by pollen from an anther on the same flower
 B pollination between stigma and anther of different flowers on the same plant
 C pollination between an anther on one flower and a stigma of another flower on a different plant of the same species
 D pollination between an anther and a stigma of flowers of different plant species.

- 36 The diagram shows the human female reproductive system. Where is the intra-uterine device placed?



- 37 The diagram shows changes during the human menstrual cycle.



Which is the fertile period?

- 38 Which is the least effective method of controlling the spread of cholera?

- A vaccination
- B boiling drinking water
- C good personal hygiene
- D adequate sanitary facilities

- 39 What causes Down's Syndrome?

- A X rays
- B mutation of a chromosome
- C cancer
- D mutation of a gene

- 40 A soil sample was weighed and placed in an oven at 105°C for 48 hrs. It was reweighed and then strongly heated. The final mass after heating was recorded. Which two soil properties could be determined from the experiment?

	1	2
A	% air content	% of micro-organisms
B	% water content	% organic content
C	drainage	water holding capacity
D	% water retention	% mineral content

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

21612

BIOLOGY

5008/1

PAPER 1 Multiple Choice

Friday 19 NOVEMBER 2004

Morning

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

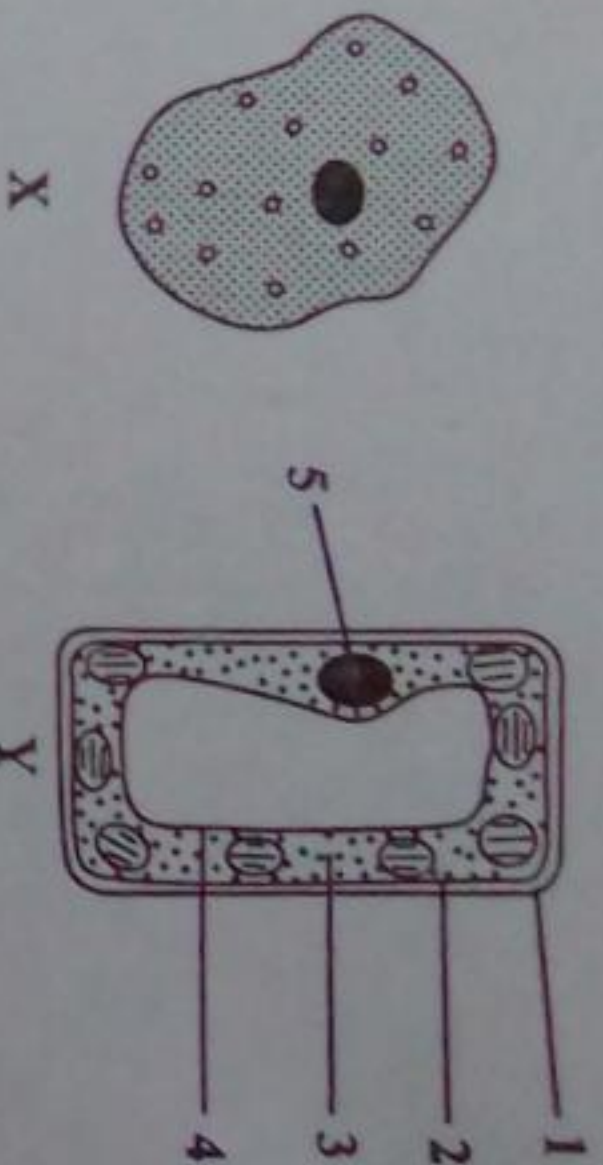
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 25 printed pages and 3 blank pages.

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- 1 The diagrams show an animal cell **X** and a plant cell **Y**.



Which features in cell **Y** are found in cell **X**?

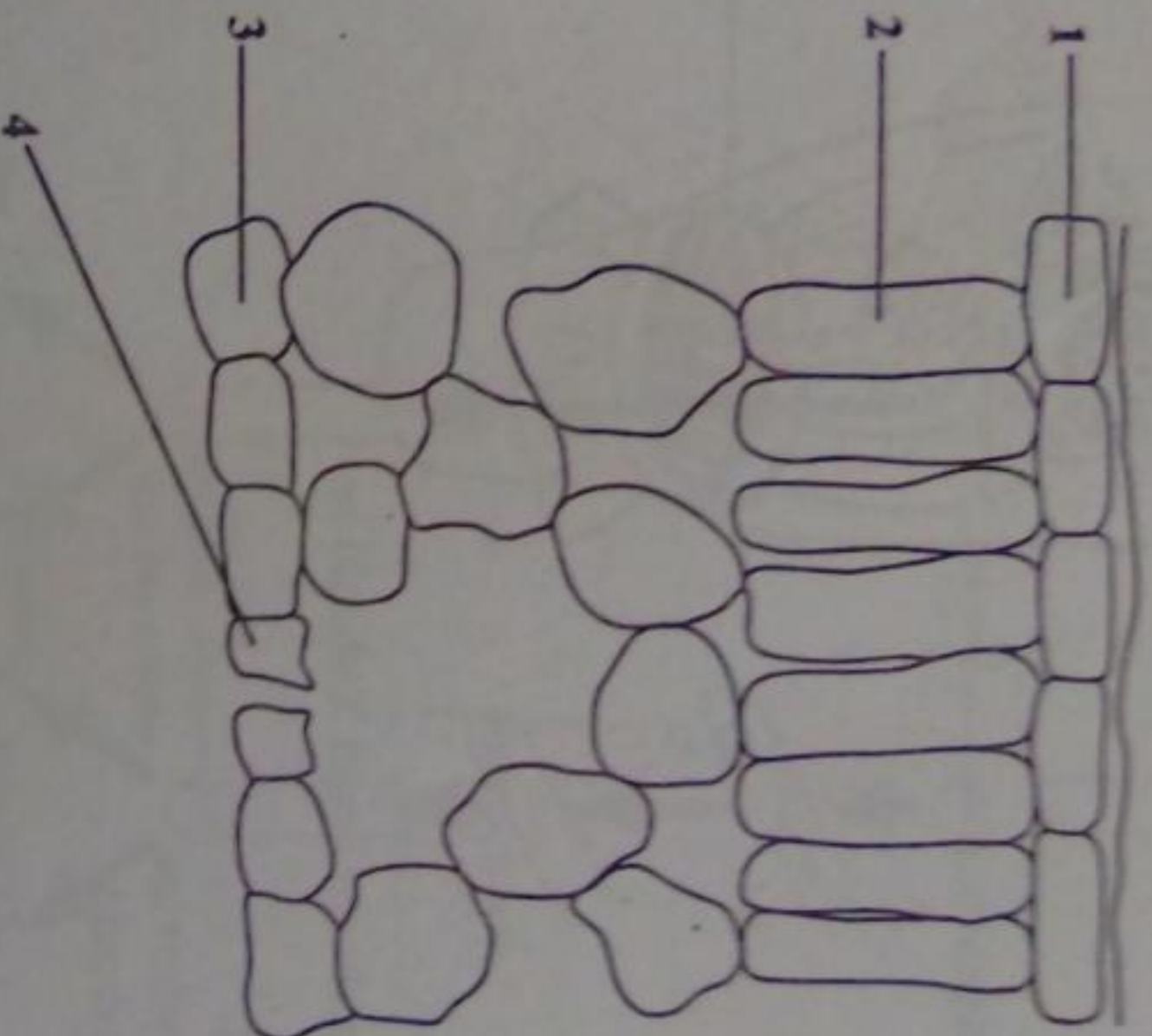
- A 1, 2 and 3
- B 1, 3 and 4
- C 2 and 3 only
- D 2, 3 and 4

2

Human blood is put into distilled water. What is the effect on the blood cells?

- A Cells swell and then burst.
- B There is no change in the cell's appearance.
- C Cells shrink slightly.
- D Cells lose their smooth rounded shape.

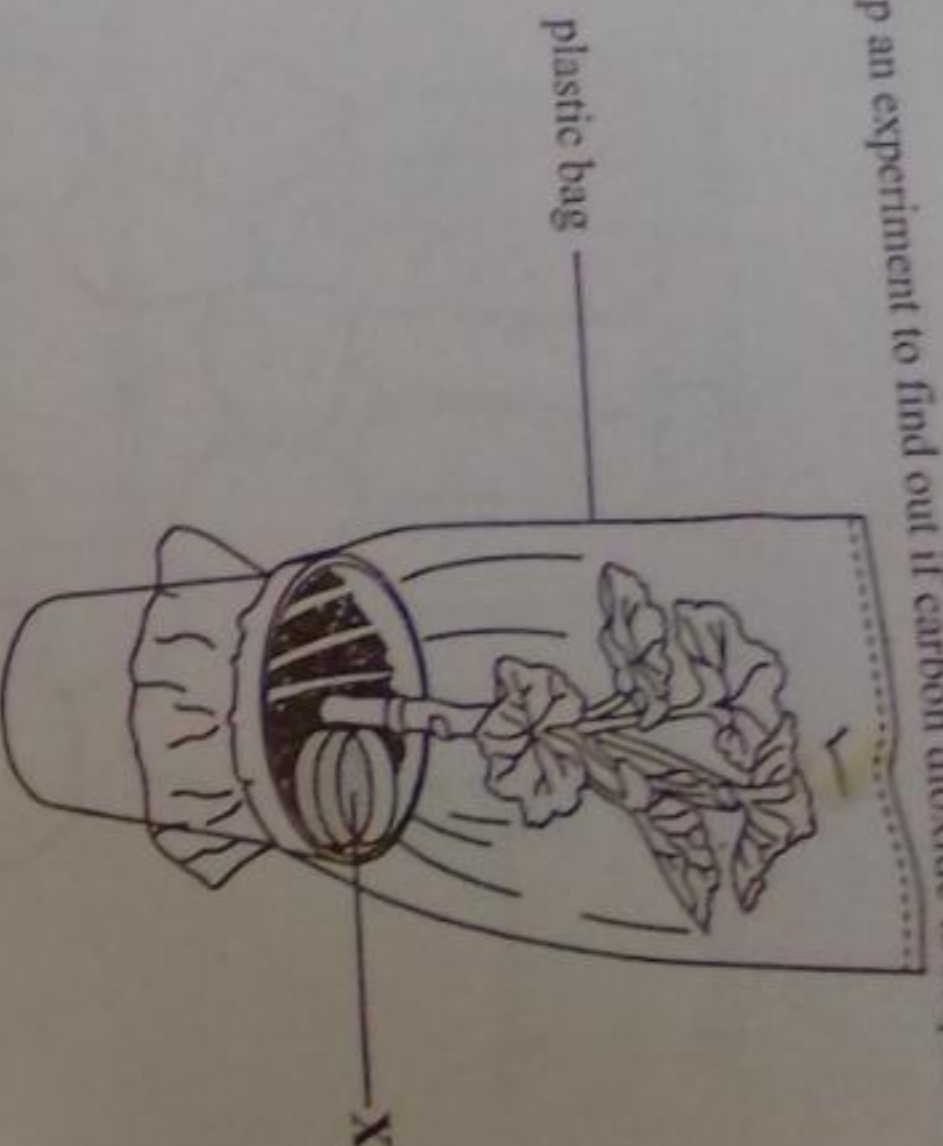
3 The diagram shows cells in a leaf.



Which cells are specialised for photosynthesis?

- A 1 and 2
- B 2 and 3
- C 2 and 4
- D 3 and 4

4 A student set up an experiment to find out if carbon dioxide is required for photosynthesis.



In the control experiment liquid X should be

- A pyrogallol.
- B calcium hydroxide.
- C sodium hydroxide.
- D water.

5 Which of the following situations numbered 1 – 4 do **not** produce carbon dioxide?

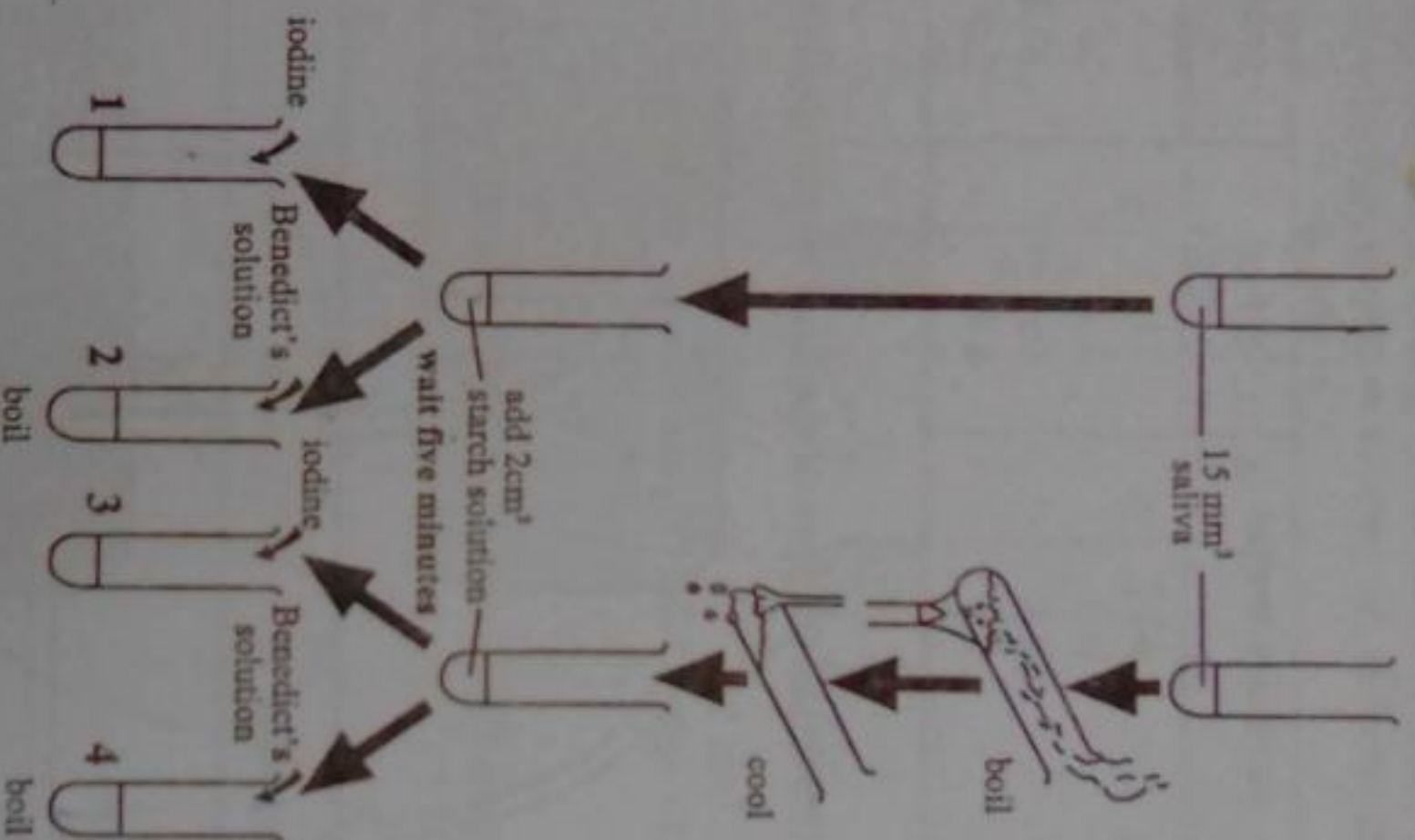
- 1. fresh green leaf
- 2. a dead cat buried for two days
- 3. a raw potato
- 4. a freshly boiled cabbage

- A 4 only
- B 2 and 4
- C 3 and 4
- D 2, 3 and 4

6 Effects of deficiency of potassium in plants are

- A stunted growth and chlorosis.
- B stunted growth of roots, sometimes purplish leaf colouring.
- C poor flowering and fruiting.
- D stunted growth of shoots and purplish leaf colouring.

The action of salivary amylase on starch was investigated as shown below.



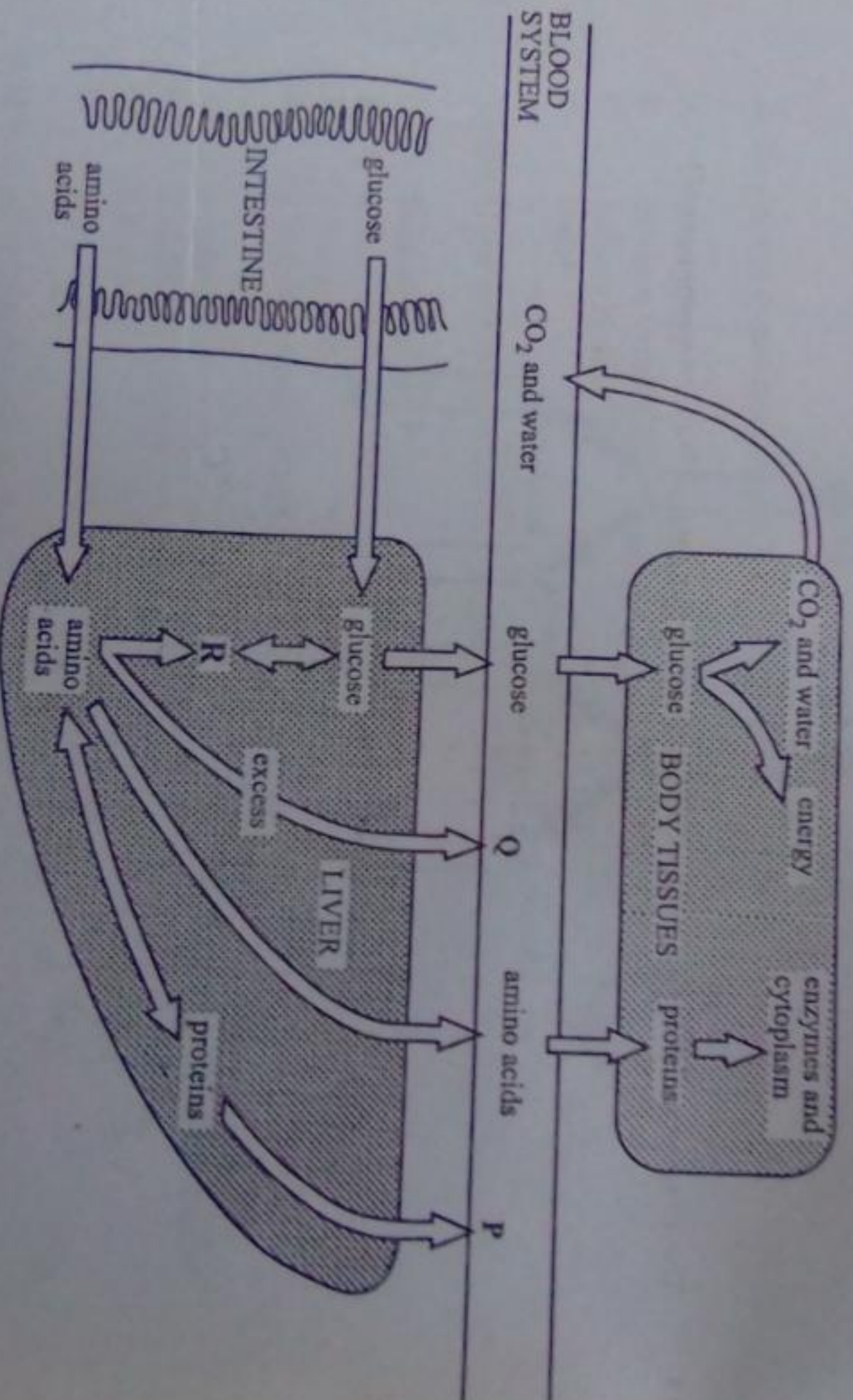
Which results were obtained?

	Test Tube			
	1	2	3	4
A	blue-black	orange	blue-black	blue
B	brown	orange	blue-black	blue
C	blue-black	blue	brown	orange
D	brown	blue	blue-black	orange

- 8 The table shows the composition of four foods in grams per 100 g samples. Which food sample would be most suitable for growth to occur?

food sample	carbohydrates /g	fats /g	proteins /g
A	68.1	0.0	0.6
B	0.0	0.8	16
C	9.8	46	30.2
D	5.2	4.4	5.6

- 9 The diagram below illustrates the role of the liver in metabolism.



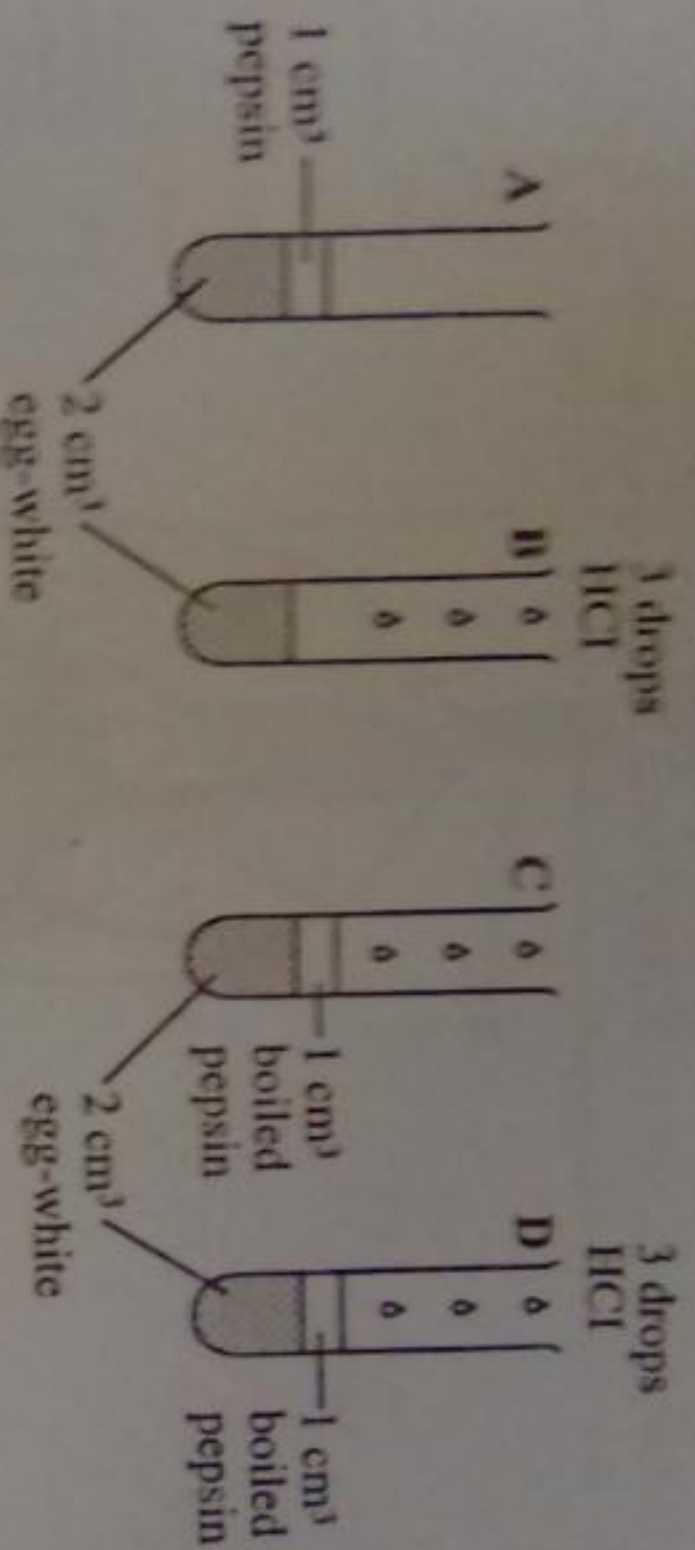
Identify the substances P, Q and R.

- P
- A urea
 - B fat
 - C uric acid
 - D plasma proteins

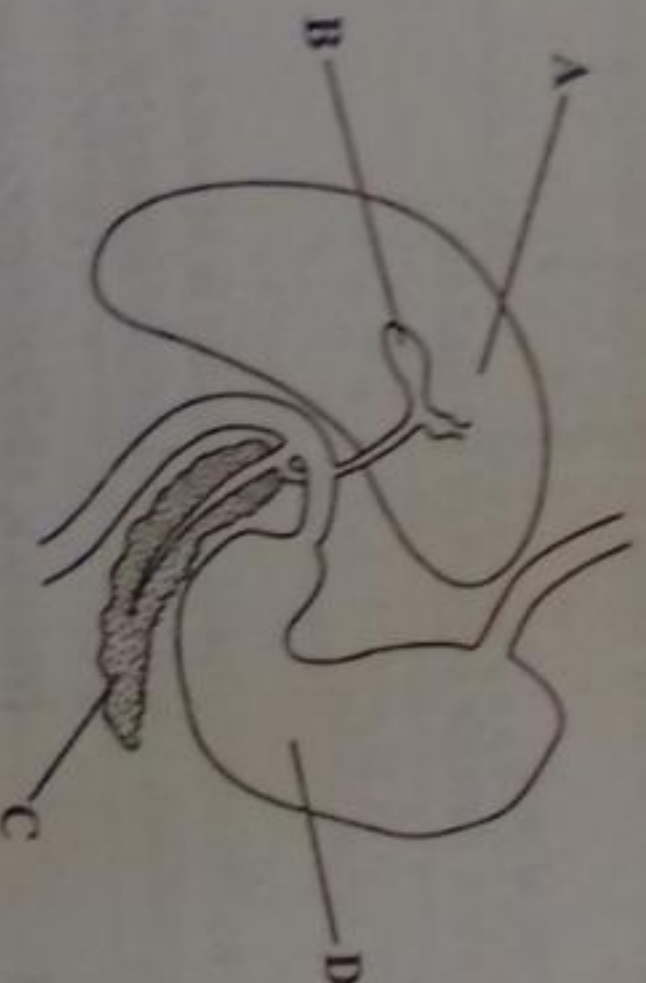
- Q
- fat
 - urea
 - glucose
 - urea

- R
- glycogen
 - glucose
 - urea
 - glycogen

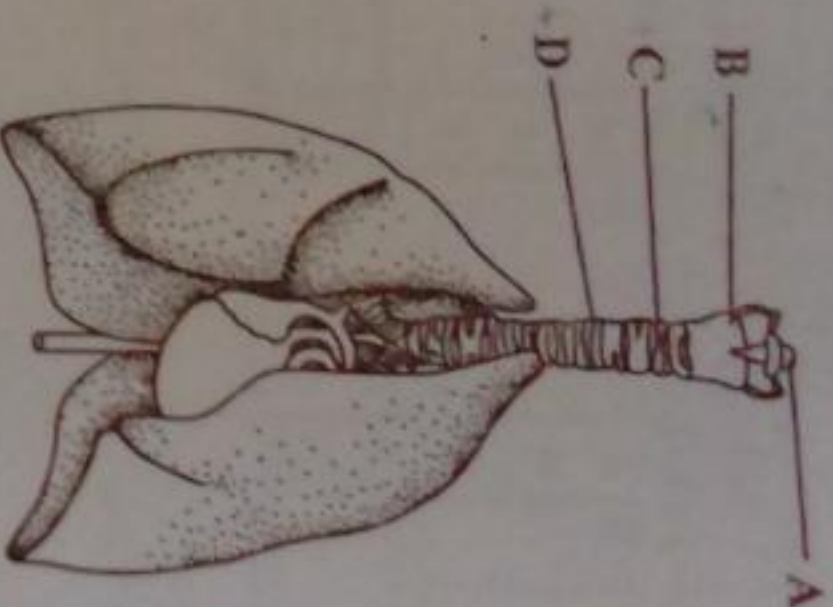
An investigation on egg-white was carried out at 35°C. Which test-tube was likely to have its contents going clear after 30 minutes?



The diagram shows part of the human alimentary canal. Which part produces bile?



- 12 The diagram shows the respiratory system. Which part is the larynx?



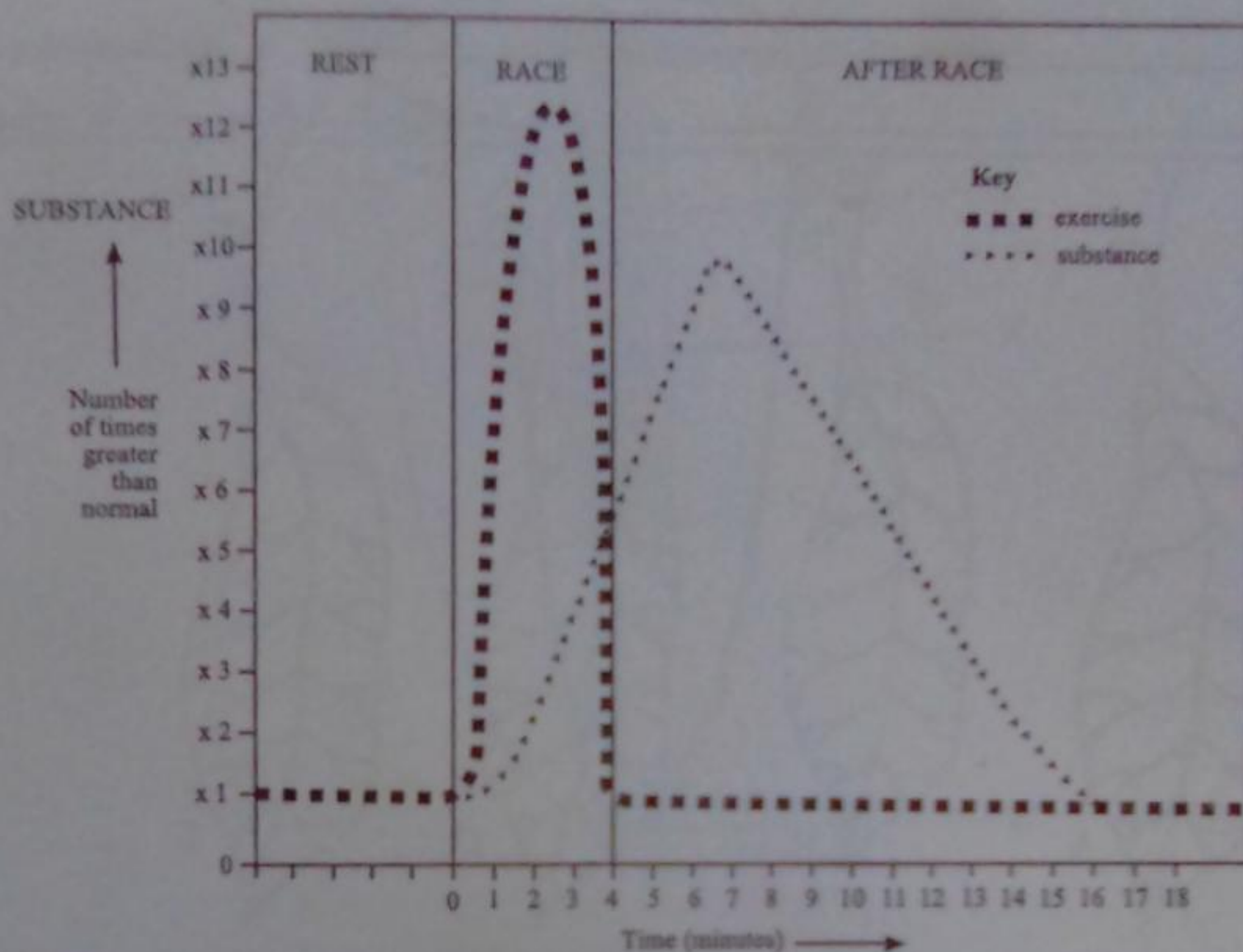
- 13 Which of the following is the **only** result of prolonged smoking of tobacco cigarettes?

- A irritation of the alveoli
- B increased carbon dioxide content of inhaled air
- C emphysema
- D damage to the bronchioles

- 14 The table below shows the energy used up each day as kilojoules per kg of body mass and per square metre of body surface. Which animal has the highest rate of metabolism?

Animal	mass /kg	kJ per day	
		per kg body mass	per m ² body surface
A	125	75	4 350
B	62.5	127	4160
C	14.2	206	4115
D	0.015	2 686	4772

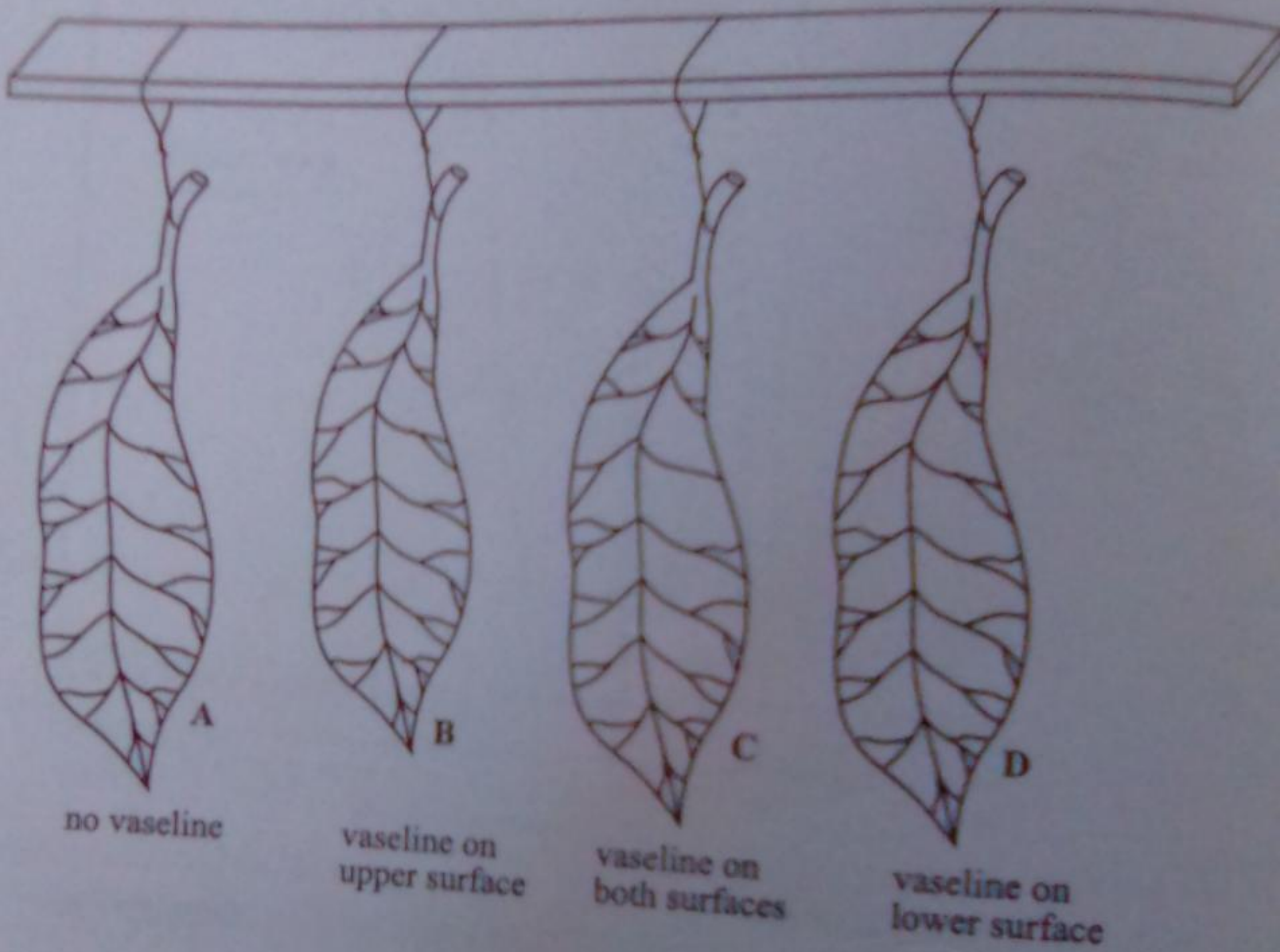
- 15 The graph shows the amount of a substance in an athlete's muscles during and immediately after a race.



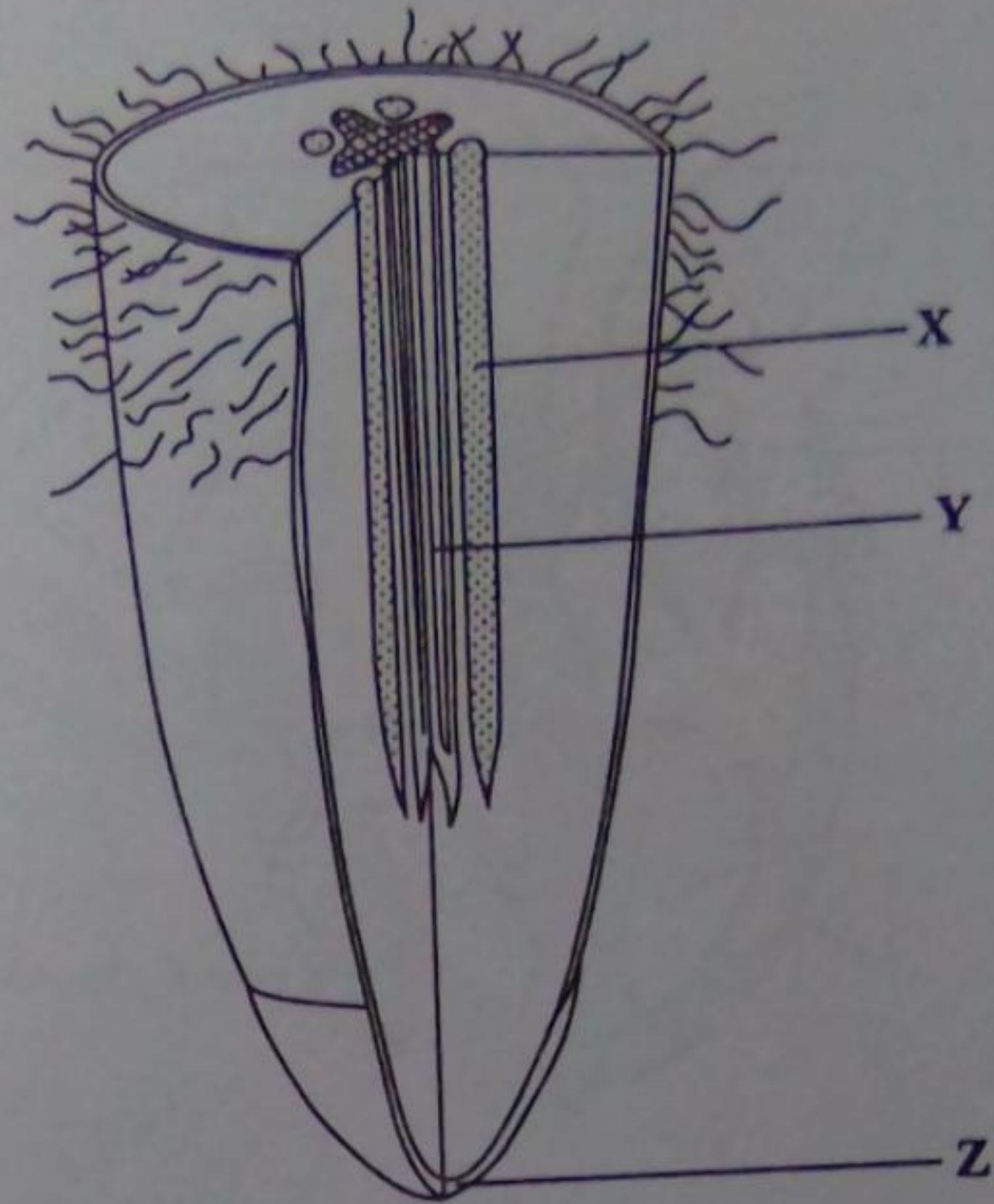
Identify the substance produced and the type of respiration involved during the race.

	substance	type of respiration
A	carbon dioxide	aerobic
B	ethanol	anaerobic
C	glucose	aerobic
D	lactic acid	anaerobic

- 16 The leaves below were left to lose water by evapo-transpiration for three days. Which leaf will dry up last?



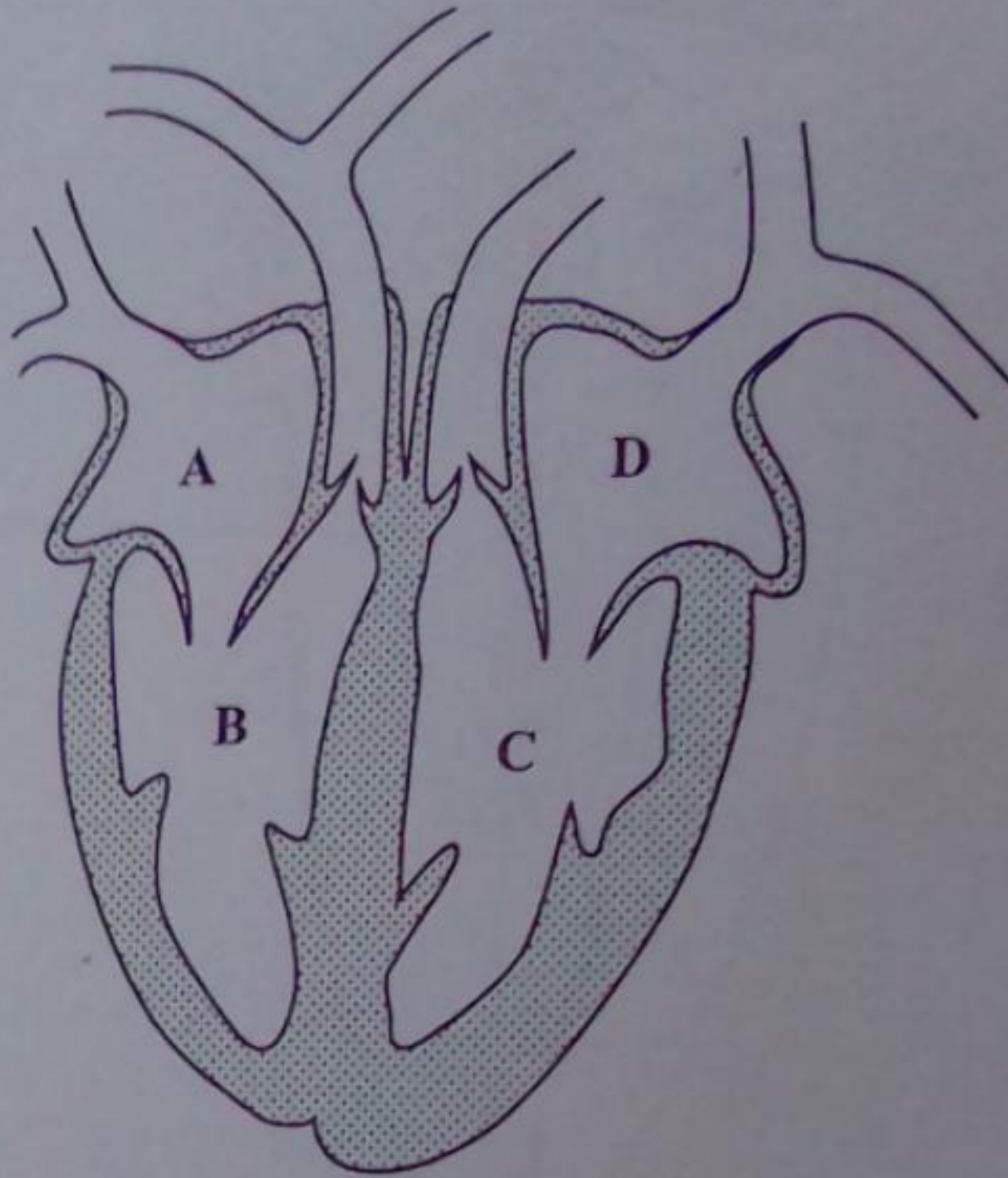
17 The diagram shows a section of a root.



What are structures X, Y and Z?

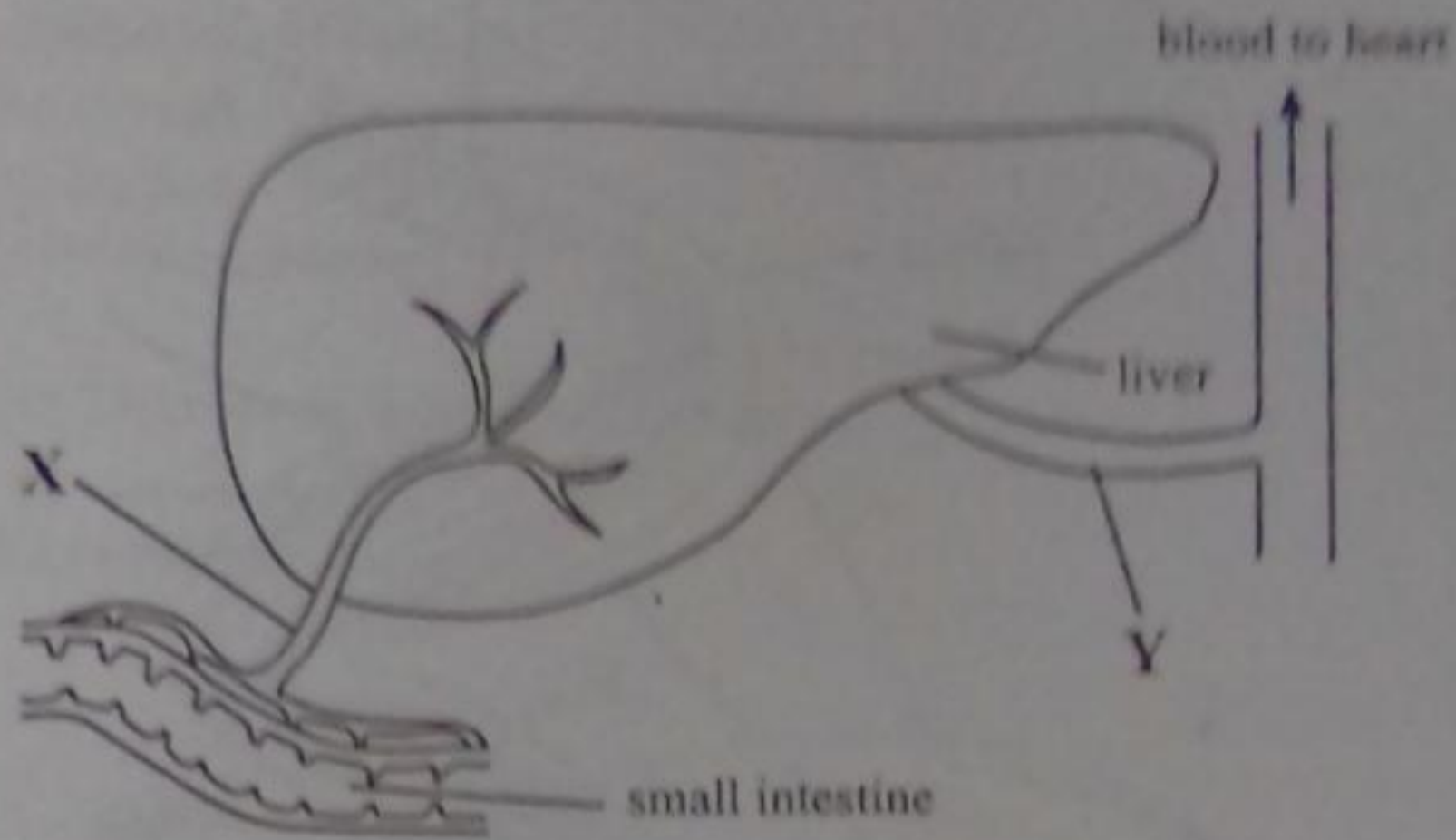
	X	Y	Z
A	pith	phloem	root cap
B	phloem	xylem	root cap
C	root cap	phloem	xylem
D	xylem	pith	phloem

- 18 The diagram shows the human heart. From which chamber does the pulmonary artery receive blood?



- 19 From which blood vessel do white blood cells escape into tissues at a wound?
- A vein
 - B capillary
 - C venule
 - D arteriole

- 20 Below is a liver and its associated blood vessels.



Identify the blood vessels **X** and **Y**.

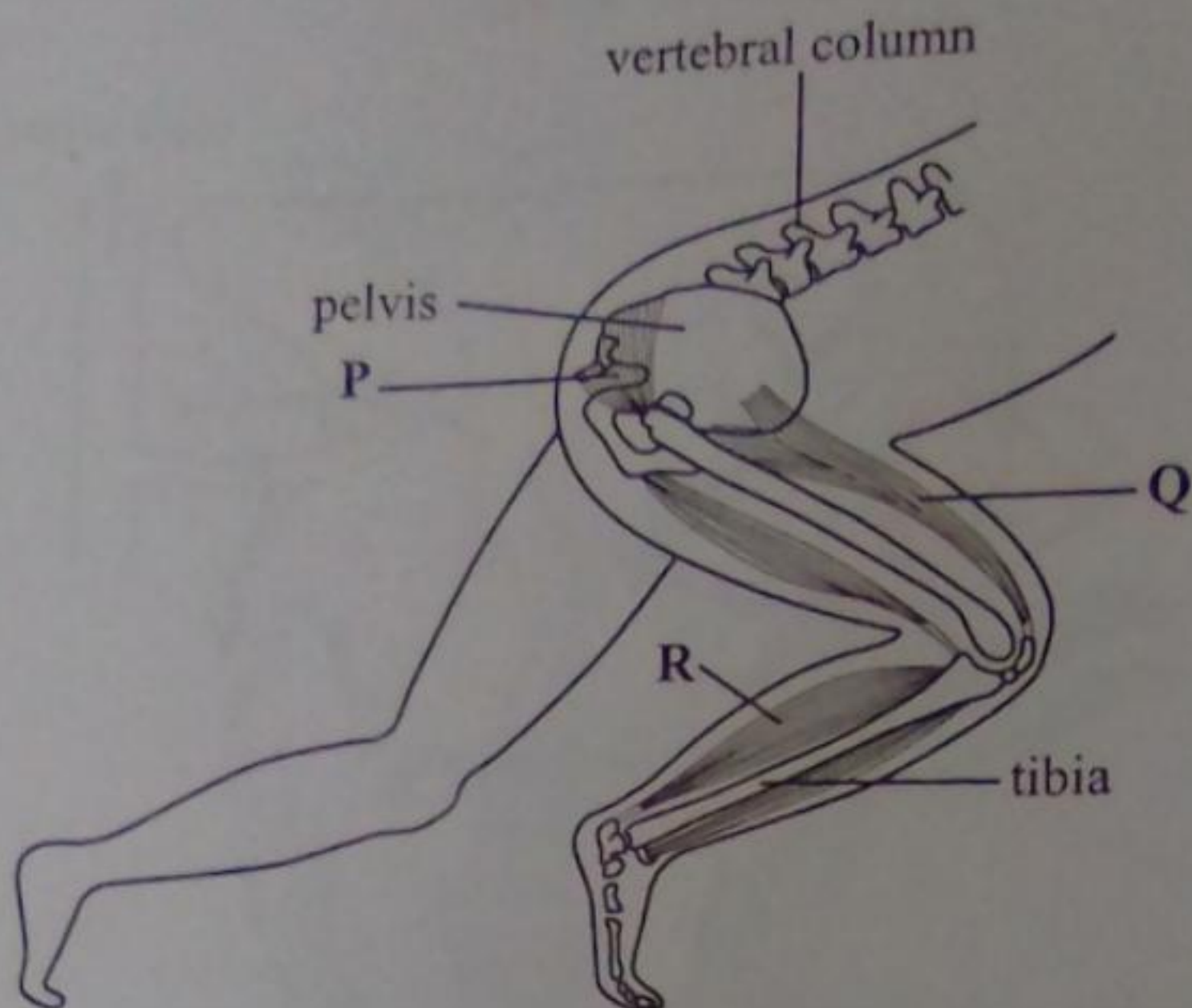
X

Y

- A** hepatic artery
- B** mesenteric artery
- C** hepatic portal vein
- D** aorta

- hepatic vein
- renal vein
- hepatic vein
- vena cava

- 21 The diagram shows muscles of the leg.



What action of muscles **P**, **Q** and **R** will result in the straightening of the leg and foot?

	P	Q	R
A	contract	relax	contract
B	relax	relax	contract
C	contract	contract	contract
D	contract	contract	relax

- 22 Which of the following is **not** present in the urine of a healthy person?

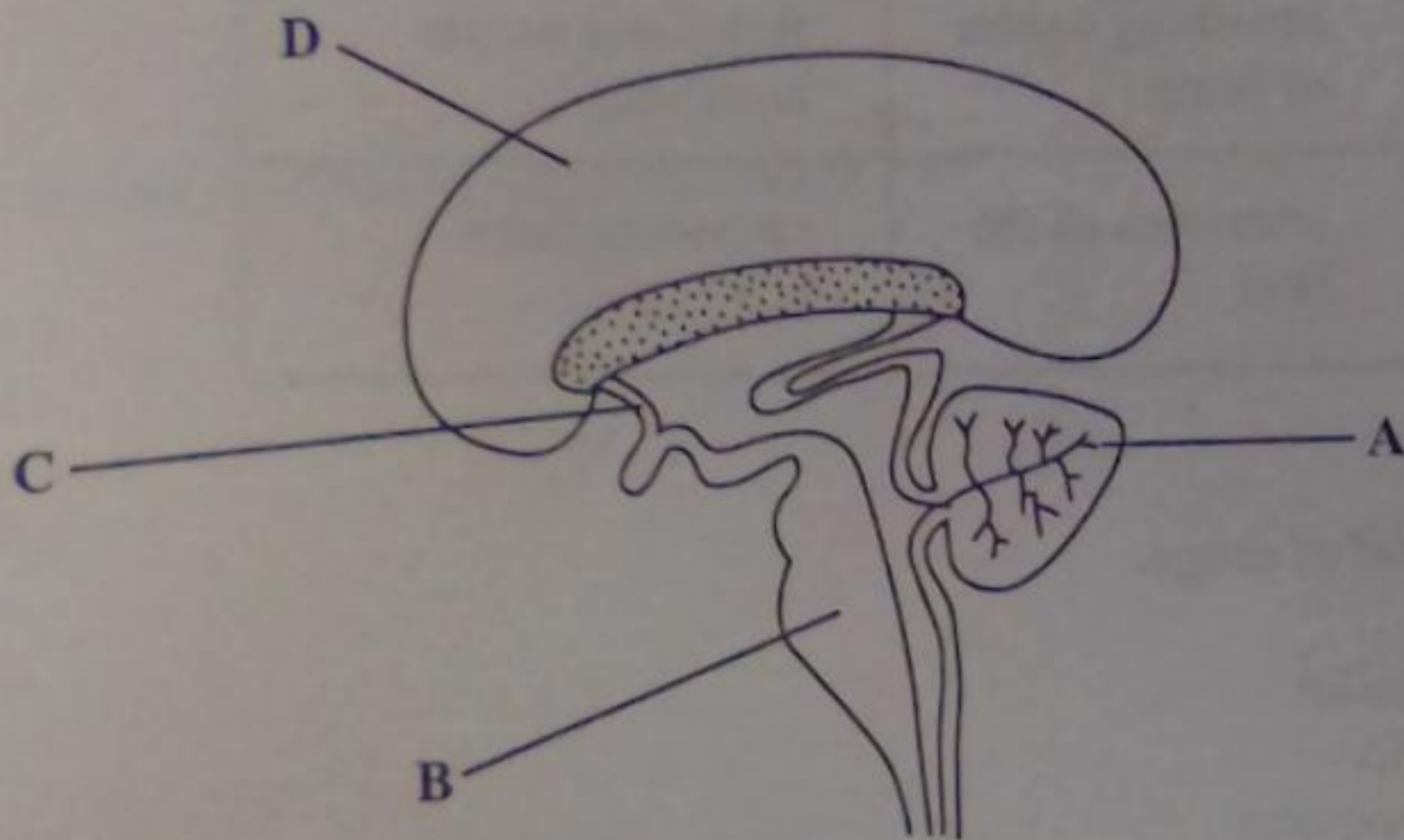
- A** urea
- B** glucose
- C** uric acid
- D** water

23 If a human body begins to lose heat faster than it can produce, which of the following does not occur?

- A Sweat production is reduced.
- B Shivering occurs.
- C Sweating begins.
- D Vasoconstriction occurs.

Questions 24 and 25 refer to the diagram below.

24 Which part is the cerebrum?



25 Which structure controls posture, balance and co-ordinated movement?

26 Coughing, mucus build-up and destruction of cilia are some effects of abuse of

- A alcohol.
- B cigarettes.
- C cannabis.
- D solvents.

27 Which of the following is **not** the correct response of the target organ to adrenaline?

	target organ	response to adrenaline
A	heart	beats faster
B	liver	conversion of glucose to glycogen
C	breathing centre of brain	faster and deeper breathing
D	arterioles of the skin	constricts them

28 Below is a list of drugs.

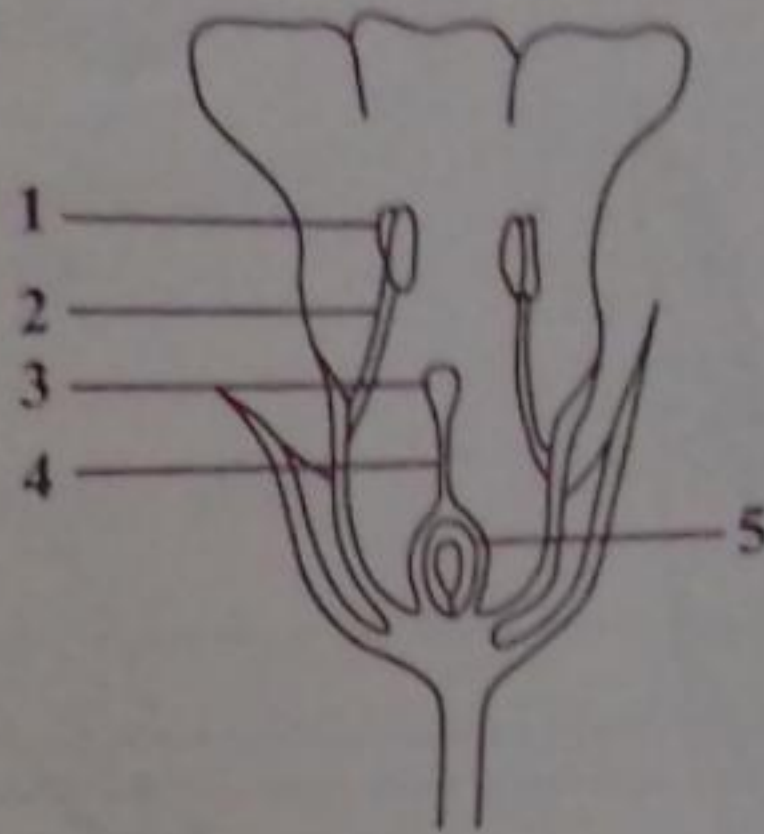
1. penicillin
2. aspirin
3. chloroquine
4. paludrine
5. paracetamol

Which drugs are analgesics?

- A** 1, 2 and 5.
B 2 and 4 only.
C 2 and 5 only.
D 4 and 5 only.

29

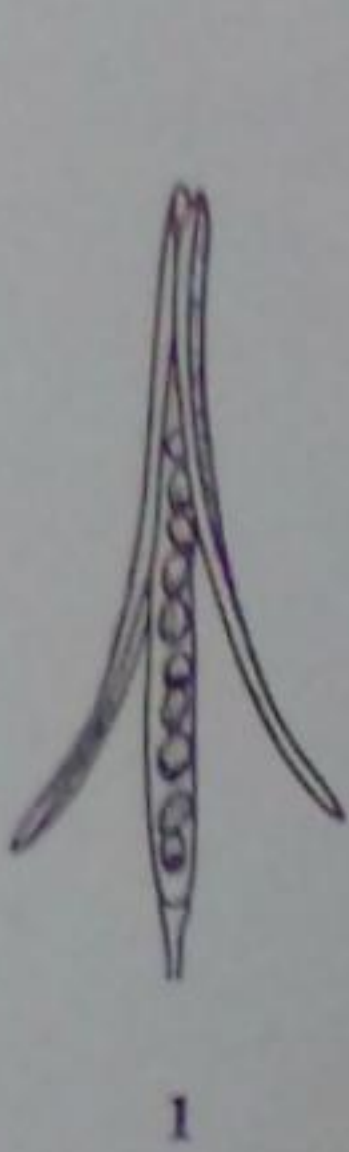
The diagram shows a half flower.



Which structures are involved in self pollination?

- A 1 and 2
- B 1 and 3
- C 2 and 4
- D 3 and 5

30 Which of the following fruits are dispersed by wind?



- A 1, 3 and 4
- B 1, 2 and 3
- C 1 and 3 only
- D 2 and 3 only

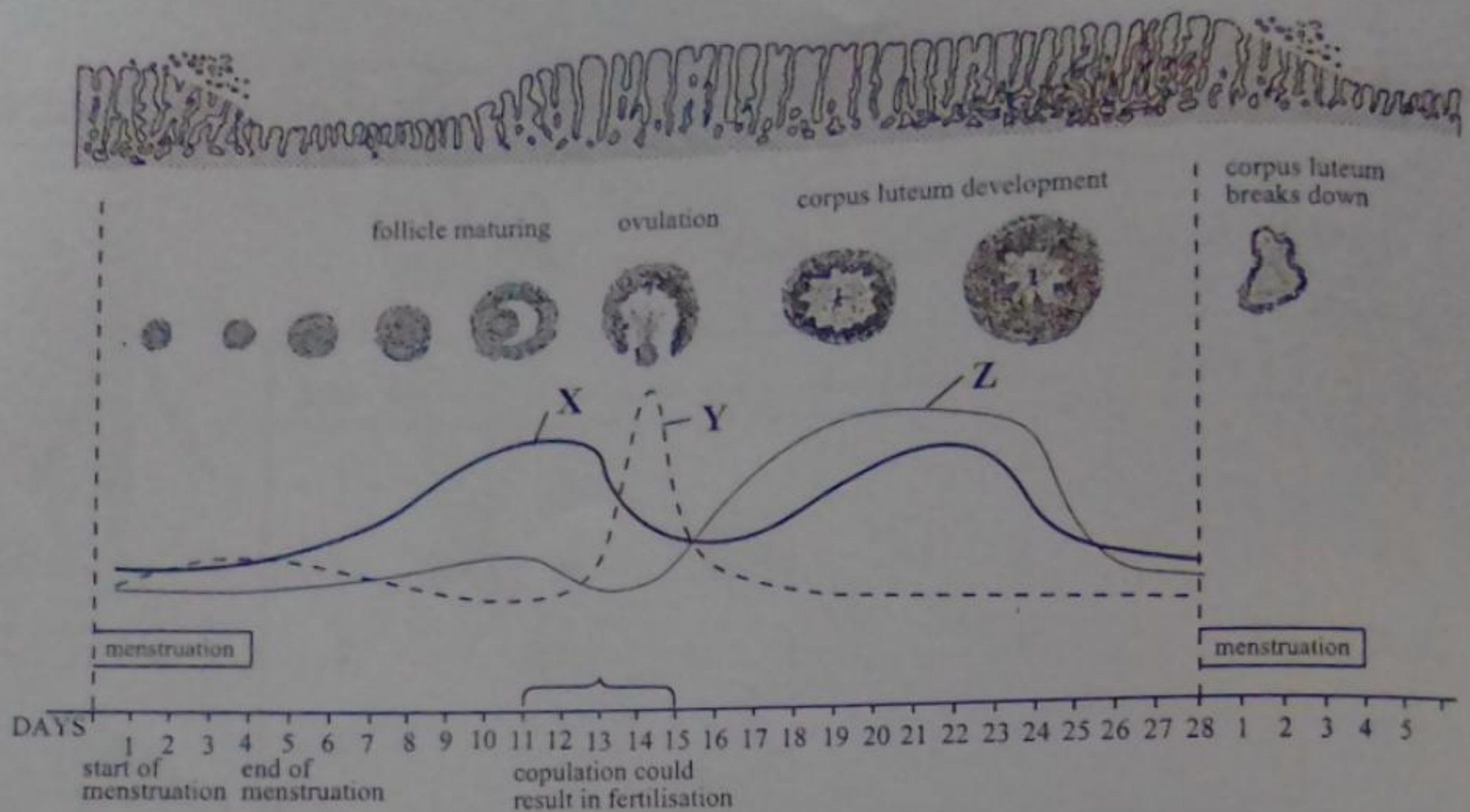
- 31 Some of the events which occur during pollination are given below.

1. bee gets dusted with pollen
2. pollen is deposited on stigma
3. bee visits flower with mature pistil
4. bee visits younger flower
5. anthers split open

In which order do these events take place?

	First  last				
A	3	4	5	1	2
B	3	5	1	4	2
C	5	4	1	3	2
D	5	4	3	1	2

The diagram below shows a human menstrual cycle.



Identify the hormones represented by graphs X, Y and Z.

	X	Y	Z
A	luteinising hormone	oestrogens	progesterone
B	oestrogens	luteinising hormone	progesterone
C	progesterone	oestrogens	luteinising hormone
D	luteinising hormone	progesterone	oestrogens

33 Which of the following statements are true about gonorrhoea?

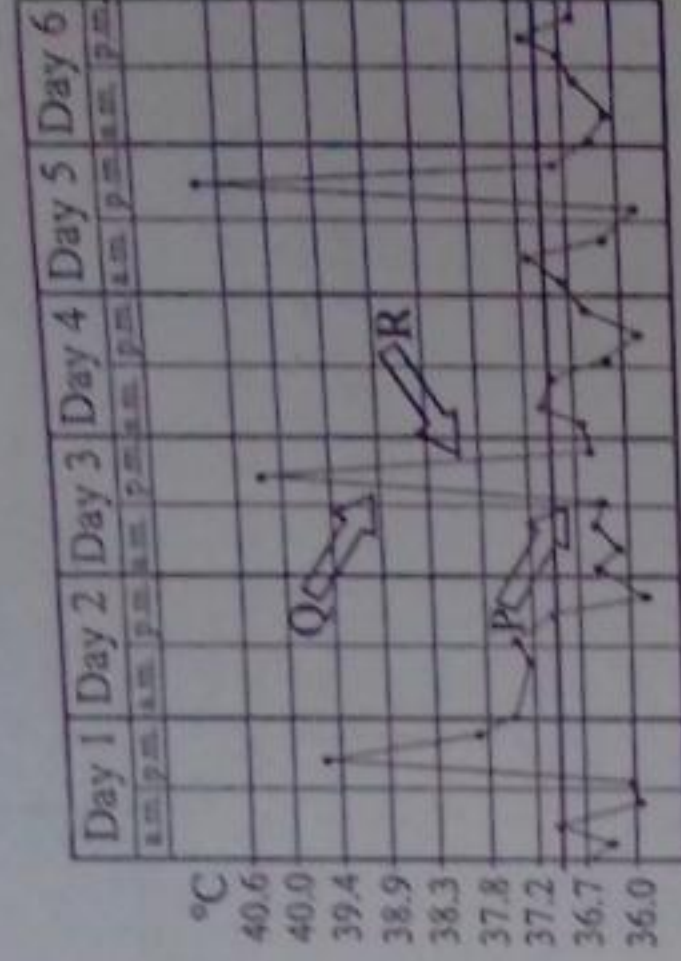
1. It is caused by a bacterium.
2. It leads to blockage of the urethra and to sterility in women.
3. In men, pain and discharge of pus occurs from the urethra.
4. It can be cured with penicillin.

- A 1 and 2 only
B 1, 3 and 4
C 2, 3 and 4
D 3 and 4 only

34 Which is a characteristic of fungi?

- A They reproduce sexually only.
B They have chlorophyll.
C They photosynthesize.
D They are saprophytes.

- 35 A temperature graph of a patient with malaria is shown below.



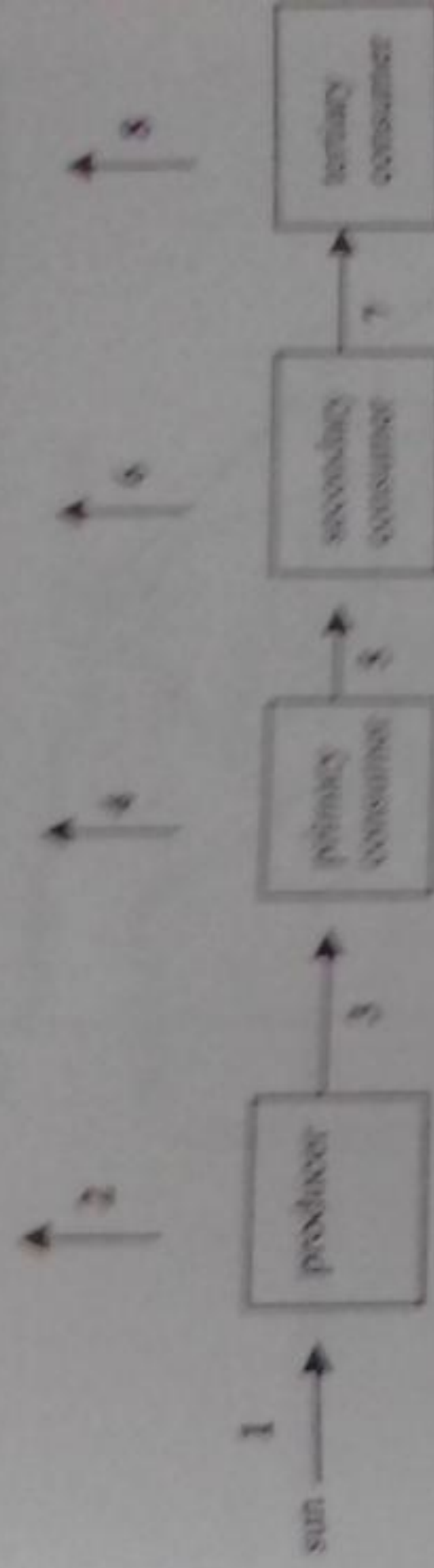
Identify the stages P, Q and R of the disease cycle.

	P	Q	R
A	sweating stage	hot stage	cold stage
B	cold stage	hot stage	hot stage
C	cold stage	hot stage	sweating stage
D	sweating stage	hot stage	sweating stage

- 36 Which statement is true about micro-organisms?

- A They are multicellular.
- B They are all parasitic.
- C They can only reproduce inside host cells.
- D They include viruses, bacteria and some fungi.

The diagram shows the flow of energy through an ecosystem.



Which arrows represent the smallest amount of energy lost to the ecosystem and the highest amount of energy transferred between organisms?

smallest amount
of energy lost

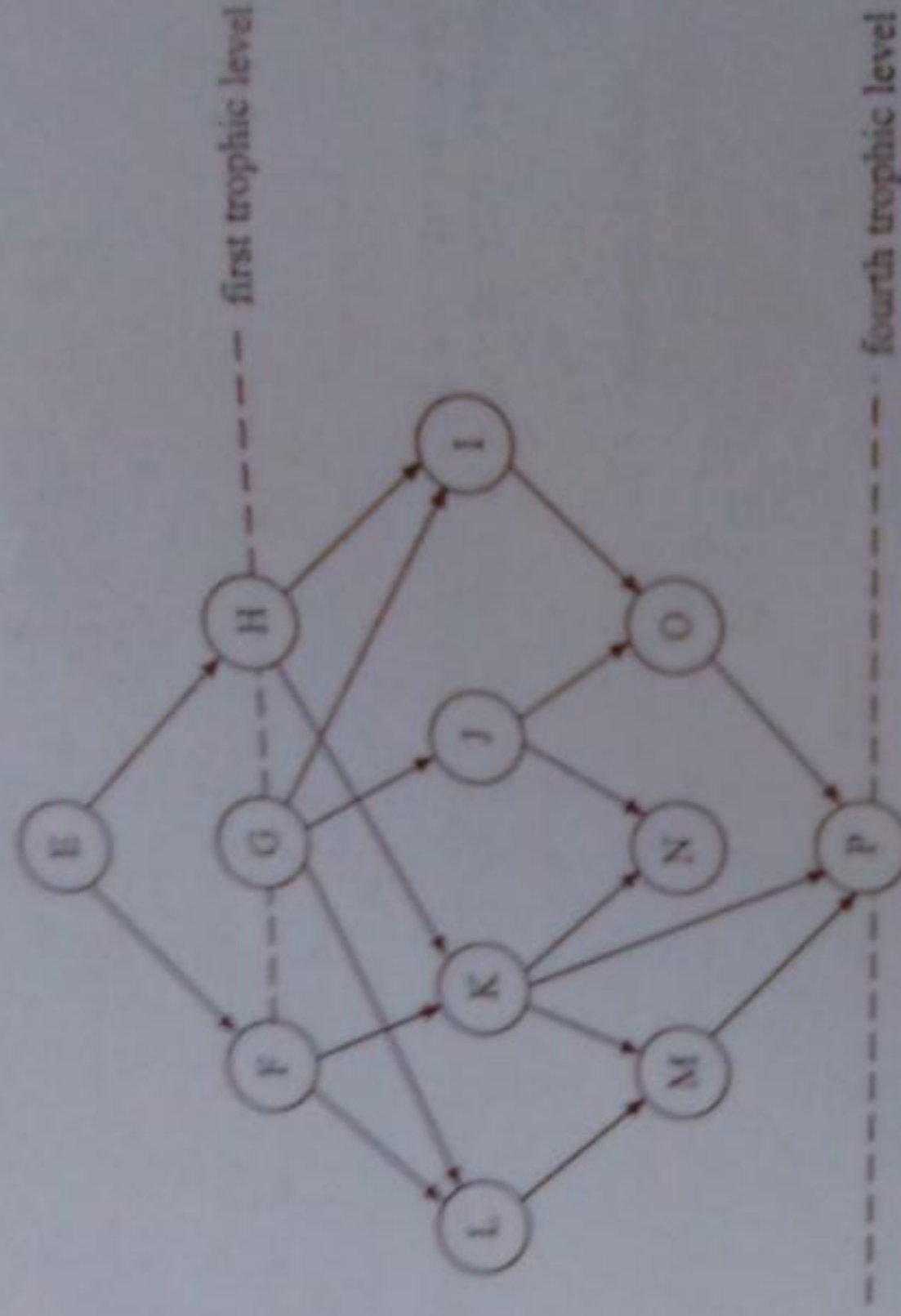
highest amount of
energy transfer

A
B
C
D

8
8
7
7

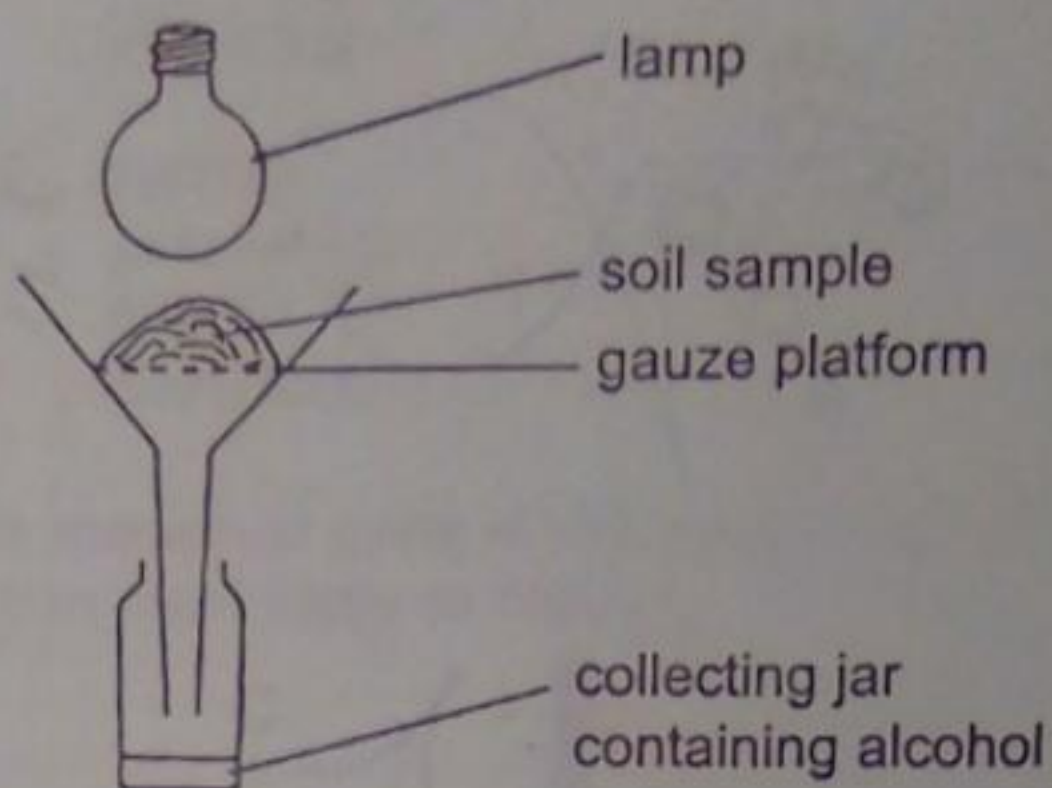
2
4
6
8

Questions 38 and 39 refer to the food web below.



- 38 P is likely to be
- the sun.
 - a secondary consumer.
 - a decomposer.
 - a producer.
- 39 What will happen if species J is suddenly affected by a disease and dies?
- The population of I and G will decrease.
 - The population of H will decrease.
 - The population of N and O will decrease.
 - The population of I will decrease.

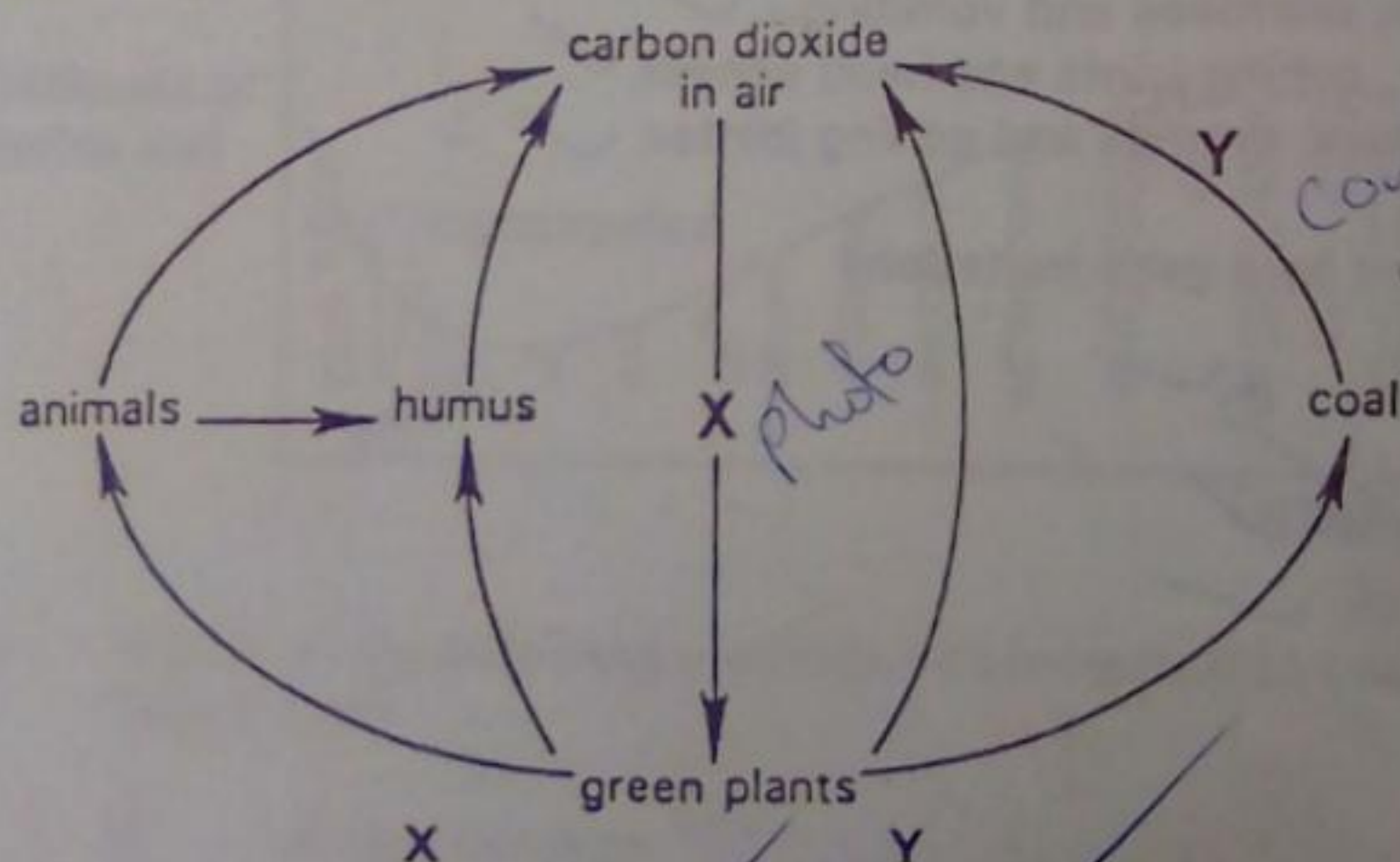
38. The diagram shows the apparatus for the extraction of micro-organisms from soil samples.



Why may only a few organisms be collected?

- A No light penetrates the soil. ☒ ☒
 B The heat applied kills the organisms. ☒ ☒
 C The gauze size is too small. ☒ ☒
 D The organisms escape from the container of alcohol. ☒ ☒

39. The chart shows the carbon cycle. Name the processes labelled X and Y.



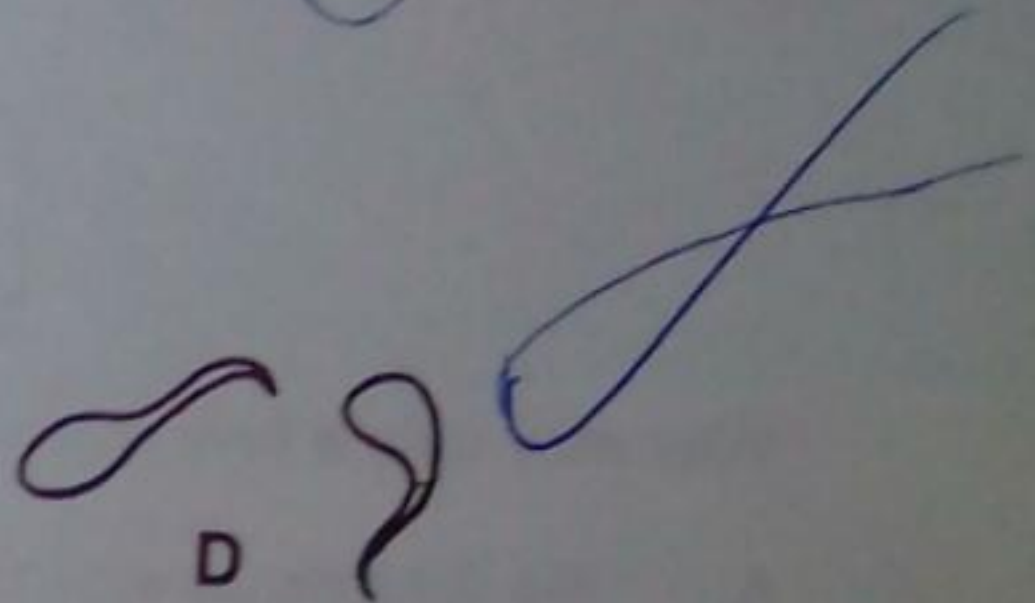
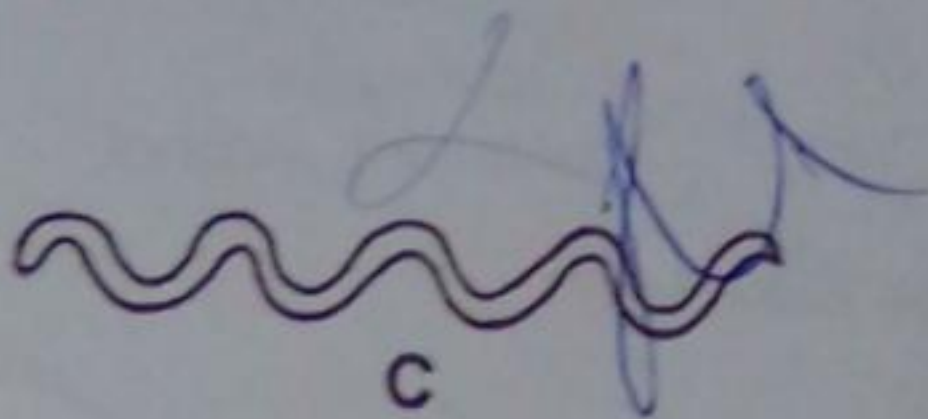
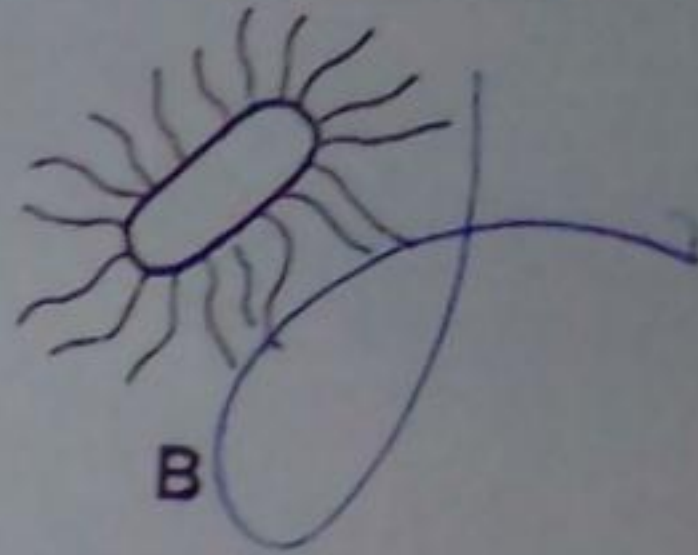
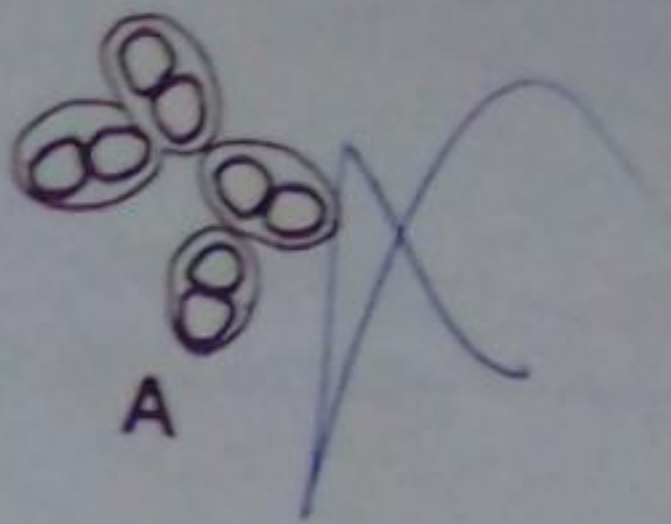
- A photosynthesis ☒ combustion ☒
 B decay ☒ respiration ☒
 C photosynthesis ☒ decay ☒
 D respiration ☒ combustion ☒

40. How long will it take for the population of a country to double if the growth rate is 2%? ☒

- A 35 years ☒
 B 17 years ☒
 C 68 years ☒
 D 140 years ☒

$$\frac{2}{100} \times \frac{5}{1}$$

The diagrams show various shapes of bacteria. Which type of bacteria causes cholera?



36. Which are the signs and symptoms of cholera?

- A Vomiting, diarrhoea, fever and cramps. ✓
- B Fever, headache, diarrhoea and vomiting. ✓
- C Diarrhoea, fever, aching joints and head aches. ✓
- D Vomiting, diarrhoea, cramps and aching joints. ✓

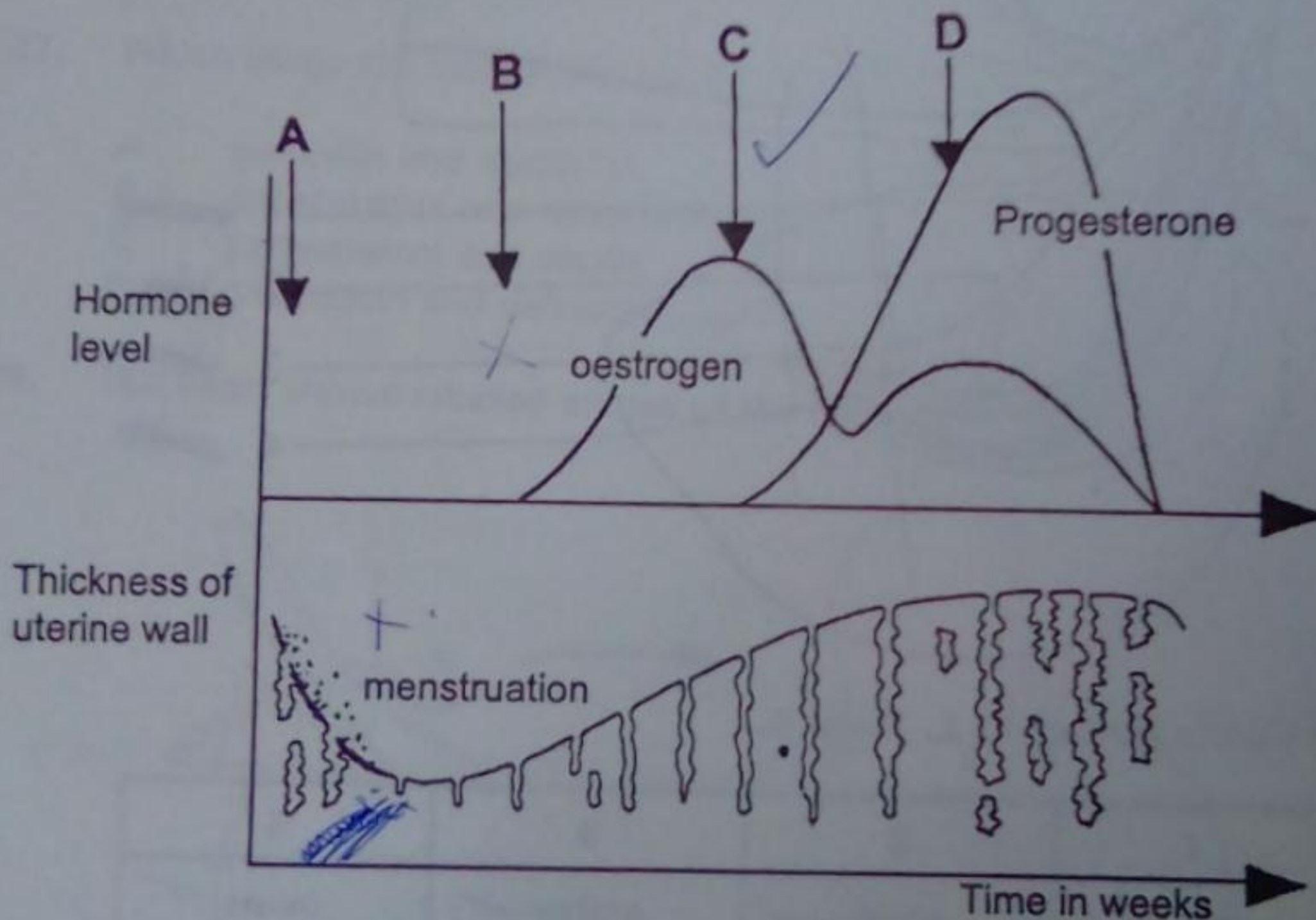
37. Which disease is caused by a gene mutation?

- A Down's syndrome ✓
- B HIV/AIDS ✓
- C sickle-cell anaemia ✓
- D goitre ✓

32. Which sexually transmitted diseases cause ulcers and discharge?

	Ulcers	Discharge
A	chancroid ✓	syphilis ✓
B	syphilis ✓	gonorrhoea ✓
C	gonorrhoea ✓	AIDS ✓
D	AIDS ✓	chancroid ✓

33. The diagram shows the menstrual cycle in the human female. At what point is ovulation most likely to occur?



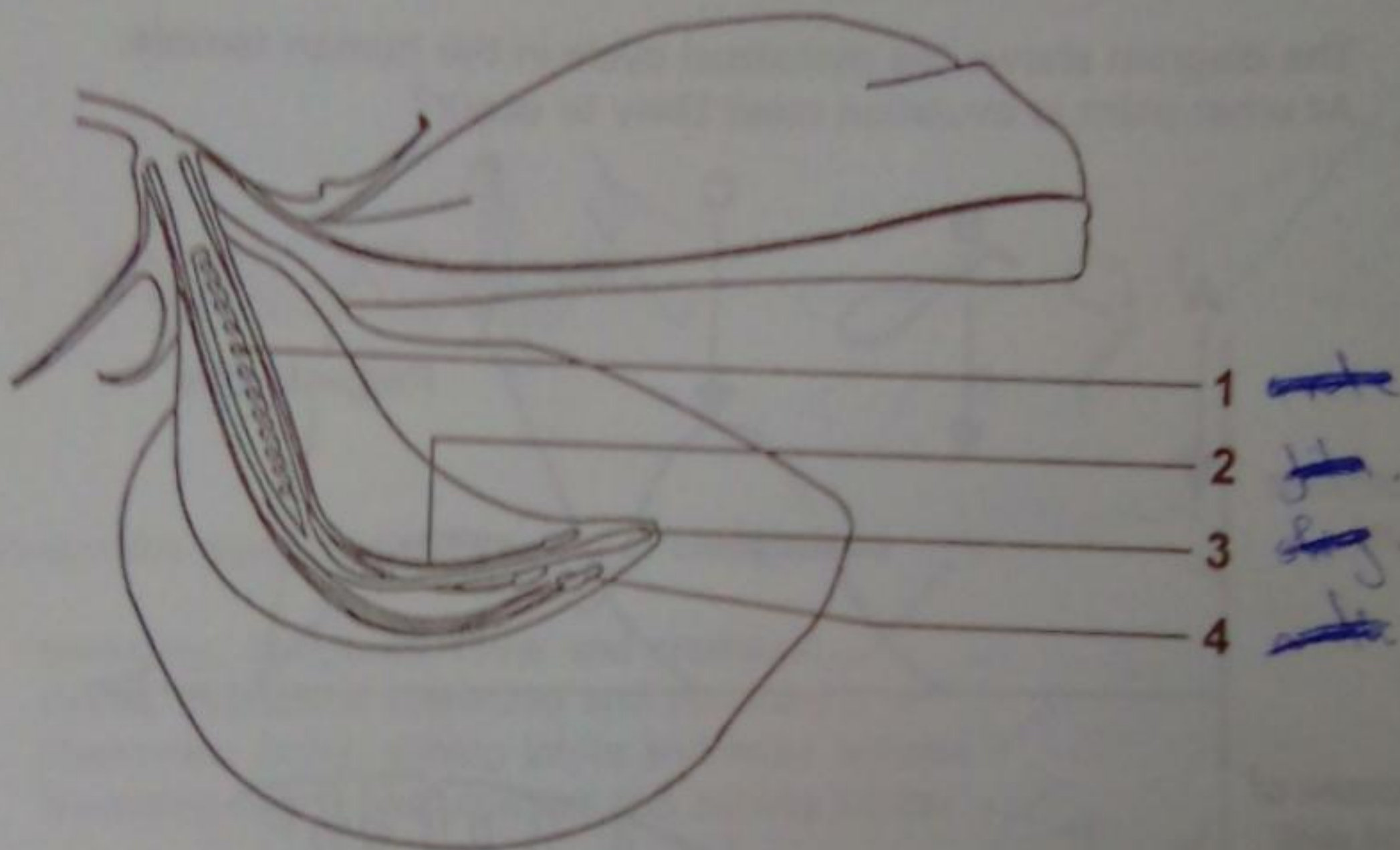
34. Which of the following methods of contraception has the highest failure rate?

- A using a condom ✓
- B rythm ✓
- C vasectomy ✓
- D the pill ✓

What is the advantage of asexual reproduction to a citrus farmer?

- A Gametes are required. ☒
- B New genotypes are produced. ☒
- C The offspring will be the same as parents. ☒
- D New characteristics will arise. ☒

30. The diagram shows an insect pollinated flower.



Identify the parts labelled 1, 2, 3 and 4.

	1	2	3	4
A	stigma ☒	style ☒	anther ☒	ovule ☒
B	stigma ☒	filament ☒	ovary ☒	anther ☒
C	style ☒	stigma ☒	ovule ☒	anther ☒
D	ovule ☒	filament ☒	stigma ☒	anther ☒

31. Which of the following plants can be propagated by suckers?

- A sweet potato ☒
- B banana ☒
- C sugar cane ☒
- D lawn grass ☒

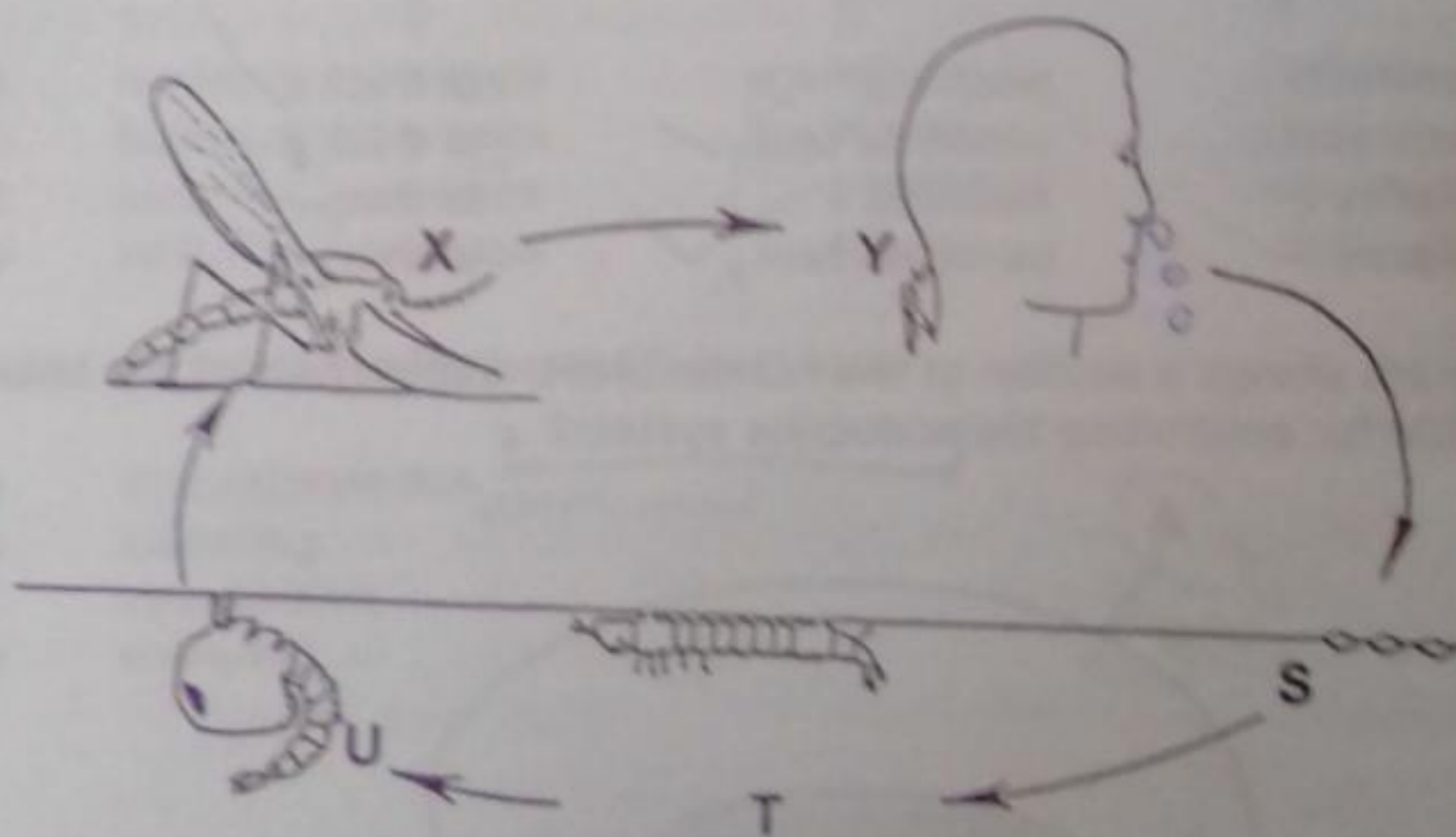
26. Which set matches hormones and their functions?

	Regulate glucose level	Increase heart beat	Ovulation
A	adrenaline	progesterone	insulin
B	oestrogen	insulin	adrenaline
C	insulin	adrenaline	oestrogen
D	glucagon	oestrogen	progesterone

27. Which drugs are used to kill malaria parasites?

- A penicillin and aspirin
 B paracetamol and chloroquin
 C paracetamol and aspirin
 D chloroquin and deltaprim

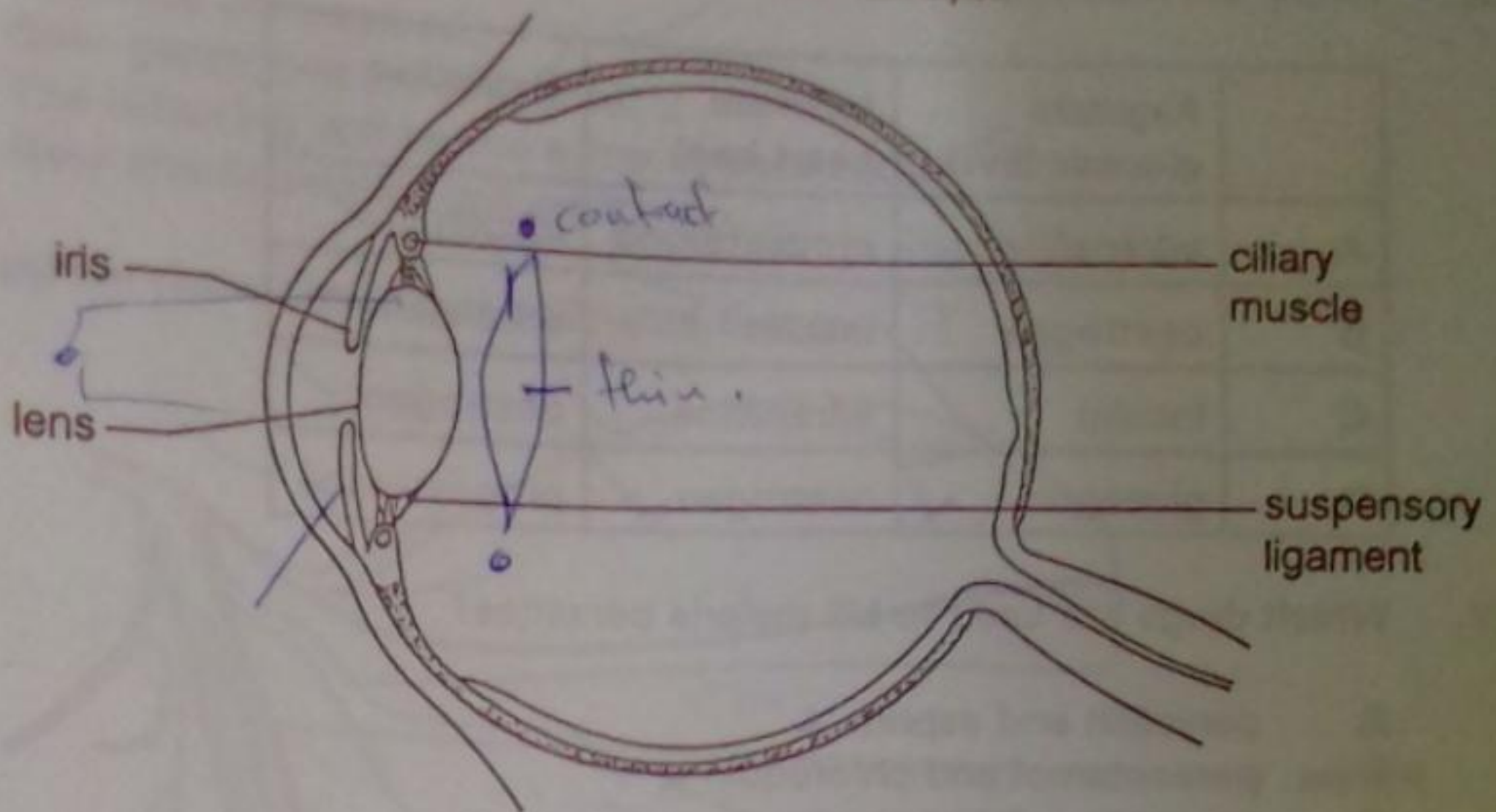
28. The chart shows labelled stages of the life cycle of the mosquito.



Which stages of the life cycle are controlled by the methods listed below?

Method of control	Spraying walls	Draining water	Use of small fish	Using insect repellent
A	X	Y	U	S
B	Y	U	S	T
C	U	Y	T	X
D	X	S	T	Y

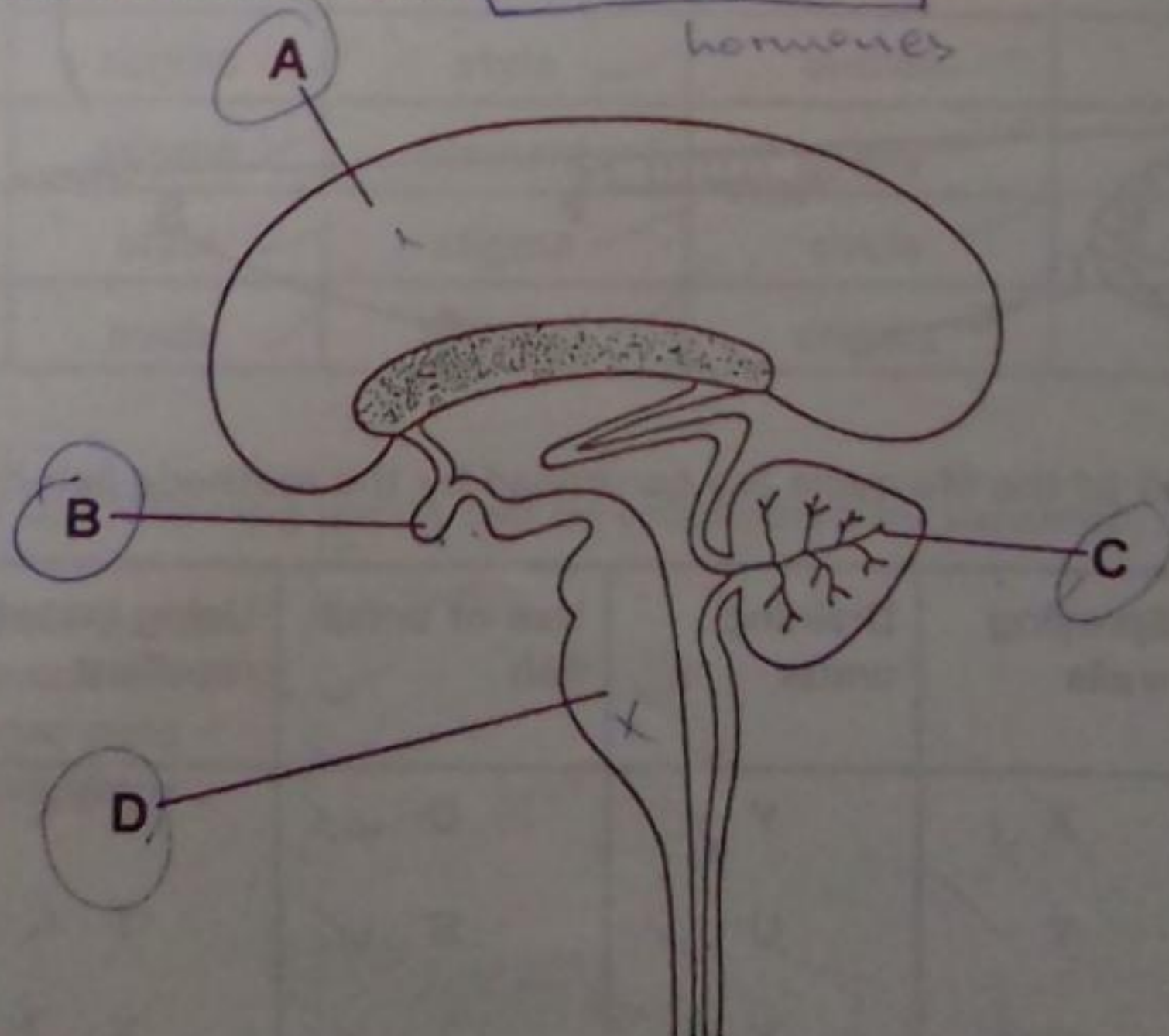
shows a section through the human eye.



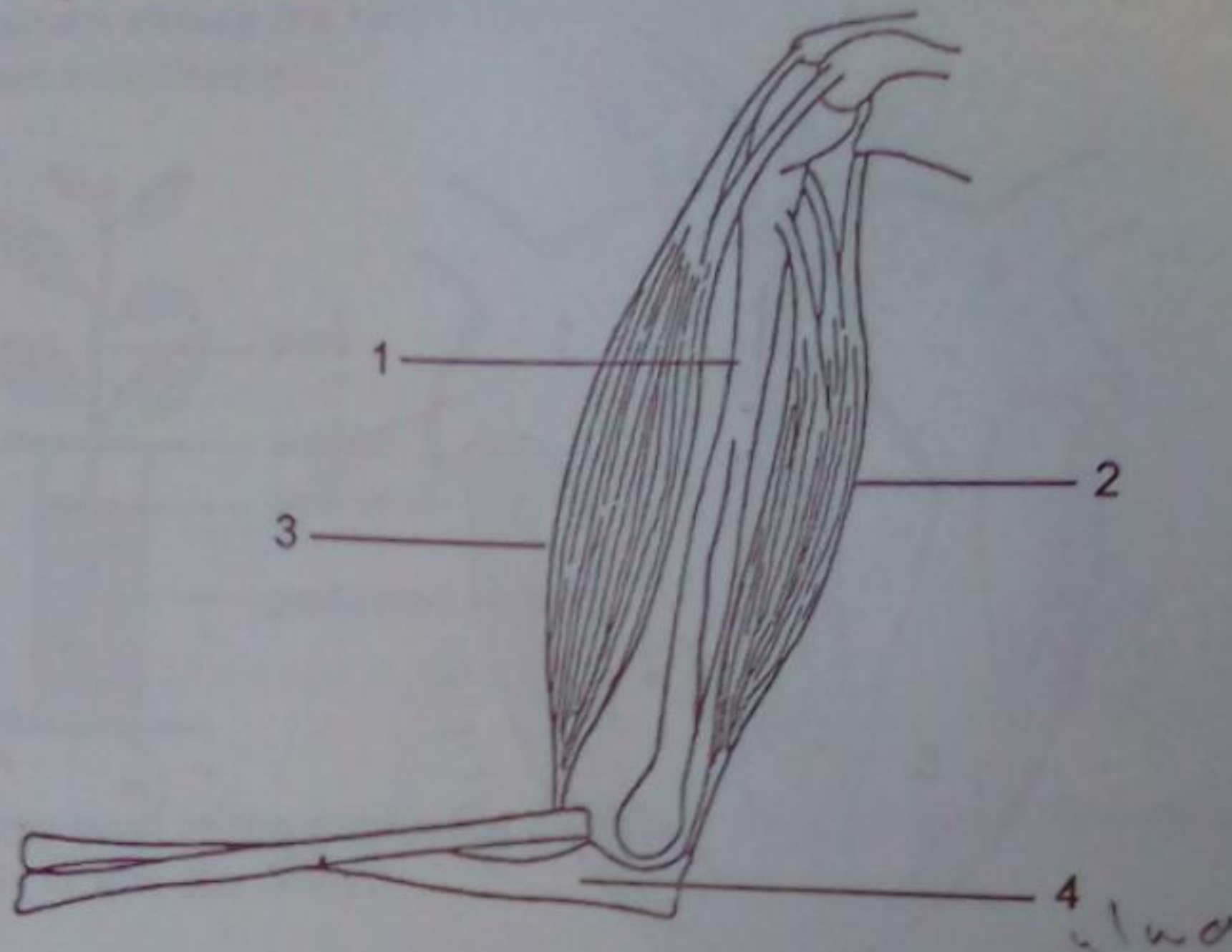
What happens in order for the eye to focus on an object that is near?

	ciliary muscle	suspensory ligament	lens
A	contracts ☒	slackens ☒	is thick ☒
B	contracts ☒	becomes taut ☒	is thick ☒
C	relaxes ☒	slackens ☒	is thin ☒
D	relaxes ☒	becomes taut ☒	is thin ☒

25. The diagram shows a section of the human brain. Which part of the brain is responsible for controlling the endocrine system?



22. The diagram shows the bones and muscles of the fore limb.



Name the parts labelled 1, 2, 3 and 4.

	1	2	3	4
A	humerus	triceps	biceps	radius
B	humerus	triceps	biceps	ulna
C	ulna	biceps	triceps	radius
D	radius	biceps	triceps	ulna

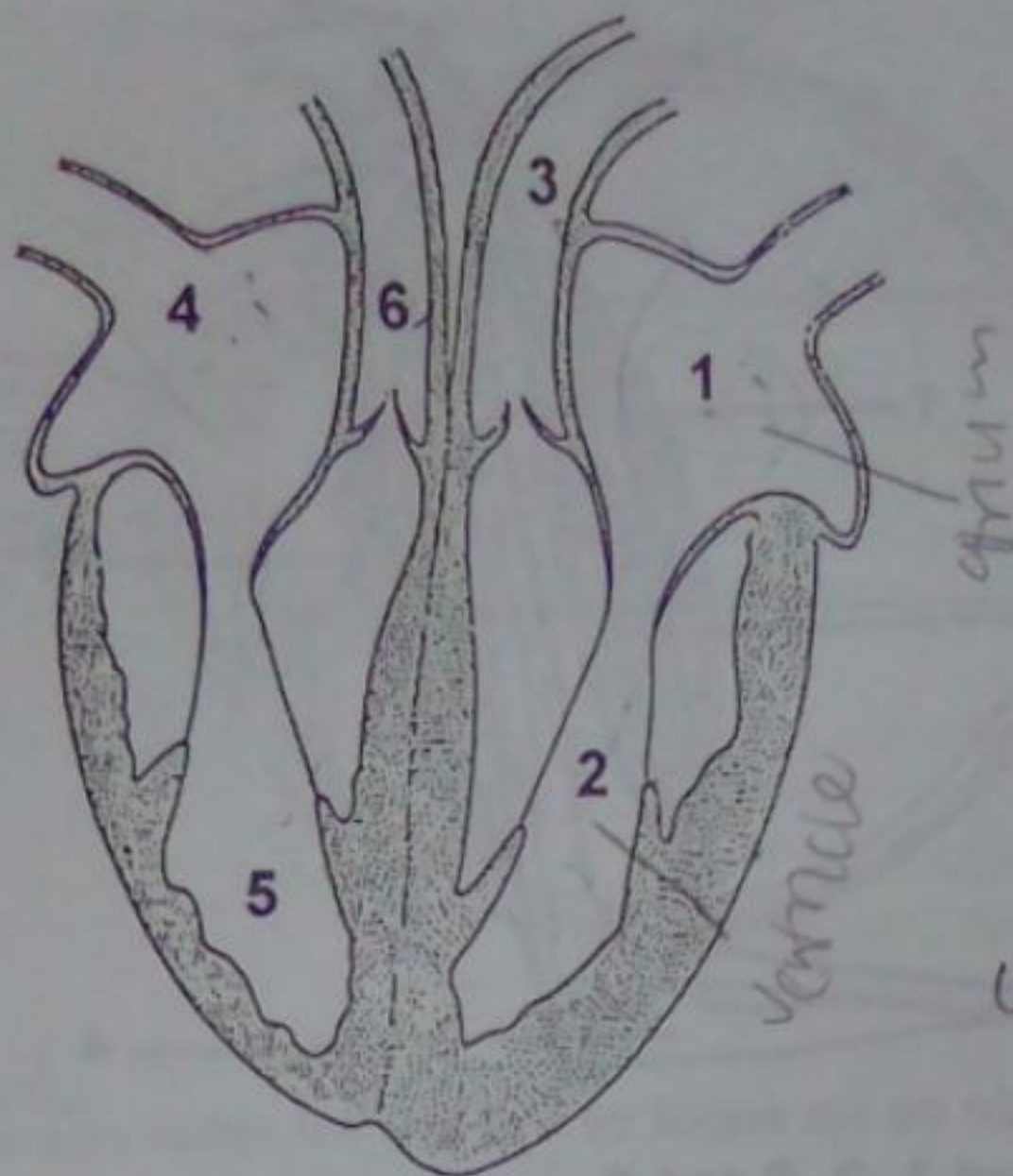
23. Which of the following prevents heat loss?

A	vasoconstriction
B	sweating
C	vasodilation
D	shivering

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4 violet
G E D Z A
H V R O R E T
20000 ©

20. The diagram shows a section of the human heart.



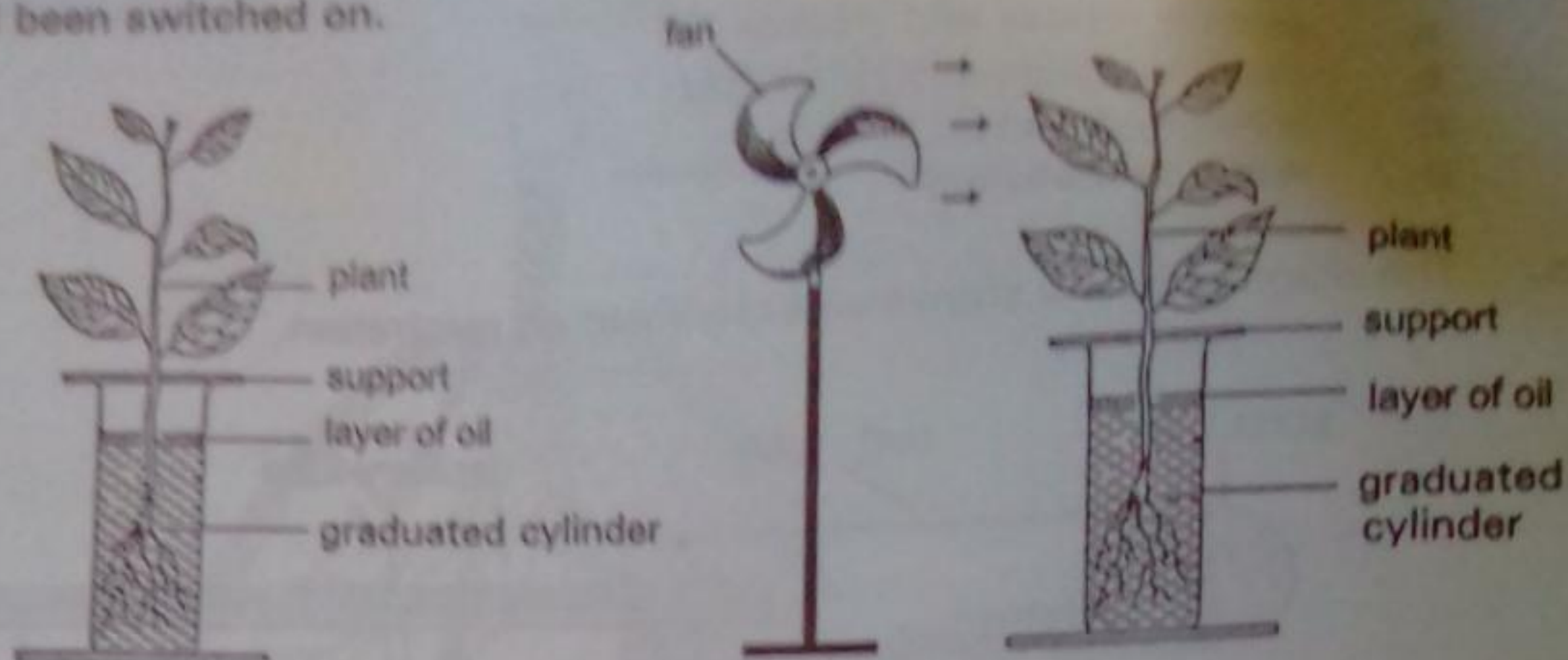
Which is the correct path followed by blood coming from the head entering the heart until it finally leaves the heart to return to the head?

- A 4 → 5 → 6 → 3 → 2 → 1
 B 6 → 5 → 4 → 1 → 2 → 3
 C 4 → 5 → 6 → 1 → 2 → 3
 D 1 → 2 → 3 → 6 → 5 → 4

21. What happens to atria, ventricles and semilunar valves during the time when blood is pumped out of the heart?

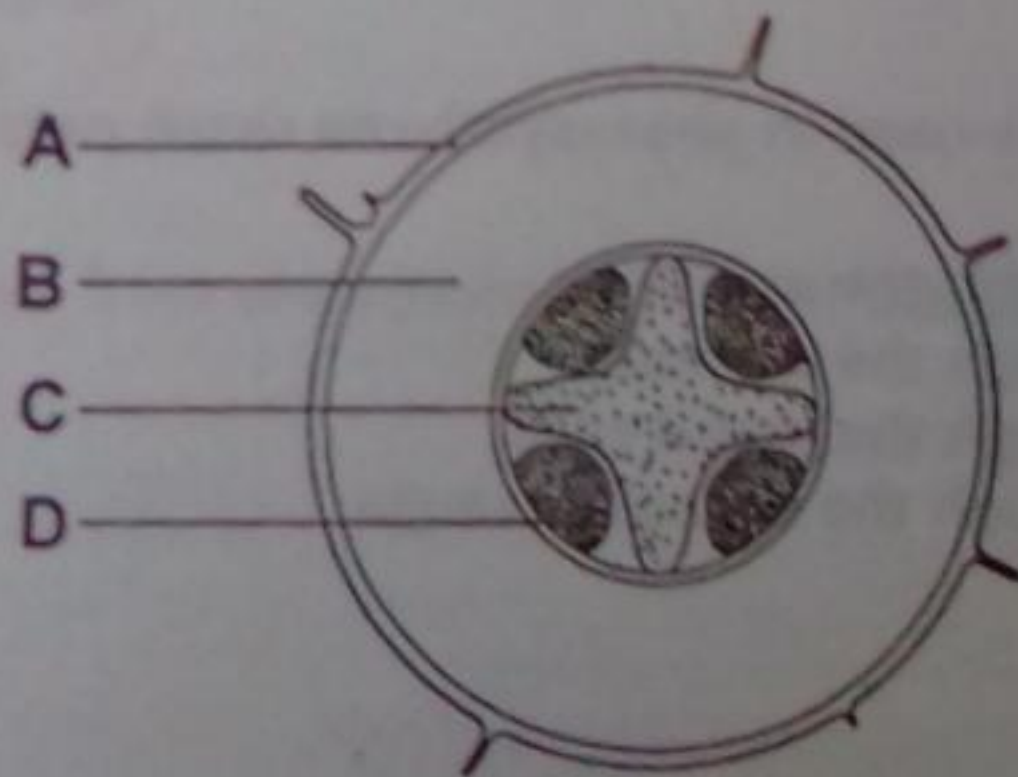
	Atria	Ventricles	Semi lunar valves
A	<u>relax</u>	<u>contract</u>	<u>open</u>
B	contract	relax	close
C	relax	relax	open
D	contract	<u>contract</u>	close

17. The diagram shows the same plant set up in still air and then later near a fan that had been switched on.



The water level in the graduated cylinder fell faster when the plant was near the fan than in still air. Why?

- A The fan caused evaporation of water from the cylinder.
 - B The moving air caused more evaporation of water from the plant.
 - C The temperature of the air decreased.
 - D The humidity of still air was low.
18. The diagram shows a section of a root of a green plant that had been placed in a red dye for a few hours. In which tissue will the dye appear?



19. Which of the following are functions of blood?

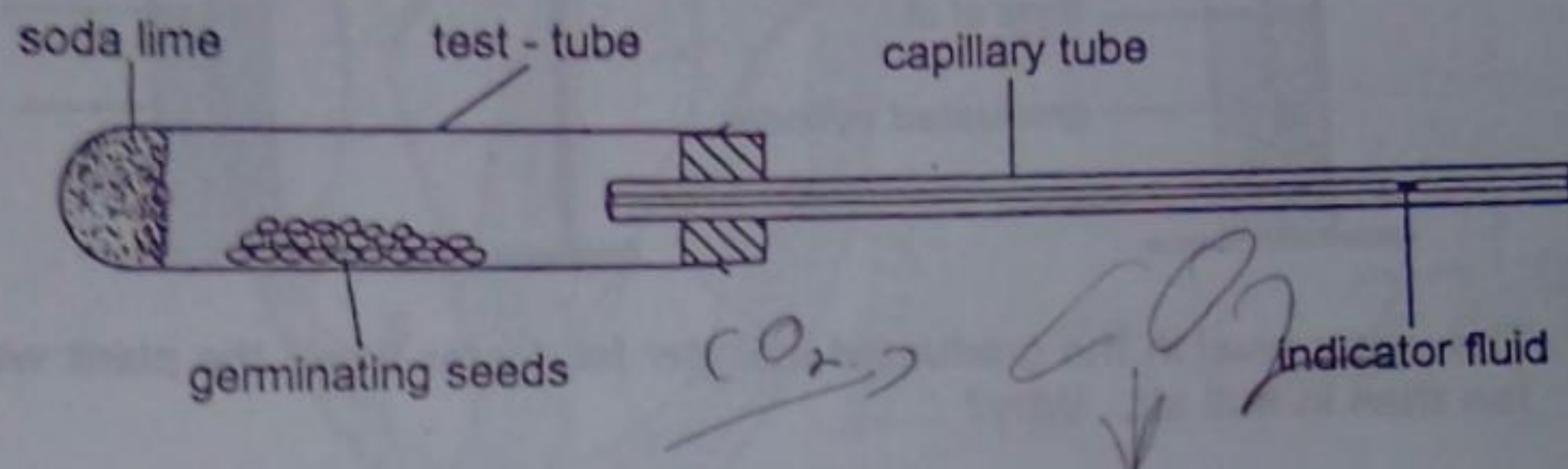
- 1. transport of oxygen and carbon dioxide ✓
- 2. production of hormones
- 3. distribution of heat round the body ✓
- 4. immunity to the body. ✓

- A 2 and 4 ✓
- B 3 and 4
- C 1, 2 and 3 x
- D 1, 3 and 4 ✓

14. What are the products of anaerobic respiration in yeast cells?

- A carbon dioxide and alcohol
- B carbon dioxide and lactic acid
- C water and carbon dioxide
- D water and alcohol

15. The diagram below shows an experiment on respiration.



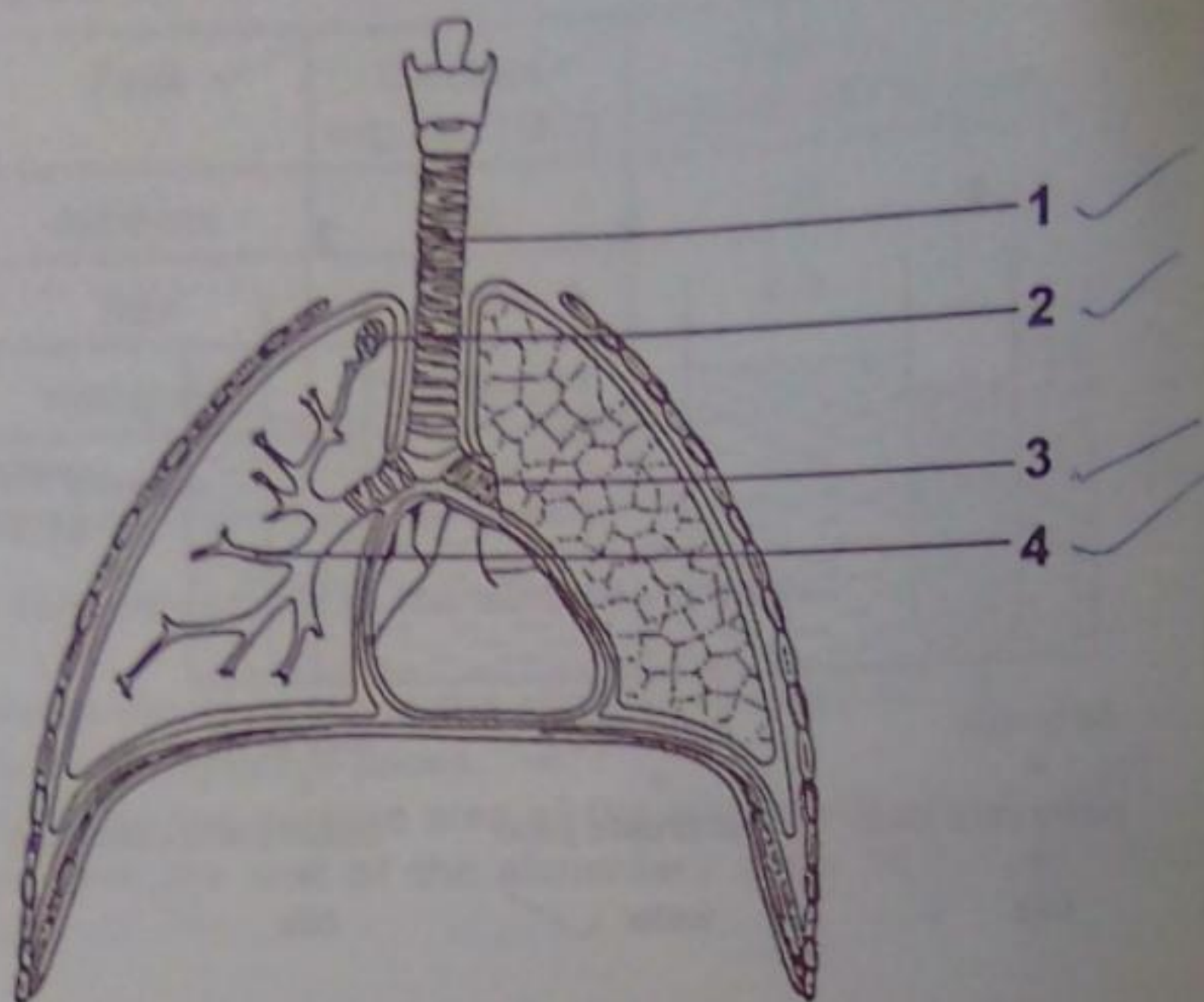
What will happen to the indicator fluid after some time?

- A It moves away from the test tube because of oxygen output by the seeds.
- B It does not move because carbon dioxide intake and oxygen output are equal.
- C It moves towards the test tube because of oxygen intake by the seeds.
- D It moves towards the test tube because of carbon dioxide intake by the seeds.

16. How does carbon dioxide used in photosynthesis reach palisade cells in a leaf?

- A By diffusion through the epidermis.
- B By diffusion through the stomata.
- C By diffusion through the guard cells.
- D By diffusion through the vascular bundle.

Questions 11 and 12 refer to the diagram of a human lung.



11. What is the part labelled 3?

- A trachea ✓
- B bronchiole ✓
- C bronchus ✓
- D alveoli ✓

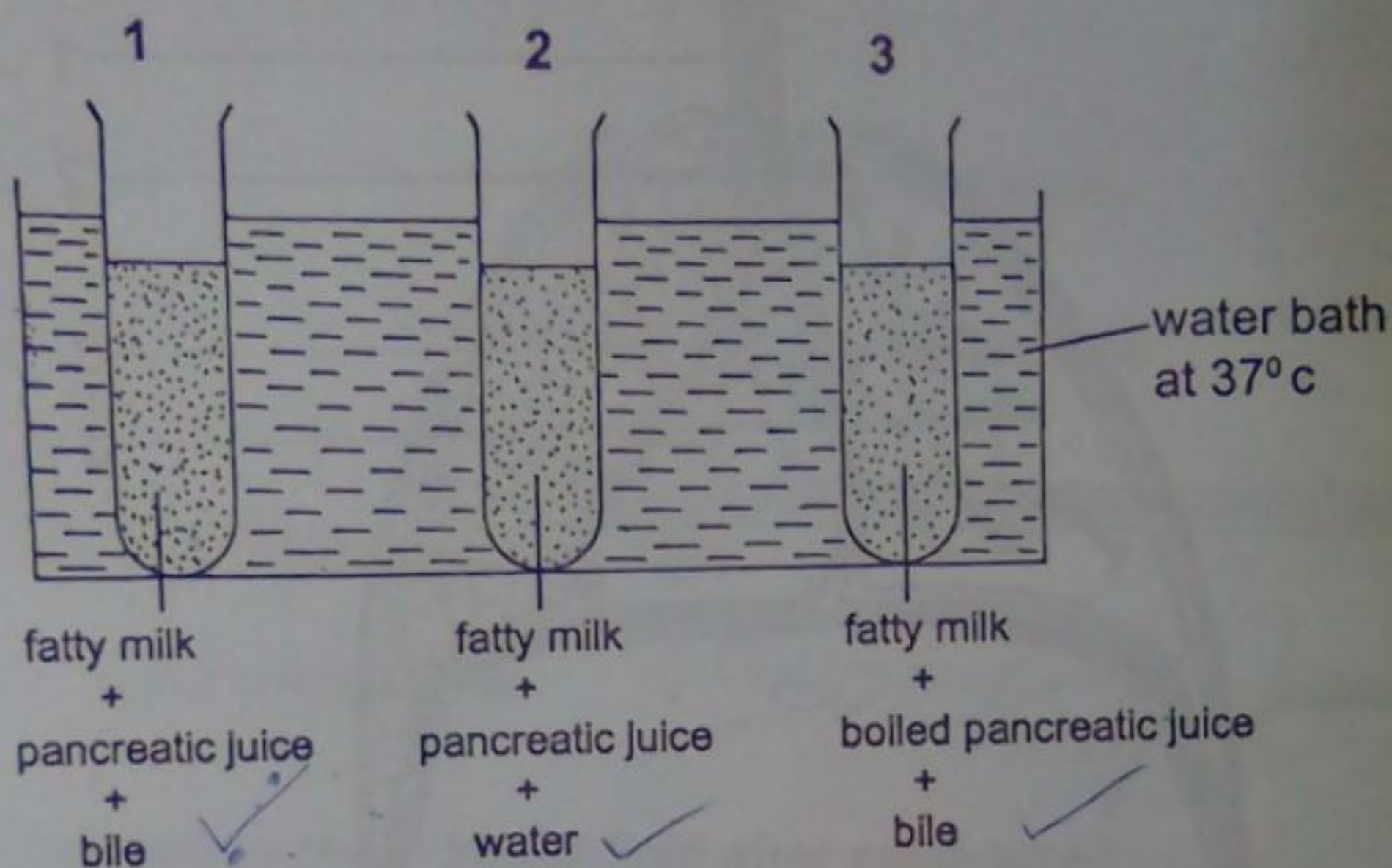
12. When one suffers from emphysema which of the labelled parts is affected?

- A 1 ✓
- B 2 ✓
- C 3 ✓
- D 4 ✓

13. When the amount of carbon dioxide in the blood increases what happens to the breathing rate and pulse rate?

	Breathing rate	Pulse rate
A	increases ✓	increases ✓
B	increases ✓	decreases ✓
C	decreases ✓	increases ✓
D	decreases ✓	decreases ✓

The diagram below shows an experiment to investigate the digestion of fats. After one hour the contents of each tube were tested for the presence of fatty acids.



Which were the correct results obtained in test tubes 1, 2, and 3?

	1	2	3
A	fatty acids present ✓	no fatty acids	fatty acids present ✓
B	no fatty acids ✓	fatty acids present ✓	fatty acids present but less than in 2
C	no fatty acids ✓	rich in fatty acids	no fatty acids
D	rich in fatty acids ✓	fatty acids present but less than in 1	no fatty acids ✓

8. The table shows some of the vitamins and minerals present in four fruits. Which fruit is best for development of strong teeth and bones?

	Fruit	Calcium mg per 100g	Iron mg per 100g	Vitamin C mg per 100g
A	apricots	73	3.4	1
B	figs	230	4.9	1
C	mangoes	12	0.7	37
D	prunes	34	2.6	0

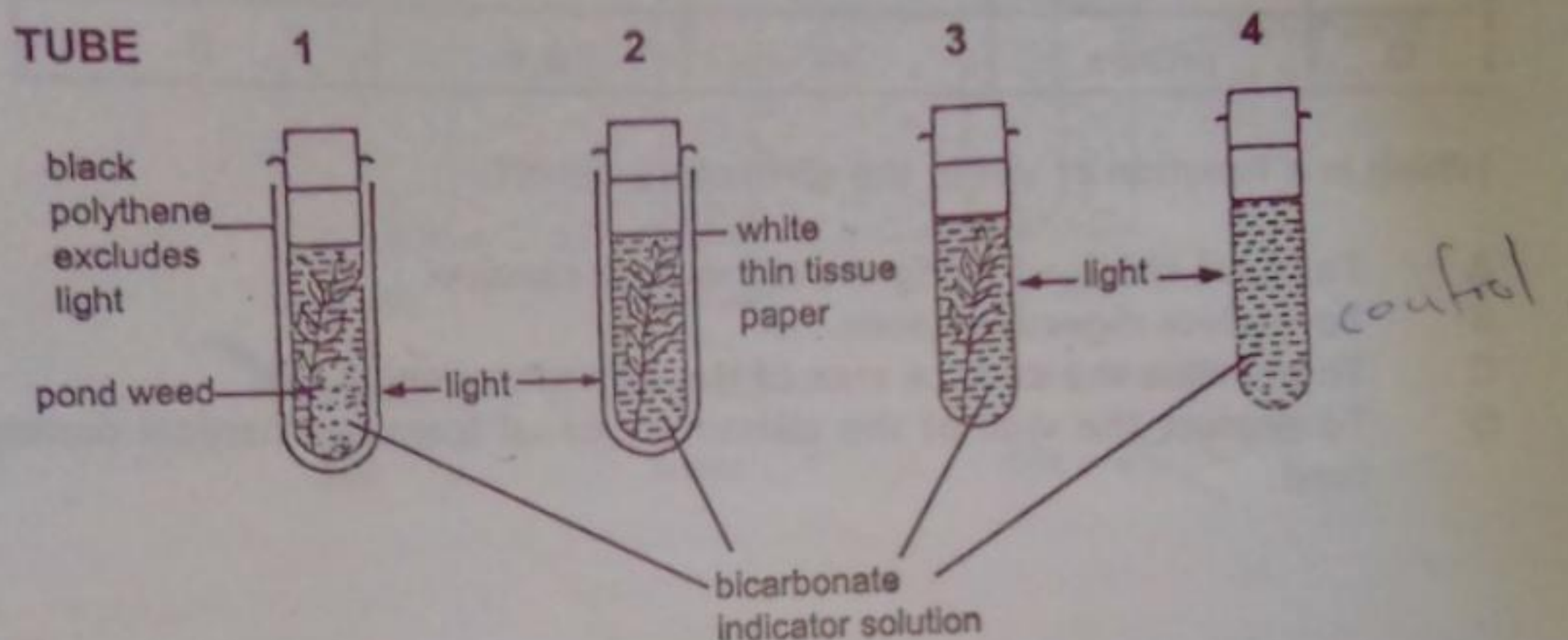
9. Which is a function of villi in the alimentary canal?

- A To move the food along the alimentary canal.
- B To secrete digestive juices.
- C To increase the surface area of the alimentary canal.
- D To protect the wall of the alimentary canal from mechanical damage by food.

4. Which of the following helps a plant to absorb carbon dioxide rapidly?

- A arrangement of leaves on the stem ✓
- B green colour ✓
- C large surface area ✓
- D thickness of the leaf ✓

Questions 5 and 6 refer to the diagram showing apparatus used in an experiment on photosynthesis.



5. What was the purpose of the experiment?

- A To show that light affects the rate of oxygen production by pondweed. ✓
- B To show that the uptake of carbon dioxide by pondweed varies with light intensity. ✓
- C To show that the absorption of light by pondweed varies with carbon dioxide concentration. ✓
- D To show that in pondweed the rate of photosynthesis equals the rate of respiration. ✓

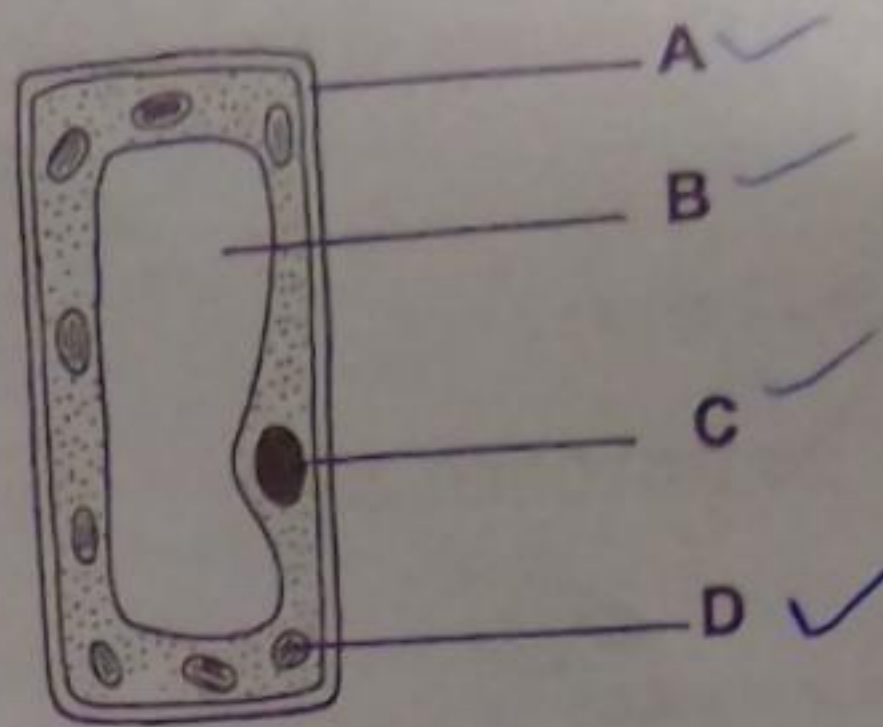
6. What was the role of tube 4?

- A To act as a control against temperature changes. ✓
- B To see if the water discoloured. ✓
- C To see if the solution affected the weed. ✓
- D To act as a colour standard for comparison. ✓

7. When testing a leaf for starch, why is it dipped in boiling water after it has been boiled in methylated spirit?

- A To kill the protoplasm. ✓
- B To soften it. ✓
- C To remove chlorophyll. ✓
- D To destroy enzymes. ✓

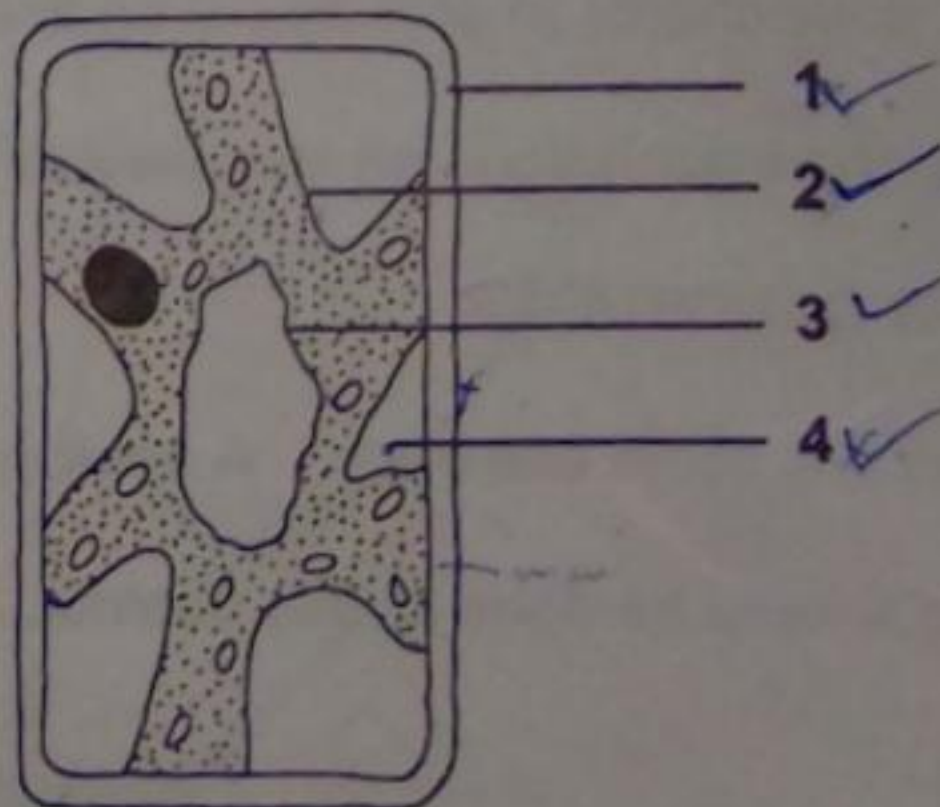
1. The diagram below shows a plant cell. Which part traps light used in photosynthesis?



2. Which of the following structures are found in plant but not animal cells?

	1	2	3
A	cell wall ✓	vacuole ✓	chloroplast ✓
B	vacuole ✓	cell membrane ✓	cell wall ✓
C	chloroplast ✓	vacuole ✓	nucleus ✓
D	cell membrane ✓	chloroplast ✓	vacuole ✓

3. The diagram shows a plant cell after being placed in a concentrated salt solution



Which of the structures are partially permeable barriers?

- A 1 and 2 ✓
 B 2 and 3 ✓
 C 3 and 4 ✓
 D 1 and 4 ✓

(44)

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple Choice

Tuesday

24 OCTOBER 2000

Afternoon

1 hour

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

020340

TIME

1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the **answer sheet**.

INFORMATION FOR CANDIDATES

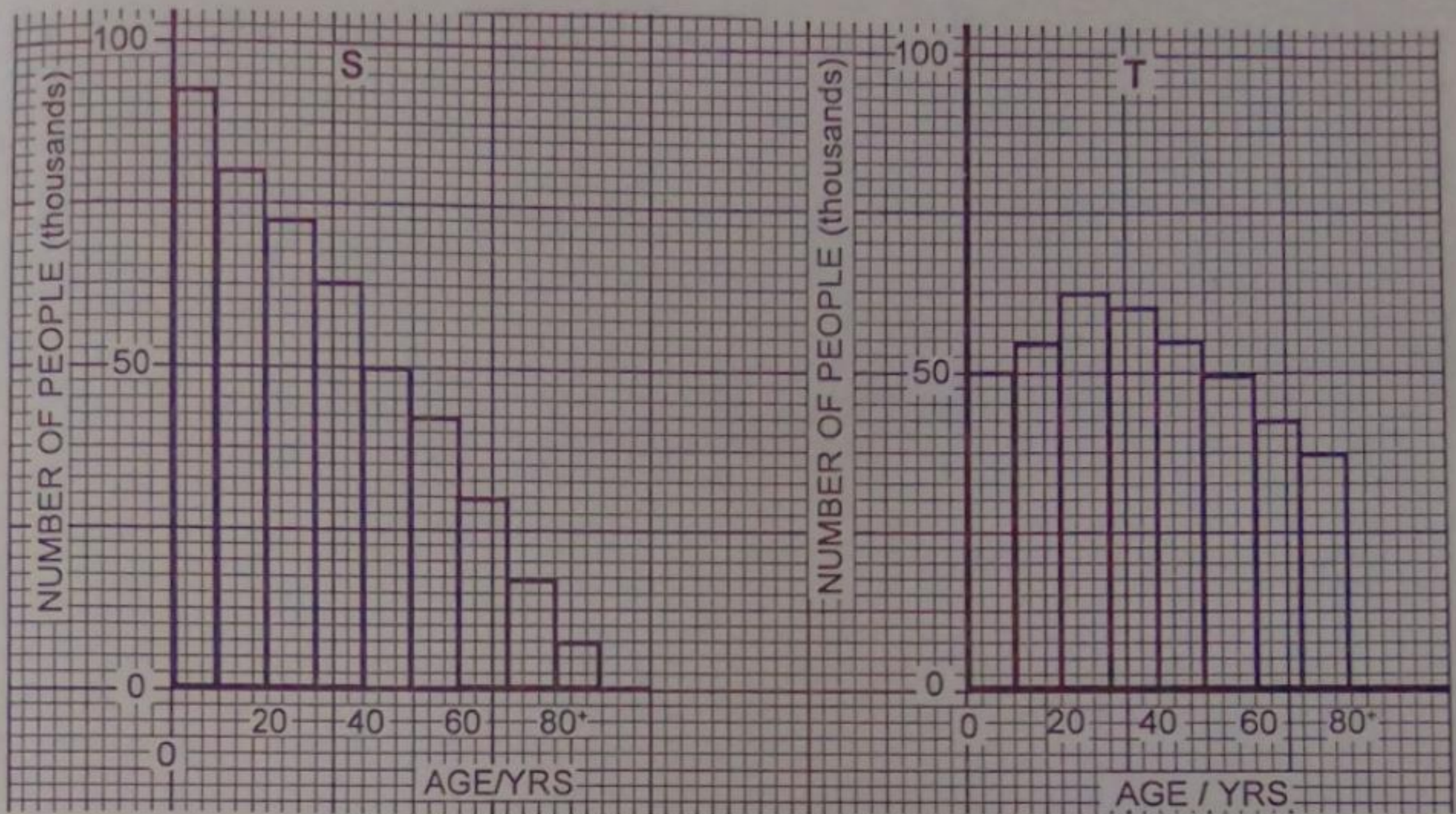
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

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40. The bar charts show the numbers of people of different ages in two countries S and T.



Which of the following statements about the two countries is true?

- A In T there are more babies and young people than in S. ✗
 B In S the birth rate is declining faster than in T. ✗
 C In S the death rate is greater in the younger people than it is in T. ✓
 D In T there is a lower proportion of older people than in S. ✓

Black or black is bliss

Black is something you can never under-rate

So tie your shoe laces And keep up the pace

Peter Tosh

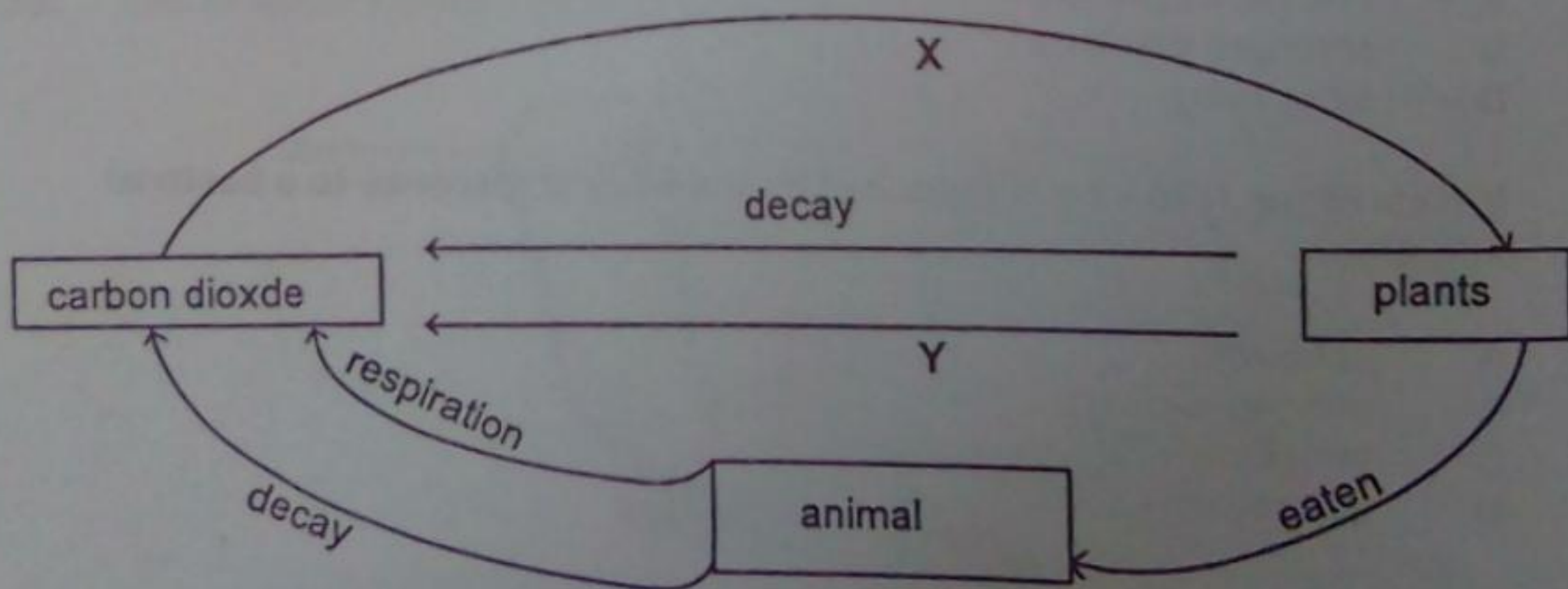
37. A patient suffers from cholera and malaria. Which type of microorganisms are found in the patient's faeces and blood?

	Faeces	Blood
A	protozoa	fungi
B	protozoa	bacteria ✓
C	fungi	bacteria
D	bacteria	protozoa

38. Which of the following food chains is the most efficient method for providing energy to humans?

- A Sun → maize → humans ✓
 B Sun → water weed → fish → humans
 C Sun → grass → cow → humans
 D Sun → grass → goats → humans ✓

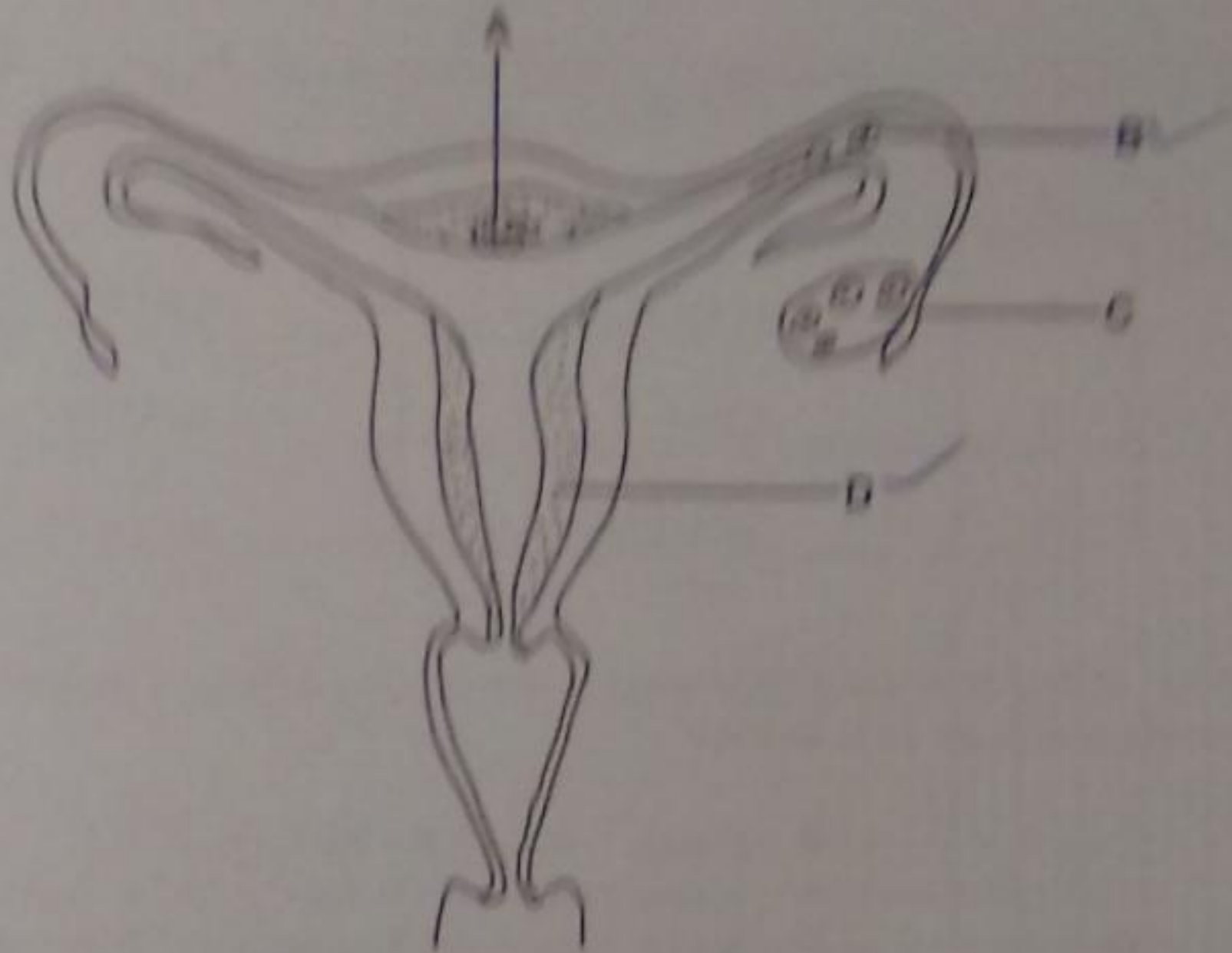
39. The diagram below represents part of a simplified carbon cycle.



What processes are represented by letters X and Y?

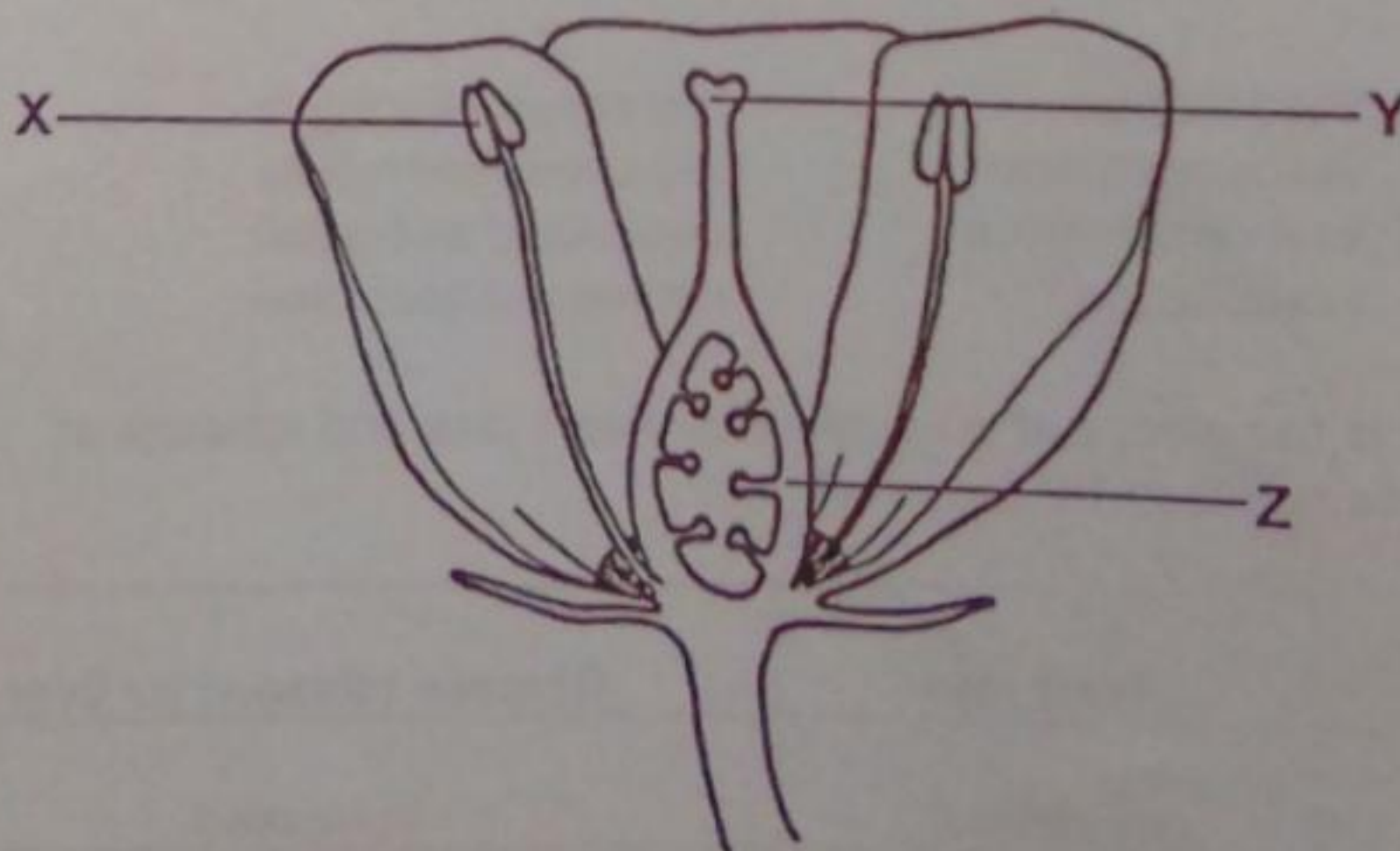
- | | | |
|---|----------------|-----------------|
| | X | Y |
| A | photosynthesis | transpiration ✓ |
| B | transpiration | respiration ✓ |
| C | absorption | photosynthesis |
| D | photosynthesis | respiration ✓ |

34. The diagram shows changes in the female reproductive system. Where is implantation of the embryo taking place?



35. What is the effect of cigarette smoking during pregnancy?
- A a short umbilical cord
 - B ✓ low blood pressure
 - C enlarged placenta
 - D ✓ large baby
36. Which of the following is produced by the body in response to a bacterial infection?
- A antibiotic
 - B antigen ✓
 - C antiseptic
 - D ✓ antibodies

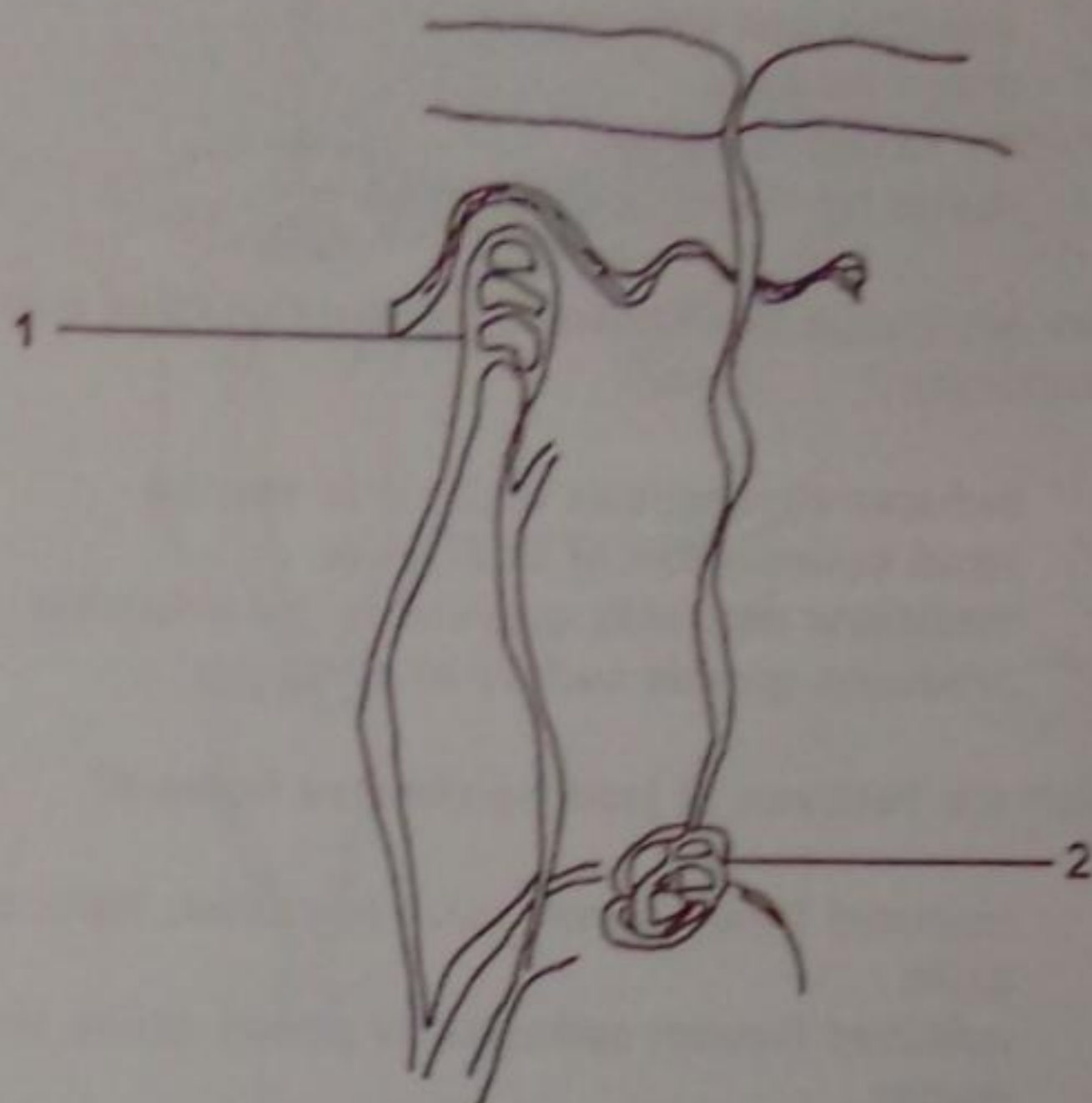
30. What may happen to a heroin addict two days after the drug is withdrawn?
- A ✓ vomiting and sweating
 - B ✓ drug tolerance
 - C ✓ the addict is cured
 - D desire for the drug is reduced
31. What advantage does sexual reproduction have over asexual reproduction?
- A ✓ reduces competition with other species
 - B ✓ rapid colonisation of new areas
 - C ✓ maintains desirable qualities in the offspring
 - D ✓ produces greater variety of offspring
32. Which are features of insect pollinated flowers?
- A coloured flower; production of nectar; light, smooth pollen grain
 - B ✓ coloured flower; spiky/sticky pollen grain; production of nectar
 - C production of nectar; small green petals; production of pollen
 - D anther and stigma inside flower; production of scent; production of large quantities of pollen
33. The diagram shows a section through a flower.



Which process is pollination?

- A X to Y ✓
- B Y to X
- C X to Z ✓
- D Z to X

28. The diagram shows a section through the skin.



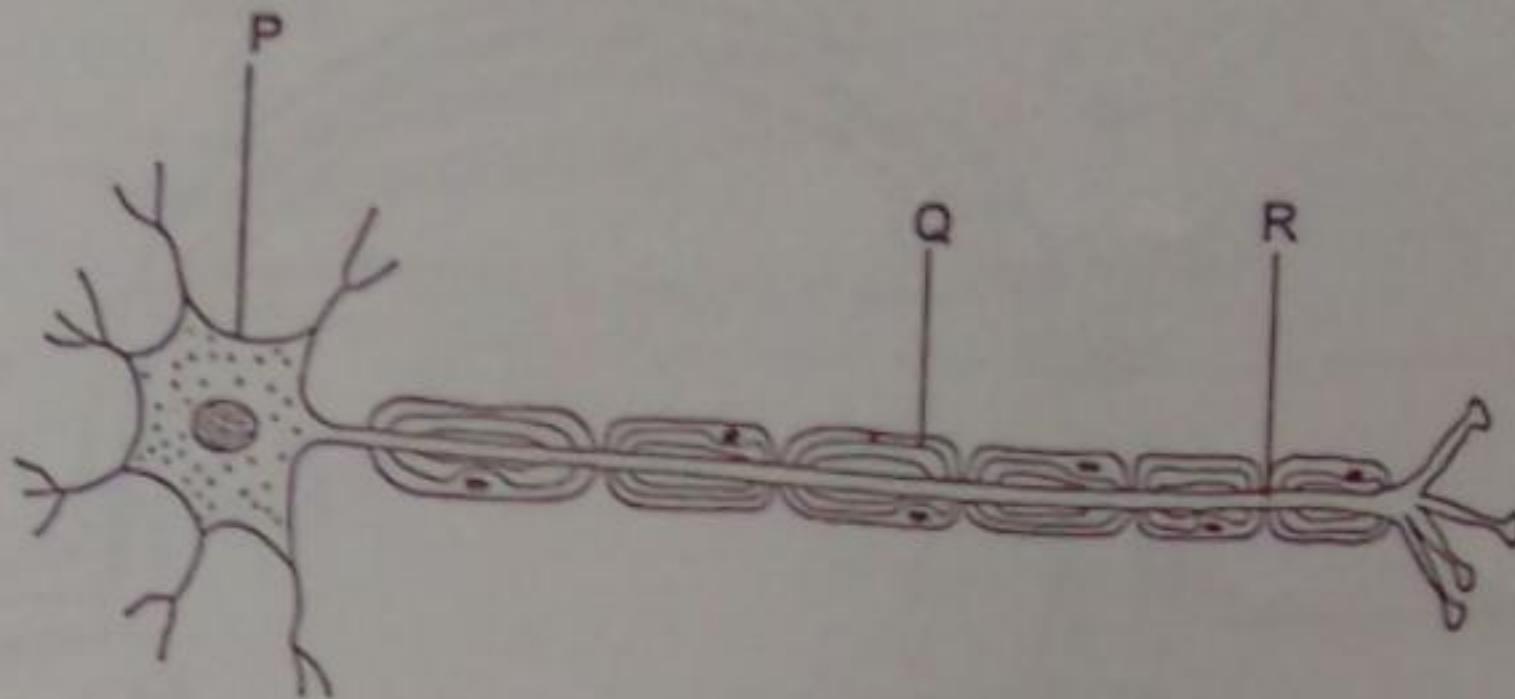
What happens in structures 1 and 2 when a healthy cook moves from the kitchen at 40°C into another room at 25°C ?

- | | 1 | 2 |
|-----|------------------|---------------------|
| A ✓ | vasodilation | increased secretion |
| B | vasoconstriction | increased secretion |
| C ✓ | vasoconstriction | decreased secretion |
| D | vasodilation | decreased secretion |

29. What is the effect of adrenaline on heart rate and amount of glucose released by the liver?

	Heart rate	Glucose released by liver
A	increased	increased
B	increased	reduced ✓
C	reduced	increased ✓
D	reduced	reduced

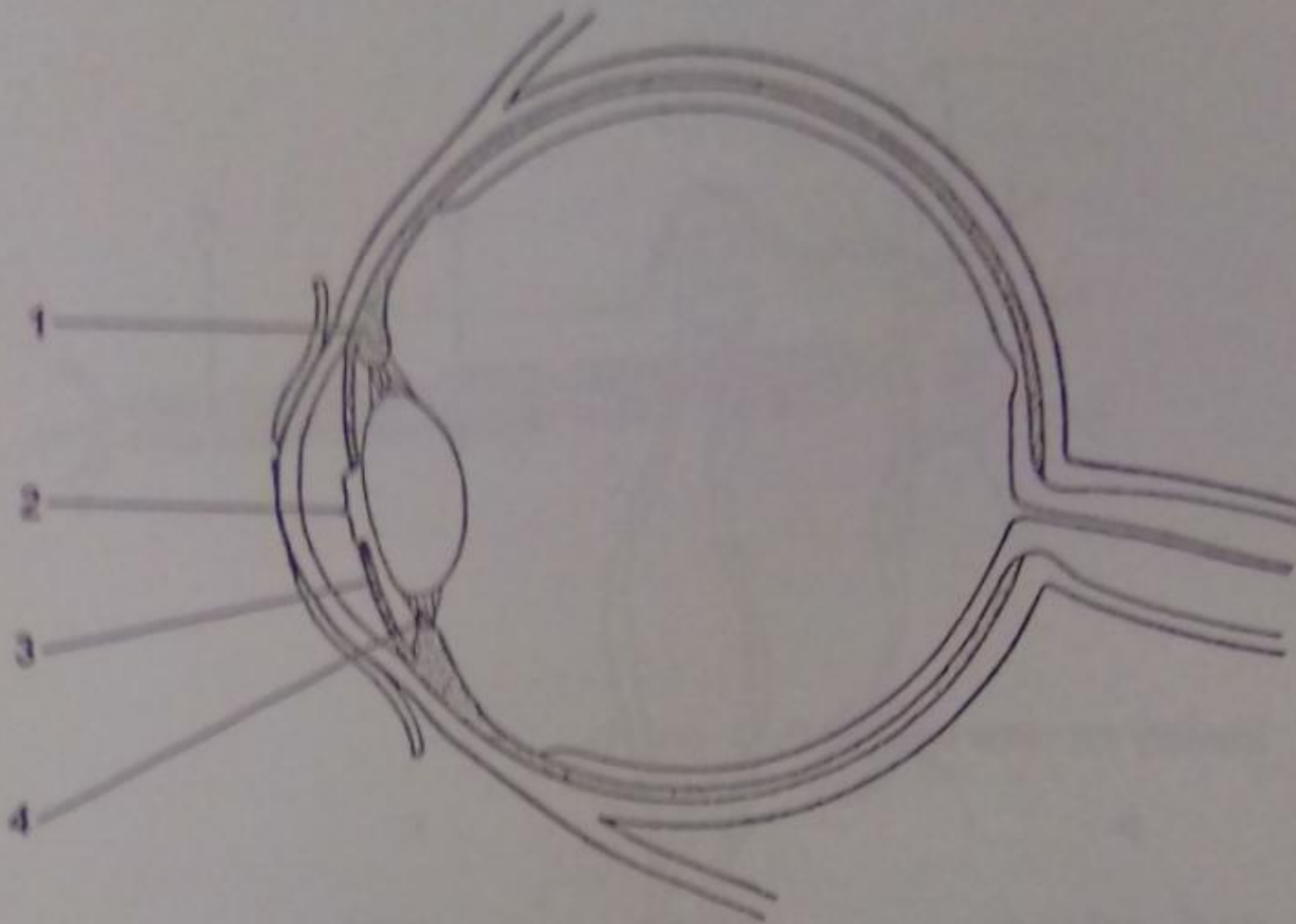
27. The diagram shows a neurone from the spinal cord.



Identify the parts labelled P, Q and R.

	P	Q	R
A ✓	cell body	sheath	axon
B	sensory receptor	dendron	axon
C ✓	motor- end- plate	cell body	dendron
D	axon	cell body	motor- end- plate

25. The diagram shows a section through the eye.



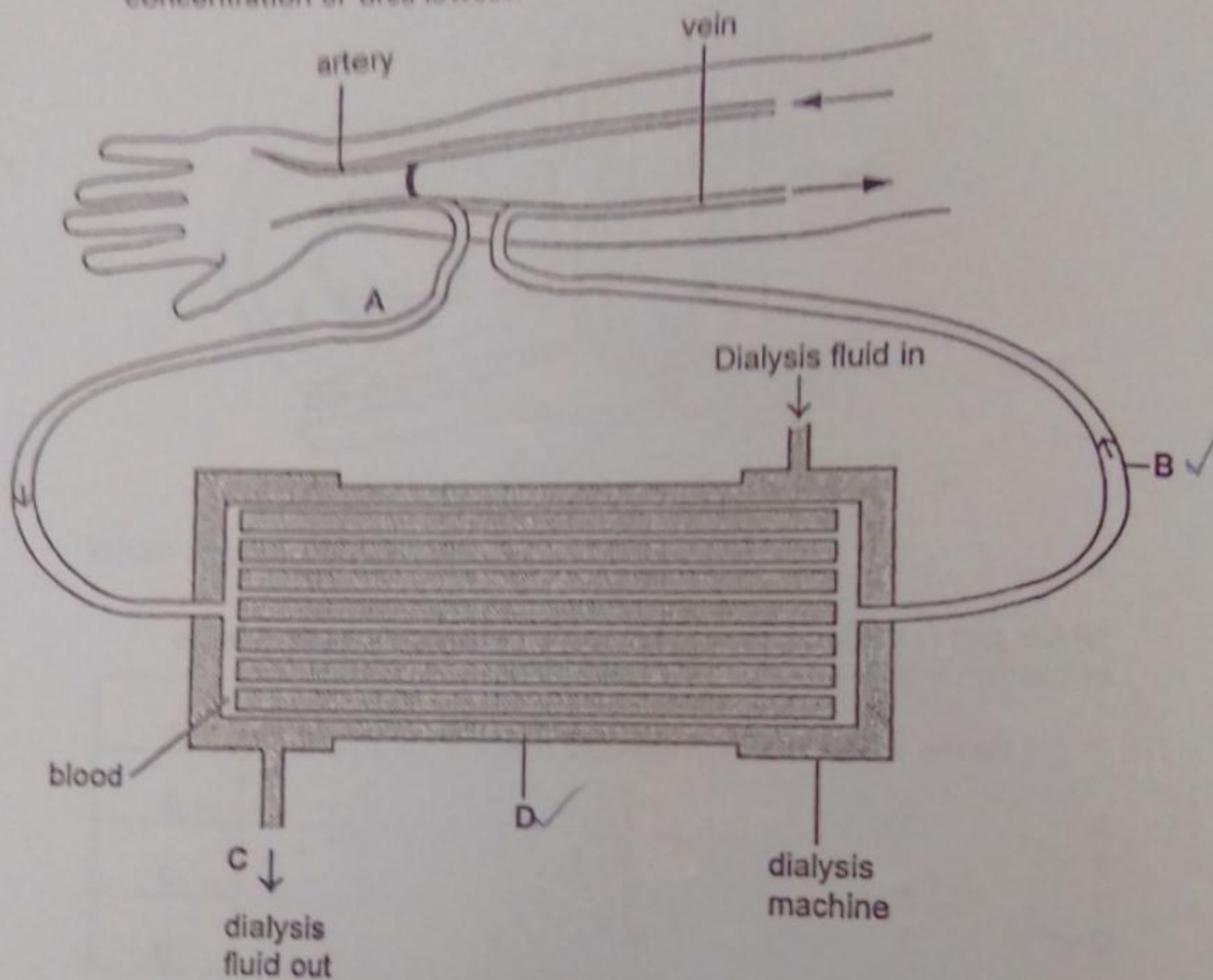
Which parts change in size when bright light is shone into the eye?

- A 1 and 2
- B 2 and 3 ✓
- C 1 and 3
- D 2 and 4 ✓

26. Which one is an example of an analgesic drug?

- A penicillin ✓
- B paracetamol
- C chloroquine ✓
- D paludrine

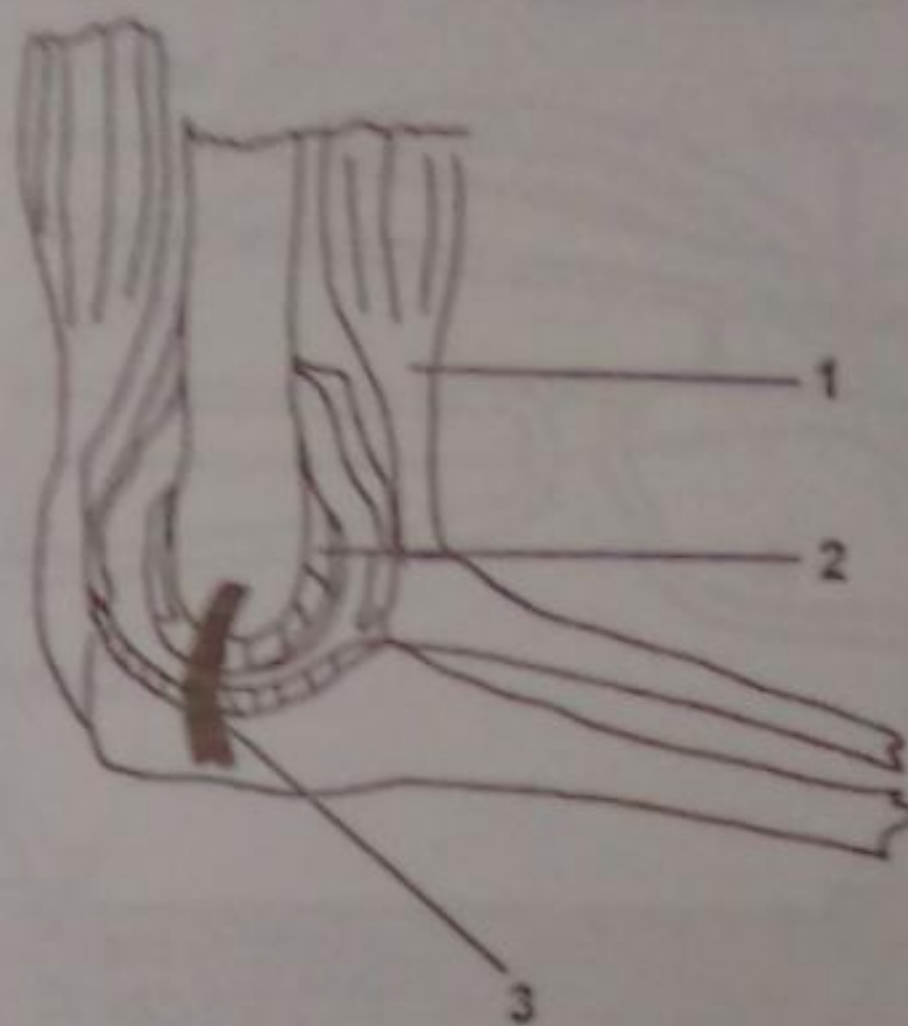
23. The diagram shows the process of kidney dialysis. At which point is the concentration of urea lowest?



24. What changes take place in volume of urine, concentration of urine and volume of sweat on a cold day?

	Volume of urine	Concentration of urine	Volume of sweat
A	increased	increased	increased
B ✓	decreased	increased	increased
C	increased	decreased	decreased
D ✓	increased	increased	decreased

- 21 The diagram shows the elbow joint.

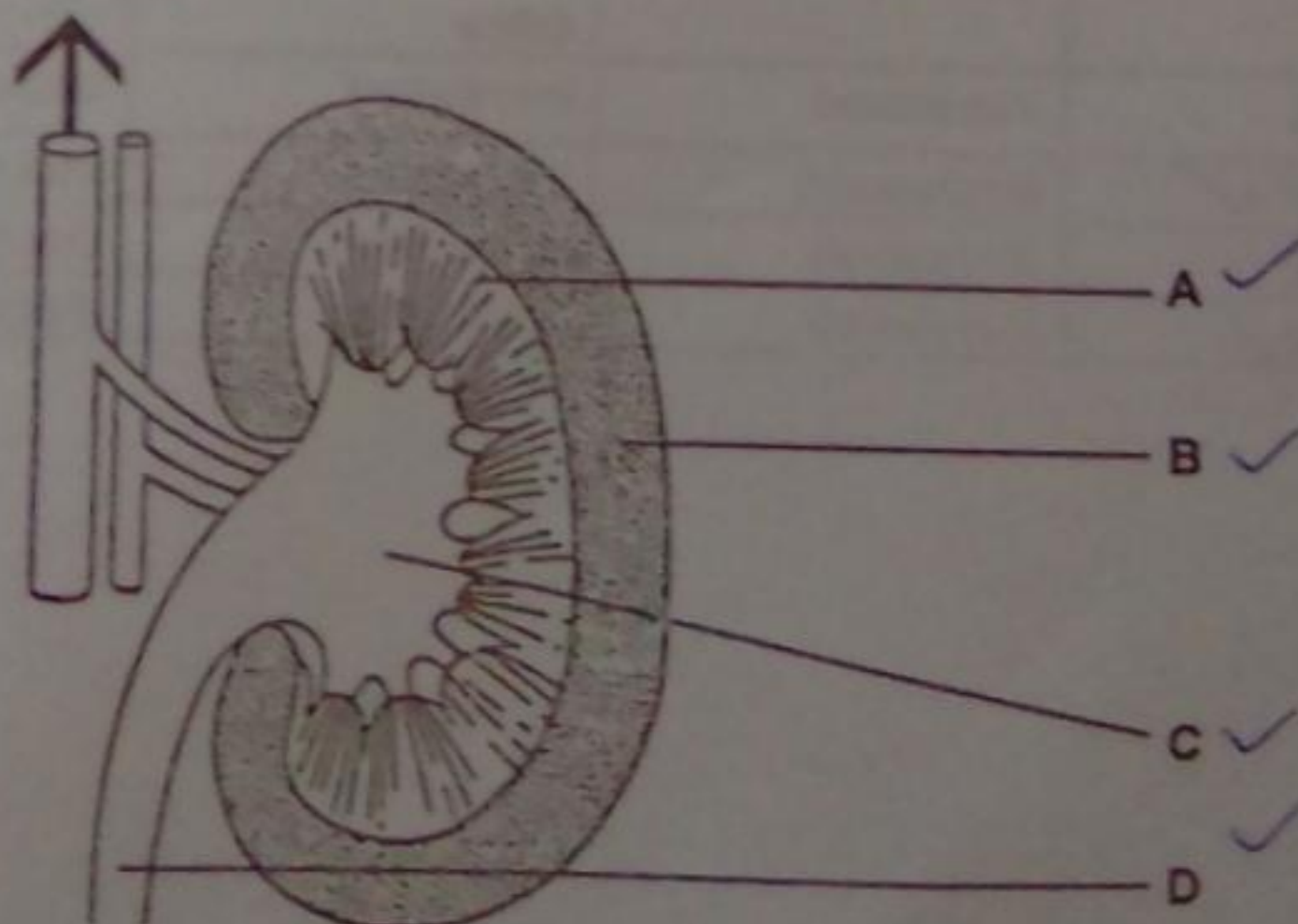


Which part is made of elastic fibrous tissue and which one is made of cushion-like tissue?

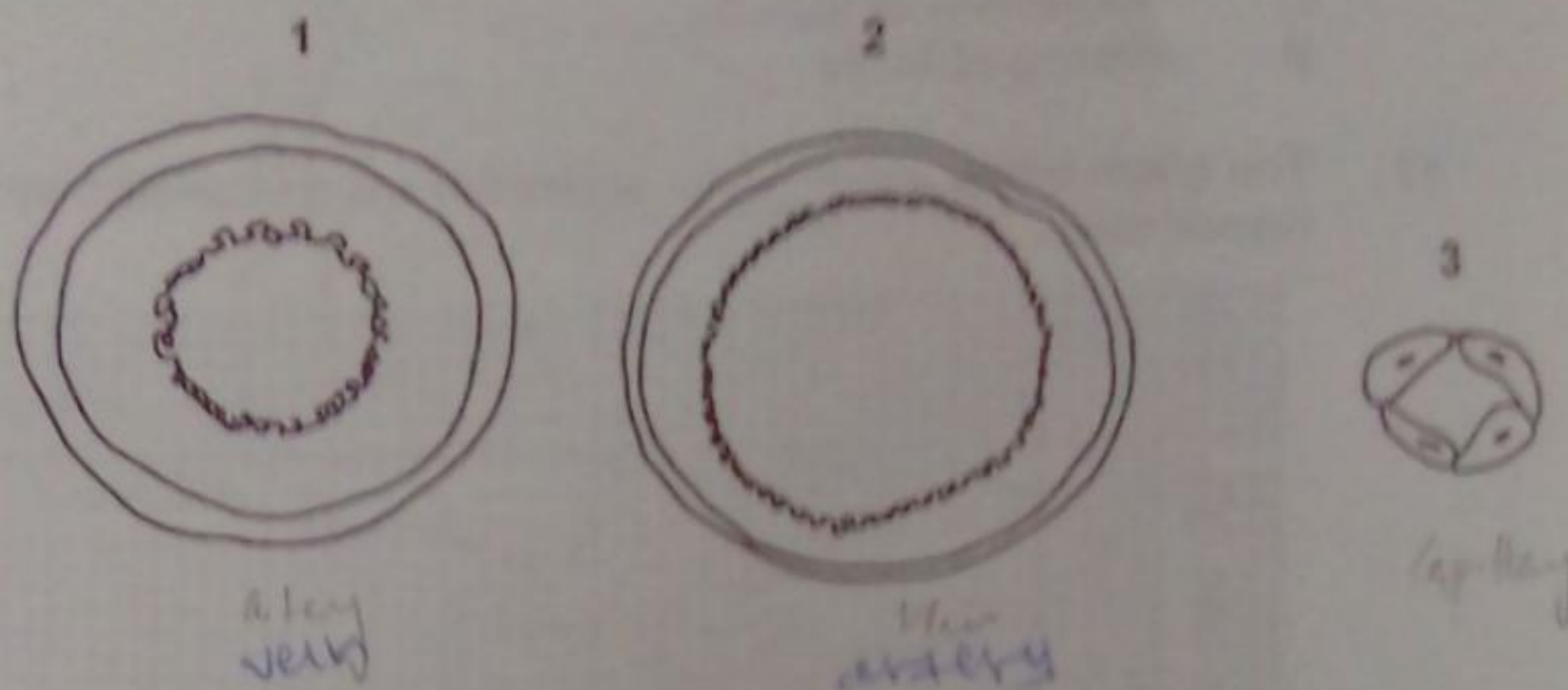
	Elastic tissue	Cushion-like tissue
A ✓	1	2
B ✓	1	3
C	2	3
D ✓	3	2

22. The diagram shows a cross-section of a kidney. In which part does filtration occur?

direction
of flow
of blood



19. The diagram shows three types of blood vessels.



Which is the correct set of features of the blood vessels?

	Fluid under high pressure	Fluid can pass through wall	Least resistance to flow of fluid
A	1	2	3
B ✓	1	3	2
C	2	3	1
D	3	1	2

20. ✱ The table shows the results of an experiment on a bone.

Mass of bone at beginning of experiment	Mass of bone after immersion in acid	Appearance after experiment
50g	5g	Same shape, bends easily.

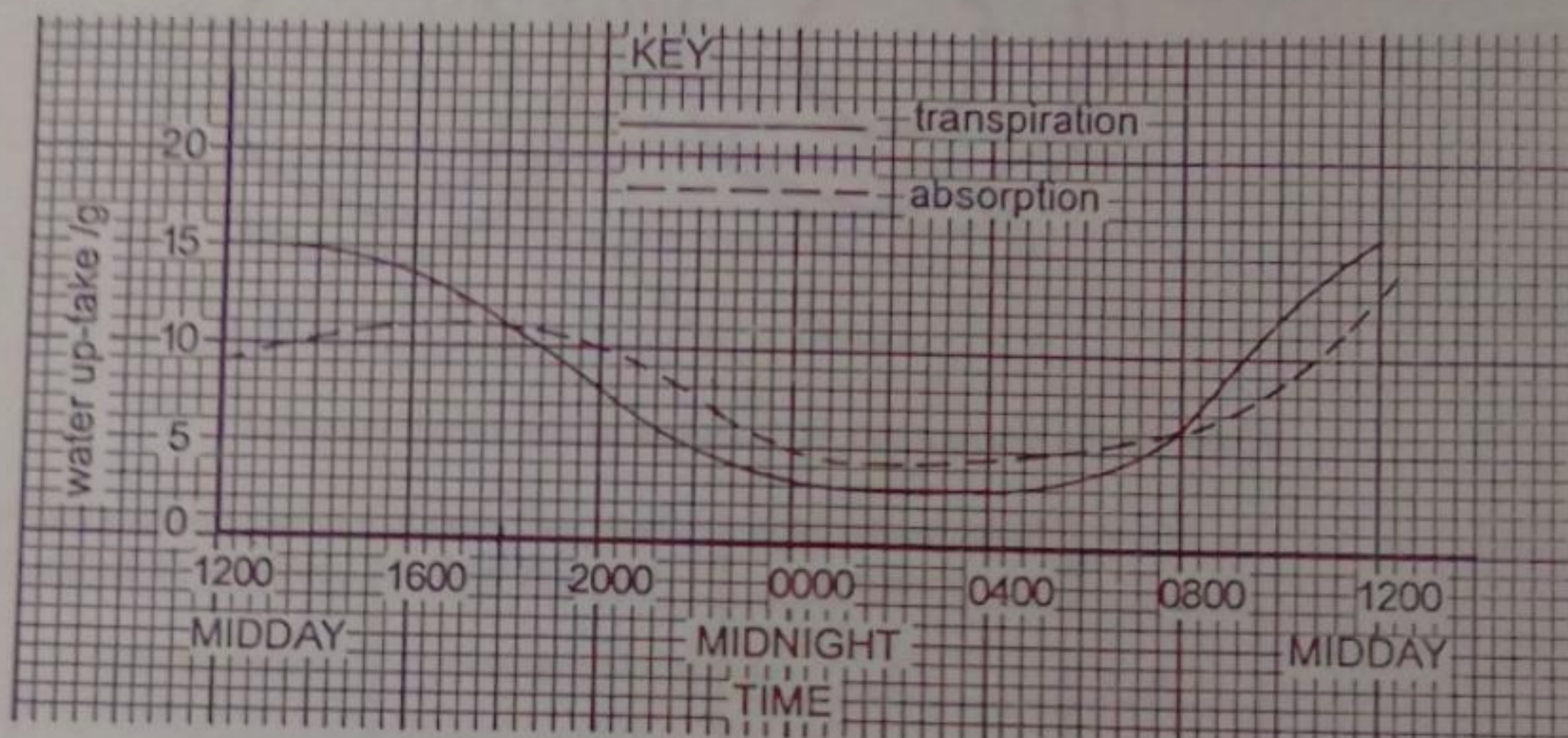
Which part of the bone reacted with the acid?

- A collagen
 B calcium ✓
 C cartilage ✓
 D protein ✓

16. Which is a function of blood plasma?

- A transport of excretory products ✓
- B production of antibodies to neutralise antigens
- C destruction of bacteria
- D clotting of blood ✓

17. The graph below shows water absorption and transpiration by a msasa tree.



At which times was transpiration equal to water absorption?

- A 1200 and 0600 ✓
- B 1200 and 0000
- C 0800 and 1800 ✓
- D 1200 and 1800

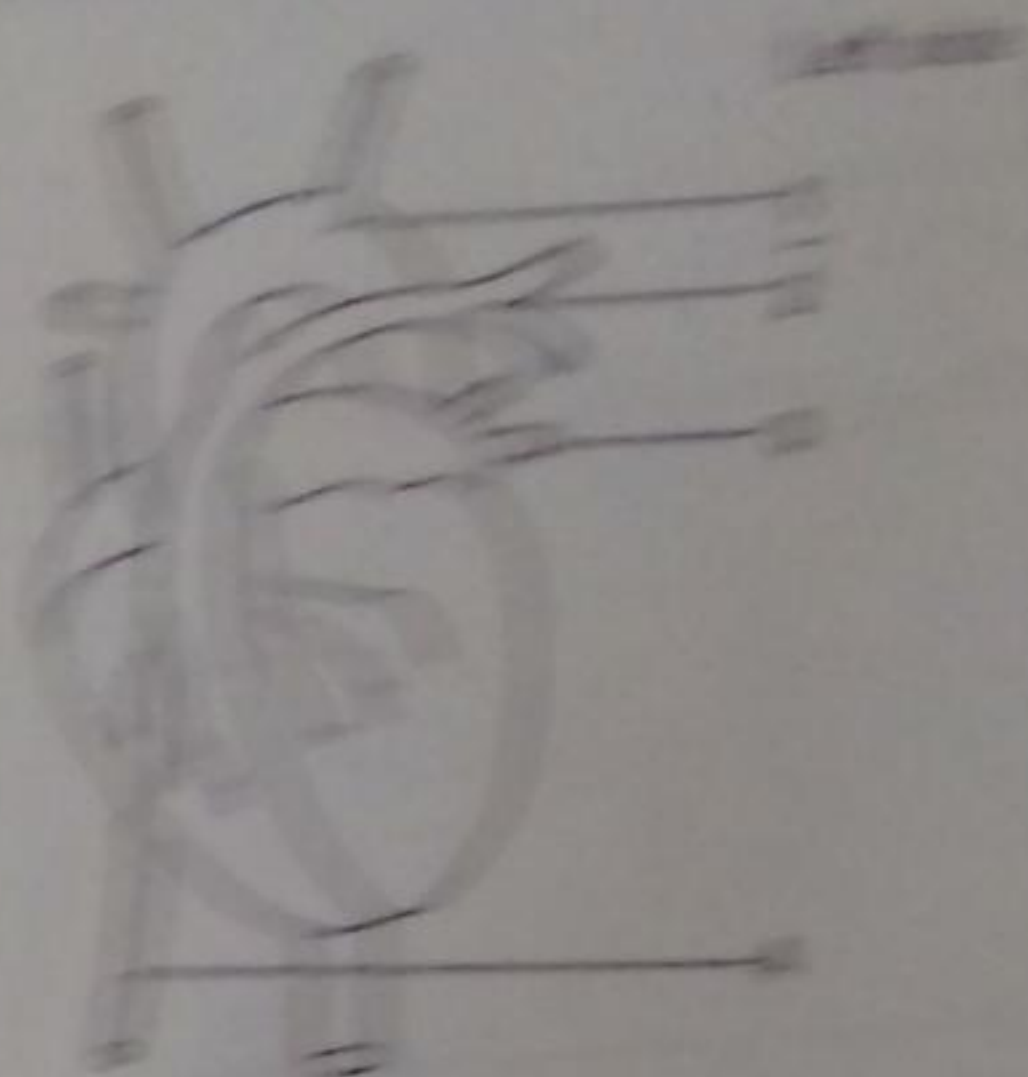
18. Which features of green leaves reduce water loss?

- 1. ✗ waxy cuticle
- 2. ✗ guard cells closing stomata ✓
- 3. ✗ large surface area

- A 1 and 2 ✓
- B 1 and 3
- C 2 and 3
- D 3 only ✓

14. The diagram shows a schematic view of the human heart.

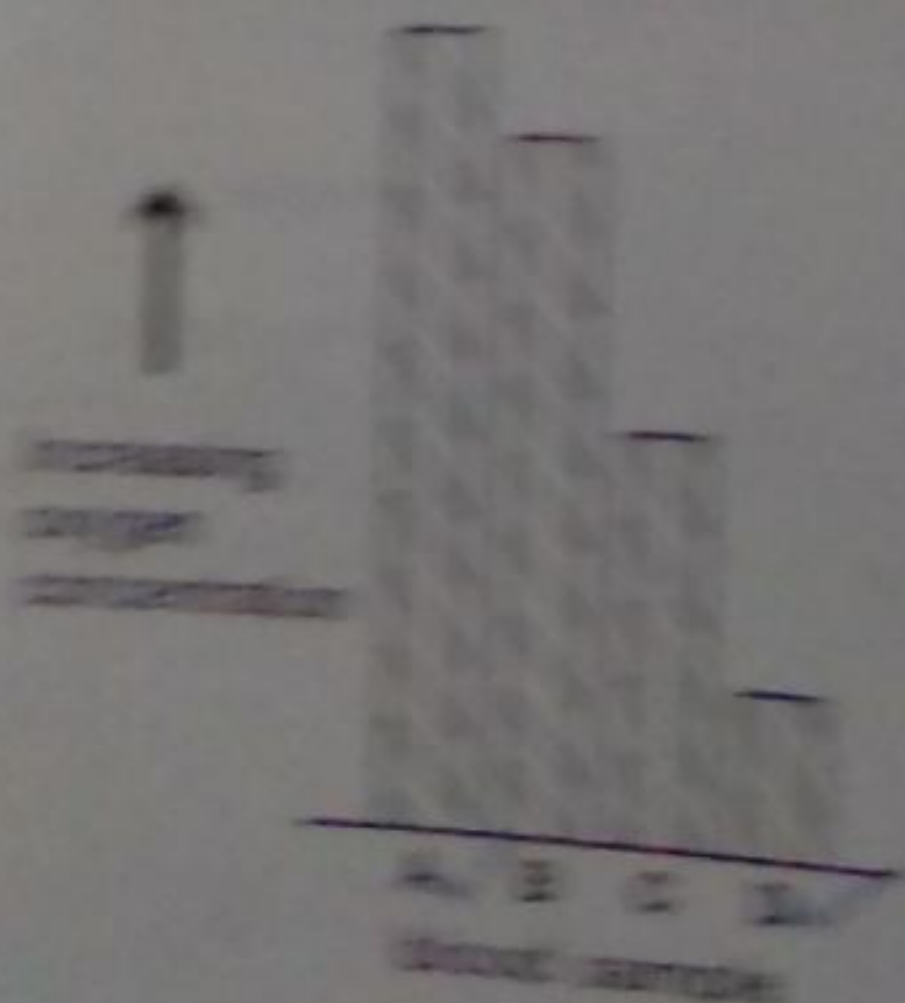
Fig. 10



Which two blood vessels have the lowest concentration of carbon dioxide?

- A 1 and 2
- B 1 and 3 ✓
- C 2 and 4
- D 3 and 4 ✓

15. The bar chart shows concentration of sugar in blood sampled from four vessels of the circulatory system. Which blood sample was taken from the pulmonary artery?



11. Which substance in cigarette smoke causes cancer?

- A carbon monoxide
- B nicotine ✓
- C smoke particles
- D tar ✓

12. The diagram below represents a transverse (cross) section through a root.



*It's got two eyes on the
border
What a weird little girl
black to white one border
for appearance sake*

What is the part labelled J?

- A phloem
- B xylem ✓
- C pith
- D cortex ✓

13. What are the adaptations of red blood cells for absorption and transport of oxygen?

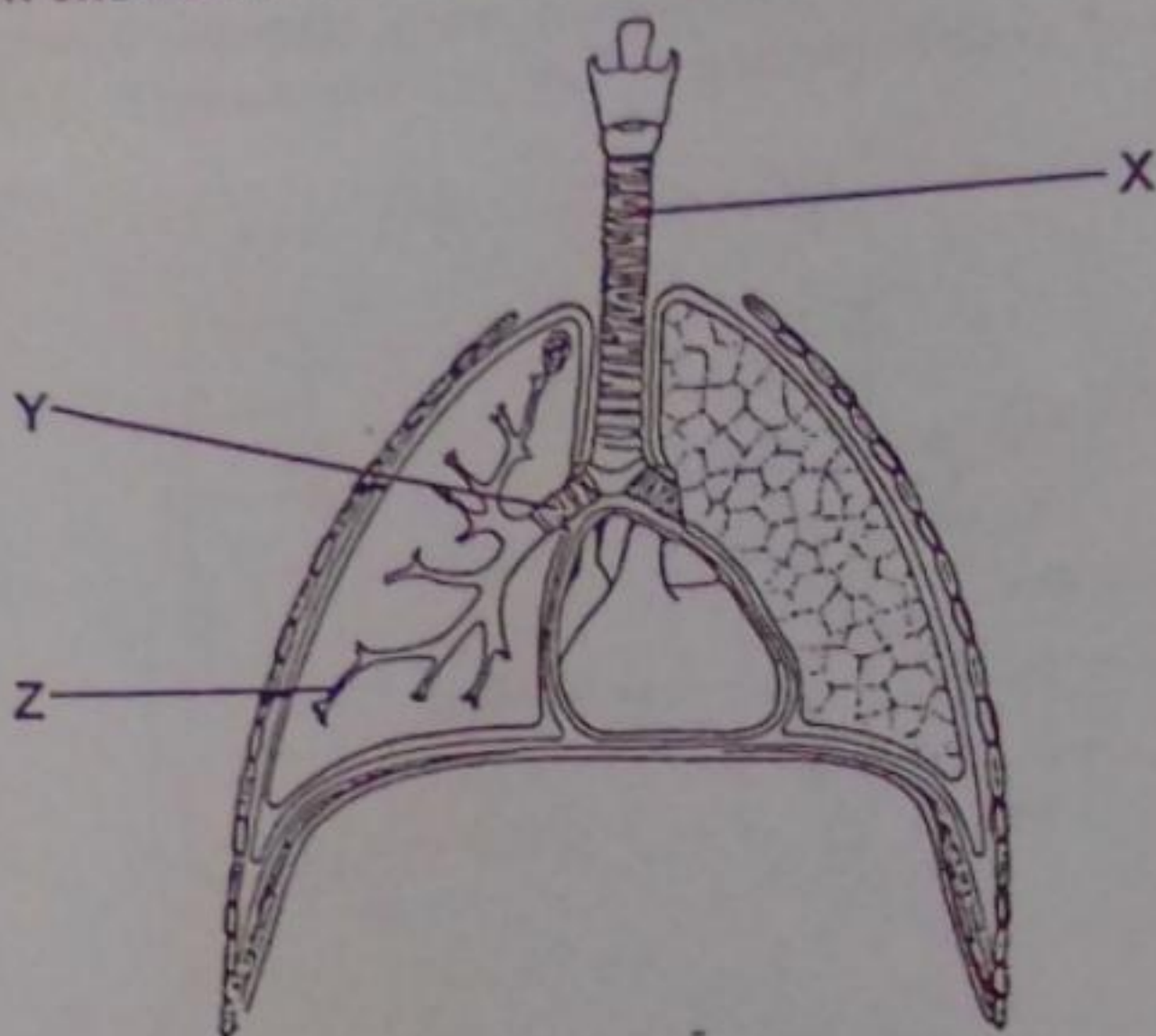
Absorption

- A large surface area
- B ability to change shape
- C non-nucleated
- D large surface area

Transport

- non-nucleated
- ability to float ✓
- large surface area
- ability to float ✓

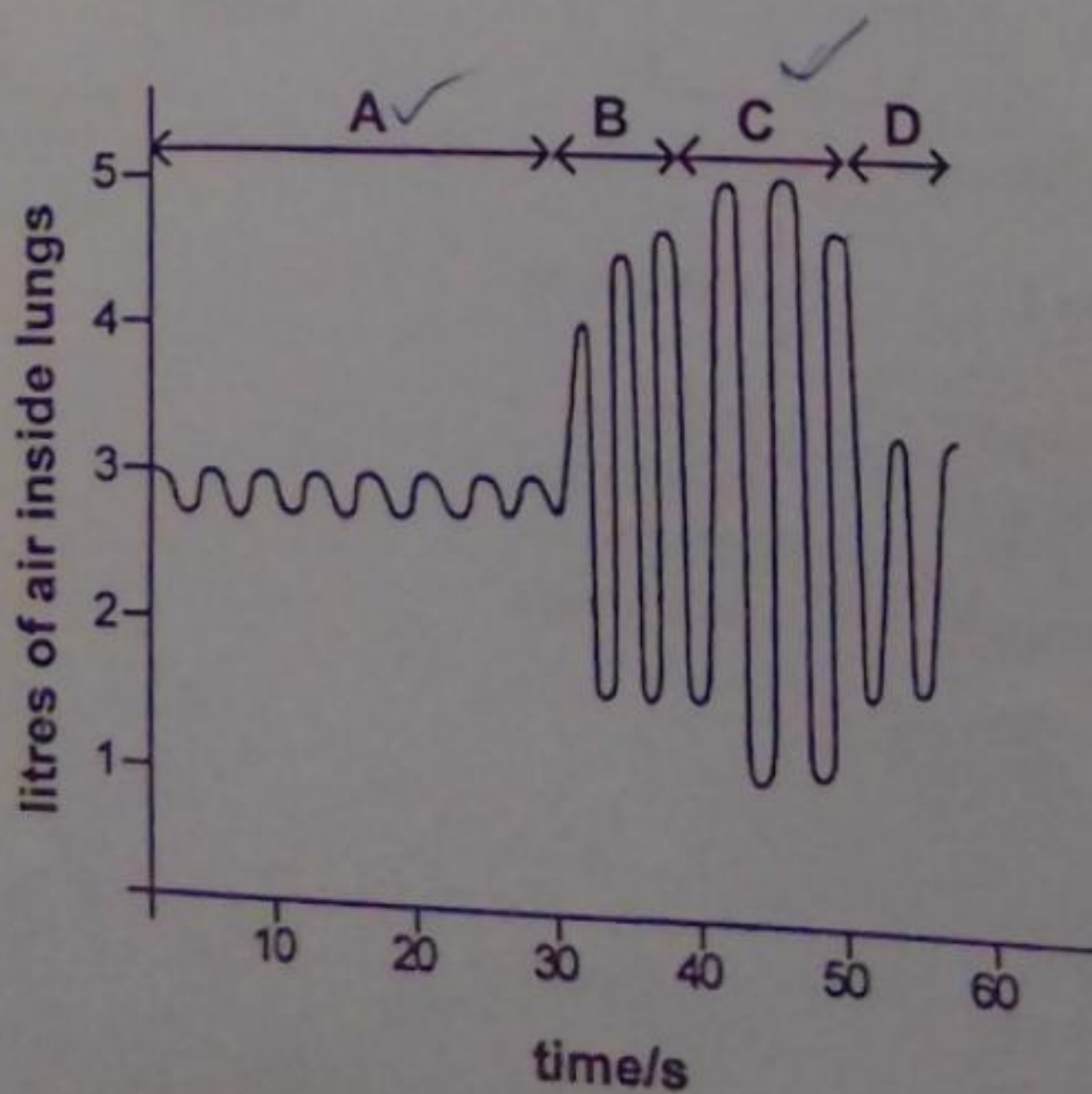
9. The diagram shows the human thoracic cavity.



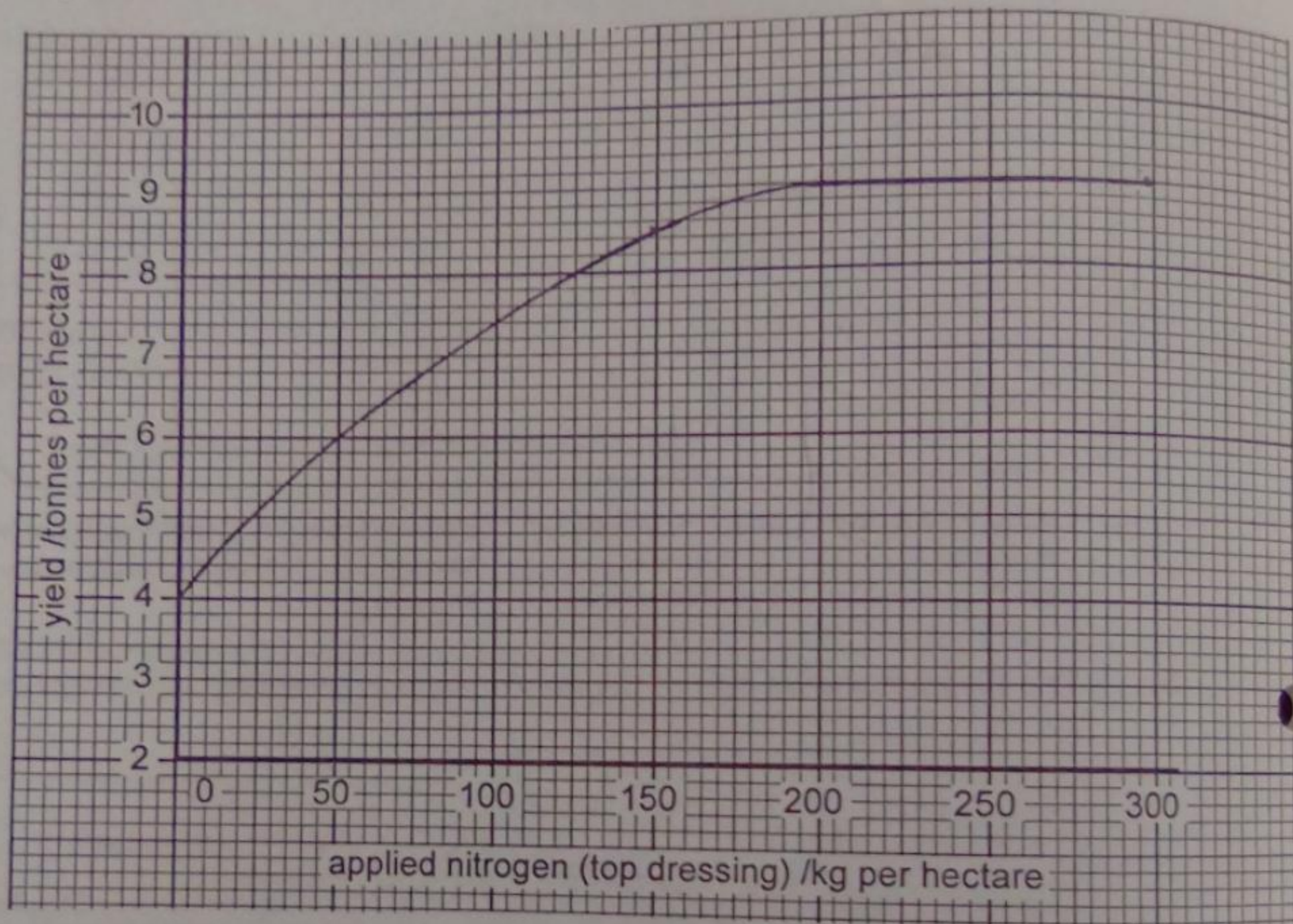
Name the structures labelled X, Y and Z.

	X	Y	Z
A ✓	trachea	bronchus	bronchiole
B	bronchus	trachea	bronchiole
C	bronchus	alveolus	trachea
D ✓	alveolus	bronchiole	bronchus

10. The graph shows a person's breathing rate and volume of air breathed. During which period was the person exercising most vigorously?



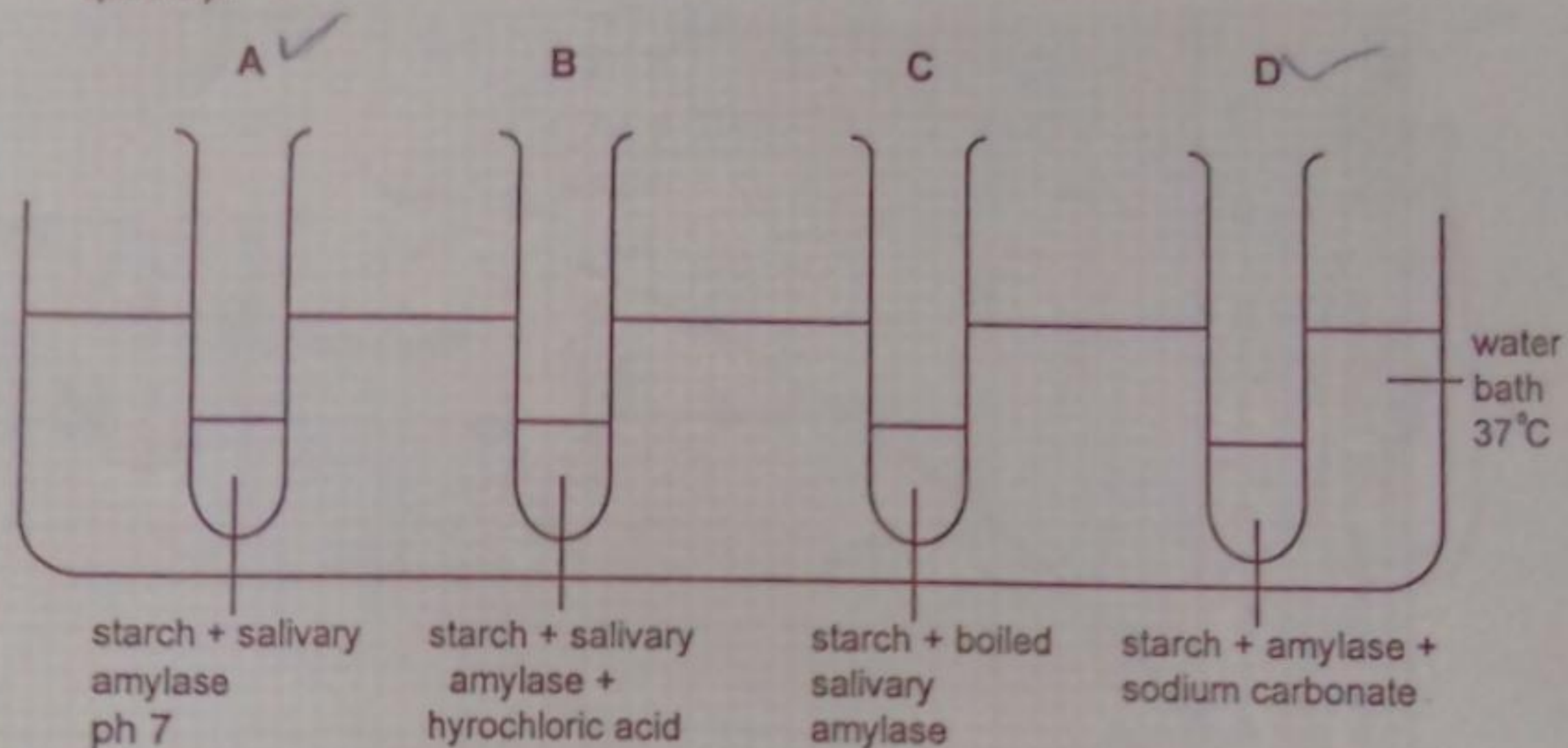
6. The graph shows the change in yield of maize in response to adding more nitrogen fertiliser.



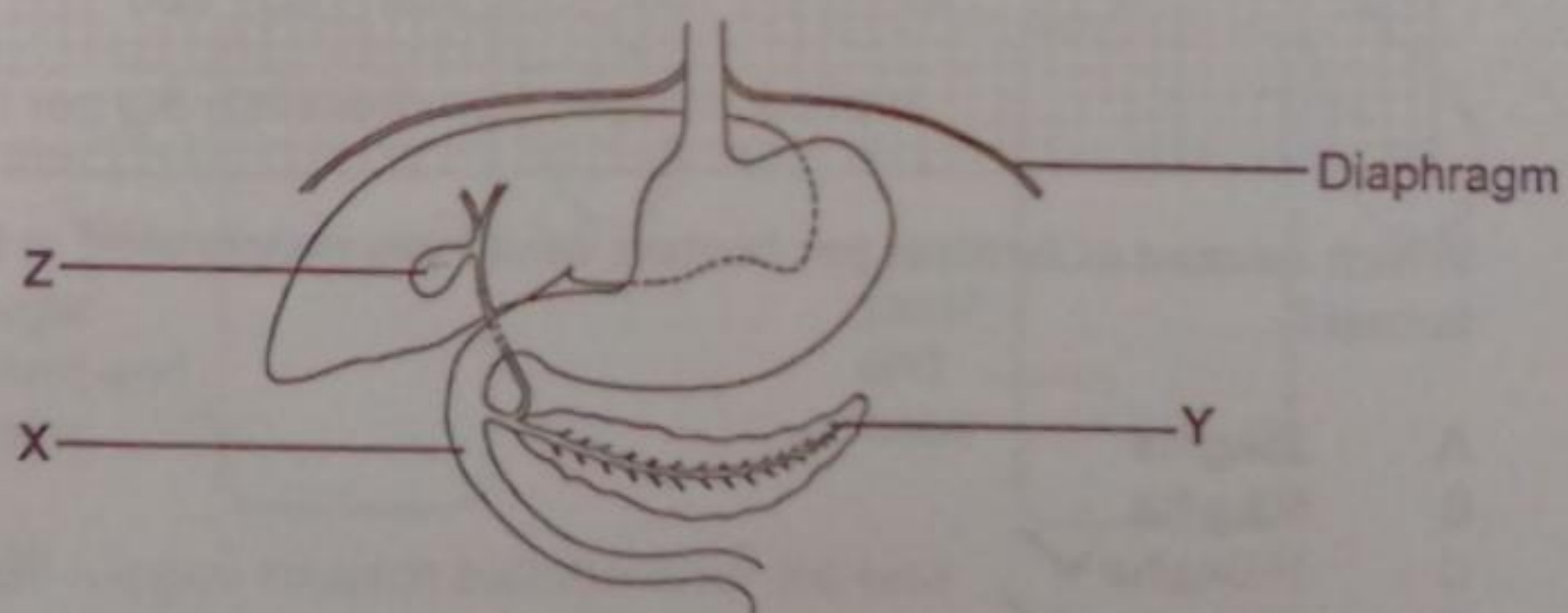
Which amount of fertilizer per hectare would you recommend to a farmer?

- A 25kg/ha
- B 50kg/ha
- C 150kg/ha ✓
- D 250kg/ha ✓

4. The diagram shows an experiment on the digestion of starch by salivary amylase. In which test tube will starch be digested most quickly?



5. The diagram shows part of the human digestive system.



Name the structures labelled X, Y and Z.

	X	Y	Z
A	<u>liver</u>	duodenum	stomach ✓
B	ileum	liver	pancreas
C	ileum	gall bladder	liver
D	<u>duodenum</u>	<u>pancreas</u>	gall bladder ✓

1. Which major elements when deficient in plants cause poor growth, yellow leaves and premature death?

	Poor growth	Yellow leaves	Premature death
A	potassium ✓	phosphorus	nitrogen
B	phosphorus	nitrogen	potassium
C	phosphorus	potassium	nitrogen
D	nitrogen ✓	phosphorus	potassium

2. Which of the following structures are only found in plant cells but not in animal cells?

- A cell membrane, mitochondria, chromosomes
 B central vacuole, cell sap, cell wall ✓
 C central vacuole, cytoplasm, nucleus
 D cell sap, cell membrane, cell wall ✓

3. The table shows the composition of four foods. Which food should you include in a diet in order to prevent constipation?

	Protein/g	Vitamin C/ mg	Vitamin A mg	Fibre/g	Iron/mg	Calcium/mg
A	27	10	1700	0	7,5	0
B	7.5	0	0	11	1.1	5 ✓
C	3.3	1	50	0	0	120
D	1.4	10	0	2	0,3	0 ✓

Teacher's plight:

She's got her eyes on her teacher
 What a naughty little girl.
 Wants to seduce her teacher

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY

PAPER 1 Multiple Choice

5008/1

Monday 26 NOVEMBER 2001

Morning

1 hour

Additional materials:

Multiple Choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

17592

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are forty questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

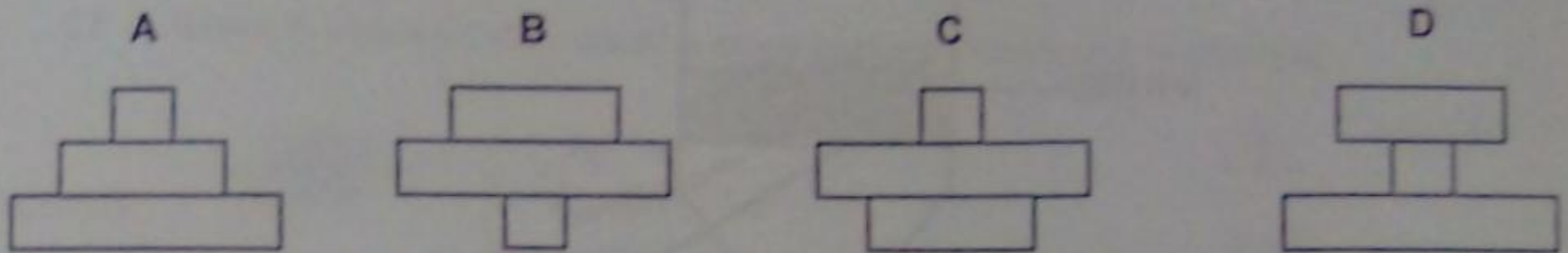
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 19 printed pages and 1 blank page.

Copyright: Zimbabwe School Examinations Council, 2001.

- 39 In the diagrams the size of each box represents the number of organisms in a food chain.

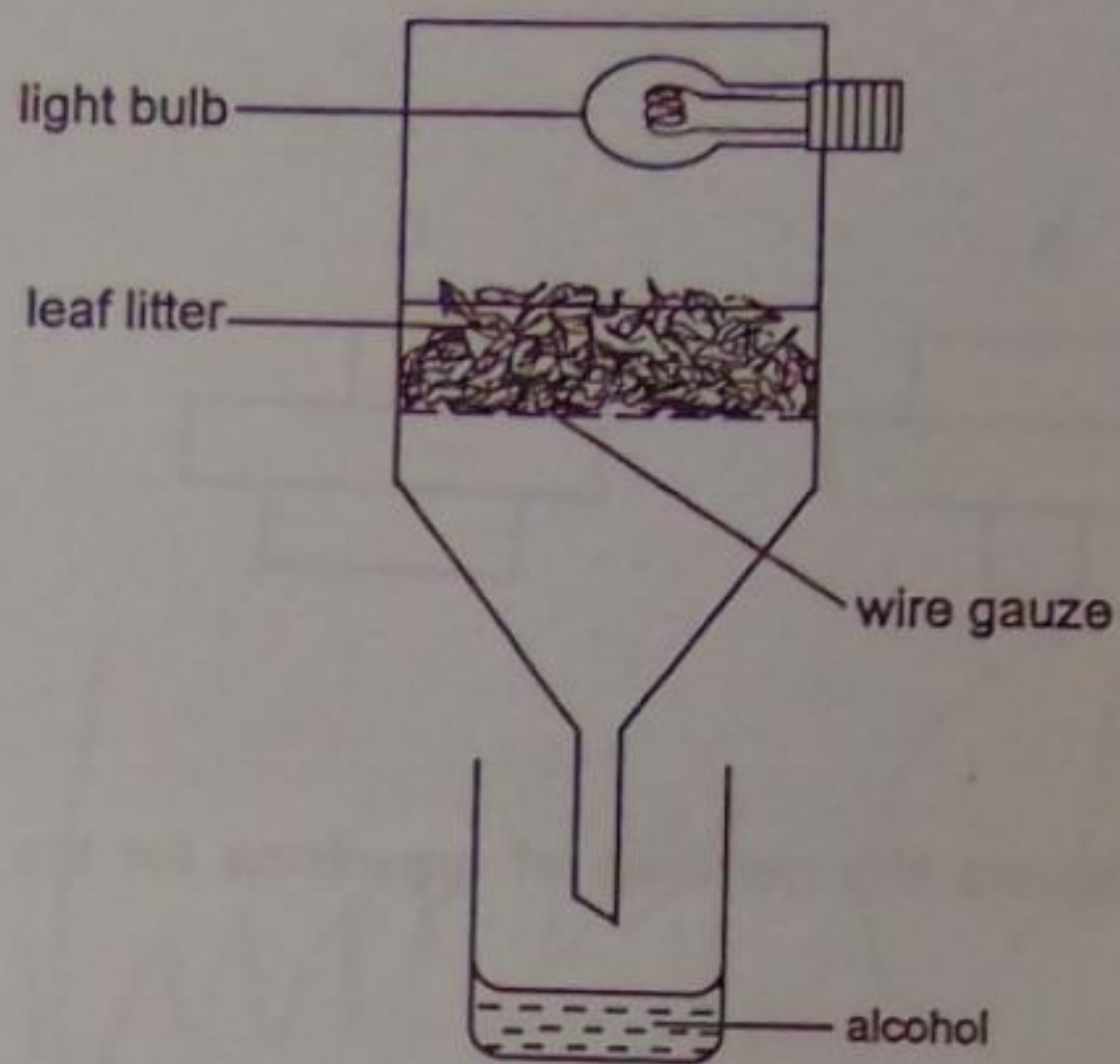


Which diagram shows the numbers of organisms for the following food chain?

grass → zebras → tsetse flies

- 40 The term population can be defined as
- A ~~the number of plants or animals in a given area.~~
 - B ~~the number of people and their domestic animals.~~
 - C the number of plants or animals of a given species present in an ecosystem.
 - D the group of organisms living together in an area.

- 38 The apparatus for trapping small animals is set up as shown below.

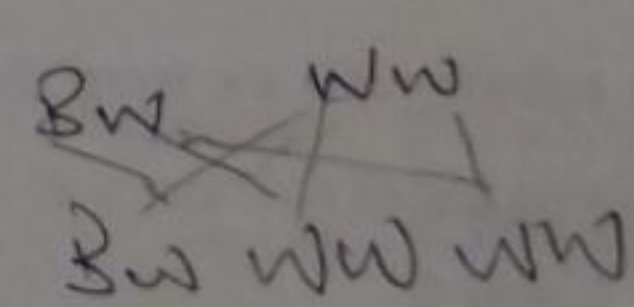


The purpose of the light bulb is to

- A attract the small animals from the leaf litter. ✓
- B to provide warmth to the small animals. ✓
- C to drive the small animals out of the leaf litter. ✓
- D to help keep the temperature of the leaf litter constant.

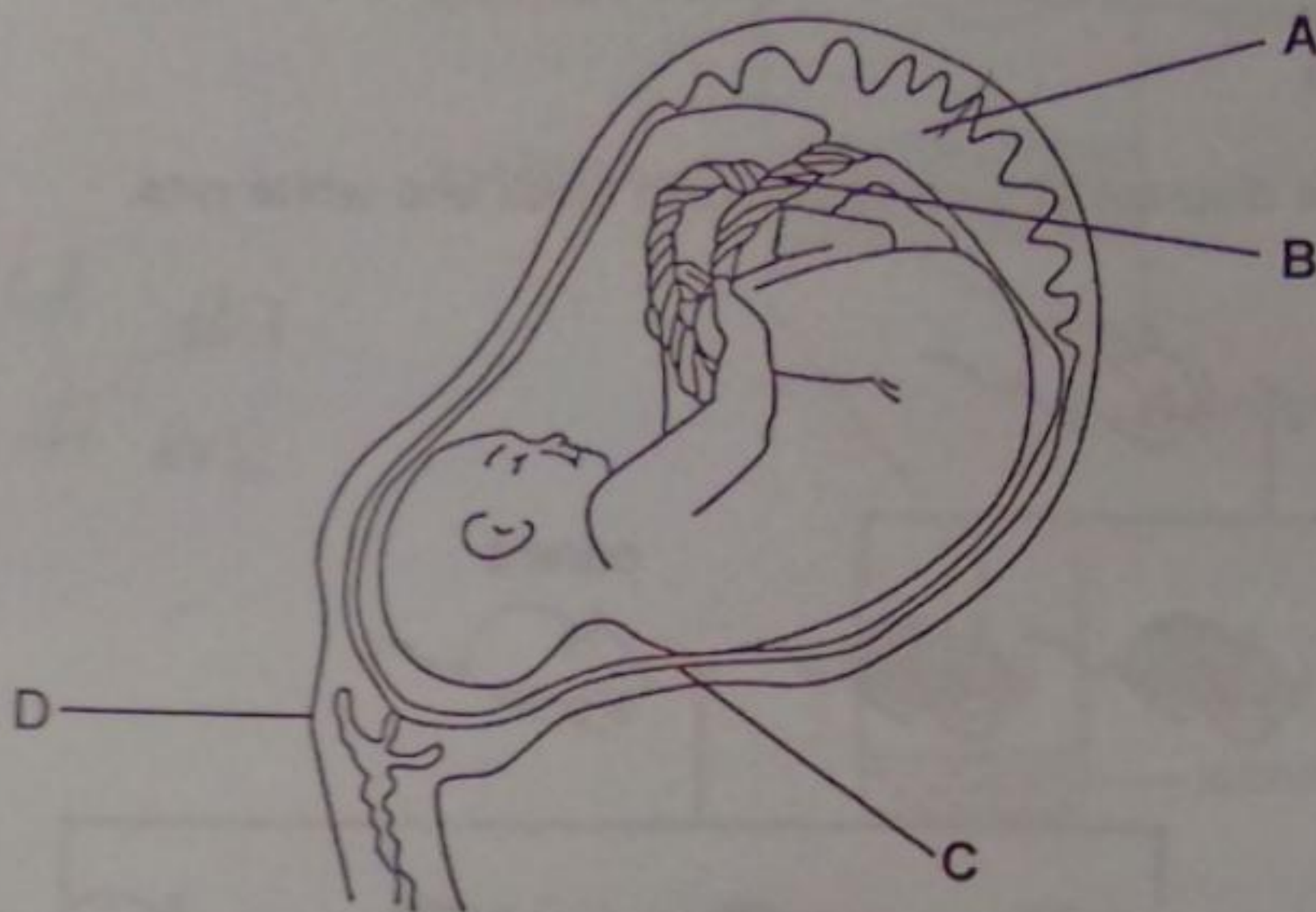
- A headaches
- B muscular cramps
- C loss of appetite
- D weak joints

BB Ww
Bw Wb Wb BW



A	4
B	2
C	8
D	6

- 33 The diagram shows a human foetus in the uterus. Which structure represents the placenta?



- 34 The following list shows substances that can pass across the placenta.

1. antibodies
2. urea ✓
3. oxygen
4. carbon dioxide ✓

Which substances are from the foetus?

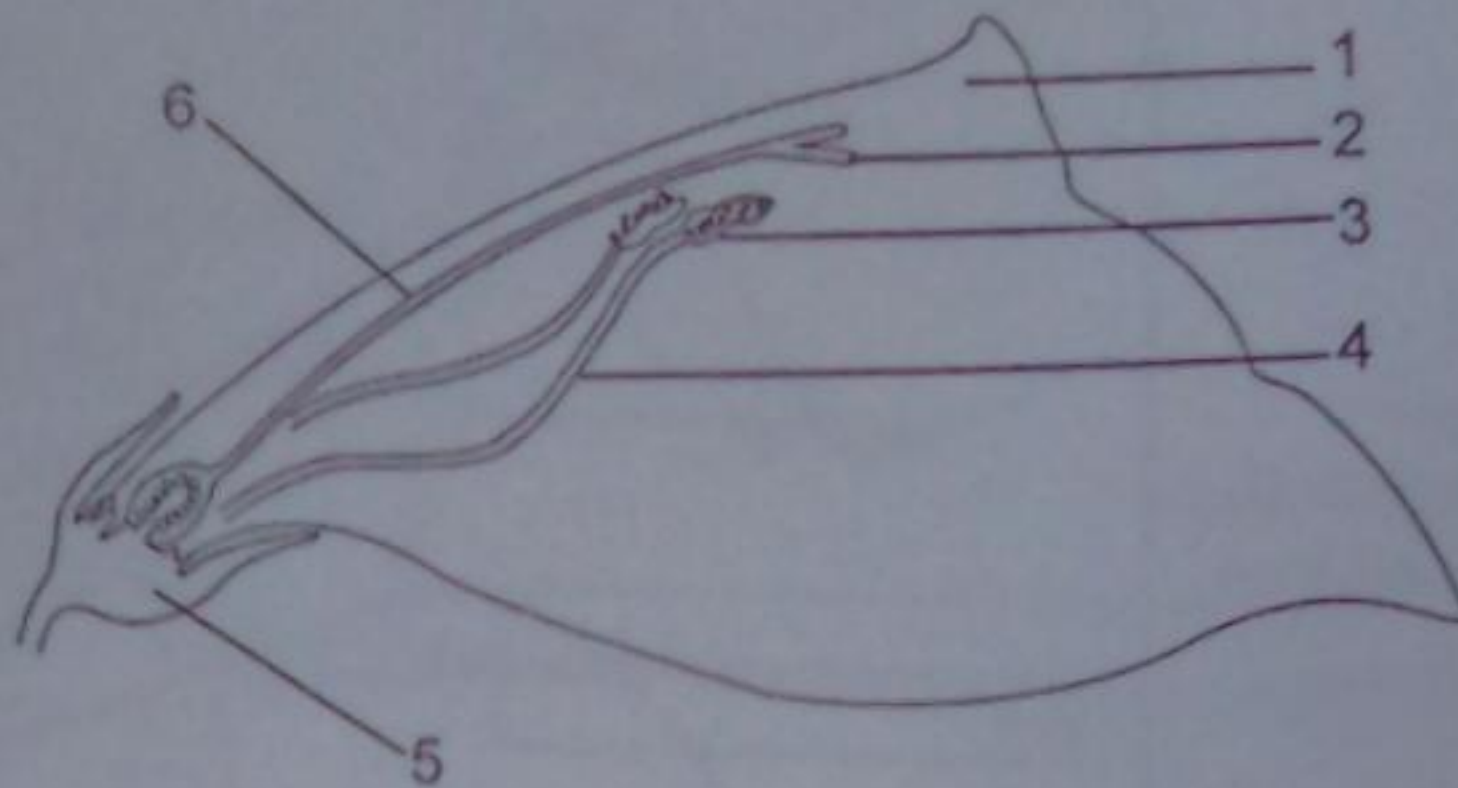
- A 1 and 3
B 2 and 4
C 3 and 4
D 1 and 2

- 35 Name the causative pathogen of malaria.

- A Anopheles mosquito
B bacterium
C virus
D Plasmodium ✓

30

The diagram shows a section through a flower.



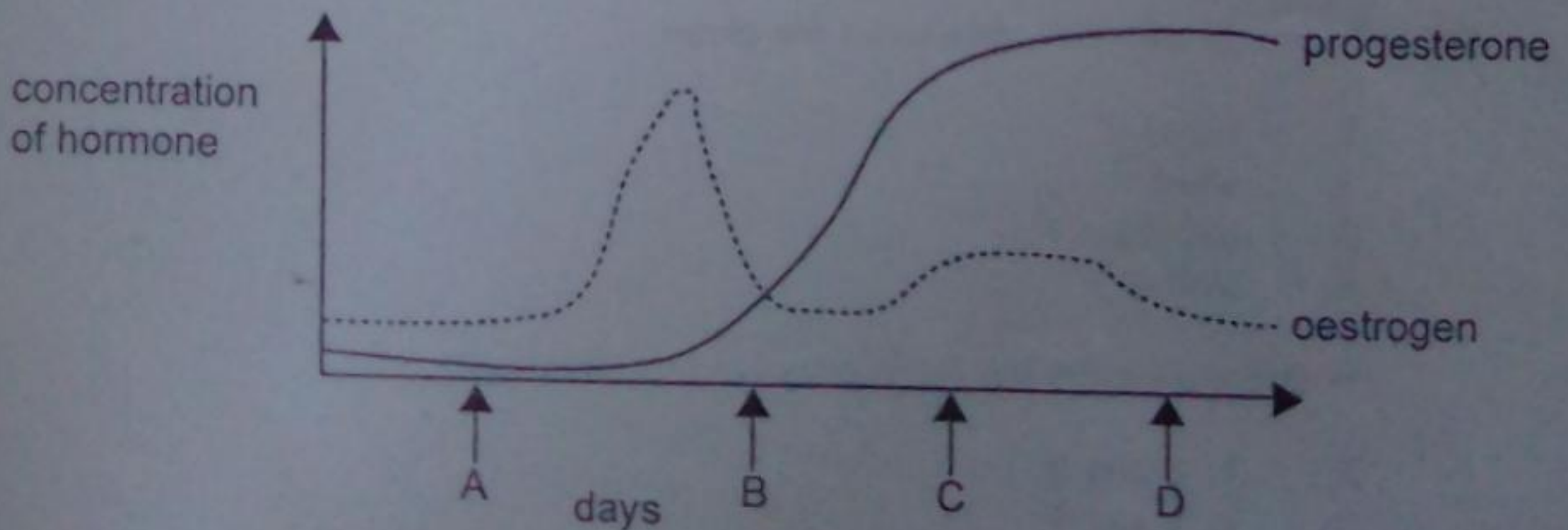
Which structures are part of the pistil?

- A 6 and 2
- B 3 and 4
- C 1 and 5
- D 2 and 3

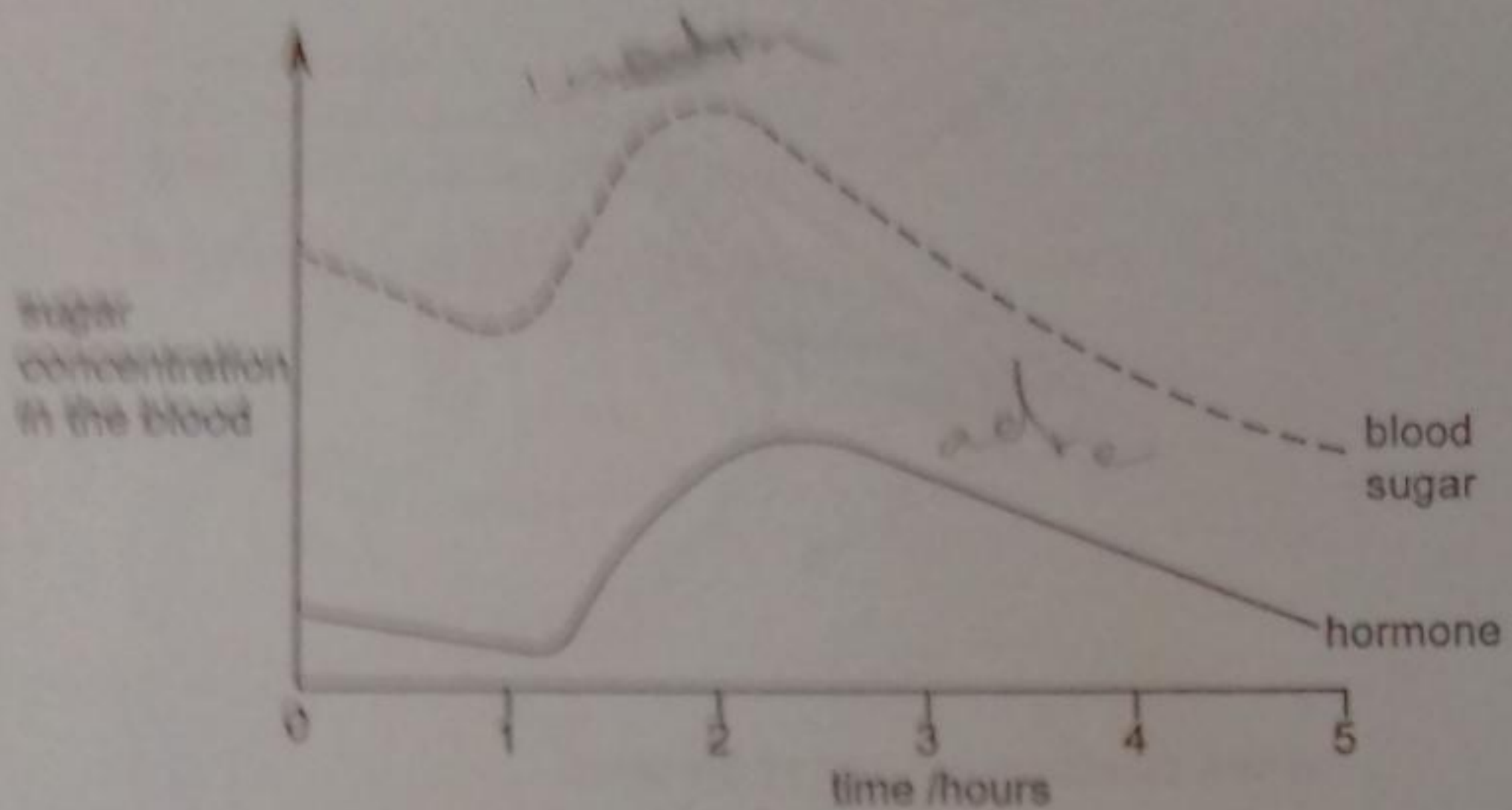
31 Which change occurs naturally without fertilisation in a flowering plant?

- A petals → wither and fall off
- B ovule → seed
- C ovary → fruit
- D zygote → embryo

32 The diagram shows the concentration of oestrogen and progesterone in a healthy young woman. On which day is ovulation most likely?



- 27 The graph below shows the effect of a hormone on blood sugar level of a healthy person.



Which hormone is it?

- A adrenaline
- B oestrogen
- C insulin
- D glucagon

- 28 Analgesics are drugs which are used to cure

- A gonorrhoea.
- B cancer.
- C pain.
- D diabetes.

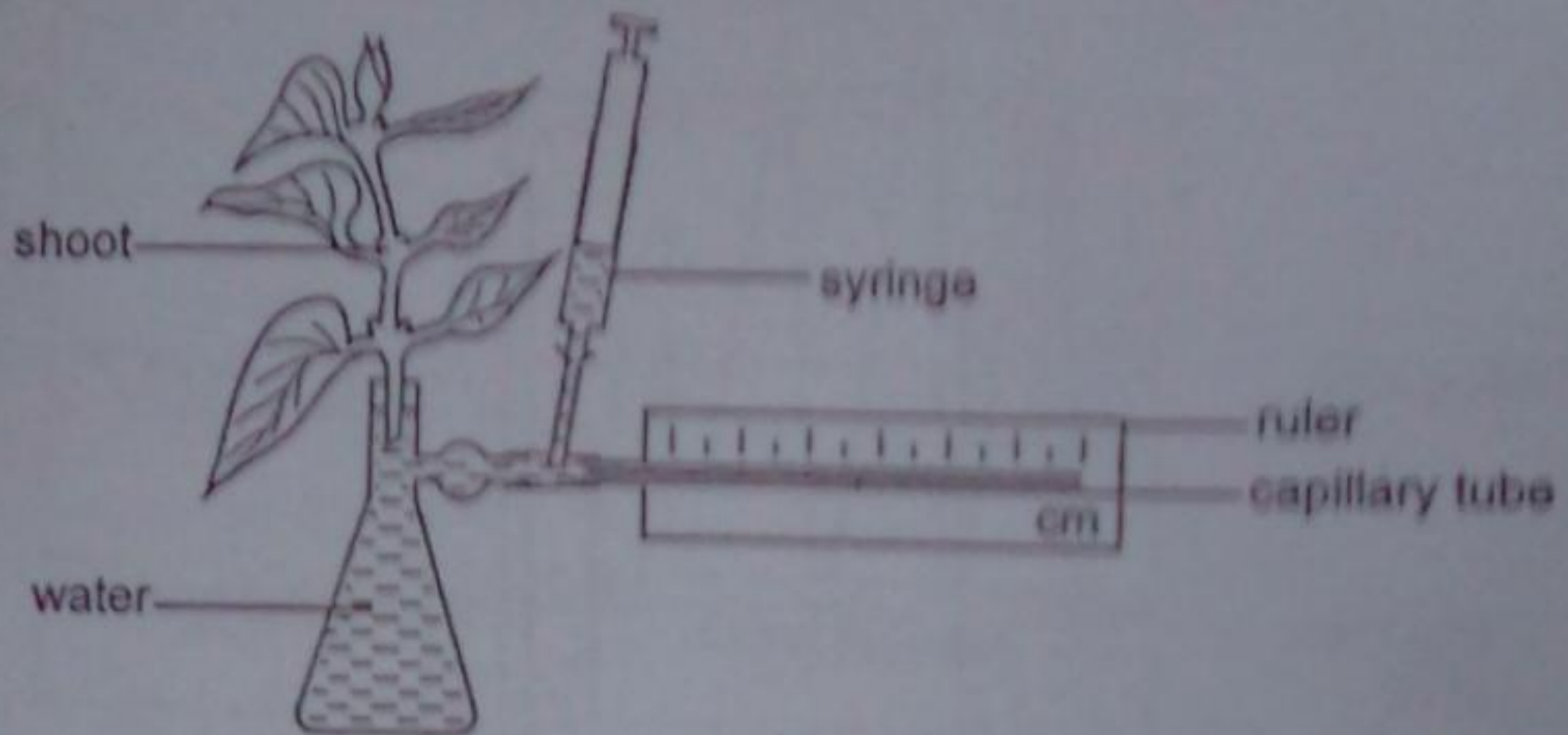
- 29 Four reproductive structures are given

- 1. tuber ✓
- 2. seed ✓
- 3. cutting ✓
- 4. sucker ✓

Which structures are for vegetative reproduction?

- A 1, 2 and 3
- B 1, 3 and 4
- C 2, 3 and 4
- D 3 and 4 only

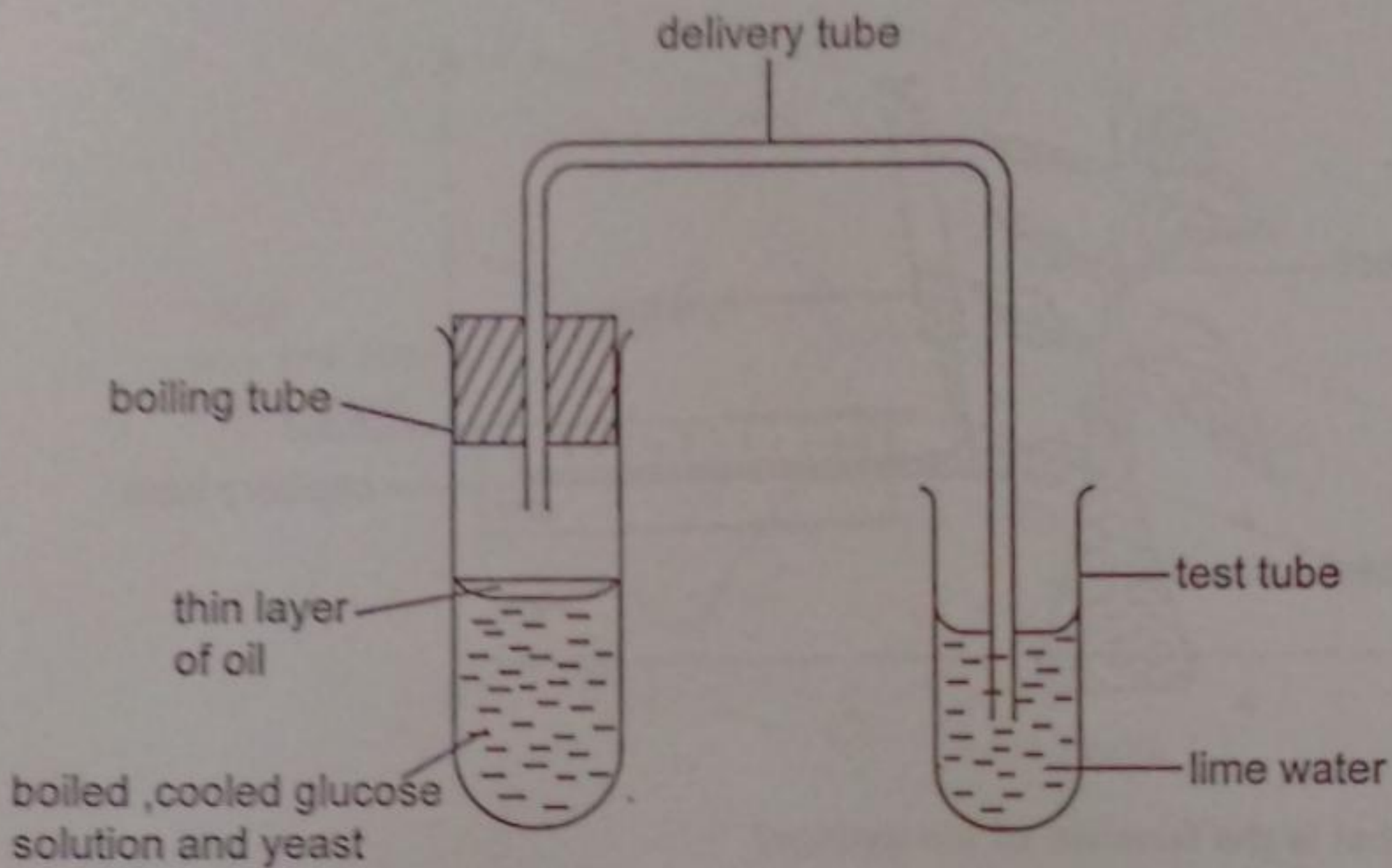
- 17 The diagram shows a potometer.



What is the function of the syringe?

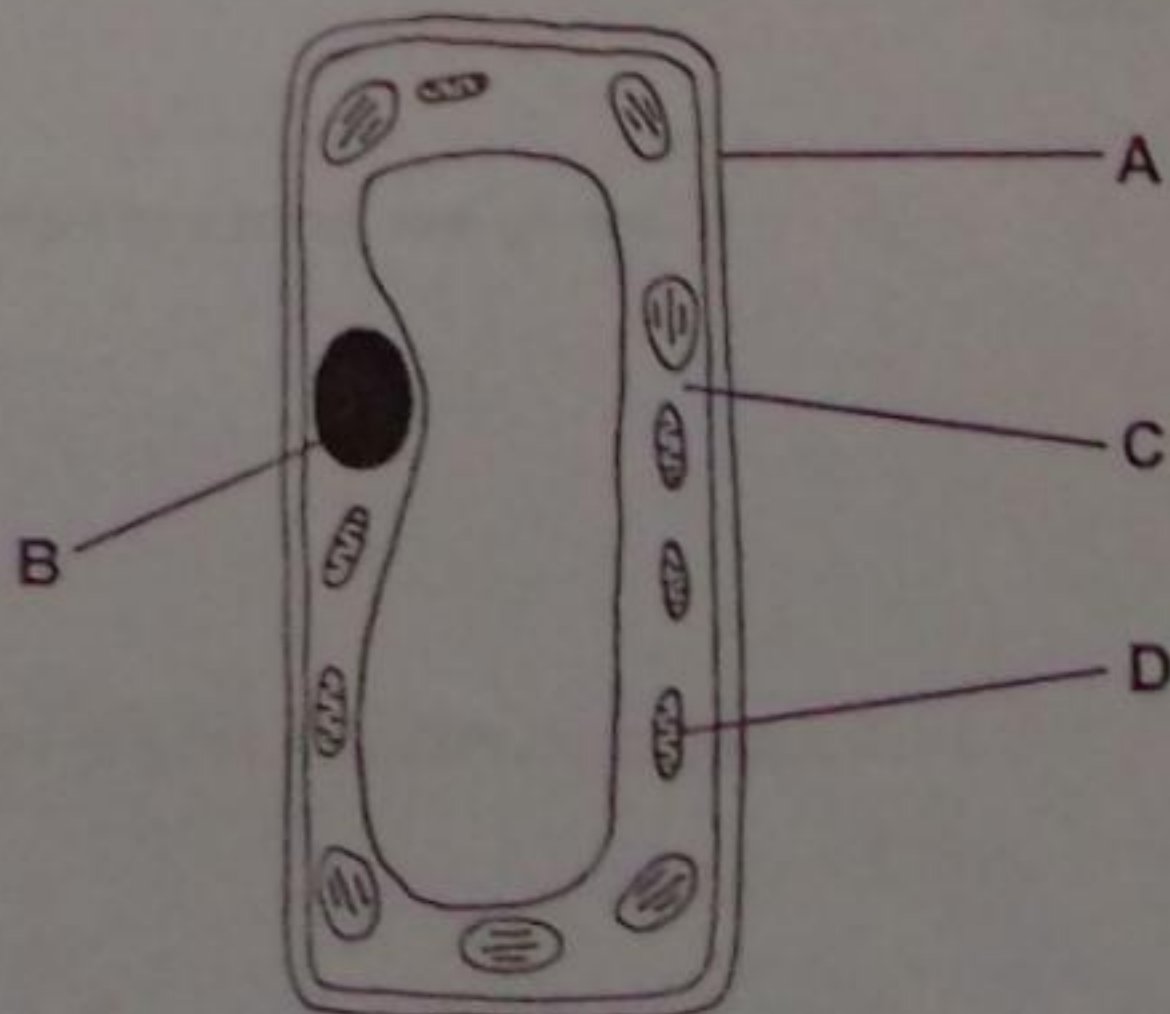
- A to pump water to the shoot for photosynthesis
 - B for measuring the volume of water absorbed by the shoot
 - C for resetting the water level in the capillary tube
 - D to suck some of the water from the capillary tube
- 18 Plant roots absorb mineral salts from the soil by
- A active transport.
 - B ion exchange.
 - C diffusion.
 - D osmosis.

- 15 An experiment to investigate whether carbon dioxide is given off during fermentation was set up as shown below.

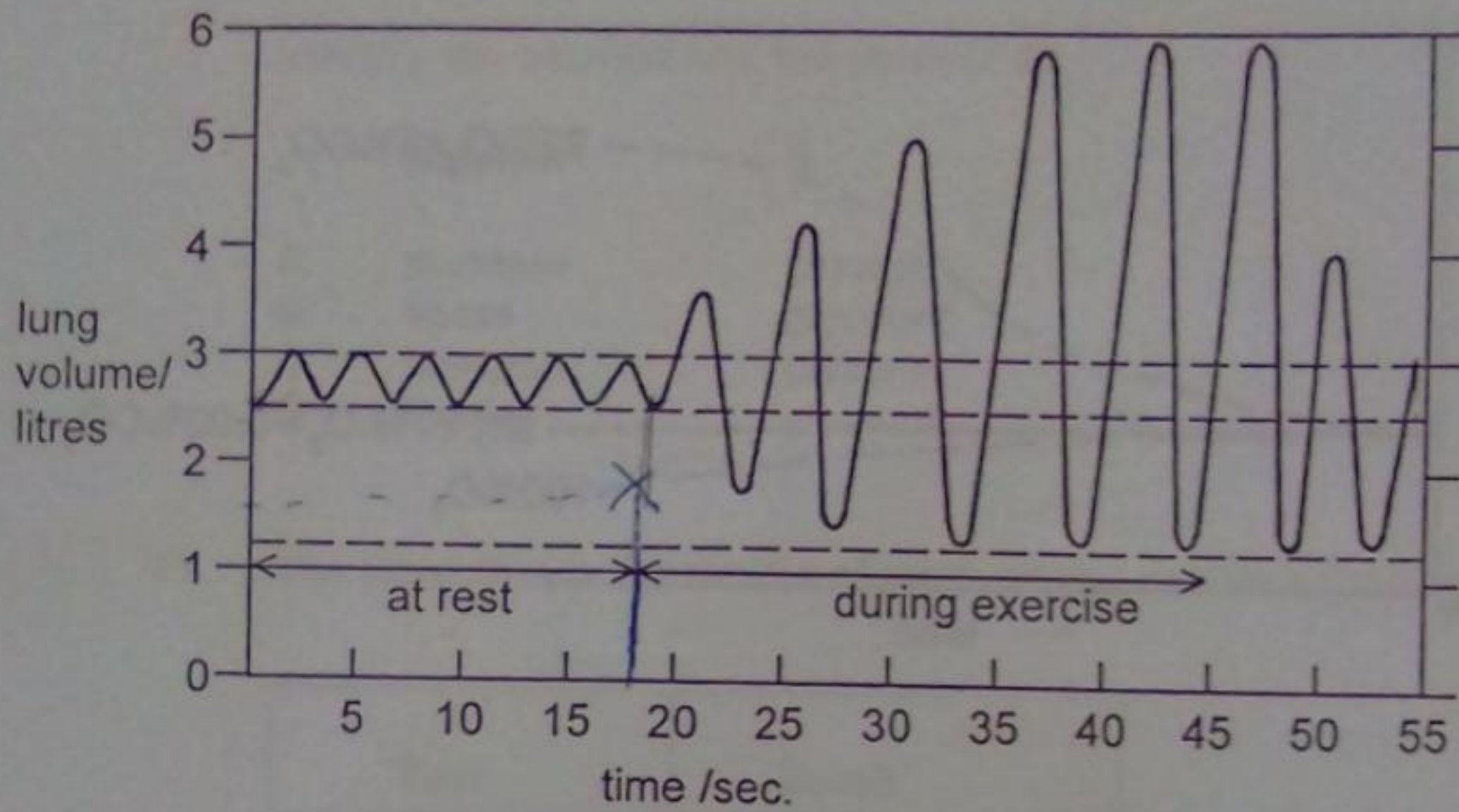


What do you expect to see after some time?

- A The layer of oil would disappear.
 - B The lime water would remain the same.
 - C The lime water would turn milky.
 - D The layer of oil sinks.
- 16 In which structure of the cell shown below does aerobic respiration take place?



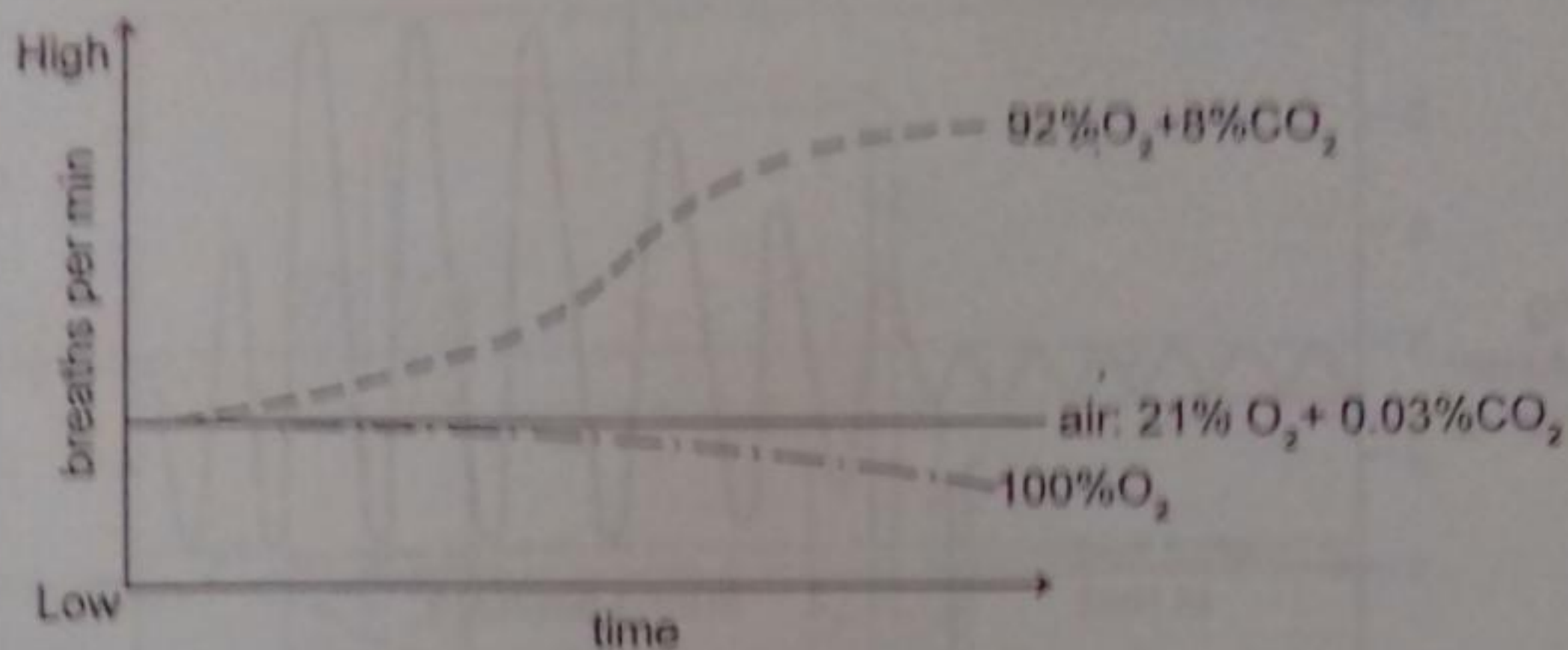
- 14 The graph shows changes in volume of air in the lungs during breathing.



Estimate the maximum volume per breath of air leaving the lungs during normal breathing.

- A 0,5 litre
- B 1,0 litres
- C 1,5 litres
- D 2 litres

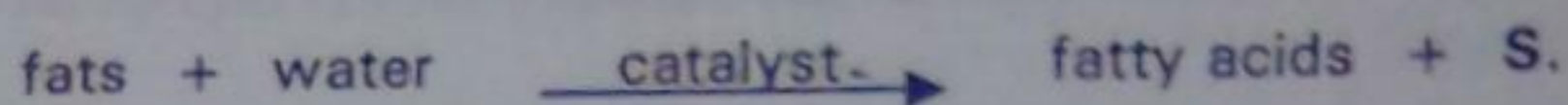
- 13 The graph shows the effect of percentage composition of oxygen and carbon dioxide in air on the breathing rate in a healthy person.



Which statement about the graph is true?

- A Breathing rate increases when the person breathes 100% oxygen.
- B Breathing rate increases when the percentage of carbon dioxide increases.
- C Breathing rate decreases when the person breathes 92% oxygen.
- D Breathing rate decreases when the percentage of carbon dioxide increases.

- 10 Fats are hydrolysed as shown by the equation:



Identify the catalyst and the product S.

	Catalyst	S
A	protease	glucose
B	lipase	glycogen
C	amylase	glycerol
D	lipase	glycerol

- 11 The table shows the results of tests carried out on a food sample.

Test	Result
Benedict's	negative
Biuret	positive
iodine	positive
emulsion	negative

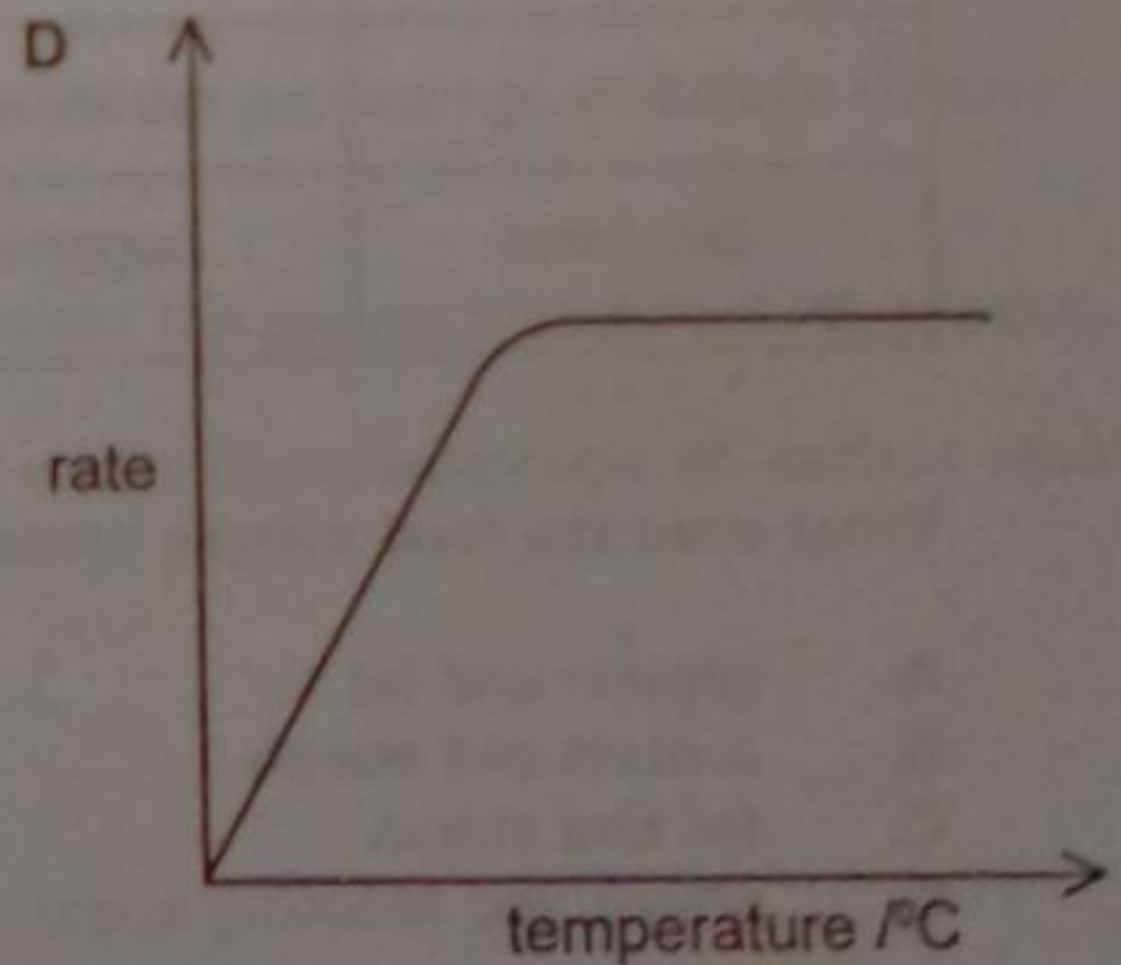
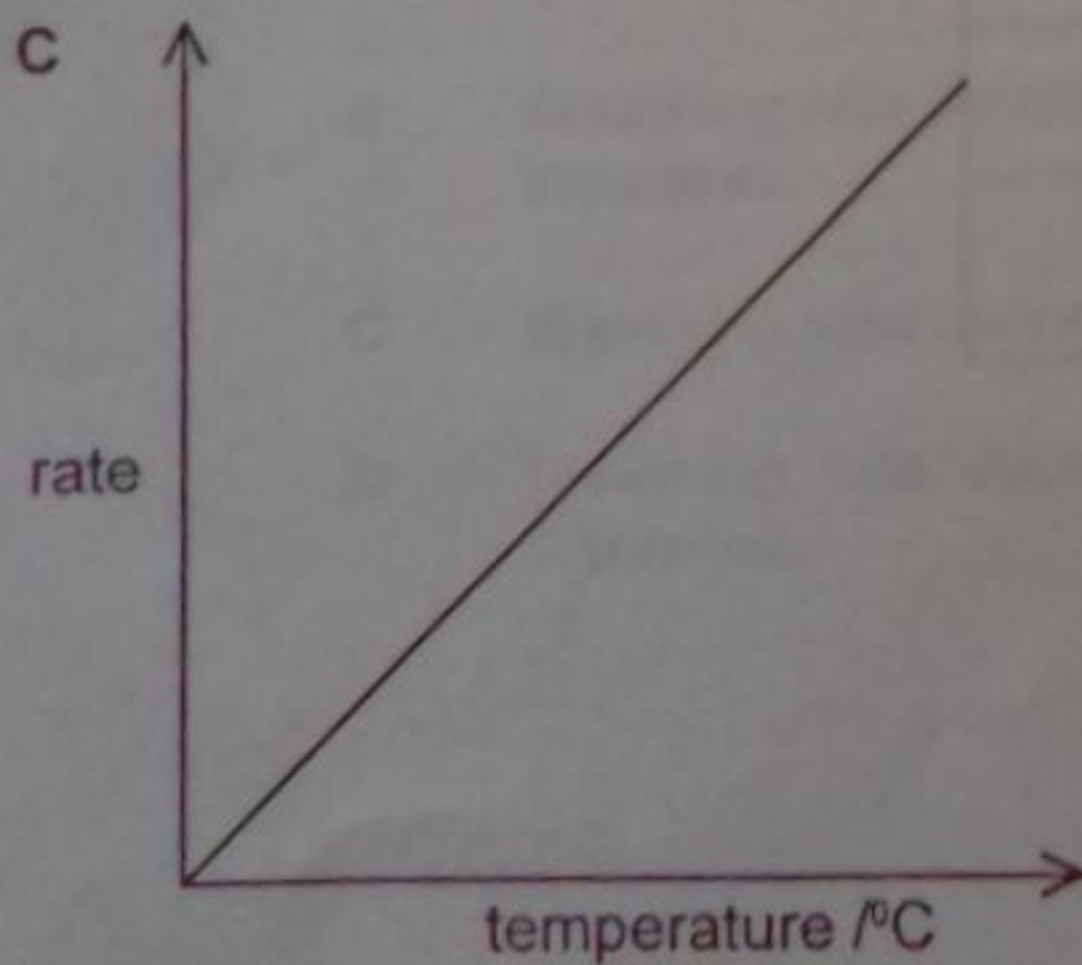
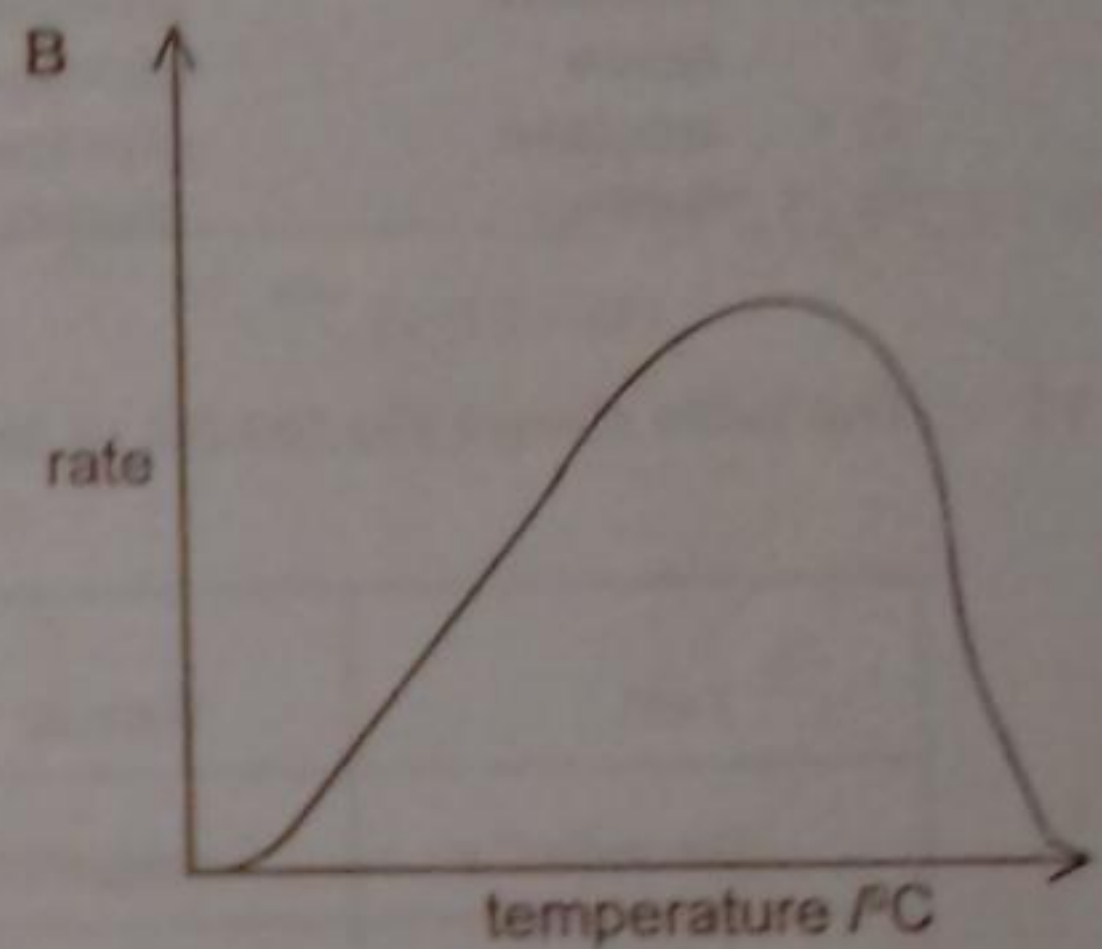
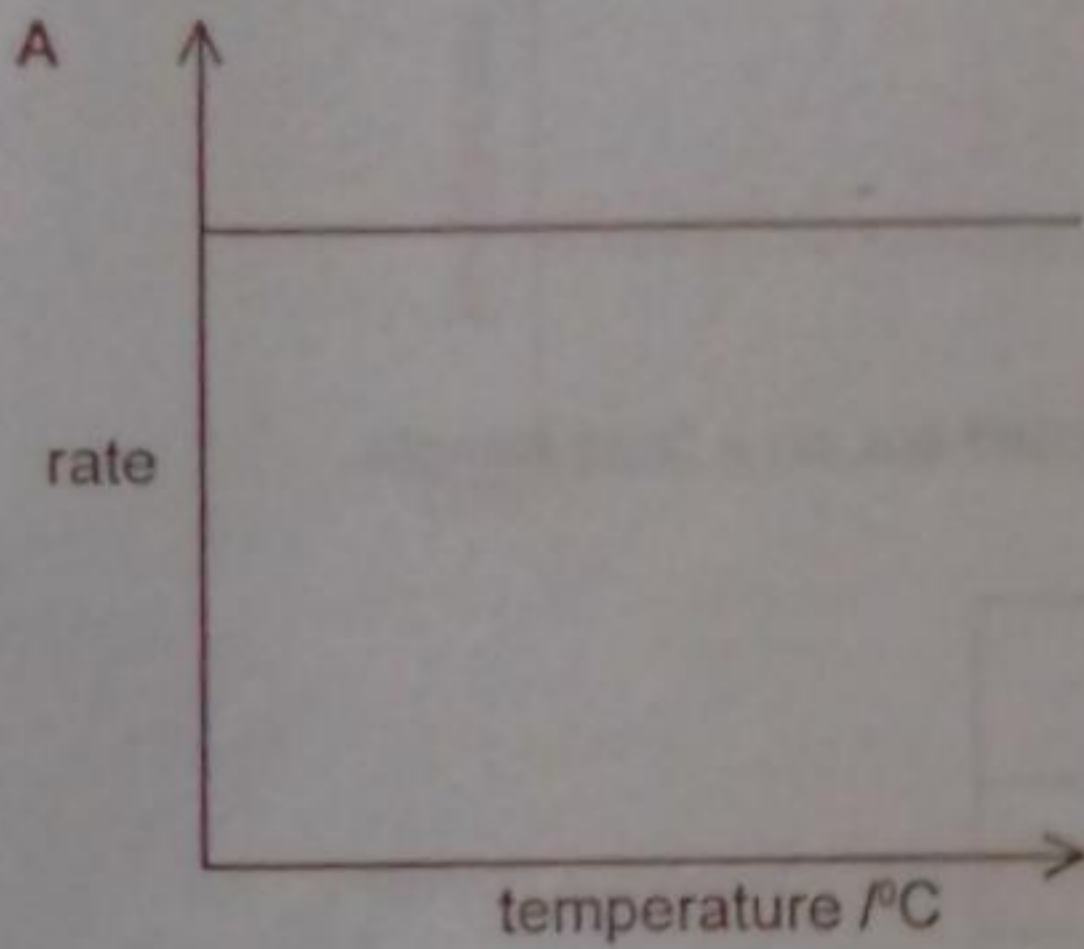
What does the food sample contain?

- A. protein and fat
- B. protein and starch
- C. fat and starch
- D. starch and reducing sugar

- 12 Which is an effect of tar from cigarette smoking?

- A lung cancer
- B emphysema
- C slurred speech
- D coronary disease

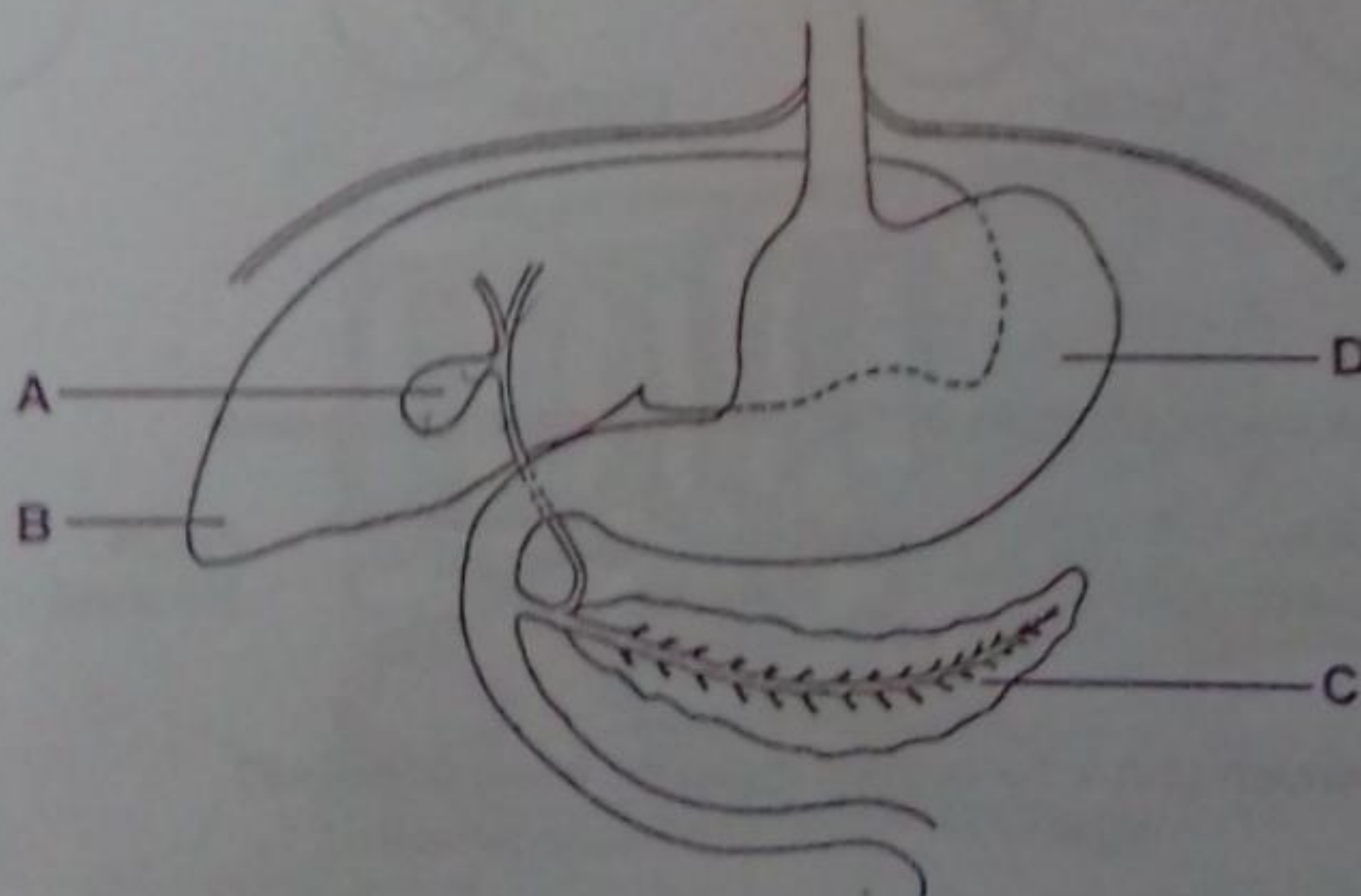
- 9 Which graph represents the effect of temperature on the rate of an enzyme catalysed reaction?



7 Which comparison about respiration and photosynthesis is incorrect?

	Respiration	Photosynthesis
A	Energy is evolved.	Energy is stored in the form of carbohydrates.
B	Oxygen is used up.	Oxygen is given off.
C	It occurs at all times in cells.	Only occurs in cells containing chlorophyll.
D	Results in a gain in dry mass.	Results in a loss in dry mass.

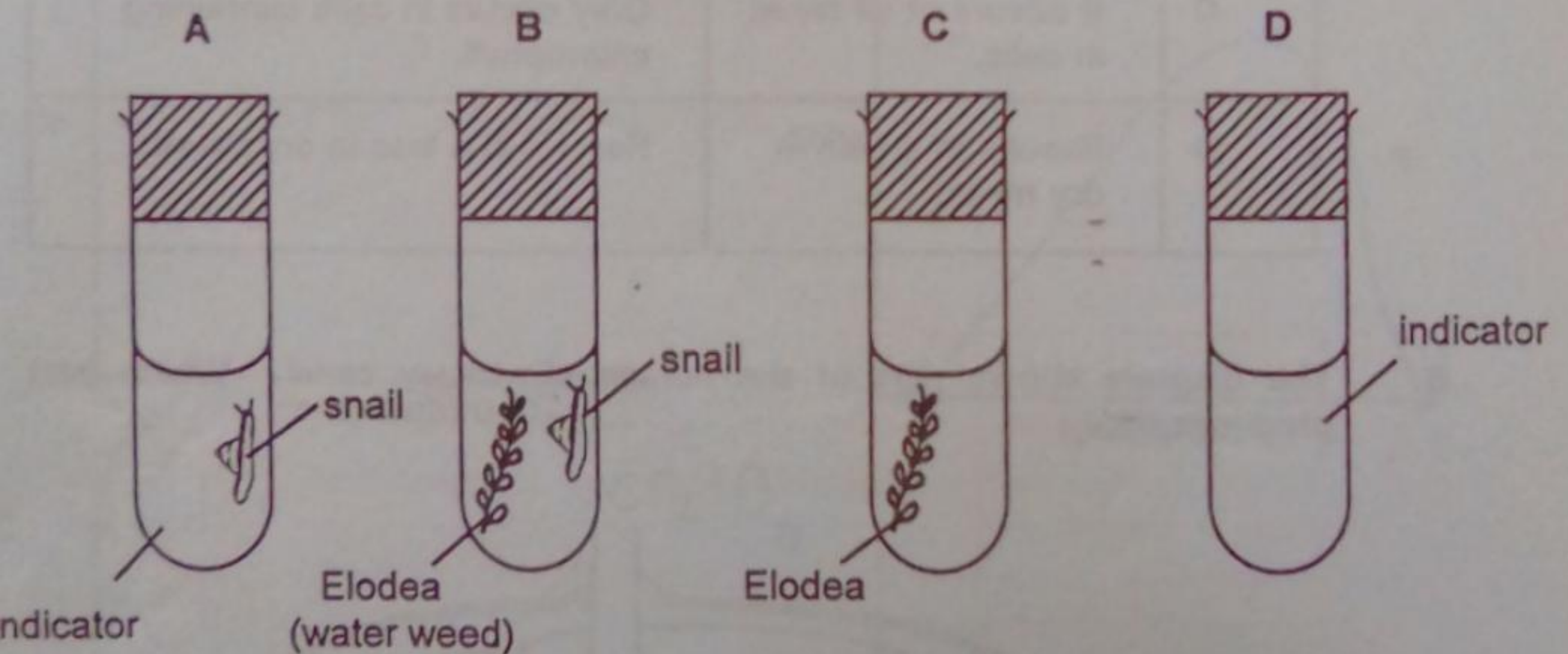
8 The diagram shows part of the human alimentary canal. Which part produces bile?



5 Which element is poisonous to plants?

- A potassium
- B phosphorus
- C nitrogen
- D copper

6 Four test tubes are set up as shown below. All contain the same amount of a red indicator and are left standing in bright light for a few hours.



The colour changes of the indicator are shown below

Yellow (acidic) ← Red (neutral) → Purple (alkaline)

In which test tube will the indicator become yellow?

1 Which is a function of water in living organisms?

- A essential constituent of protoplasm
- B essential constituent of the cell wall
- C source of energy
- D acts as a catalyst

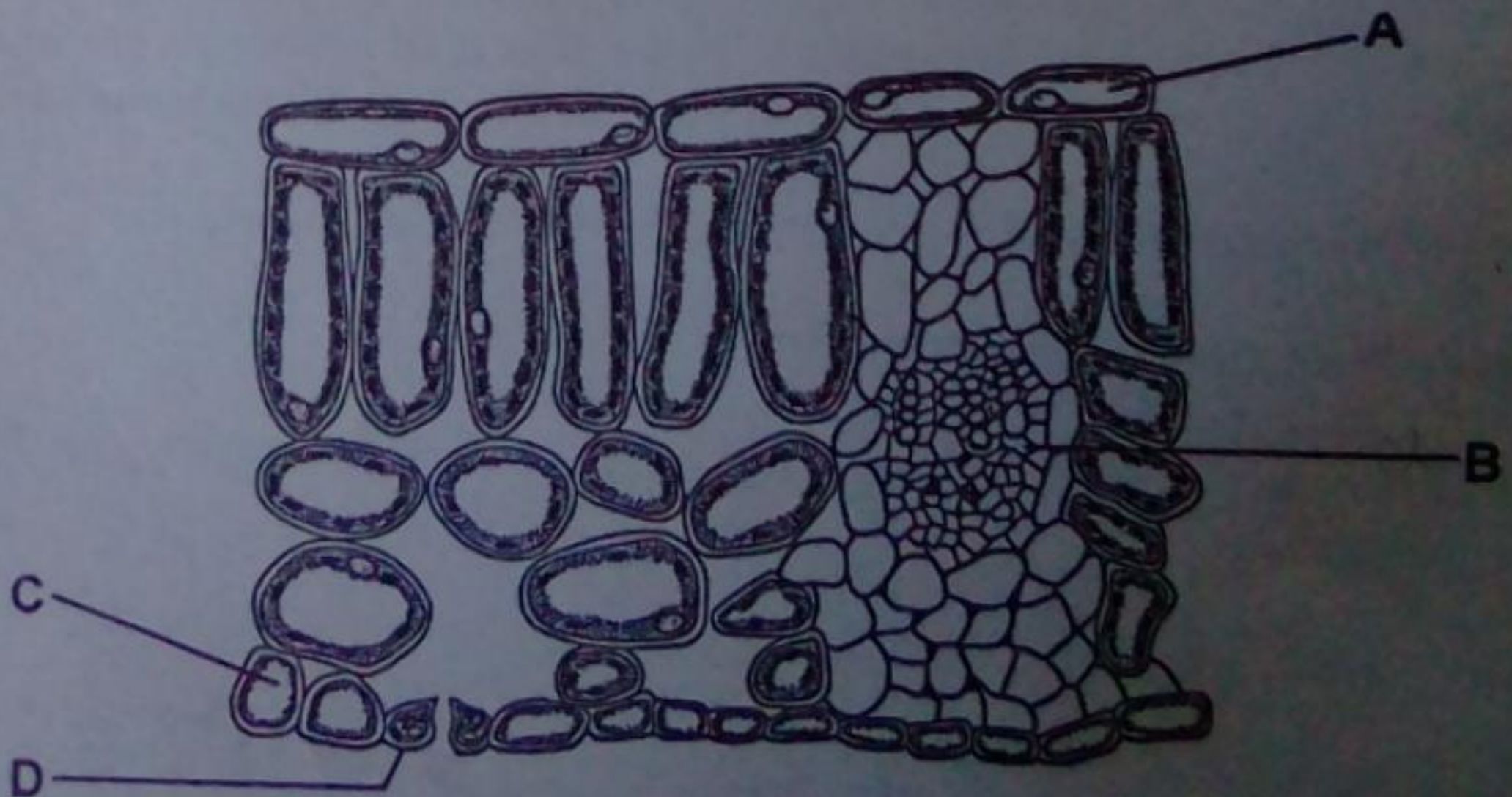
2 Which is an adaptation of a root hair cell for absorption of water?

- A It is short and narrow.
- B It is long and narrow.
- C It is long and wide.
- D It is short and wide.

3 Which cell structure is **not** common to both plants and animals?

- A cytoplasm
- B cell wall
- C nucleus
- D cell membrane

4 The diagram shows the arrangement of cells inside a leaf. In which cell does photosynthesis take place?



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple choice

Friday

22 NOVEMBER 2002

Morning

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

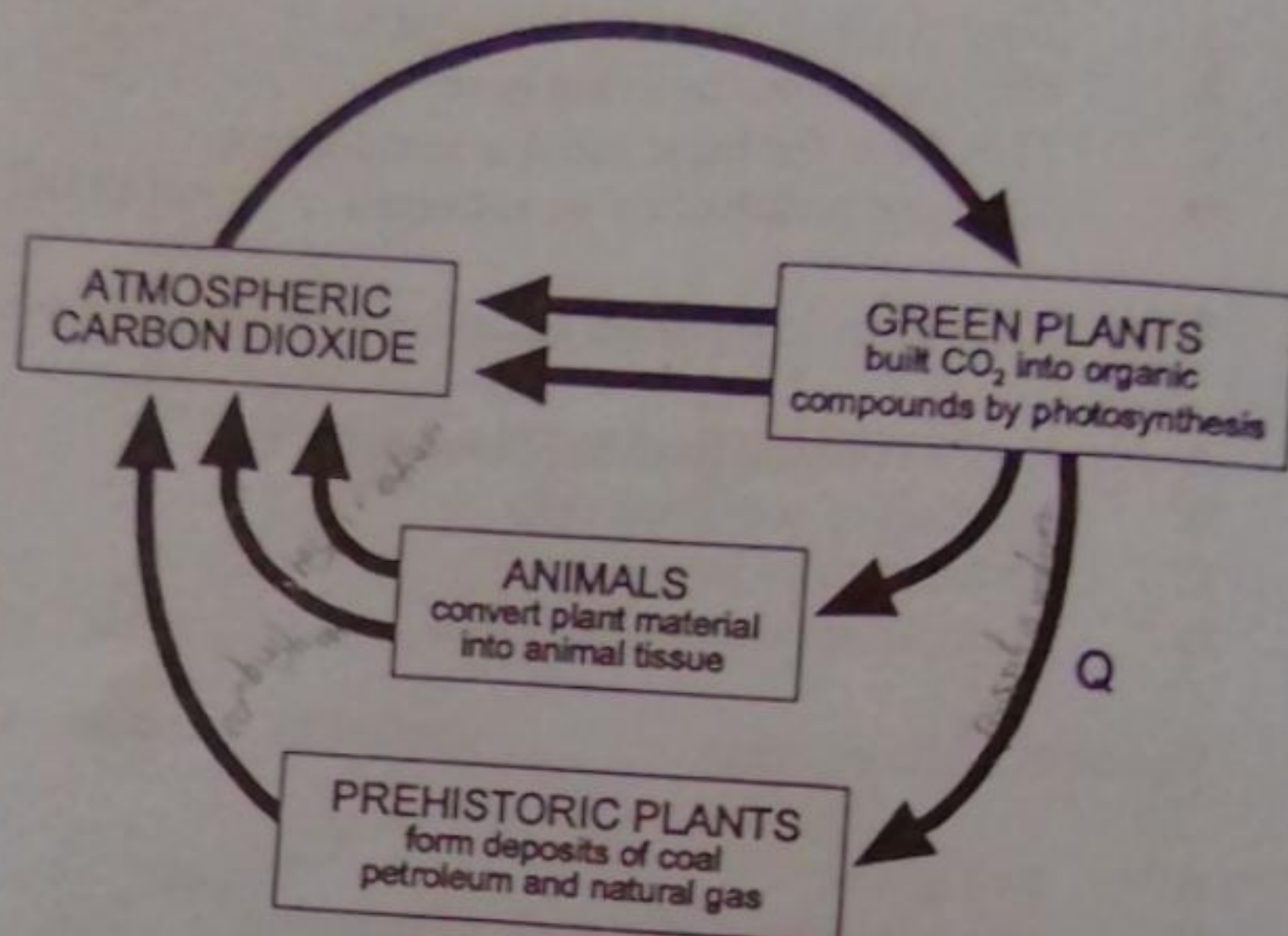
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

PAUL TAFF 2002
CHING CHOO

This question paper consists of 20 printed pages and 4 blank pages.

- 39 The diagram shows the carbon cycle.

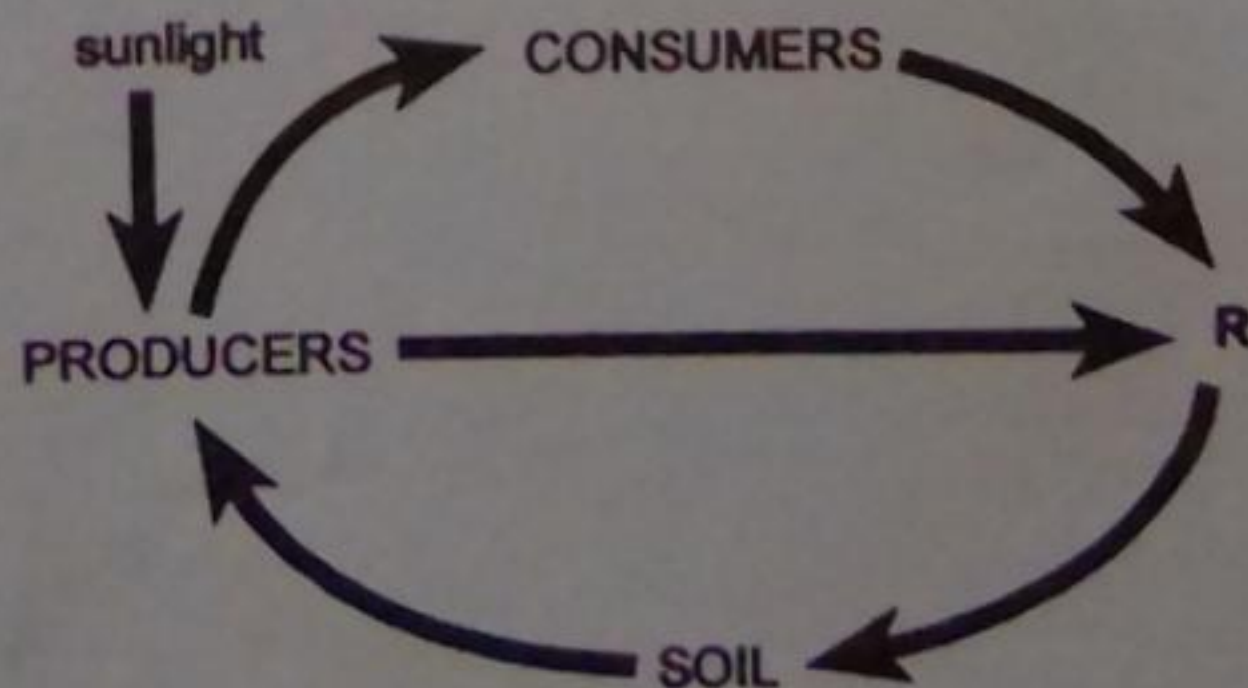


Identify process Q.

- A respiration
 - B combustion
 - C fossilisation
 - D decaying
- 40 The formula for determining the time it takes for human population to double is

- A $\% \text{ growth} \times \text{dependency ratio}.$
- B $\frac{\text{dependency ratio}}{\% \text{ growth}}.$
- C $\frac{70}{\% \text{ growth}}.$
- D $\frac{70}{\text{dependency ratio}}.$

- 34 Which statement is true about viruses?
- A They produce enzymes to digest food.
 - B They reproduce in living host.
 - C They have the basic cellular components.
 - D They can be killed by an antibiotic.
- 35 What causes fever in a patient suffering from malaria?
- A toxins released from red blood cells
 - B bursting of red blood cells
 - C destruction of liver cells
 - D multiplication of plasmodium
- 36 The term allele refers to
- A two genes.
 - B genes produced as a result of mutation.
 - C genes of true breeding organisms.
 - D alternative forms of the same gene.
- 37 Nitrifying bacteria change
- A nitrates to nitrogen.
 - B organic remains to ammonia.
 - C ammonium compounds to nitrates.
 - D nitrogen to nitrates.
- 38 The diagram shows a cycle of nutrients in an ecosystem.



What does R represent in the cycle?

- A secondary consumers
- B tertiary consumers
- C decomposers
- D plants

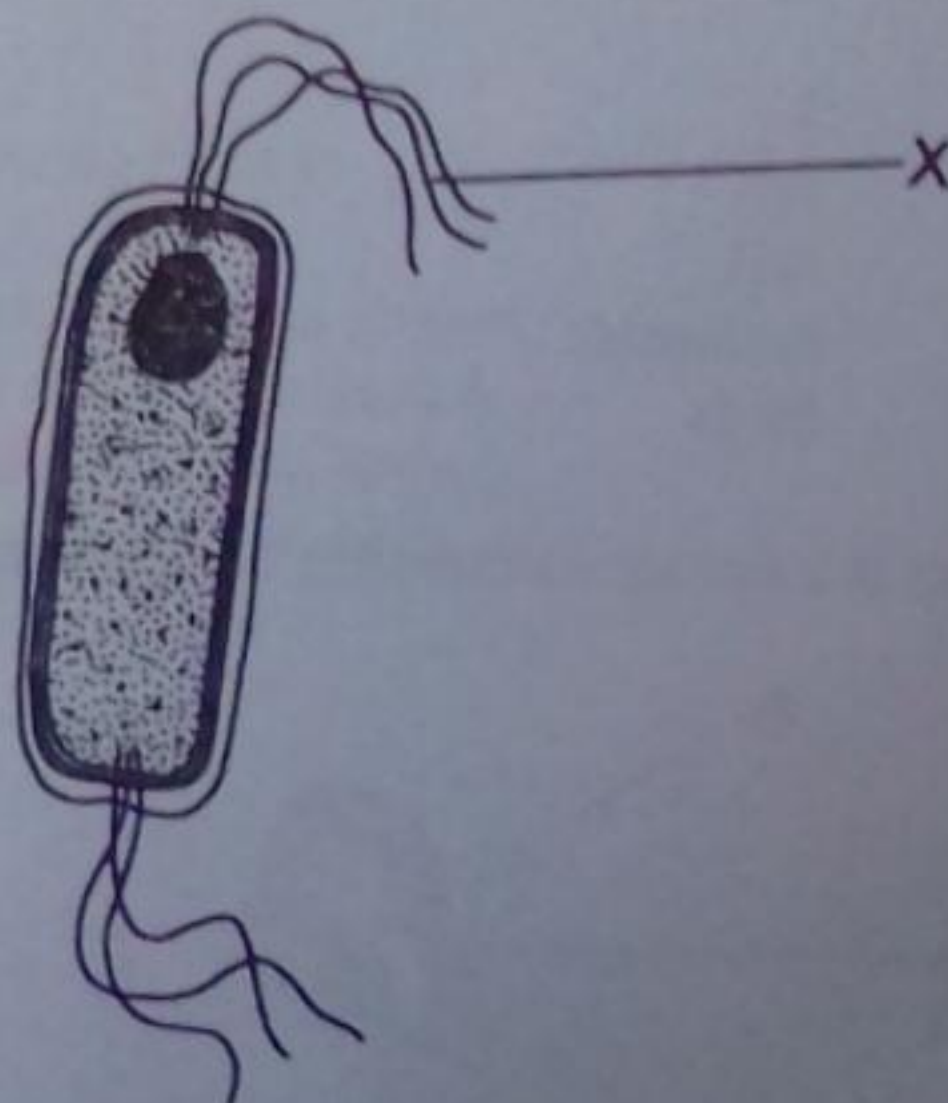
- 32 The following are symptoms of a sexually transmitted disease:

pus from penis
sterility
burning sensation when urinating

Which disease causes the symptoms?

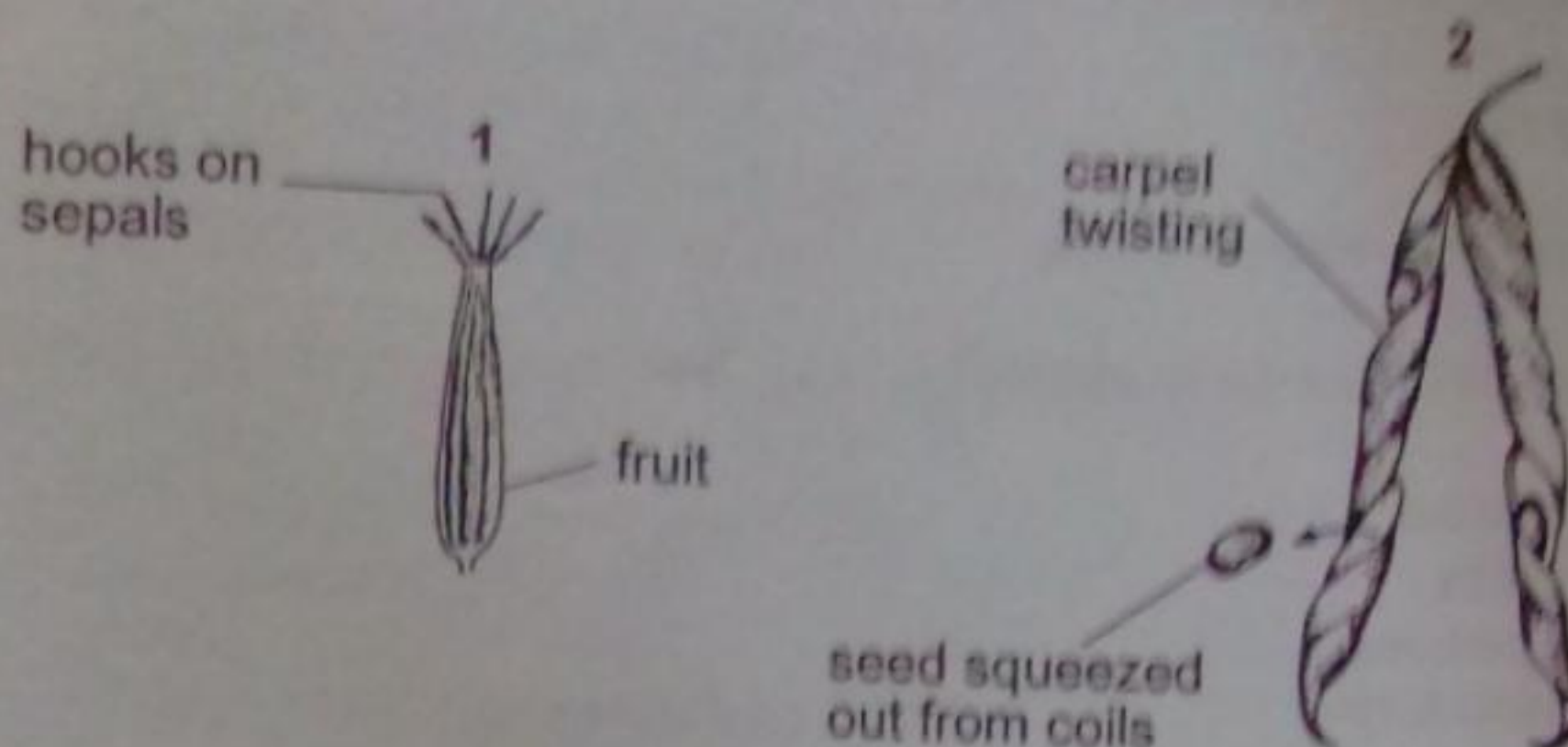
- A chancroid
- B syphilis
- C gonorrhoea
- D genital herpes

- 33 The diagram shows the structure of a bacterium.



What is the function of the part labelled X?

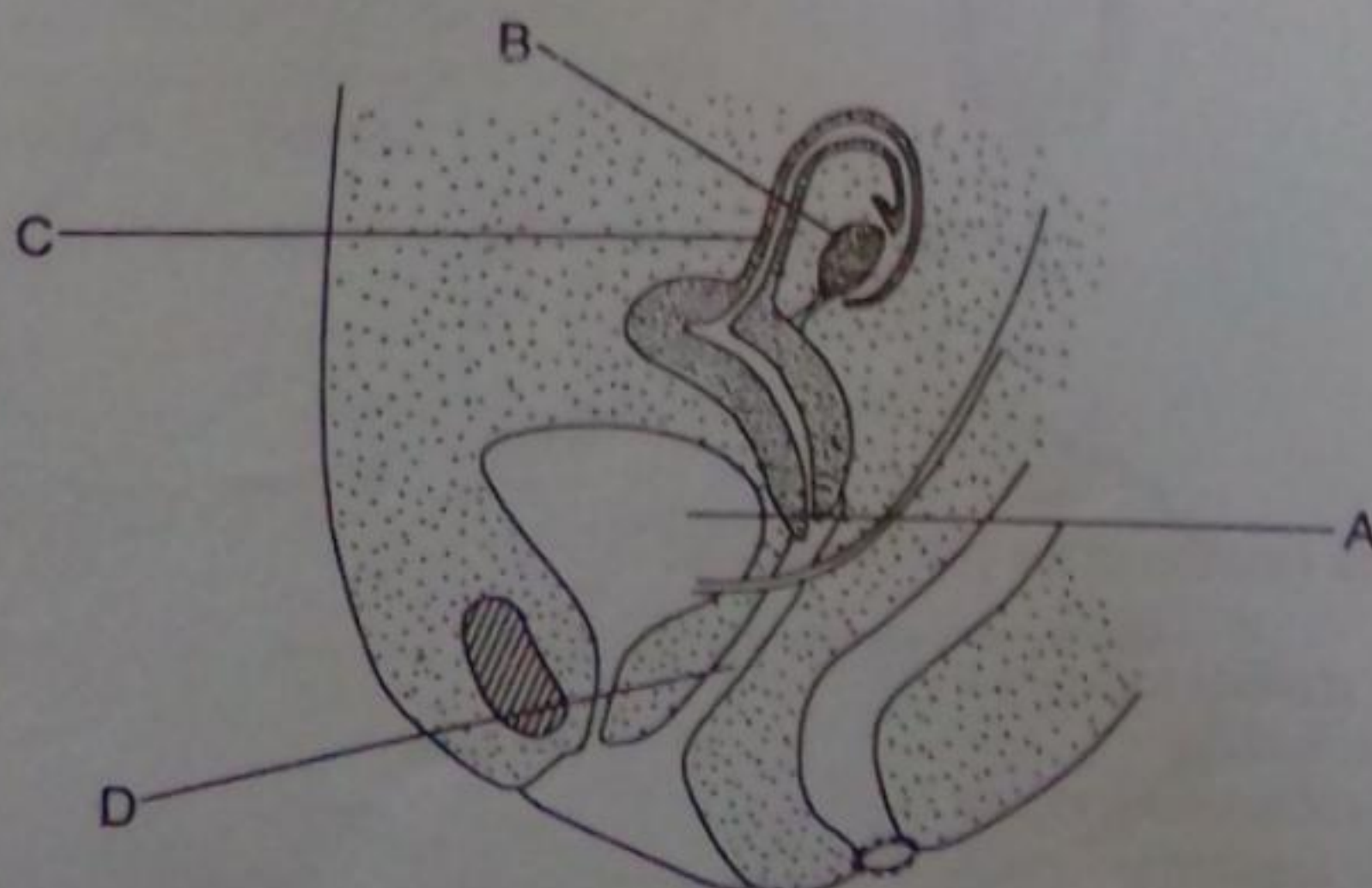
- A to produce movement
- B for feeding
- C for attachment to host
- D for sensitivity



What are the agents of dispersal of seeds of these fruits?

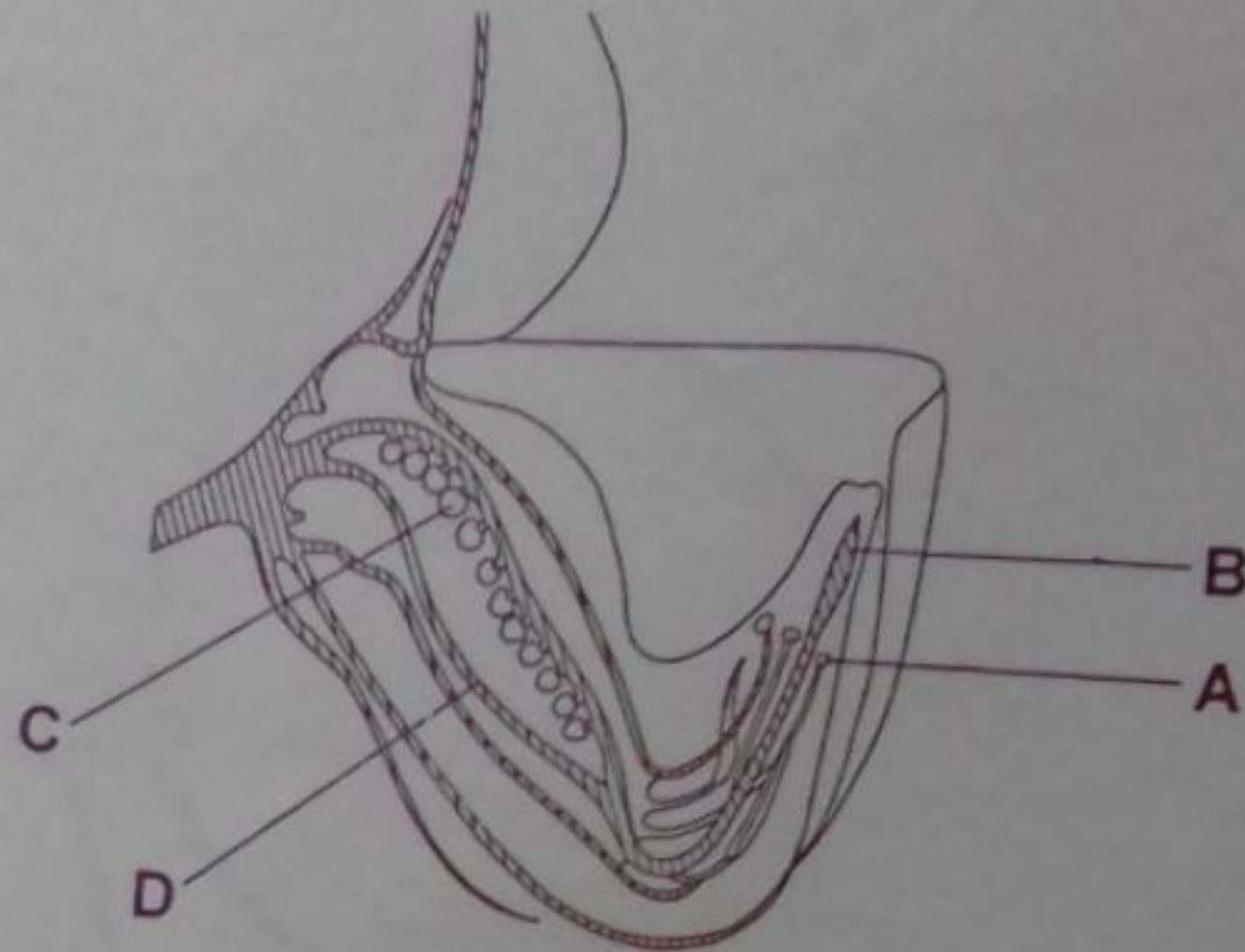
- | | 1 | 2 |
|---|---------|---------|
| A | mammals | self |
| B | wind | self |
| C | birds | mammals |
| D | mammals | birds |

31 The diagram shows part of the female mammalian urinogenital system.



Where are sperms deposited during sexual intercourse?

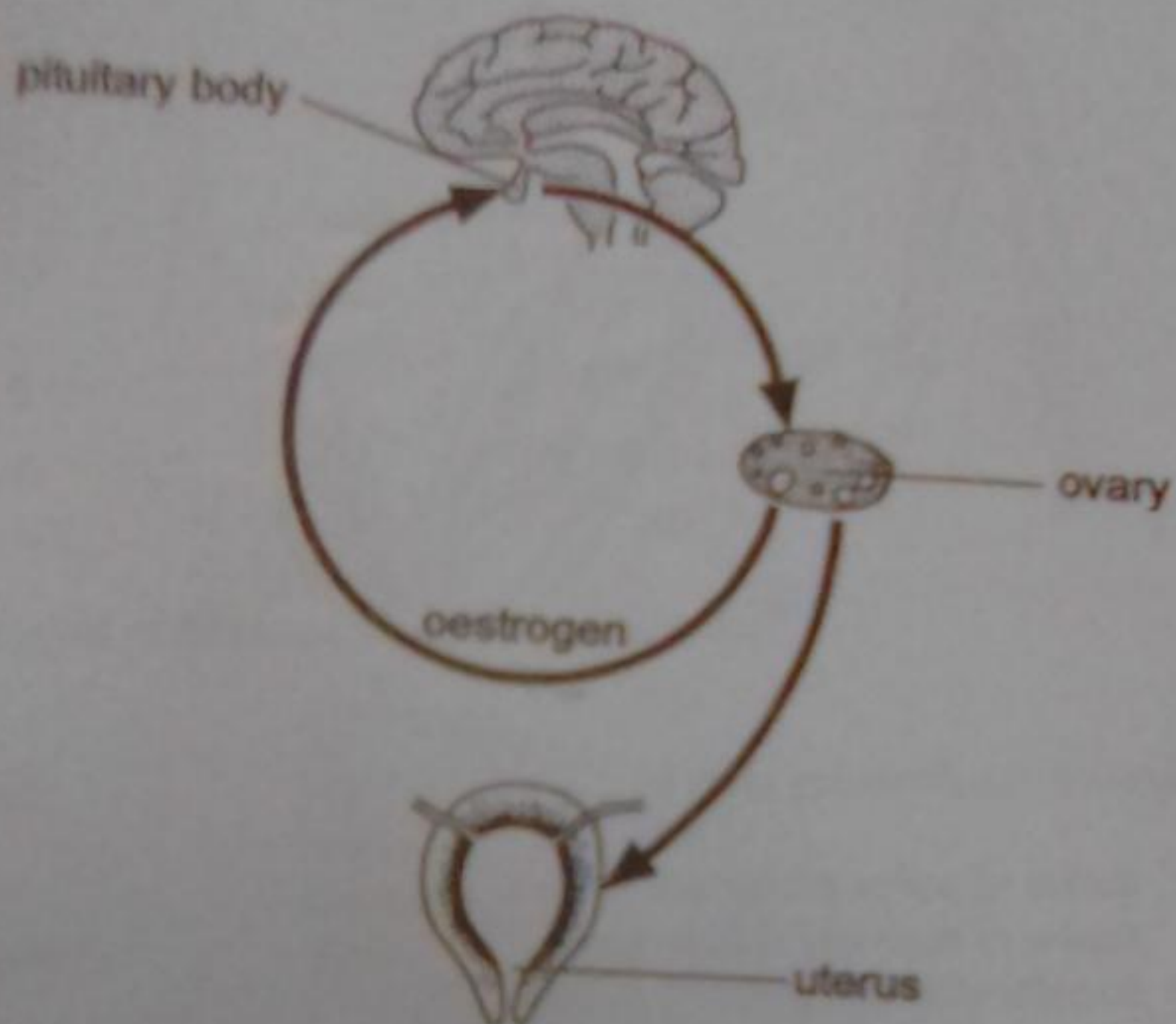
- 28 The diagram shows a flower cut in half. Which part must receive pollen grains first before fertilisation can take place?



- 29 Fertilisation in a flowering plant is

- A fusion of pollen grains and ovules.
- B fusion of male and female nuclei.
- C transfer of pollen grains to stigma.
- D fusion of pollen grain and ovum.

26 The diagram below shows part of the endocrine system.



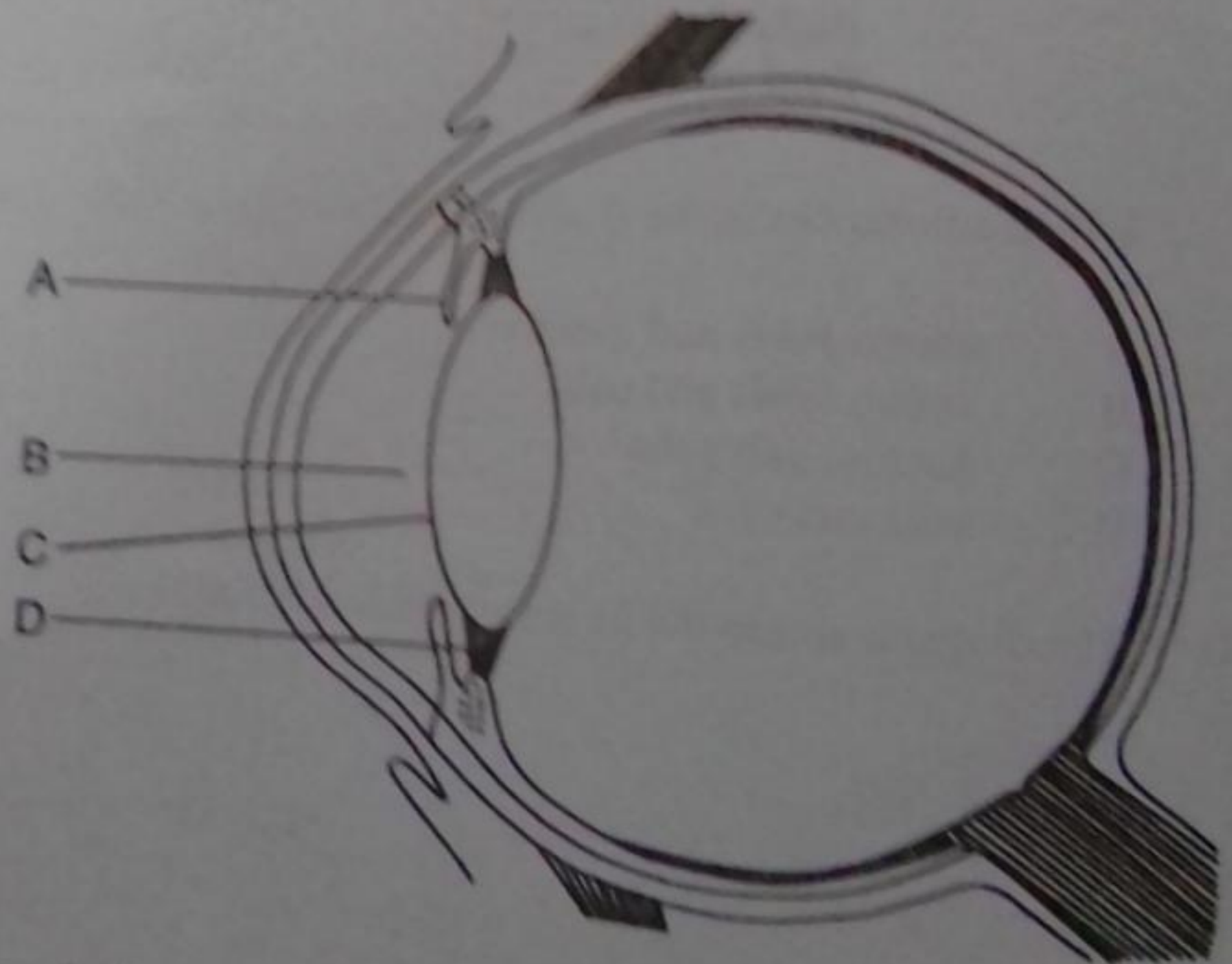
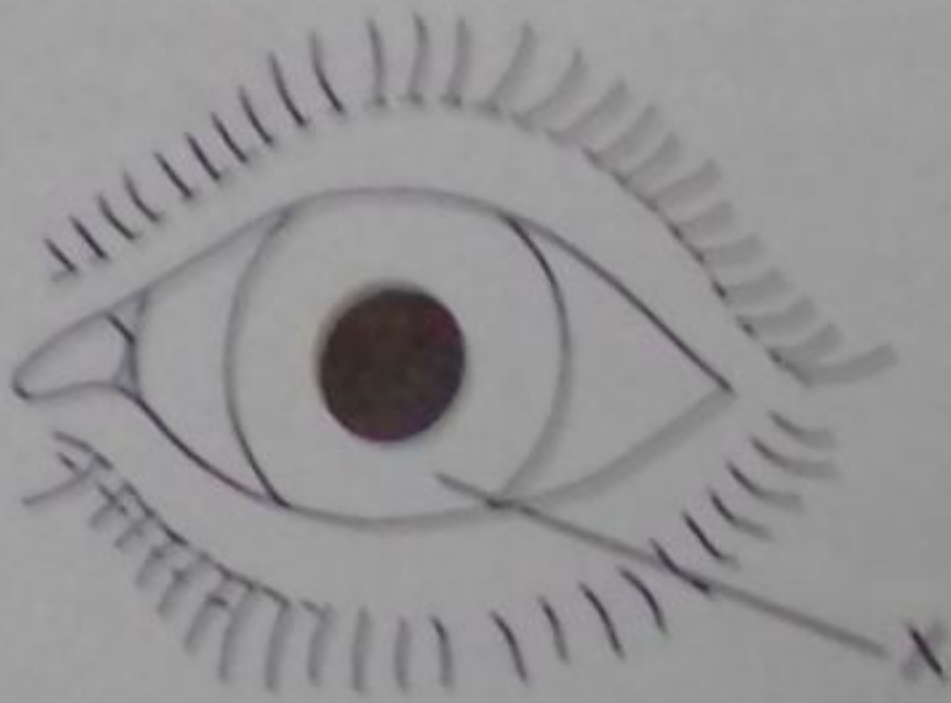
What is the effect of oestrogen on the pituitary?

- A thickening of pituitary
- B activates the pituitary
- C shrinking of pituitary
- D suppresses activity of pituitary

27 Which form of drug abuse exposes an individual to the highest risk of infection with the HIV (AIDS) virus?

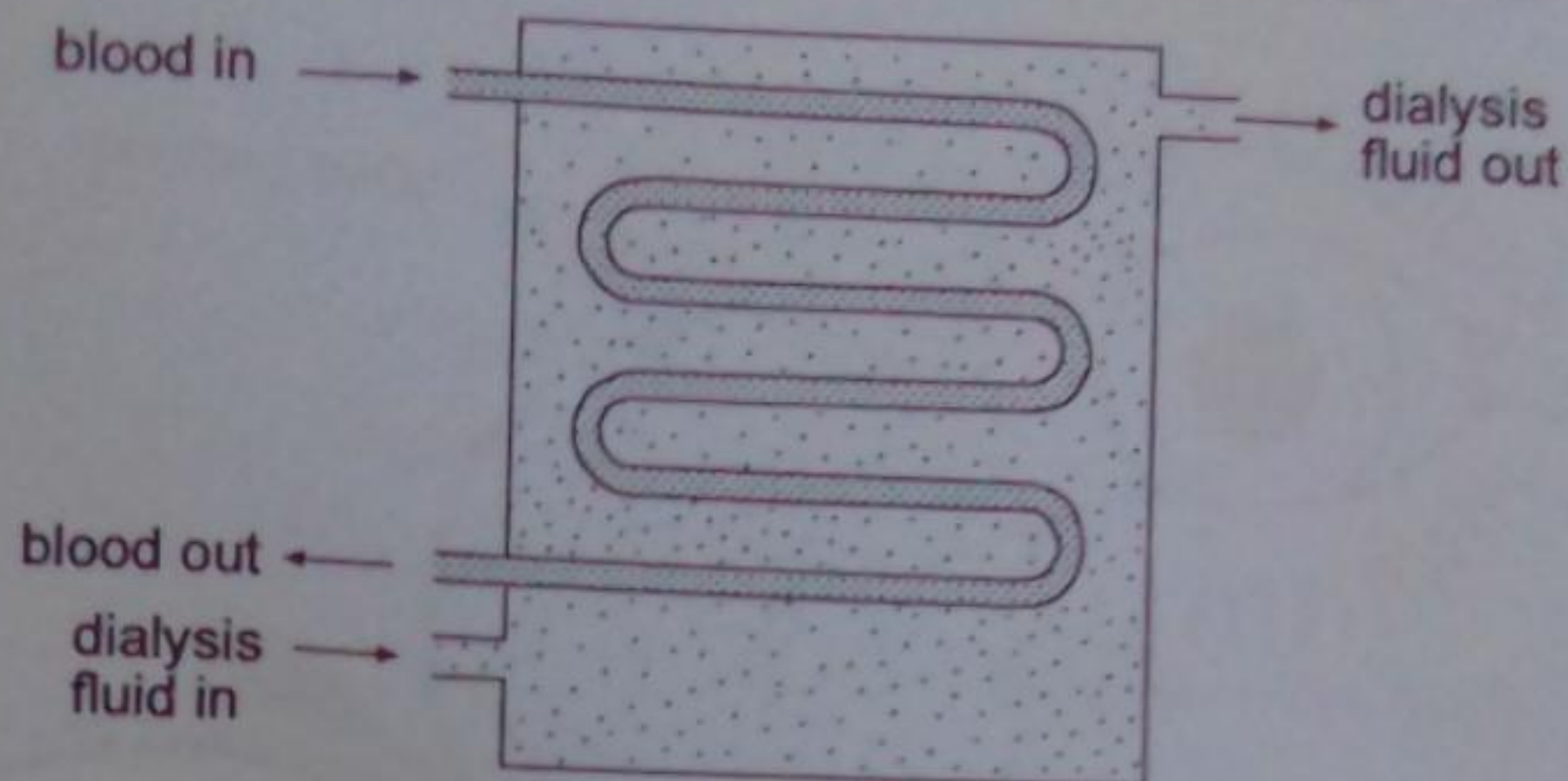
- A injection of heroin
- B drinking alcohol
- C cigarette-smoking
- D inhaling solvents

- 25 The diagrams show the front view and horizontal section of the eye respectively.



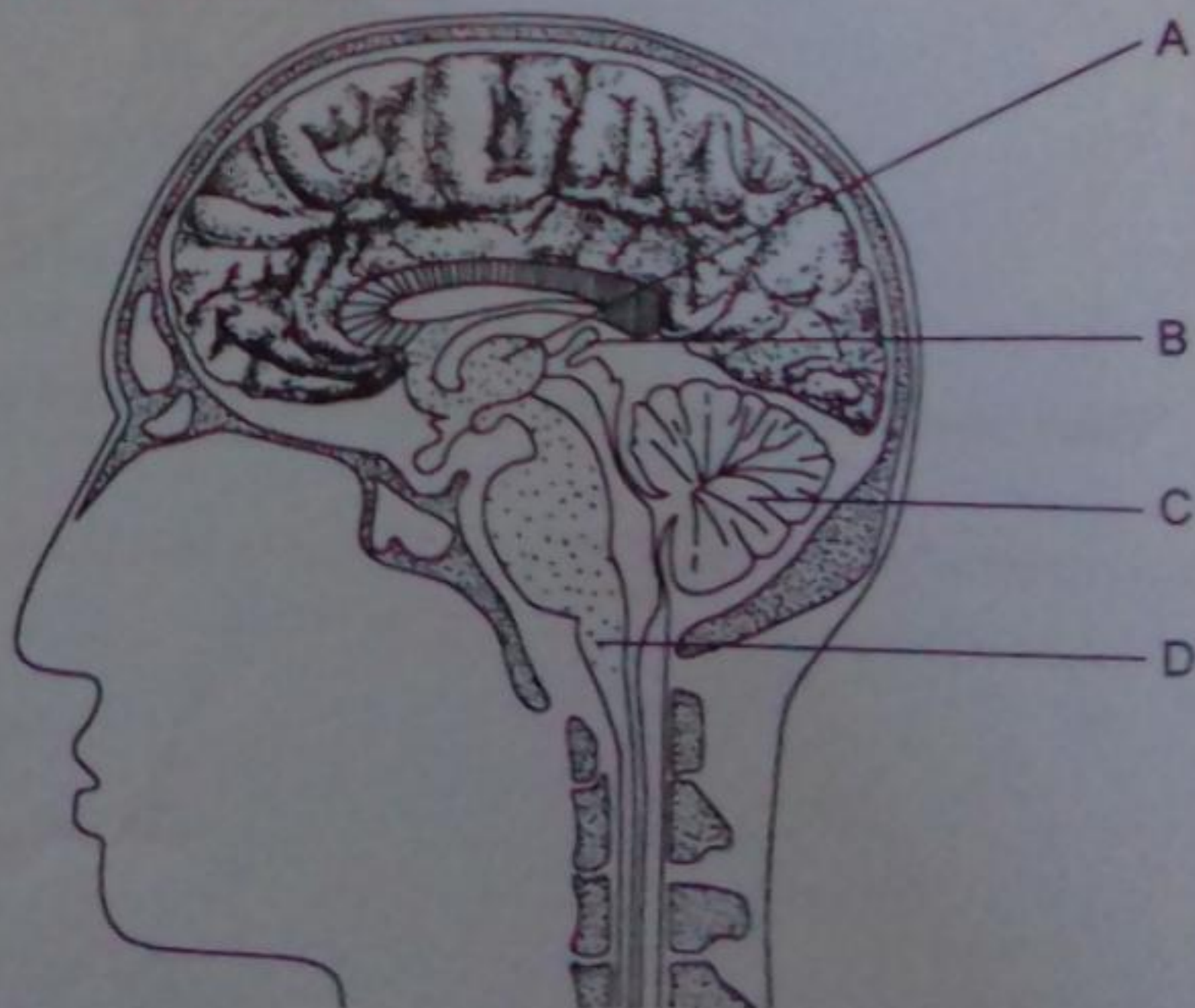
Which labelled structure on the horizontal section is the same as X in the front view?

- 23 The diagram represents a dialysis machine (kidney machine).



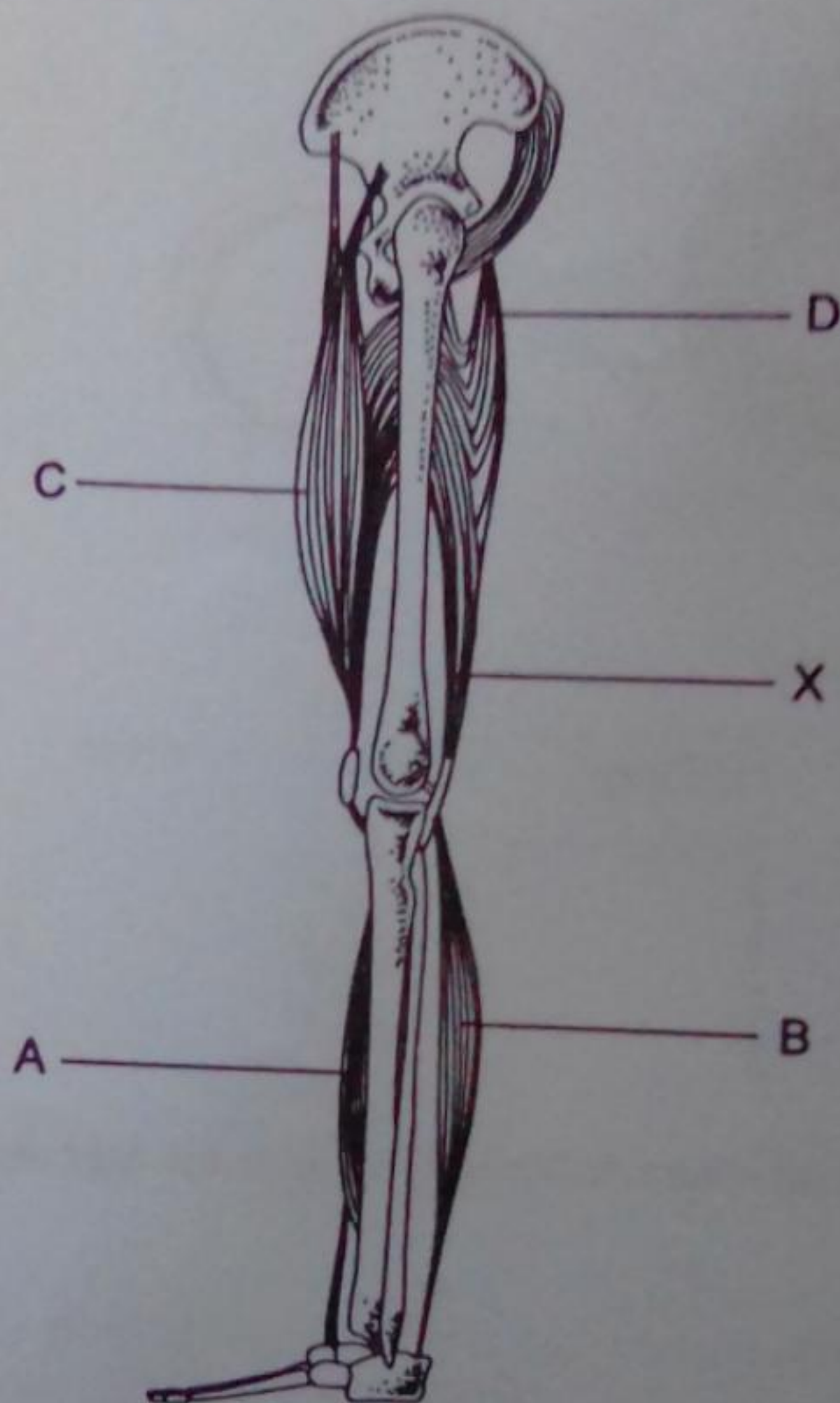
Which substances in the dialysis fluid should be at the same concentration as in the blood?

- A amino acids and glucose
 - B amino acids and salts
 - C glucose and salts
 - D salts and urea
- 24 The diagram shows the human skull and brain.



Which part is responsible for controlling balance and muscular co-ordination?

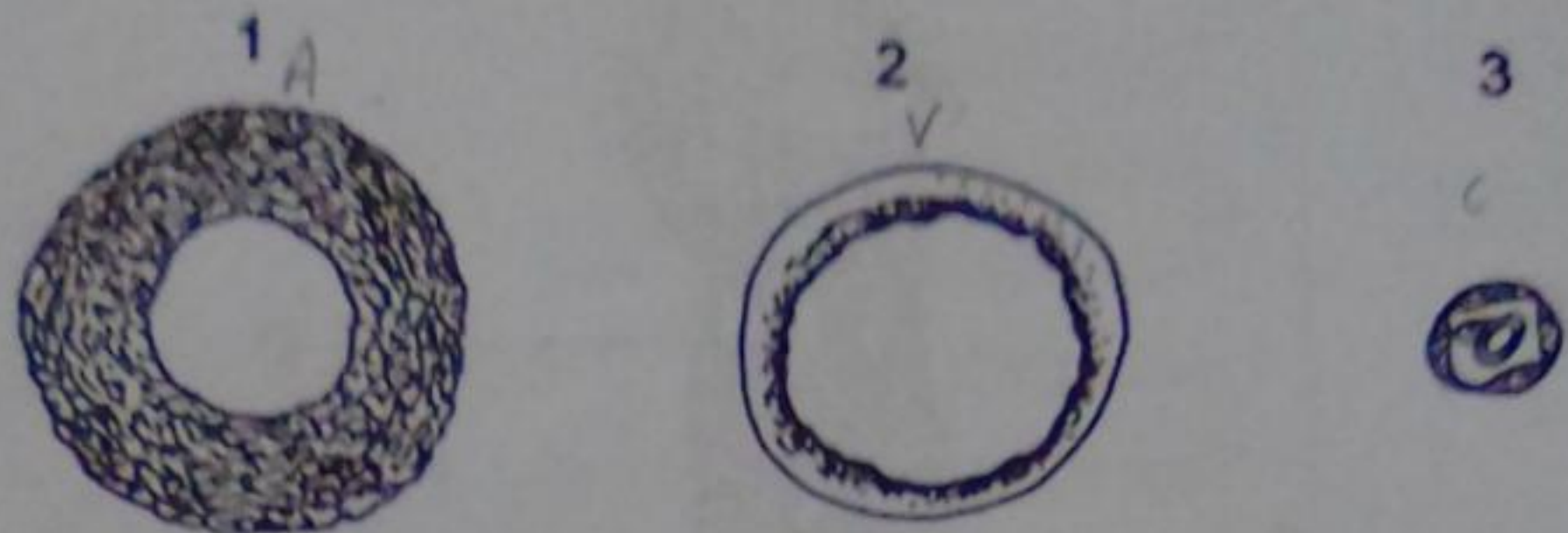
- 21 The diagram shows some of the muscles used in standing and walking.



Which muscle is antagonistic to muscle X?

- 2 Which of the following structures of the skin does **not** play a part in temperature regulation in a mammal?
- A hair erector muscle
 - B fat in the dermis
 - C malpighian layer
 - D nerve endings

- 18 The diagrams show sections of three types of blood vessels.



What are these vessels?

	artery	capillary	vein
A	1	2	3
B	2	3	1
C	3	2	1
D	1	3	2

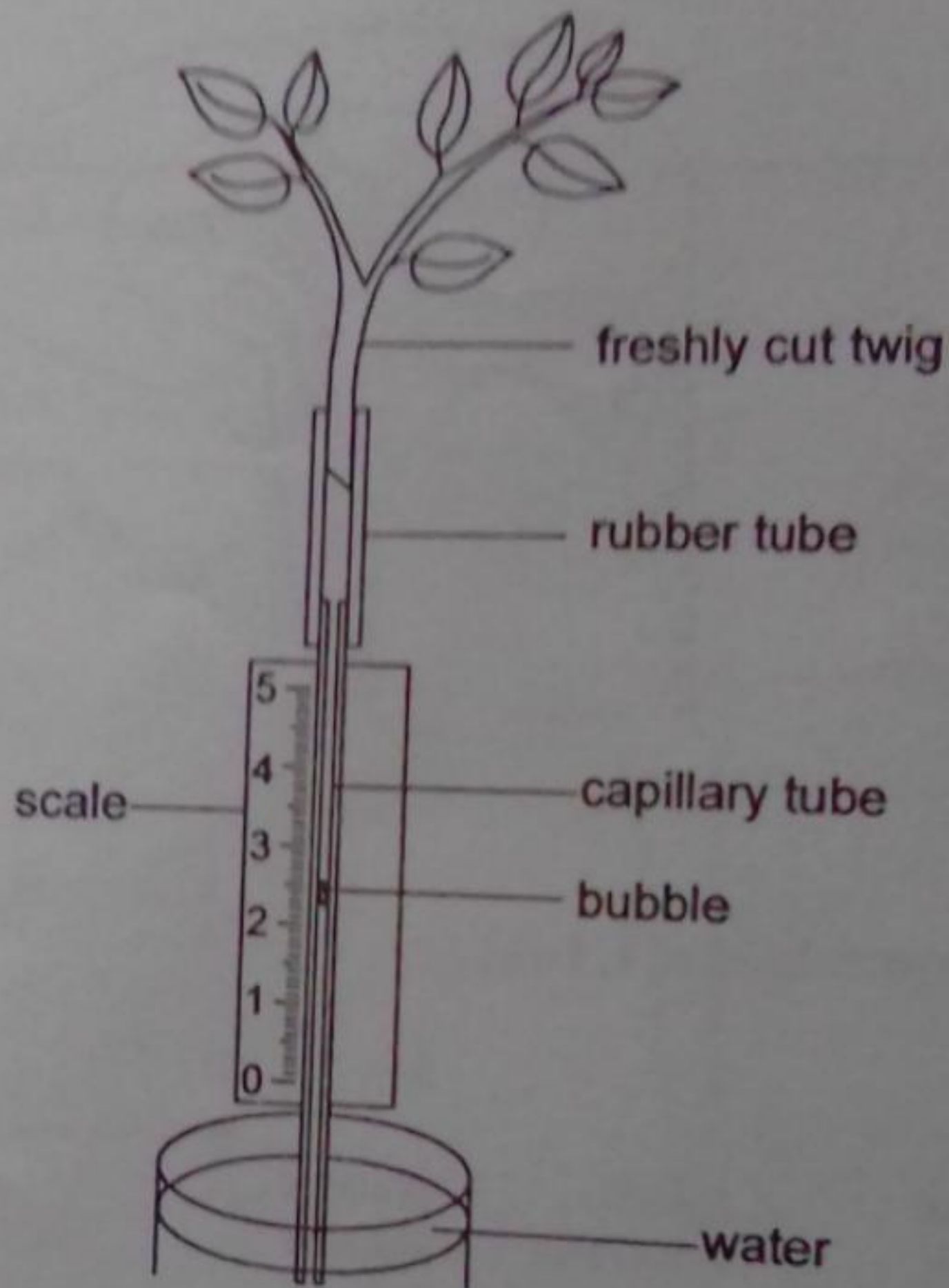
- 19 When the right atrium contracts, blood flows from the right atrium into the

- A aorta.
- B right ventricle.
- C left atrium.
- D pulmonary artery.

- 20 Which of the following components of blood can pass into tissue fluid?

- A protein
- B red blood cells
- C white blood cells
- D platelets

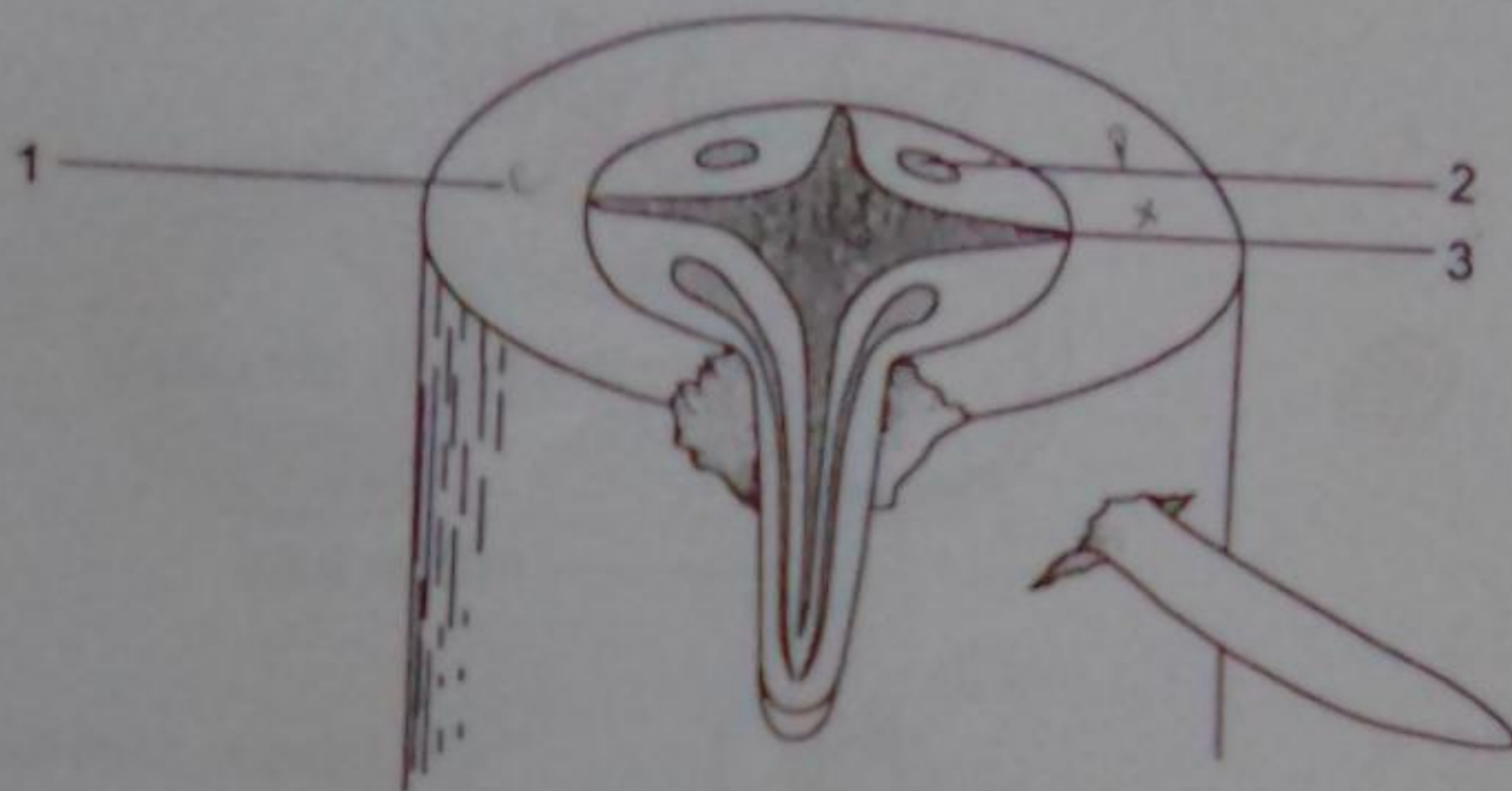
- 17 The diagram shows apparatus set up to investigate transpiration rate.



It is necessary to keep the twig stem in water from the moment it is cut from the plant when setting up the apparatus. Why is this necessary?

- A to prevent air entering its xylem vessels
- B to keep the twig alive
- C to prevent wilting
- D to introduce an air bubble into the capillary tube

- 15 The diagram shows the internal structure of a dicotyledonous root.



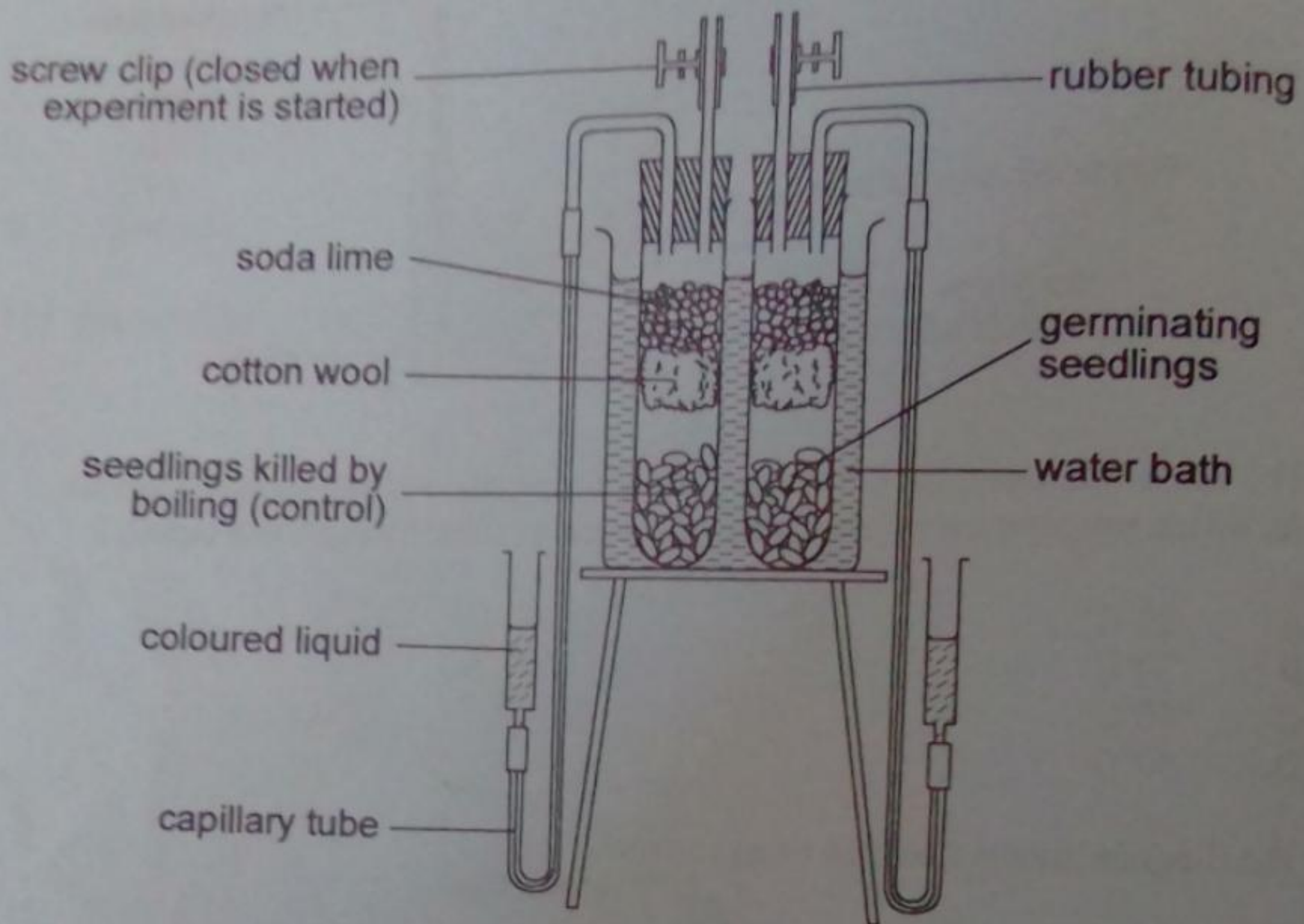
What are the parts labelled 1, 2 and 3?

	1	2	3
A	cortex	xylem	phloem
B	cortex	epidermis	phloem
C	cortex	phloem	xylem
D	epidermis	cortex	xylem

- 16 Water enters the vacuole of the root hair cell by

- A active transport.
- B diffusion.
- C osmosis.
- D transpiration.

- 13 The diagram shows an experiment to see if oxygen is taken up in respiration.



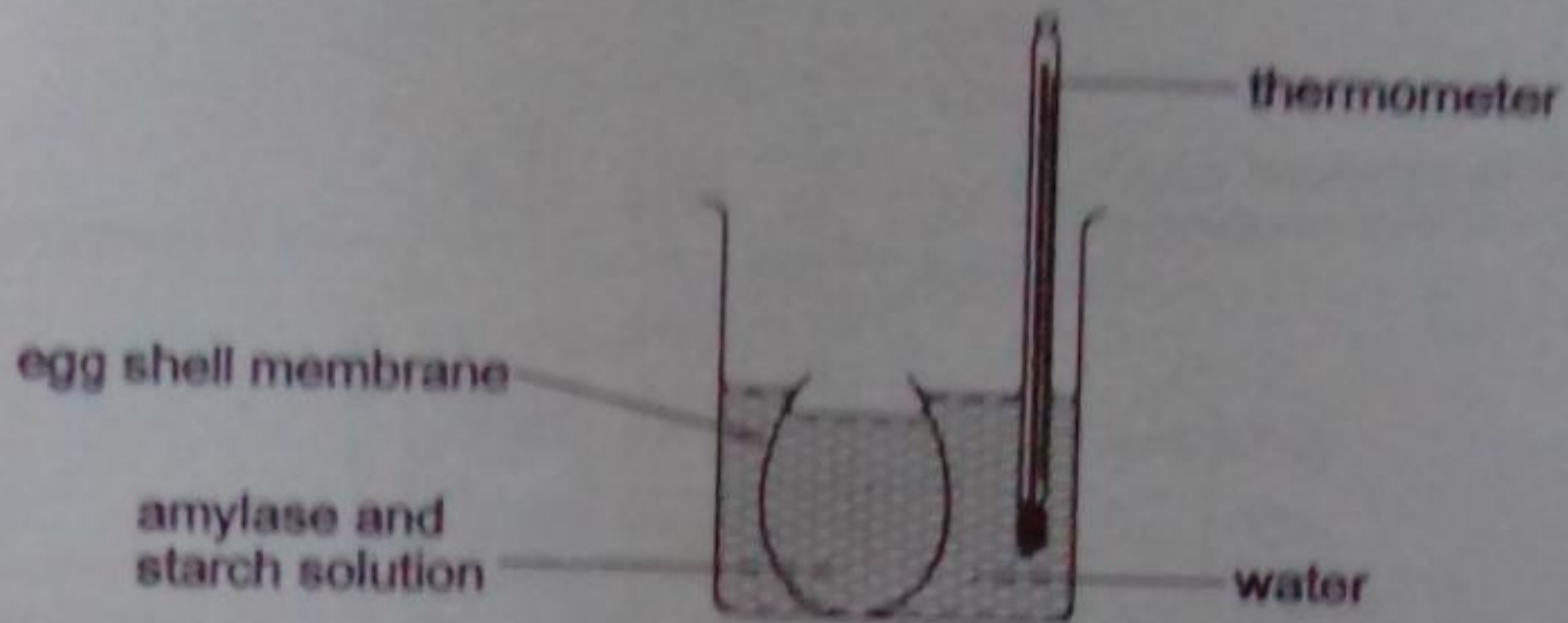
If soda lime is excluded in the set up, the level of the coloured liquid remains stationary because

- A the seeds will not get oxygen.
- B oxygen used in respiration is replaced by carbon dioxide.
- C concentration of carbon dioxide decreases.
- D seeds are killed by micro-organisms.

- 14 Which are products of anaerobic respiration in plants?

- A carbon dioxide and alcohol
- B carbon dioxide and lactic acid
- C alcohol and lactic acid
- D alcohol and water

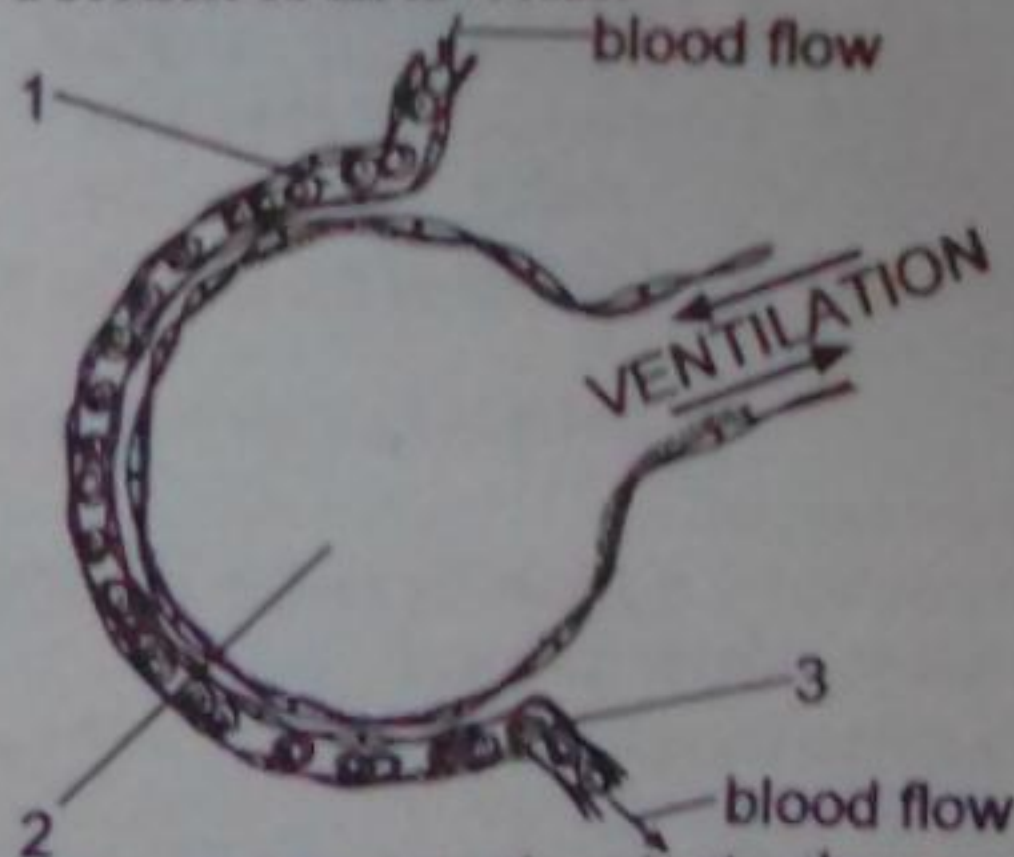
- 11 The diagram shows an experiment to investigate effect of temperature on enzyme activity.



The water was tested for the presence of reducing sugar at half-minute intervals. At which temperature would the concentration of reducing sugar rise rapidly?

- A 20°C
- B 30°C
- C 40°C
- D 50°C

- 12 The diagram shows a section of an alveolus.



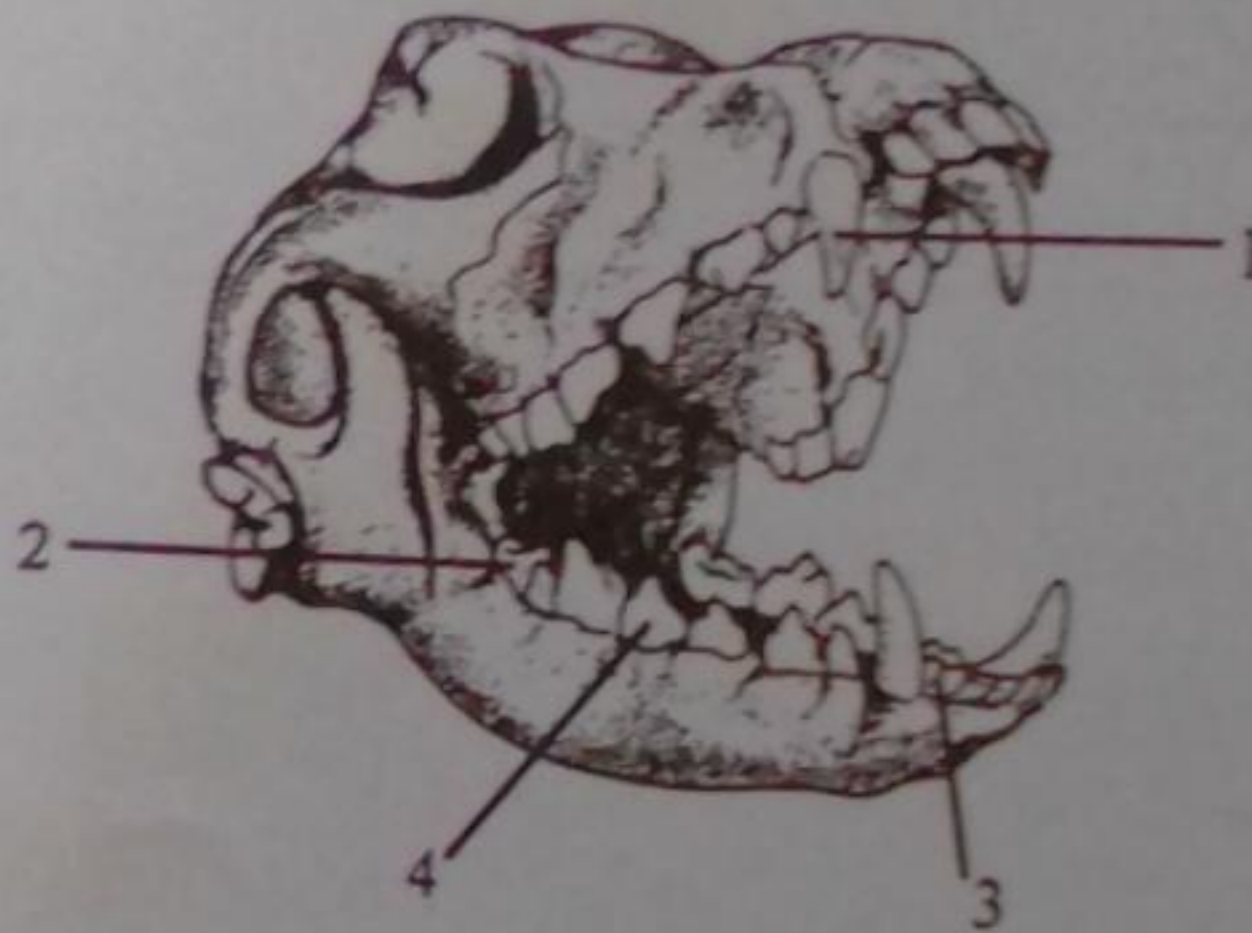
What are the relative oxygen concentrations in the three parts labelled 1, 2 and 3?

- | | 1 | 2 | 3 |
|---|------|------|------|
| A | low | low | high |
| B | low | high | low |
| C | high | low | low |
| D | low | high | high |

9 Where is insulin produced and where does it act?

	site of production	site of action
A	adrenal glands	body muscles
B	adrenal glands	liver
C	liver	pancreas
D	pancreas	liver

10 The diagram shows a skull and teeth of a dog.



What types of teeth are 1, 2, 3 and 4?

	1	2	3	4
A	canine	molar	incisor	premolar
B	canine	incisor	molar	premolar
C	canine	incisor	premolar	molar
D	incisor	canine	molar	premolar

The table below shows the chemical elements present in each of four substances.

Key

✓ = present X = absent

substance	carbon	hydrogen	nitrogen	oxygen
A	✓	✓	✓	X
B	✓	✓	X	✓
C	✓	✓	✓	✓
D	✓	X	✓	✓

Which substance could be protein?

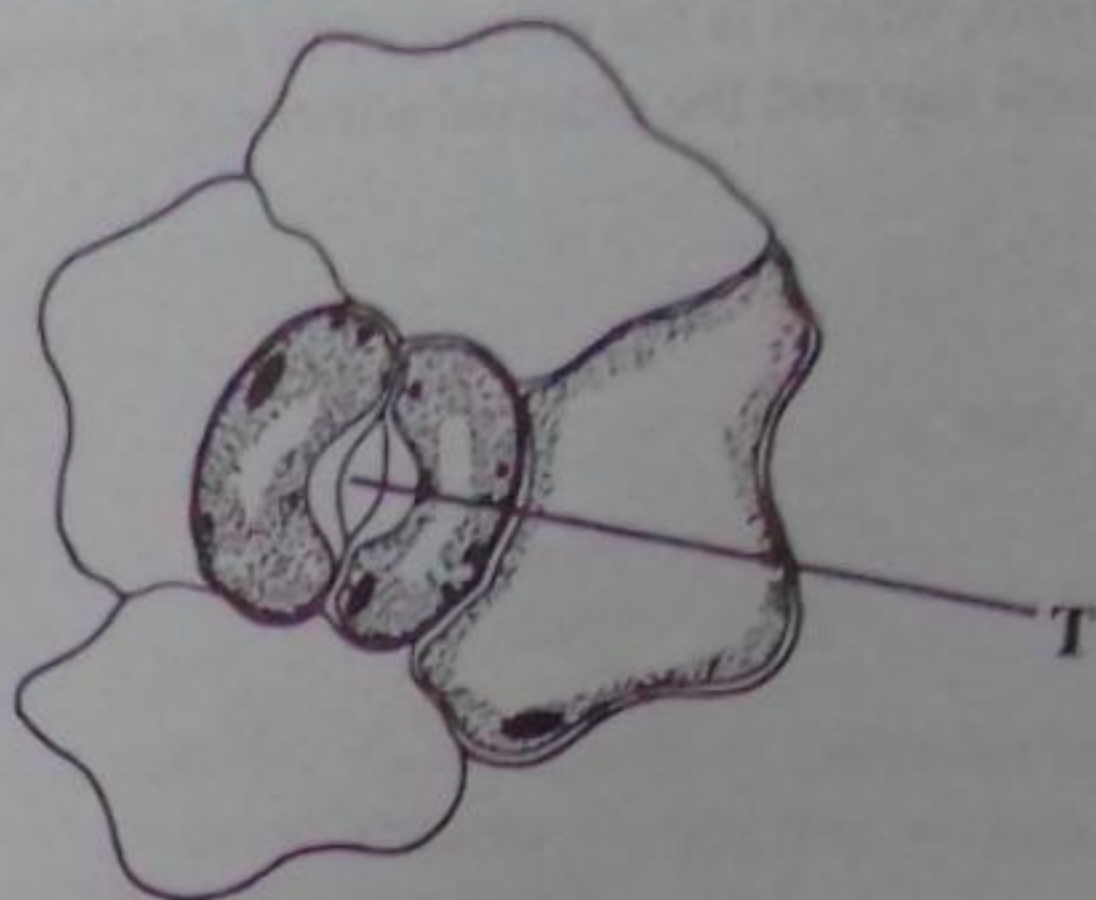
8 The photograph shows a person suffering from a deficiency disease.



The disease is due to deficiency of

- A vitamin C.
- B iodine.
- C iron.
- D calcium.

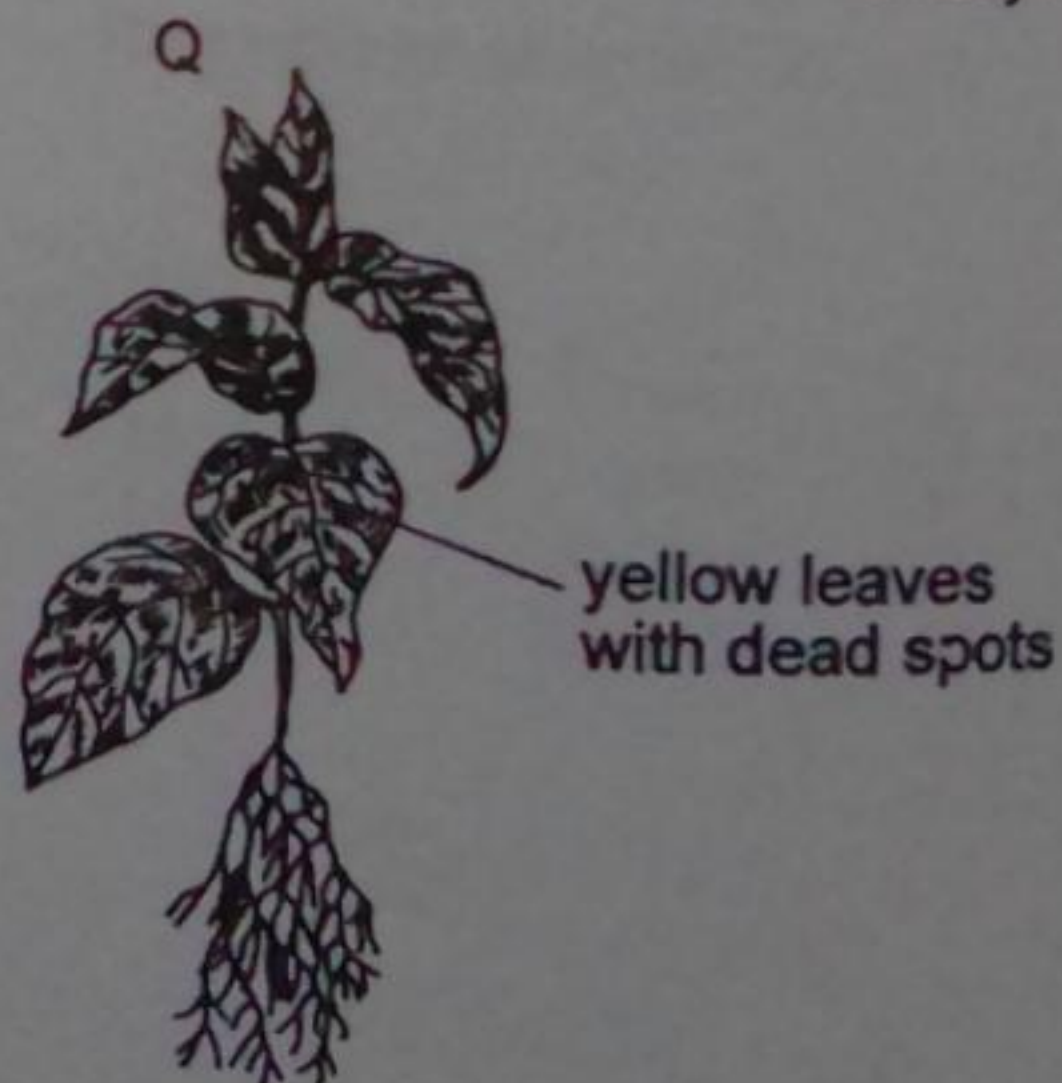
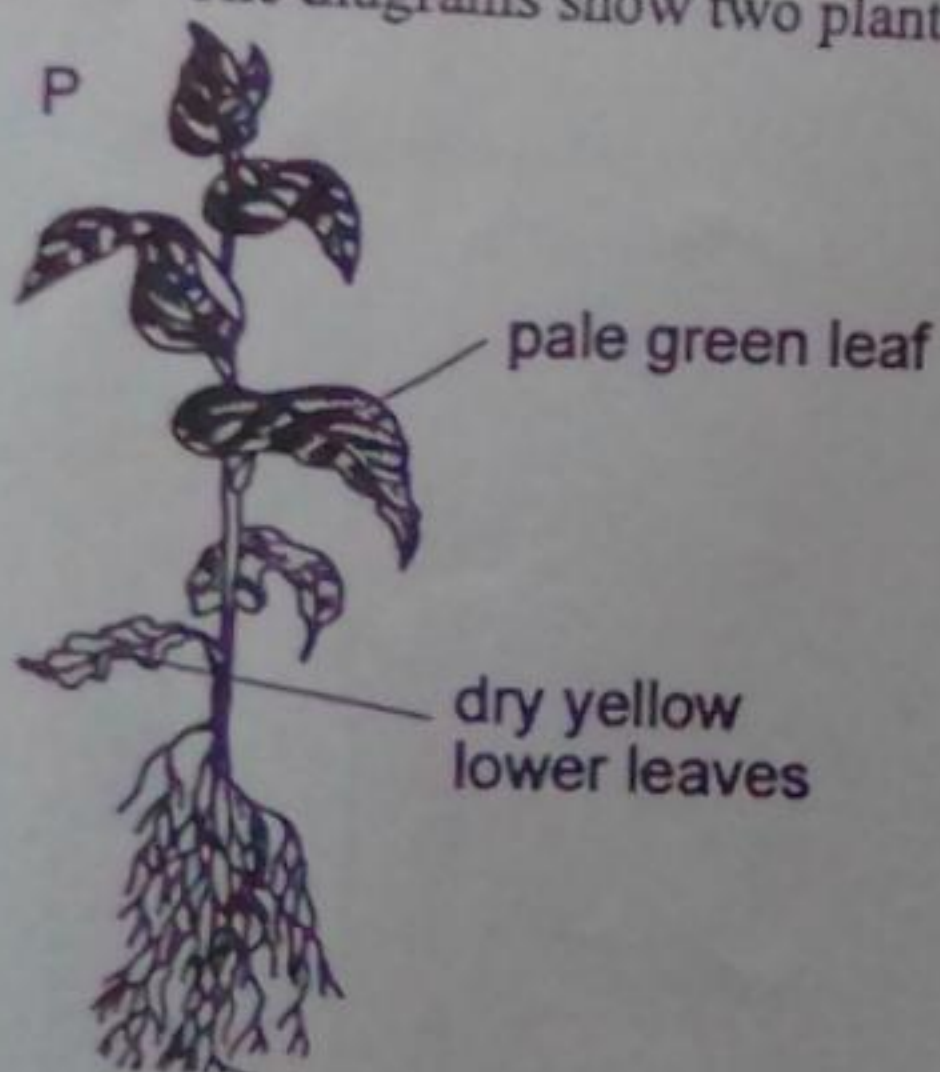
- 5 The diagram shows part of the surface view of a dicotyledonous leaf.



What is T?

- A vascular bundle
- B guard cell
- C epidermal cell
- D stoma

- 6 The diagrams show two plants, P and Q, with symptoms of some mineral deficiency.



What mineral is lacking in plant P and Q?

- | | P | Q |
|---|-----------|------------|
| A | nitrogen | potassium |
| B | nitrogen | phosphorus |
| C | potassium | phosphorus |
| D | nitrogen | nitrogen |

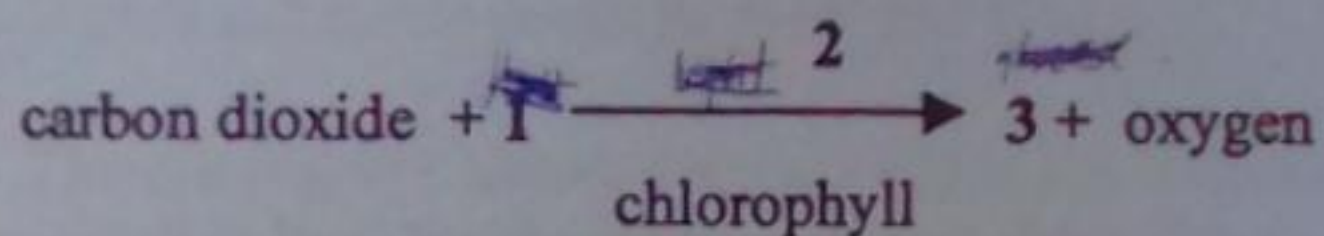
- 3 During plasmolysis, which is the correct order of concentration of water molecules in the cytoplasm, cell sap and the external solution?

Key

> greater than

= equal to

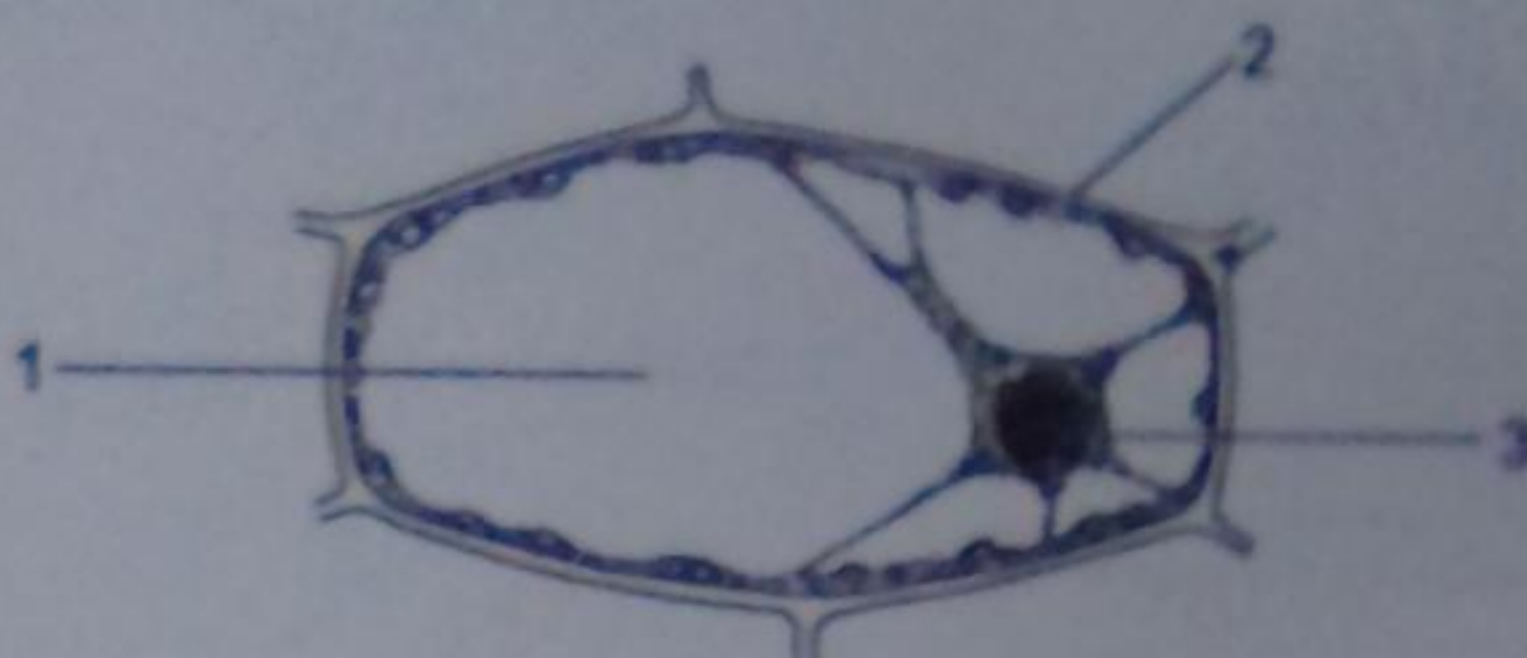
- A external solution > cytoplasm > cell sap
 B cytoplasm = cell sap > external solution
 C cell sap > cytoplasm > external solution
 D cell sap > cytoplasm = external solution
- 4 The following equation for photosynthesis is incomplete.



What do the numbers represent?

	1	2	3
A	water	glucose	sunlight
B	water	sunlight	sugar
C	oxygen	sunlight	sugar
D	sunlight	glucose	water

- 1 The diagram below shows a plant cell.



Which structures are made of cellulose?

- A 1 and 2
- B 3 only
- C 2 and 3
- D 2 only

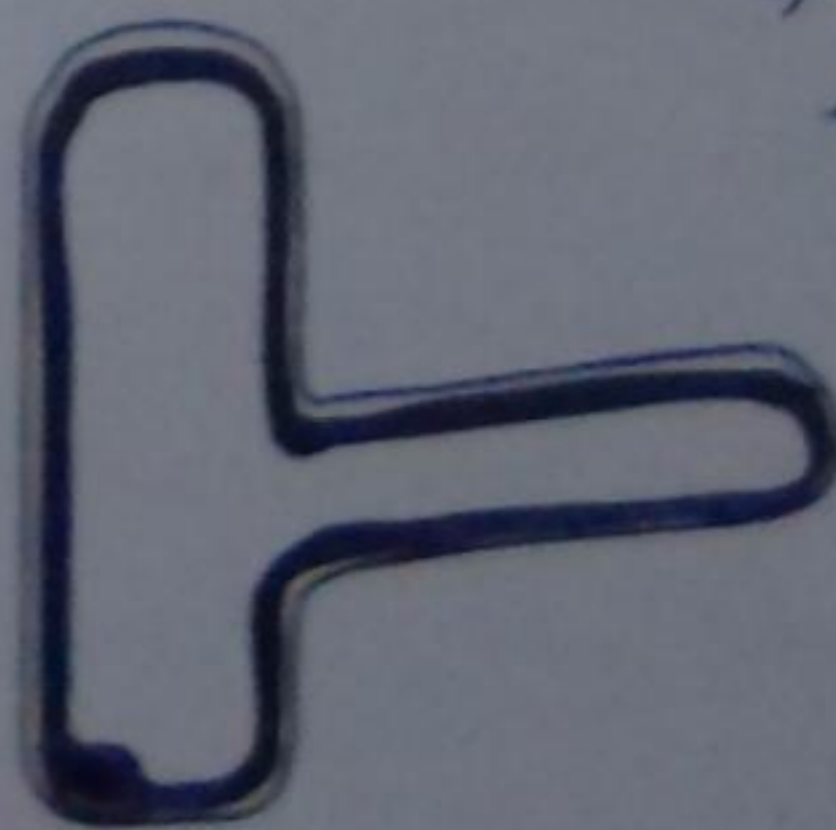
- 2 The diagrams below show four different types of cells.



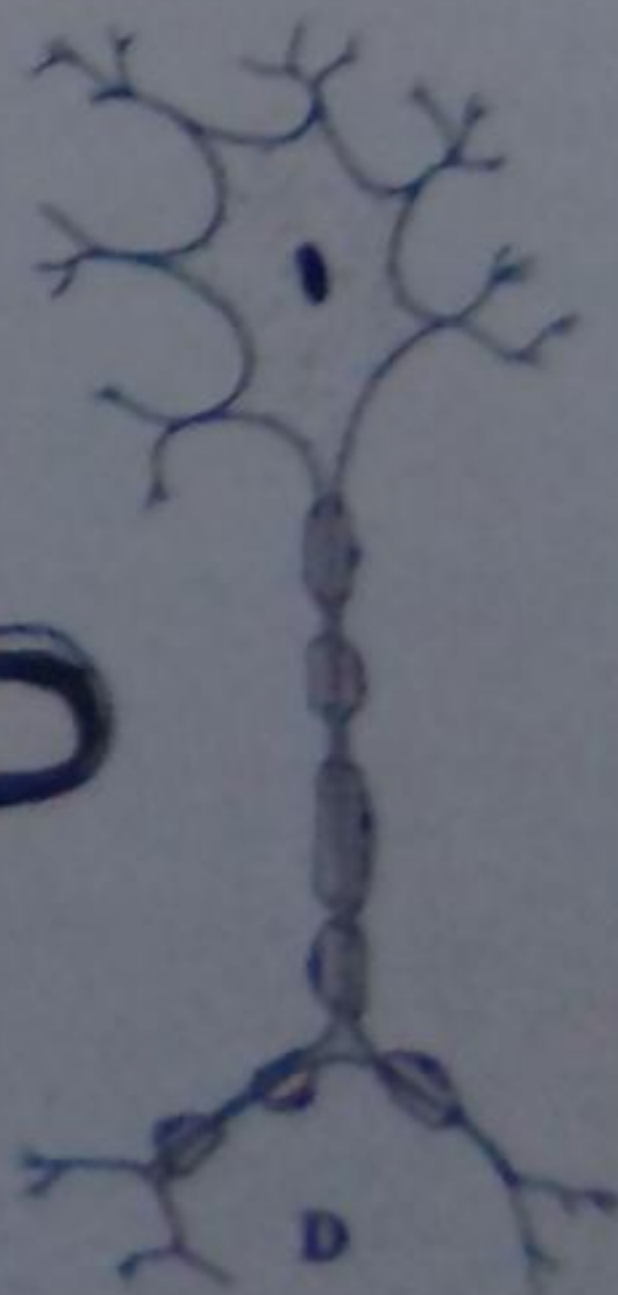
A



B



C



D

Which cell functions by changing shape?

00019974

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

BIOLOGY

5008/1

PAPER 1 Multiple Choice

Wednesday

19 NOVEMBER 2003

Morning

1 hour

Additional materials:

Multiple choice answer sheet

Soft clean eraser

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C and D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

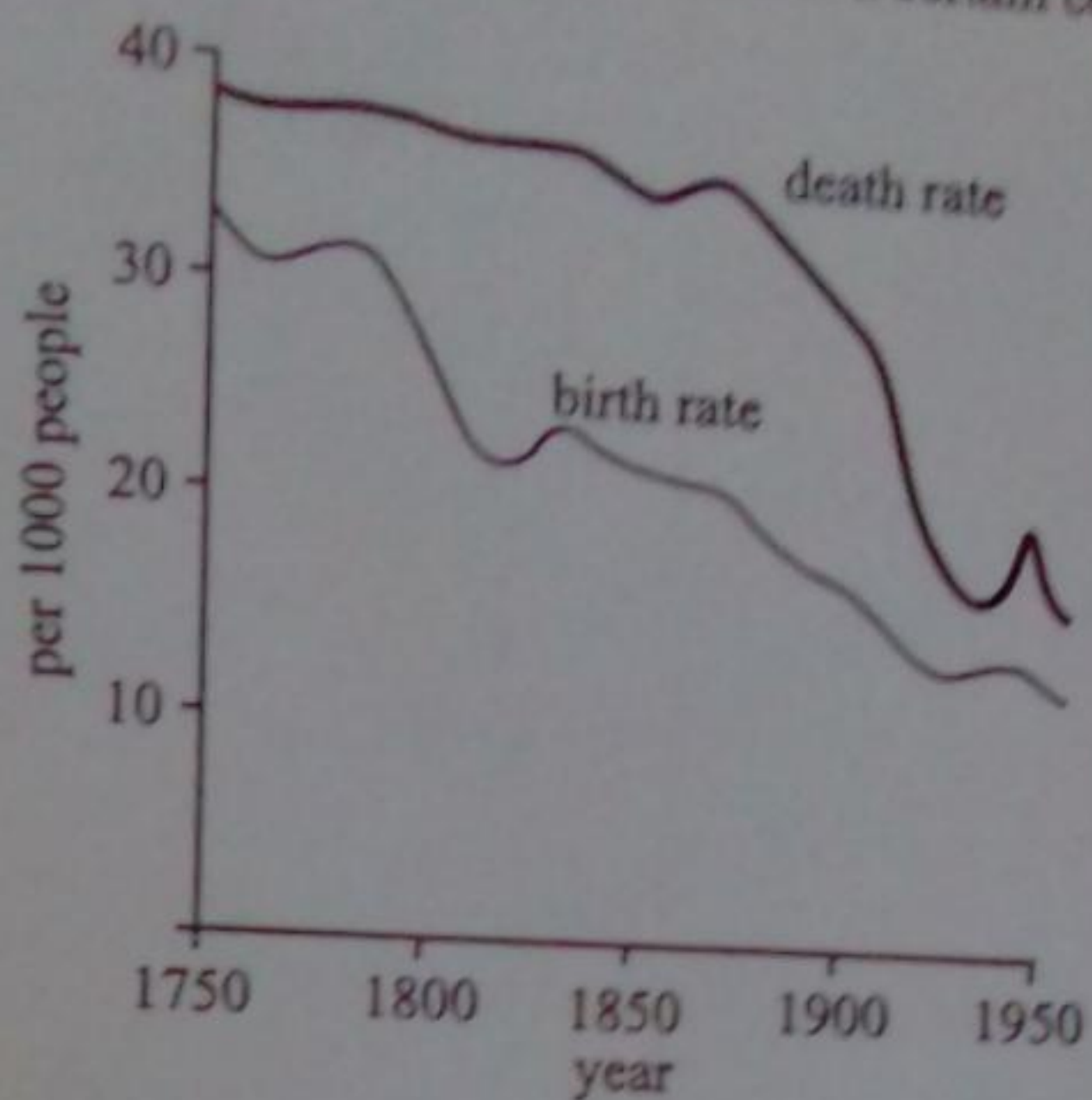
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 20 printed pages.

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- The graph below shows the death rate and birth rate of a certain country.



Which graph shows the resultant change in population?

